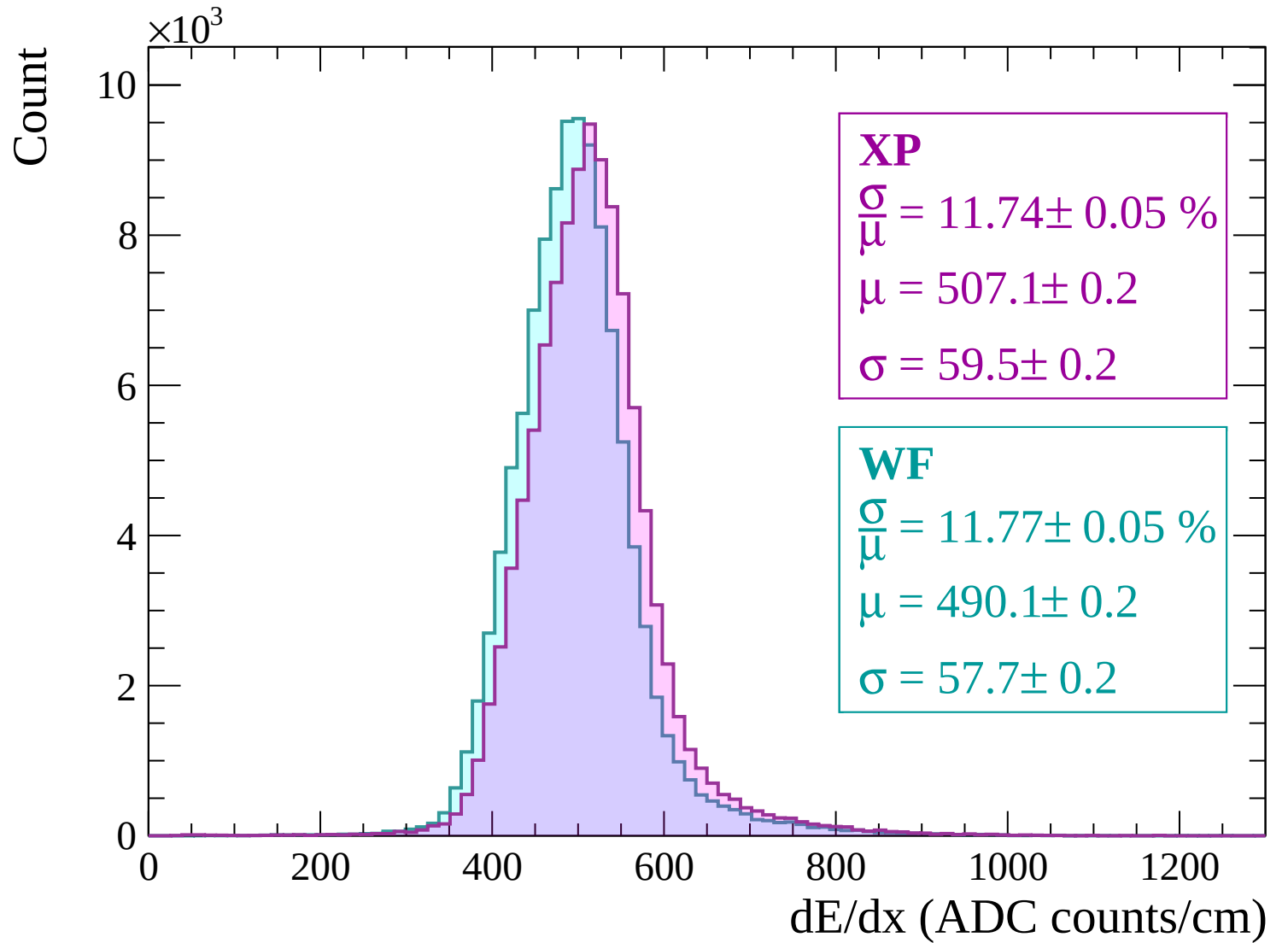
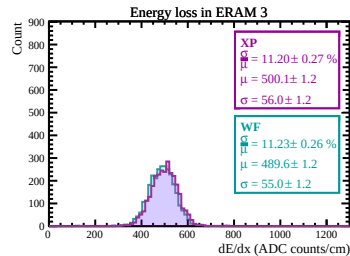
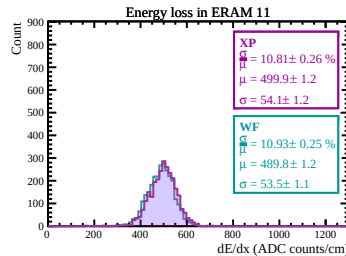
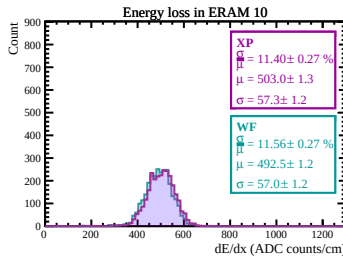
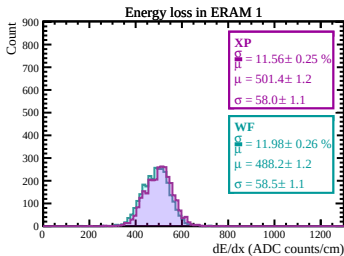
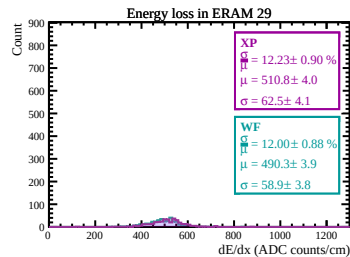
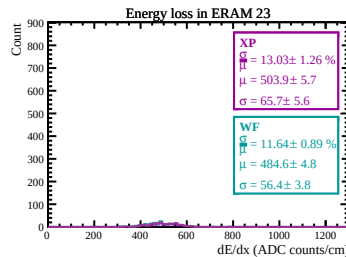
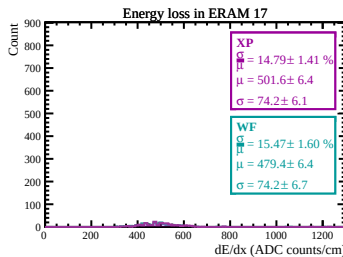
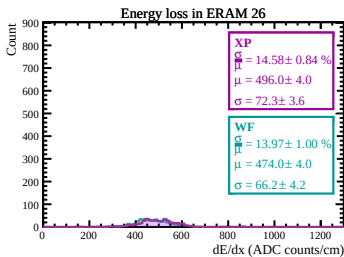
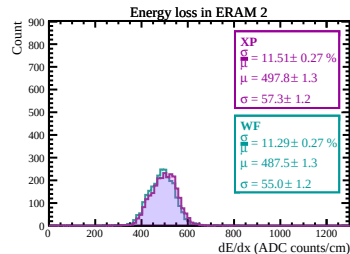
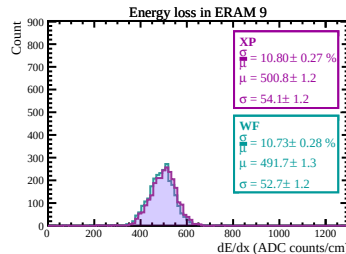
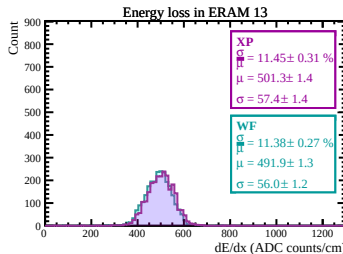
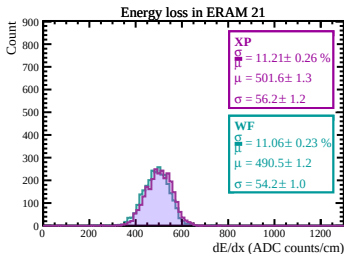
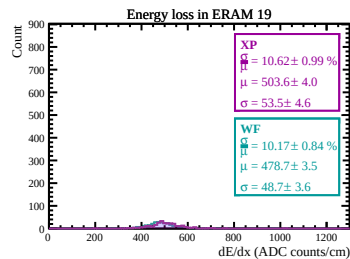
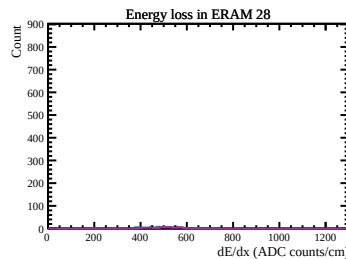
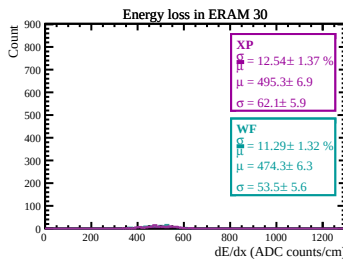
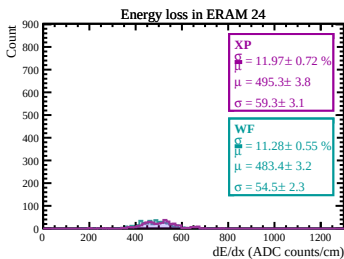
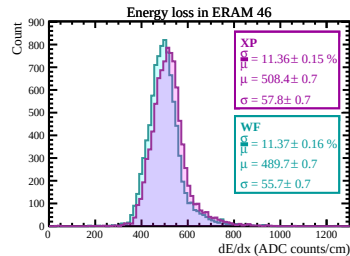
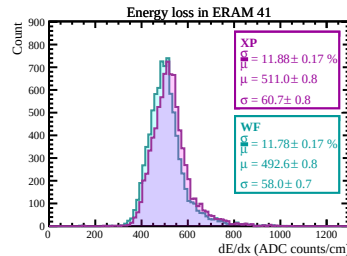
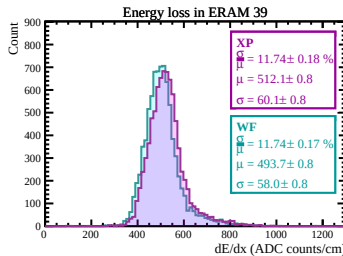
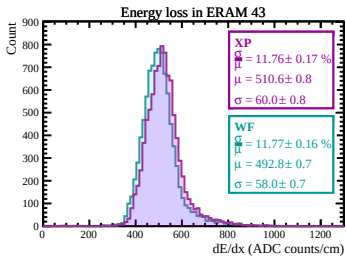
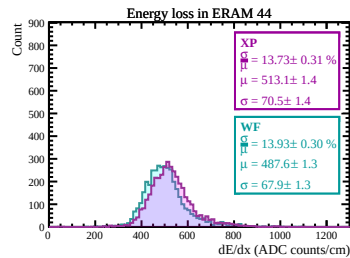
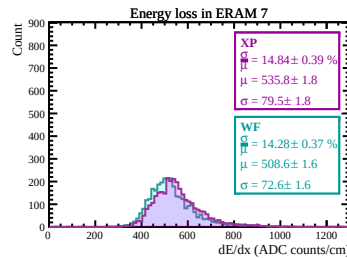
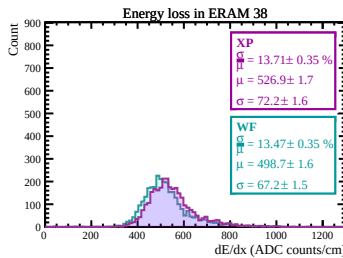
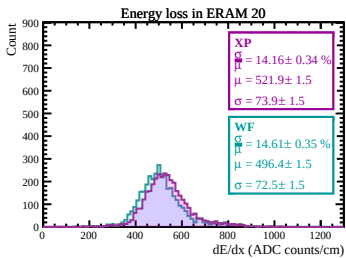
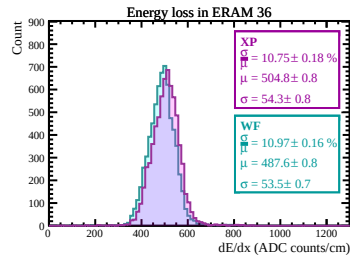
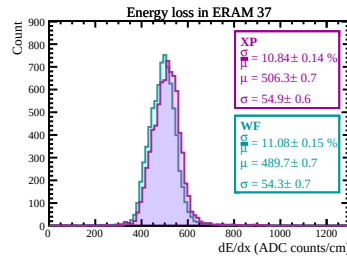
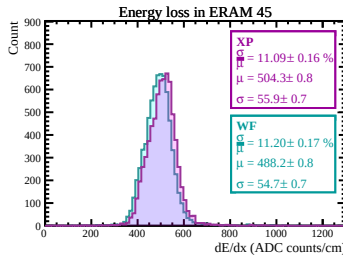
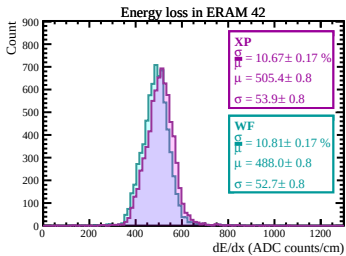
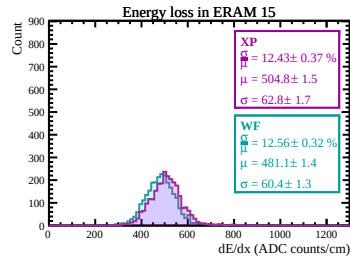
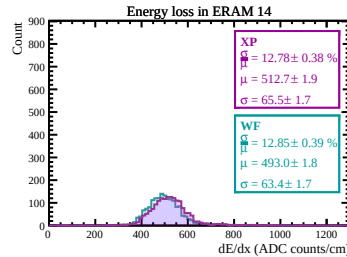
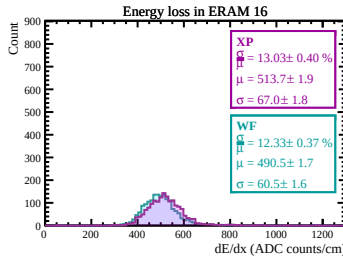
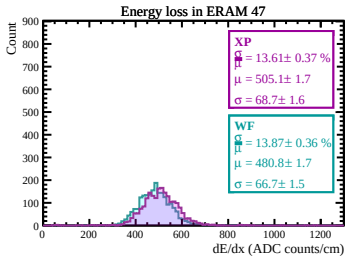
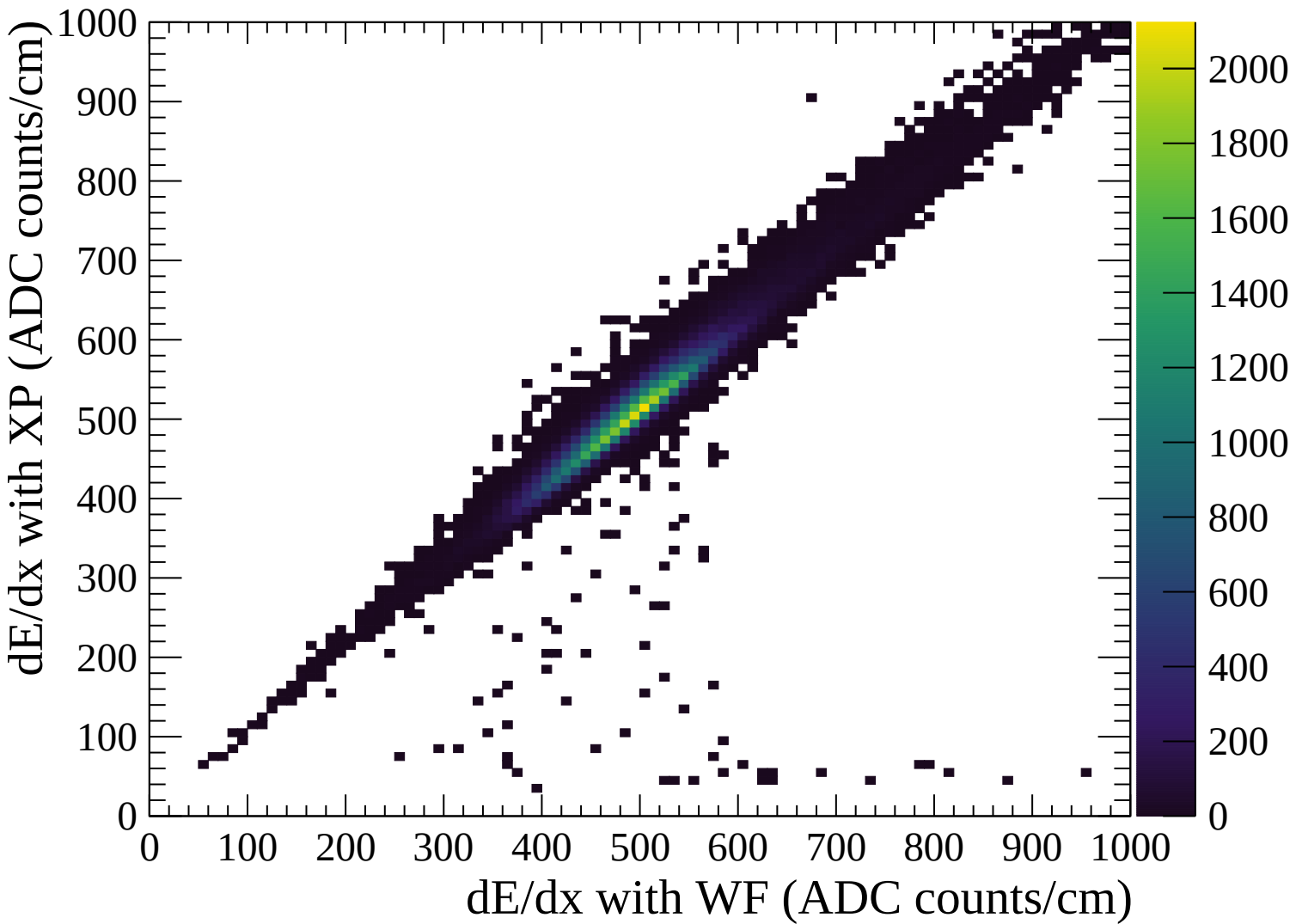
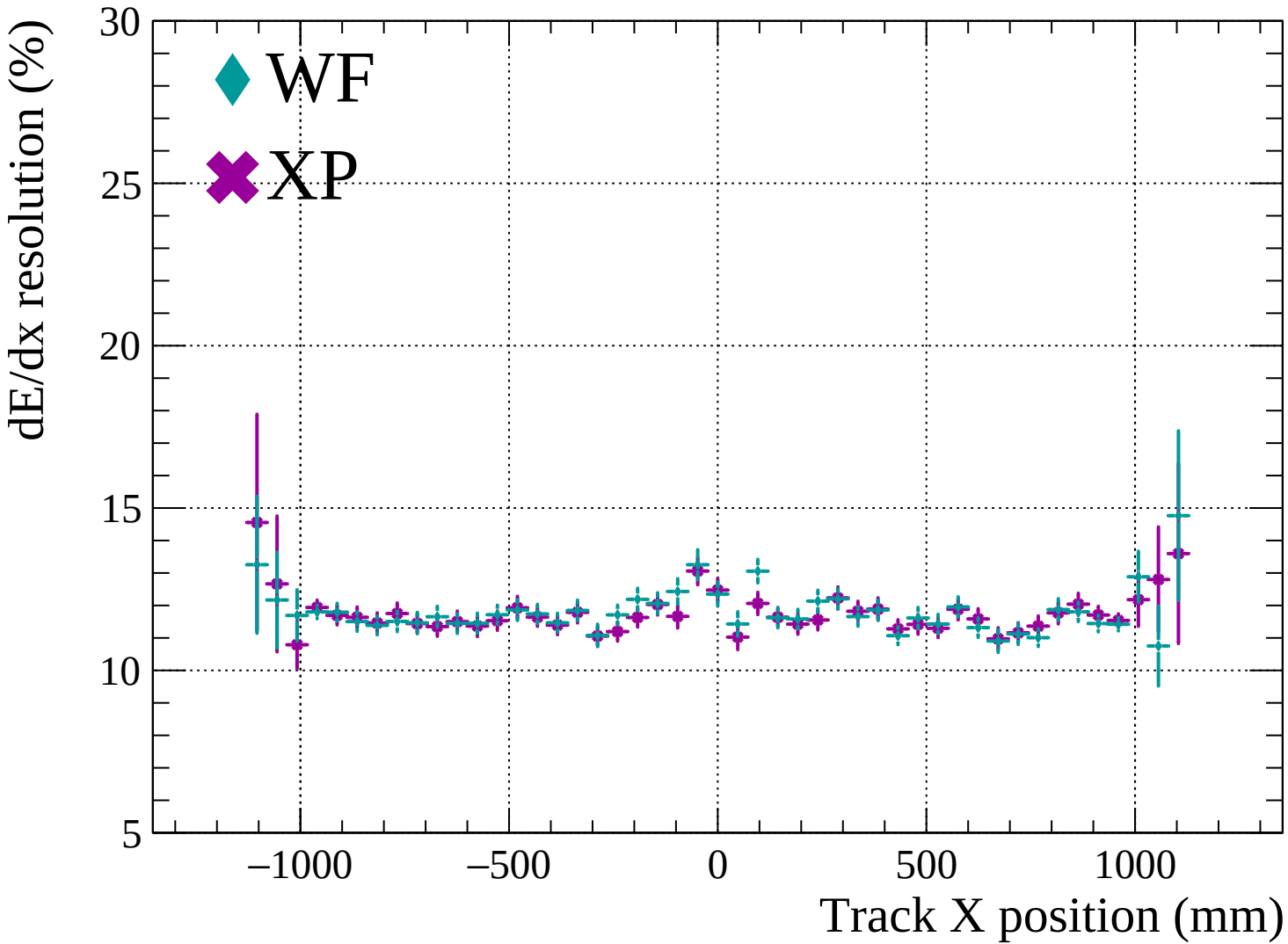


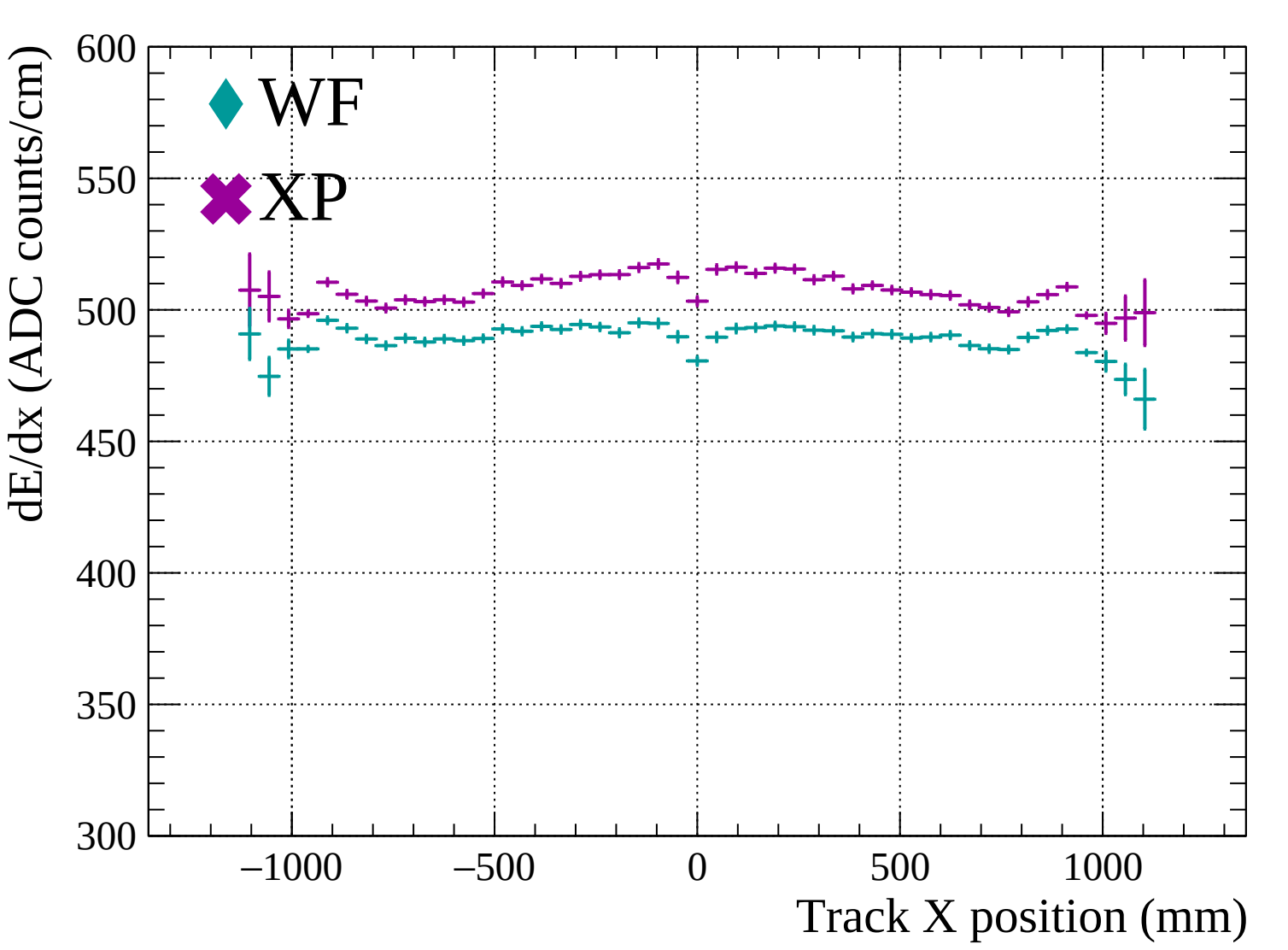


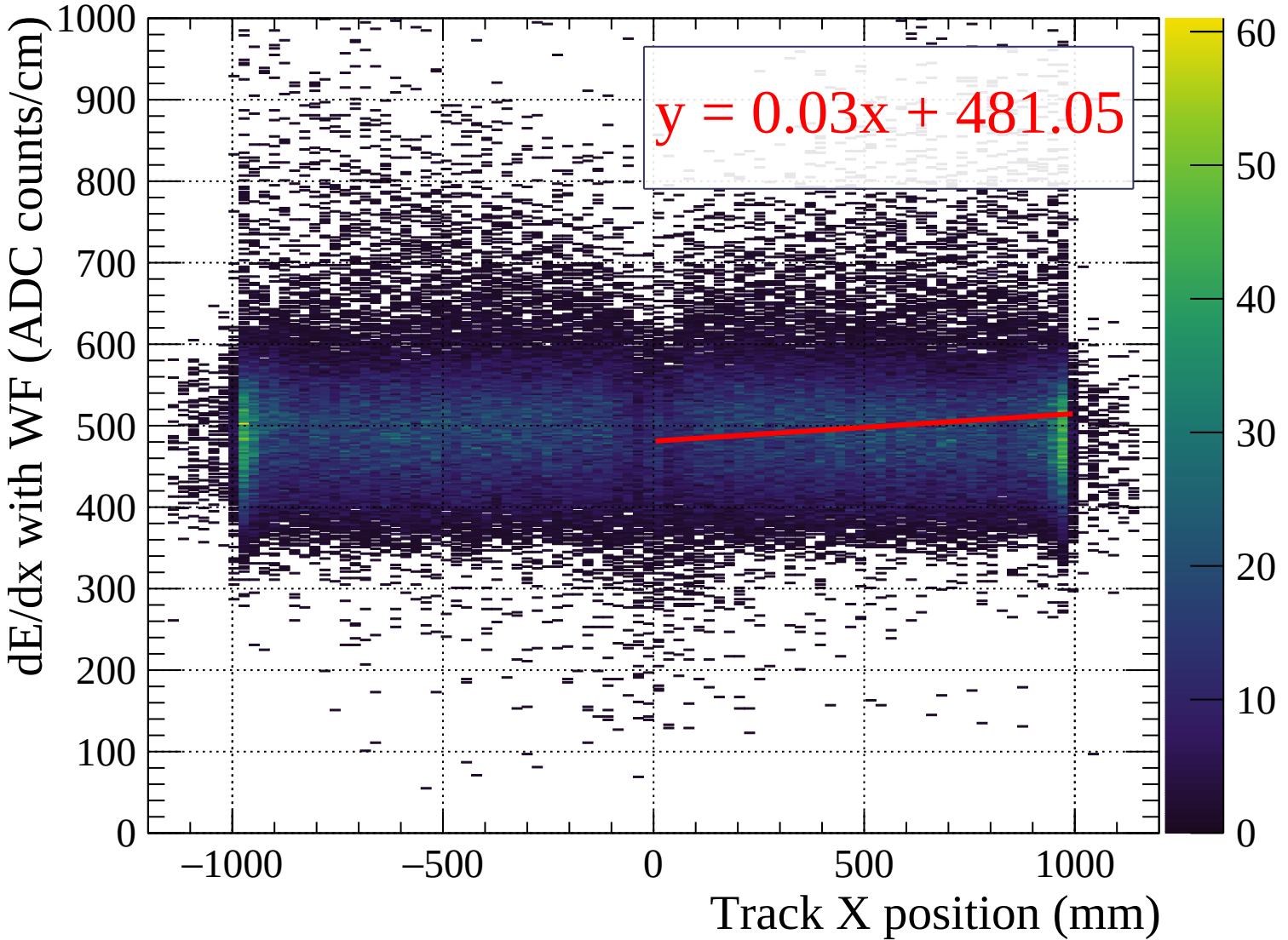


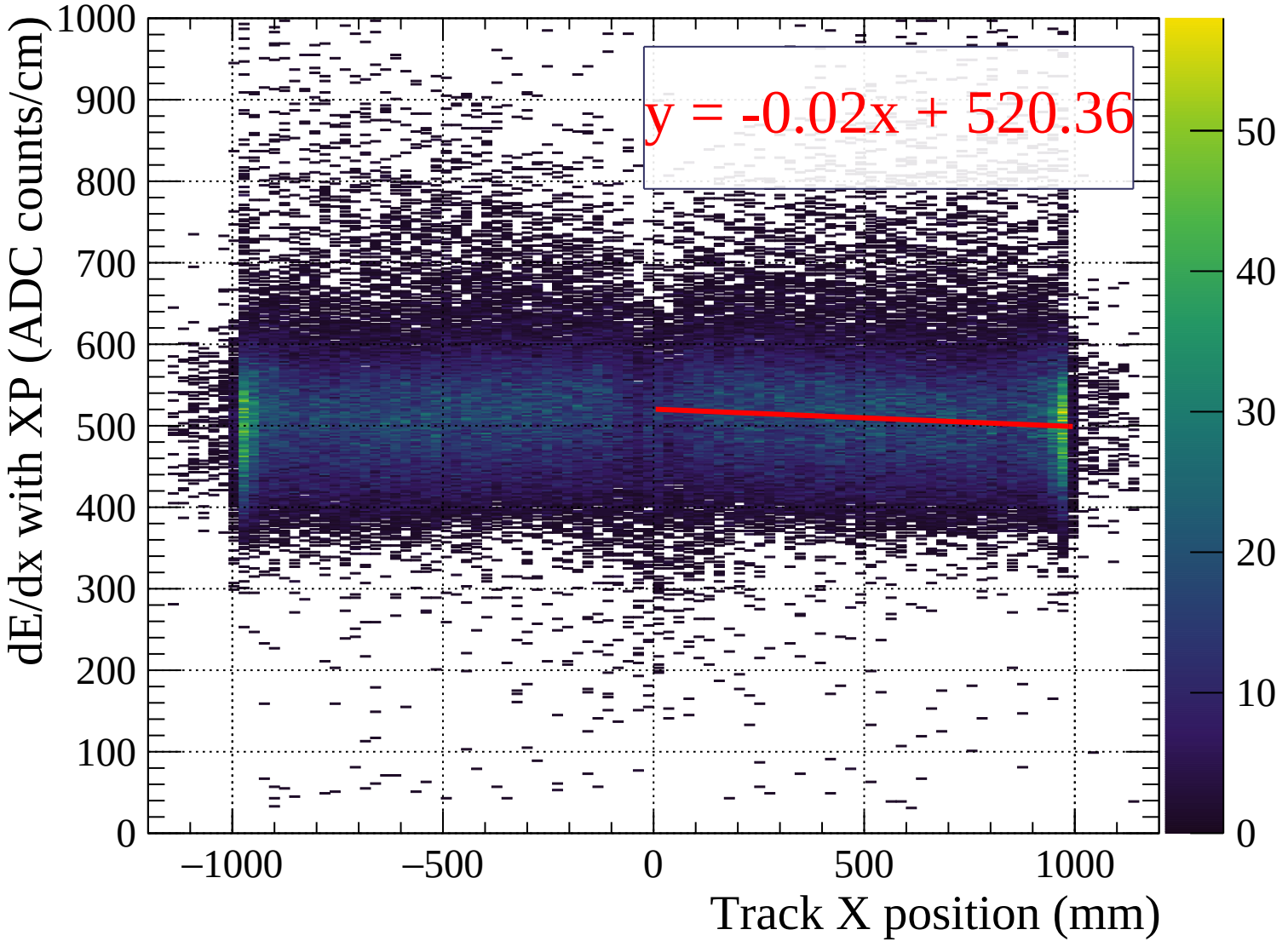


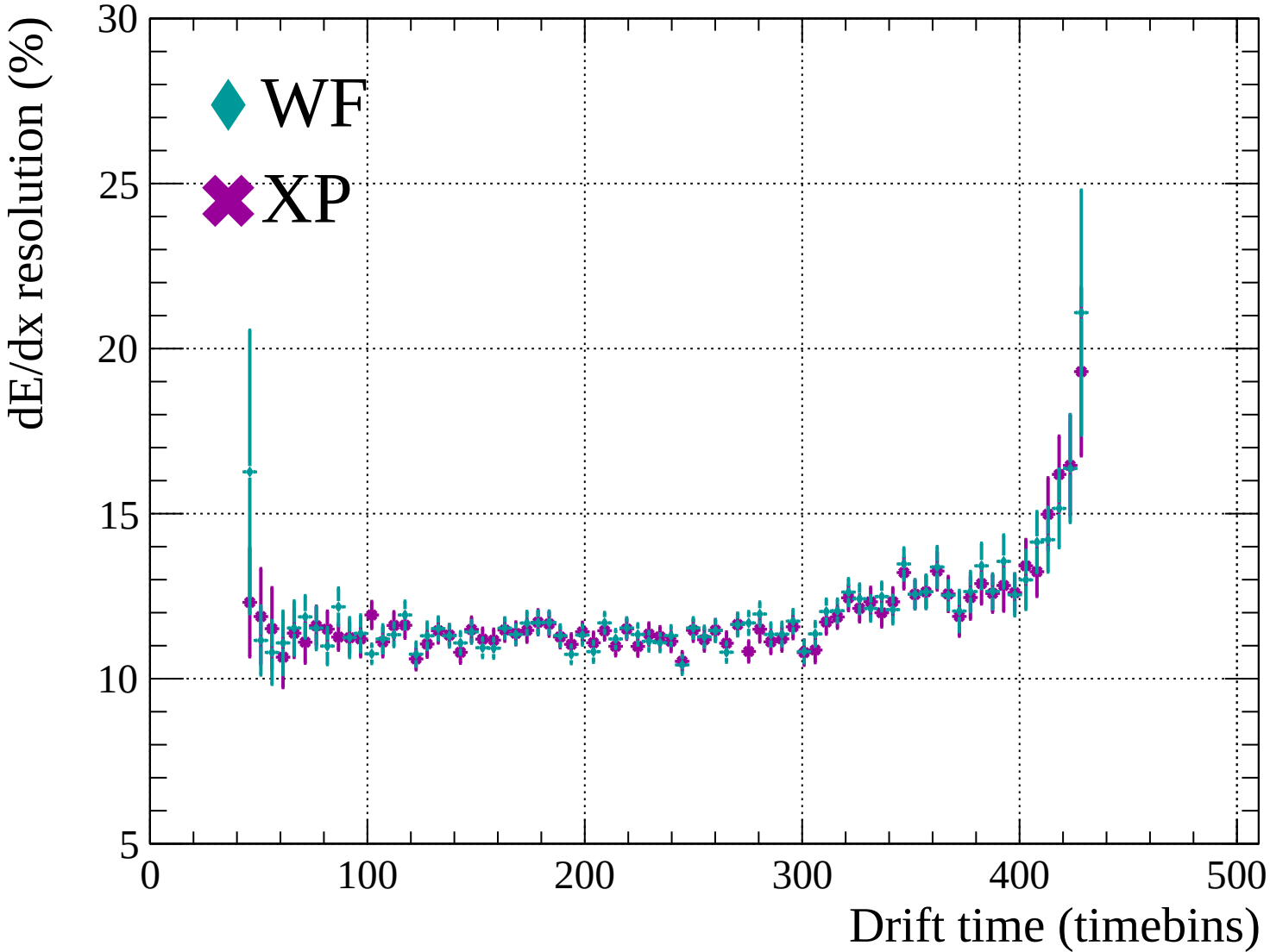


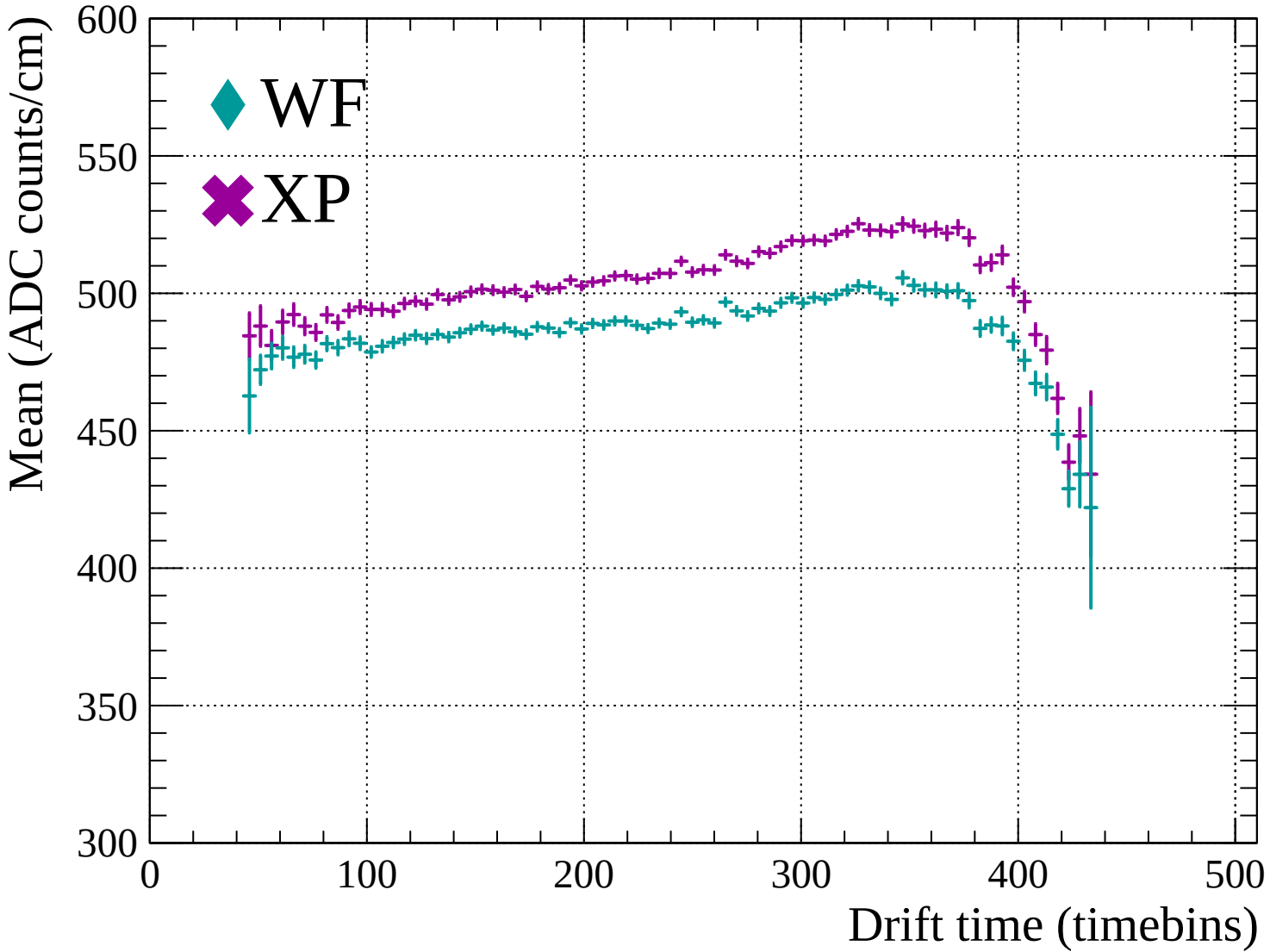


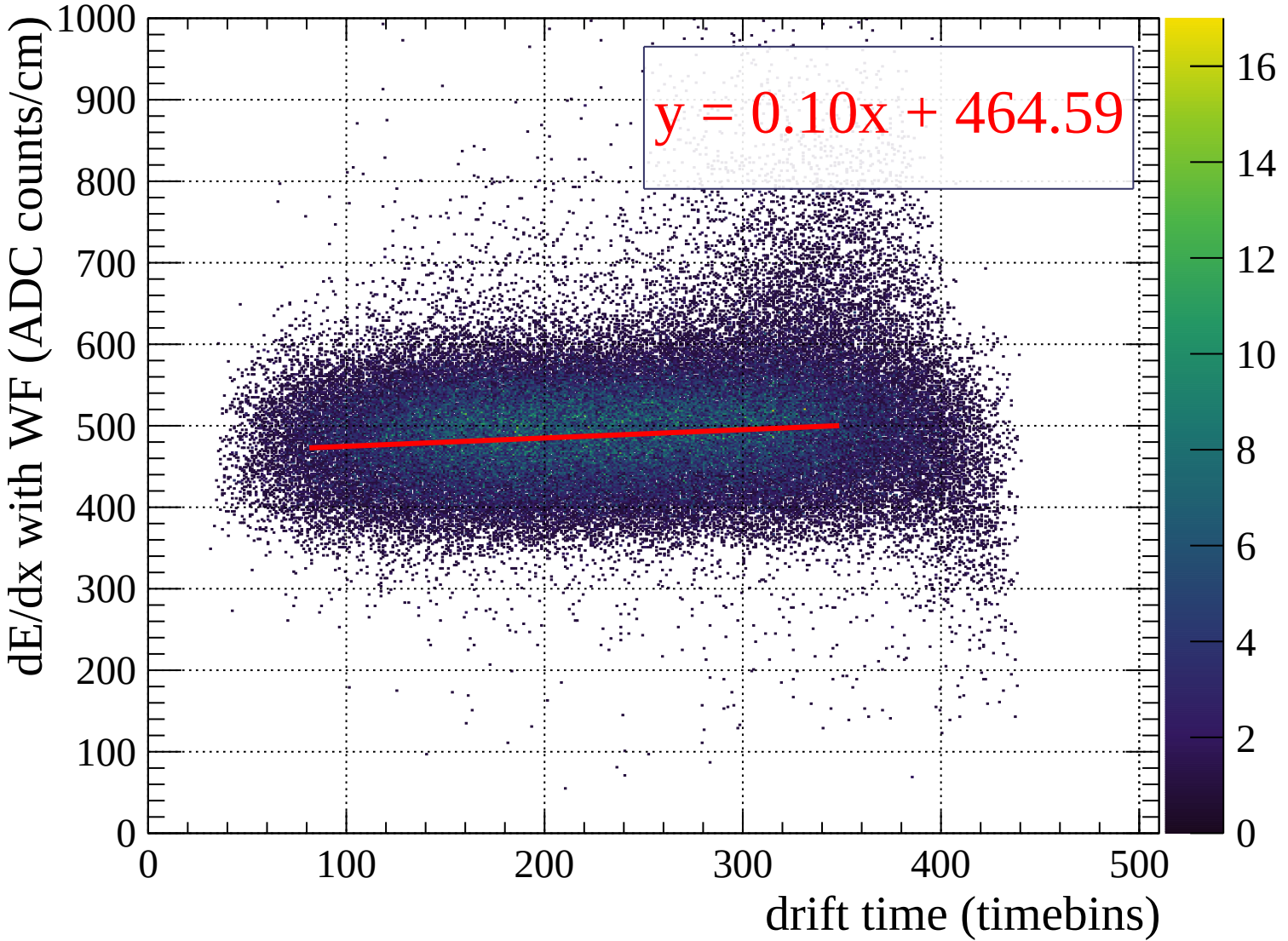


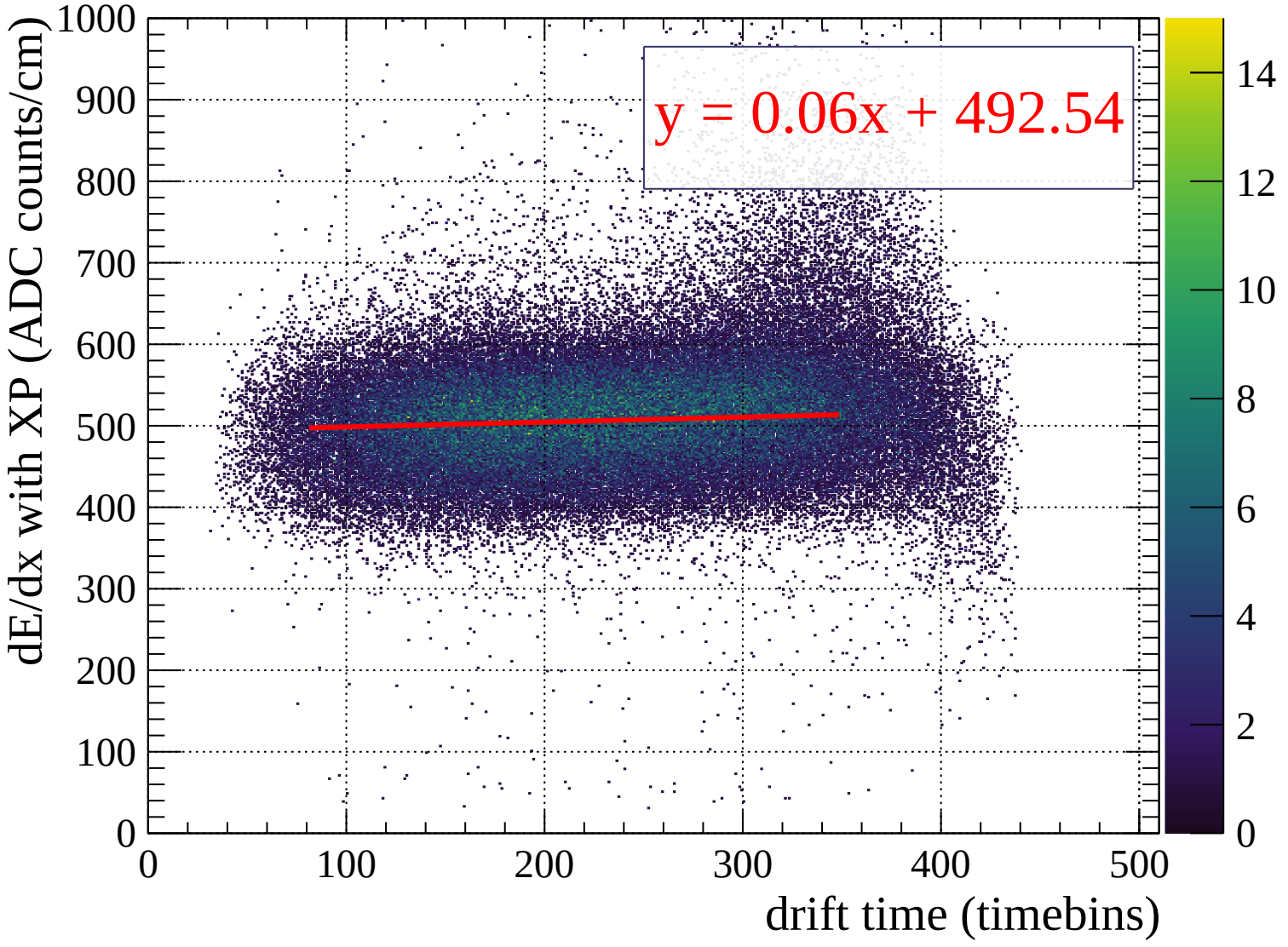


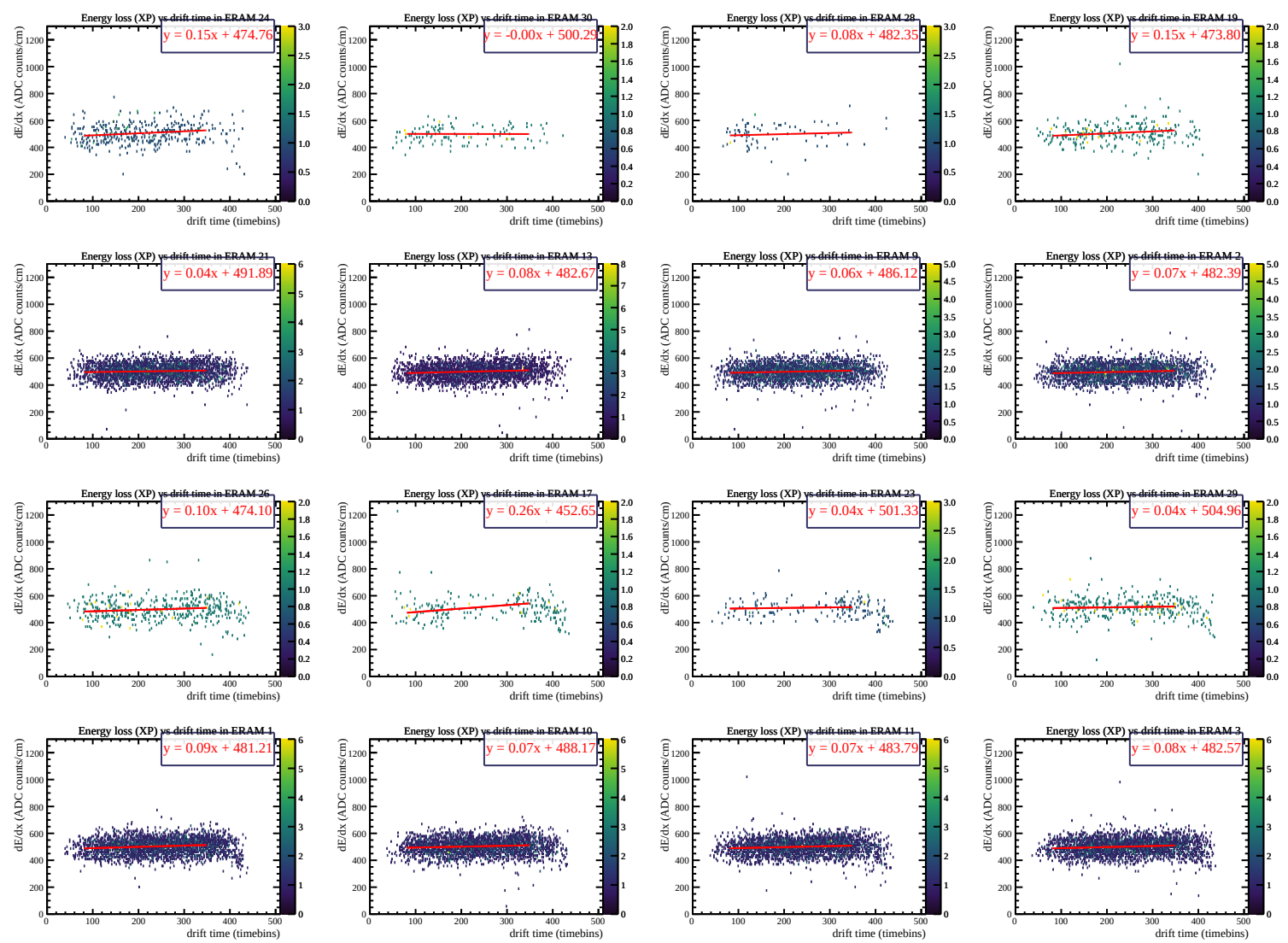


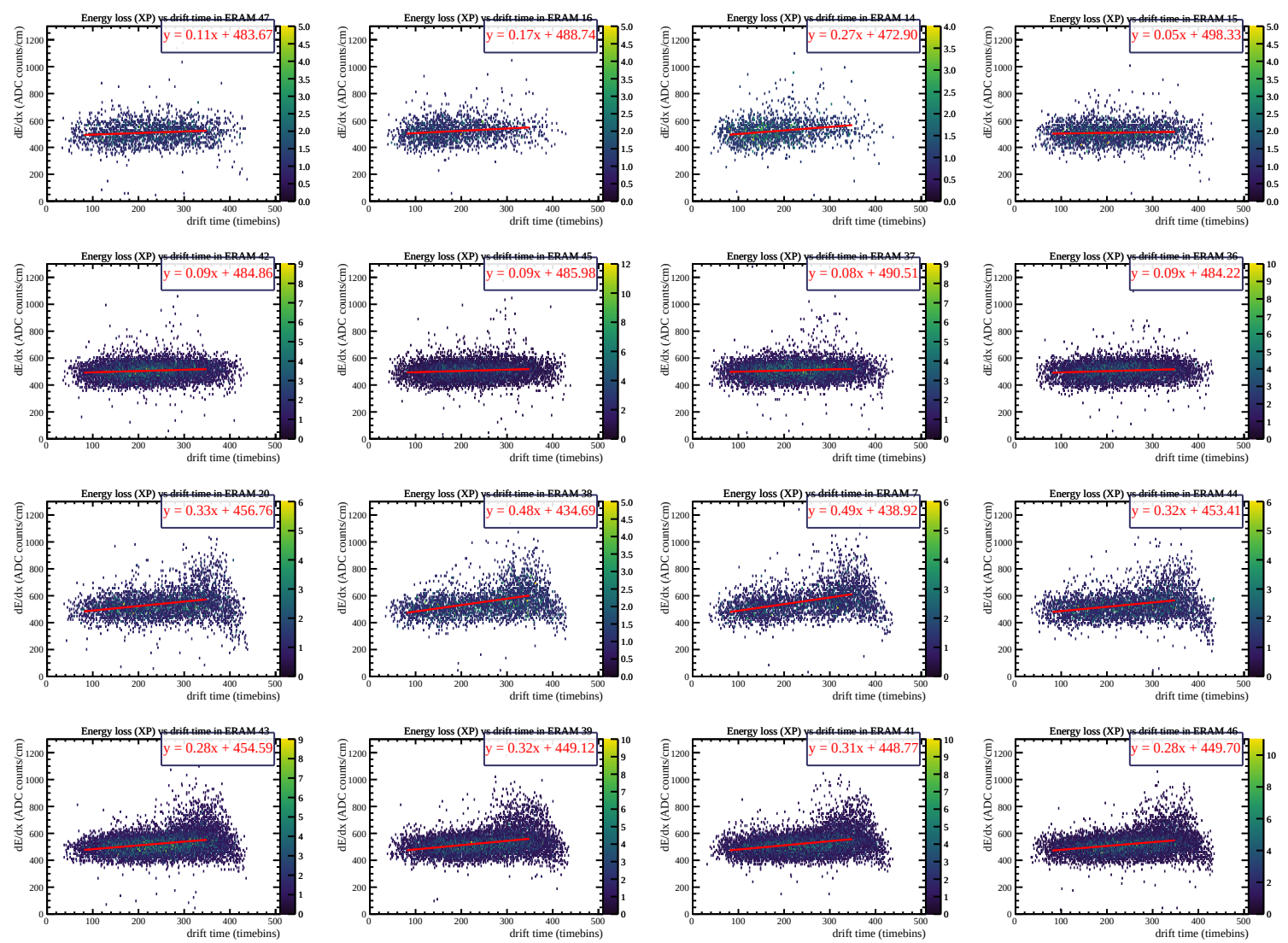


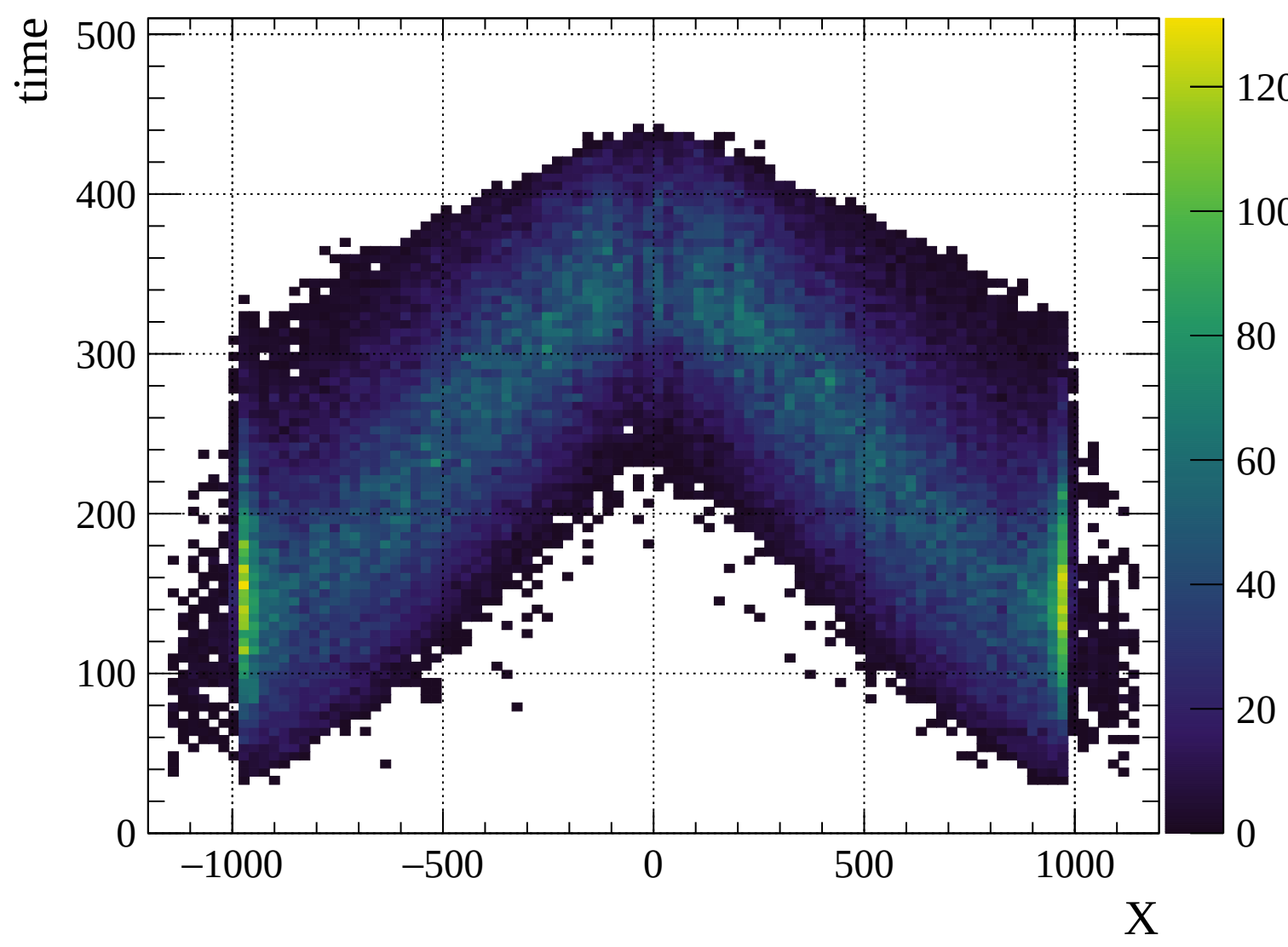


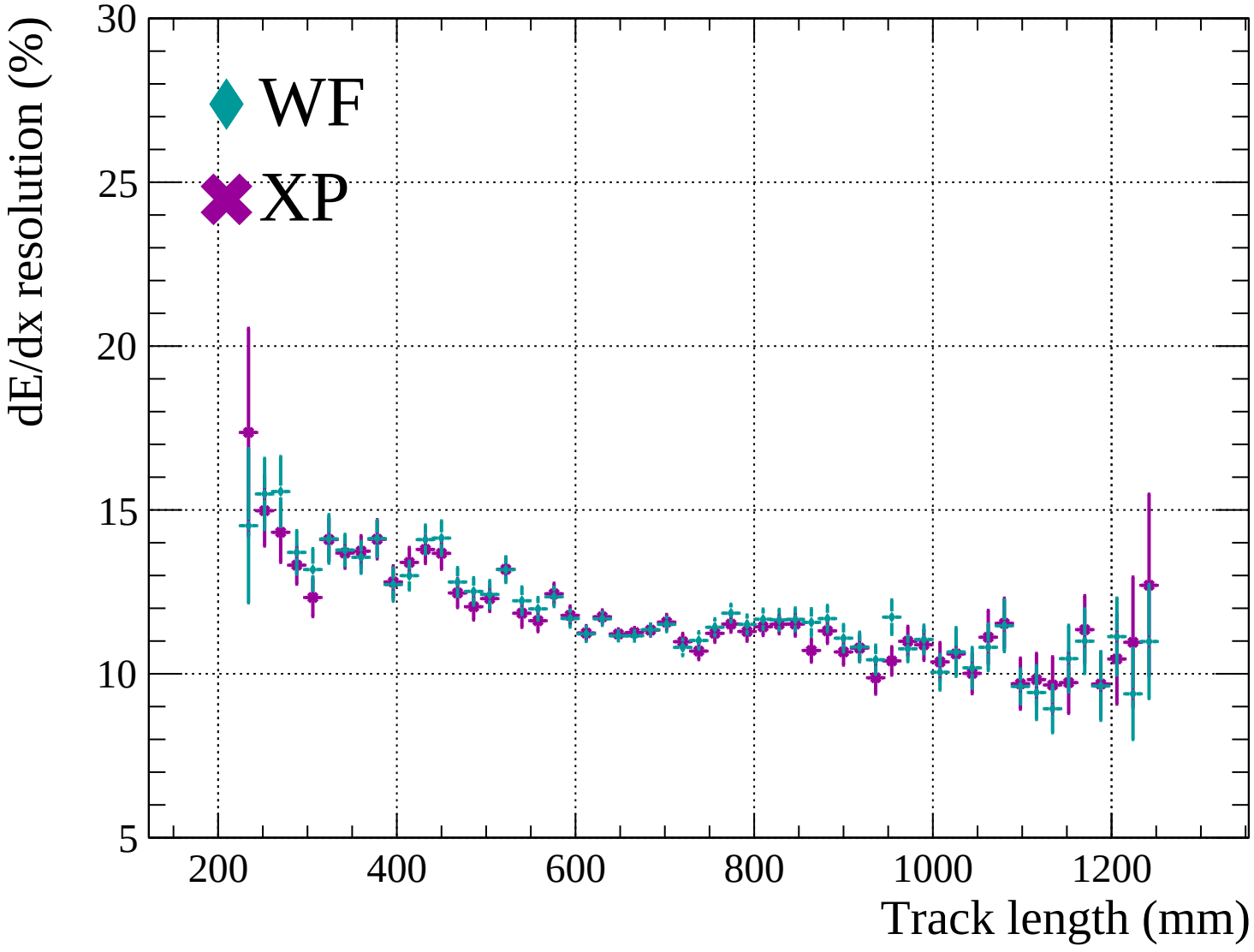


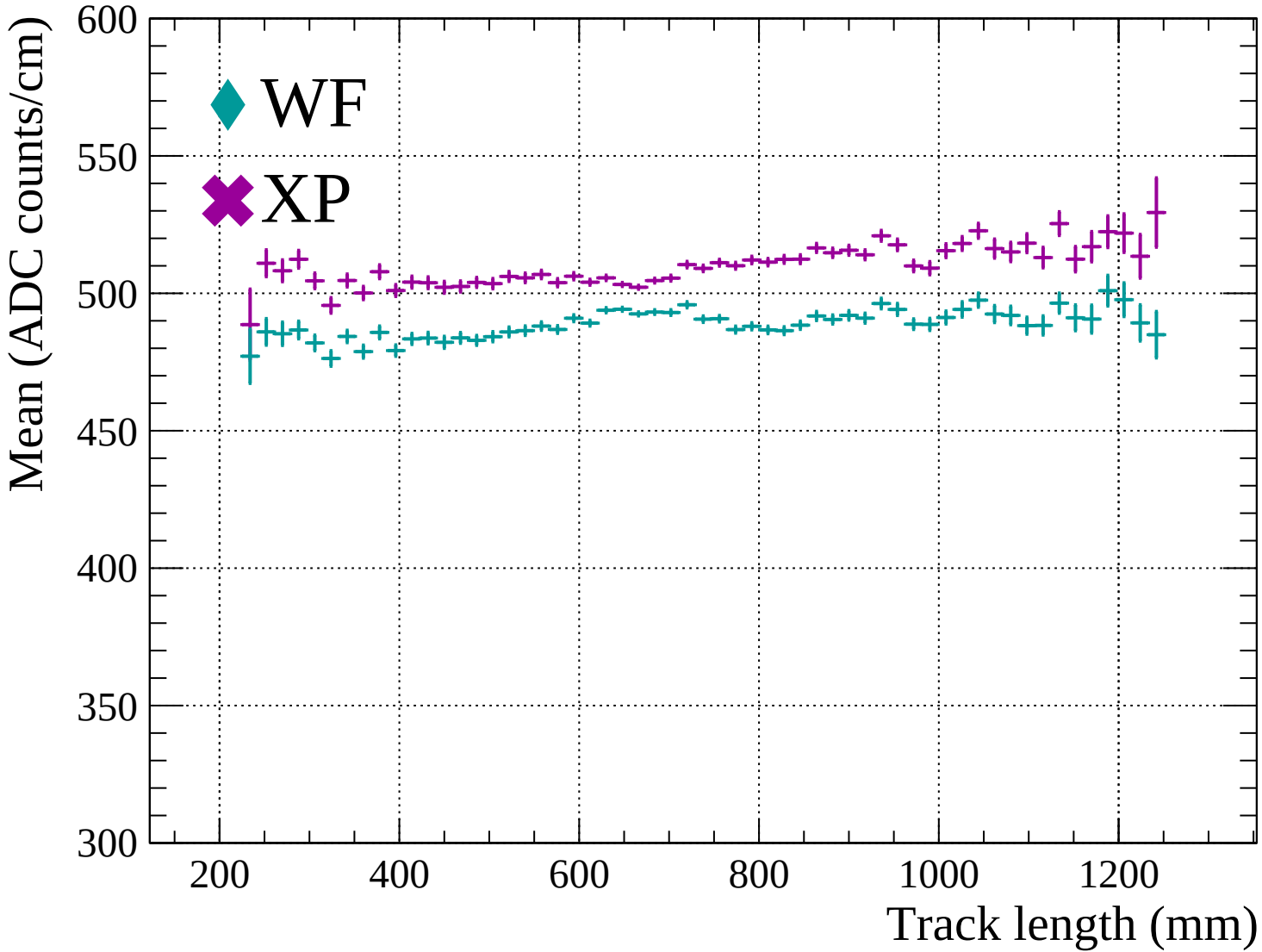


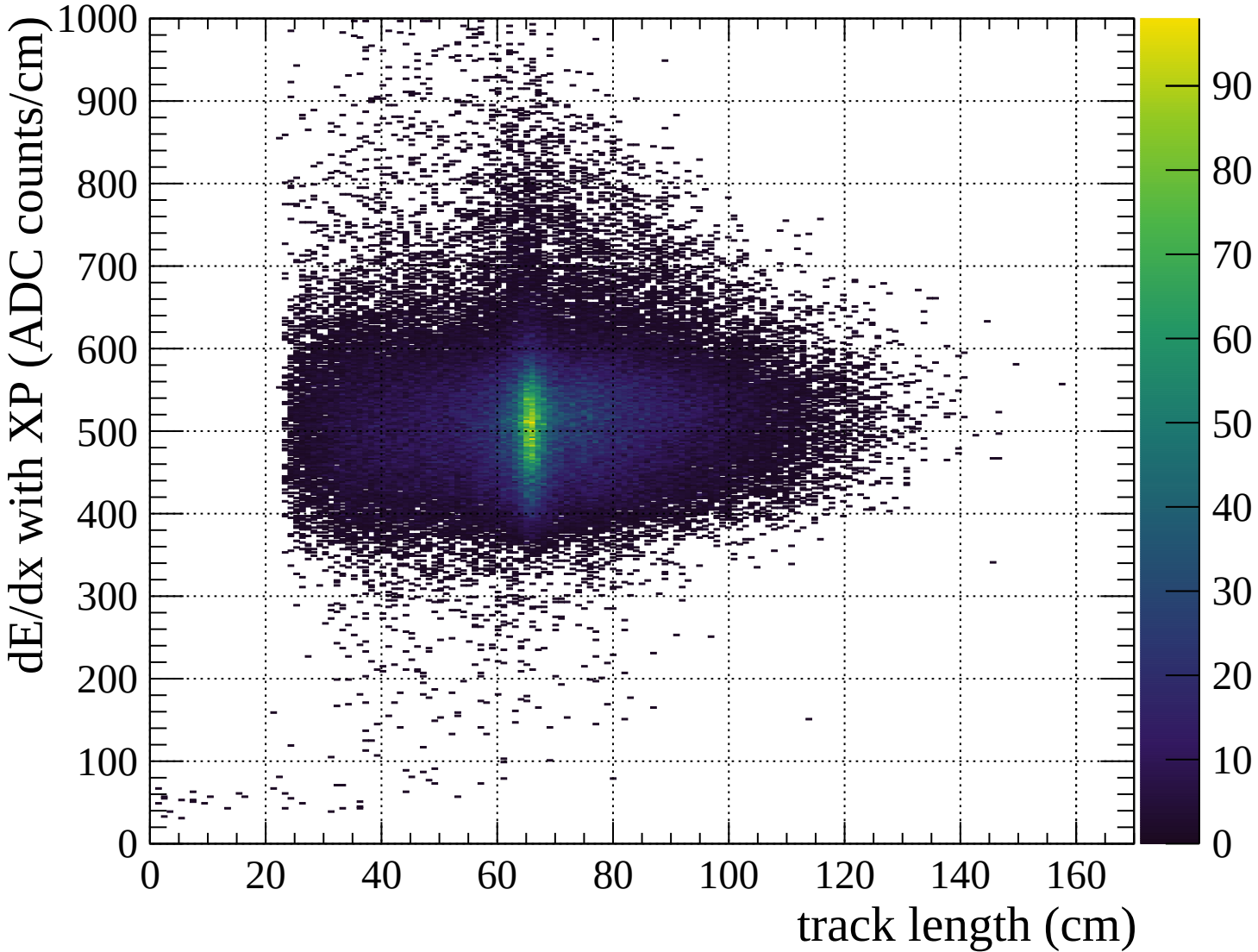


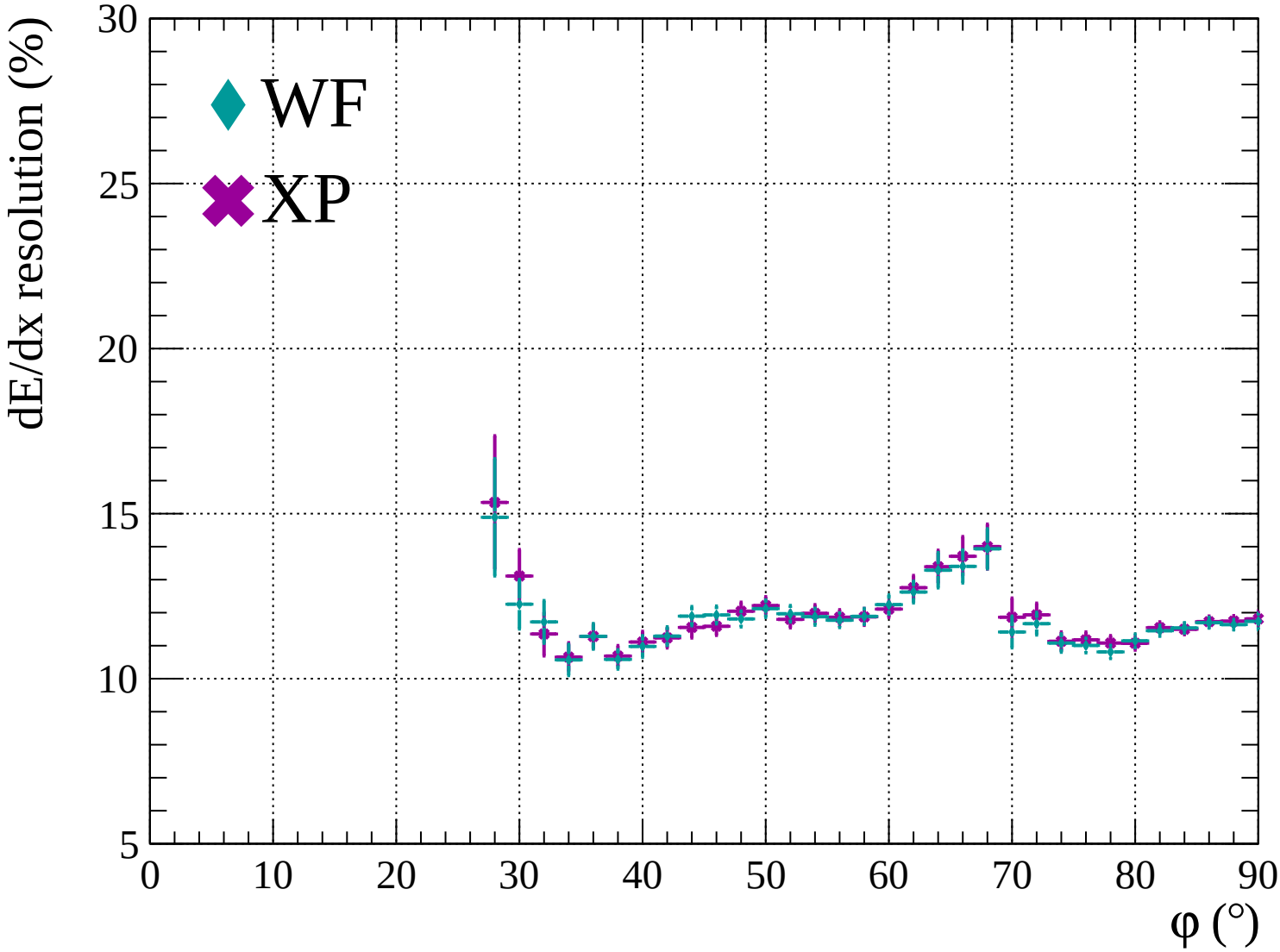


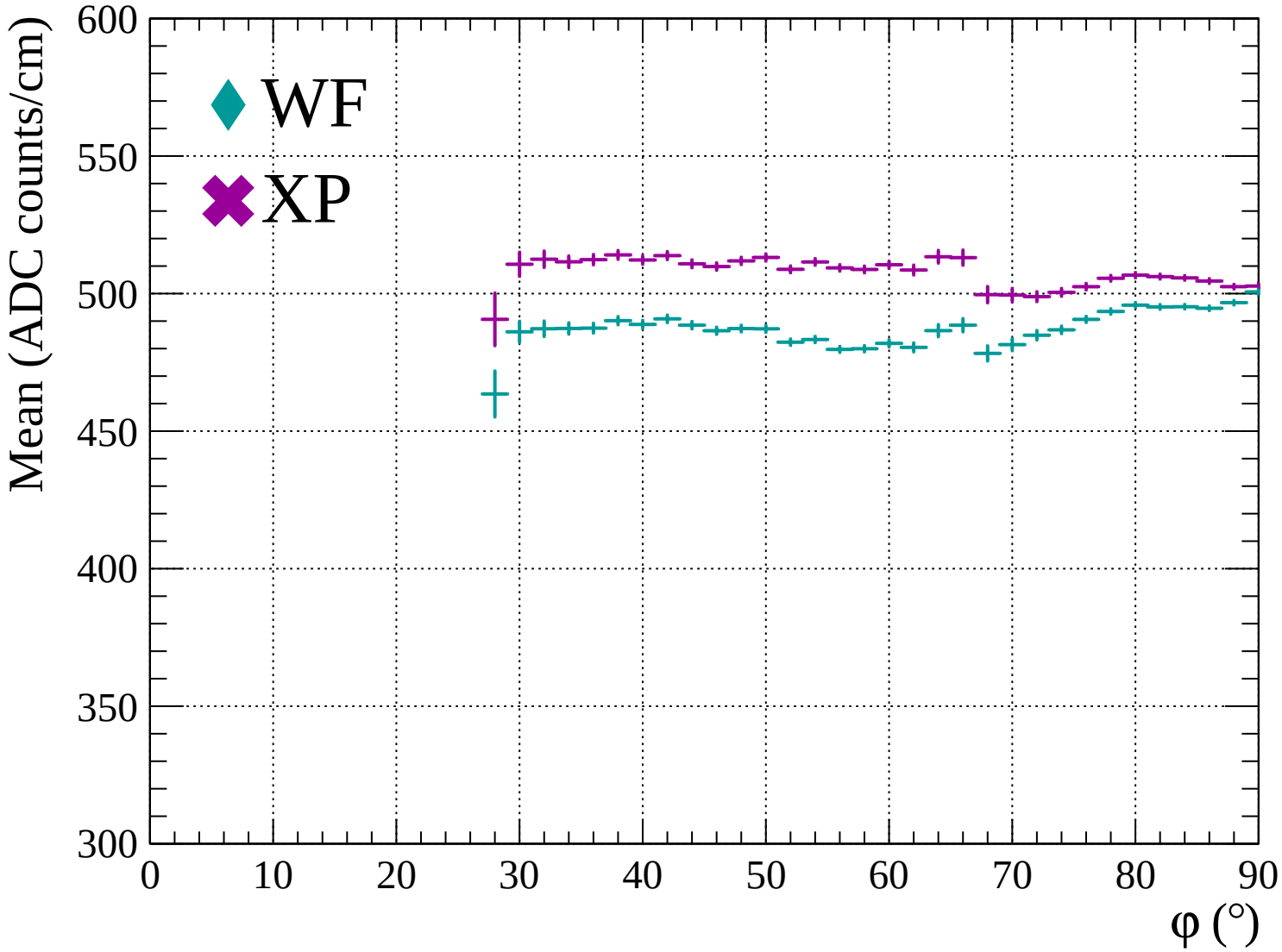


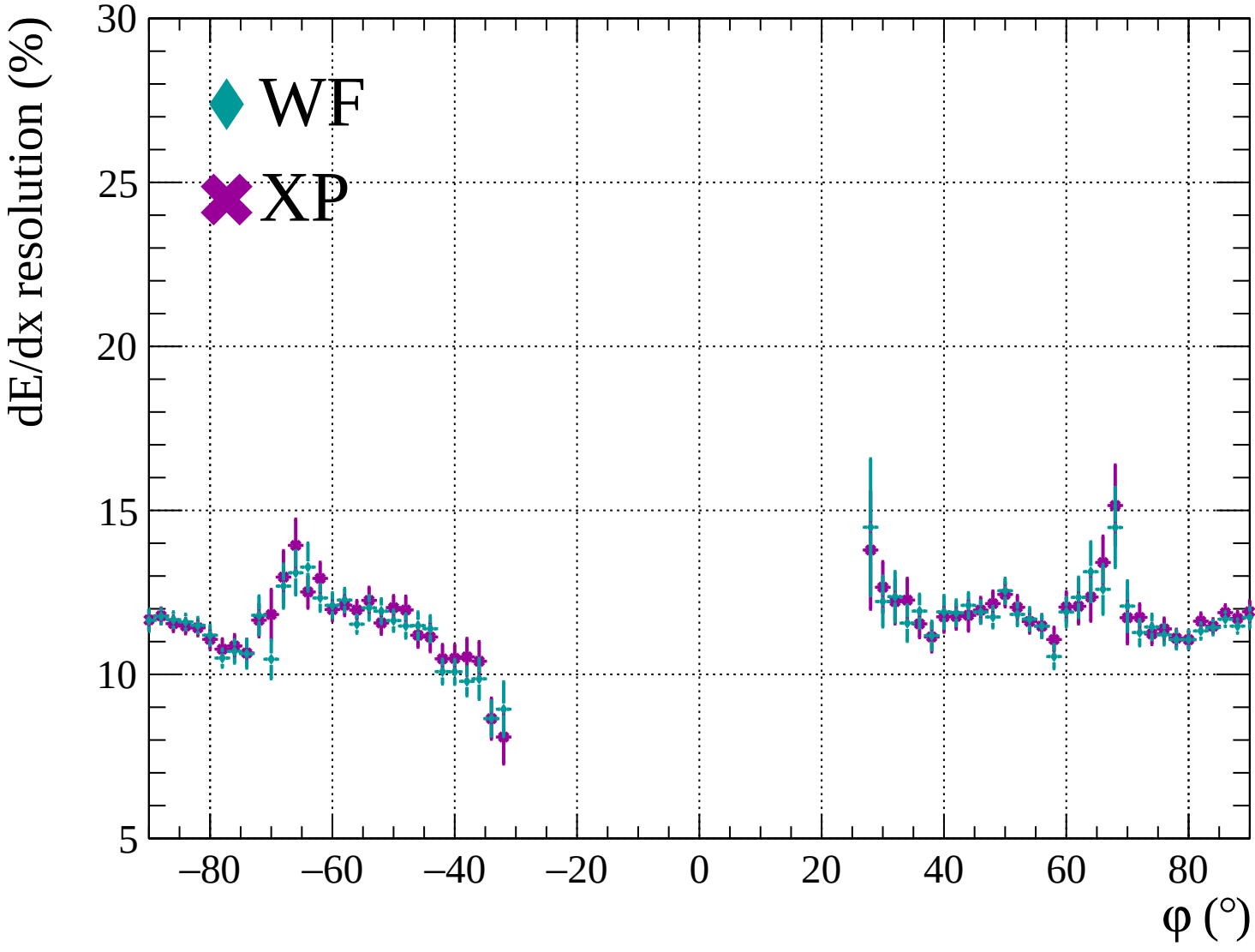


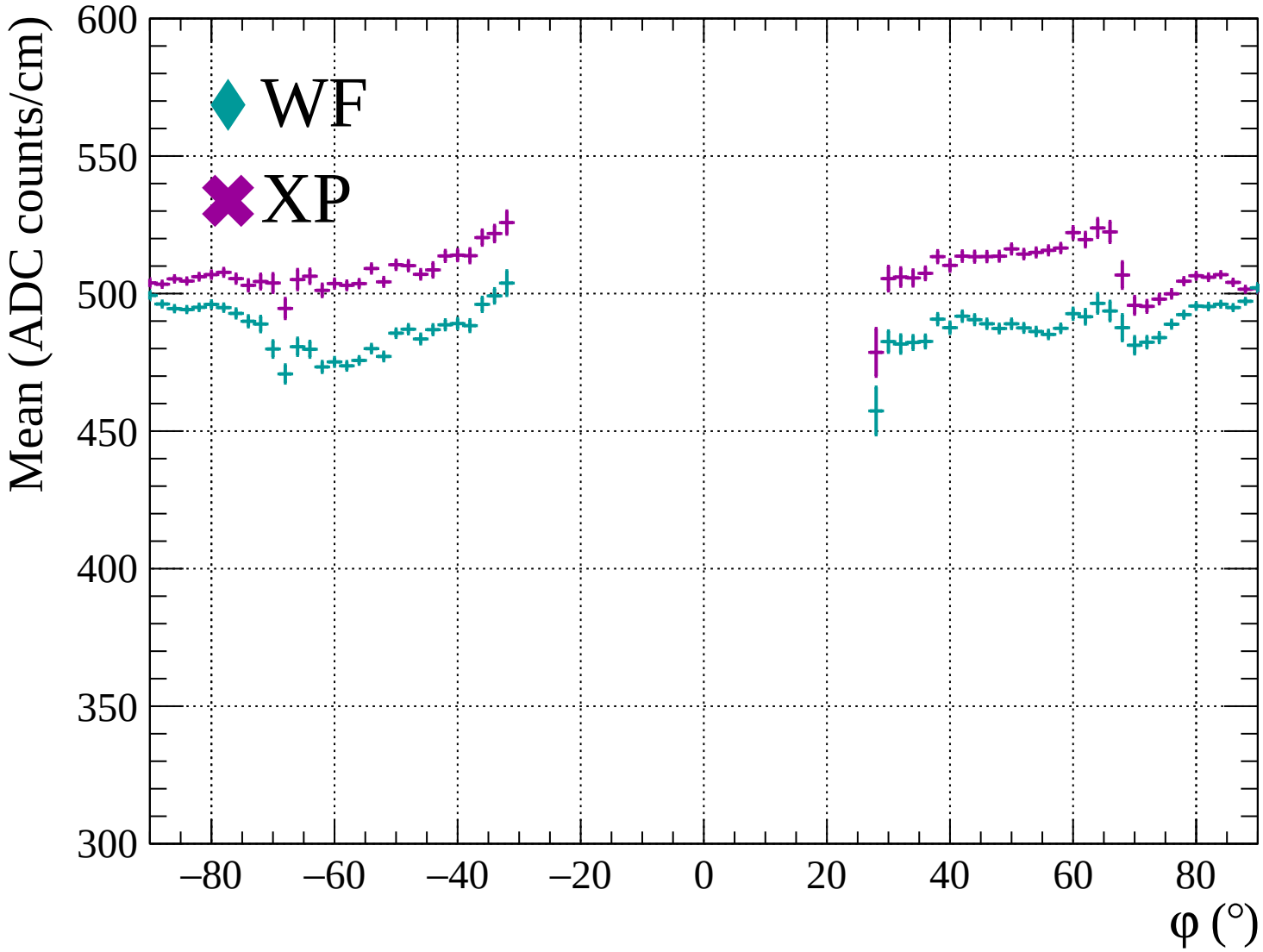


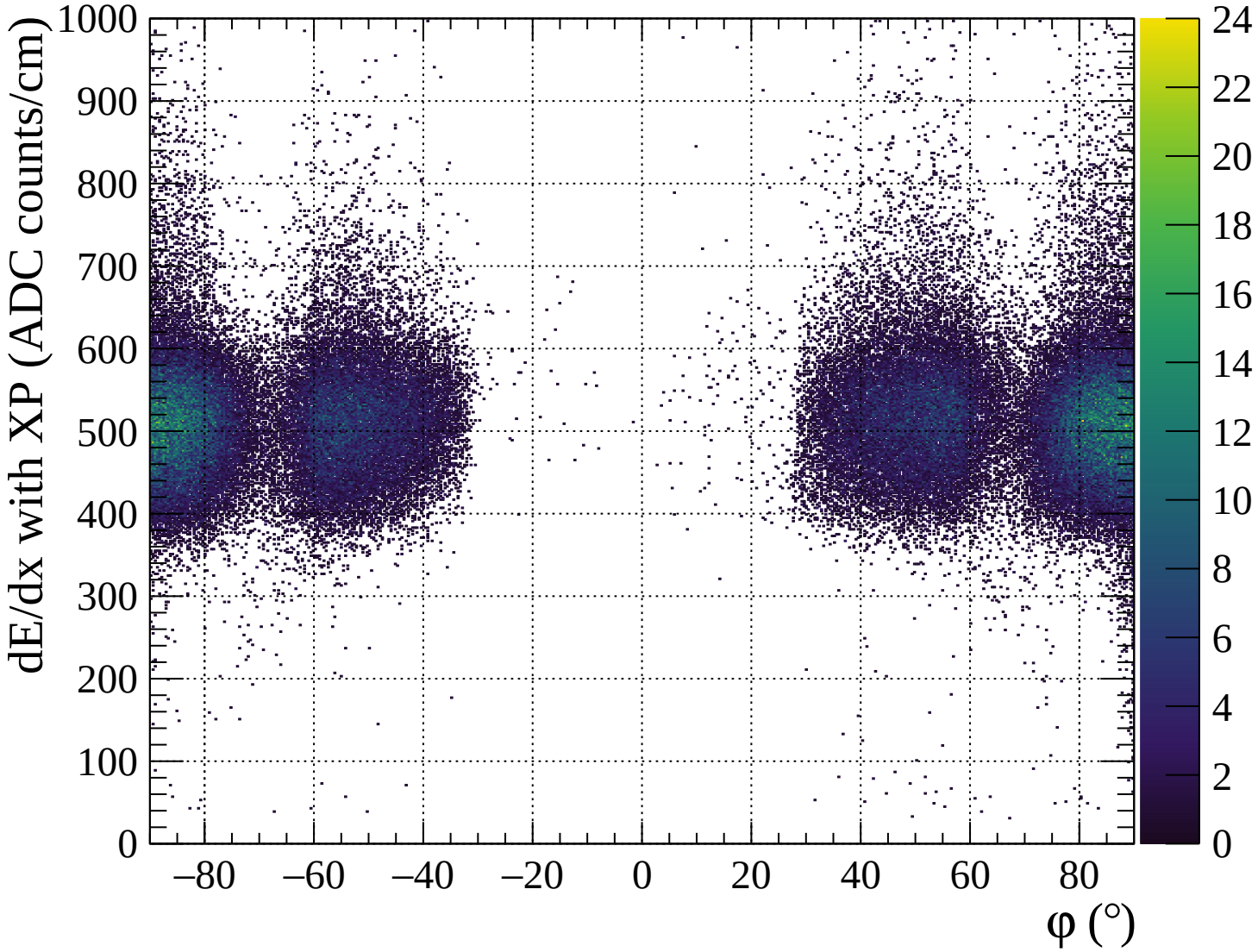


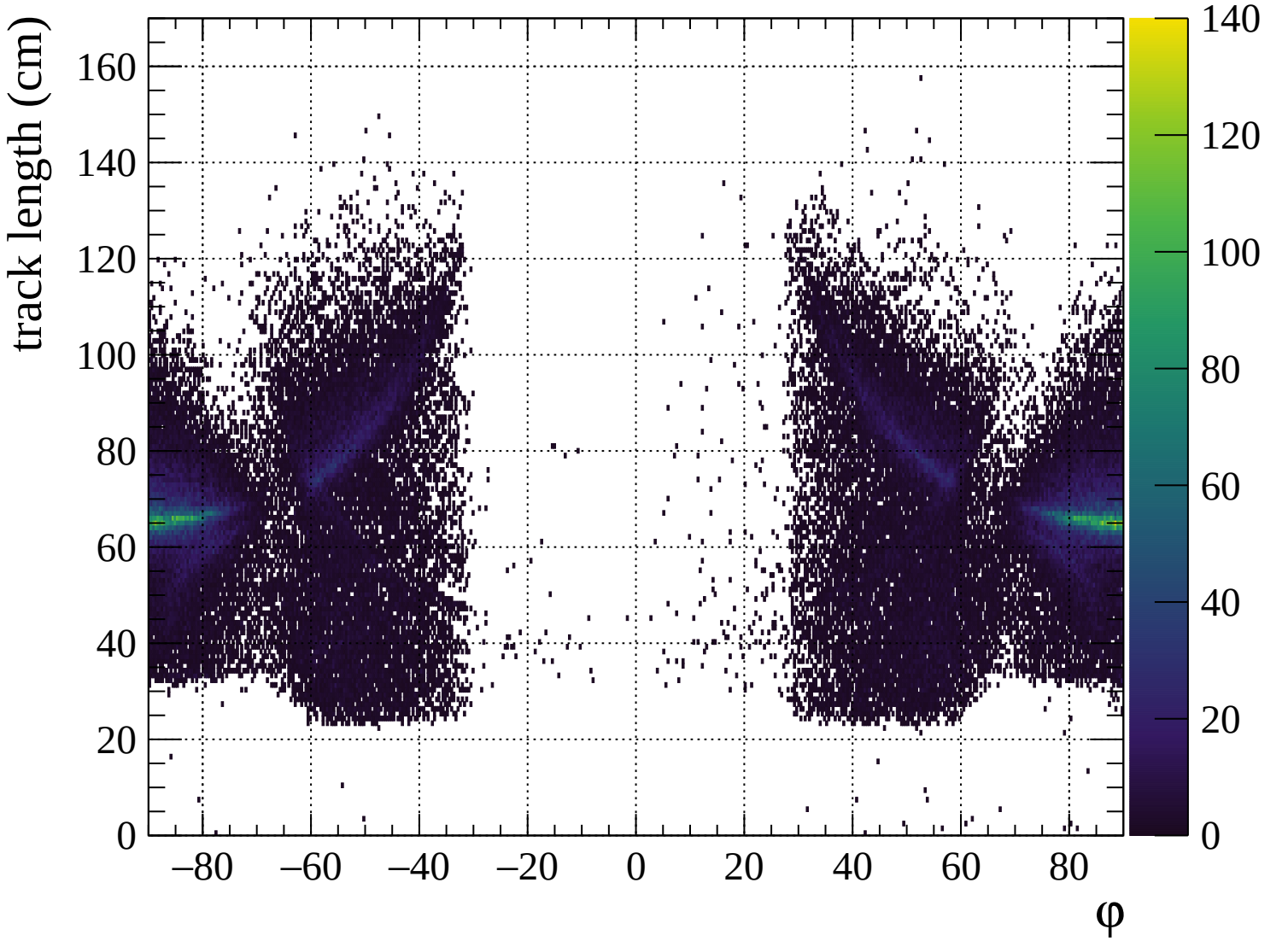


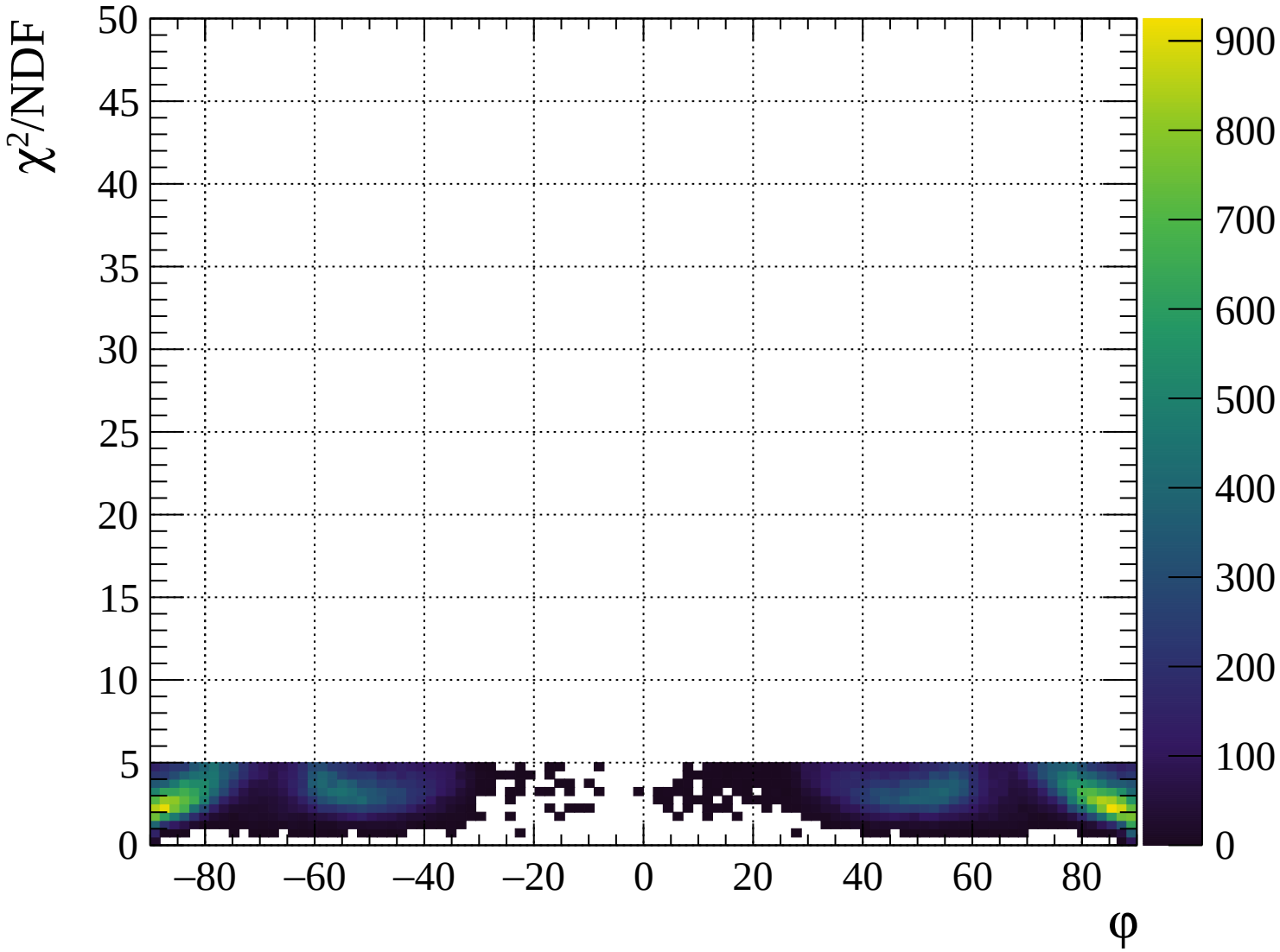


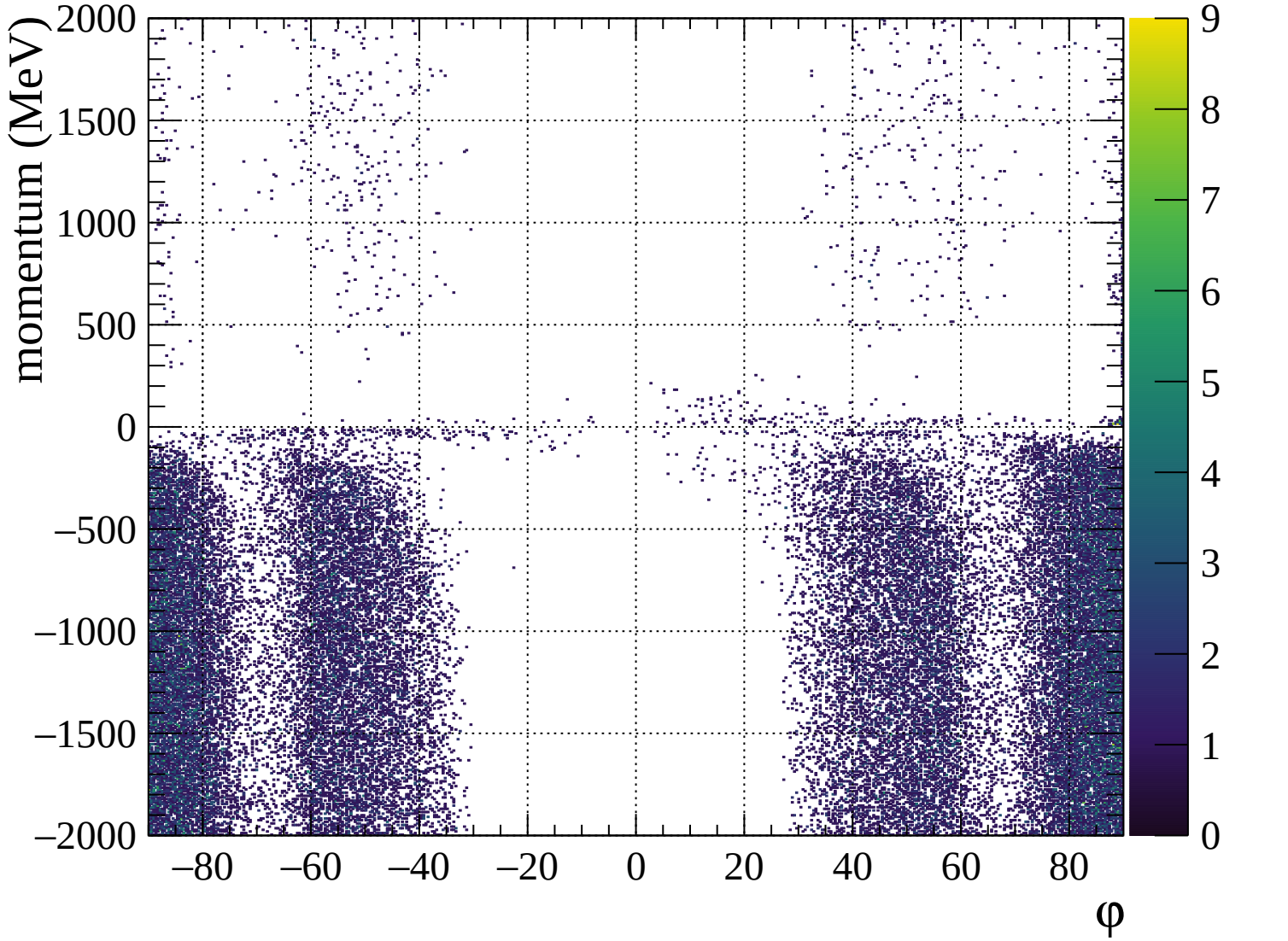


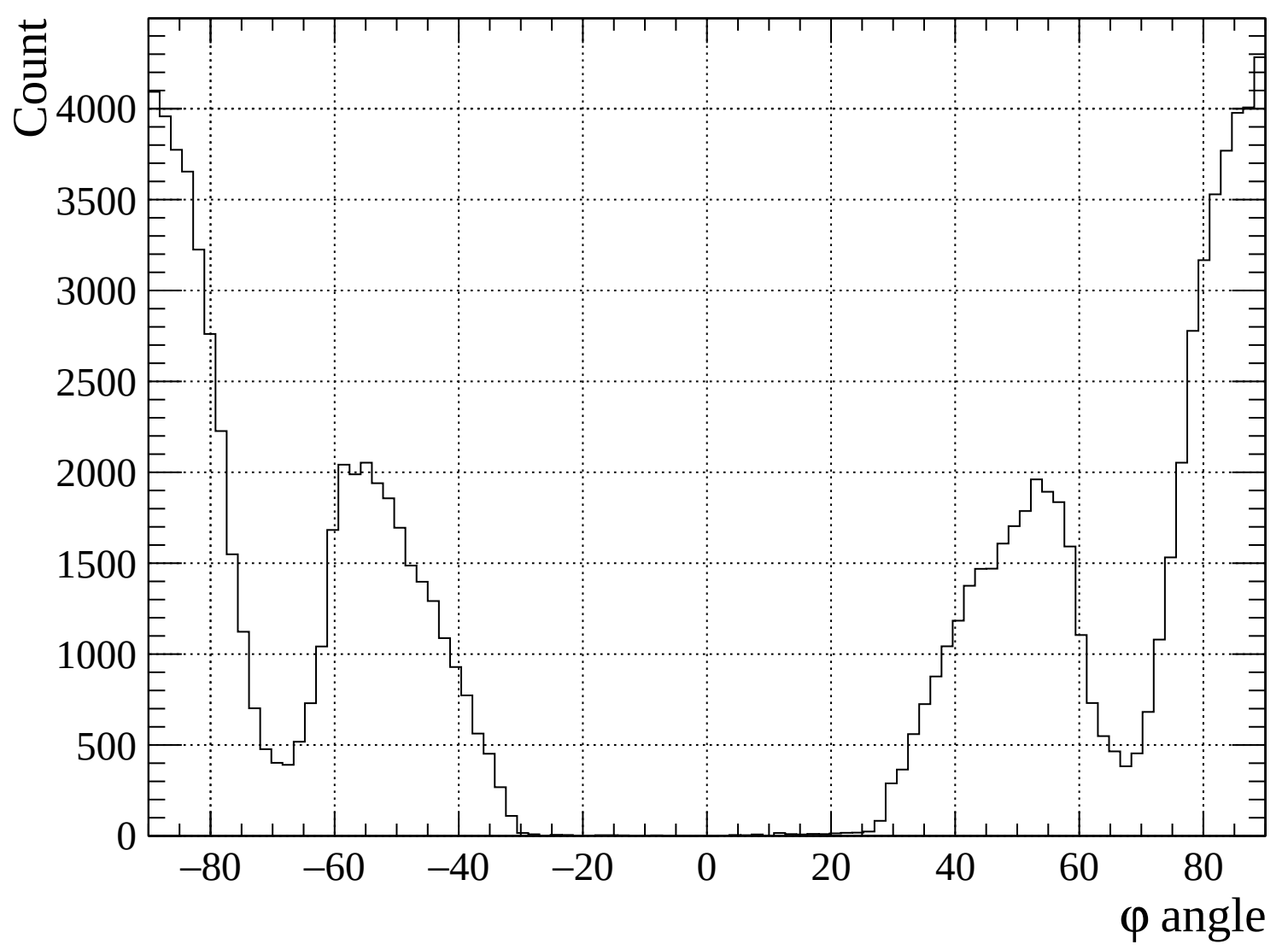


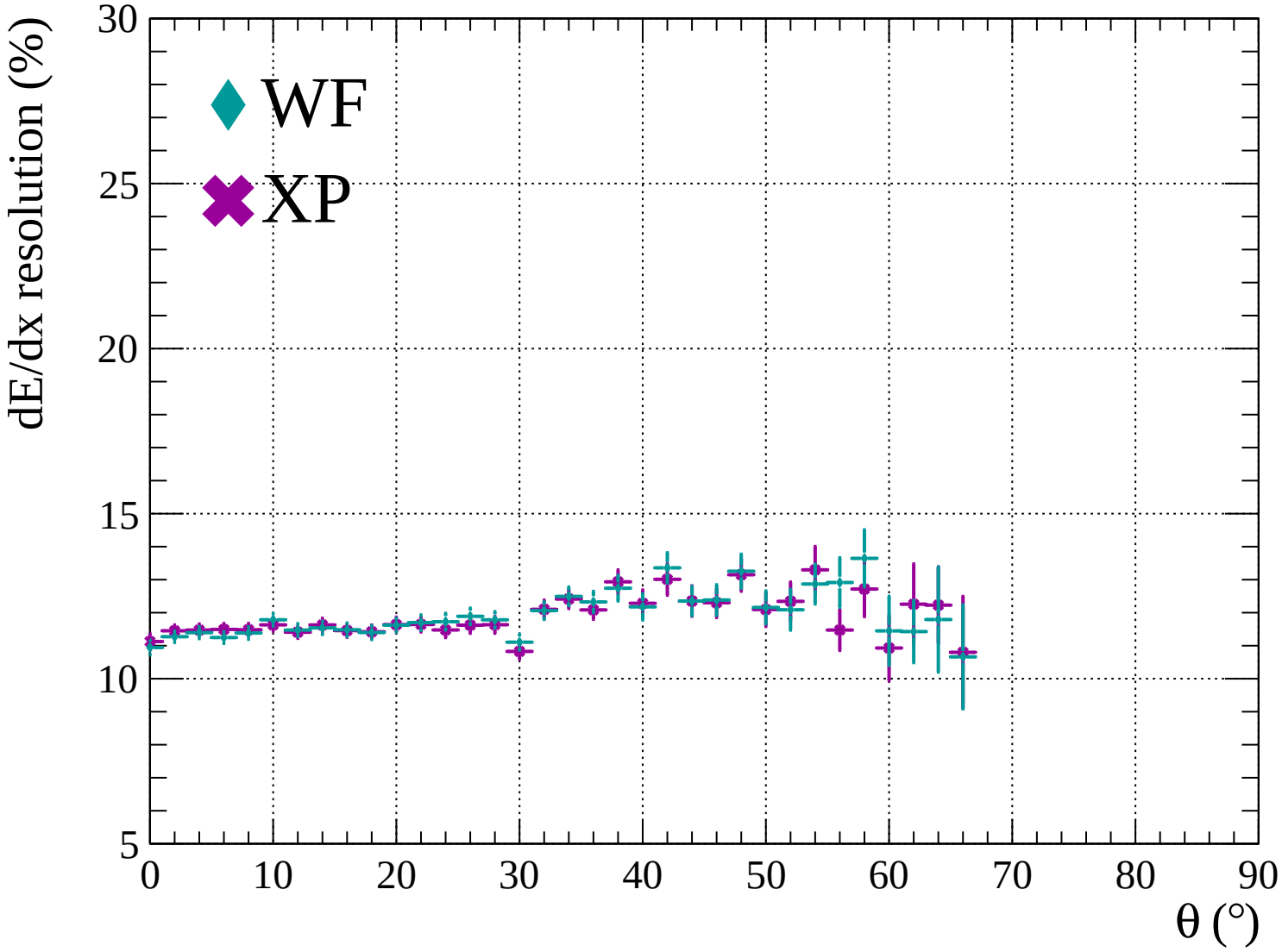


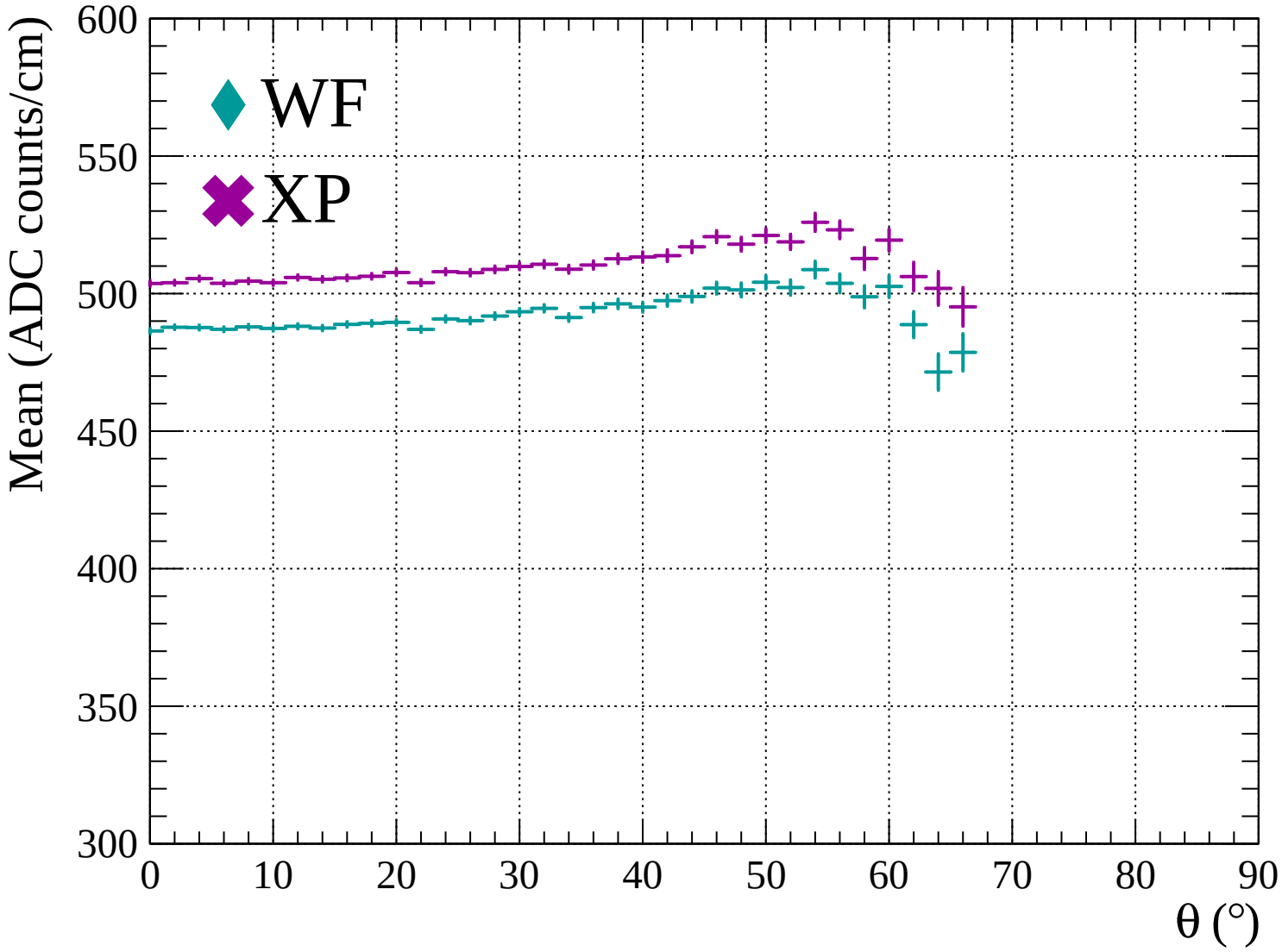


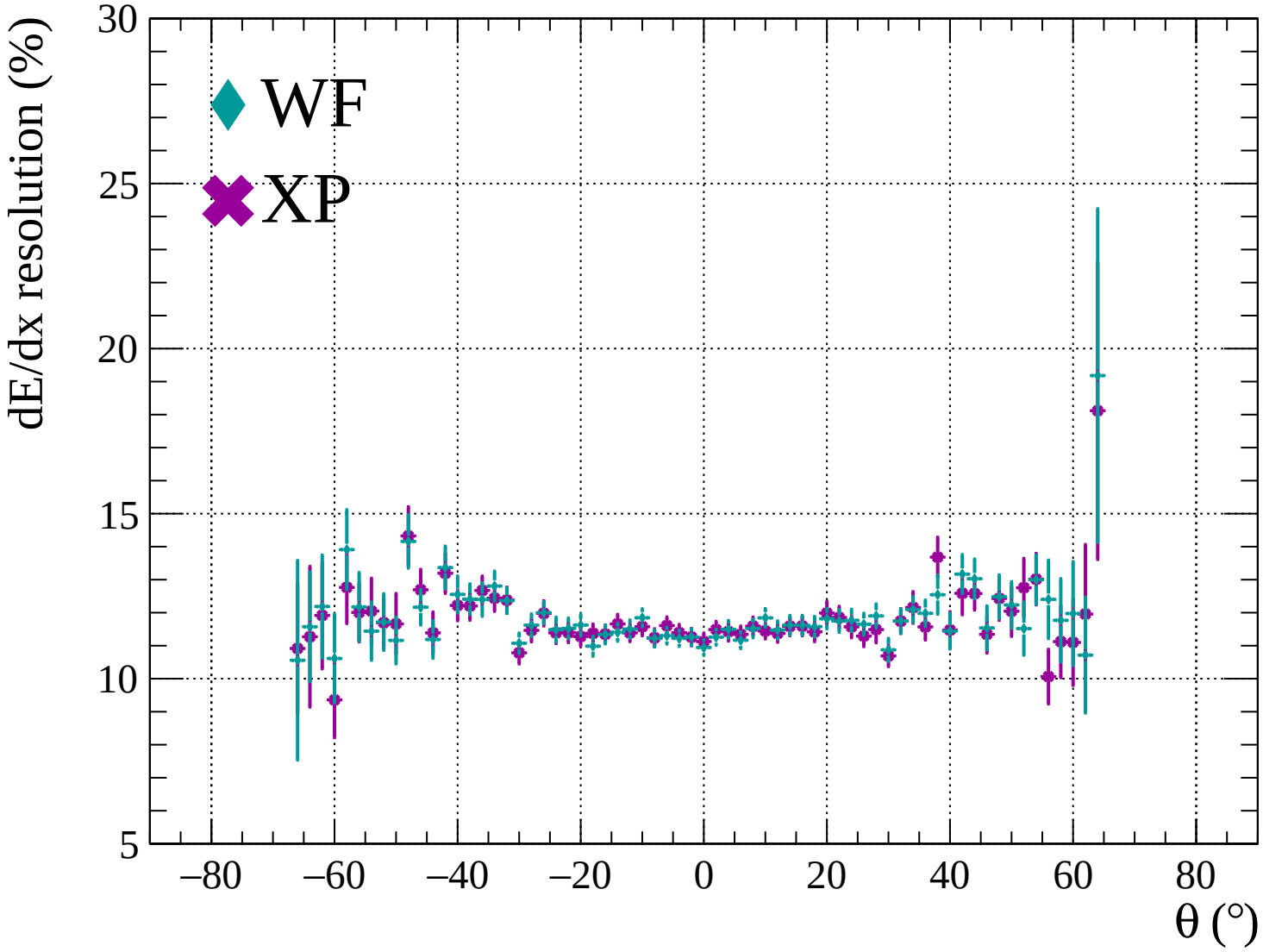


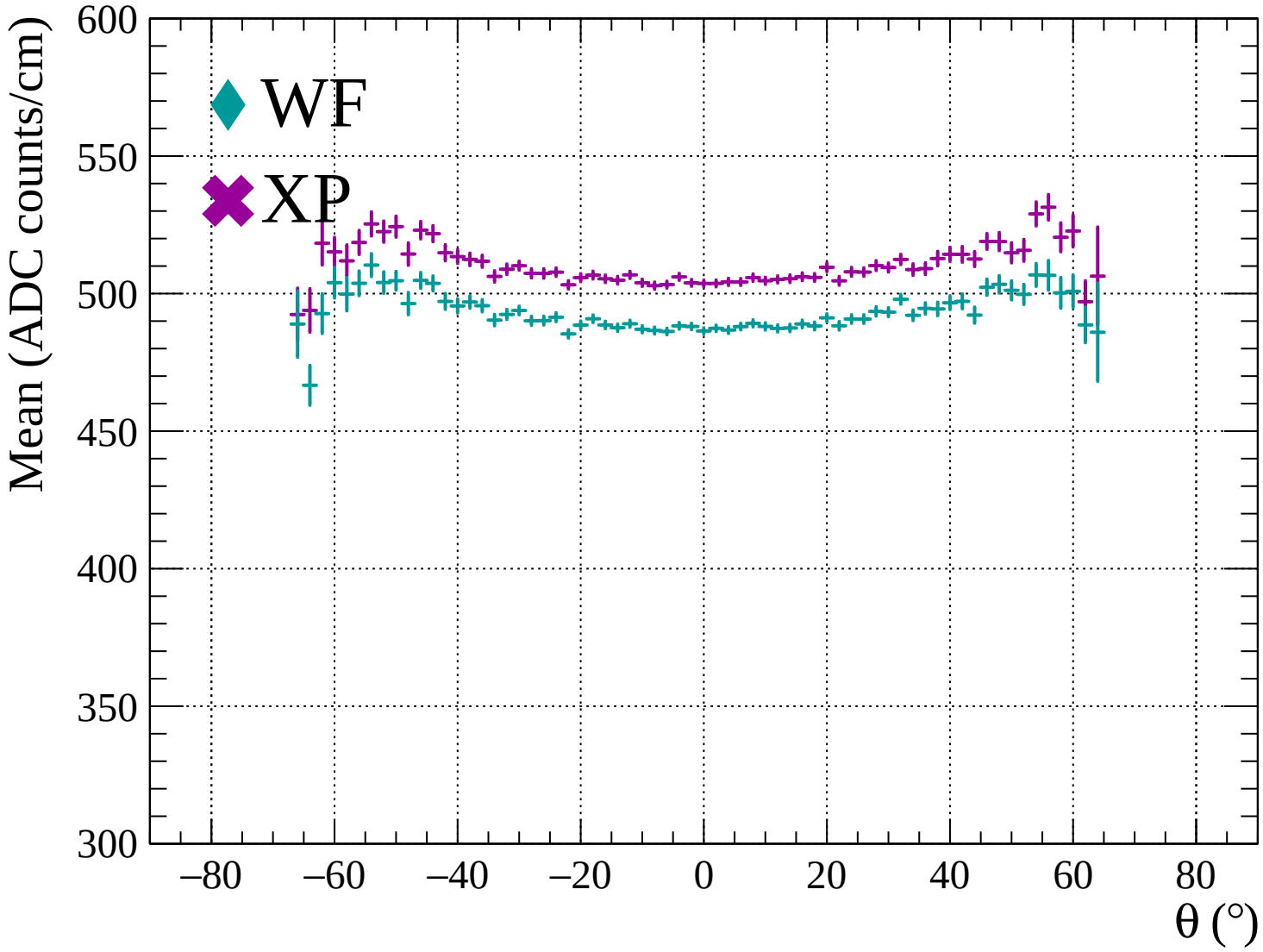


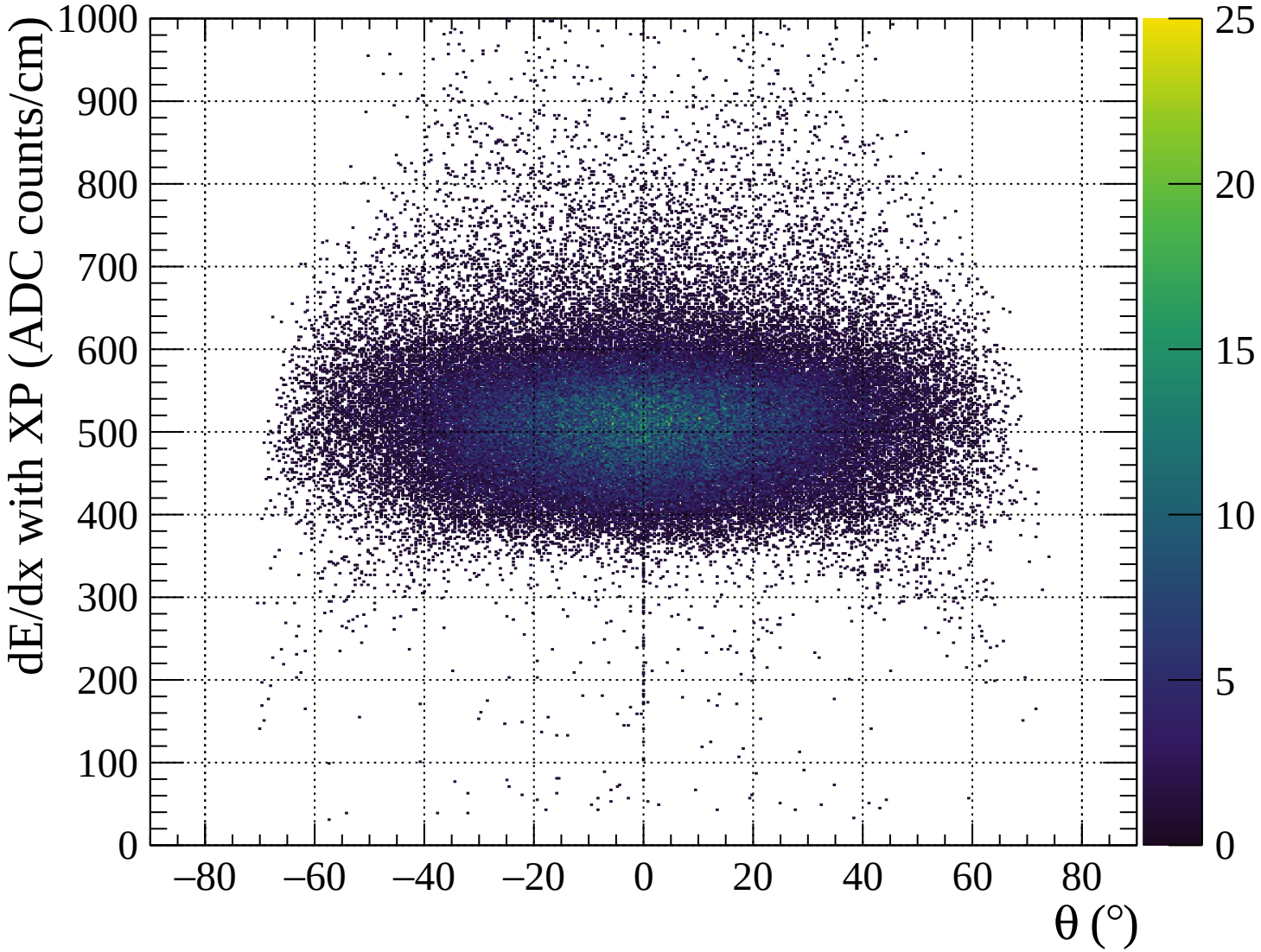


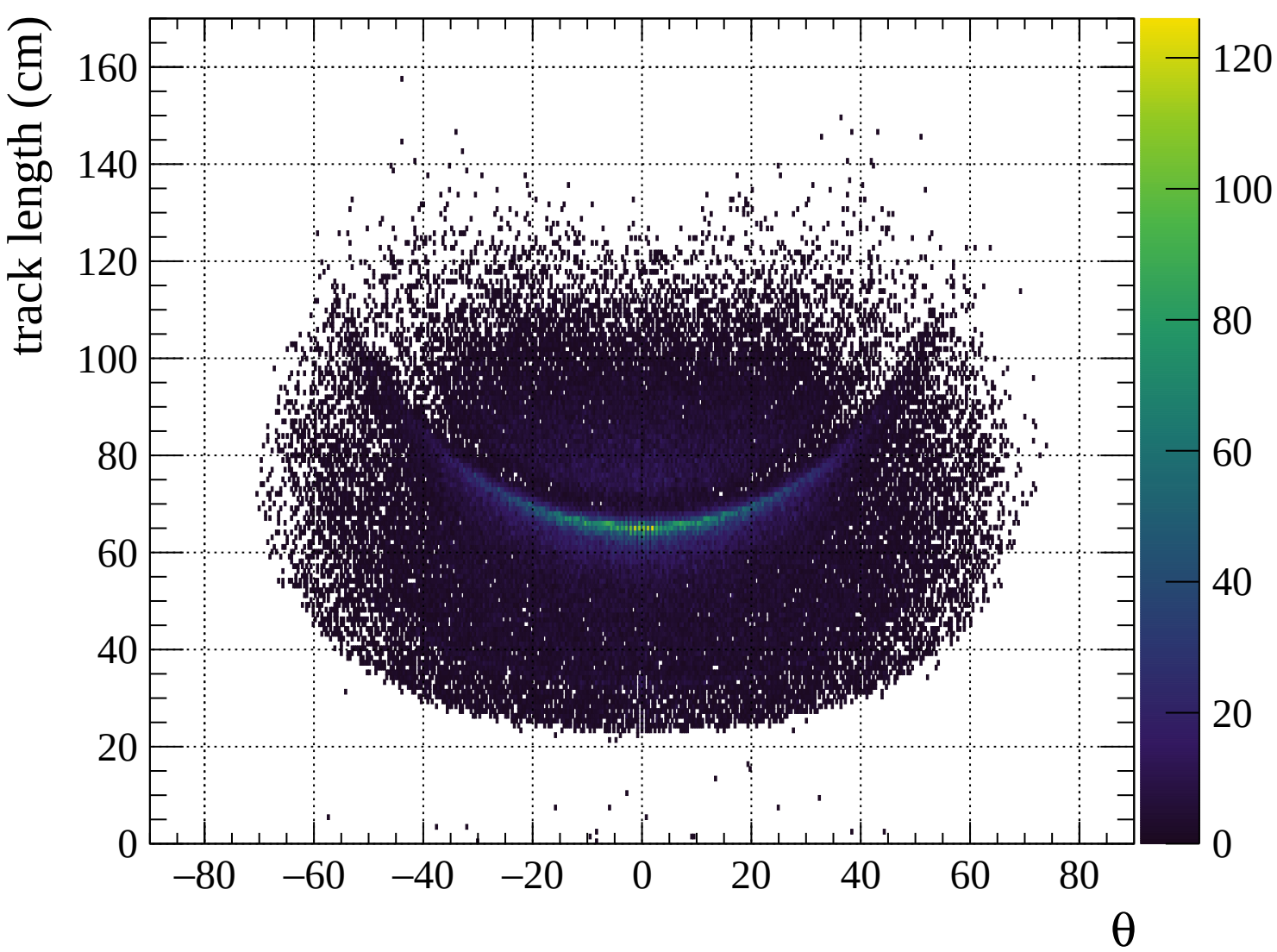


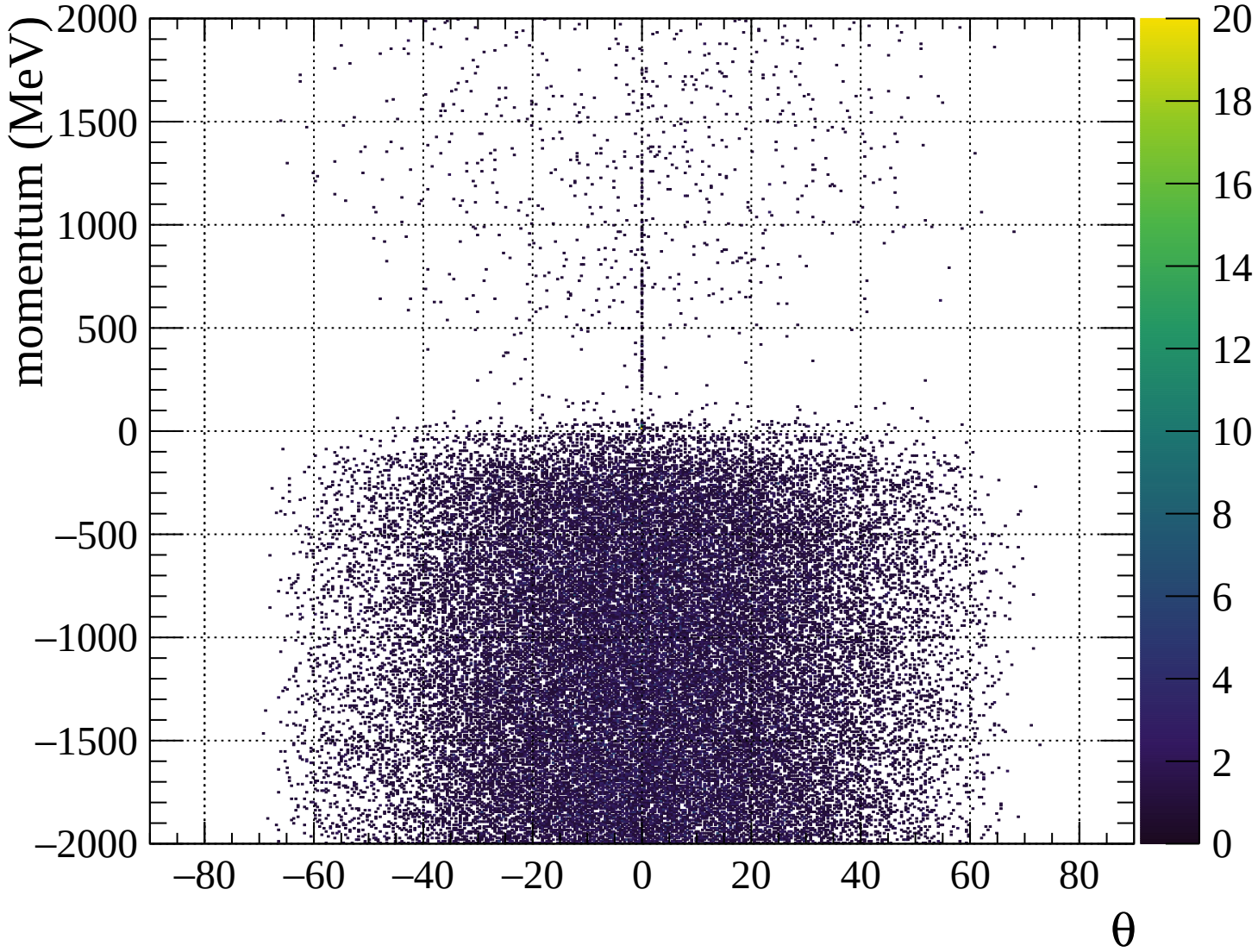


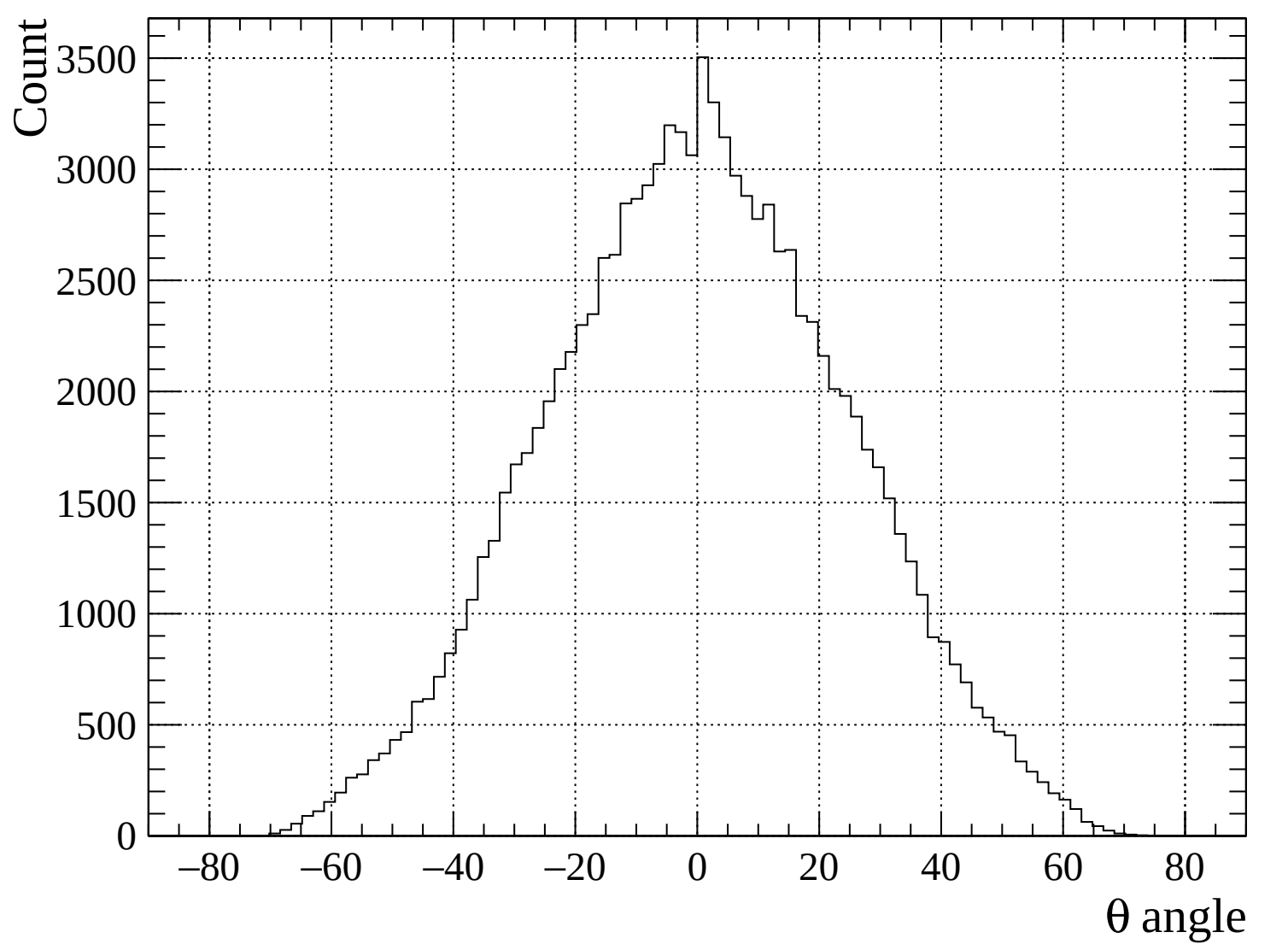


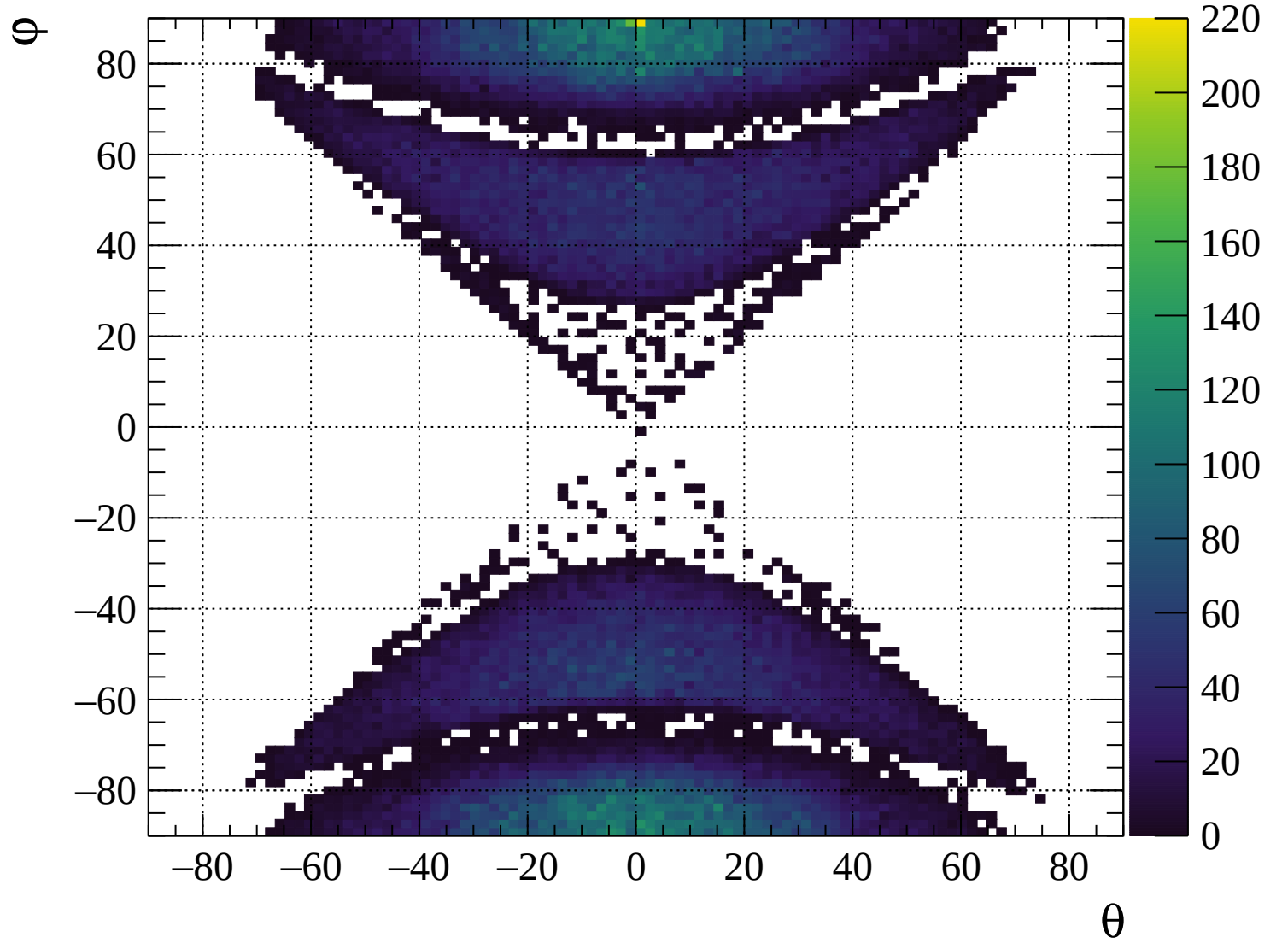


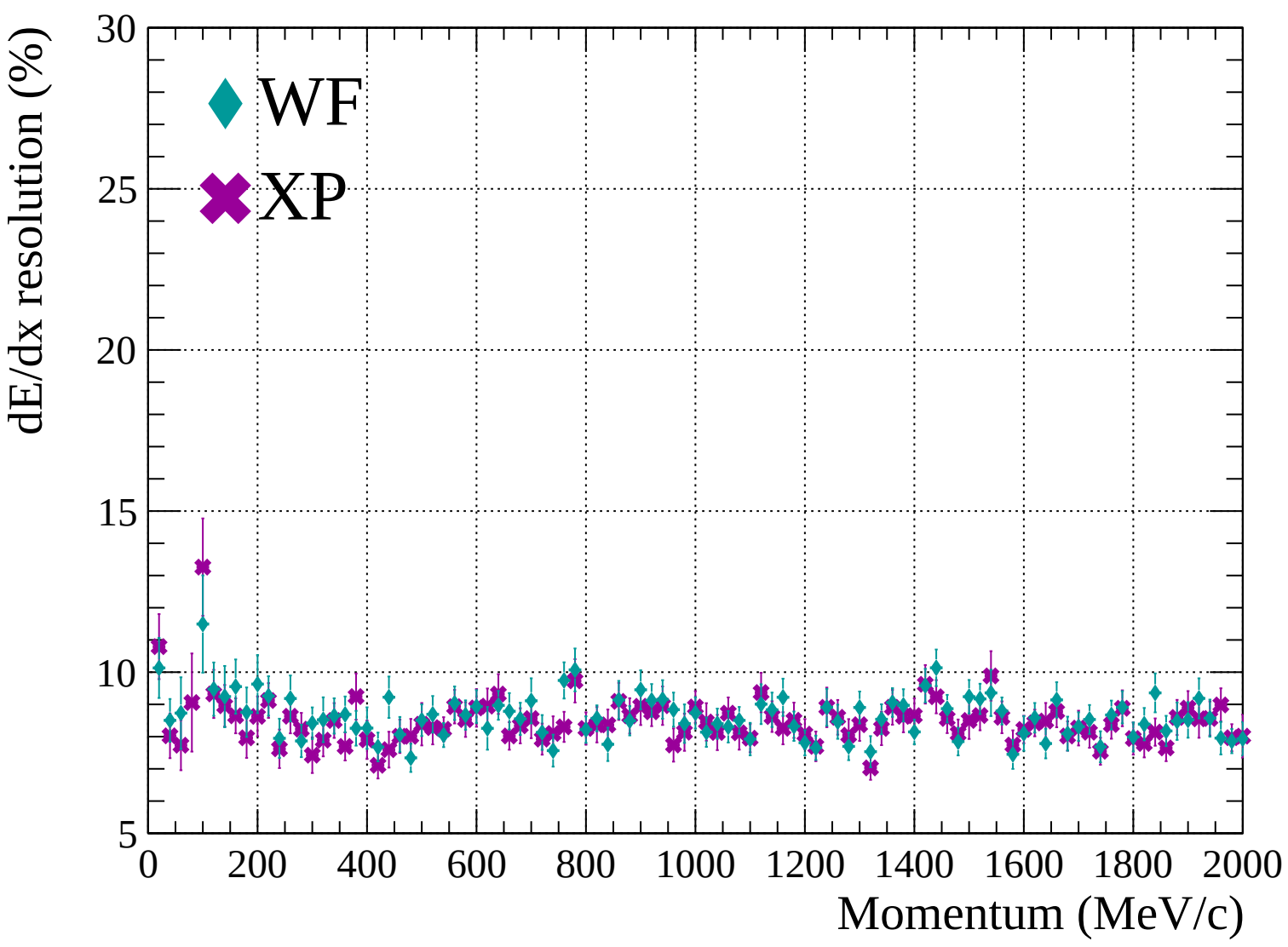


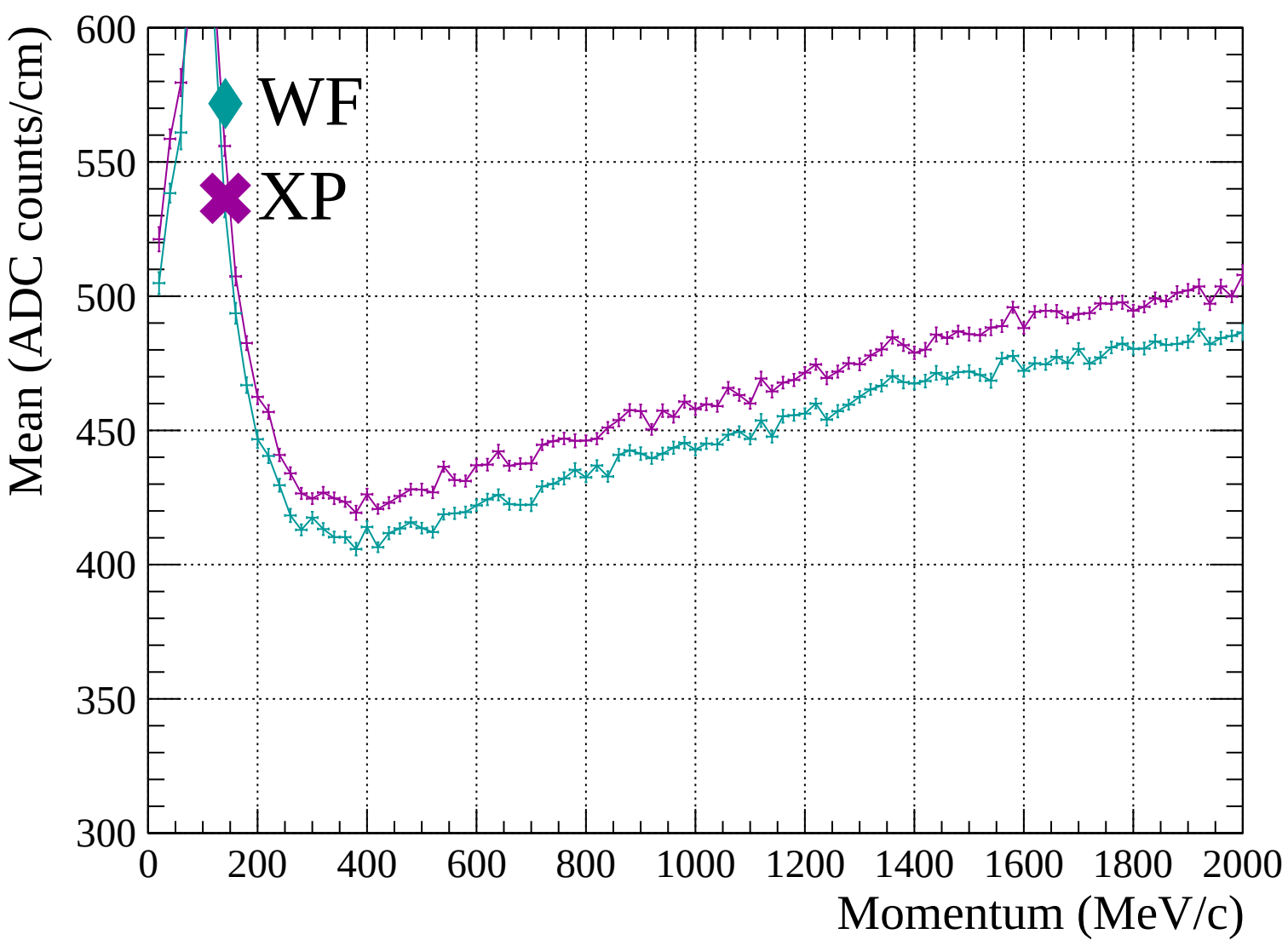


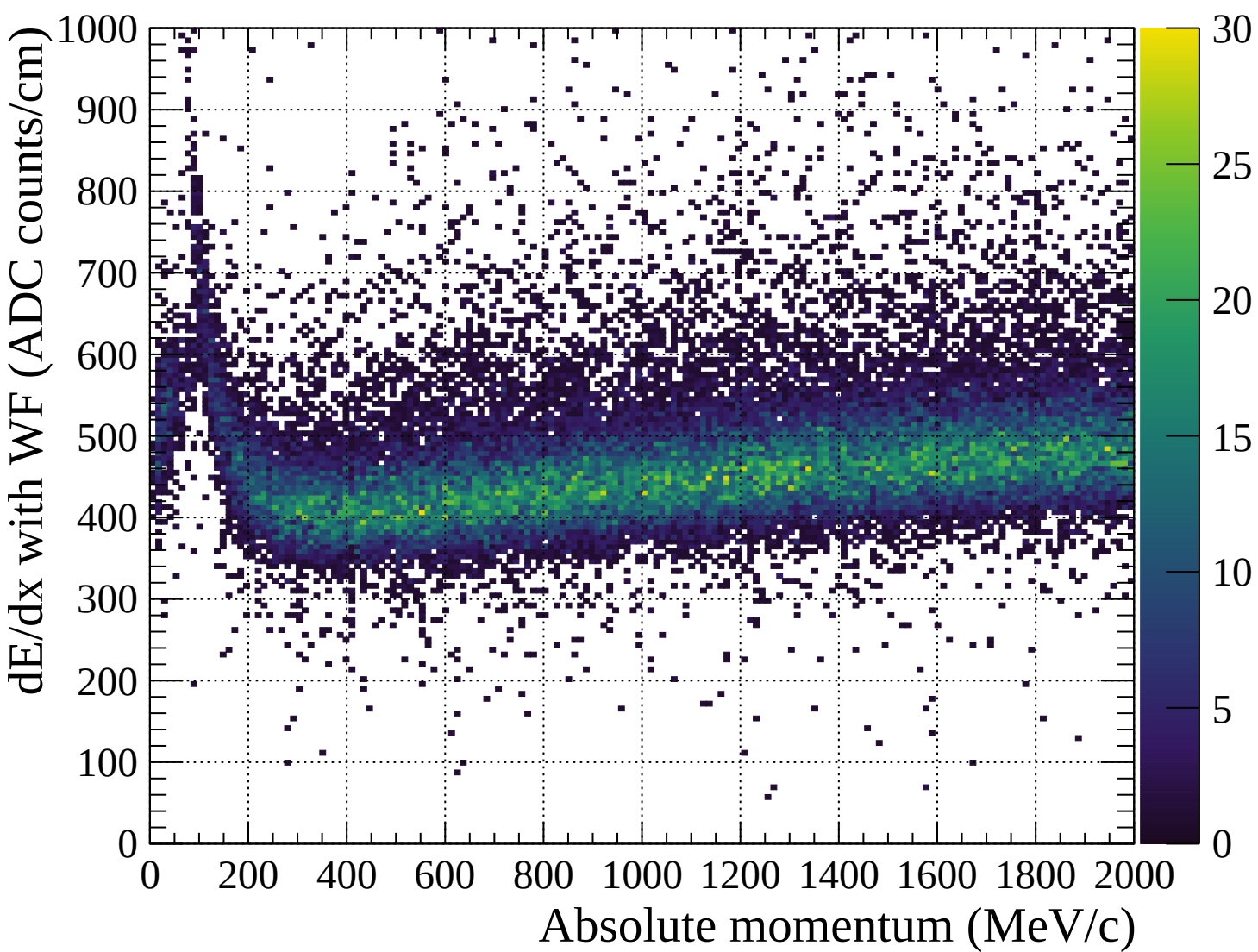


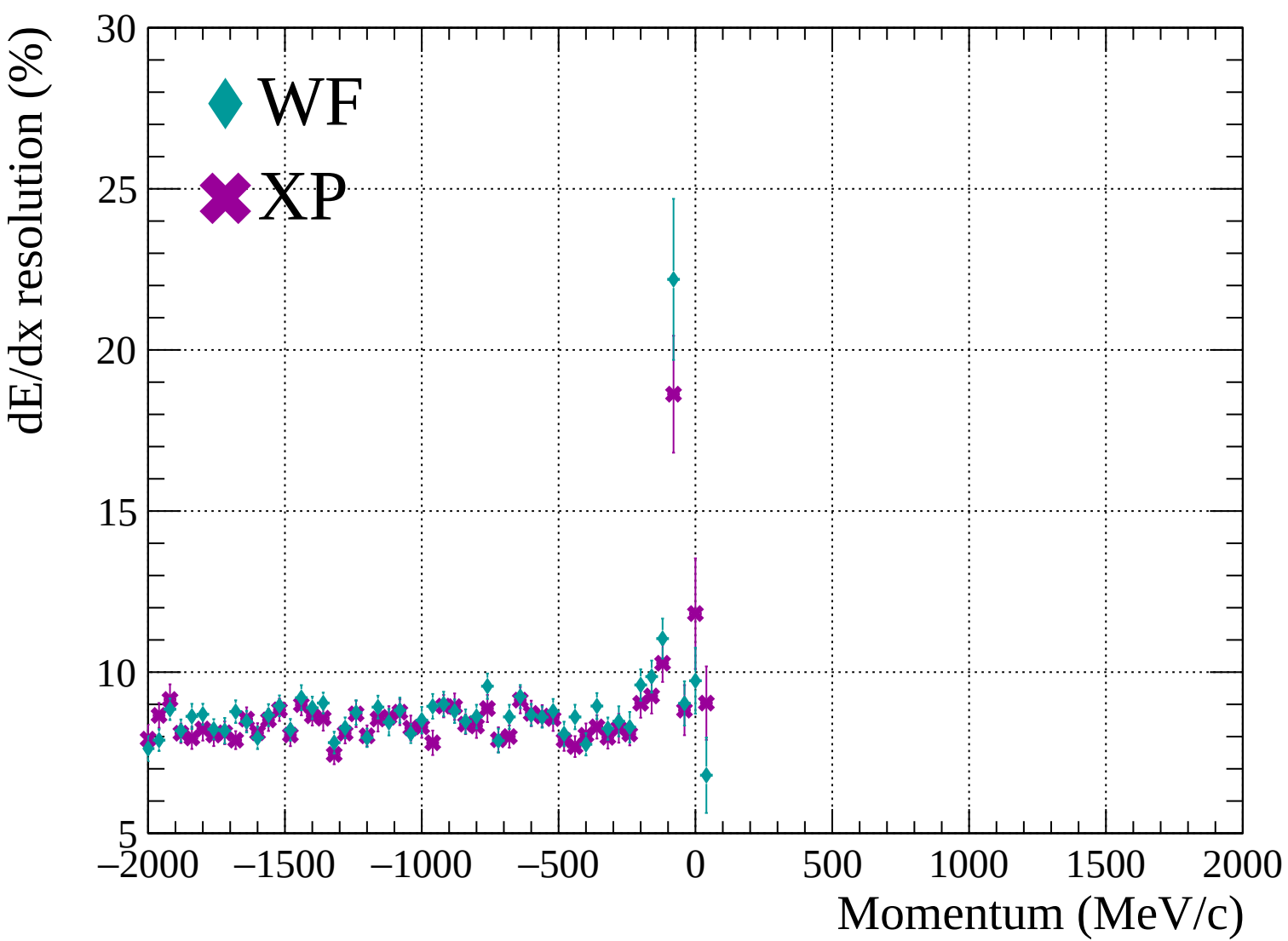


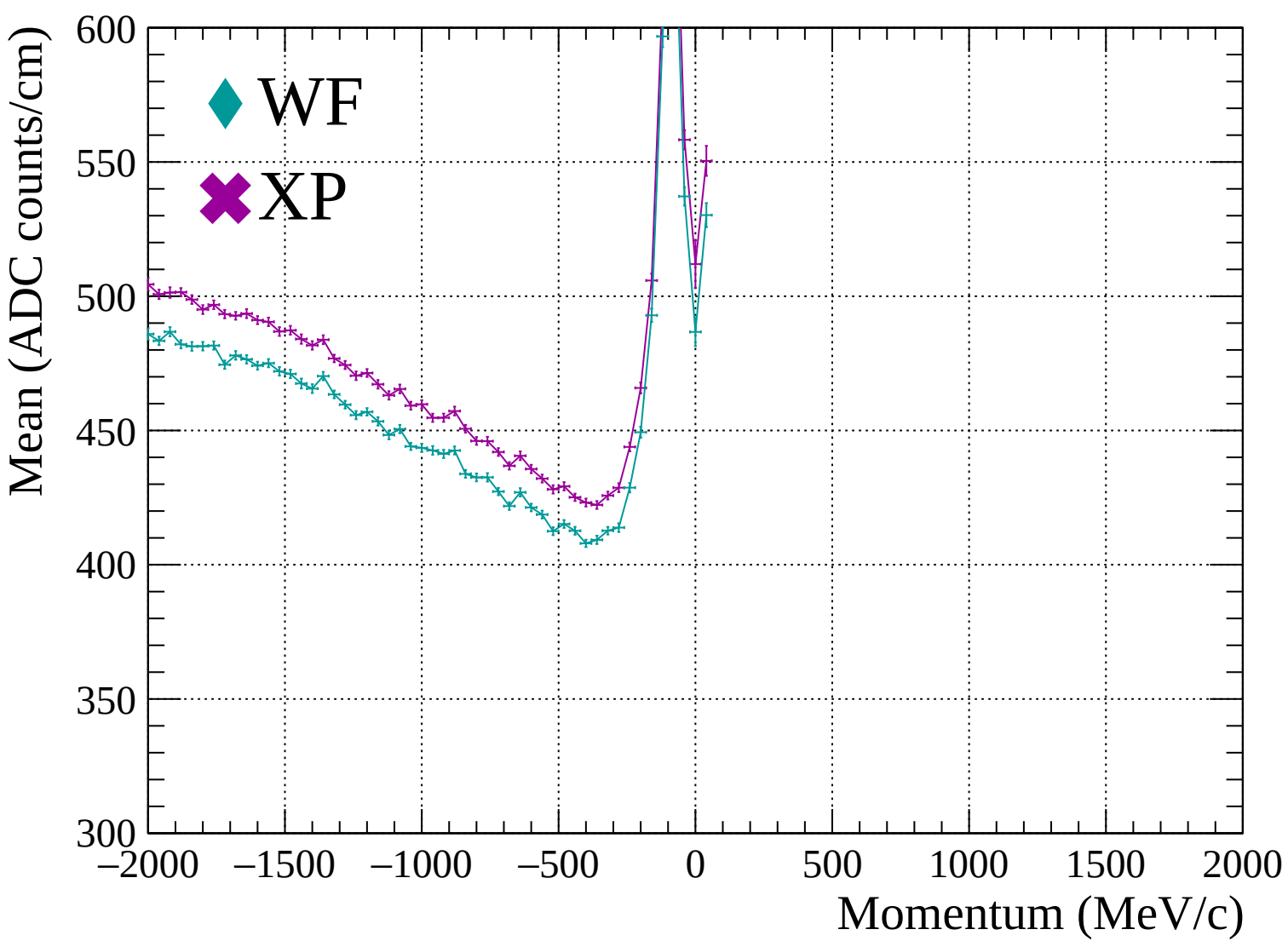


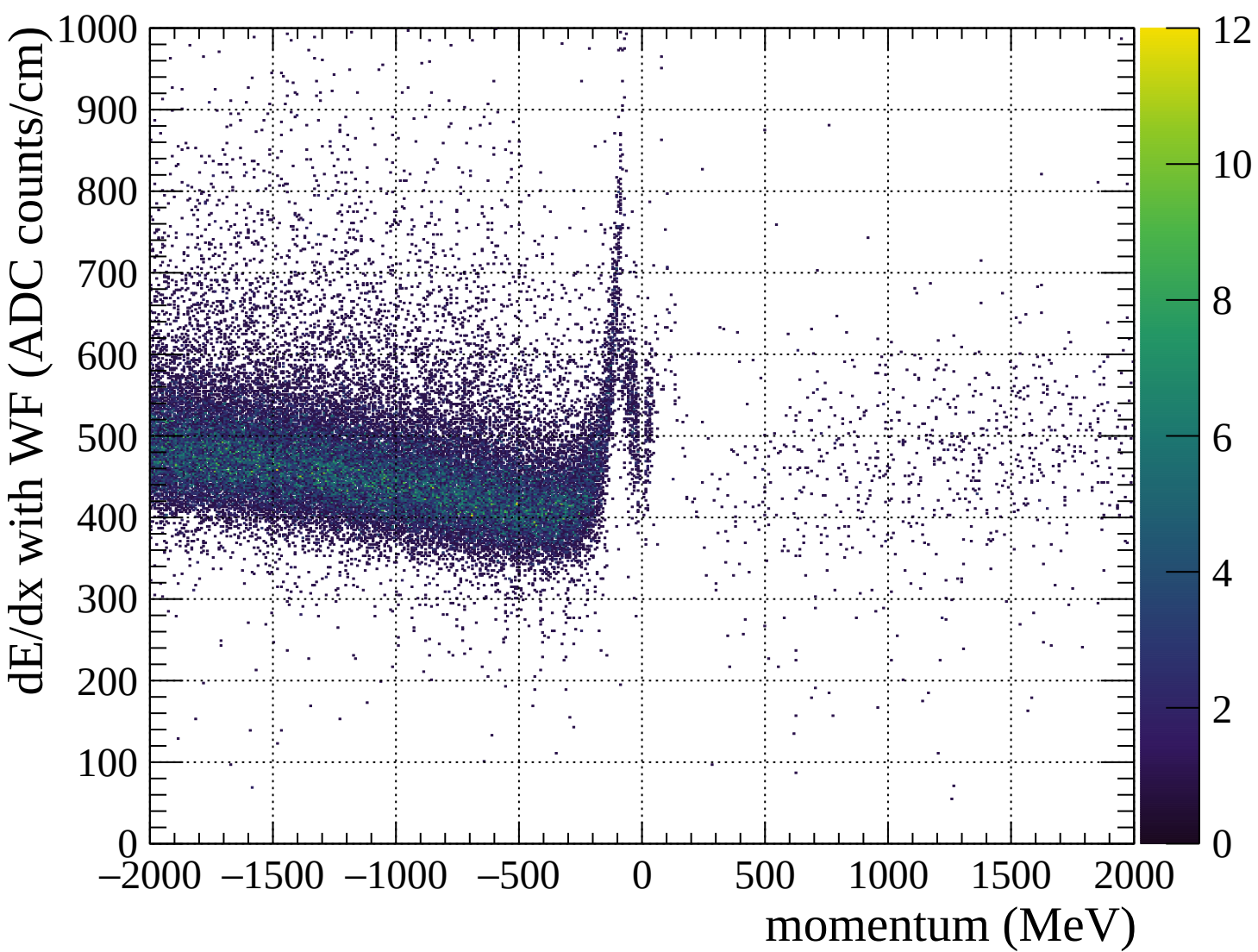


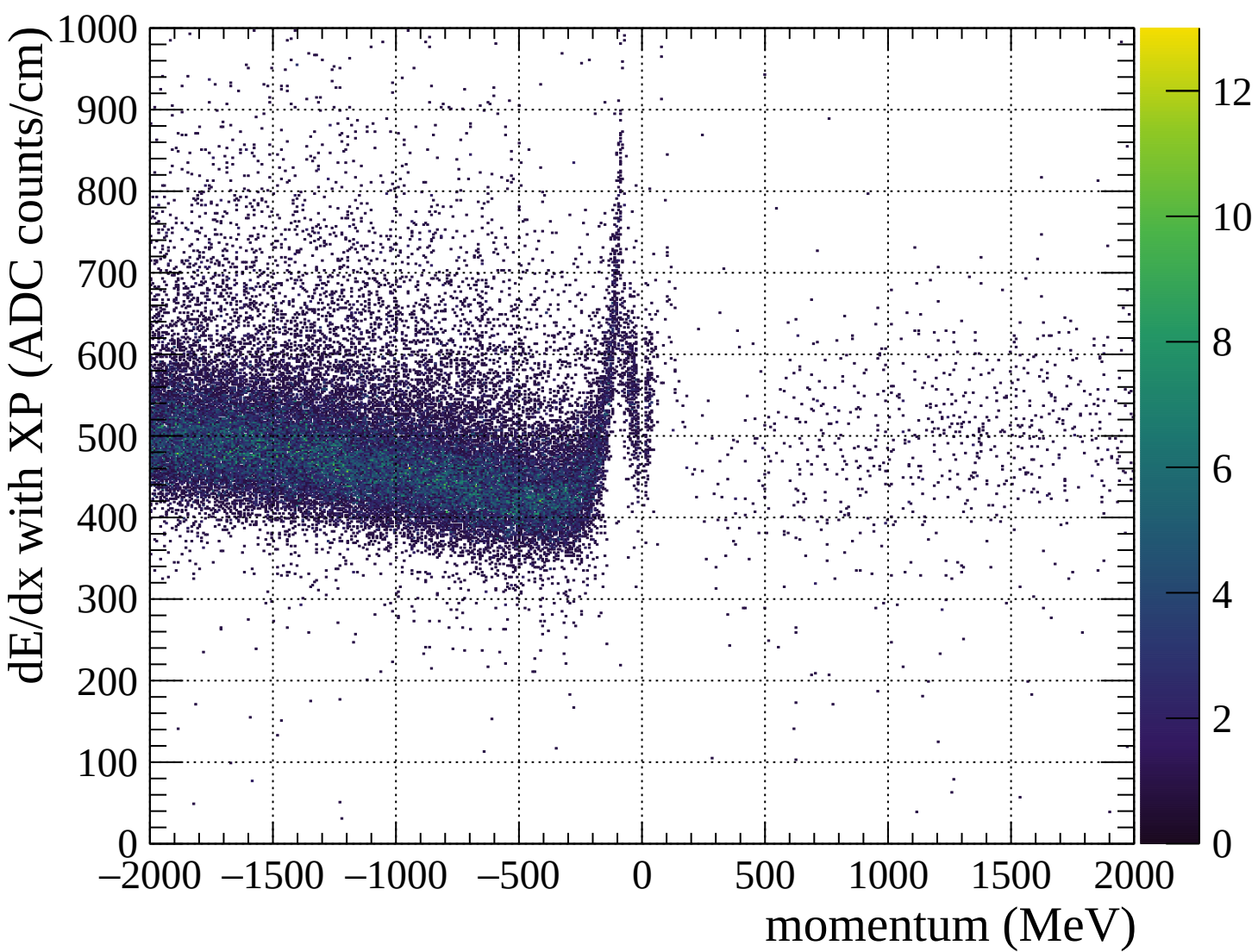


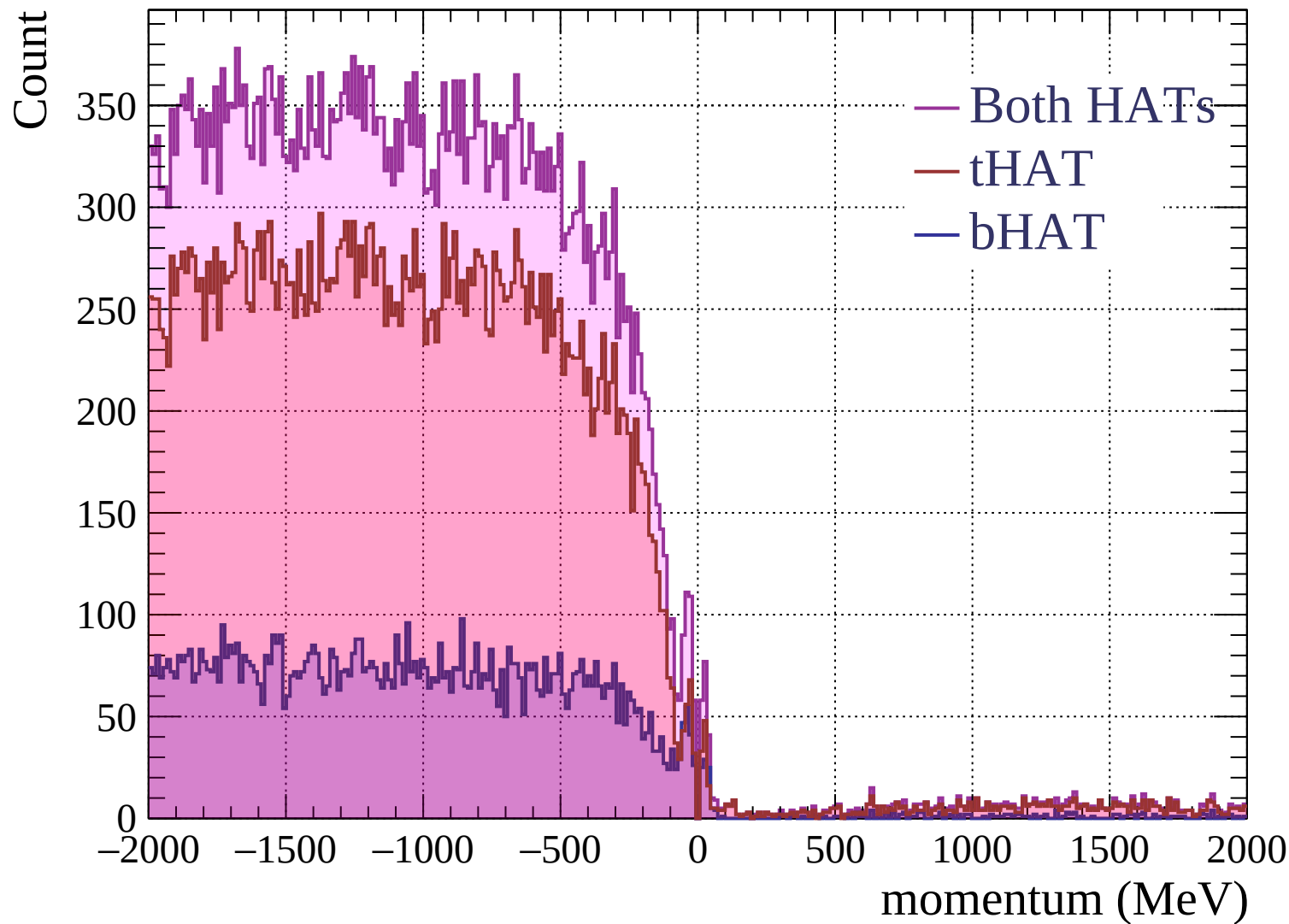


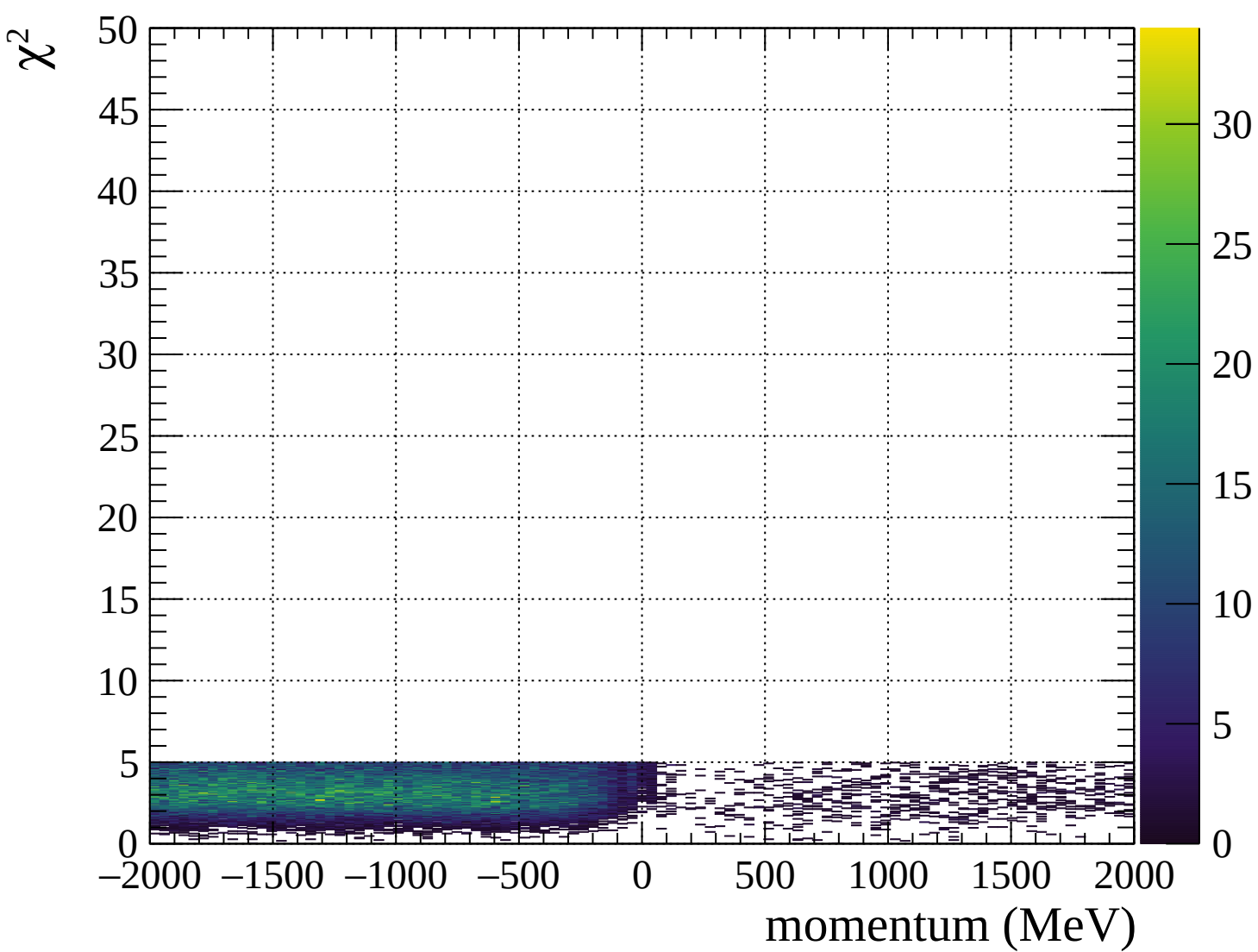




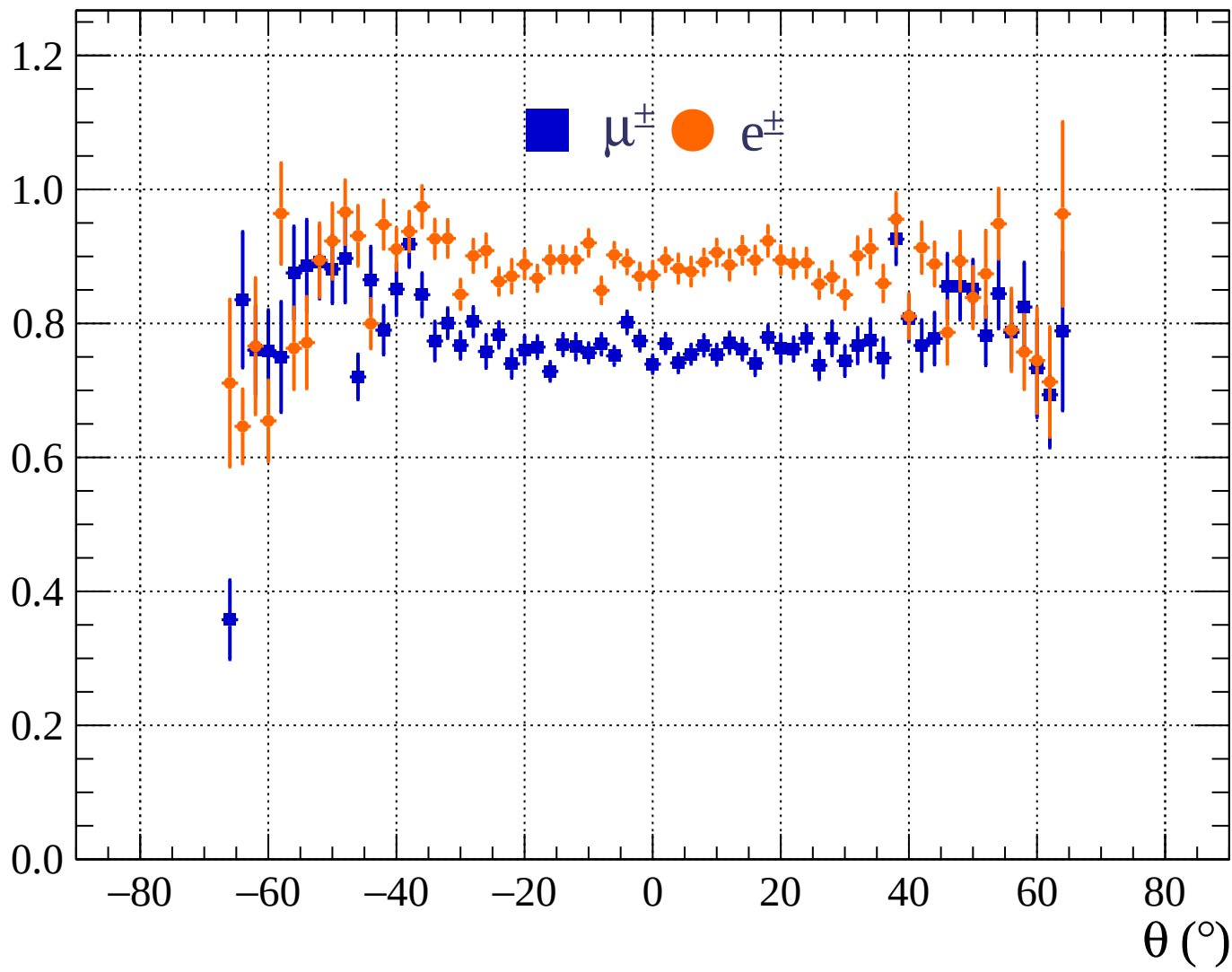


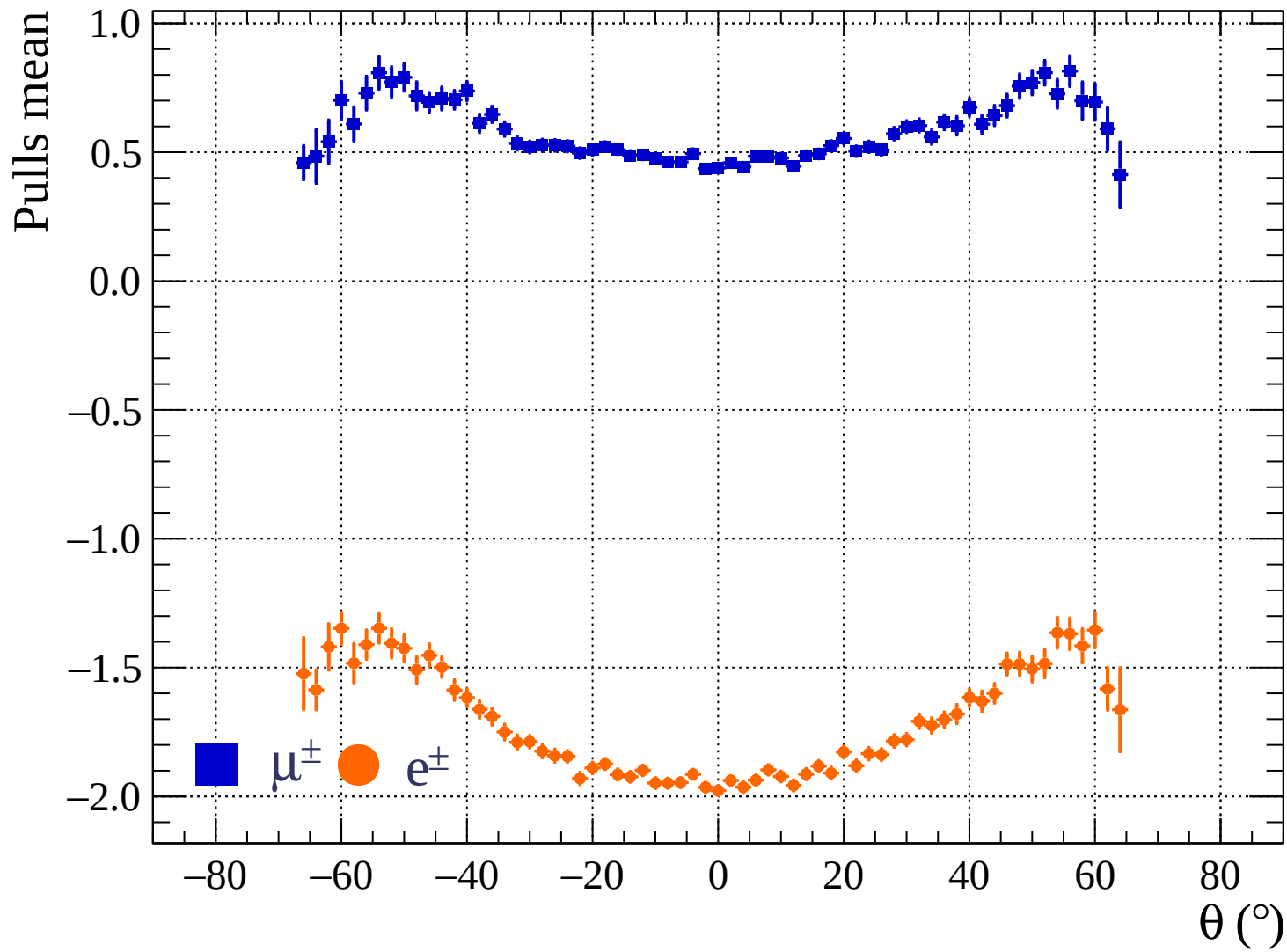




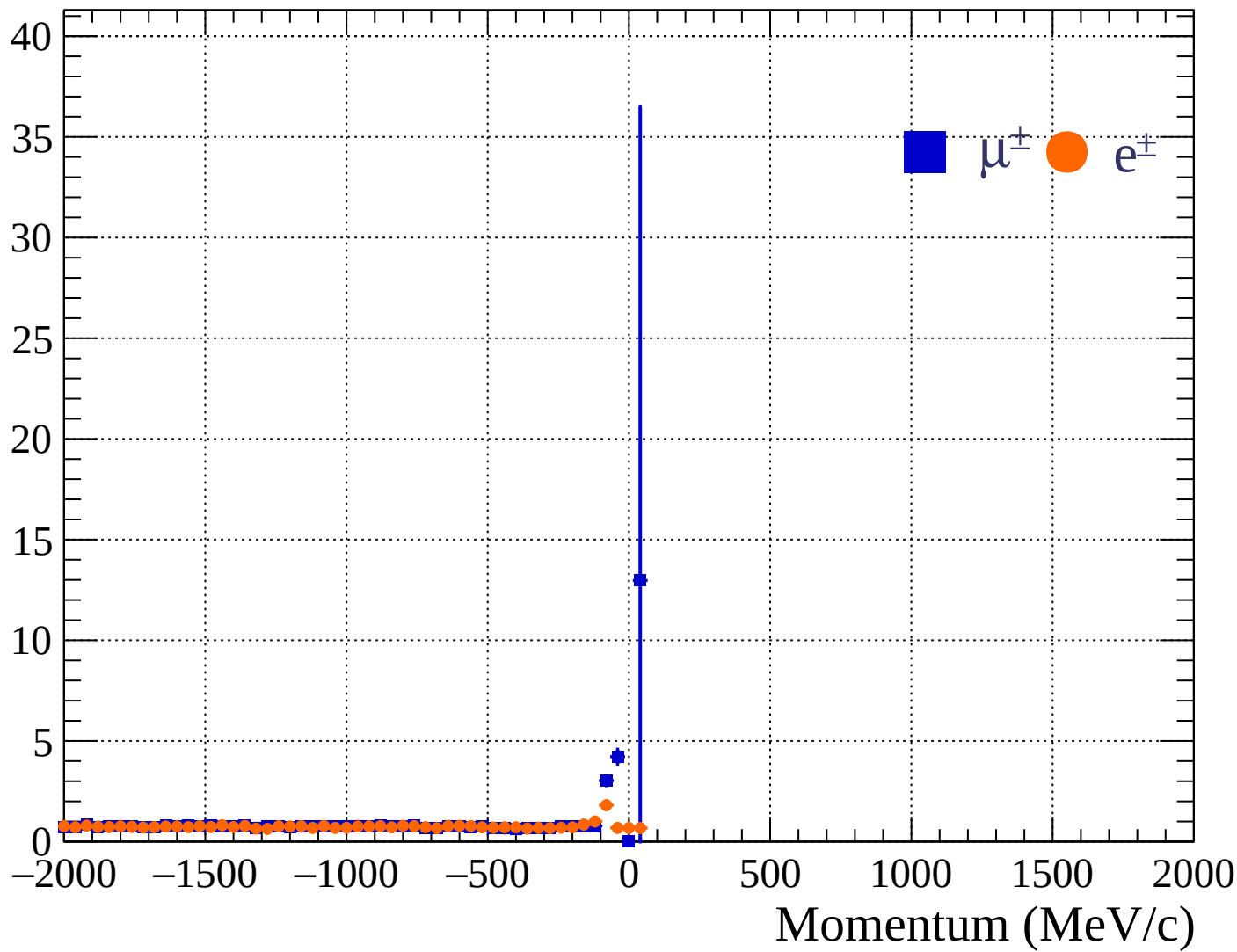


Pulls standard deviation

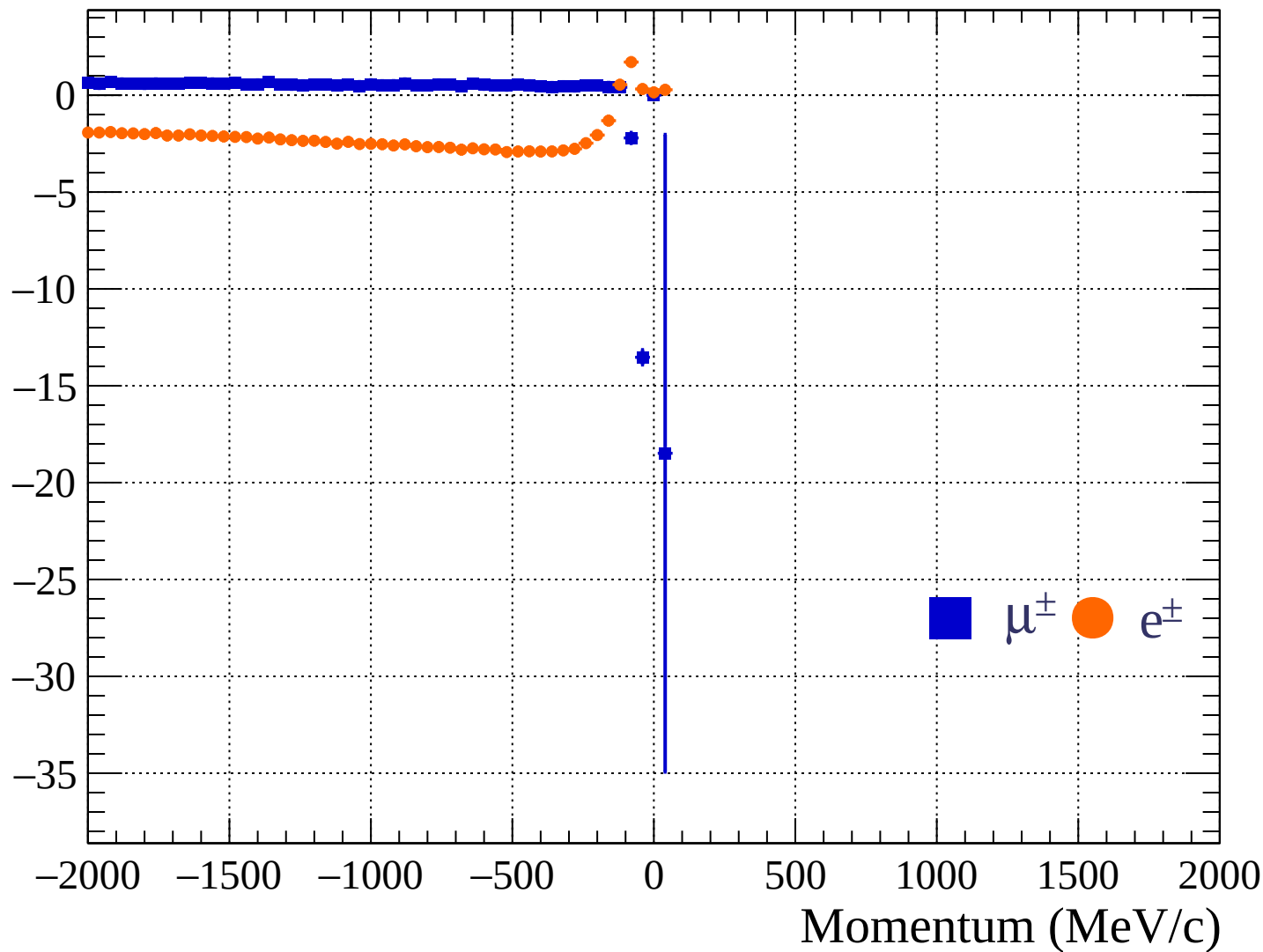


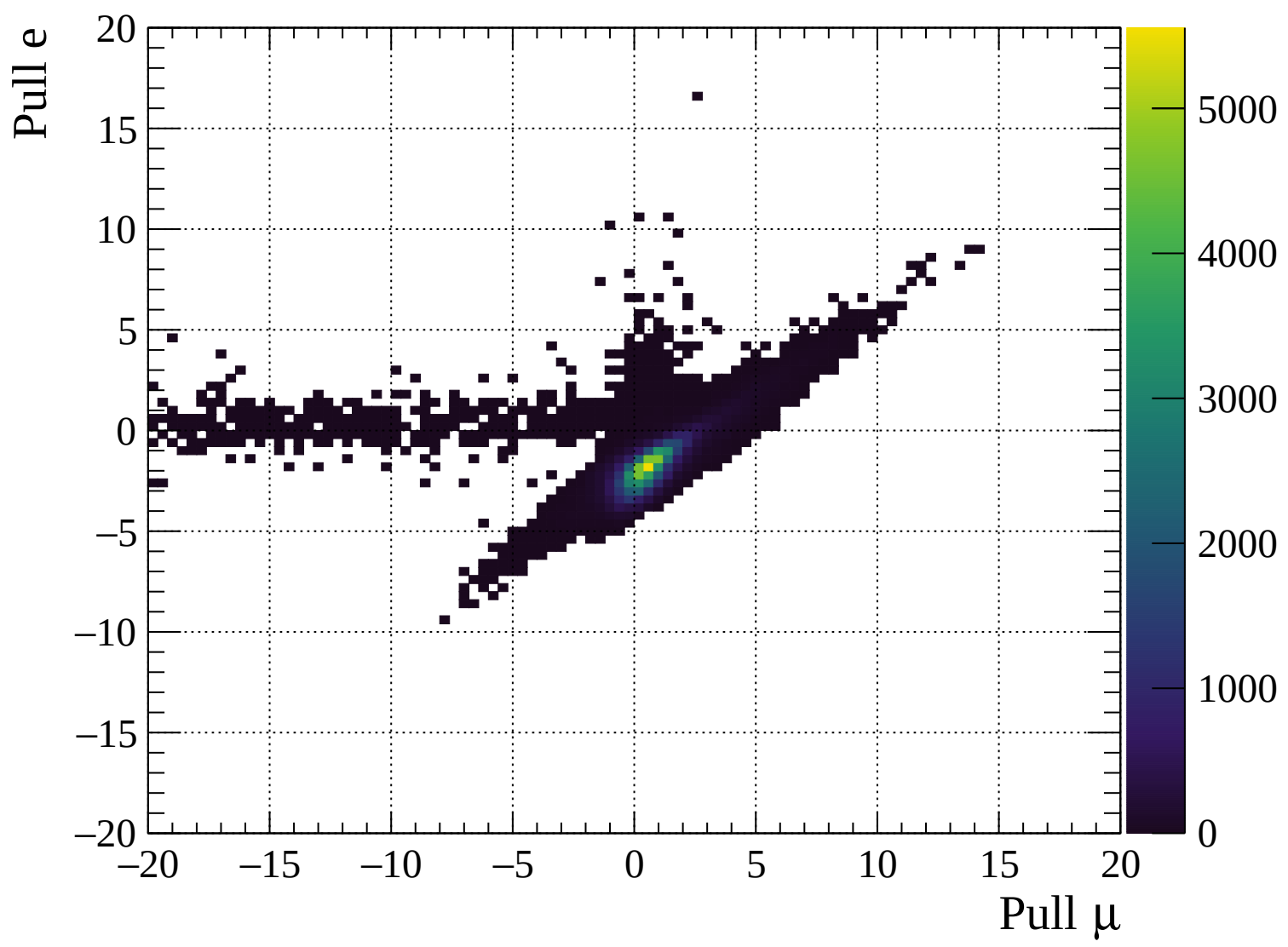


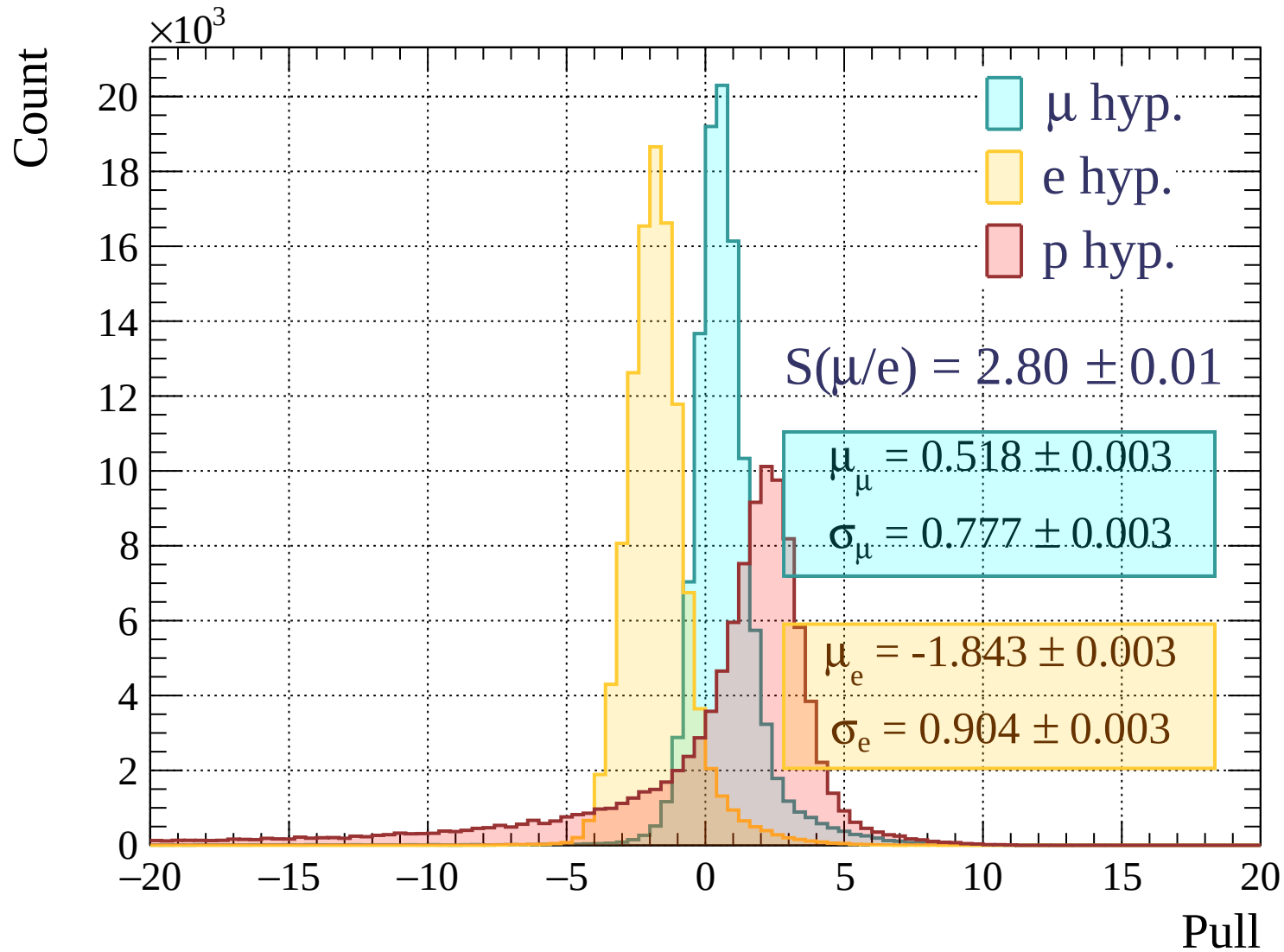
Pulls standard deviation

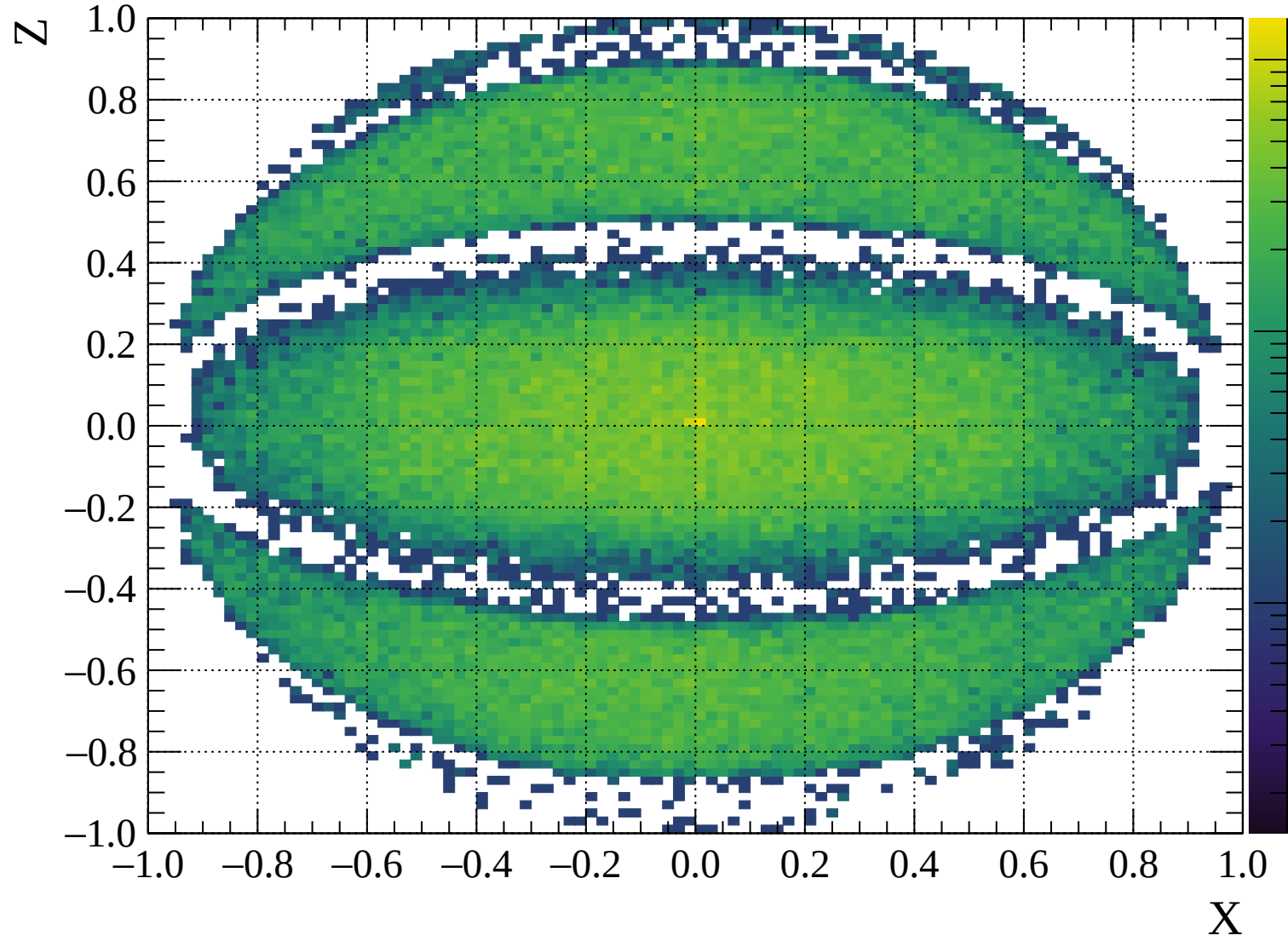


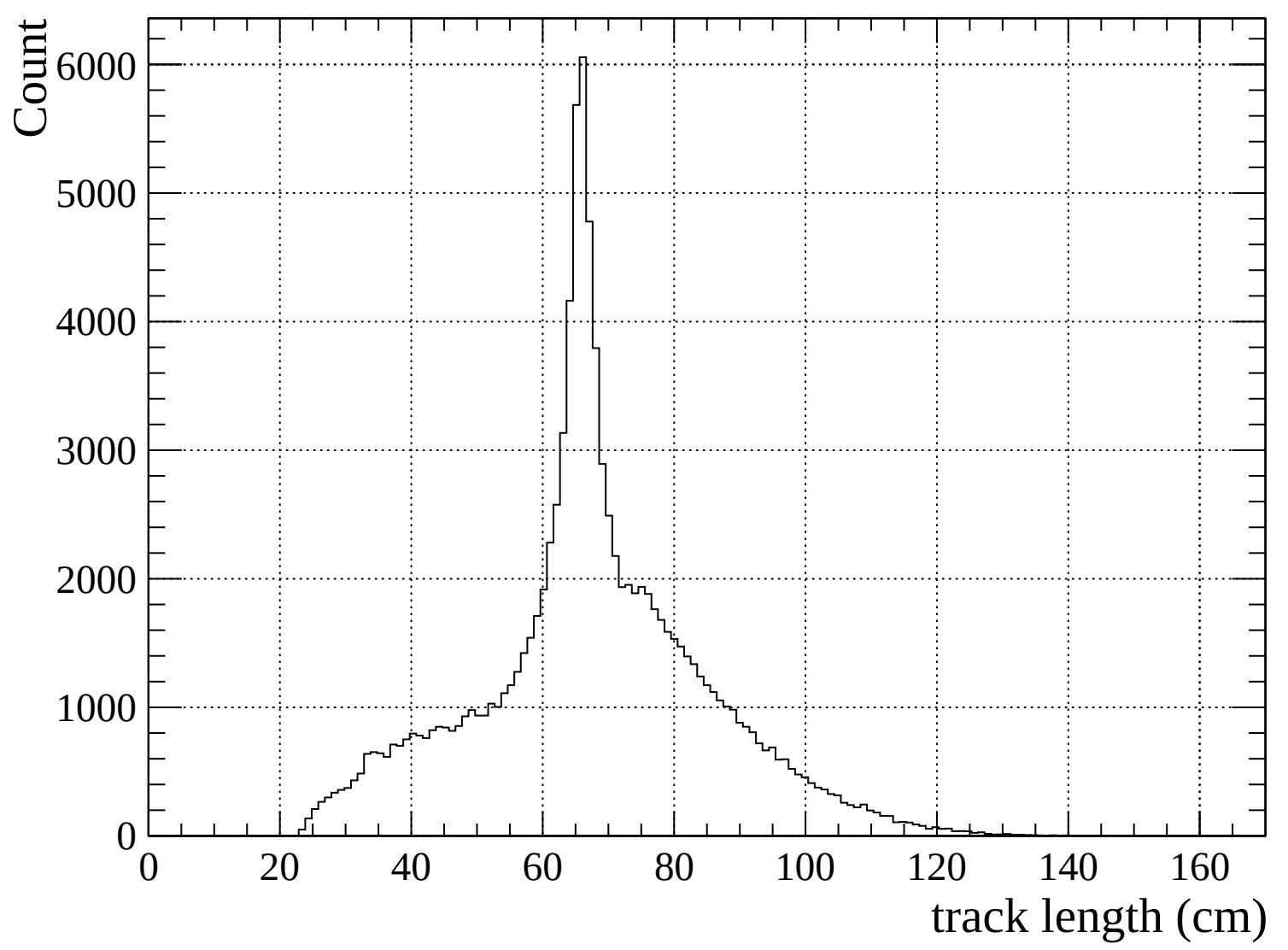
Pulls mean

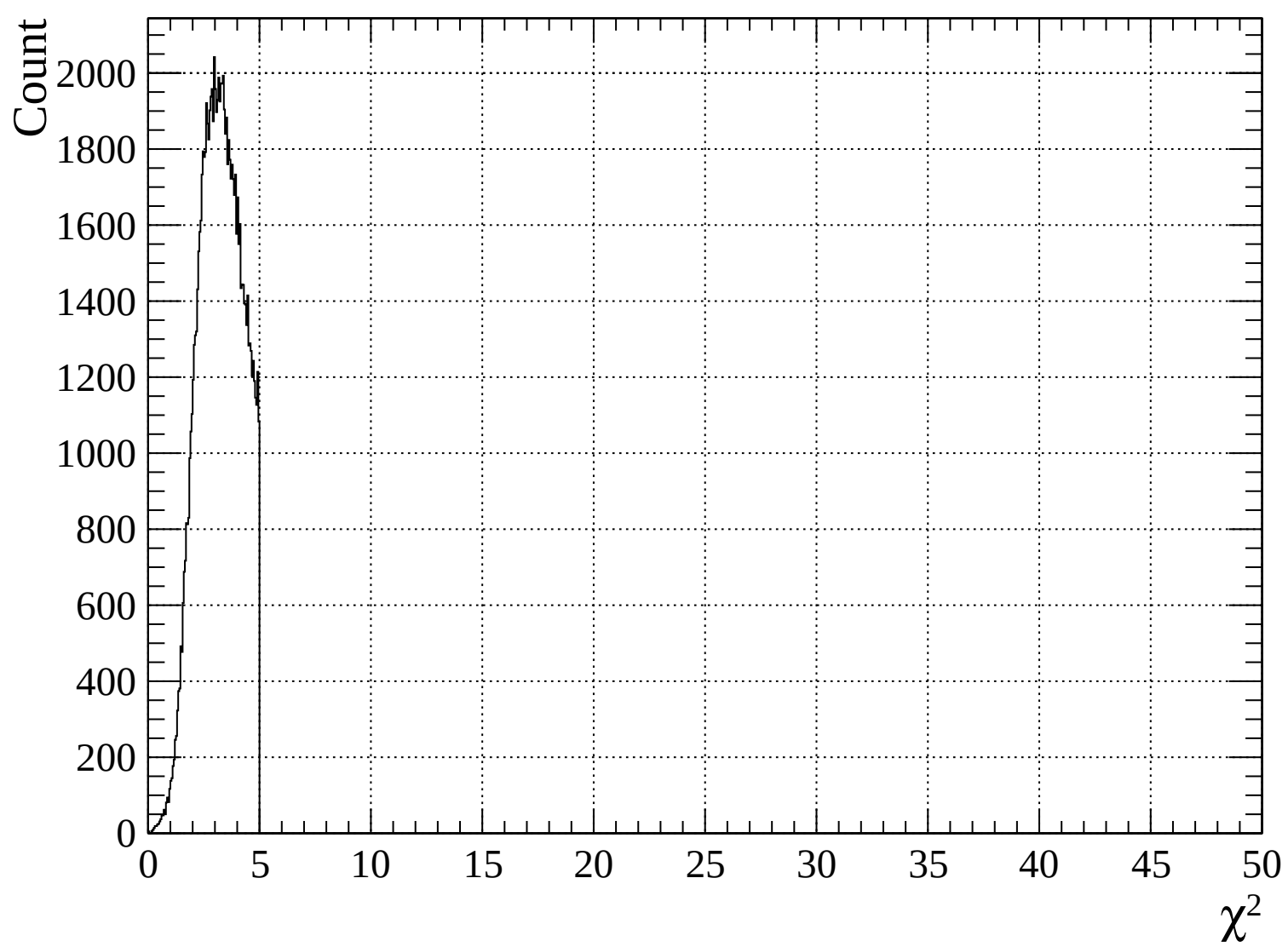


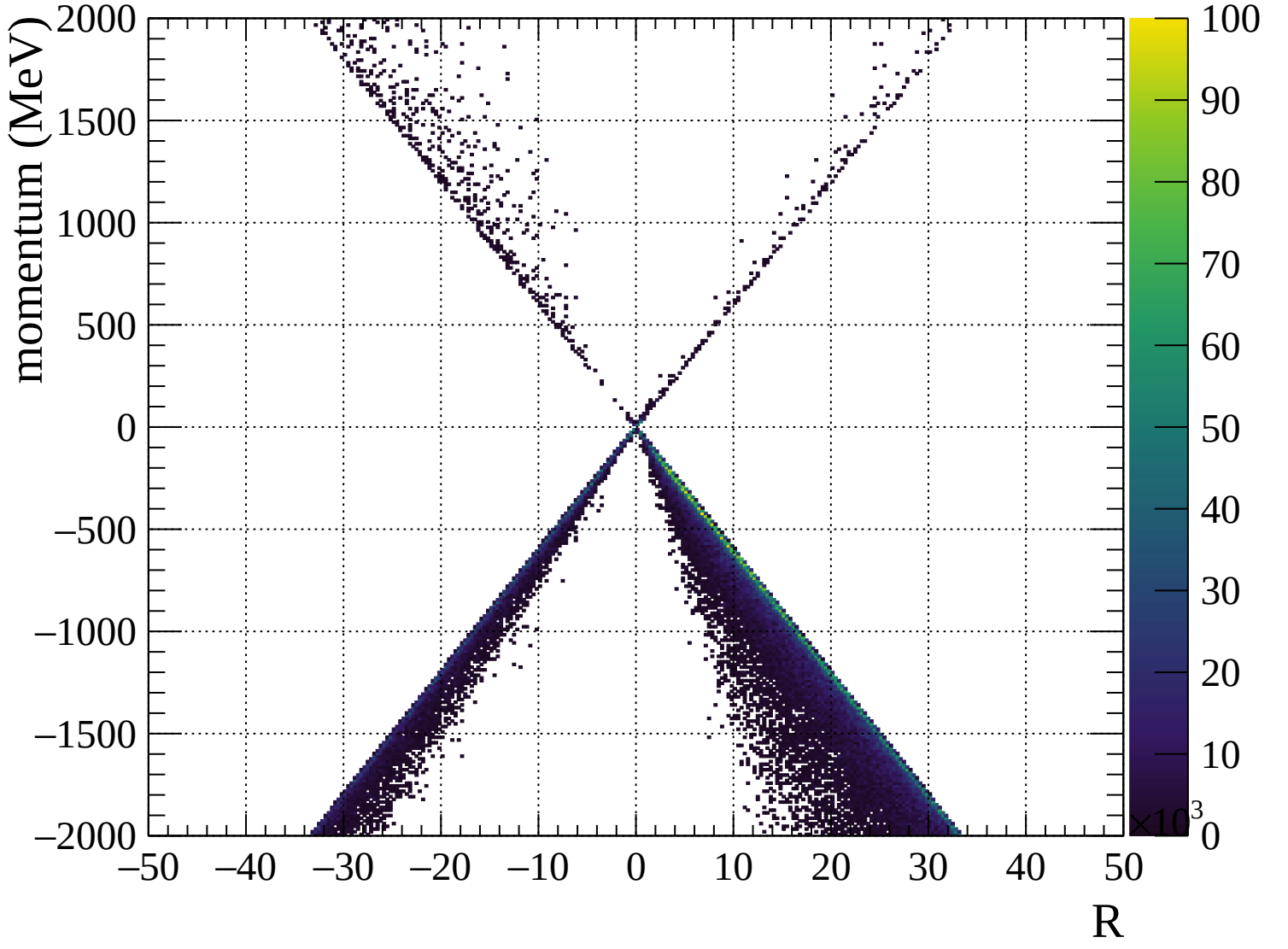


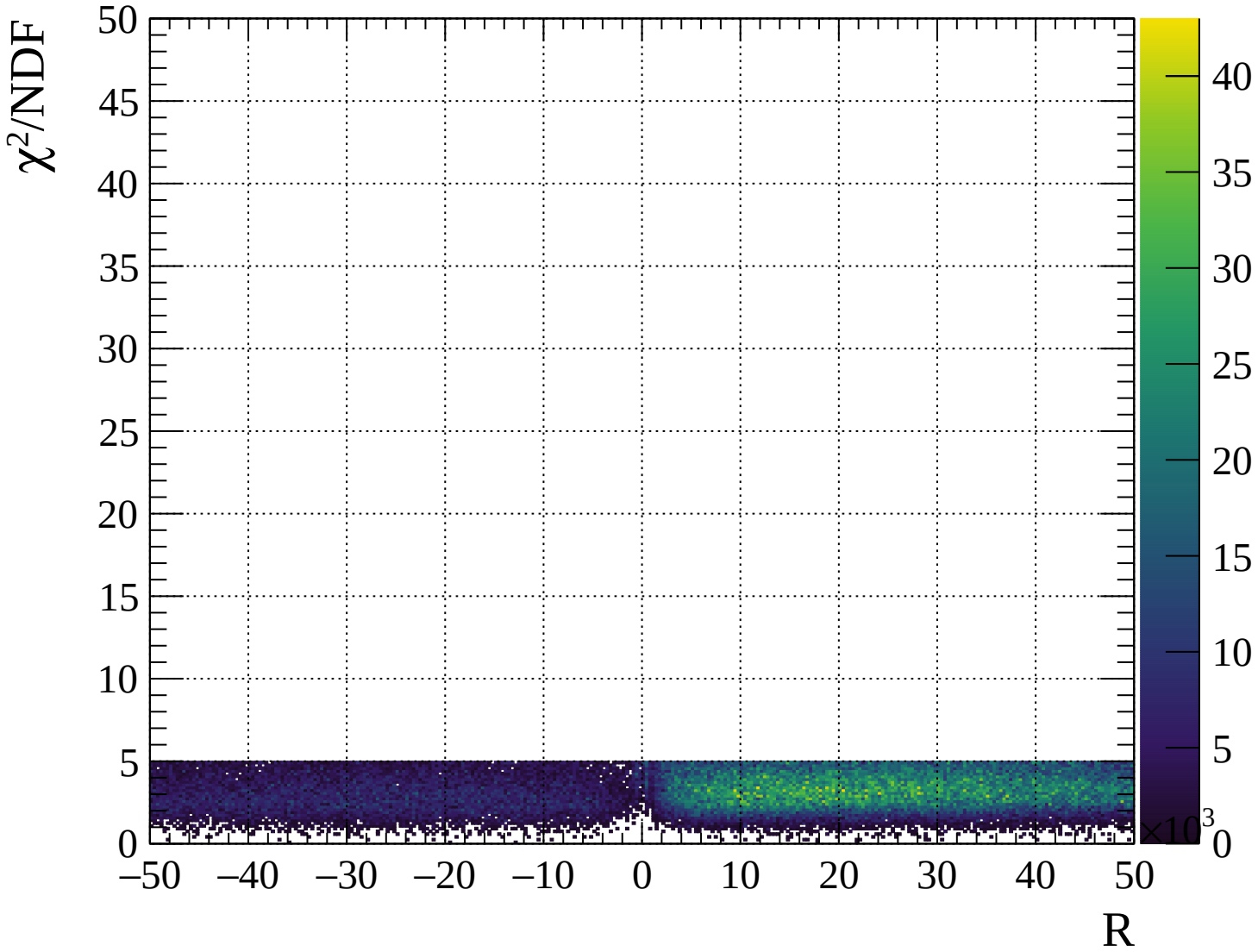






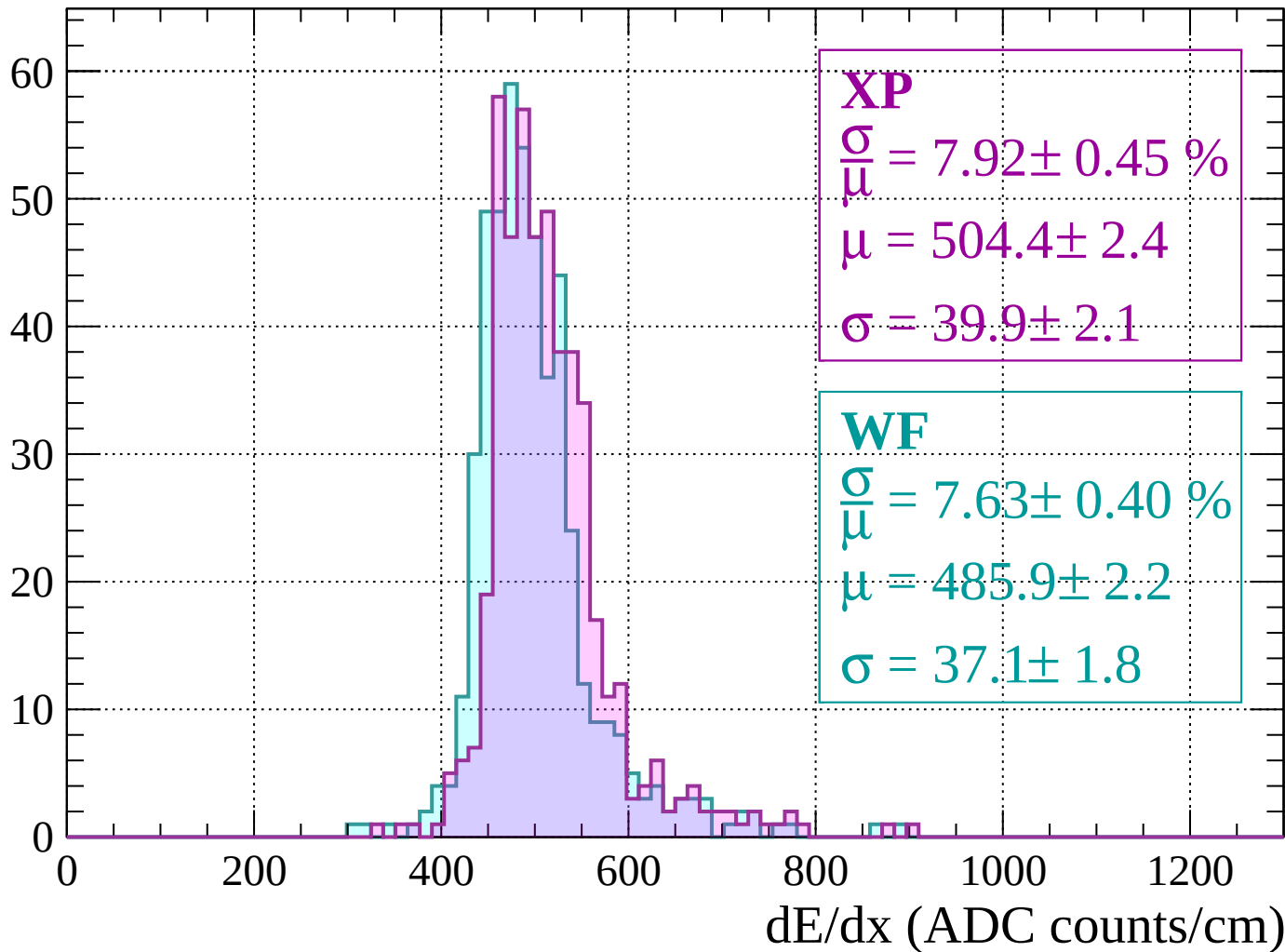






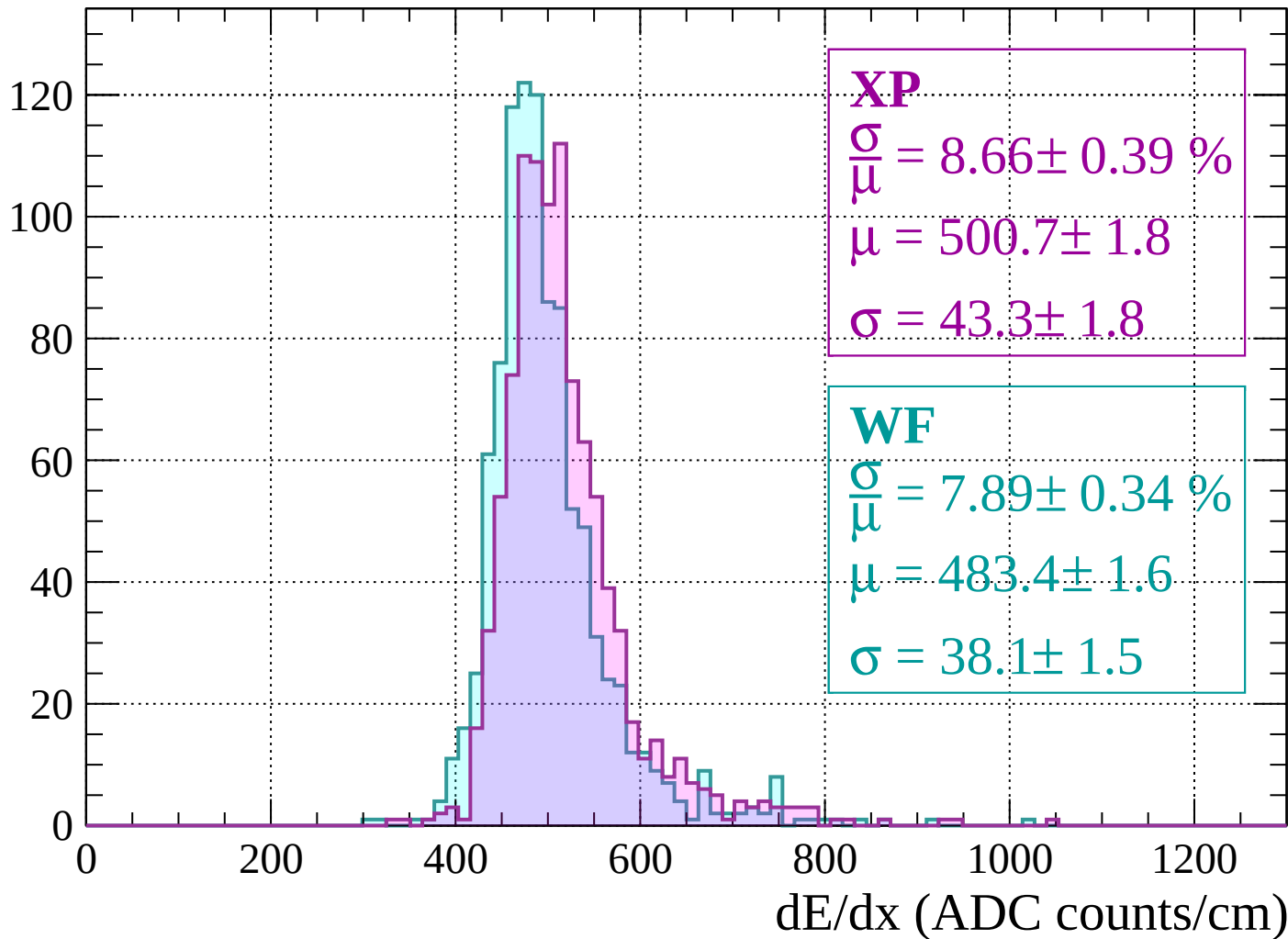
Energy loss | $-2000 < p < -1960$

Count



Energy loss | -1960 < p < -1920

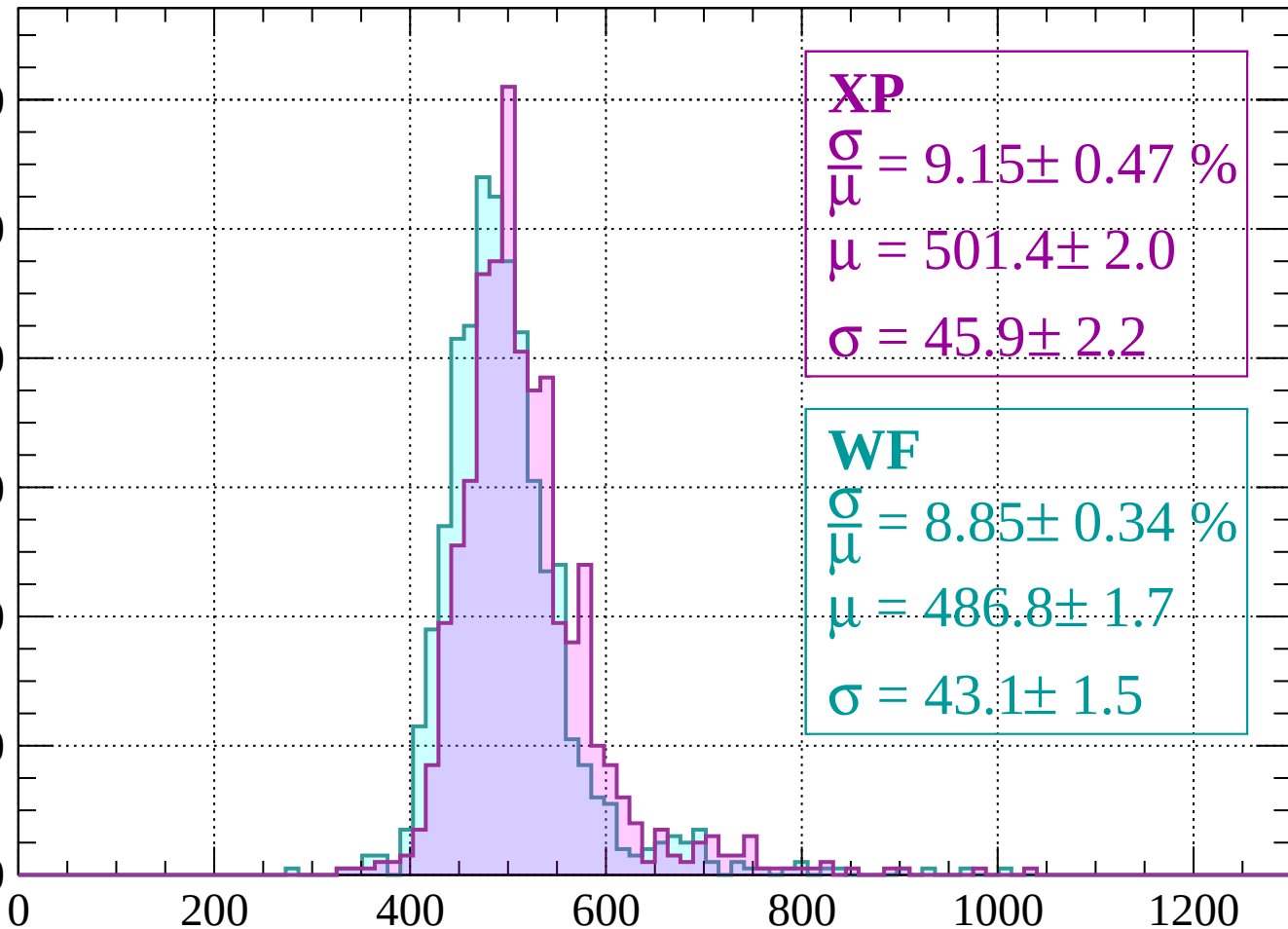
Count



Energy loss | -1920 < p < -1880

Count

120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 9.15 \pm 0.47 \%$$

$$\mu = 501.4 \pm 2.0$$

$$\sigma = 45.9 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 8.85 \pm 0.34 \%$$

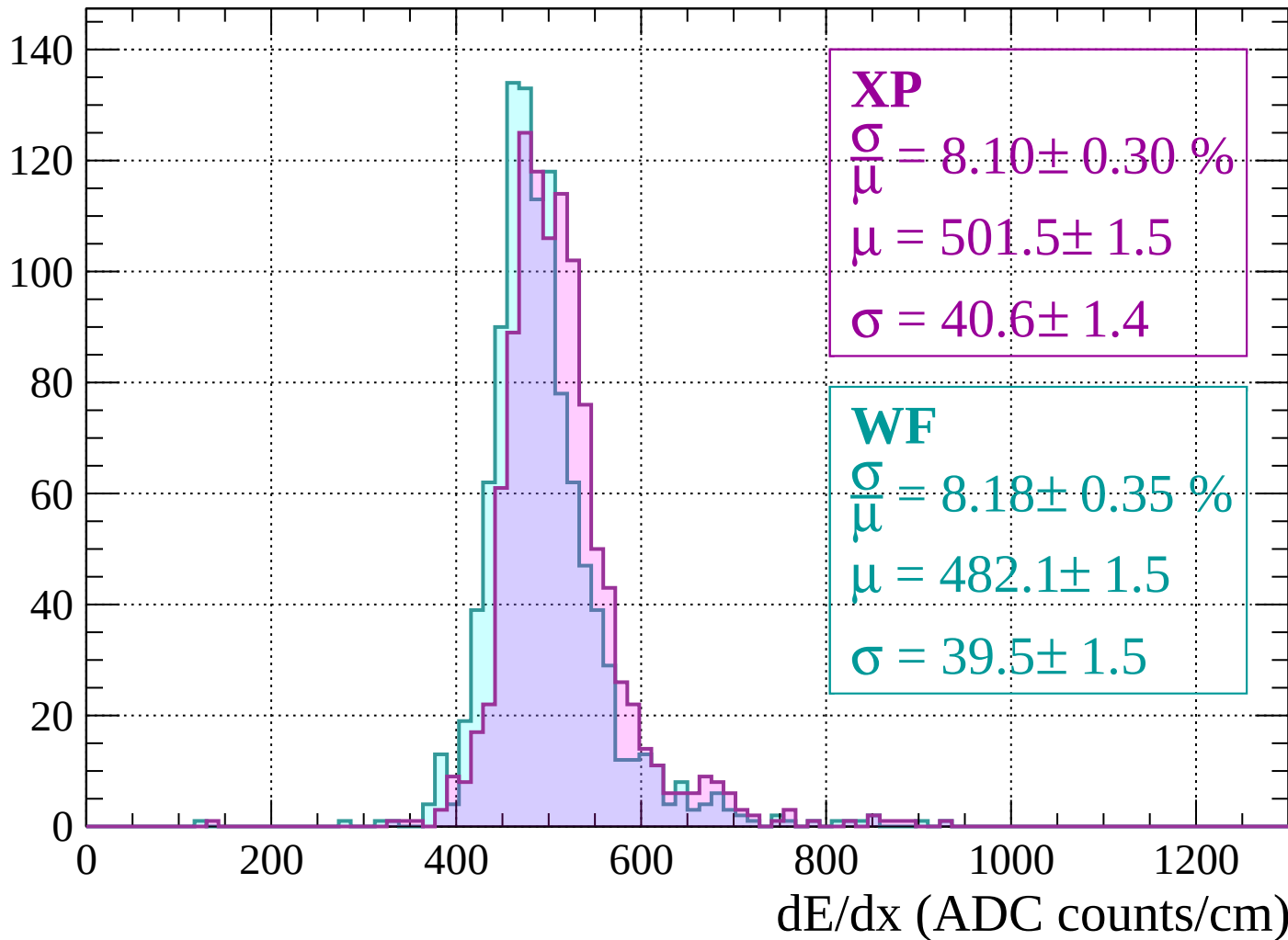
$$\mu = 486.8 \pm 1.7$$

$$\sigma = 43.1 \pm 1.5$$

dE/dx (ADC counts/cm)

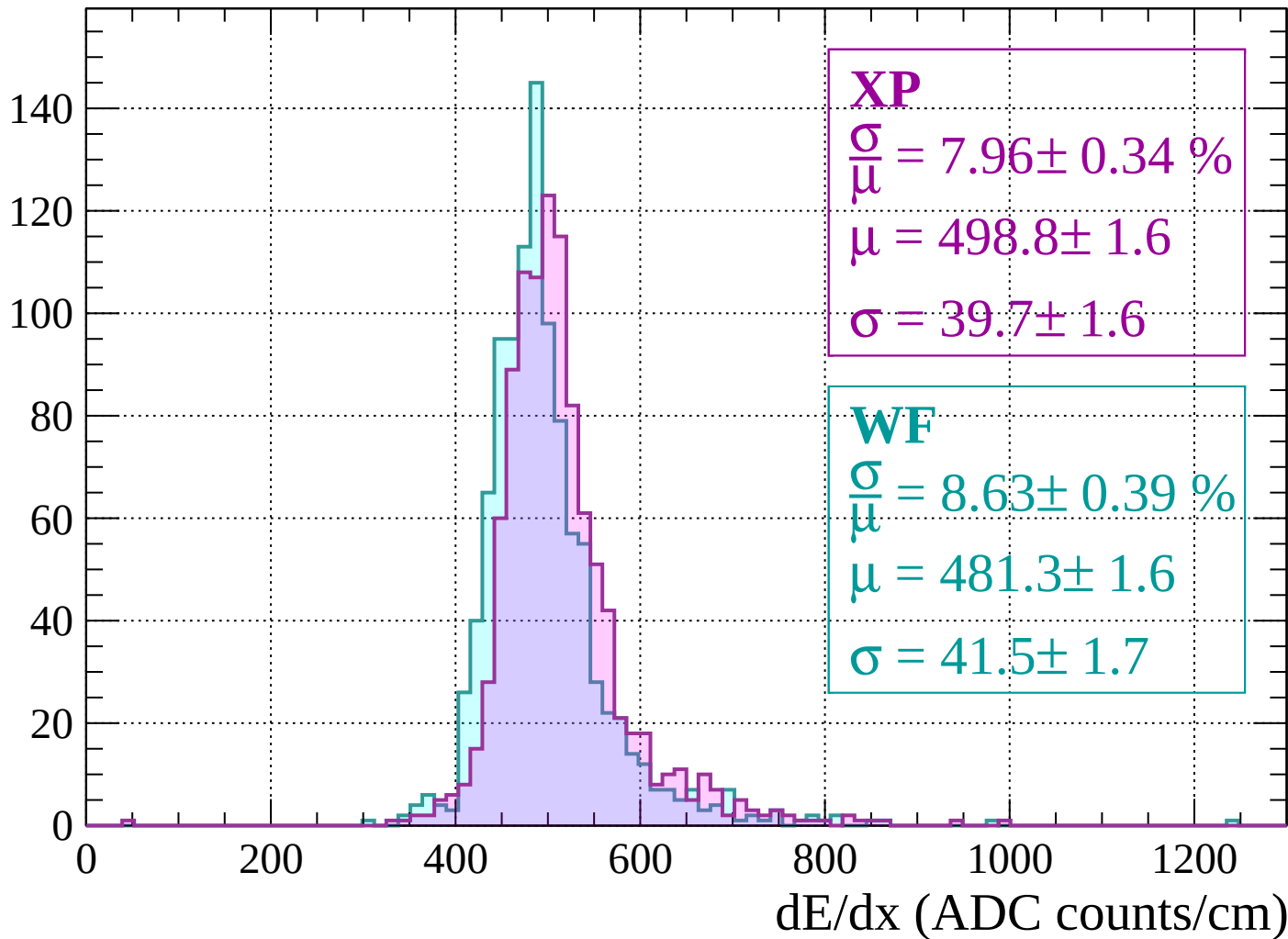
Energy loss | $-1880 < p < -1840$

Count



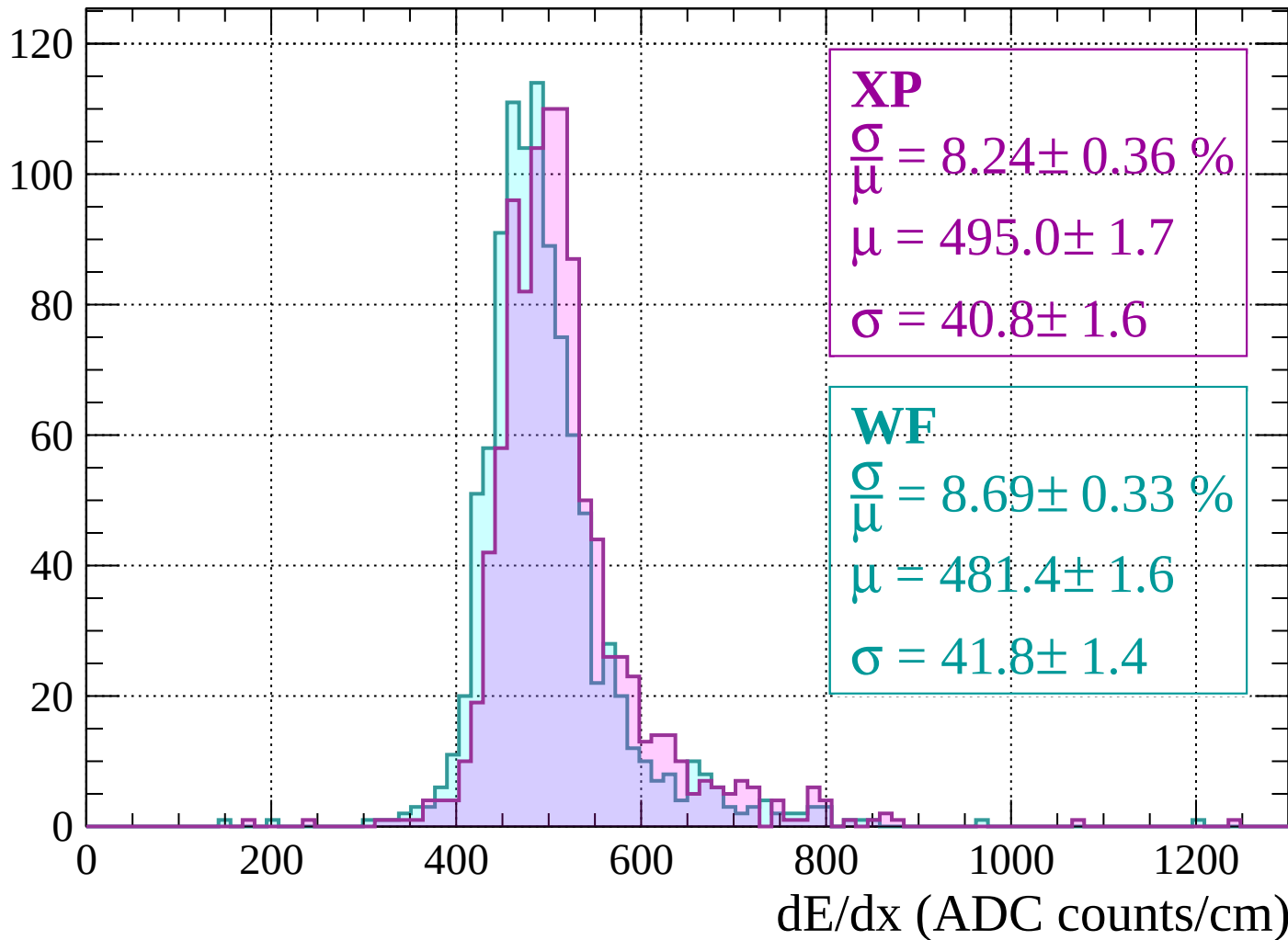
Energy loss | $-1840 < p < -1800$

Count



Energy loss | -1800 < p < -1760

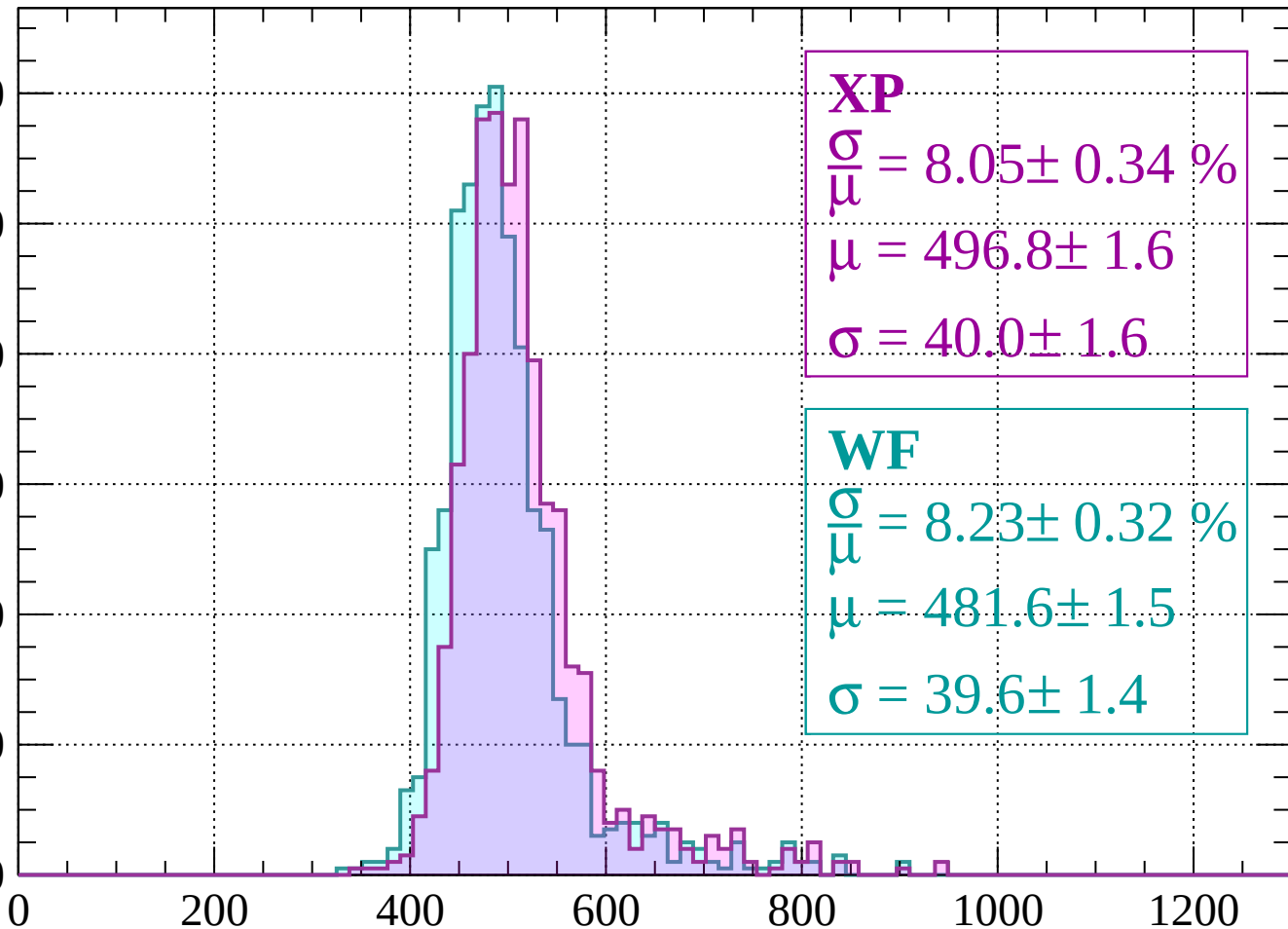
Count



Energy loss | $-1760 < p < -1720$

Count

120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 8.05 \pm 0.34 \%$$

$$\mu = 496.8 \pm 1.6$$

$$\sigma = 40.0 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 8.23 \pm 0.32 \%$$

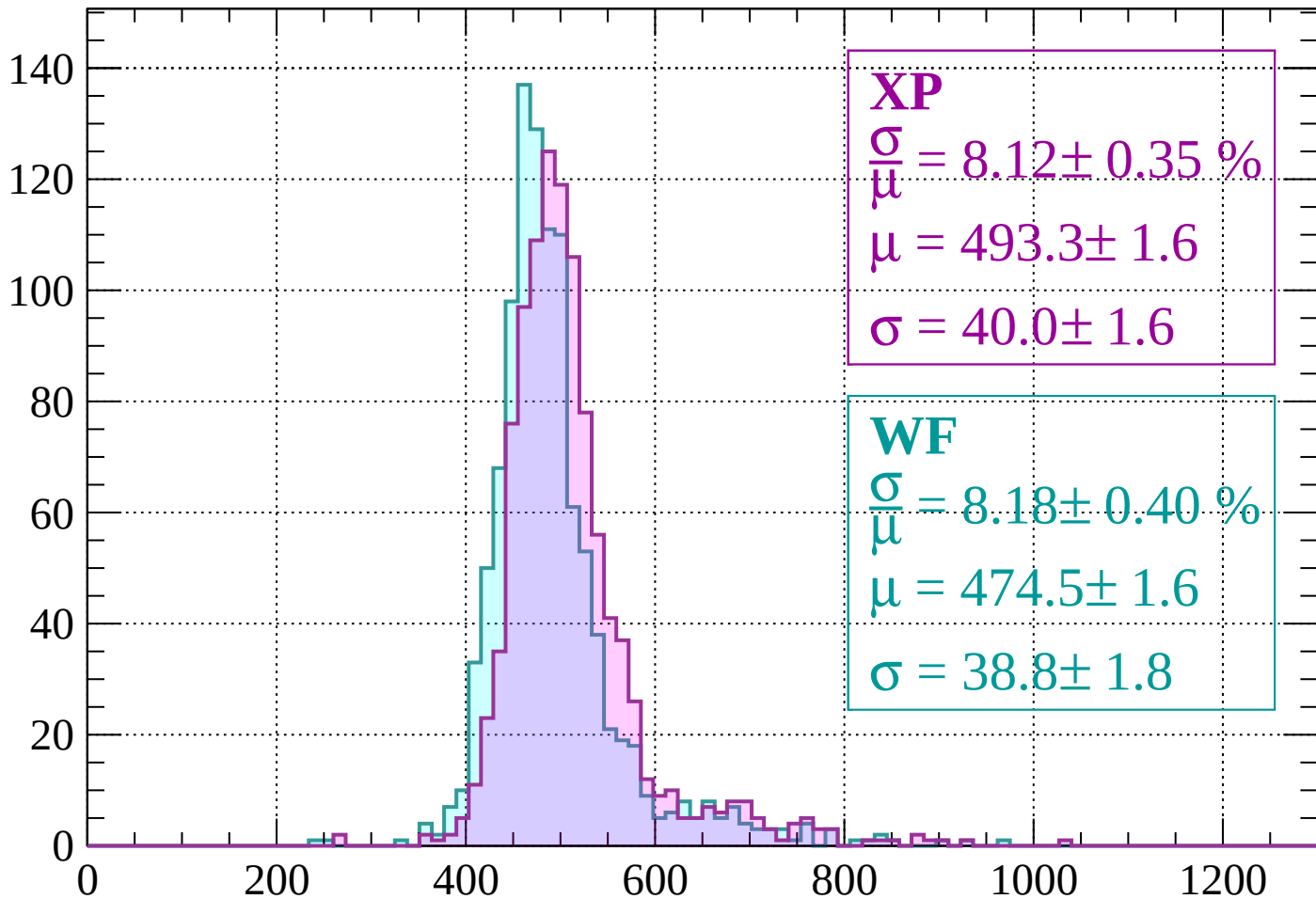
$$\mu = 481.6 \pm 1.5$$

$$\sigma = 39.6 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-1720 < p < -1680$

Count



XP

$$\frac{\sigma}{\mu} = 8.12 \pm 0.35 \%$$

$$\mu = 493.3 \pm 1.6$$

$$\sigma = 40.0 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 8.18 \pm 0.40 \%$$

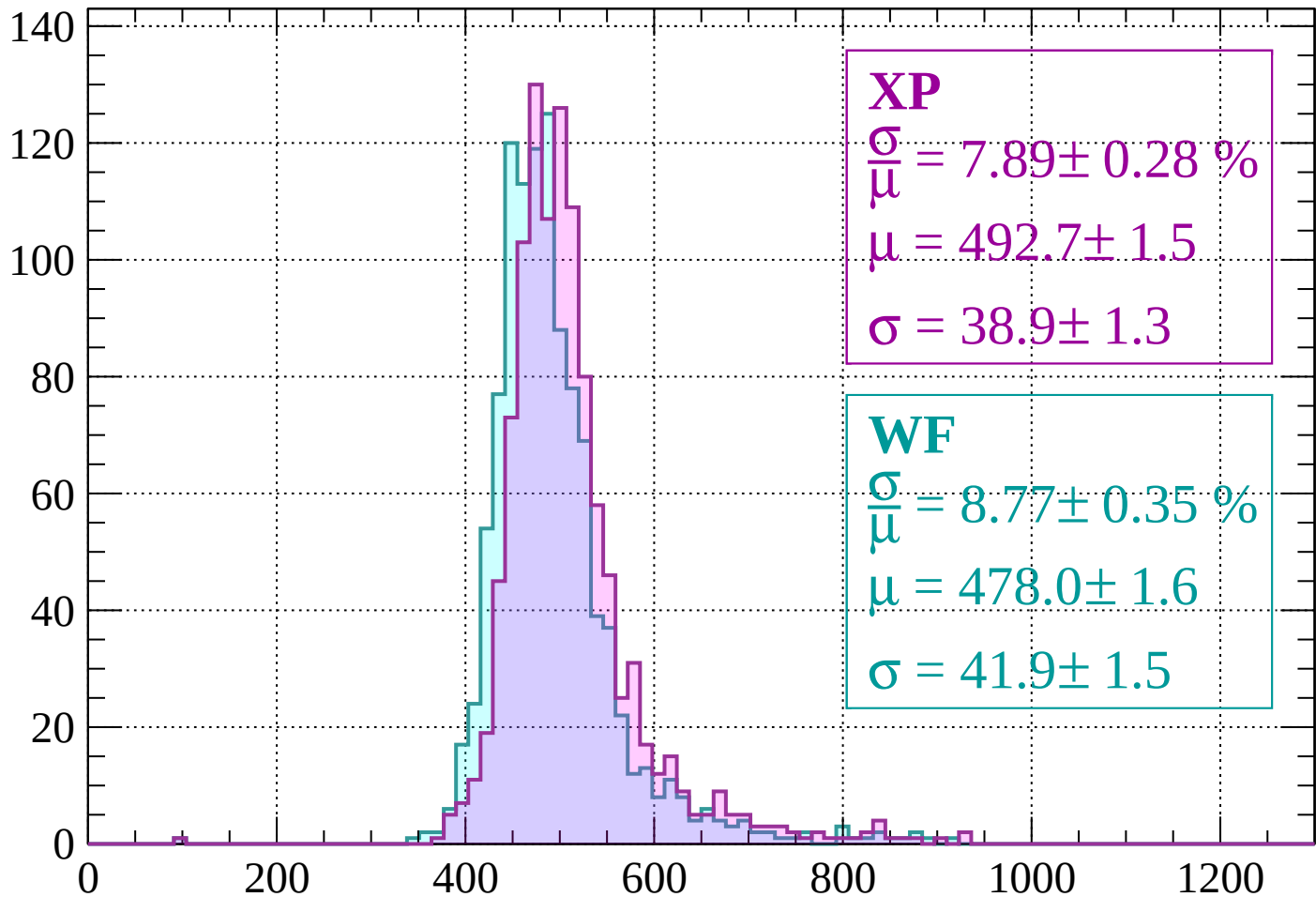
$$\mu = 474.5 \pm 1.6$$

$$\sigma = 38.8 \pm 1.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1680 < p < -1640$

Count



XP

$$\frac{\sigma}{\mu} = 7.89 \pm 0.28 \%$$

$$\mu = 492.7 \pm 1.5$$

$$\sigma = 38.9 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 8.77 \pm 0.35 \%$$

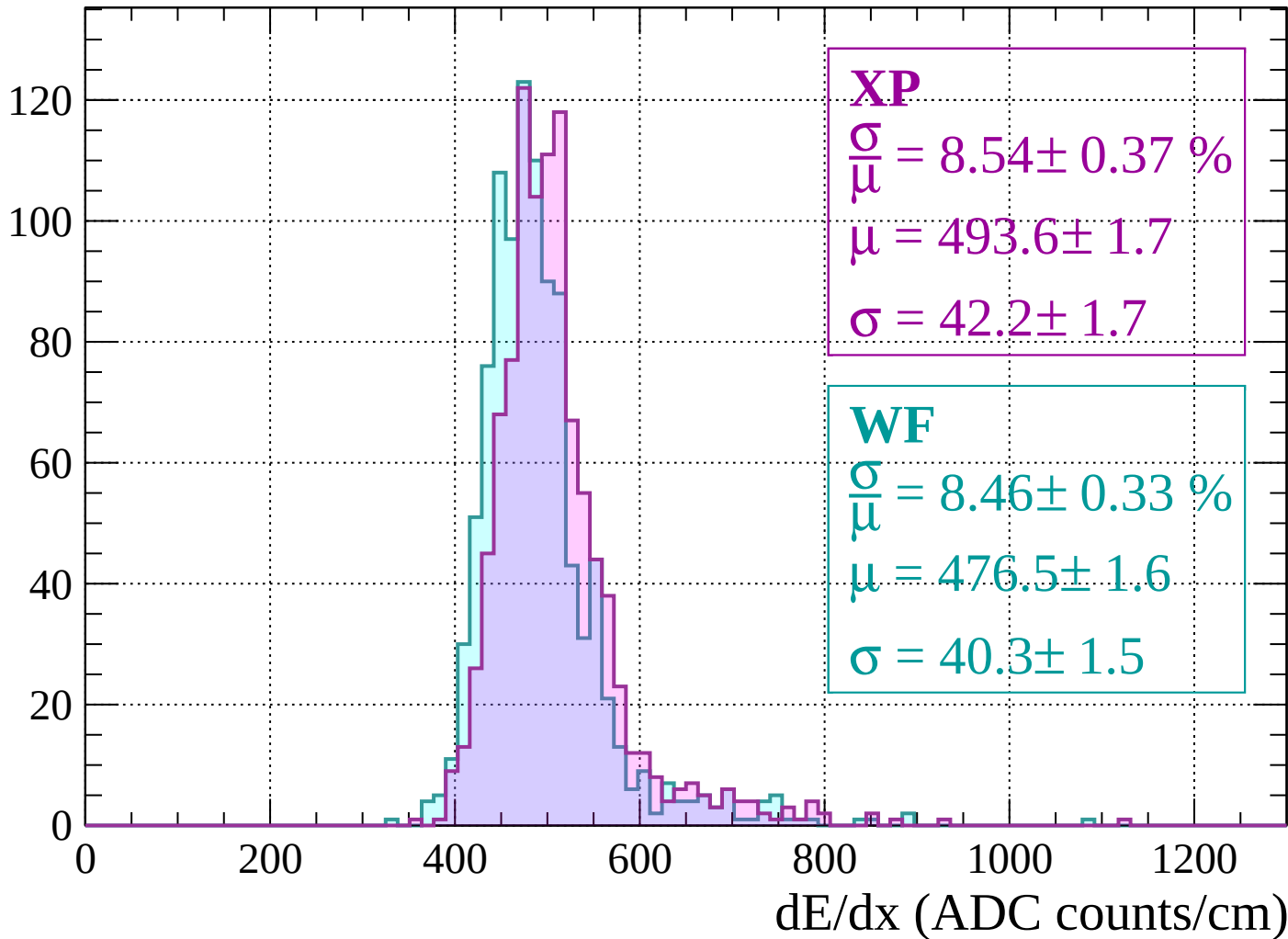
$$\mu = 478.0 \pm 1.6$$

$$\sigma = 41.9 \pm 1.5$$

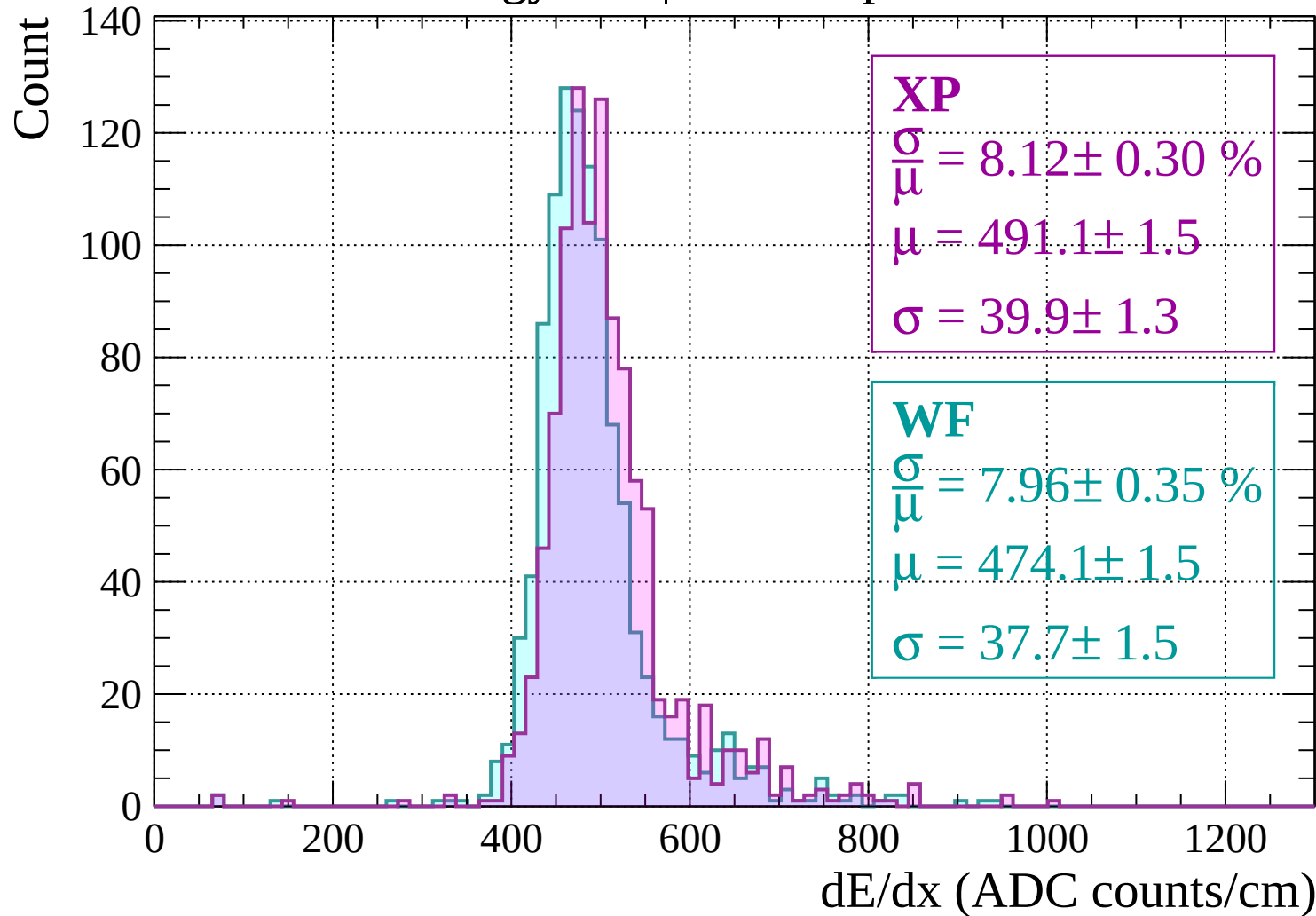
dE/dx (ADC counts/cm)

Energy loss | $-1640 < p < -1600$

Count

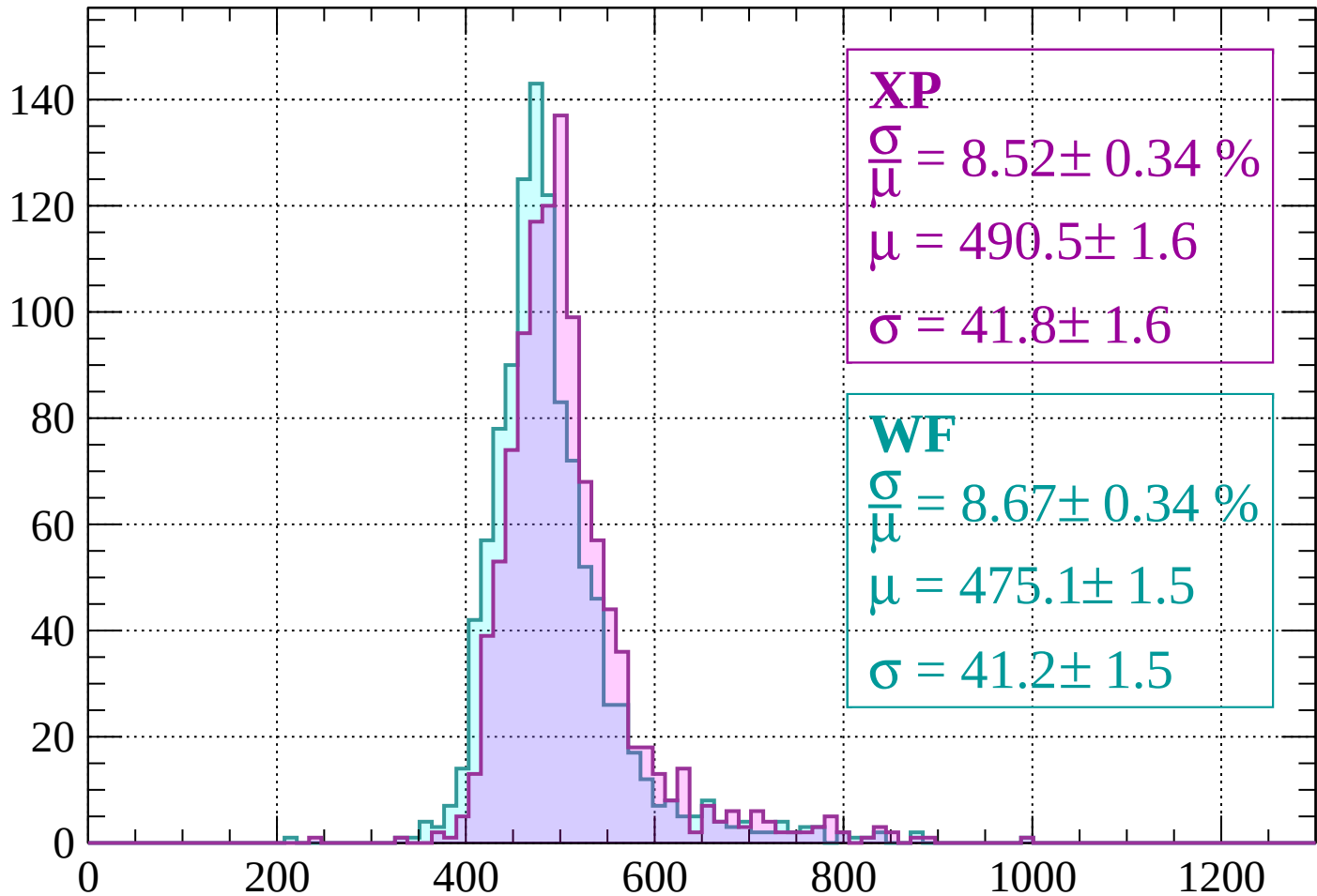


Energy loss | $-1600 < p < -1560$



Energy loss | $-1560 < p < -1520$

Count



XP

$$\frac{\sigma}{\mu} = 8.52 \pm 0.34 \%$$

$$\mu = 490.5 \pm 1.6$$

$$\sigma = 41.8 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 8.67 \pm 0.34 \%$$

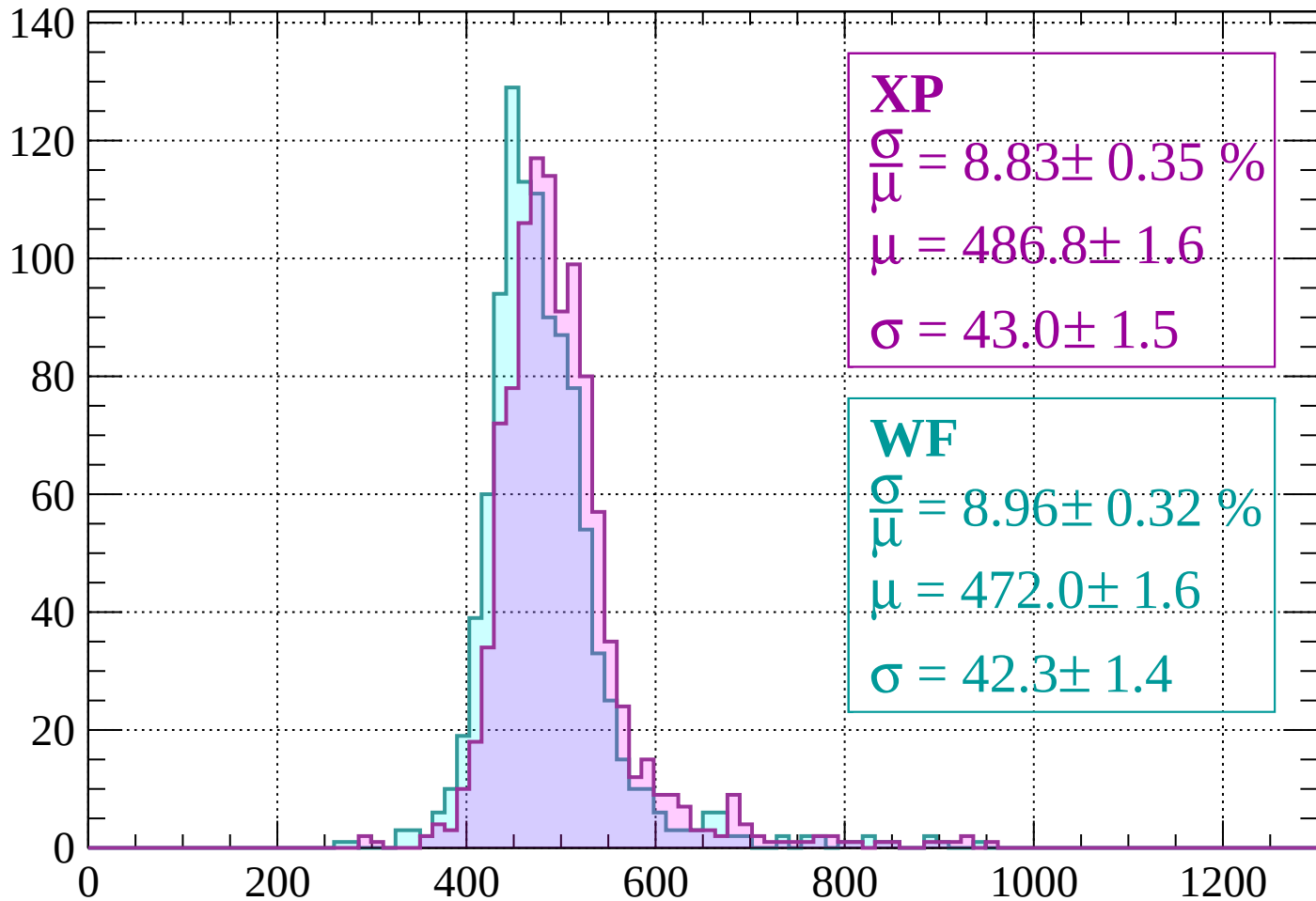
$$\mu = 475.1 \pm 1.5$$

$$\sigma = 41.2 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1520 < p < -1480$

Count



XP

$$\frac{\sigma}{\mu} = 8.83 \pm 0.35 \%$$

$$\mu = 486.8 \pm 1.6$$

$$\sigma = 43.0 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 8.96 \pm 0.32 \%$$

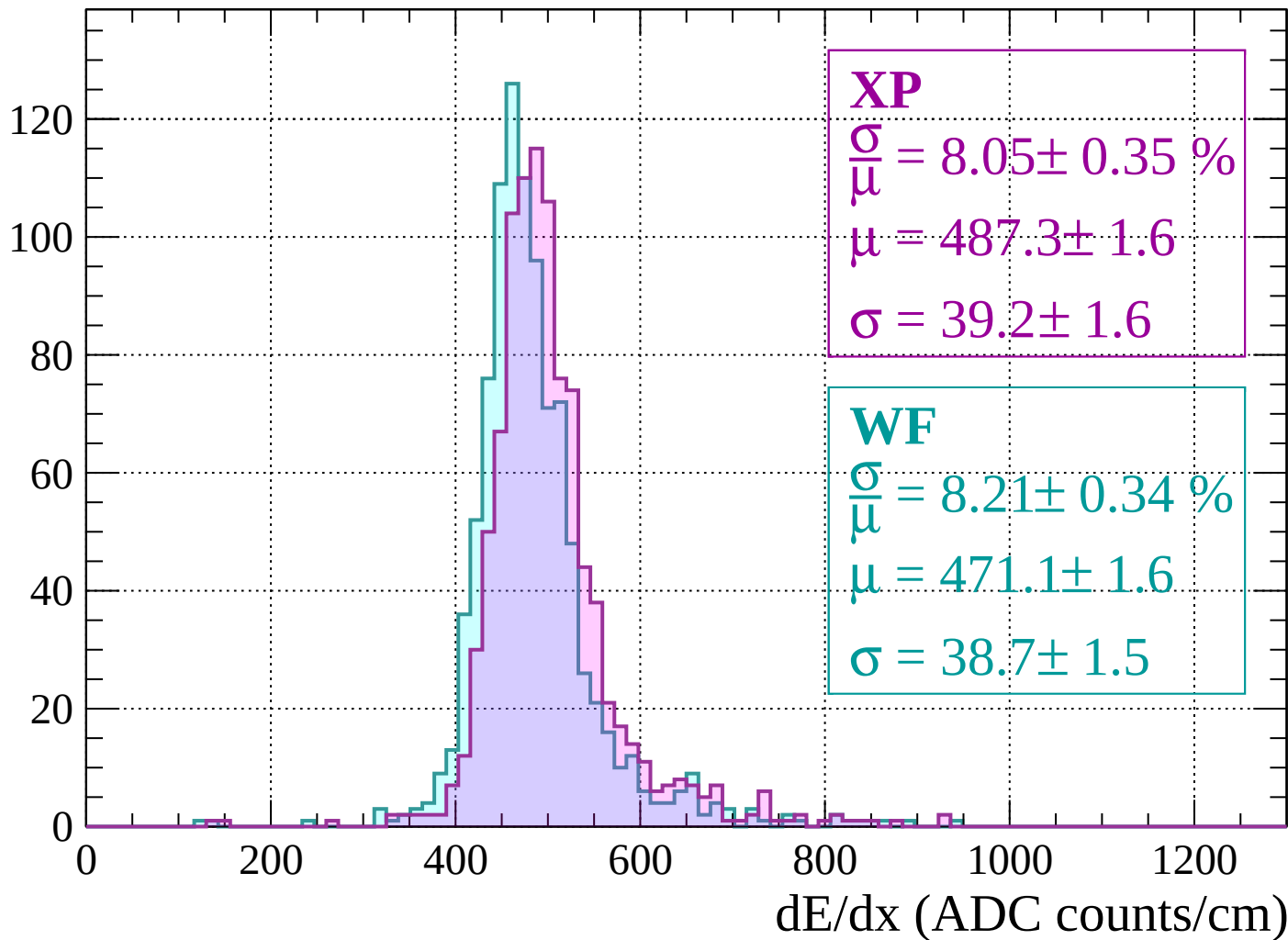
$$\mu = 472.0 \pm 1.6$$

$$\sigma = 42.3 \pm 1.4$$

dE/dx (ADC counts/cm)

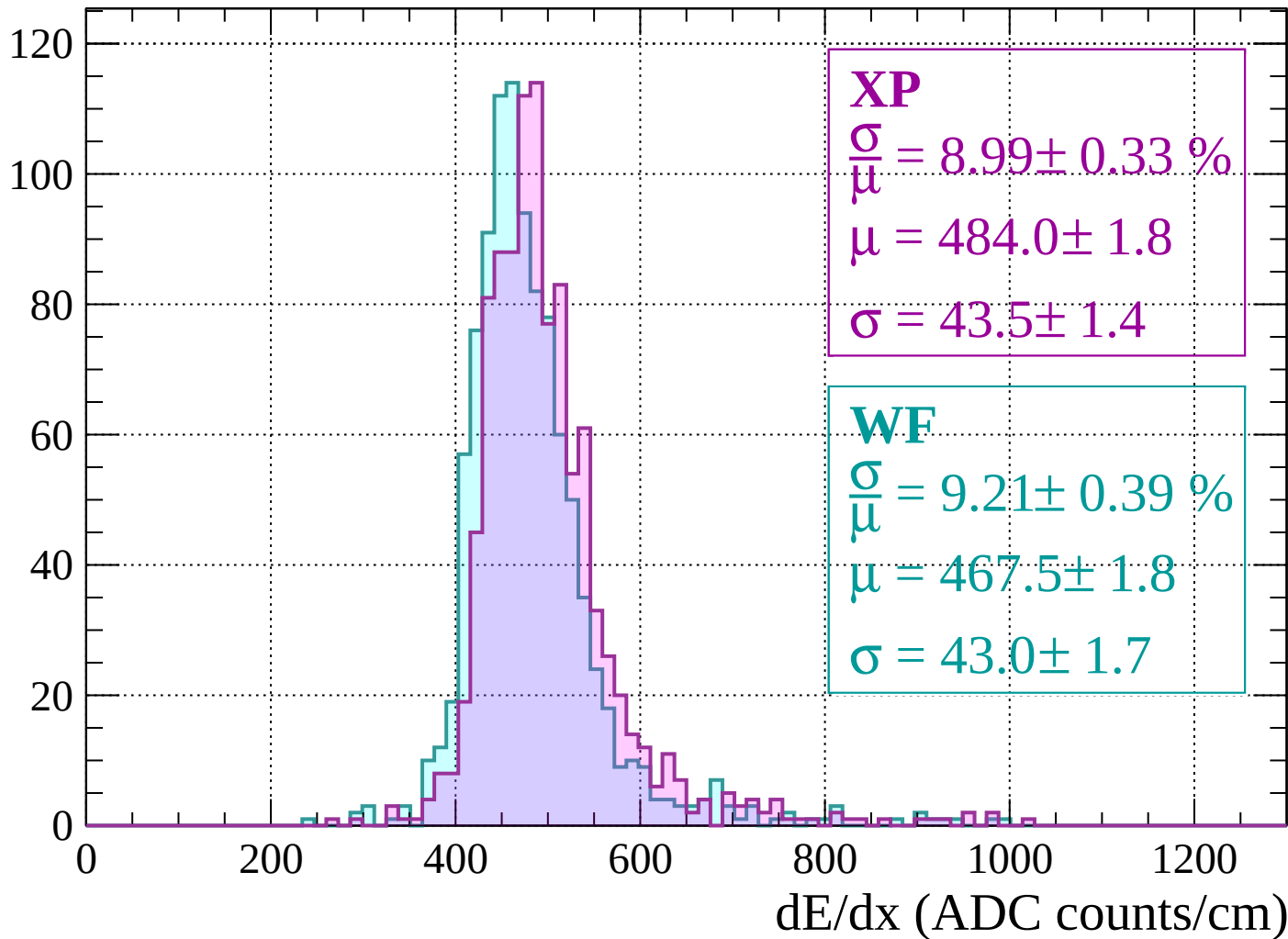
Energy loss | $-1480 < p < -1440$

Count



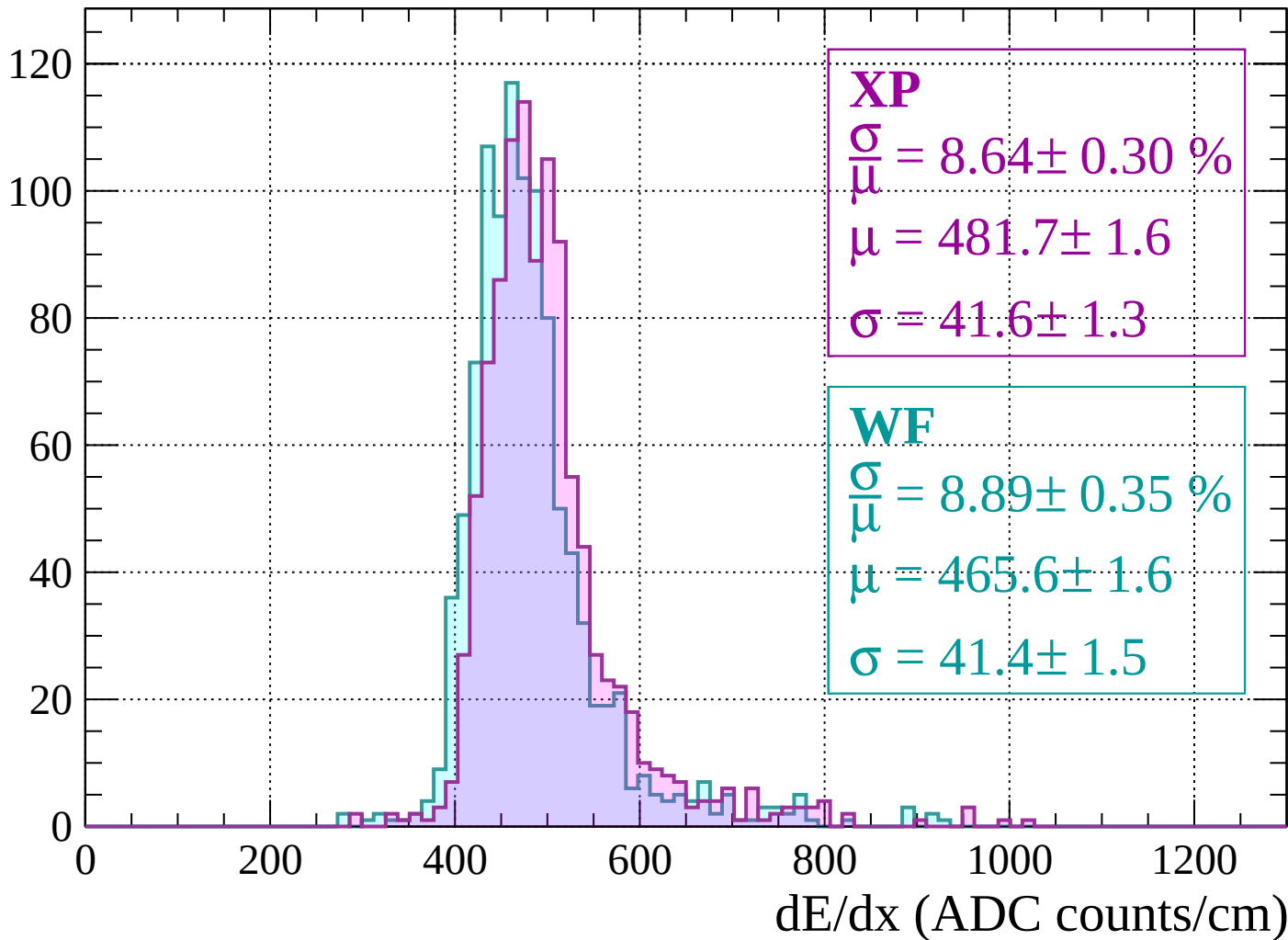
Energy loss | $-1440 < p < -1400$

Count



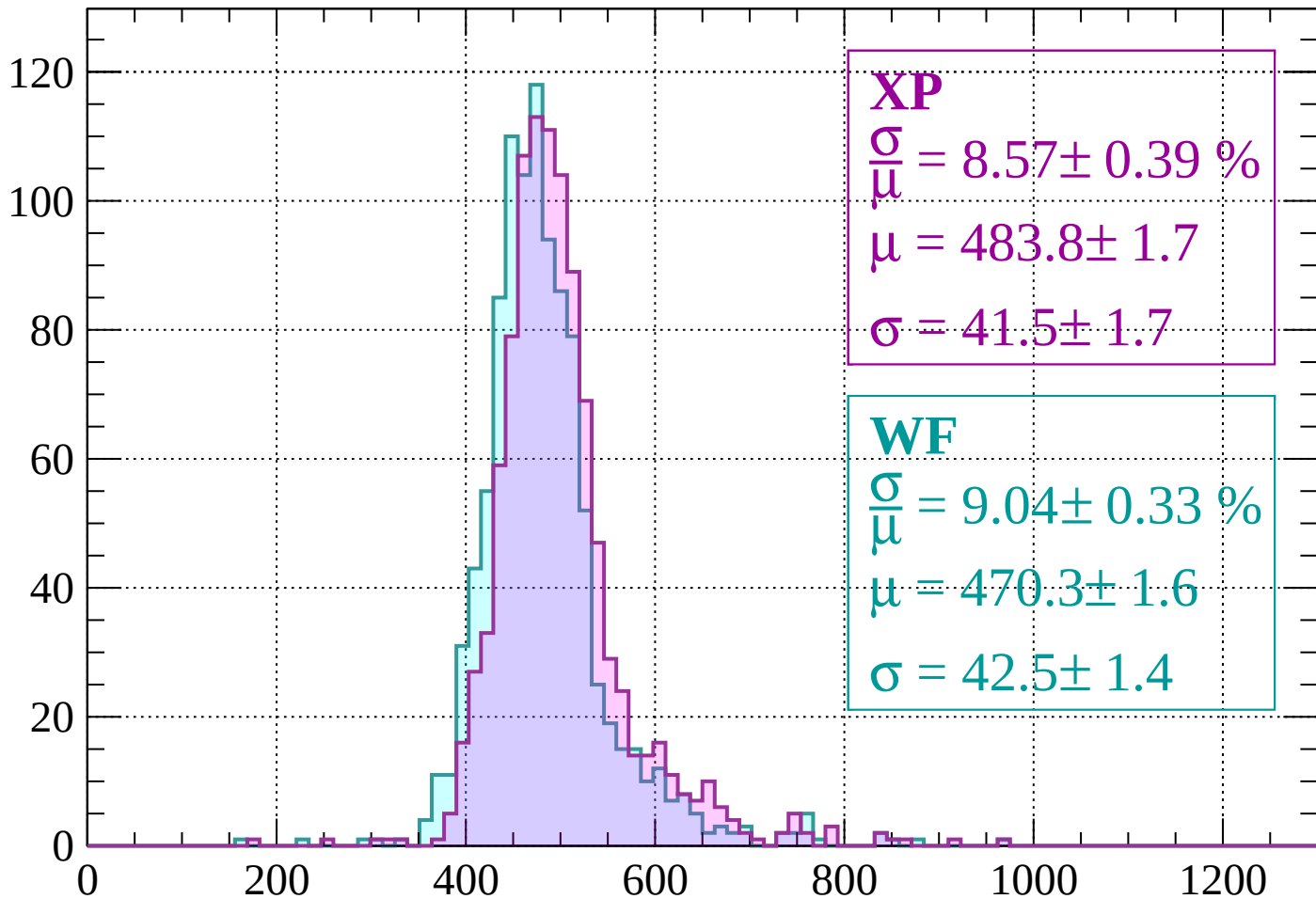
Energy loss | $-1400 < p < -1360$

Count



Energy loss | $-1360 < p < -1320$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 8.57 \pm 0.39 \%$$

$$\mu = 483.8 \pm 1.7$$

$$\sigma = 41.5 \pm 1.7$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.04 \pm 0.33 \%$$

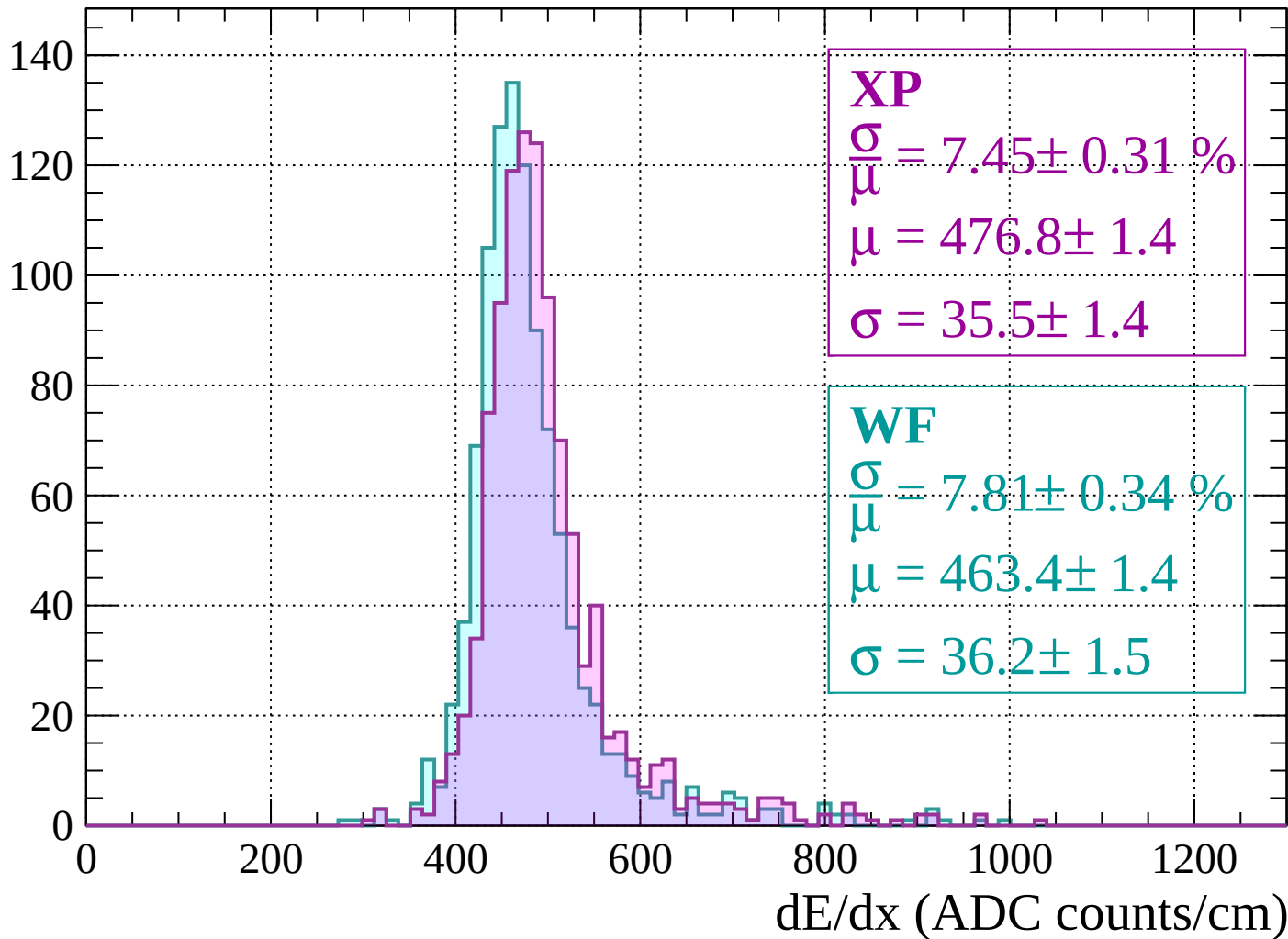
$$\mu = 470.3 \pm 1.6$$

$$\sigma = 42.5 \pm 1.4$$

dE/dx (ADC counts/cm)

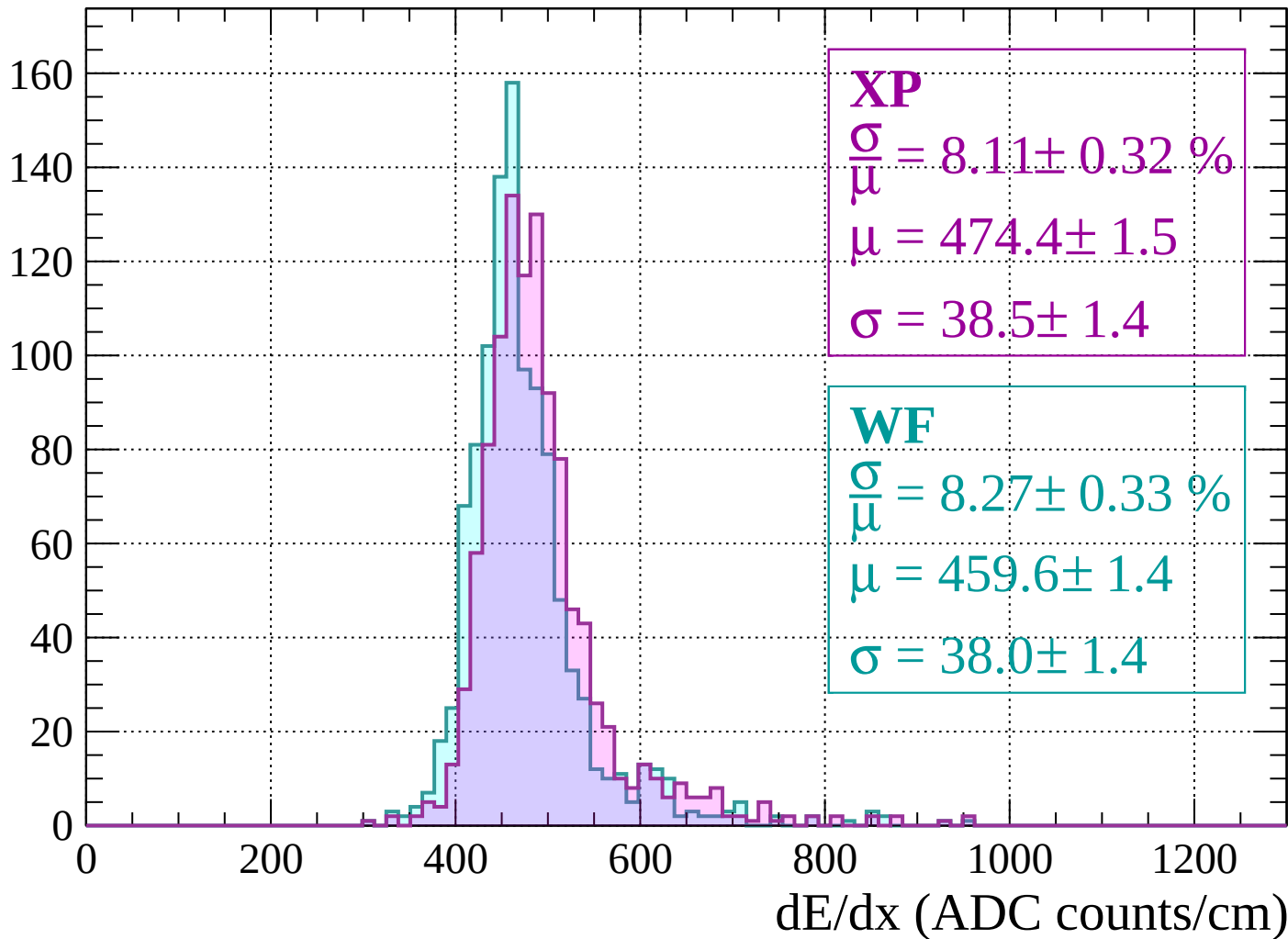
Energy loss | $-1320 < p < -1280$

Count



Energy loss | $-1280 < p < -1240$

Count



Energy loss | $-1240 < p < -1200$

Count

140

120

100

80

60

40

20

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 8.70 \pm 0.41 \%$$

$$\mu = 470.4 \pm 1.6$$

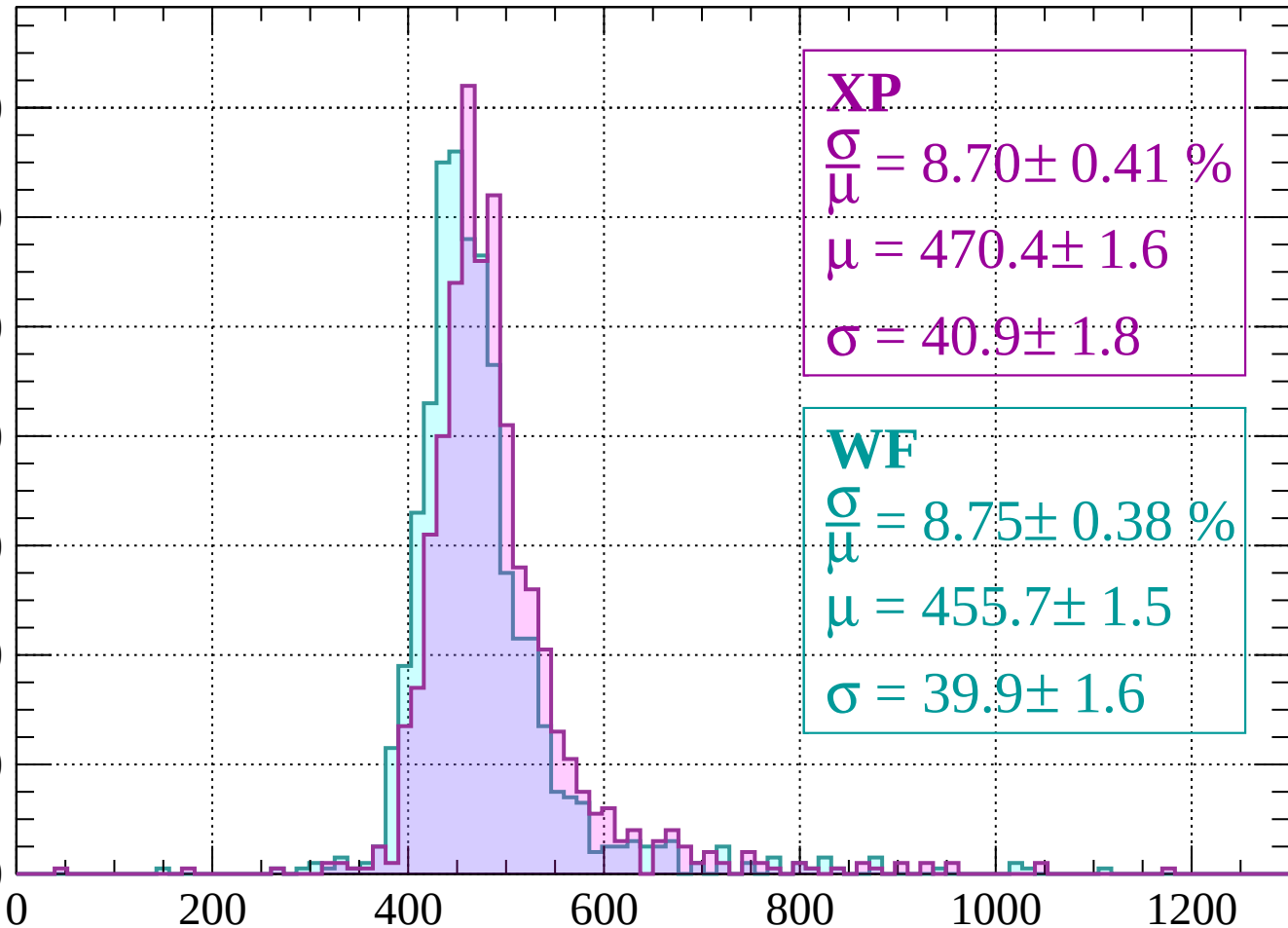
$$\sigma = 40.9 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 8.75 \pm 0.38 \%$$

$$\mu = 455.7 \pm 1.5$$

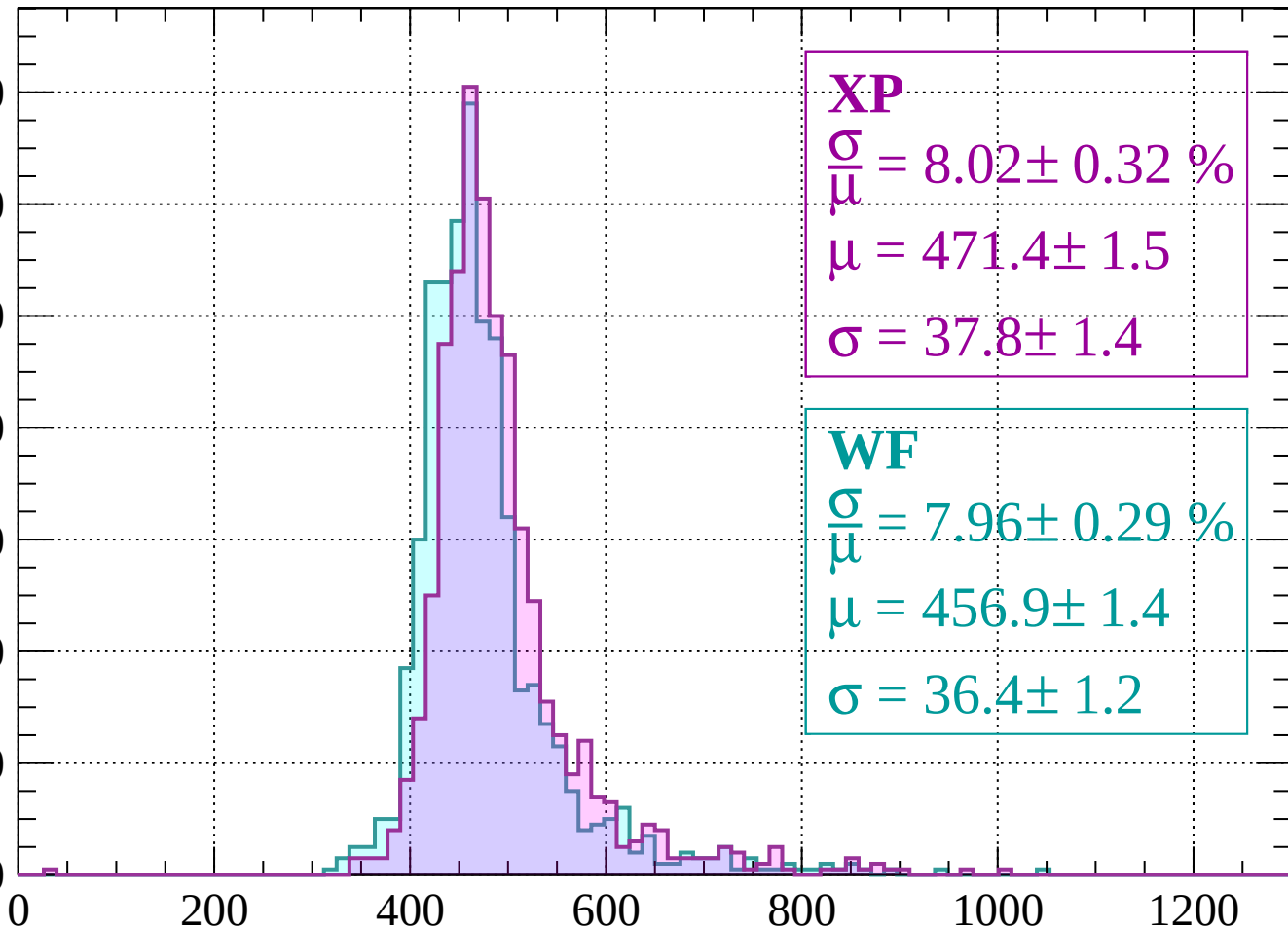
$$\sigma = 39.9 \pm 1.6$$



Energy loss | $-1200 < p < -1160$

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 8.02 \pm 0.32 \%$$

$$\mu = 471.4 \pm 1.5$$

$$\sigma = 37.8 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 7.96 \pm 0.29 \%$$

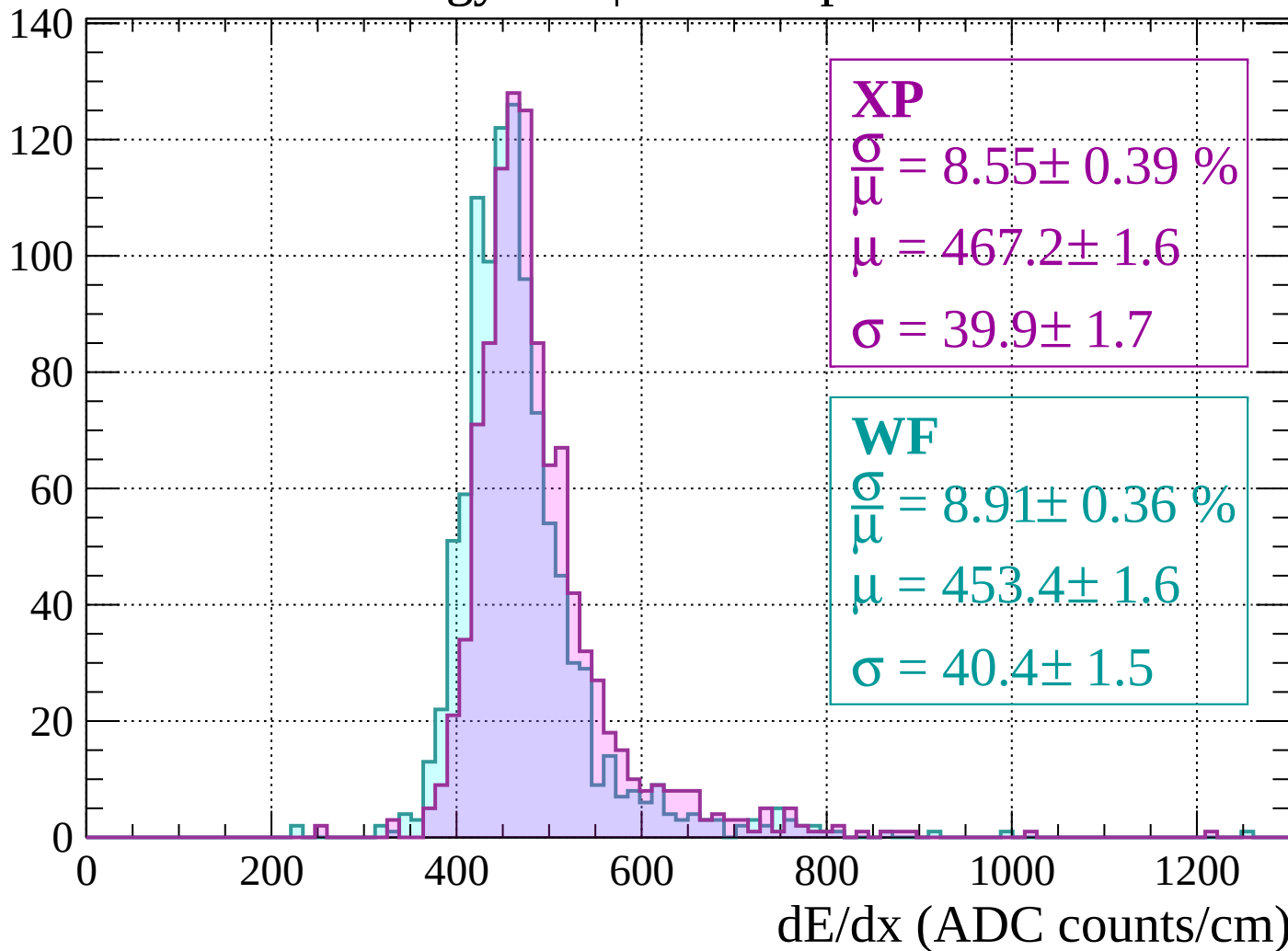
$$\mu = 456.9 \pm 1.4$$

$$\sigma = 36.4 \pm 1.2$$

dE/dx (ADC counts/cm)

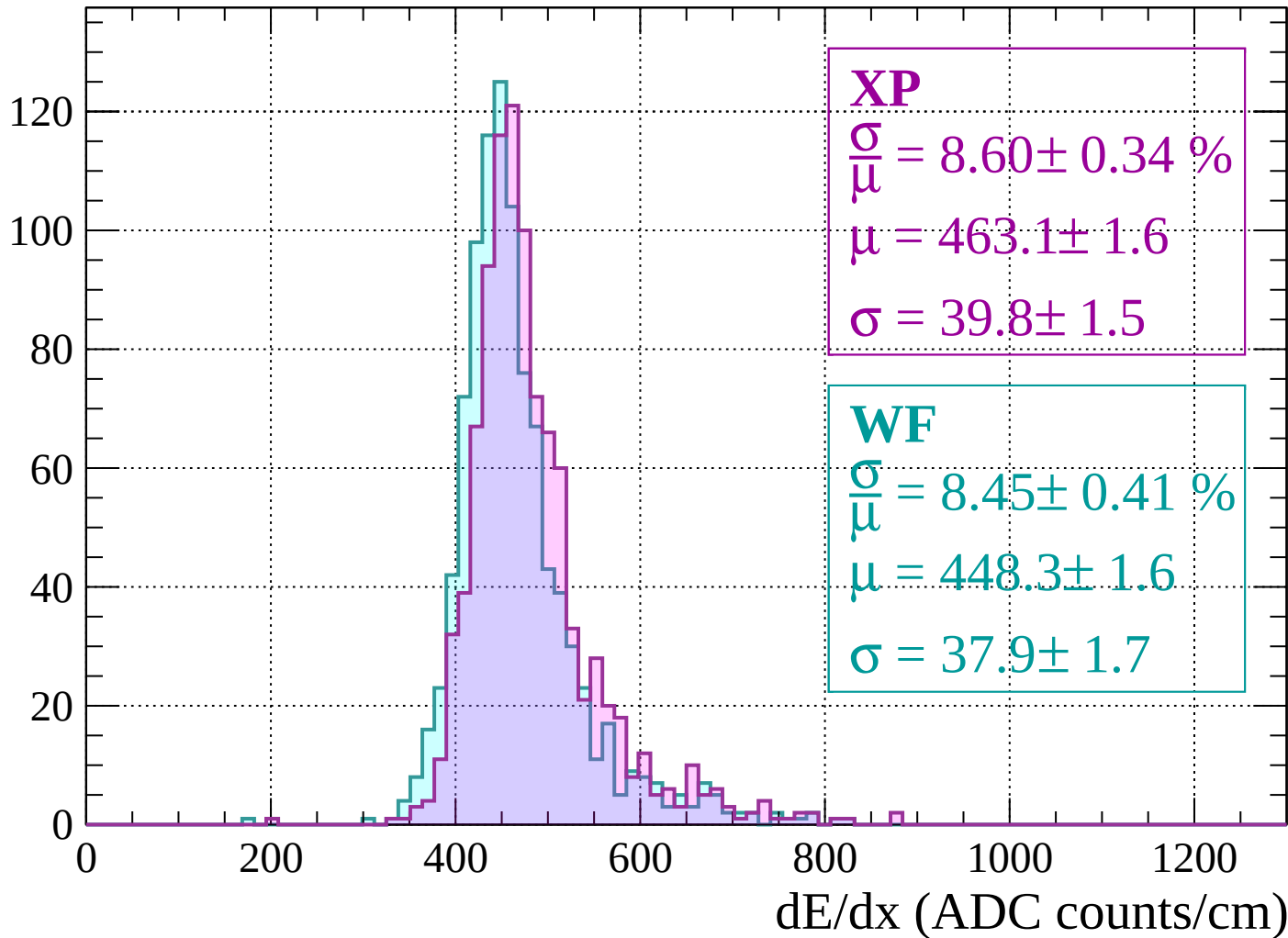
Energy loss | $-1160 < p < -1120$

Count



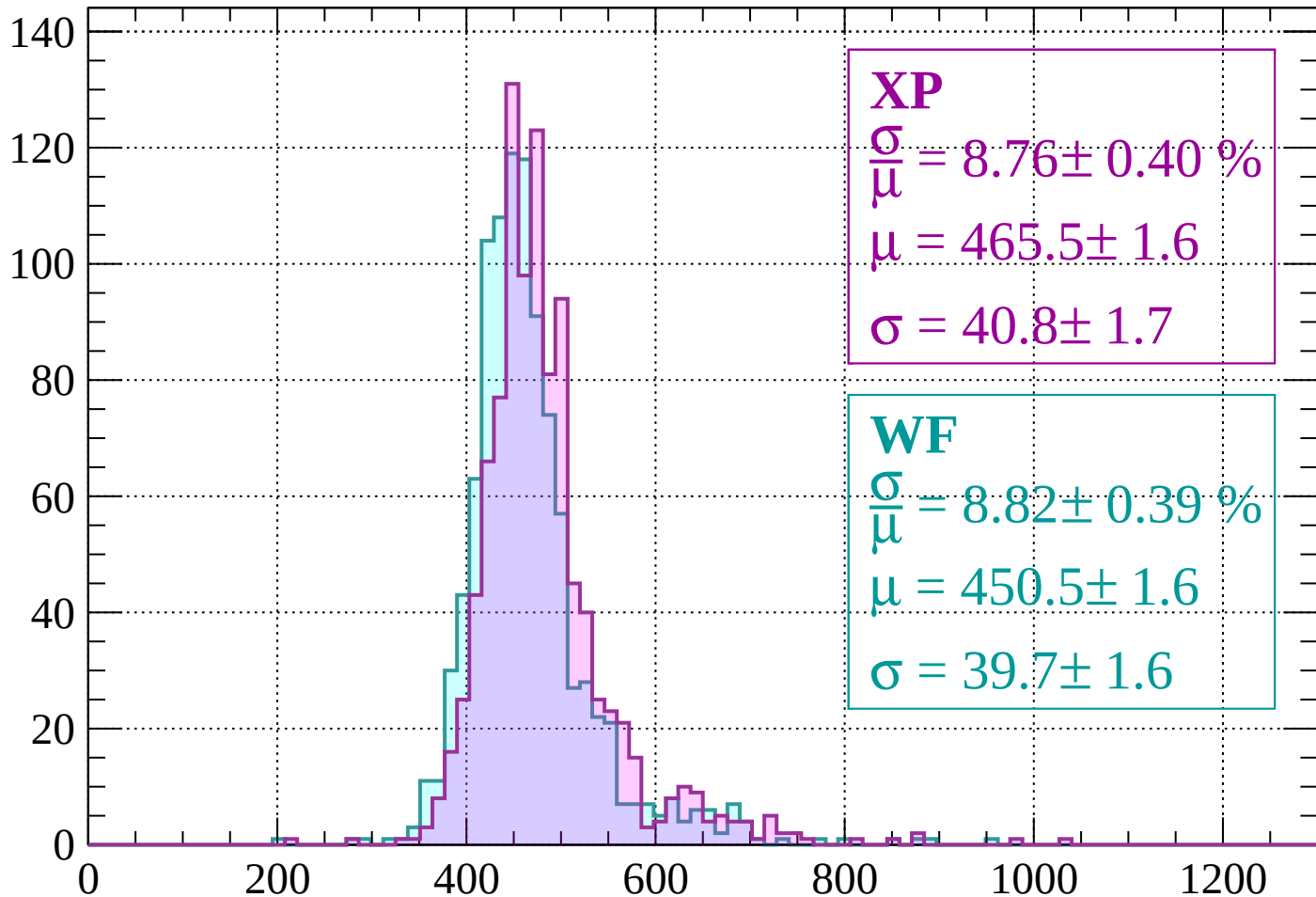
Energy loss | -1120 < p < -1080

Count



Energy loss | $-1080 < p < -1040$

Count



XP

$$\frac{\sigma}{\mu} = 8.76 \pm 0.40 \%$$

$$\mu = 465.5 \pm 1.6$$

$$\sigma = 40.8 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 8.82 \pm 0.39 \%$$

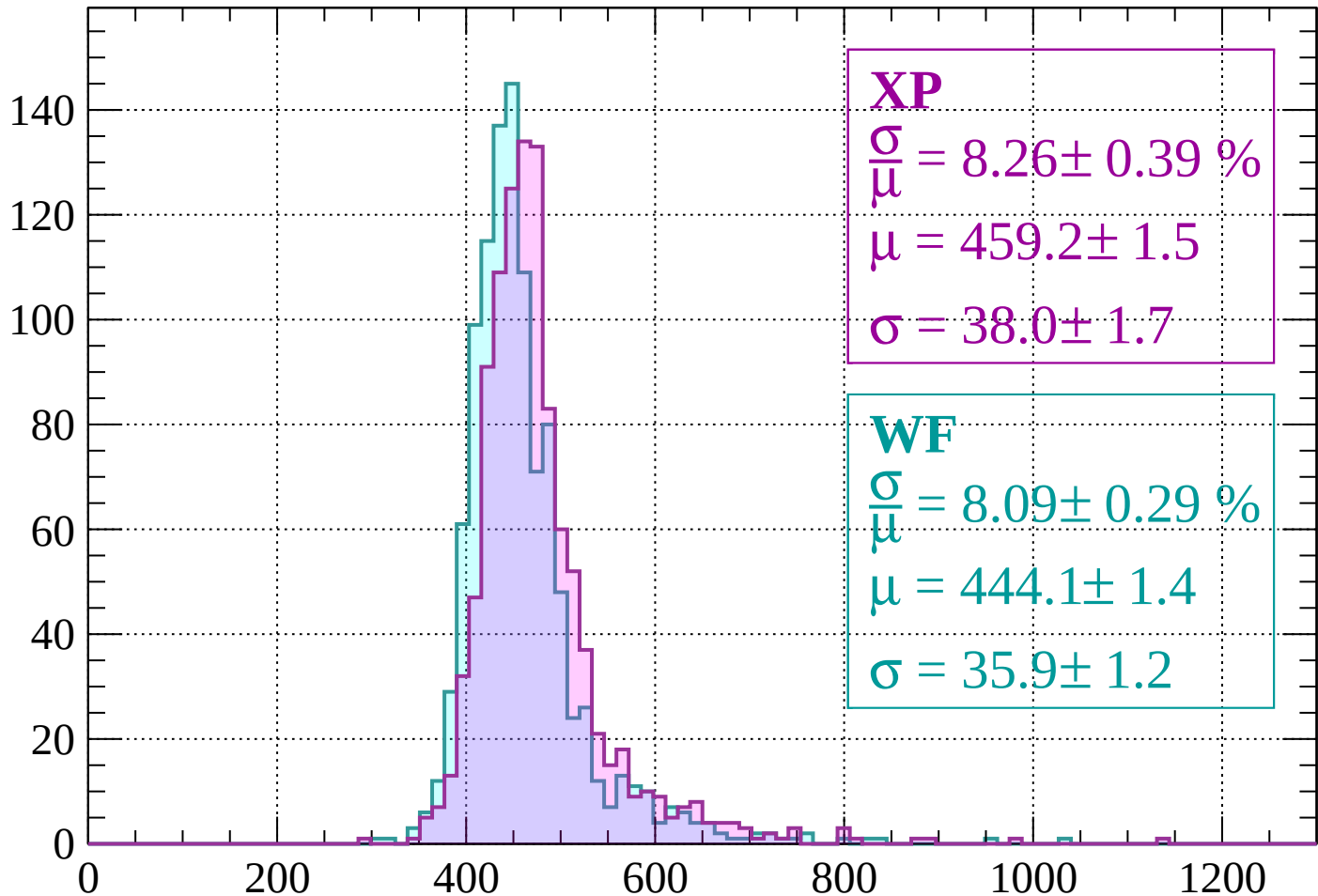
$$\mu = 450.5 \pm 1.6$$

$$\sigma = 39.7 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $-1040 < p < -1000$

Count



XP

$$\frac{\sigma}{\mu} = 8.26 \pm 0.39 \%$$

$$\mu = 459.2 \pm 1.5$$

$$\sigma = 38.0 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 8.09 \pm 0.29 \%$$

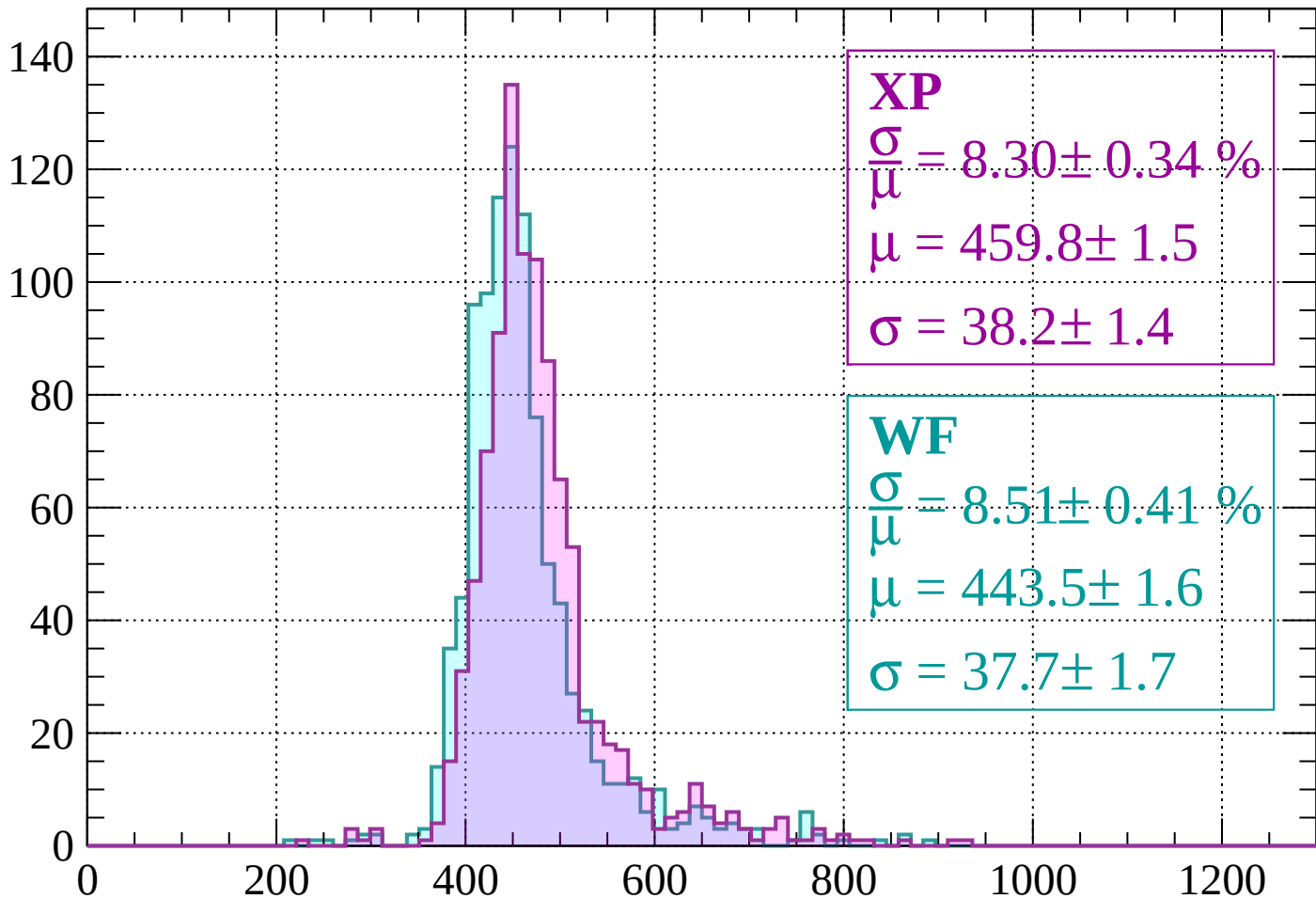
$$\mu = 444.1 \pm 1.4$$

$$\sigma = 35.9 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $-1000 < p < -960$

Count



XP

$$\frac{\sigma}{\mu} = 8.30 \pm 0.34 \%$$

$$\mu = 459.8 \pm 1.5$$

$$\sigma = 38.2 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.51 \pm 0.41 \%$$

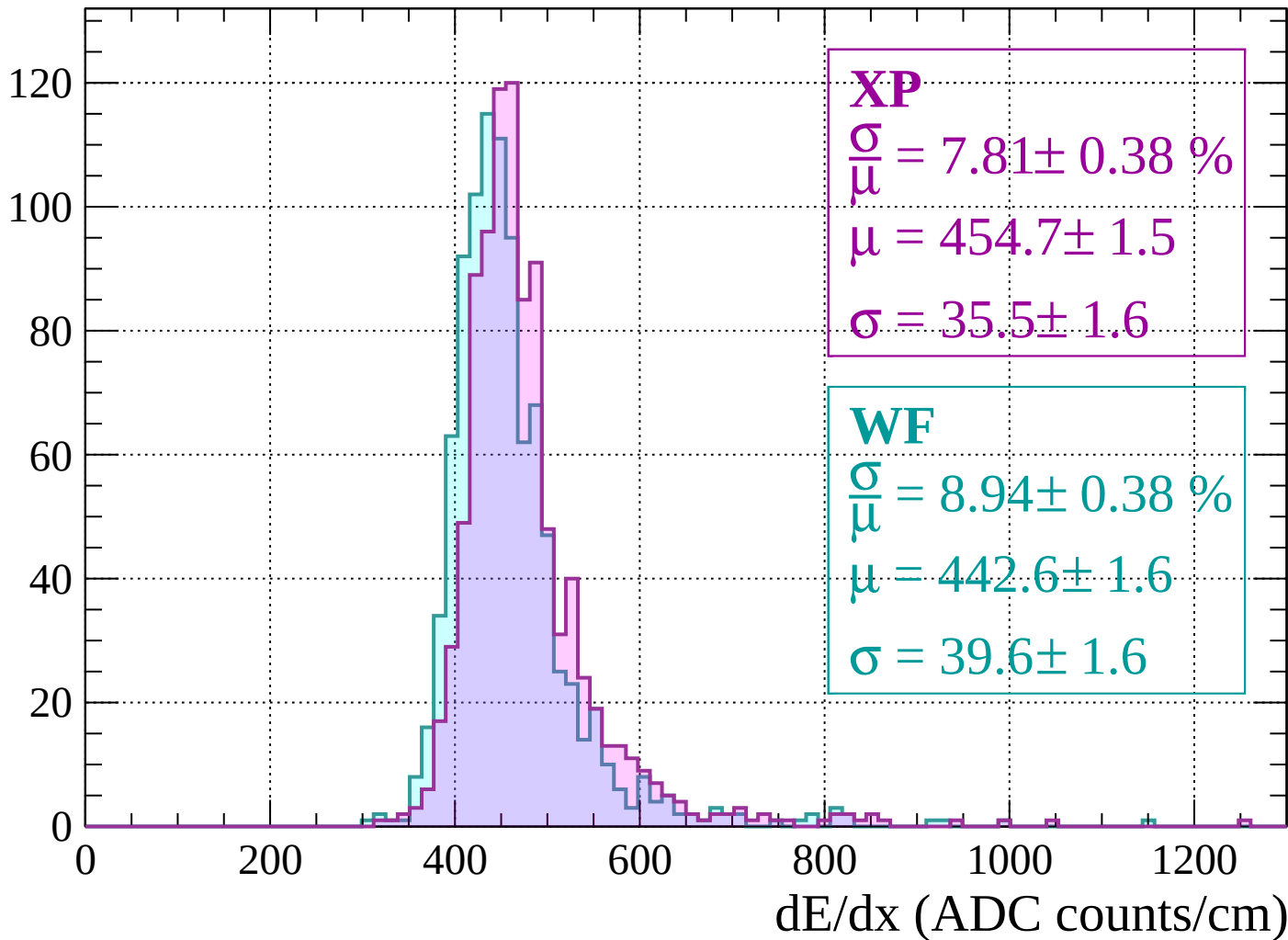
$$\mu = 443.5 \pm 1.6$$

$$\sigma = 37.7 \pm 1.7$$

dE/dx (ADC counts/cm)

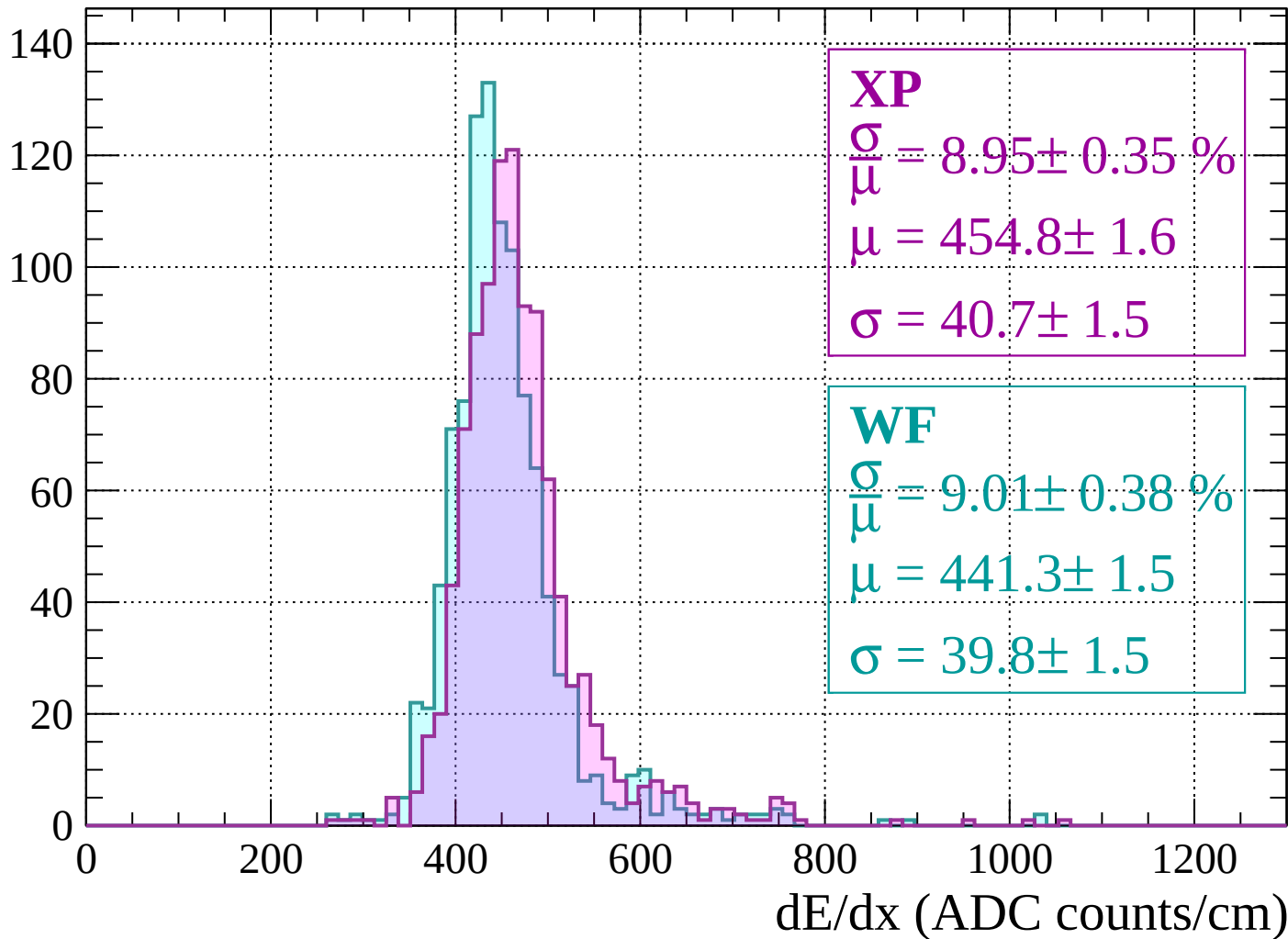
Energy loss | $-960 < p < -920$

Count



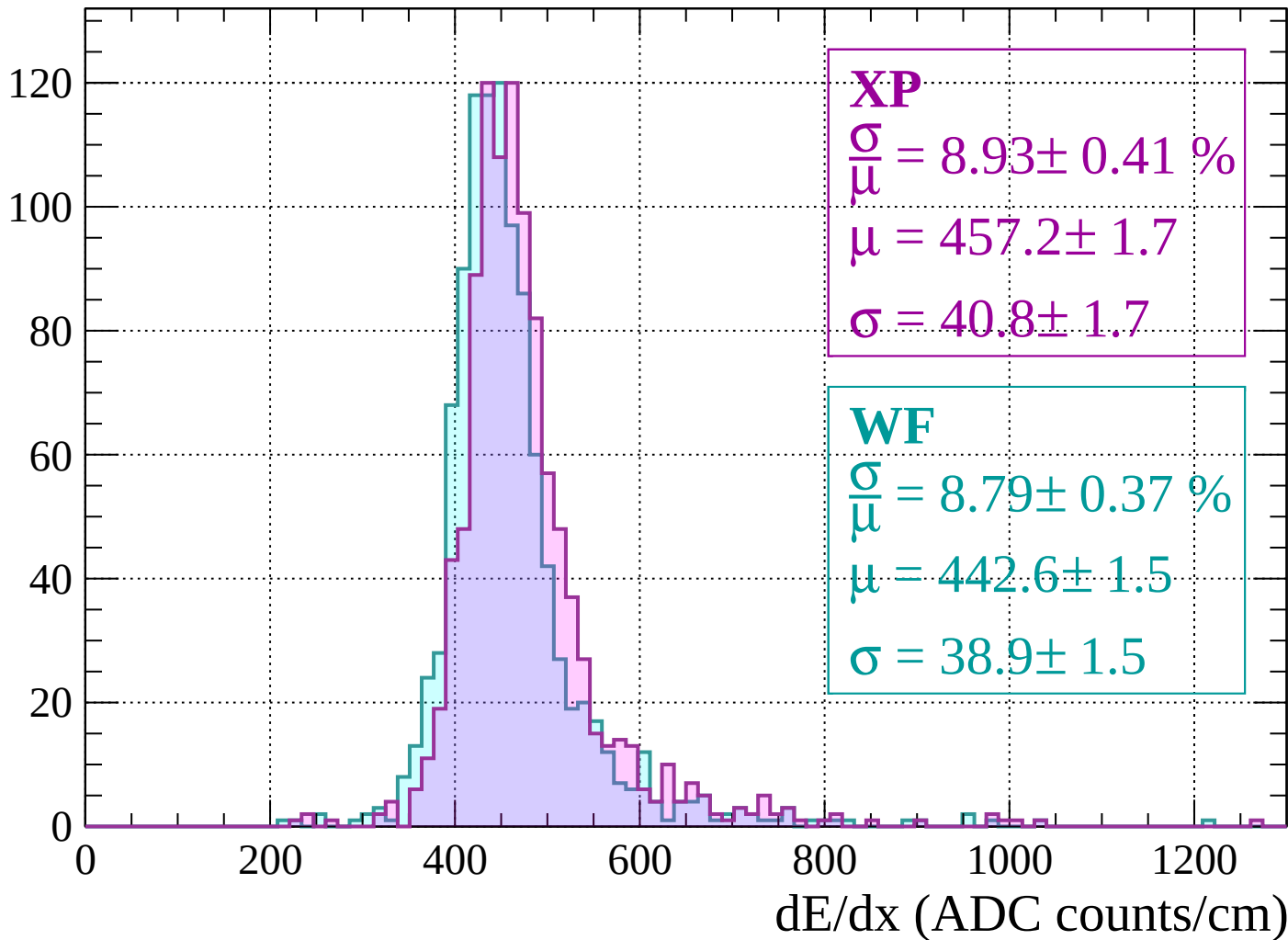
Energy loss | $-920 < p < -880$

Count



Energy loss | $-880 < p < -840$

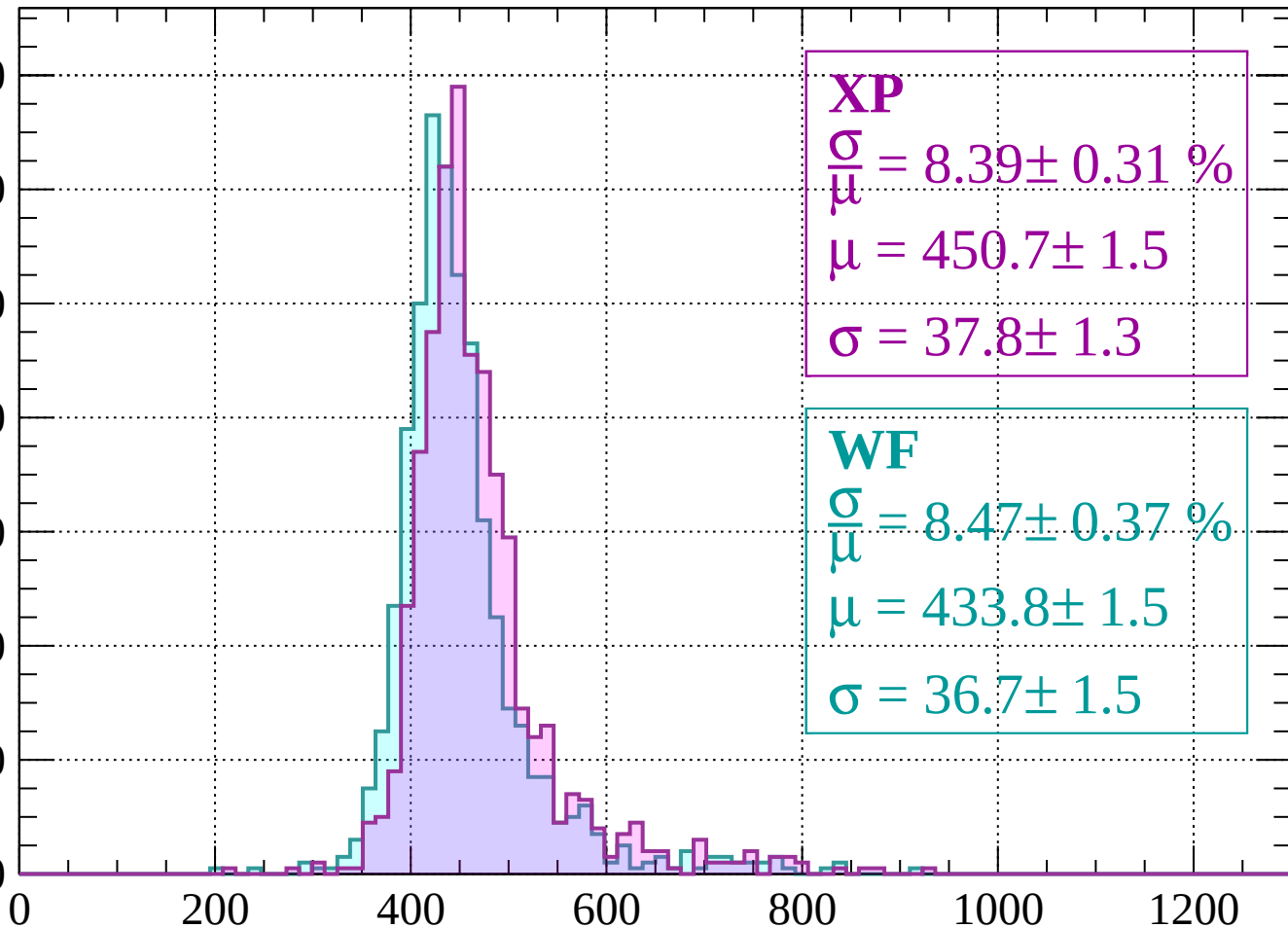
Count



Energy loss | $-840 < p < -800$

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 8.39 \pm 0.31 \%$$

$$\mu = 450.7 \pm 1.5$$

$$\sigma = 37.8 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 8.47 \pm 0.37 \%$$

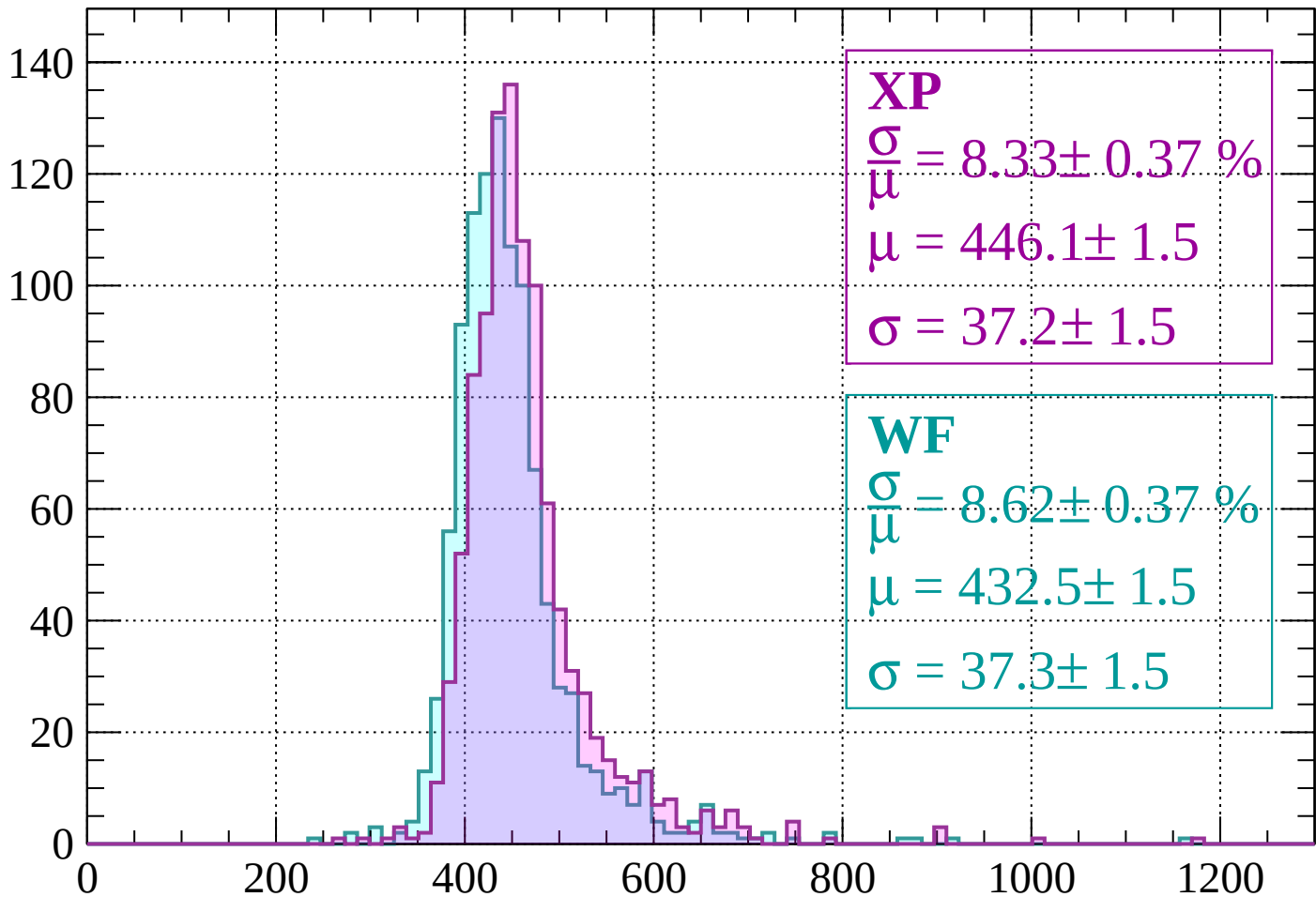
$$\mu = 433.8 \pm 1.5$$

$$\sigma = 36.7 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-800 < p < -760$

Count



XP

$$\frac{\sigma}{\mu} = 8.33 \pm 0.37 \%$$

$$\mu = 446.1 \pm 1.5$$

$$\sigma = 37.2 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 8.62 \pm 0.37 \%$$

$$\mu = 432.5 \pm 1.5$$

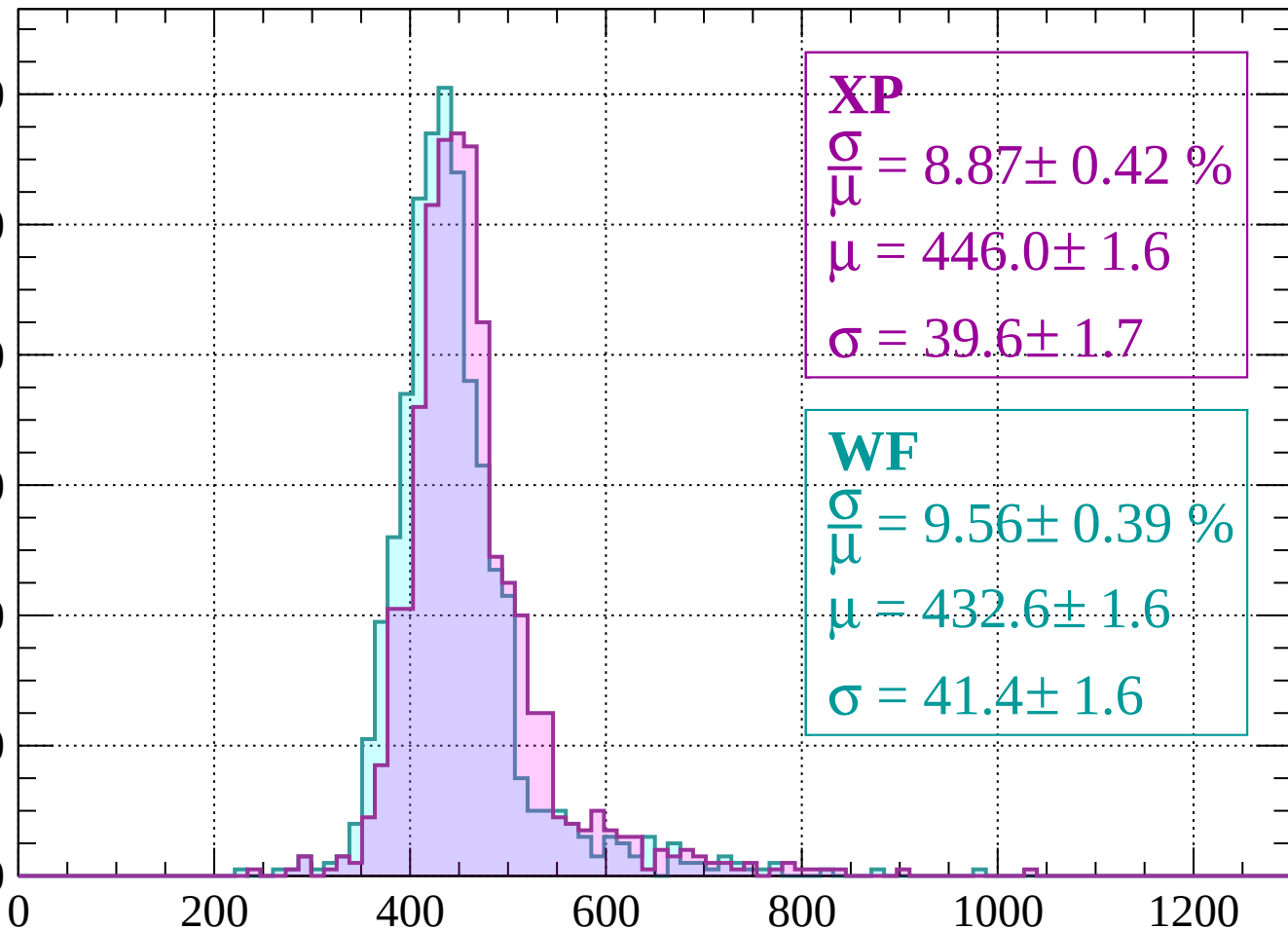
$$\sigma = 37.3 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-760 < p < -720$

Count

120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 8.87 \pm 0.42 \%$$

$$\mu = 446.0 \pm 1.6$$

$$\sigma = 39.6 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 9.56 \pm 0.39 \%$$

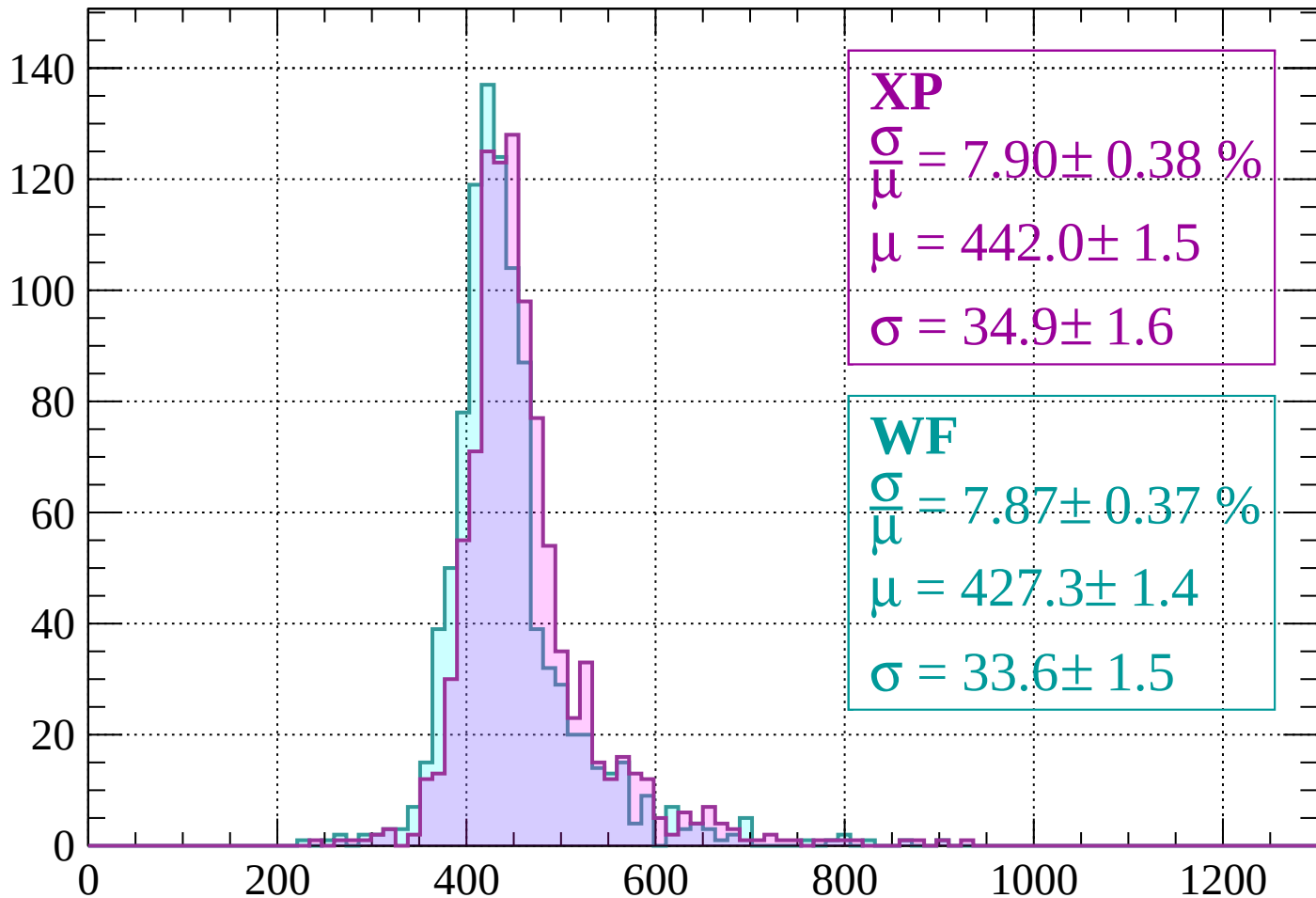
$$\mu = 432.6 \pm 1.6$$

$$\sigma = 41.4 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -680$

Count



XP

$$\frac{\sigma}{\mu} = 7.90 \pm 0.38 \%$$

$$\mu = 442.0 \pm 1.5$$

$$\sigma = 34.9 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 7.87 \pm 0.37 \%$$

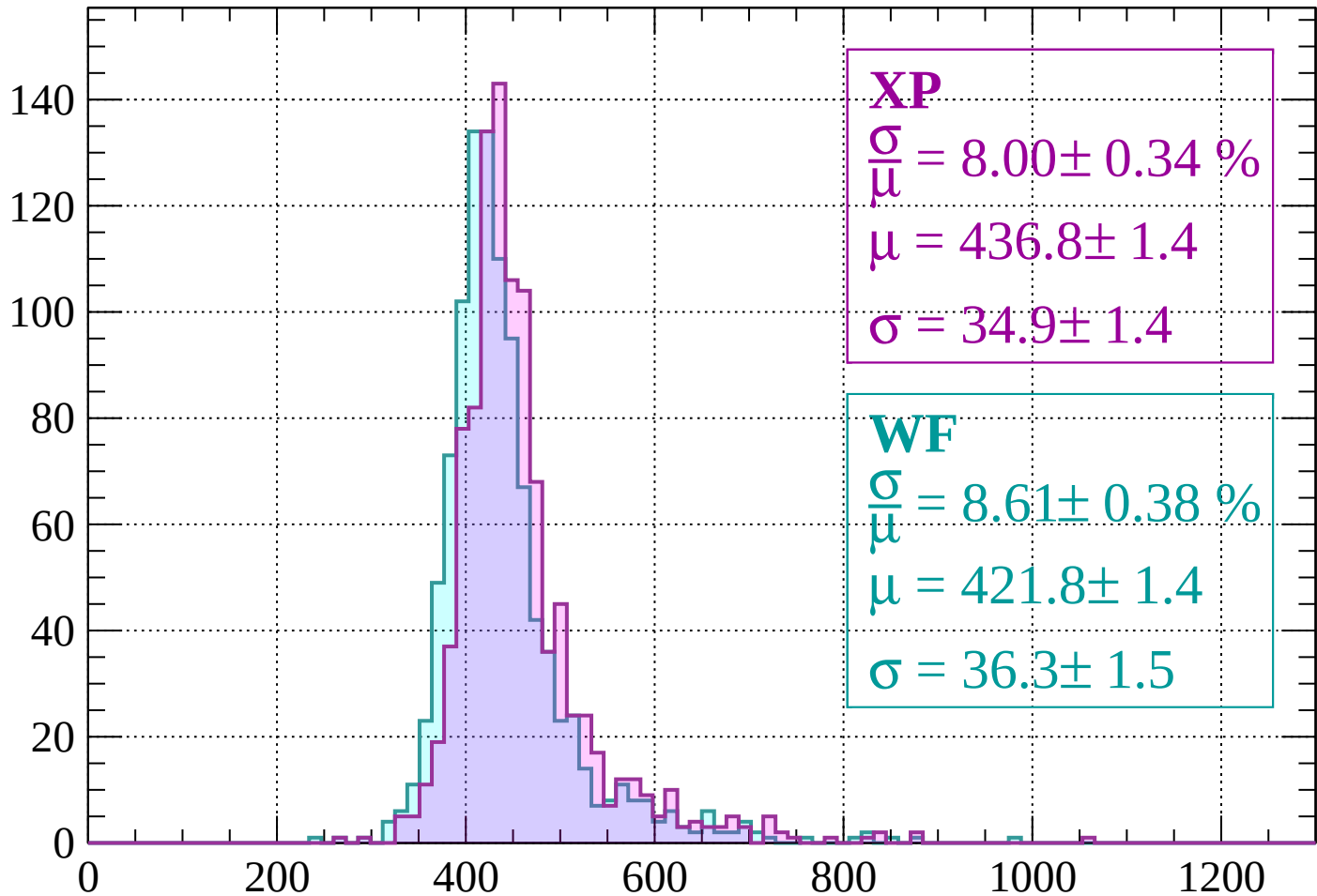
$$\mu = 427.3 \pm 1.4$$

$$\sigma = 33.6 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-680 < p < -640$

Count



XP

$$\frac{\sigma}{\mu} = 8.00 \pm 0.34 \%$$

$$\mu = 436.8 \pm 1.4$$

$$\sigma = 34.9 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.61 \pm 0.38 \%$$

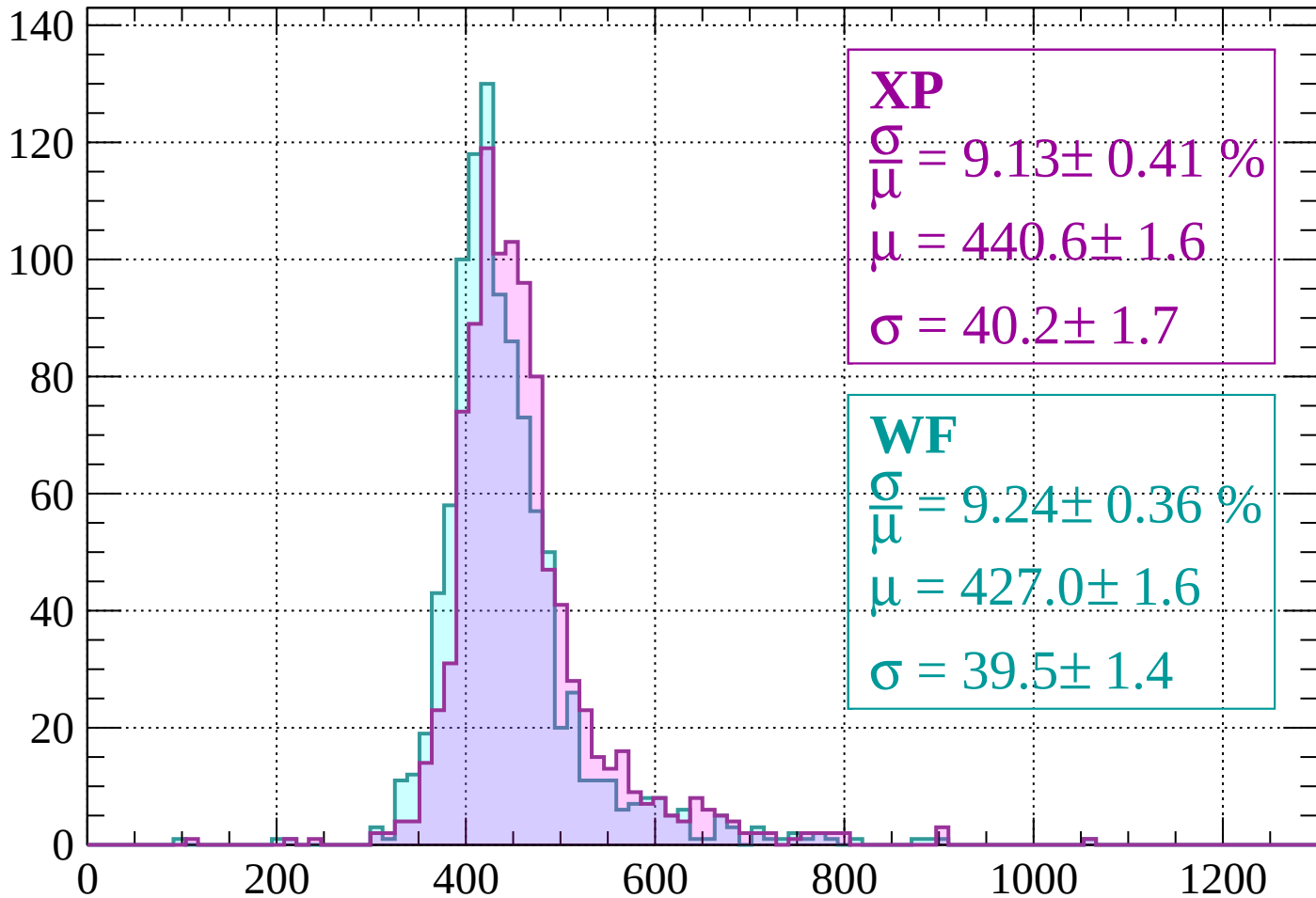
$$\mu = 421.8 \pm 1.4$$

$$\sigma = 36.3 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-640 < p < -600$

Count



XP

$$\frac{\sigma}{\mu} = 9.13 \pm 0.41 \%$$

$$\mu = 440.6 \pm 1.6$$

$$\sigma = 40.2 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 9.24 \pm 0.36 \%$$

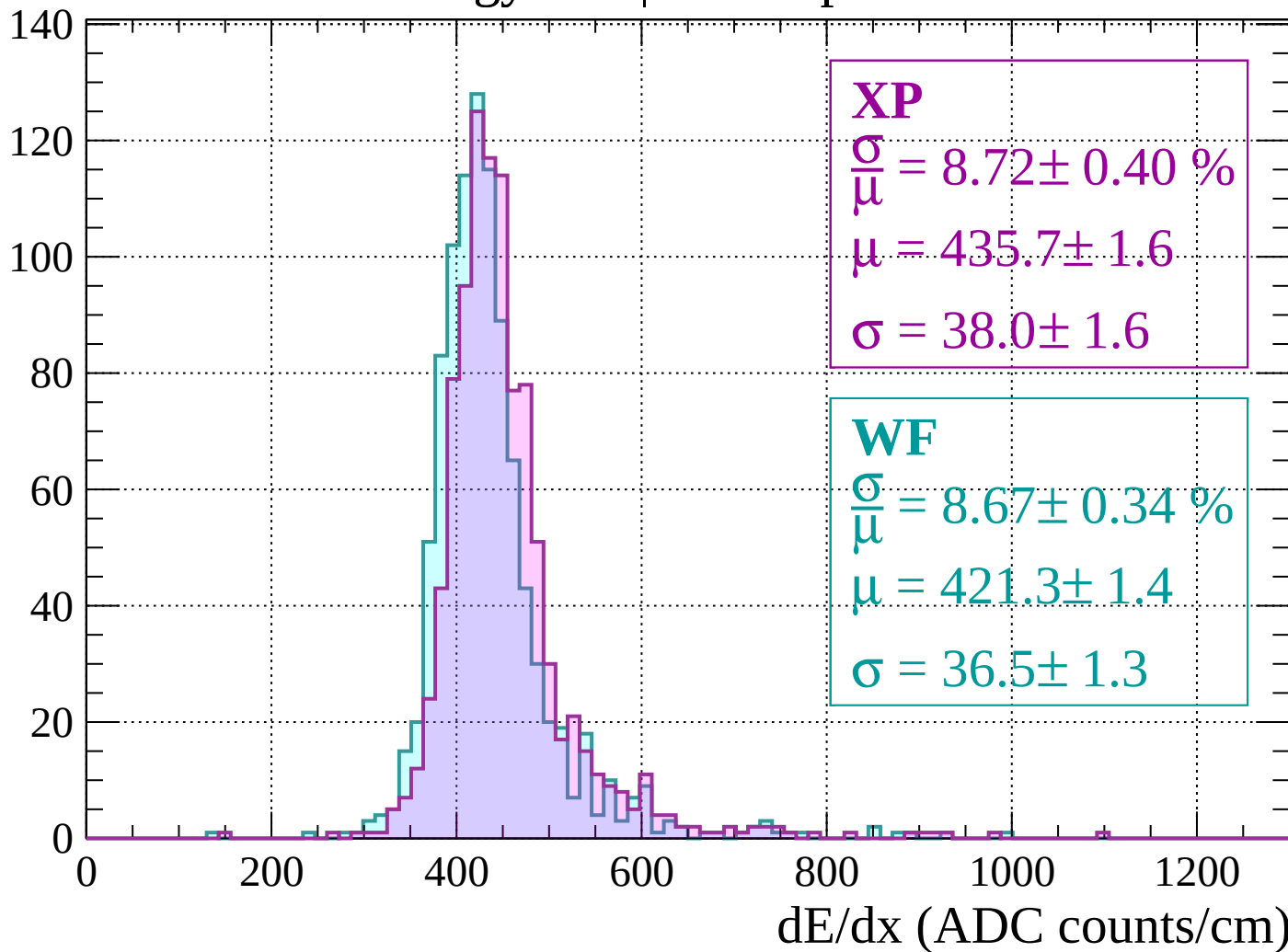
$$\mu = 427.0 \pm 1.6$$

$$\sigma = 39.5 \pm 1.4$$

dE/dx (ADC counts/cm)

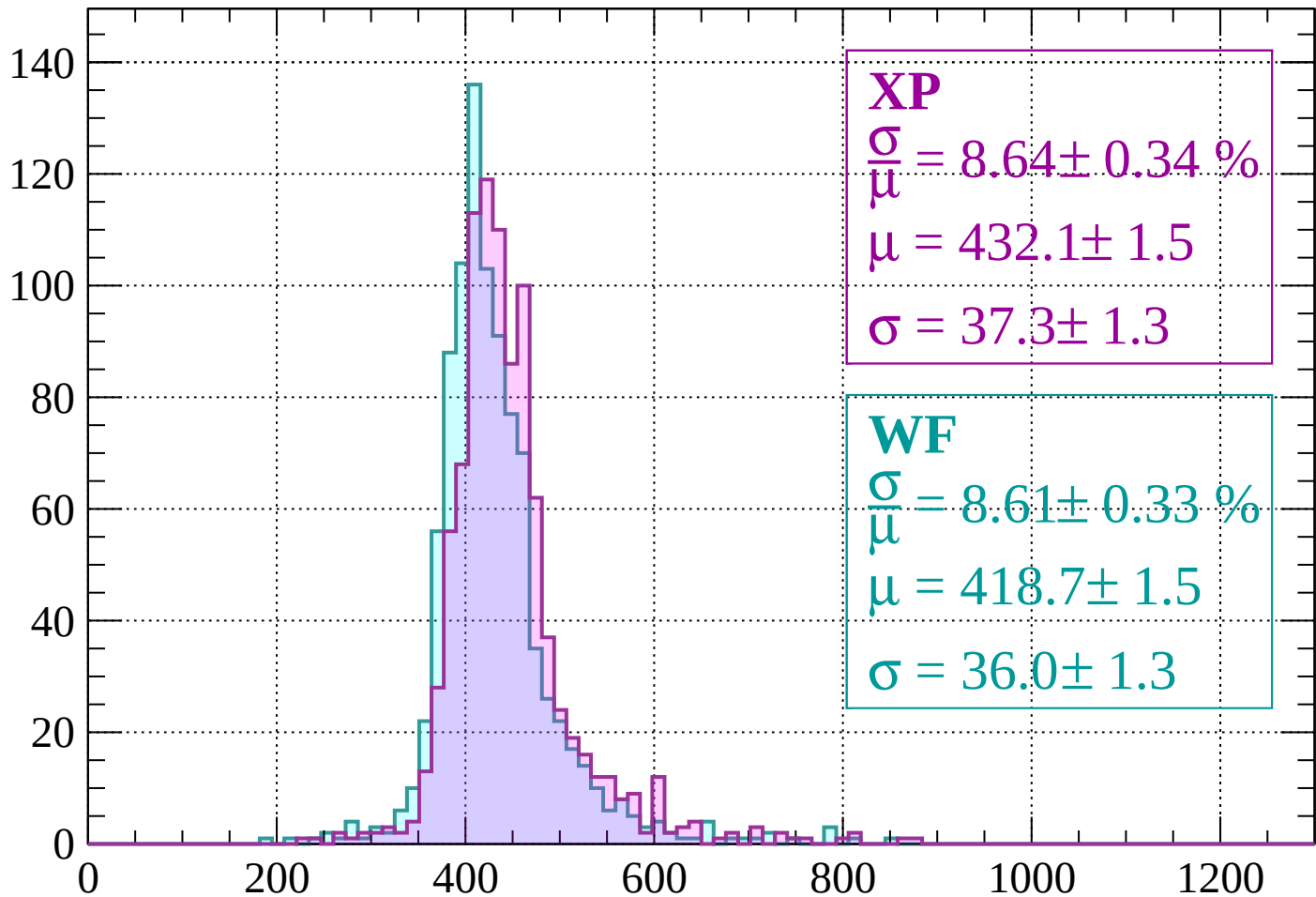
Energy loss | $-600 < p < -560$

Count



Energy loss | $-560 < p < -520$

Count



XP

$$\frac{\sigma}{\mu} = 8.64 \pm 0.34 \%$$

$$\mu = 432.1 \pm 1.5$$

$$\sigma = 37.3 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 8.61 \pm 0.33 \%$$

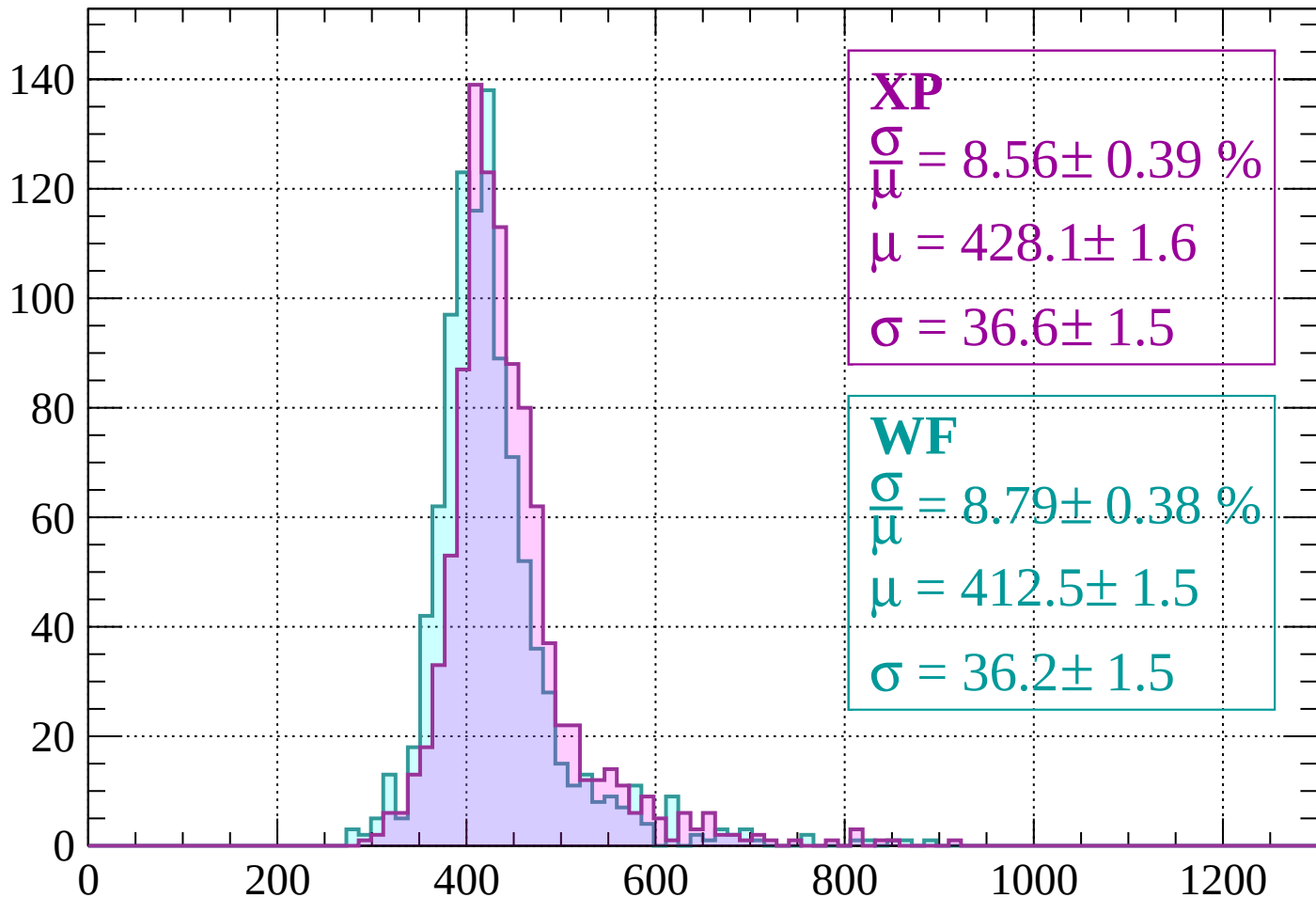
$$\mu = 418.7 \pm 1.5$$

$$\sigma = 36.0 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-520 < p < -480$

Count



XP

$$\frac{\sigma}{\mu} = 8.56 \pm 0.39 \%$$

$$\mu = 428.1 \pm 1.6$$

$$\sigma = 36.6 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 8.79 \pm 0.38 \%$$

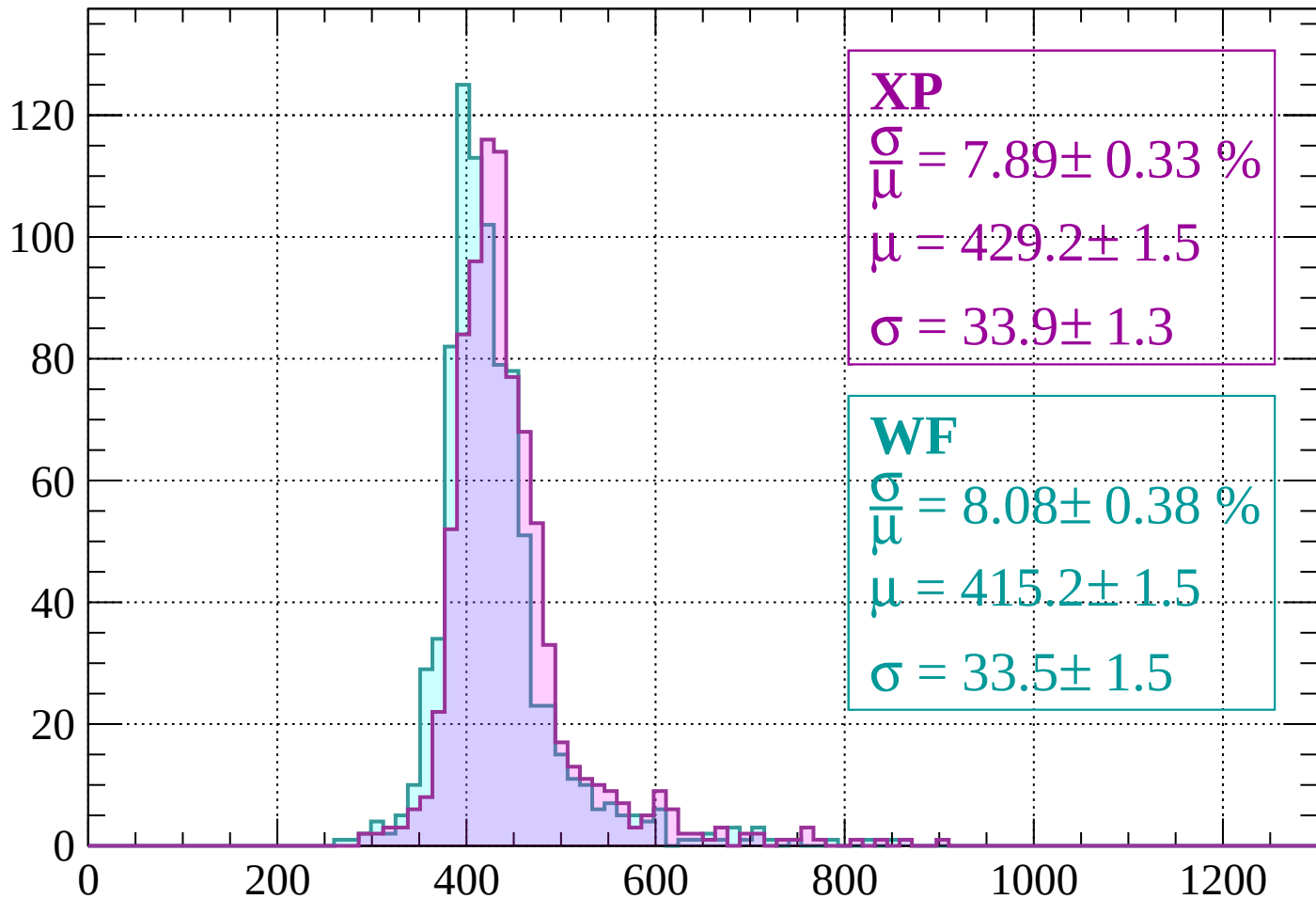
$$\mu = 412.5 \pm 1.5$$

$$\sigma = 36.2 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -440$

Count



XP

$$\frac{\sigma}{\mu} = 7.89 \pm 0.33 \%$$

$$\mu = 429.2 \pm 1.5$$

$$\sigma = 33.9 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 8.08 \pm 0.38 \%$$

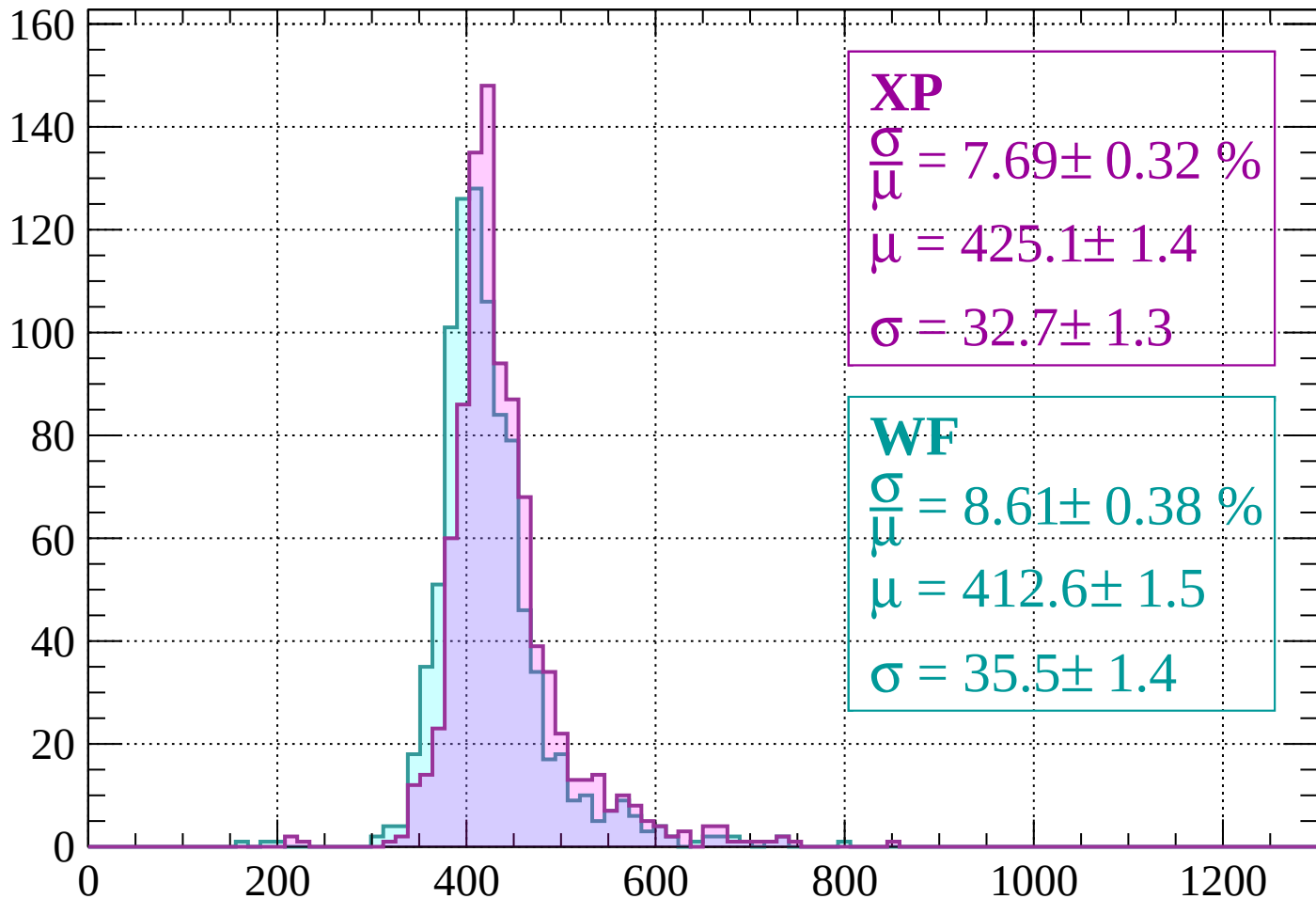
$$\mu = 415.2 \pm 1.5$$

$$\sigma = 33.5 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-440 < p < -400$

Count



XP

$$\frac{\sigma}{\mu} = 7.69 \pm 0.32 \%$$

$$\mu = 425.1 \pm 1.4$$

$$\sigma = 32.7 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 8.61 \pm 0.38 \%$$

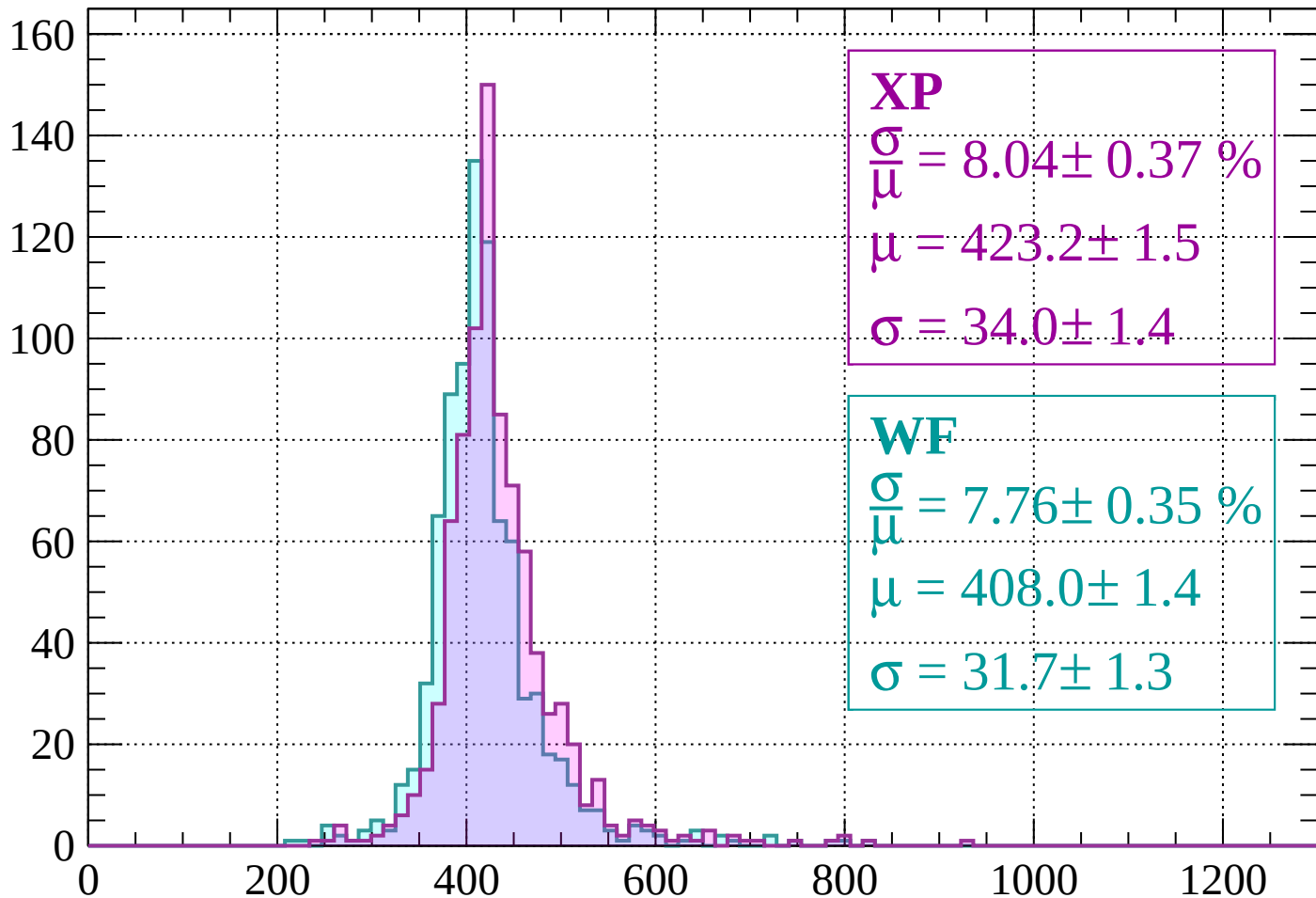
$$\mu = 412.6 \pm 1.5$$

$$\sigma = 35.5 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-400 < p < -360$

Count



XP

$$\frac{\sigma}{\mu} = 8.04 \pm 0.37 \%$$

$$\mu = 423.2 \pm 1.5$$

$$\sigma = 34.0 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 7.76 \pm 0.35 \%$$

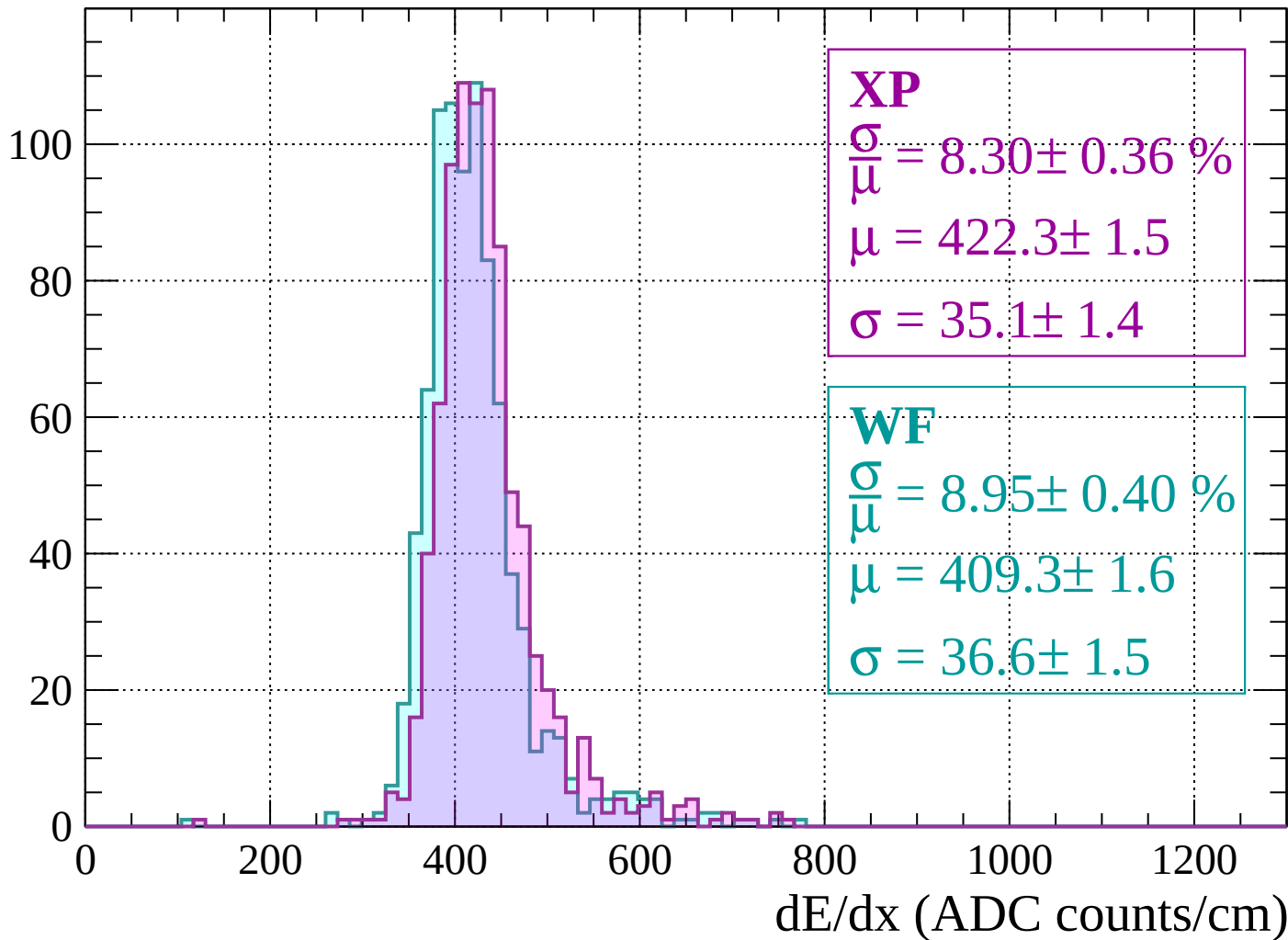
$$\mu = 408.0 \pm 1.4$$

$$\sigma = 31.7 \pm 1.3$$

dE/dx (ADC counts/cm)

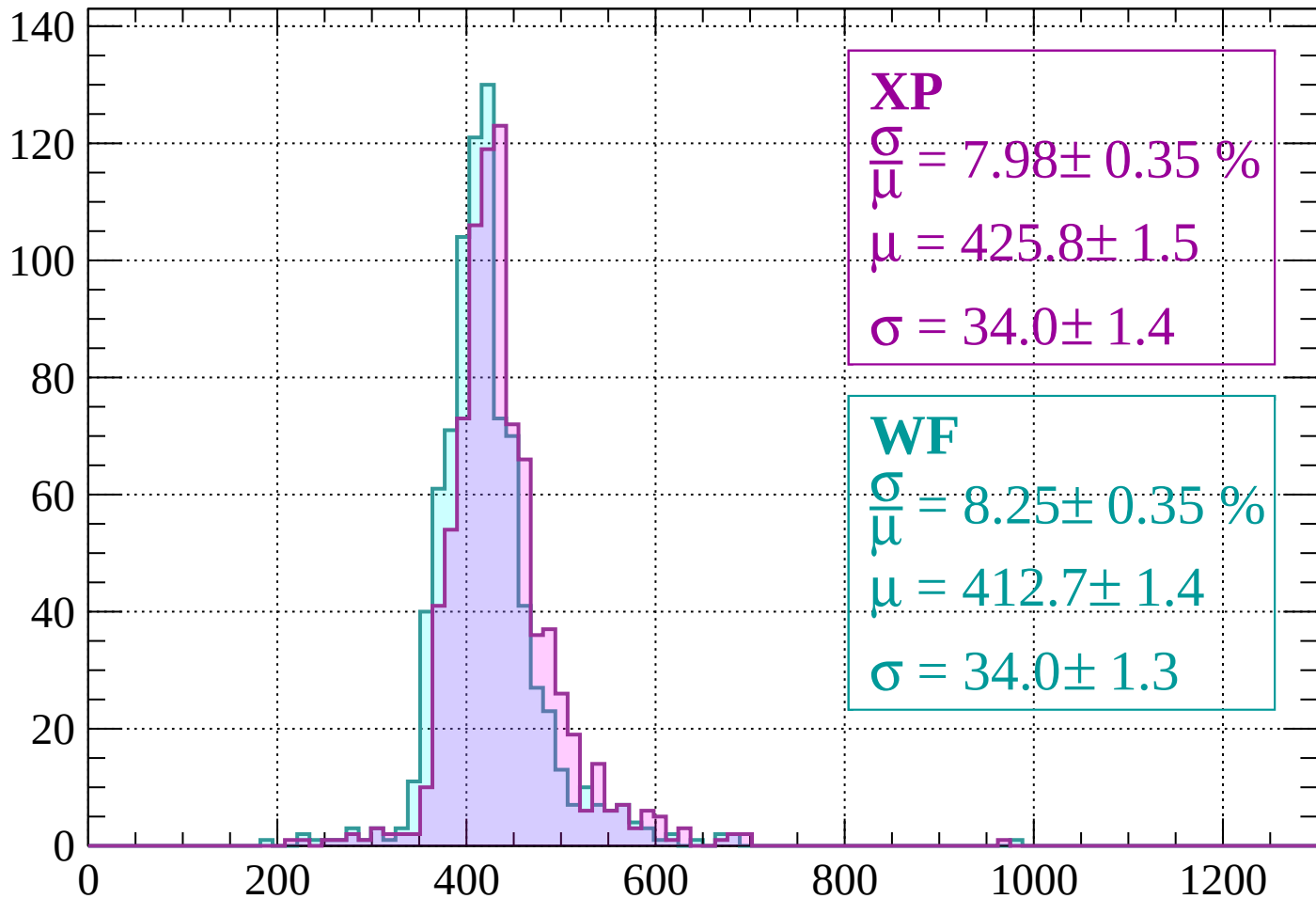
Energy loss | $-360 < p < -320$

Count



Energy loss | $-320 < p < -280$

Count



XP

$$\frac{\sigma}{\mu} = 7.98 \pm 0.35 \%$$

$$\mu = 425.8 \pm 1.5$$

$$\sigma = 34.0 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.25 \pm 0.35 \%$$

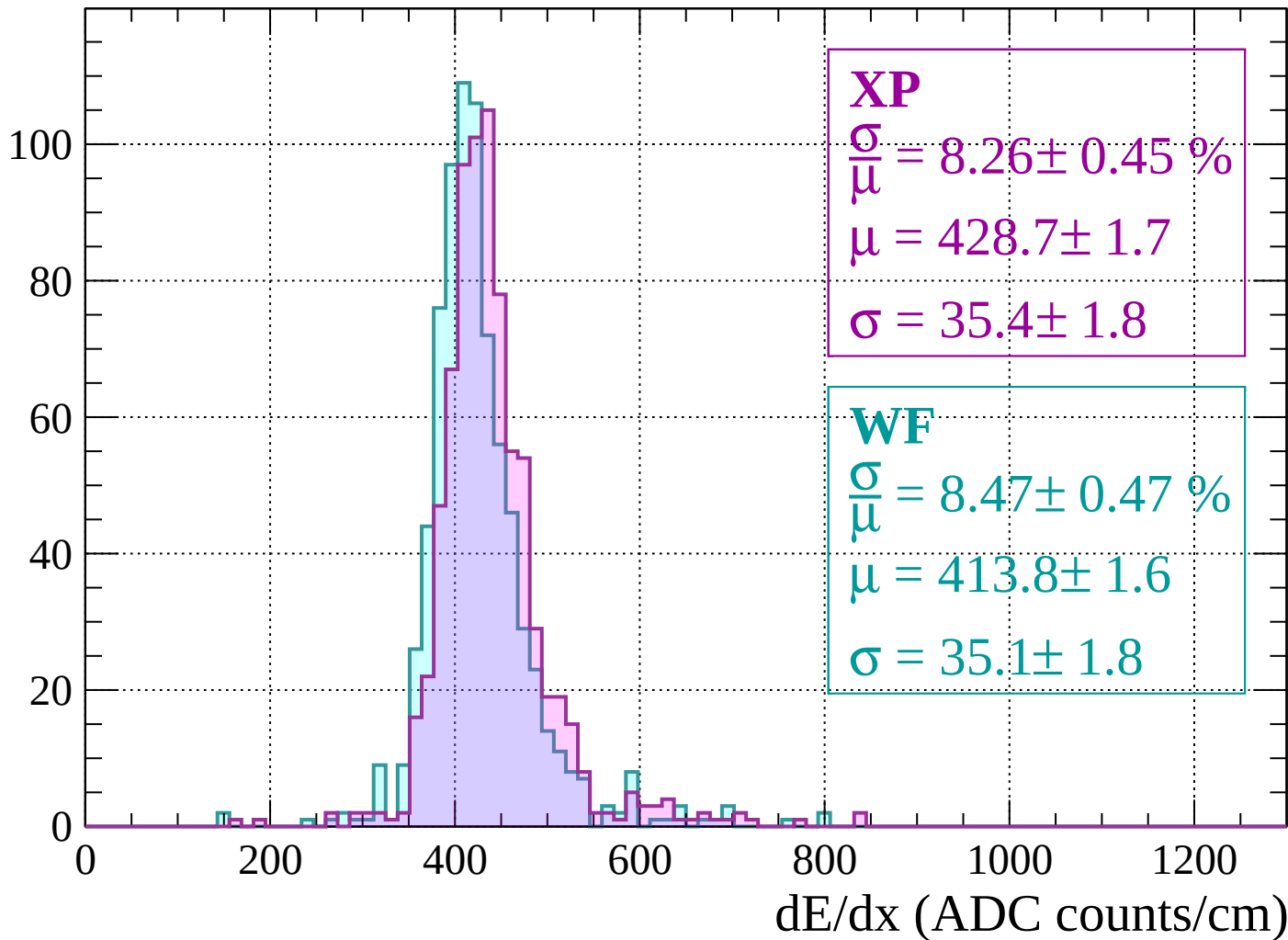
$$\mu = 412.7 \pm 1.4$$

$$\sigma = 34.0 \pm 1.3$$

dE/dx (ADC counts/cm)

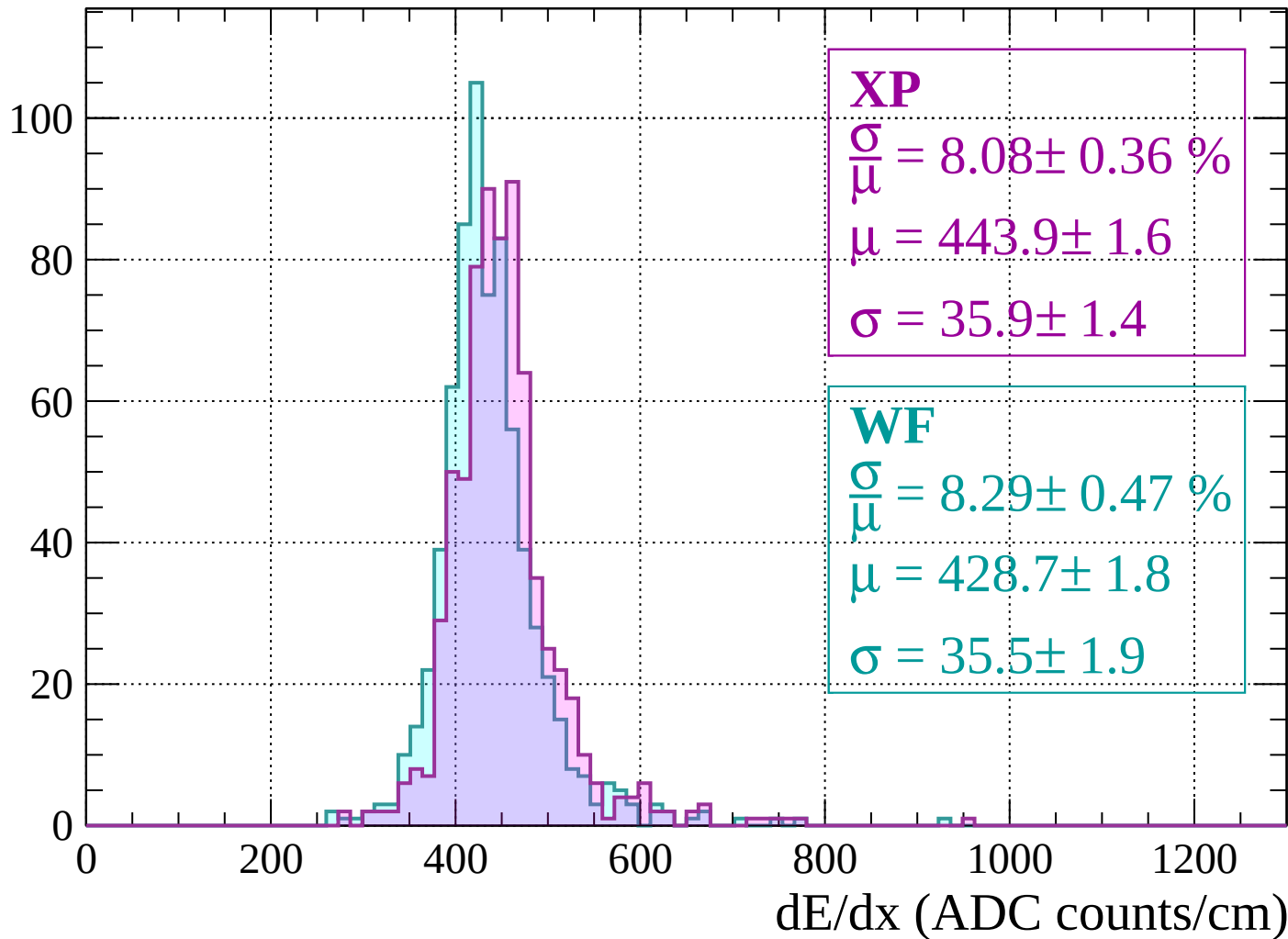
Energy loss | $-280 < p < -240$

Count



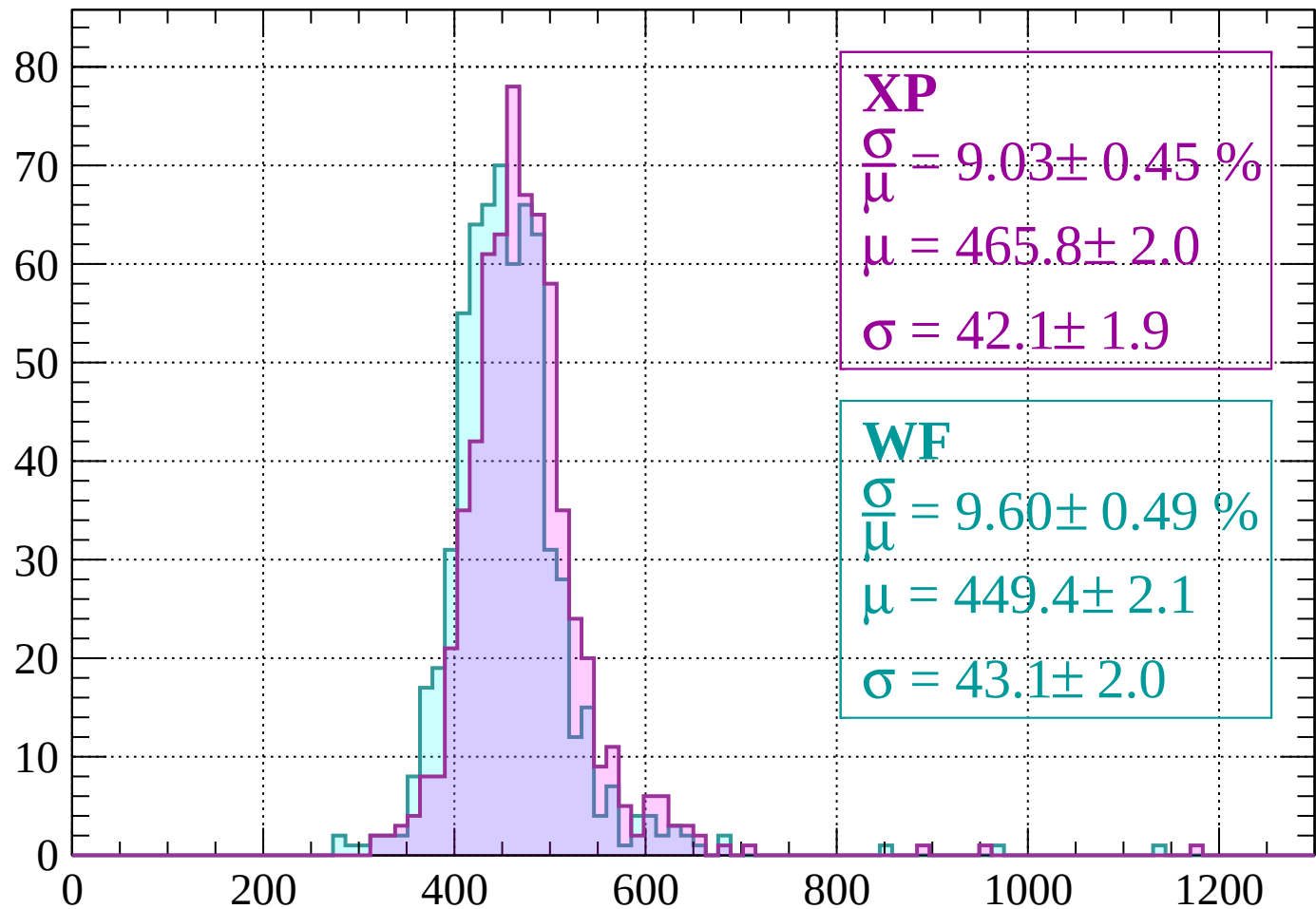
Energy loss | $-240 < p < -200$

Count



Energy loss | $-200 < p < -160$

Count



XP

$$\frac{\sigma}{\mu} = 9.03 \pm 0.45 \%$$

$$\mu = 465.8 \pm 2.0$$

$$\sigma = 42.1 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 9.60 \pm 0.49 \%$$

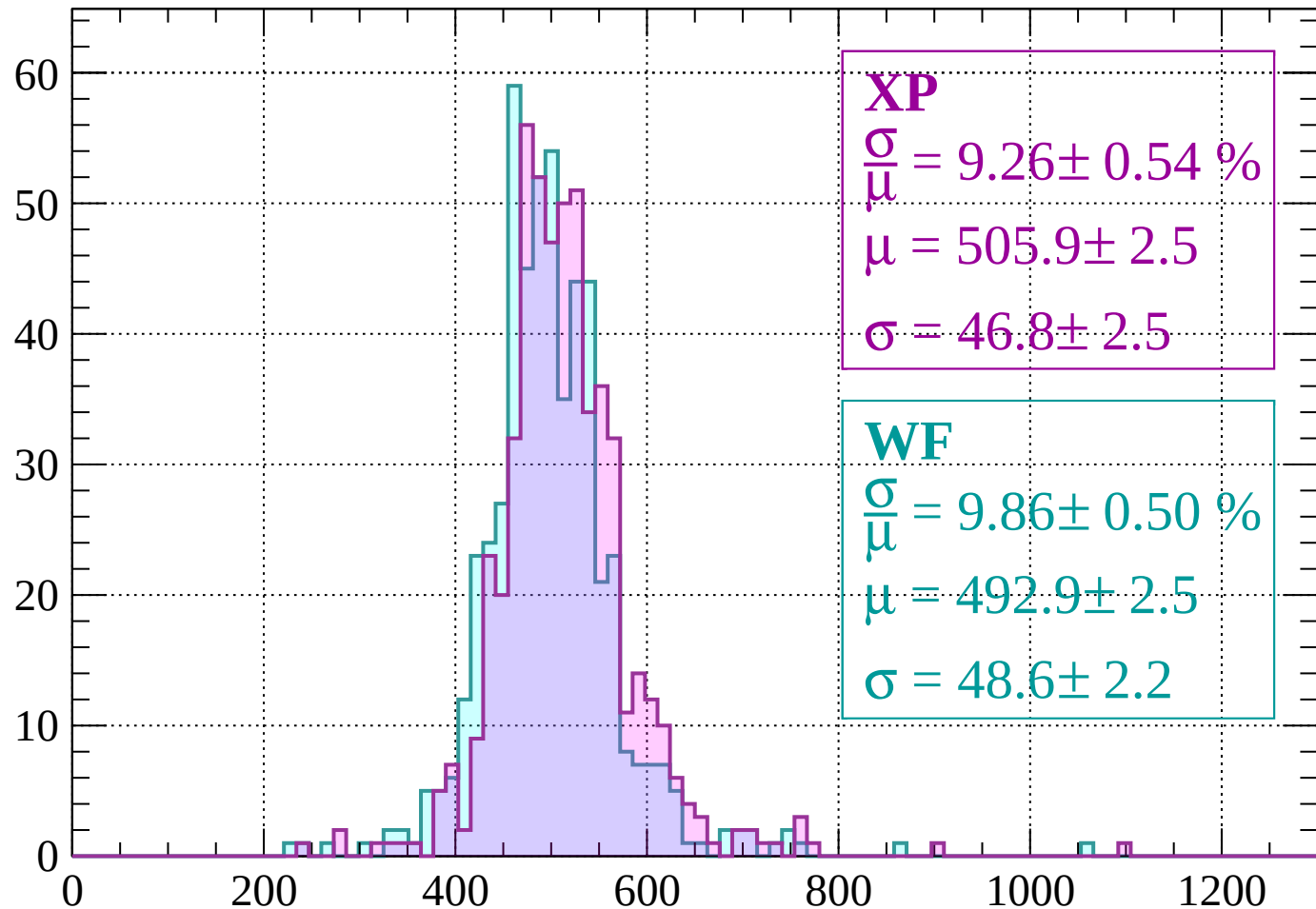
$$\mu = 449.4 \pm 2.1$$

$$\sigma = 43.1 \pm 2.0$$

dE/dx (ADC counts/cm)

Energy loss | $-160 < p < -120$

Count



XP

$$\frac{\sigma}{\mu} = 9.26 \pm 0.54 \%$$

$$\mu = 505.9 \pm 2.5$$

$$\sigma = 46.8 \pm 2.5$$

WF

$$\frac{\sigma}{\mu} = 9.86 \pm 0.50 \%$$

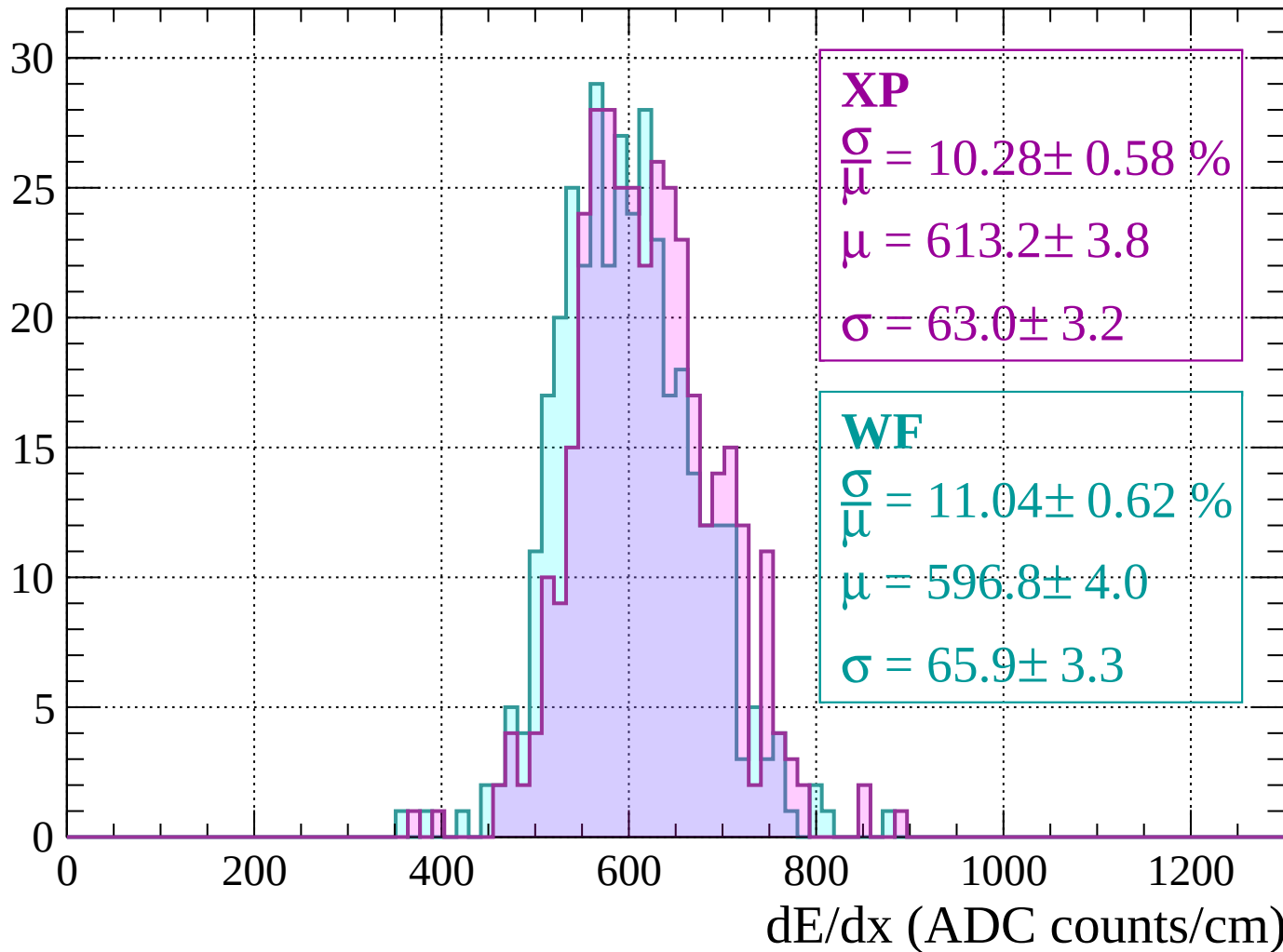
$$\mu = 492.9 \pm 2.5$$

$$\sigma = 48.6 \pm 2.2$$

dE/dx (ADC counts/cm)

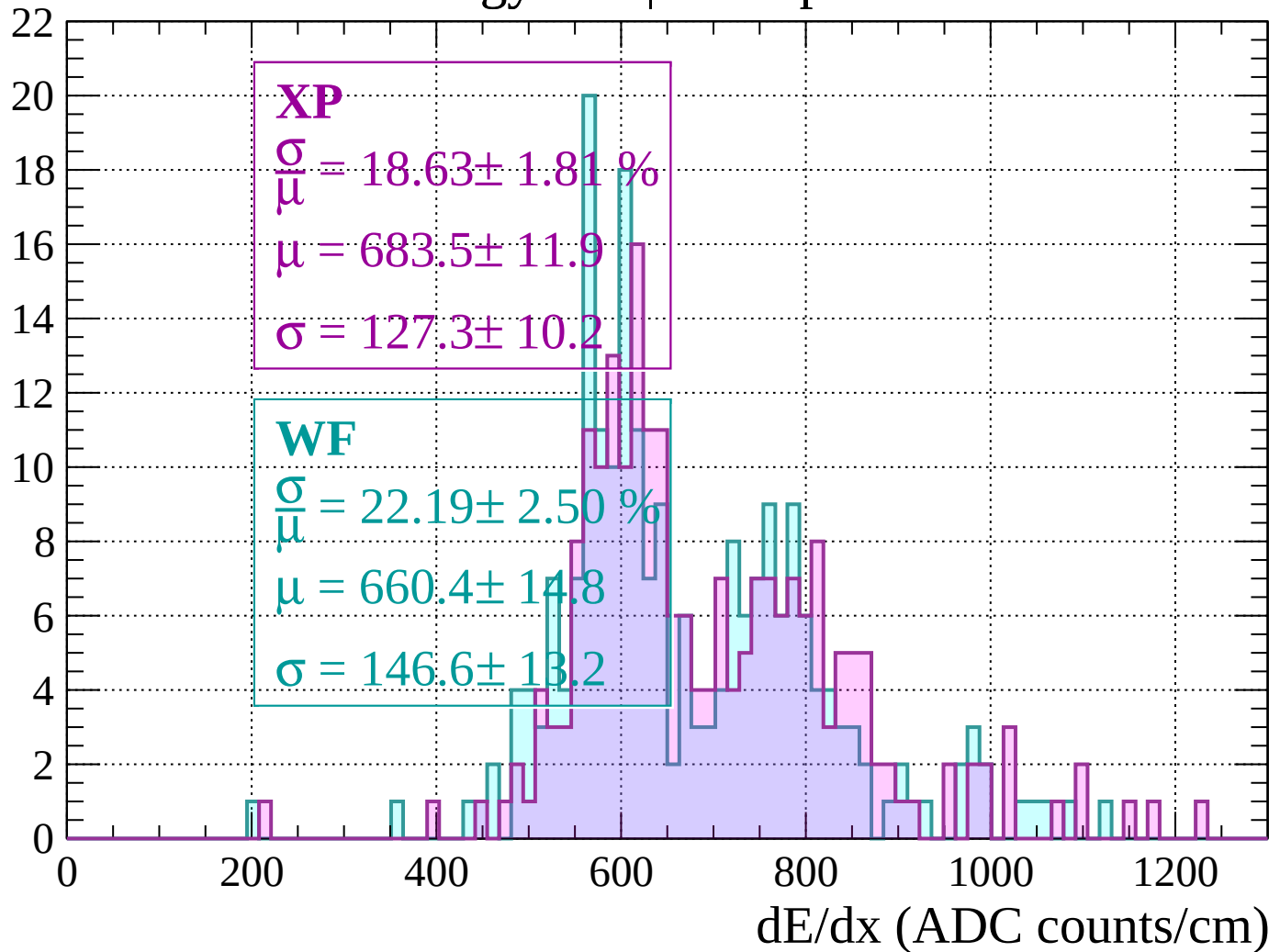
Energy loss | $-120 < p < -80$

Count



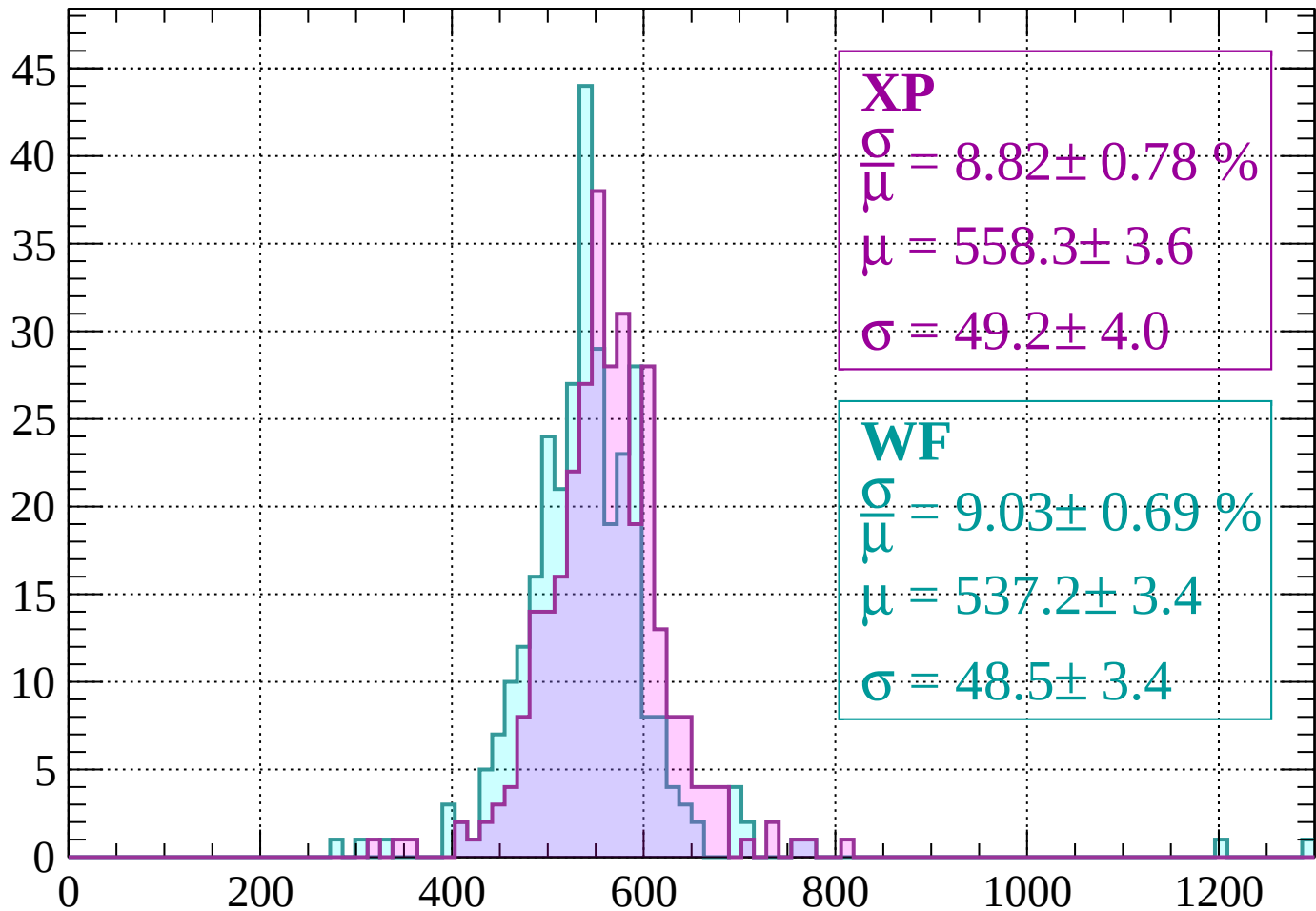
Energy loss | $-80 < p < -40$

Count



Energy loss | $-40 < p < 0$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 8.82 \pm 0.78 \%$$

$$\mu = 558.3 \pm 3.6$$

$$\sigma = 49.2 \pm 4.0$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.03 \pm 0.69 \%$$

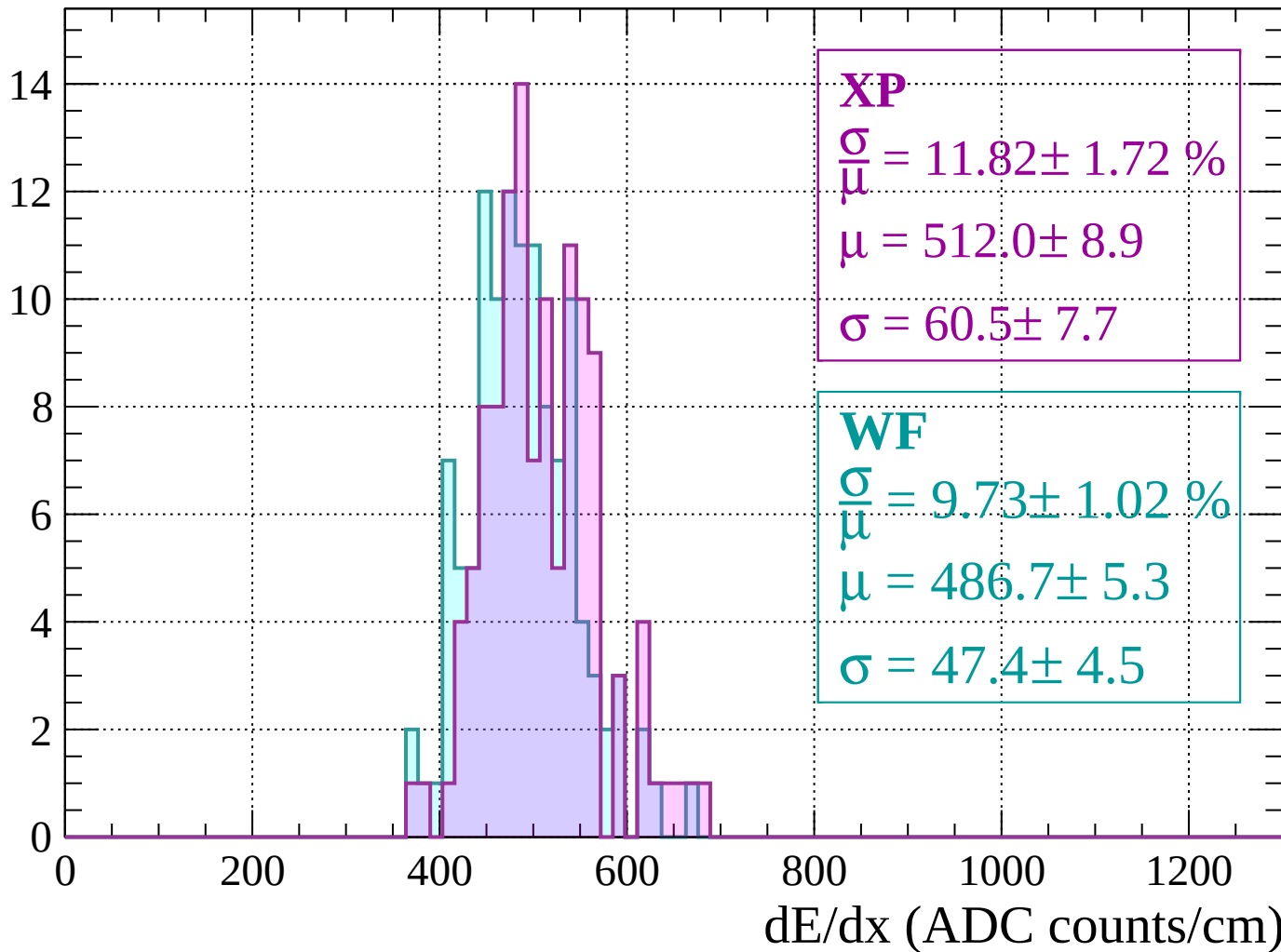
$$\mu = 537.2 \pm 3.4$$

$$\sigma = 48.5 \pm 3.4$$

dE/dx (ADC counts/cm)

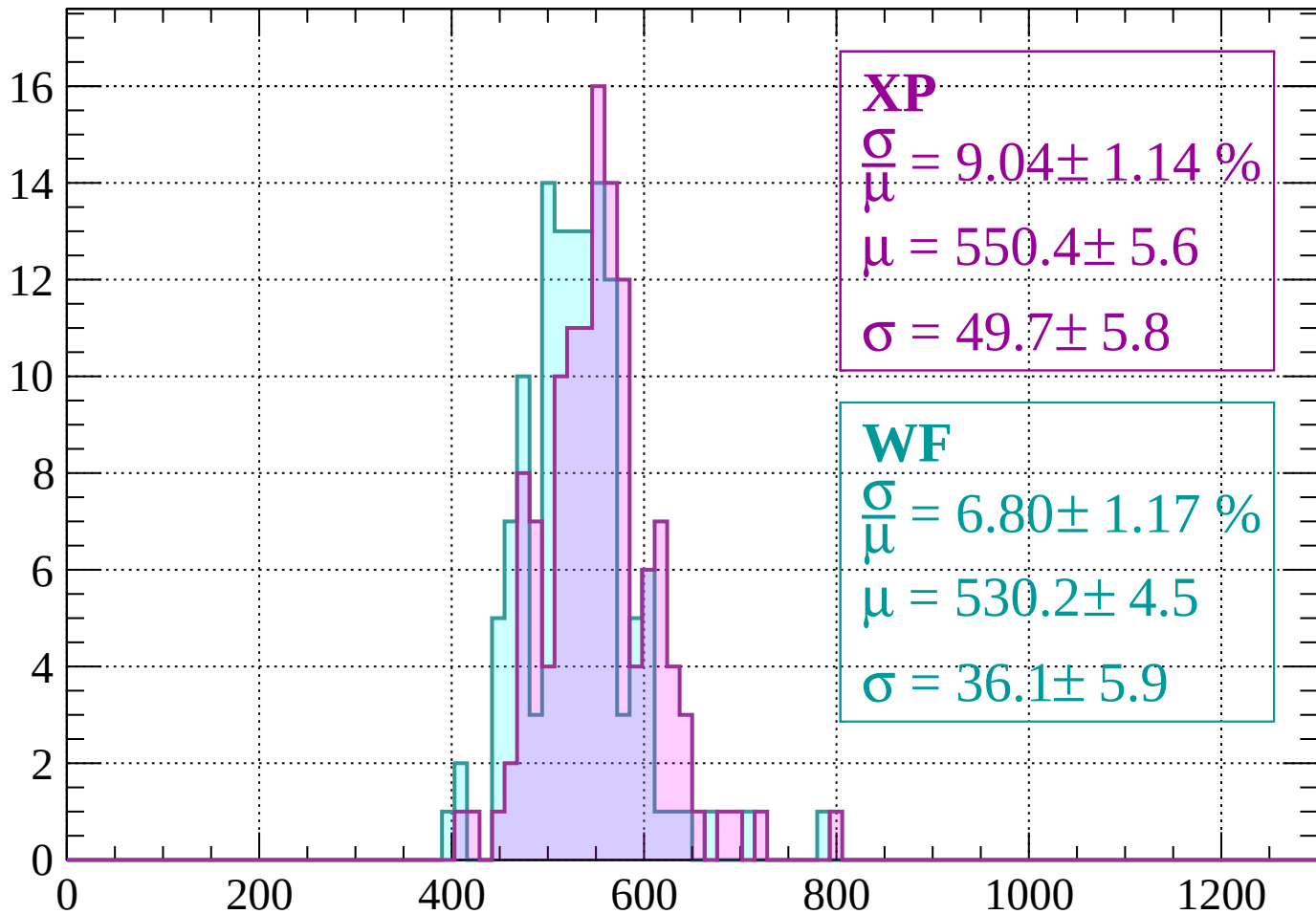
Energy loss | $0 < p < 40$

Count



Energy loss | $40 < p < 80$

Count



XP

$$\frac{\sigma}{\mu} = 9.04 \pm 1.14 \%$$

$$\mu = 550.4 \pm 5.6$$

$$\sigma = 49.7 \pm 5.8$$

WF

$$\frac{\sigma}{\mu} = 6.80 \pm 1.17 \%$$

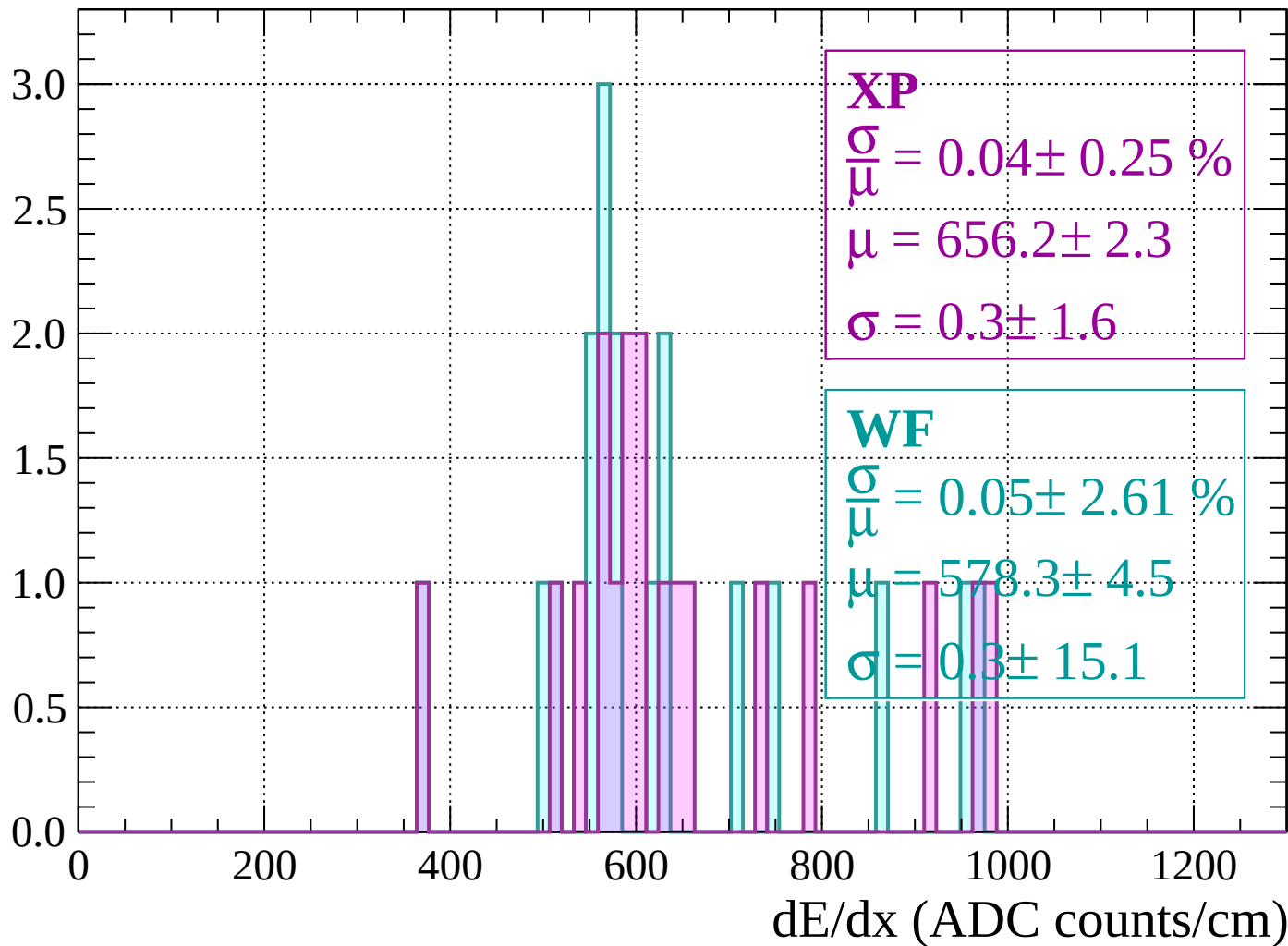
$$\mu = 530.2 \pm 4.5$$

$$\sigma = 36.1 \pm 5.9$$

dE/dx (ADC counts/cm)

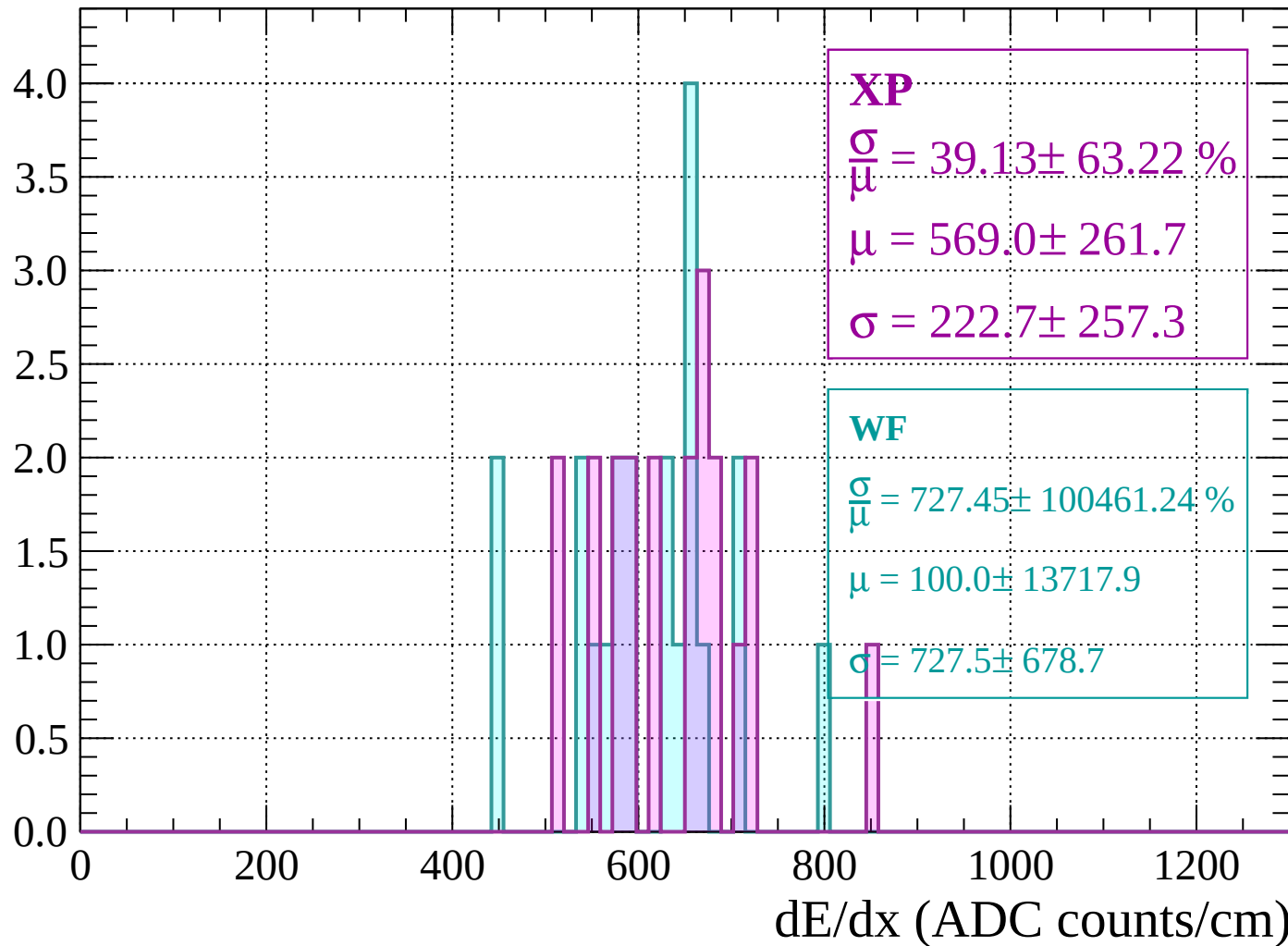
Energy loss | $80 < p < 120$

Count



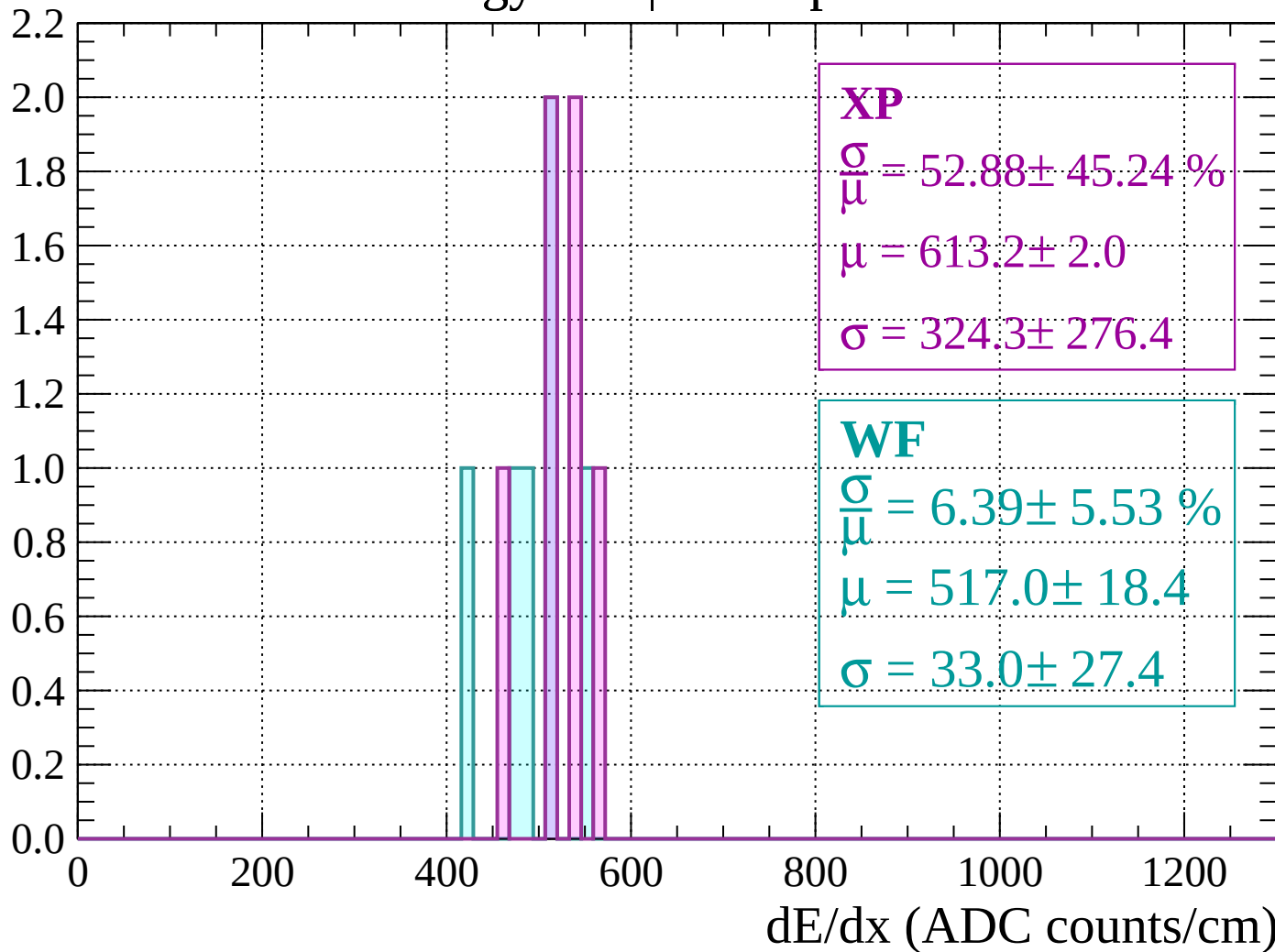
Energy loss | $120 < p < 160$

Count



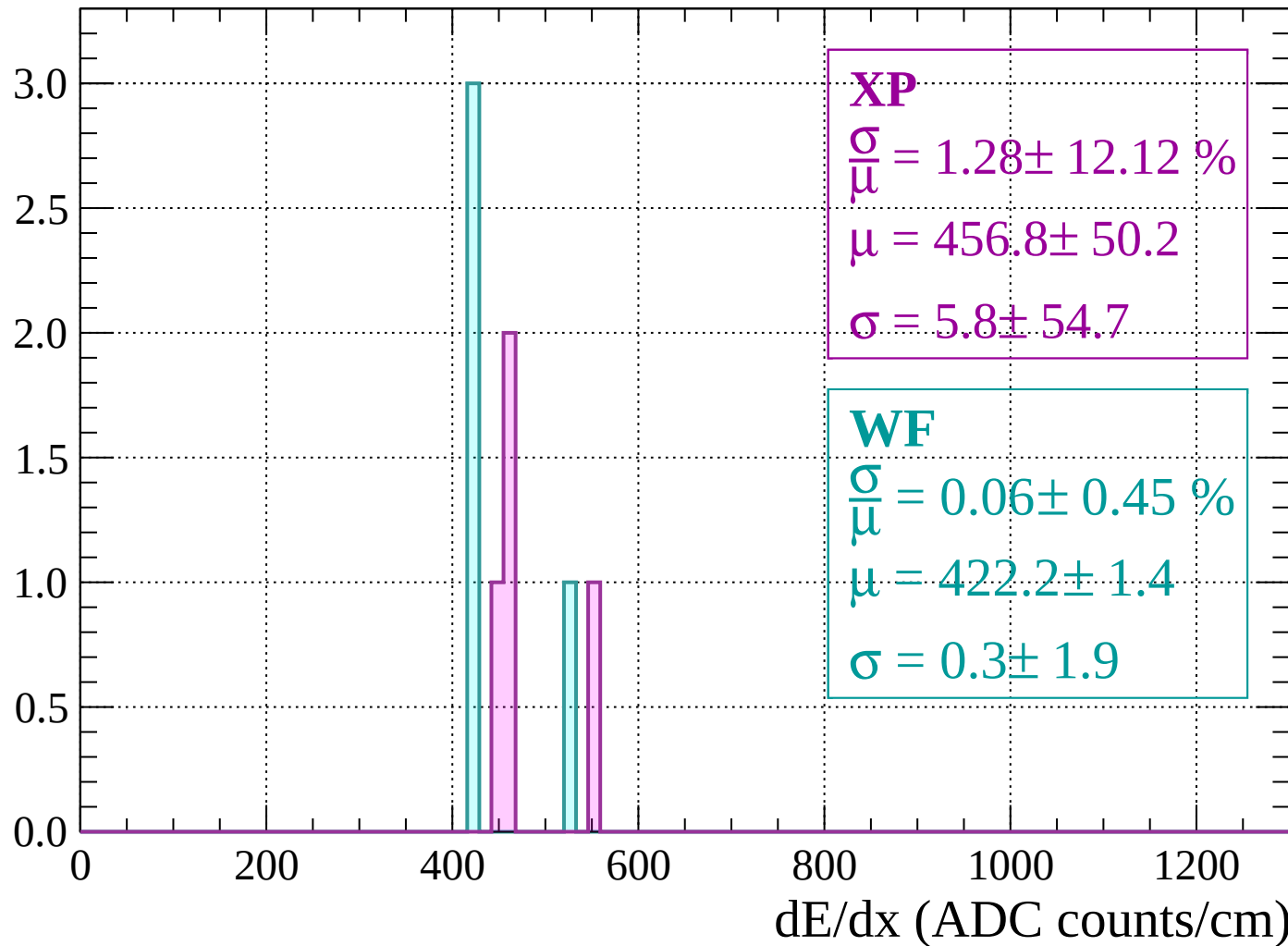
Energy loss | $160 < p < 200$

Count



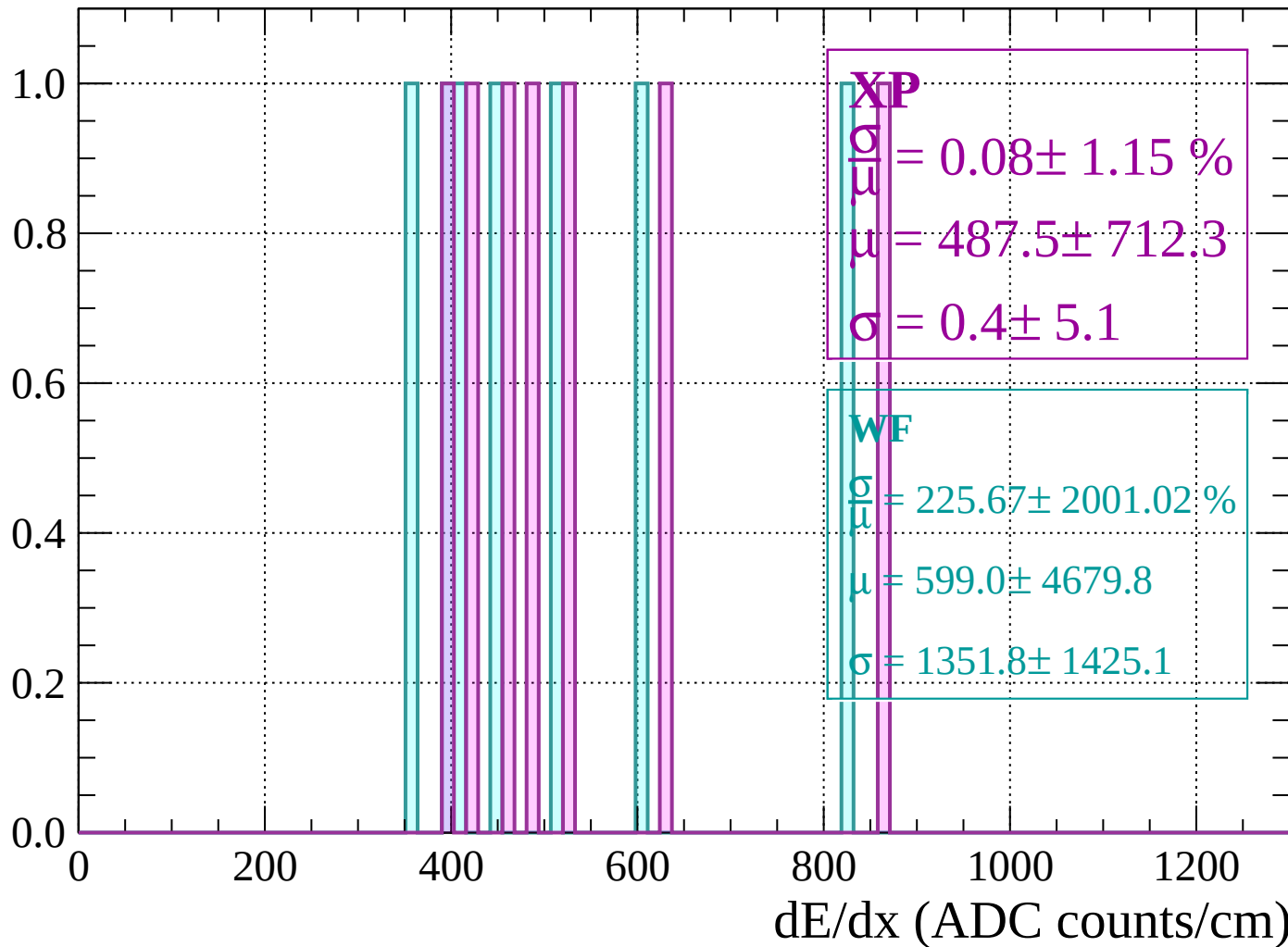
Energy loss | $200 < p < 240$

Count



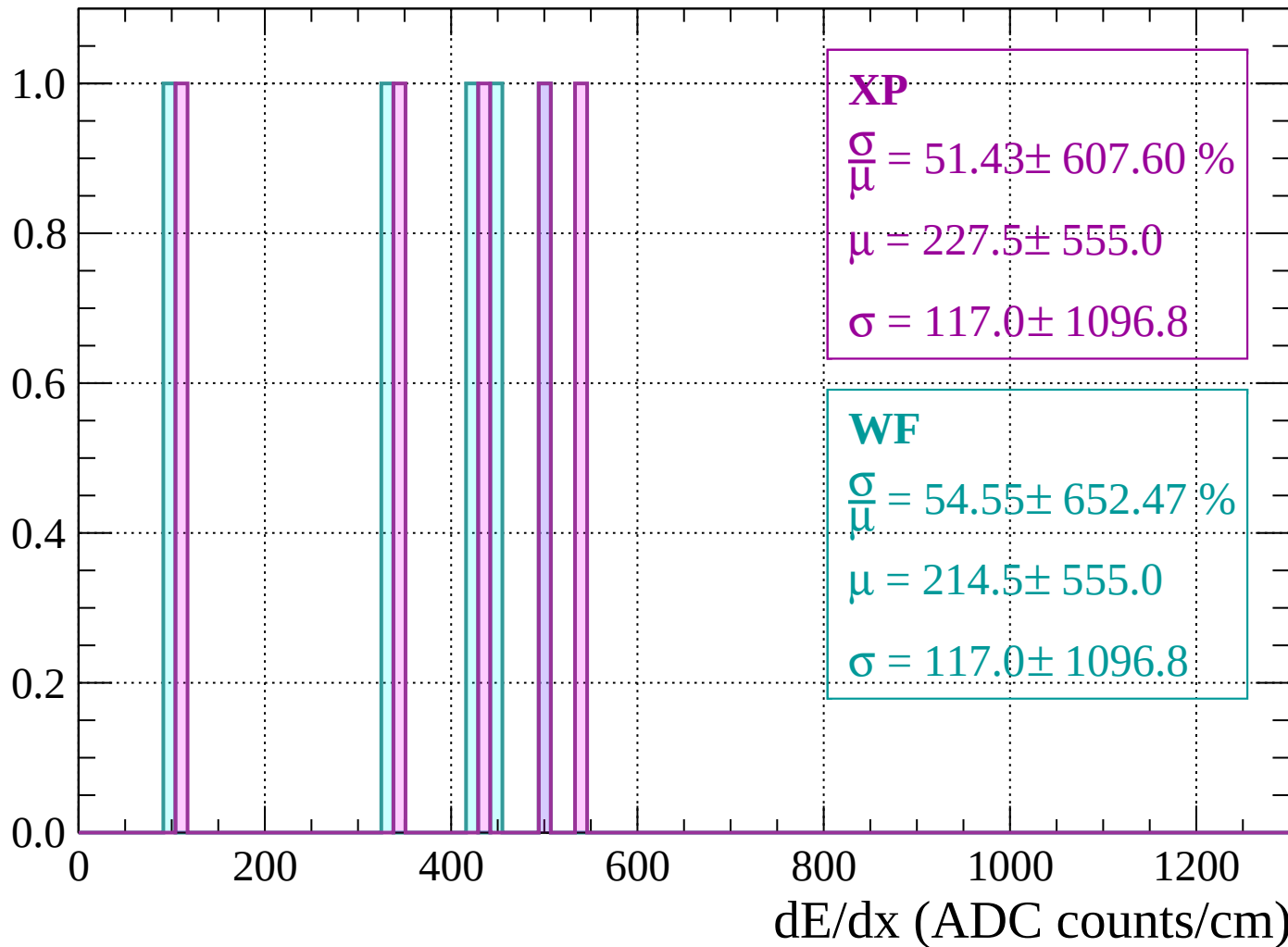
Energy loss | $240 < p < 280$

Count



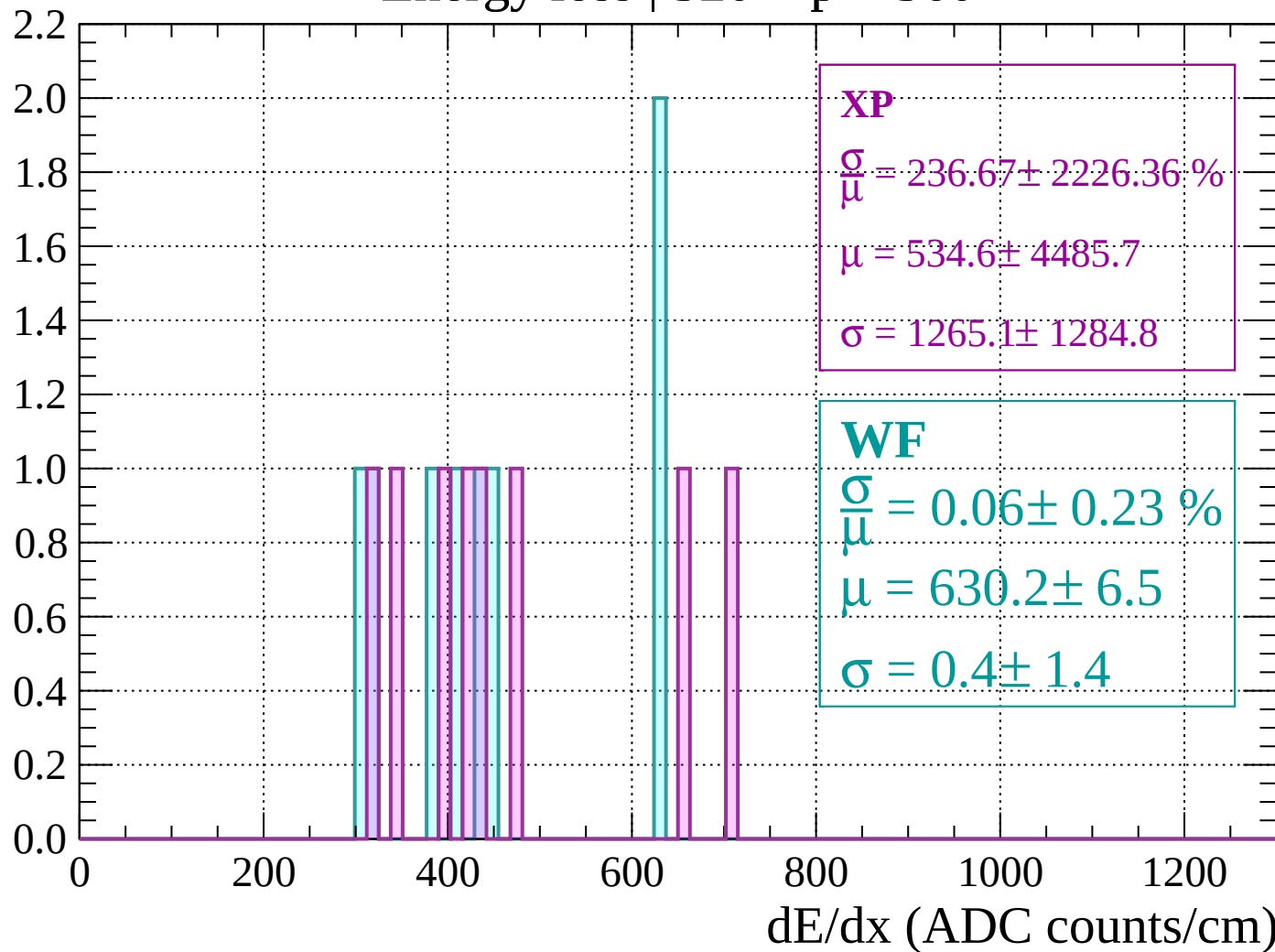
Energy loss | $280 < p < 320$

Count



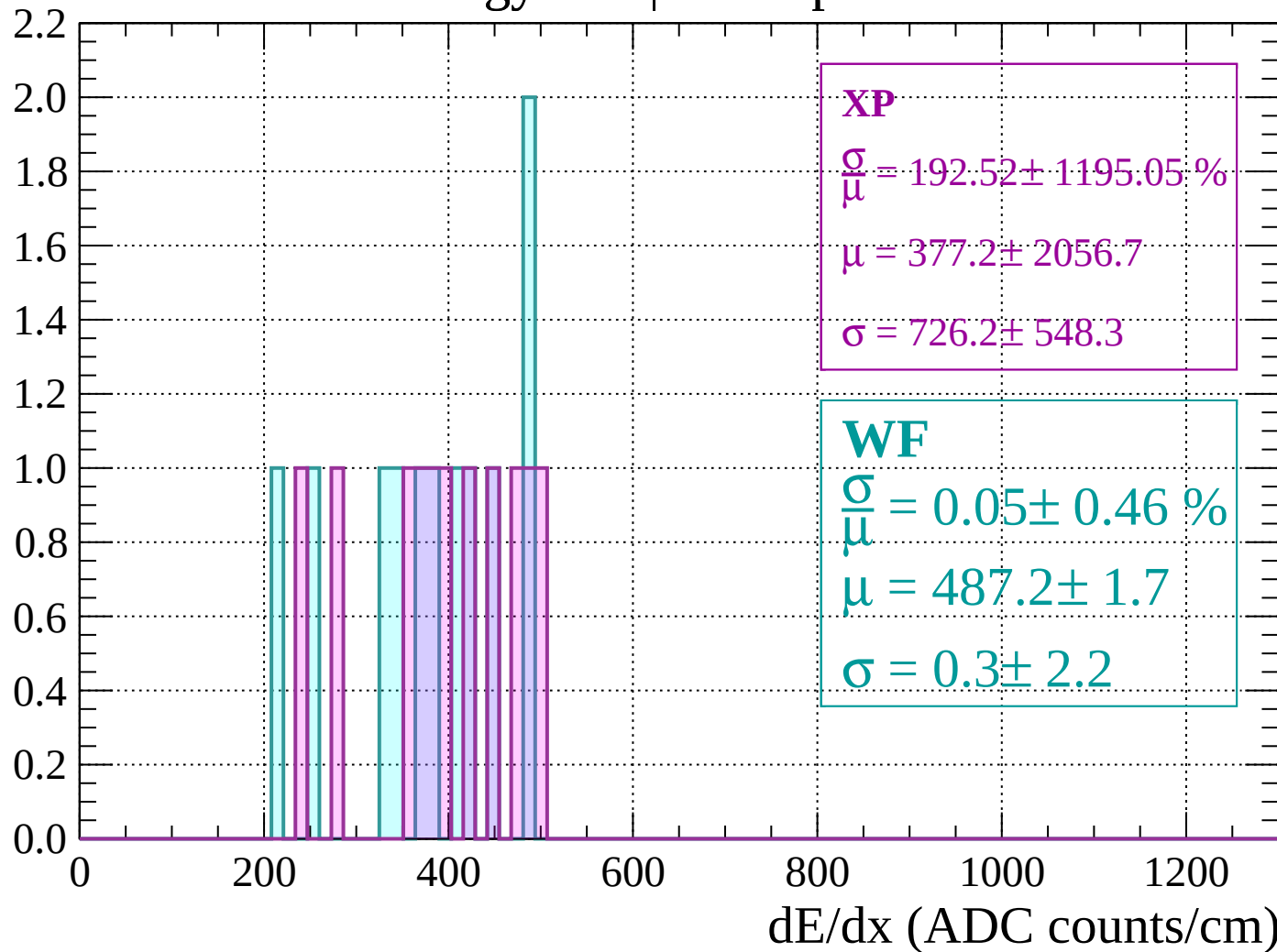
Energy loss | $320 < p < 360$

Count



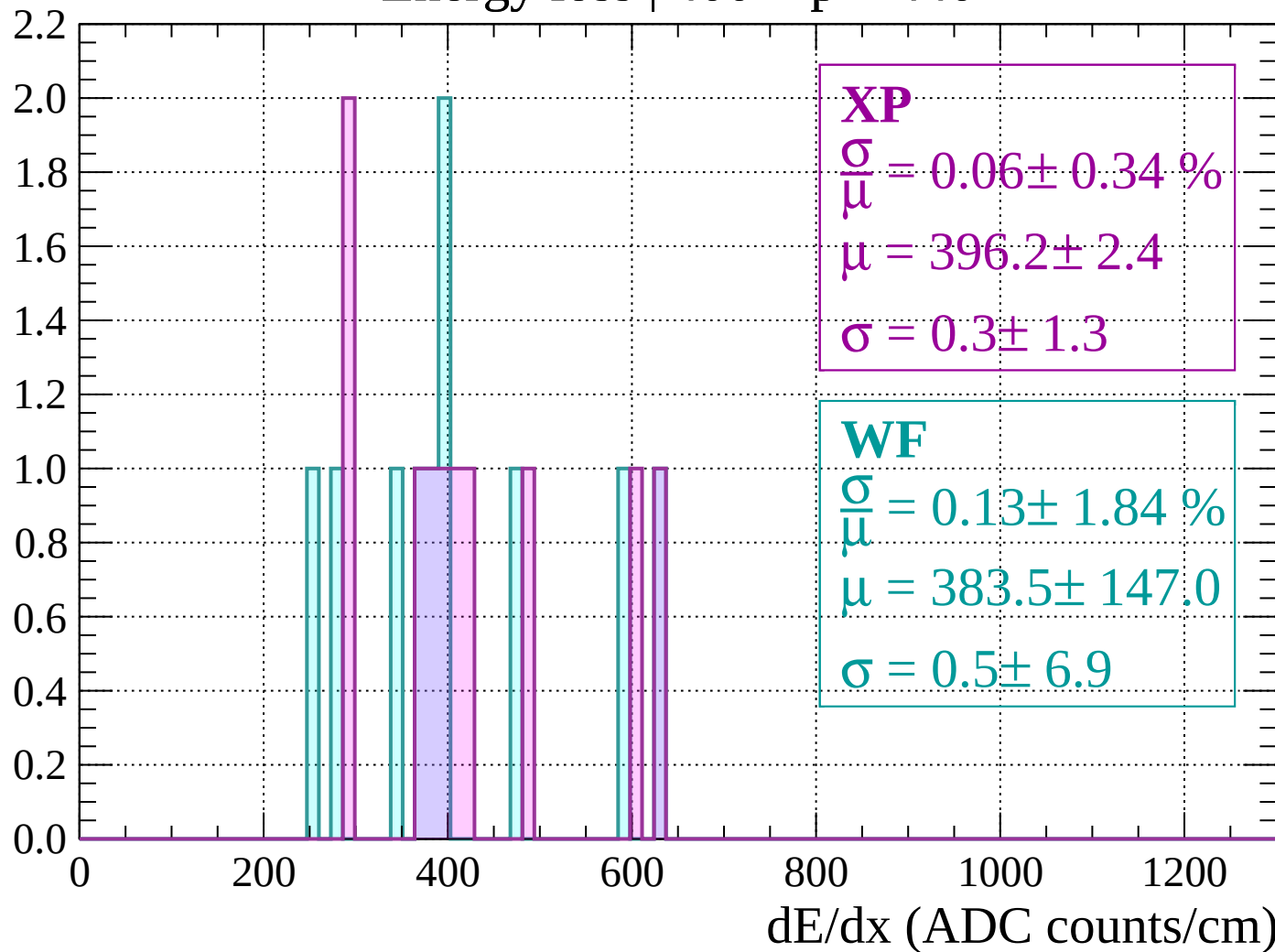
Energy loss | $360 < p < 400$

Count



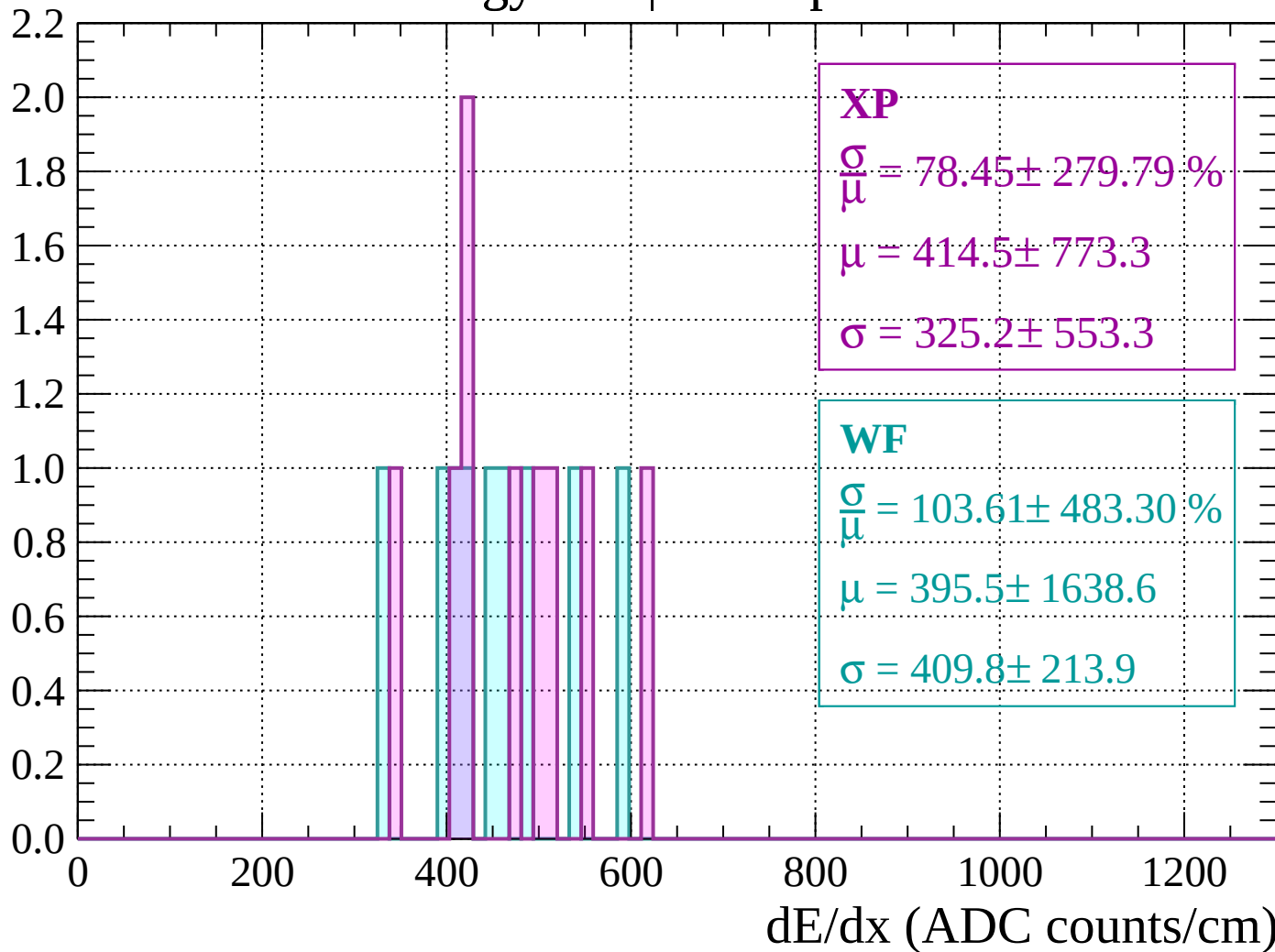
Energy loss | $400 < p < 440$

Count



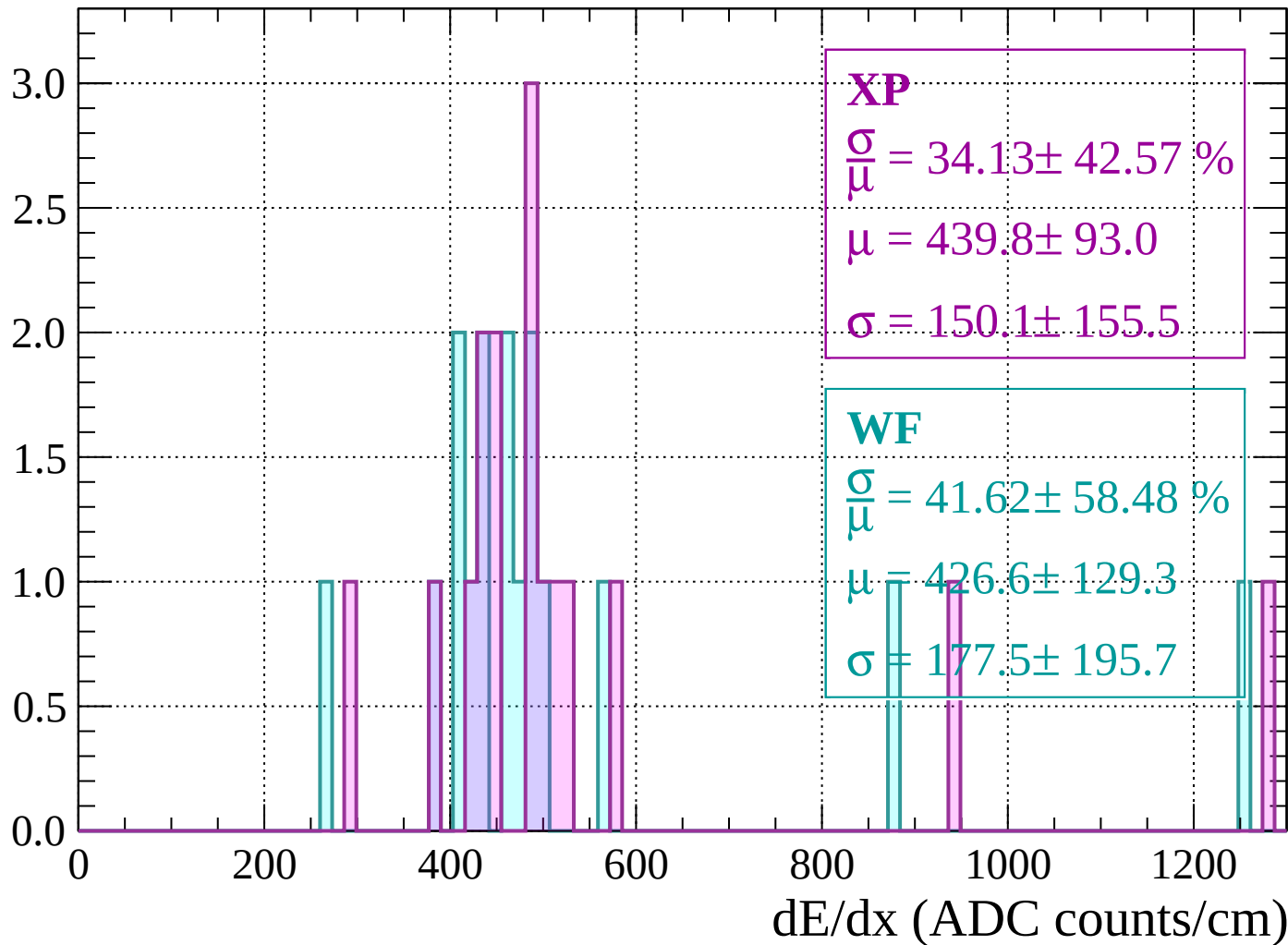
Energy loss | $440 < p < 480$

Count



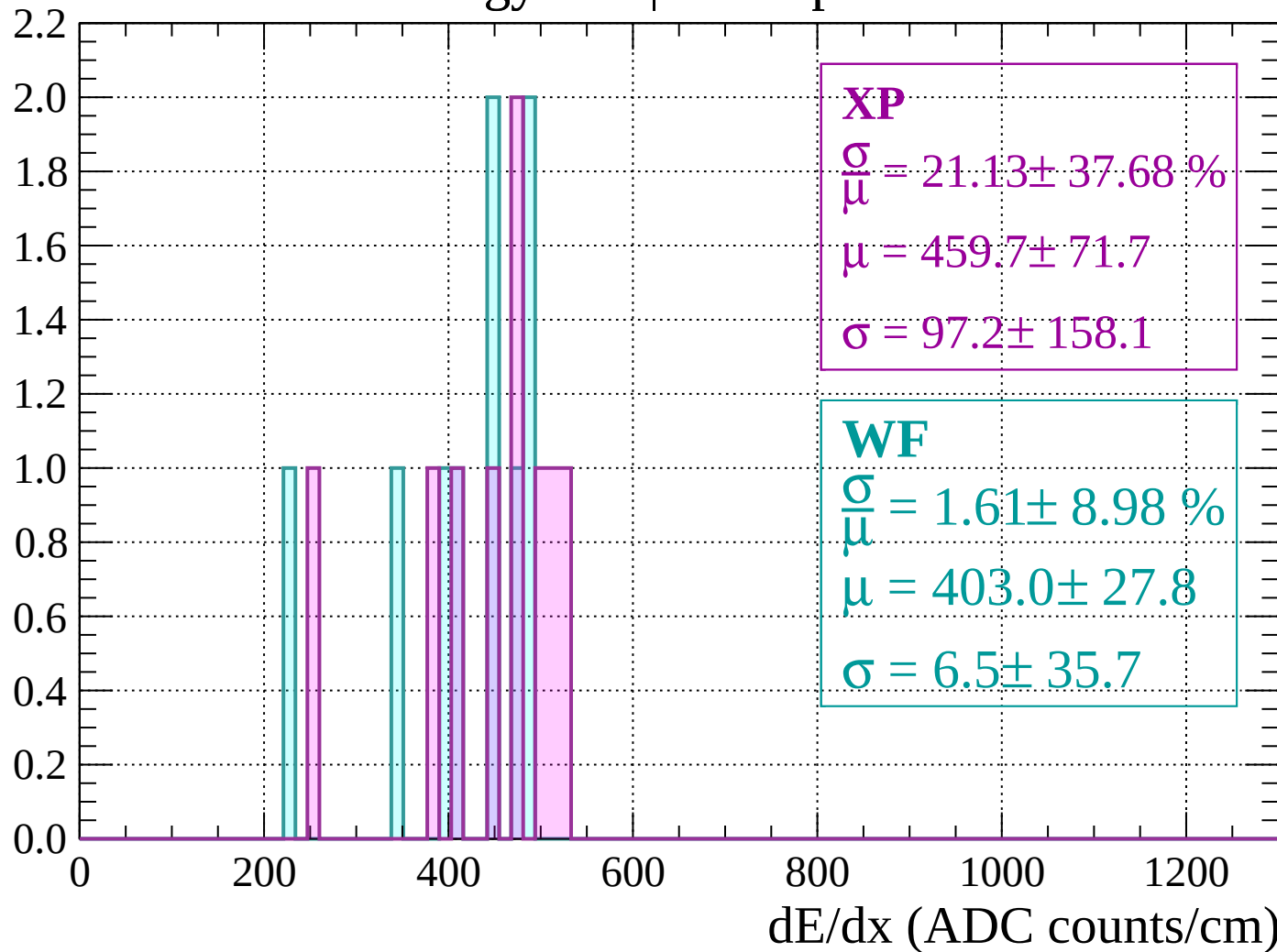
Energy loss | $480 < p < 520$

Count



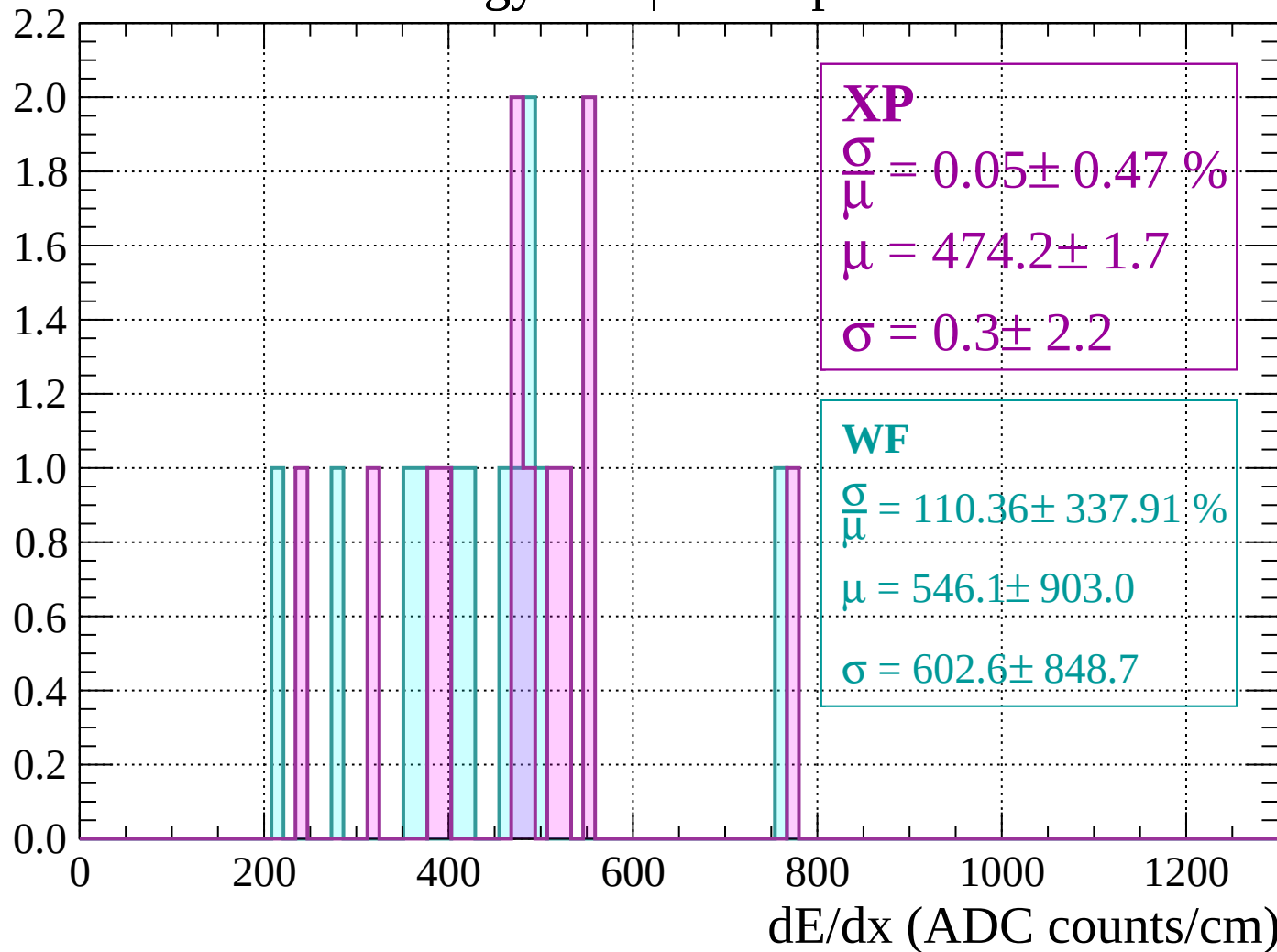
Energy loss | $520 < p < 560$

Count

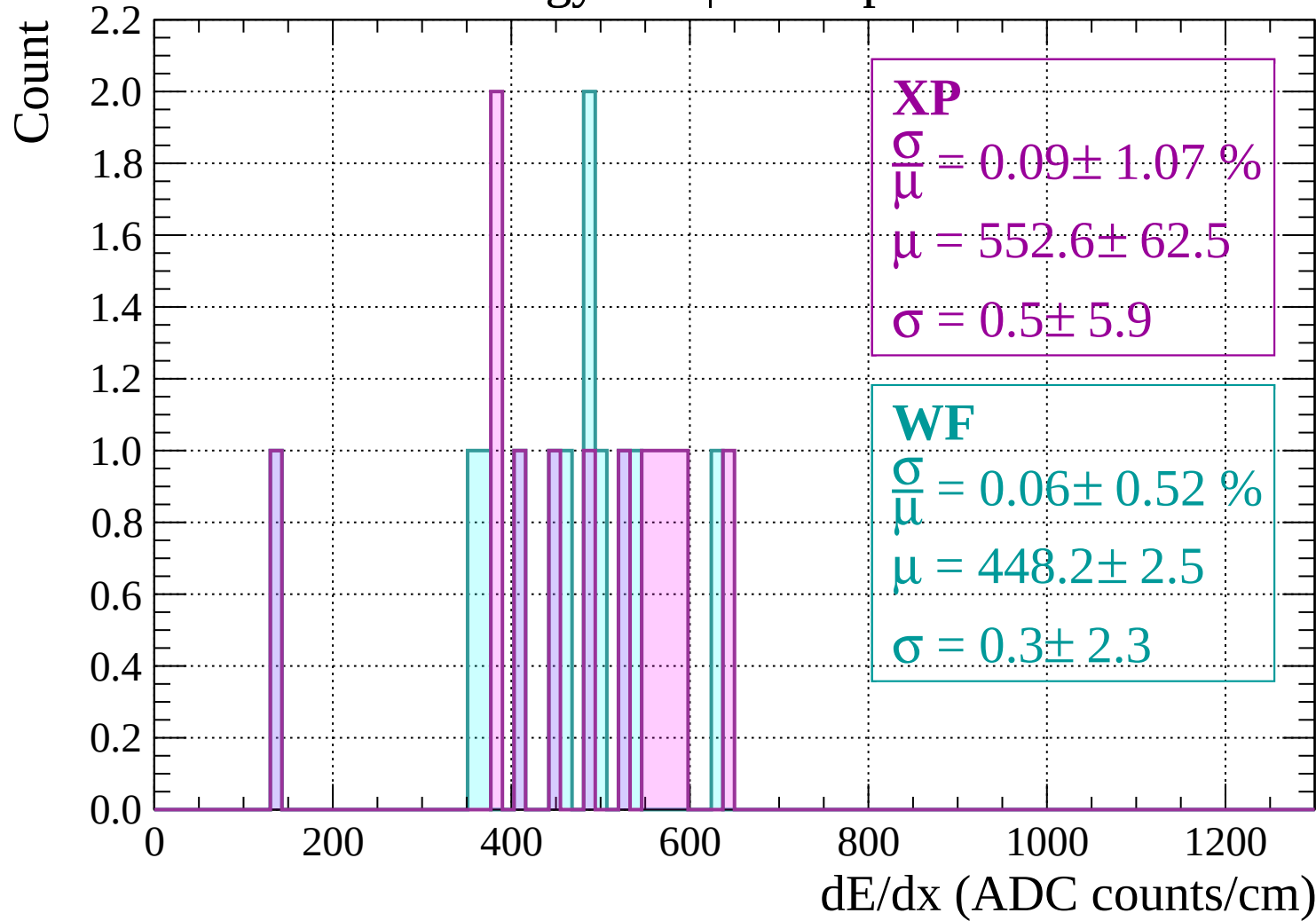


Energy loss | $560 < p < 600$

Count

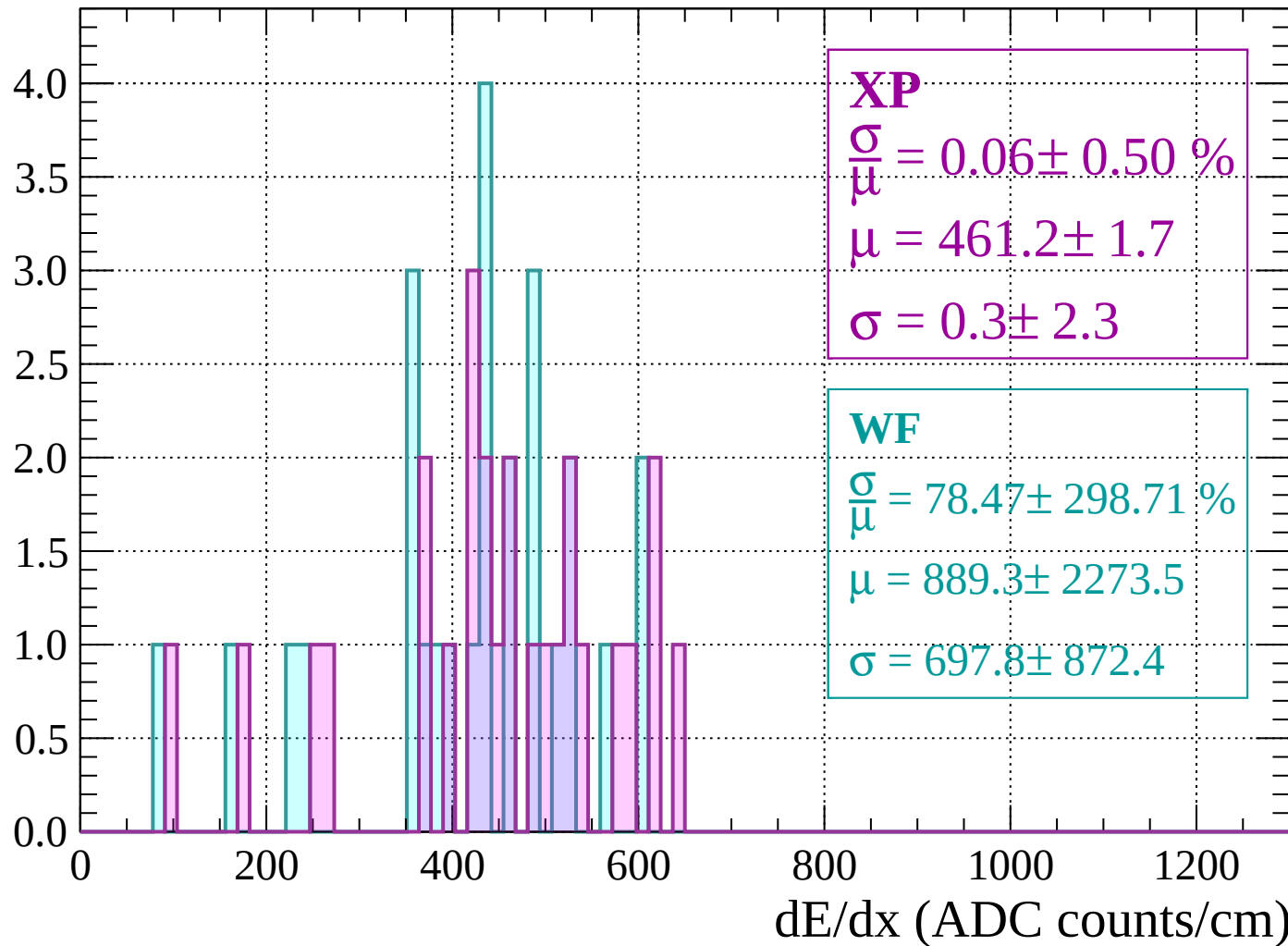


Energy loss | $600 < p < 640$



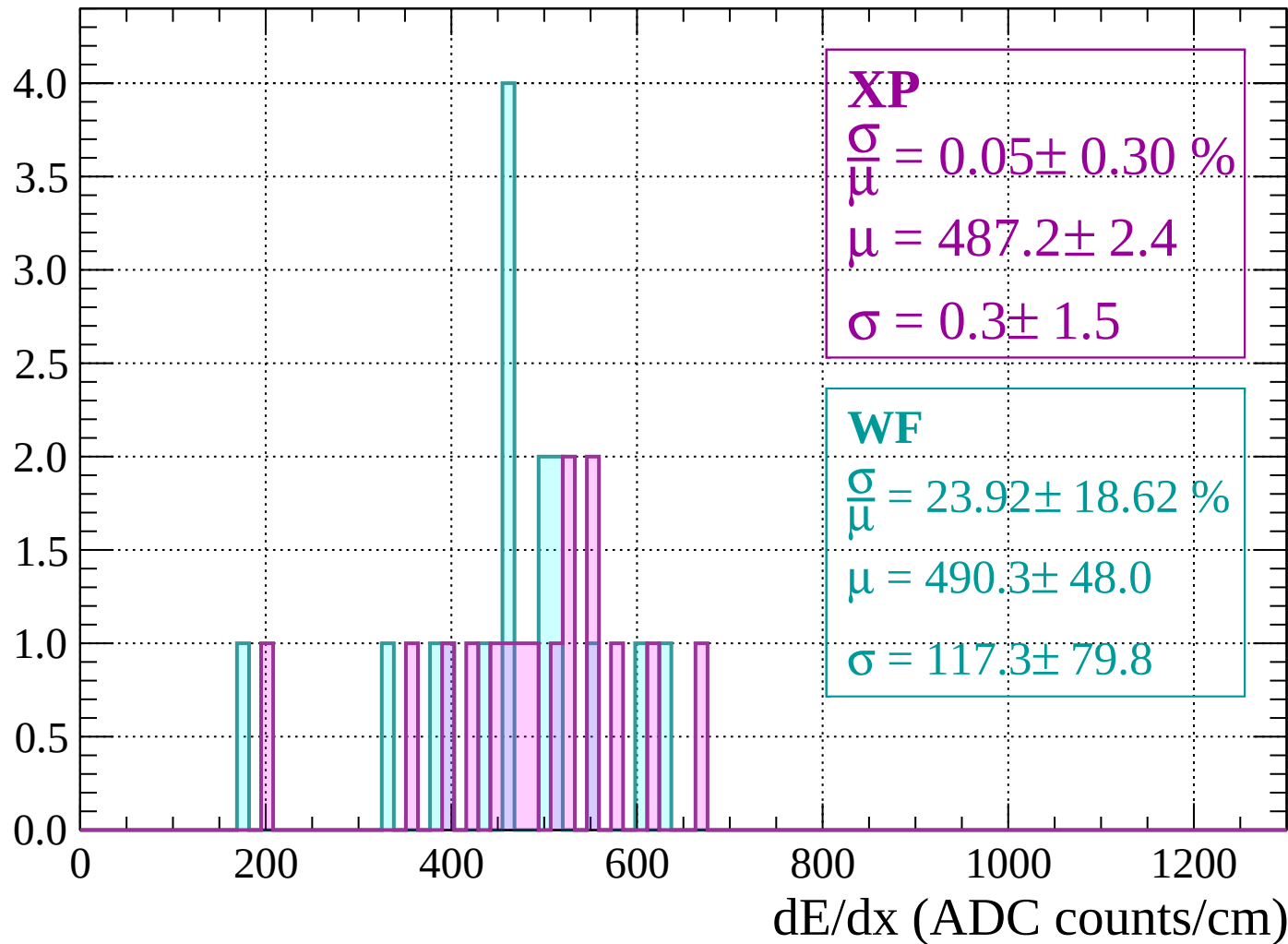
Energy loss | $640 < p < 680$

Count



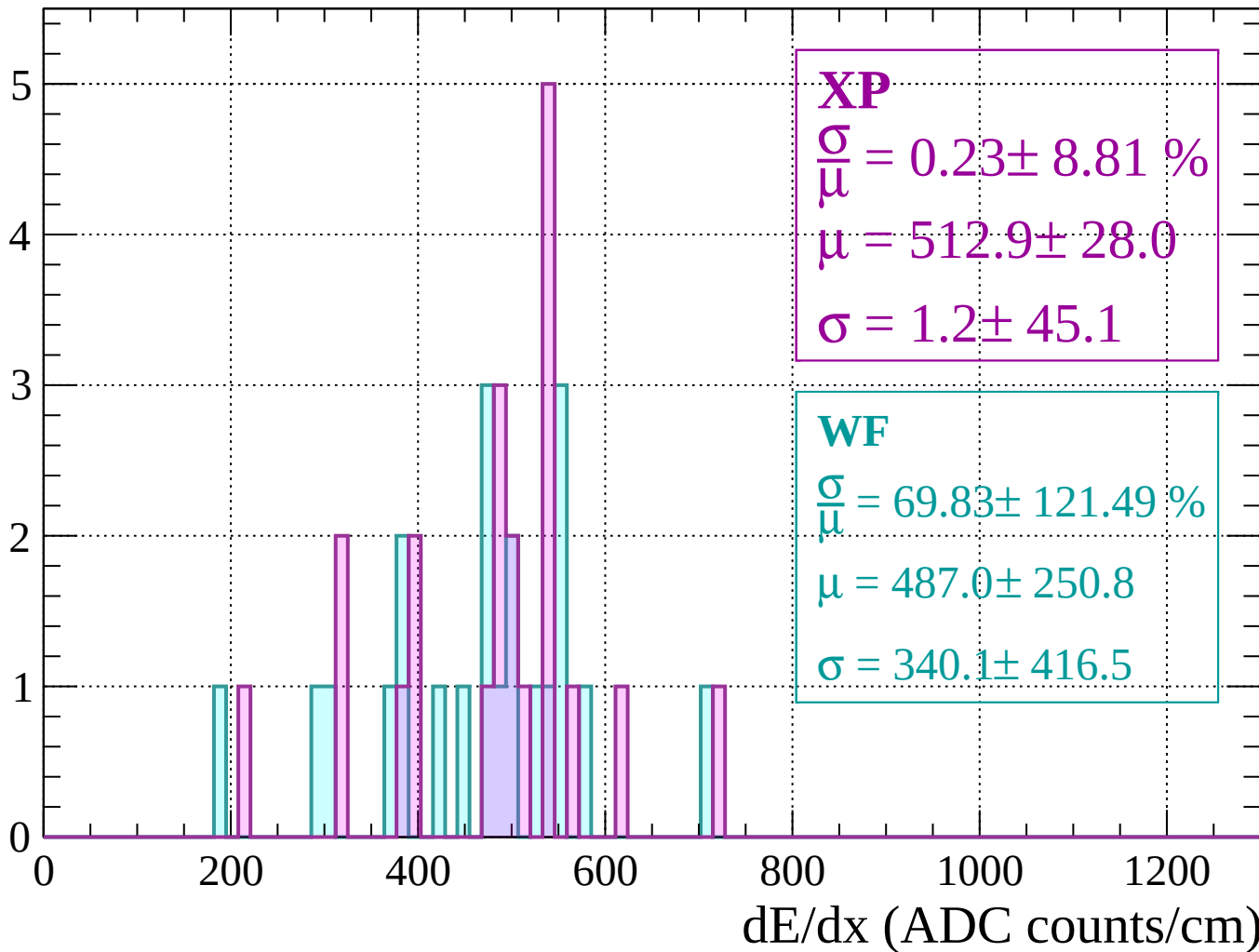
Energy loss | $680 < p < 720$

Count



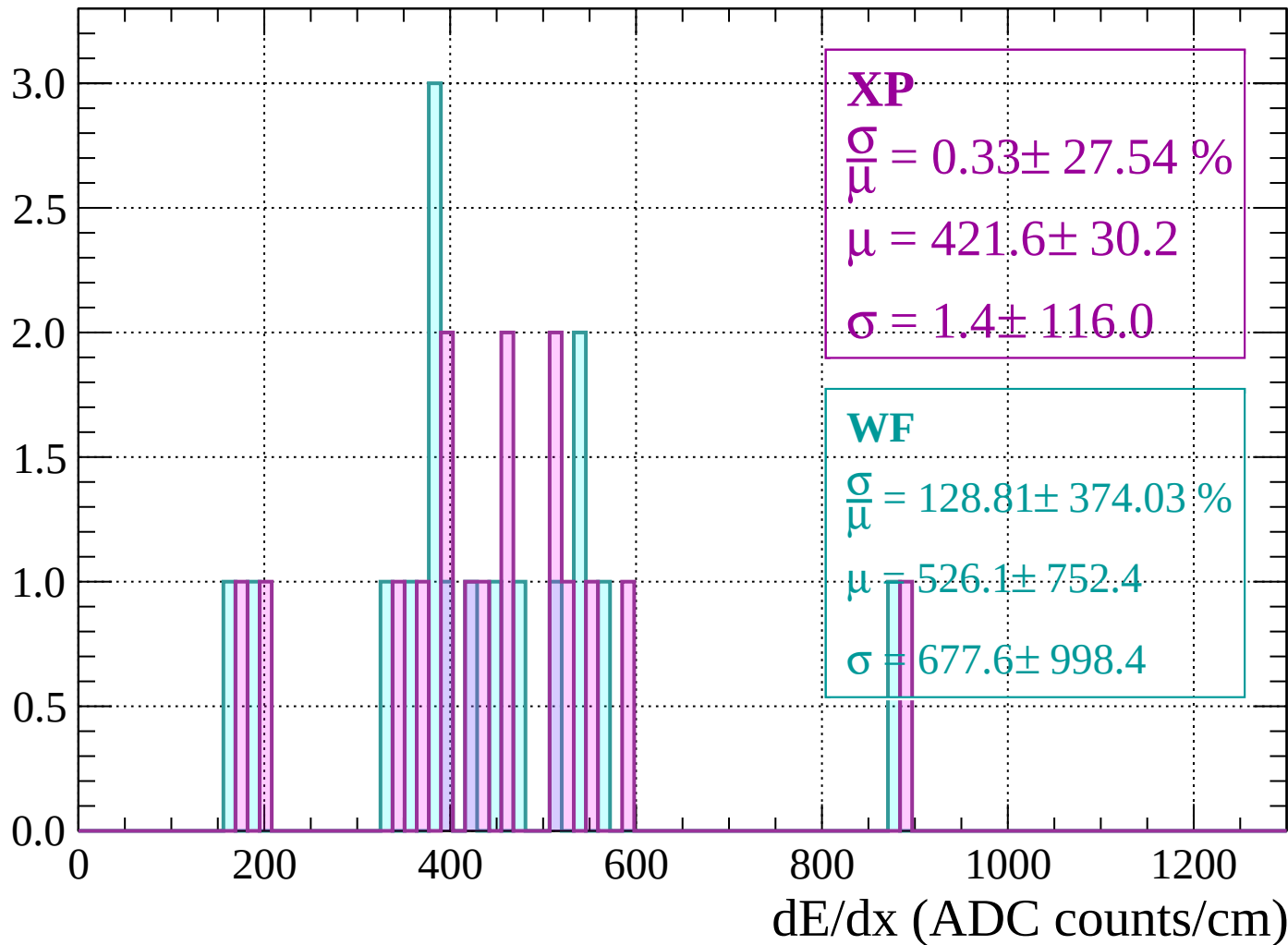
Energy loss | $720 < p < 760$

Count



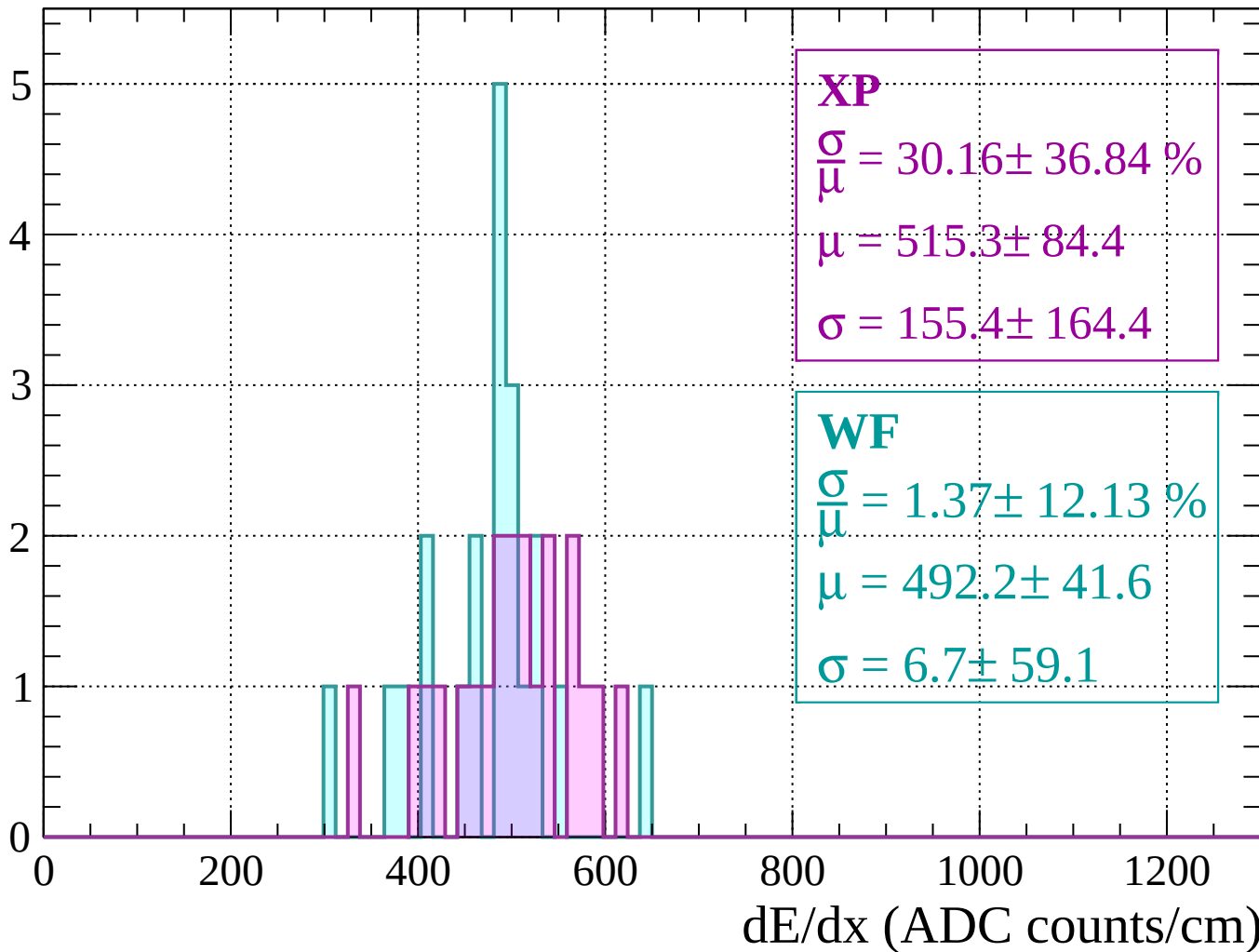
Energy loss | $760 < p < 800$

Count



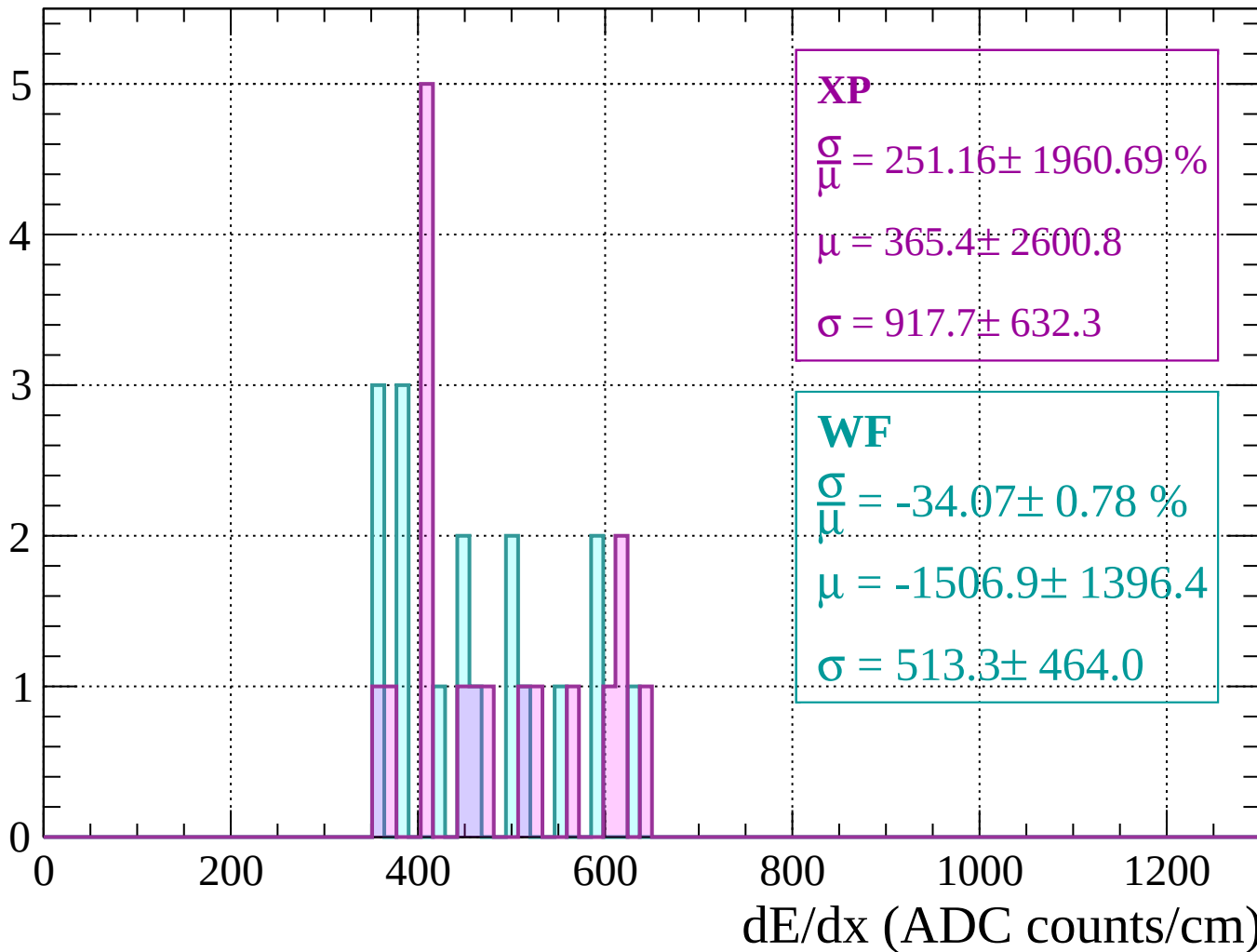
Energy loss | $800 < p < 840$

Count



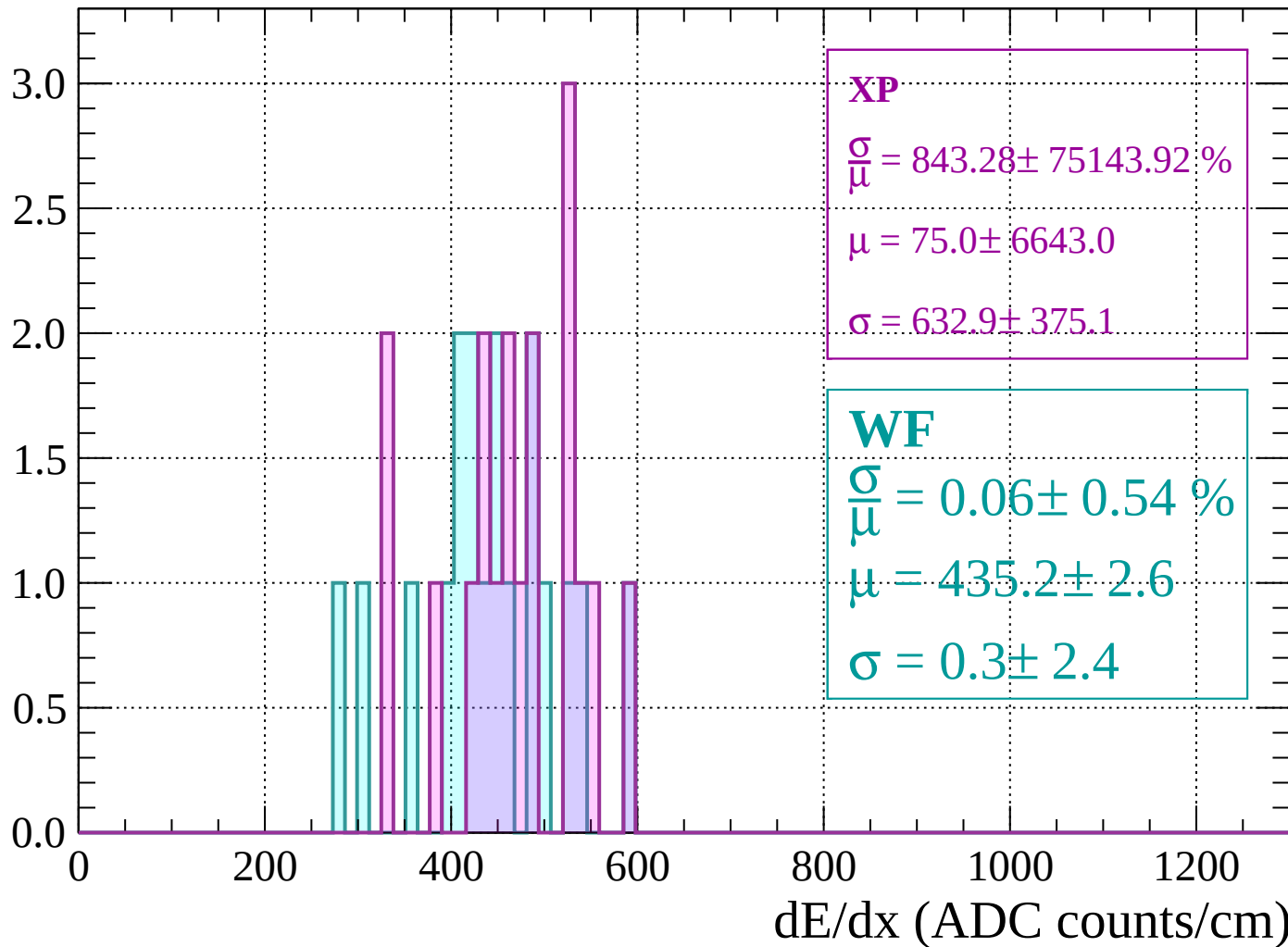
Energy loss | $840 < p < 880$

Count



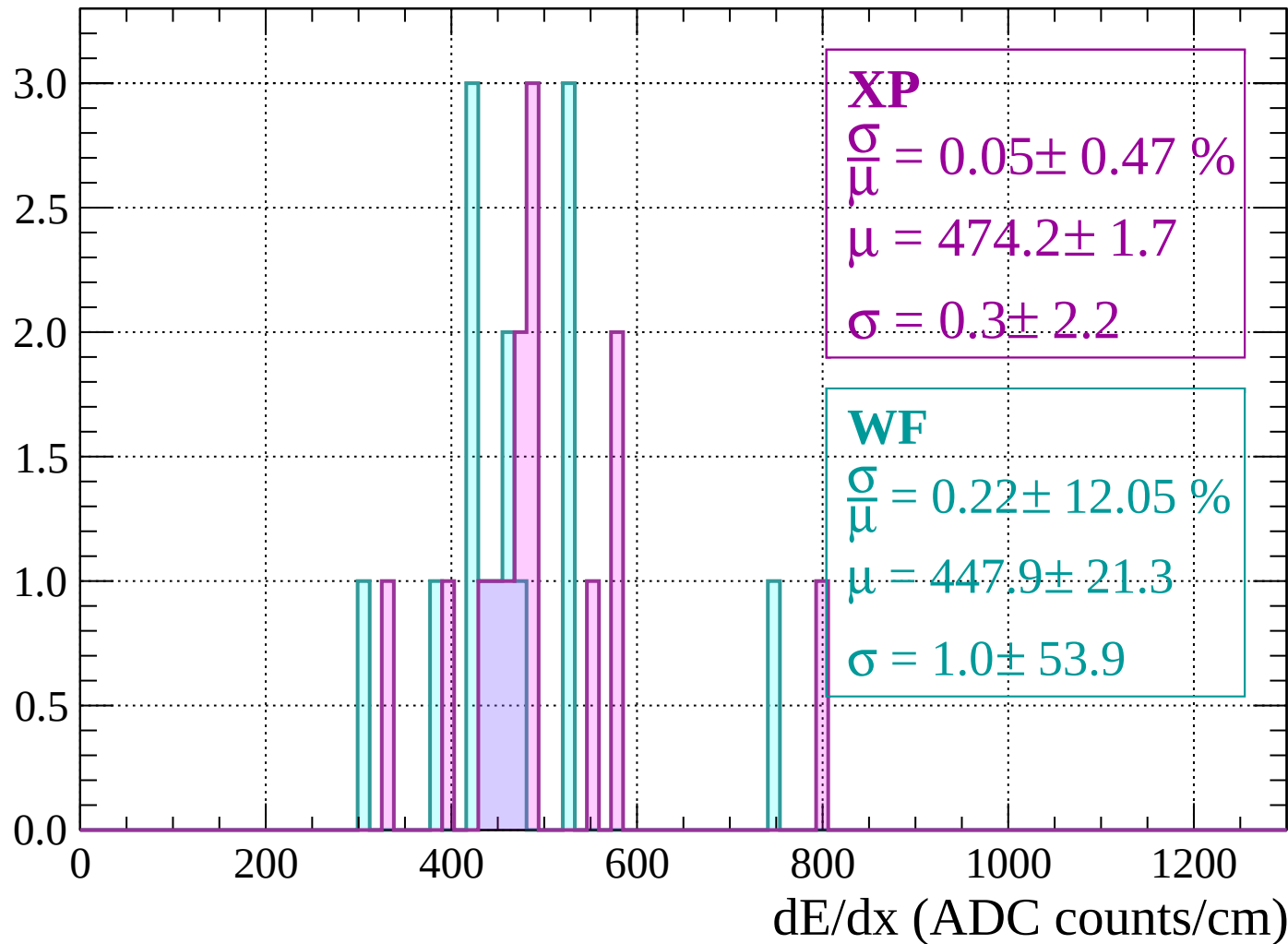
Energy loss | $880 < p < 920$

Count



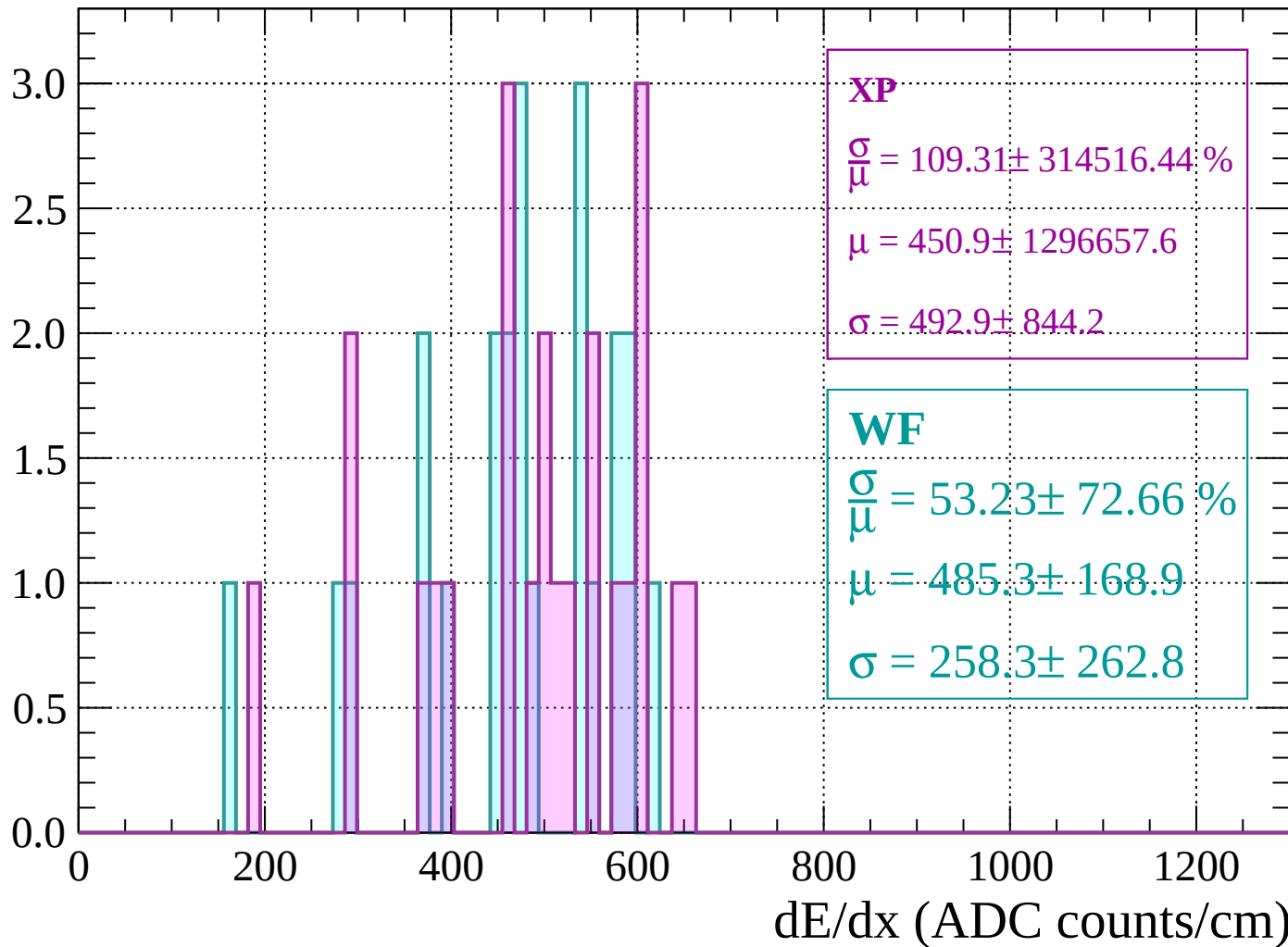
Energy loss | $920 < p < 960$

Count



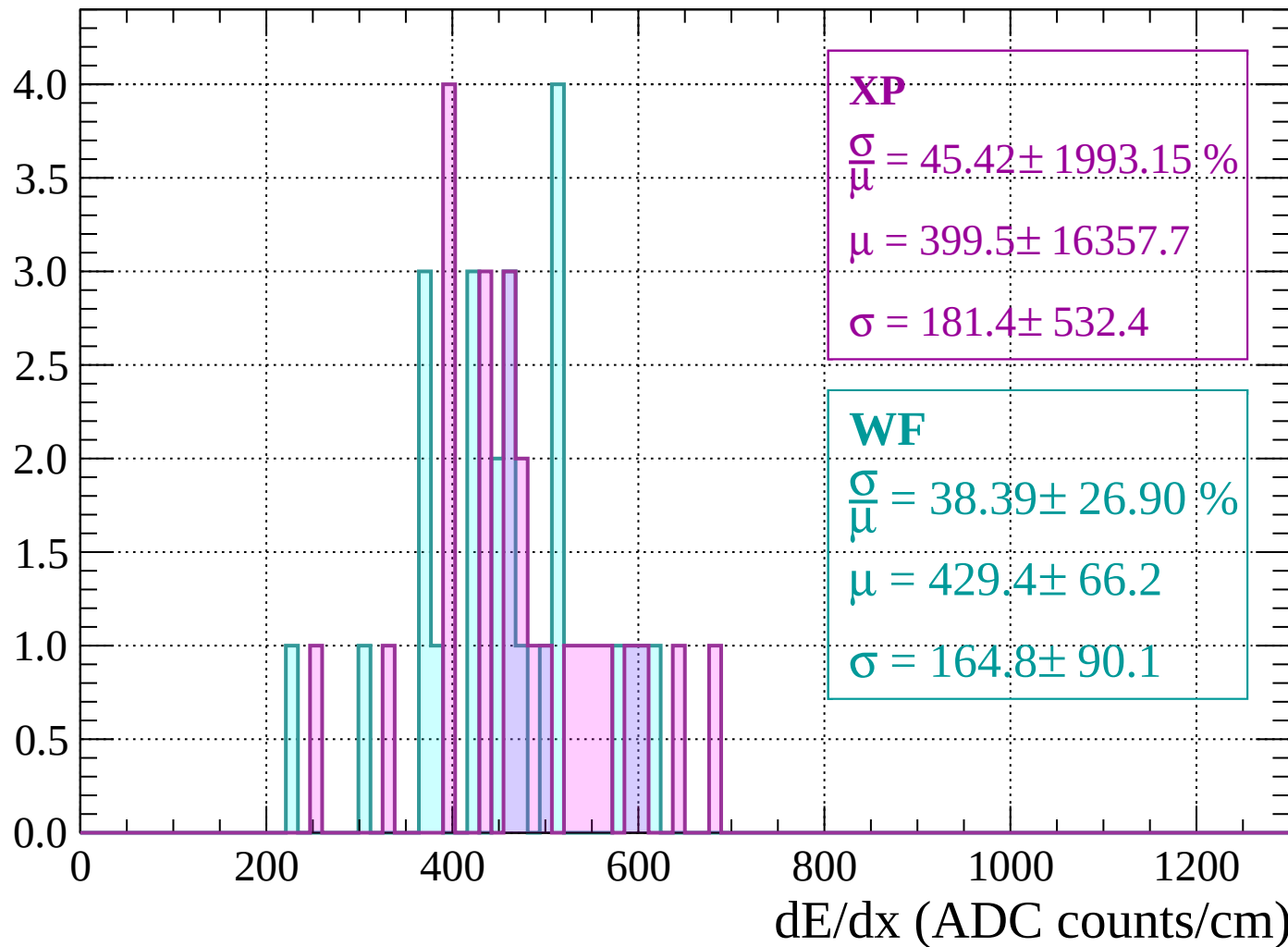
Energy loss | $960 < p < 1000$

Count



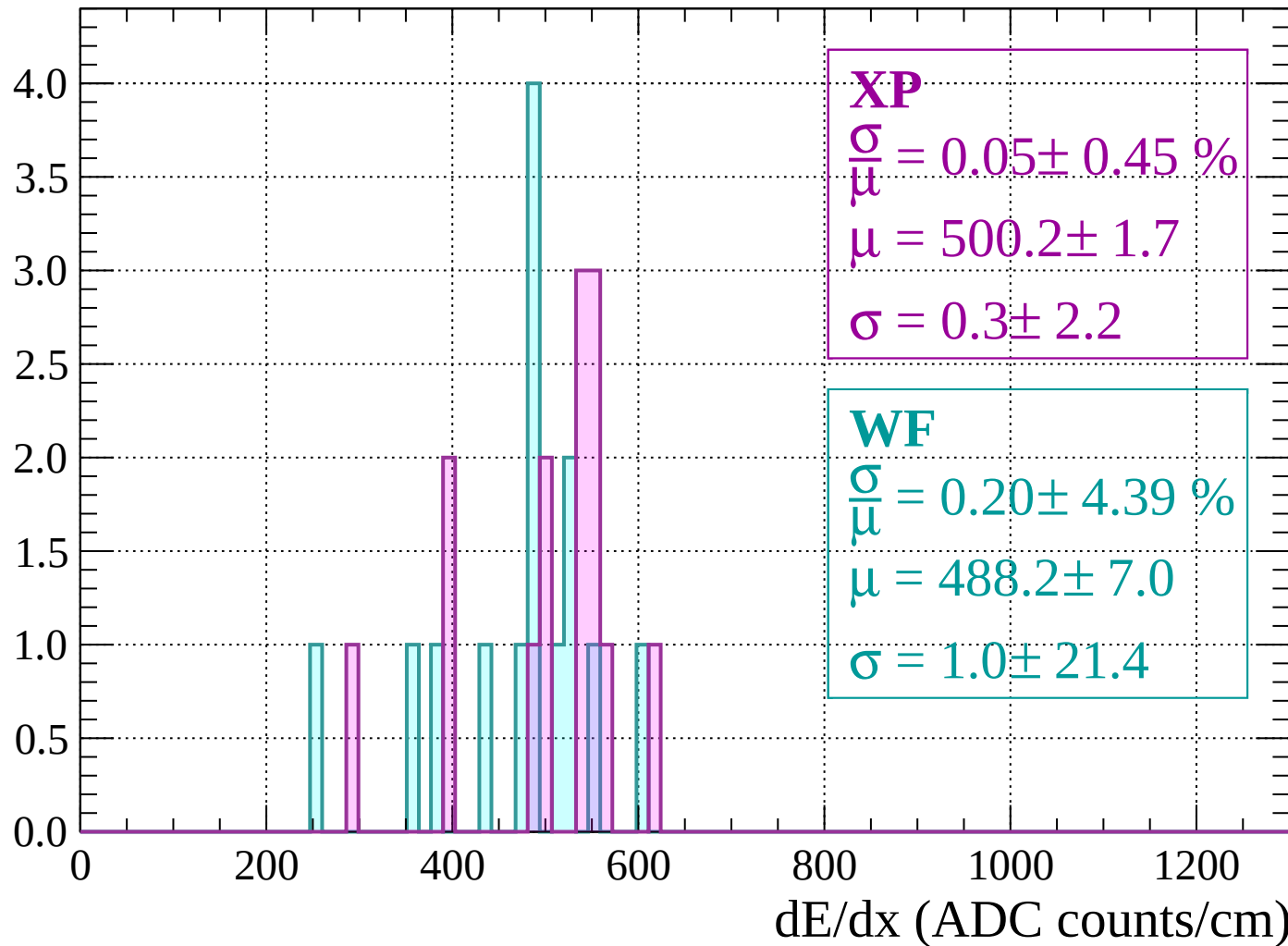
Energy loss | $1000 < p < 1040$

Count



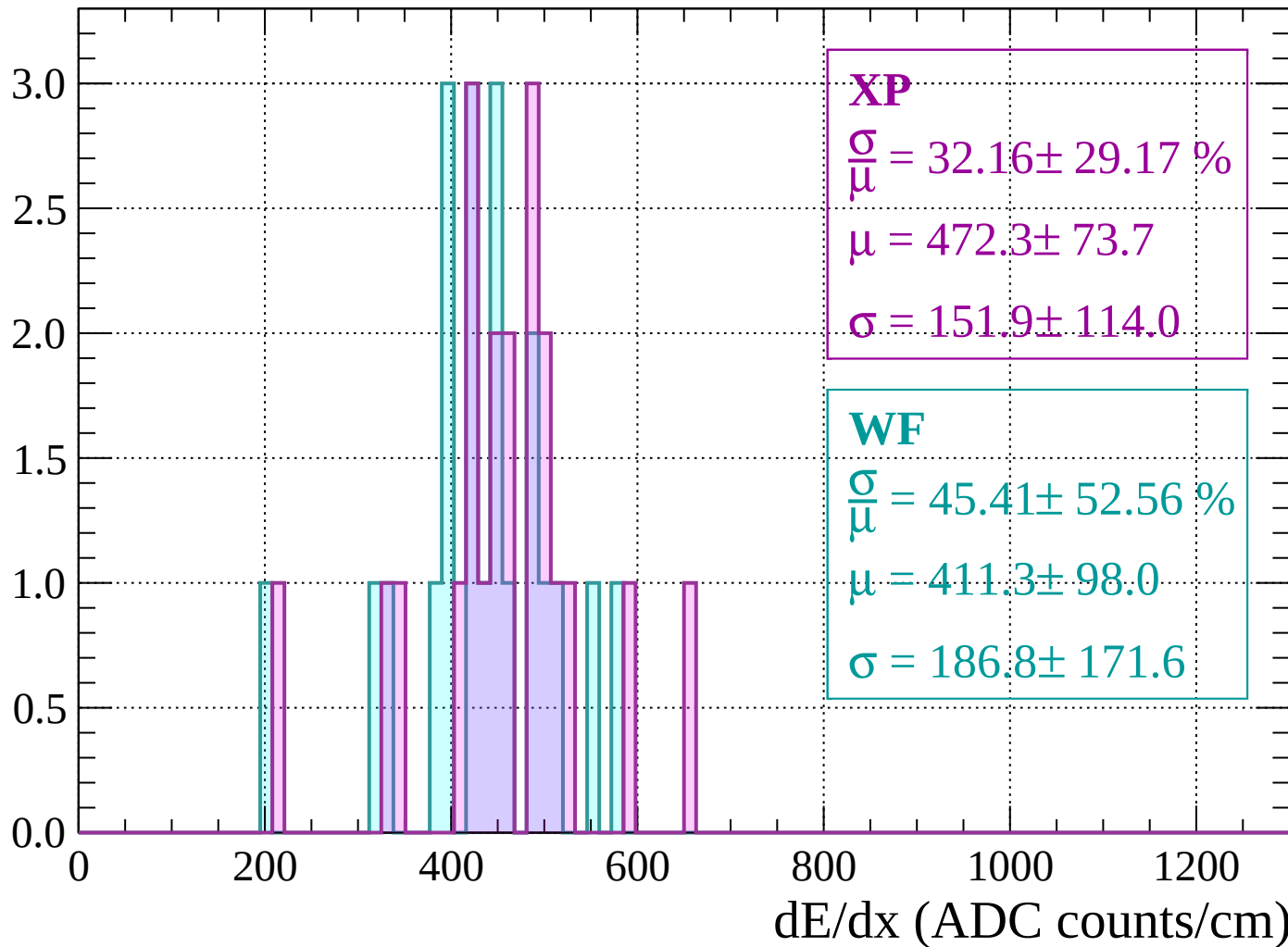
Energy loss | $1040 < p < 1080$

Count



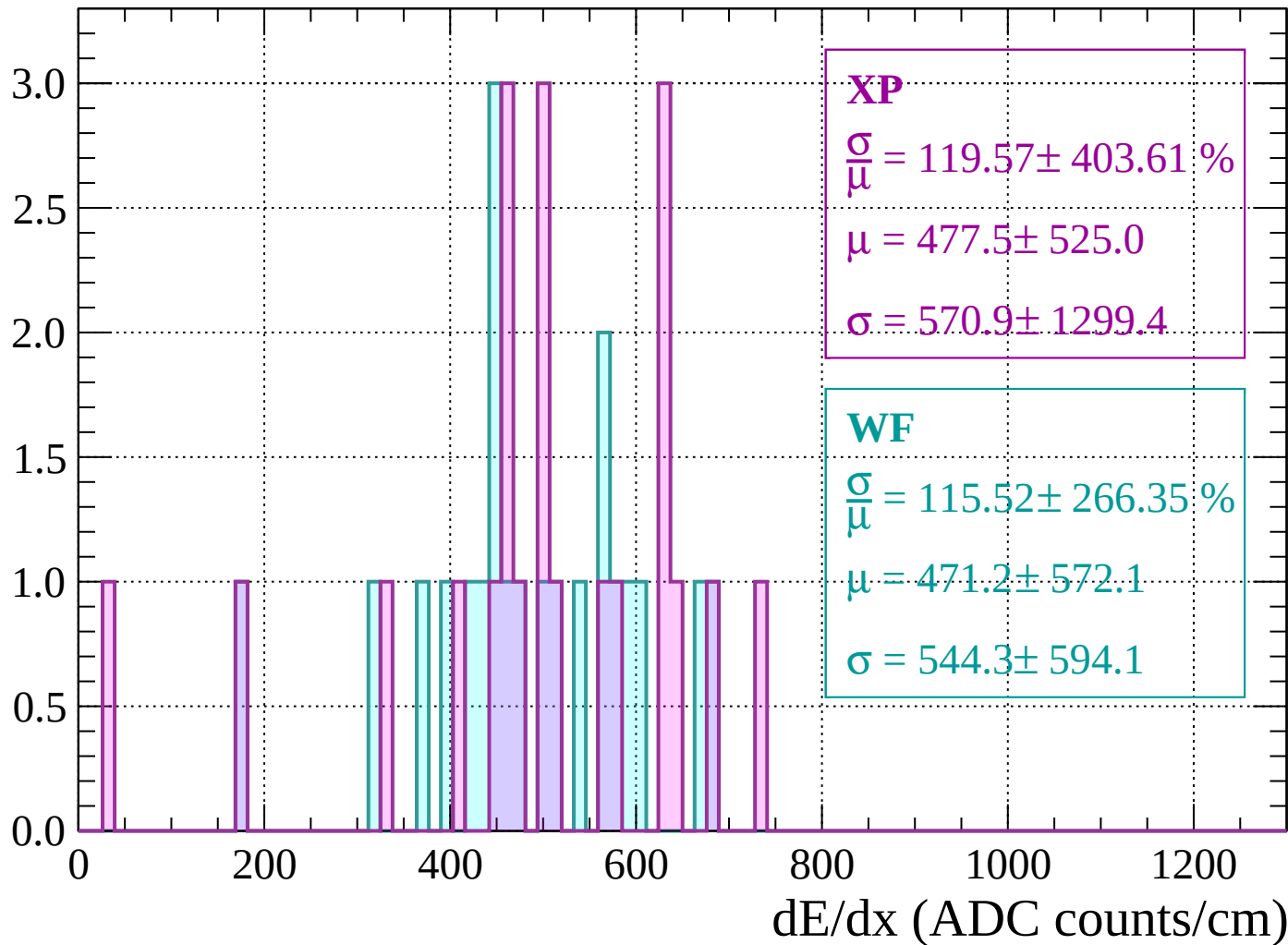
Energy loss | $1080 < p < 1120$

Count



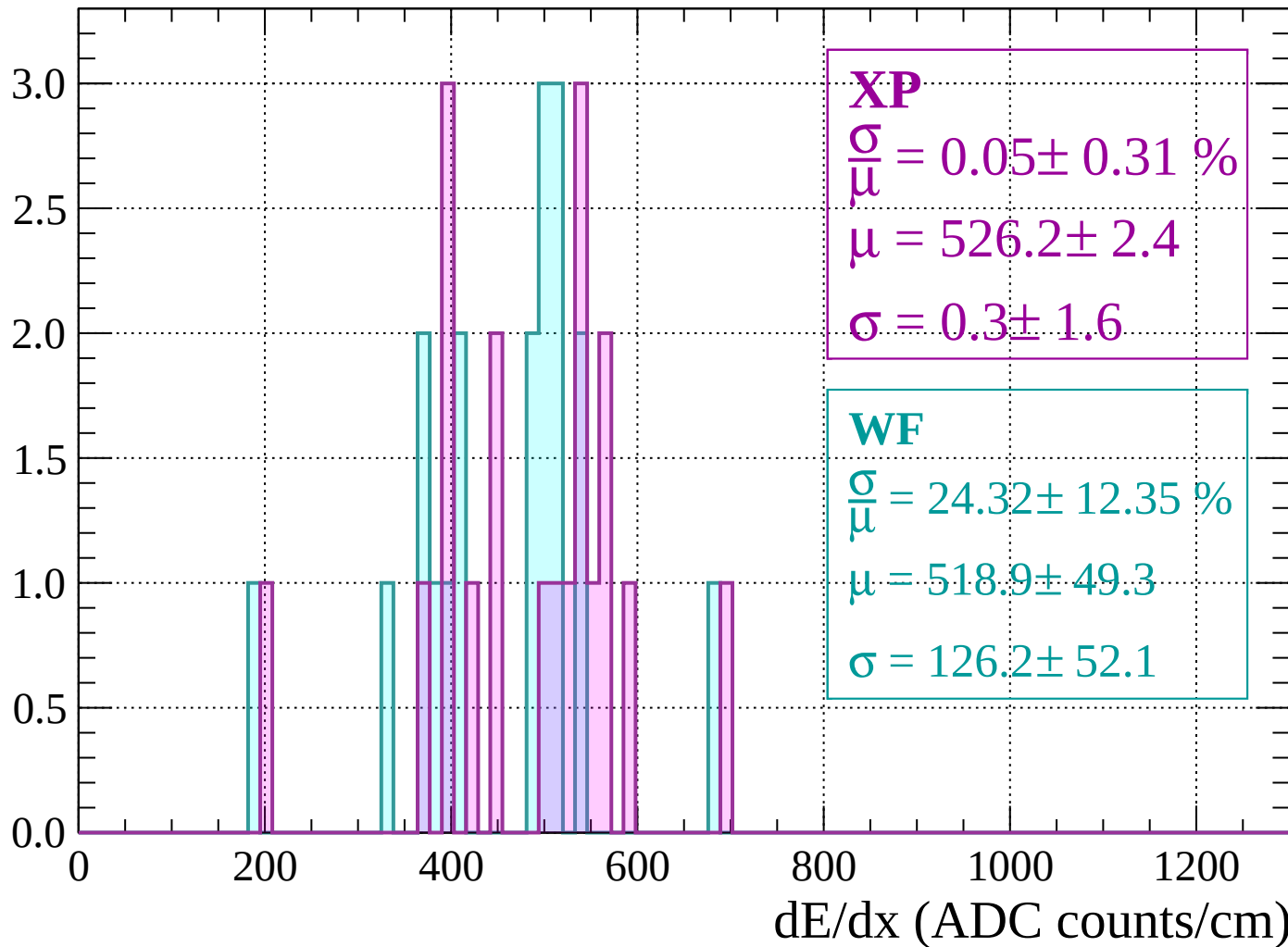
Energy loss | $1120 < p < 1160$

Count



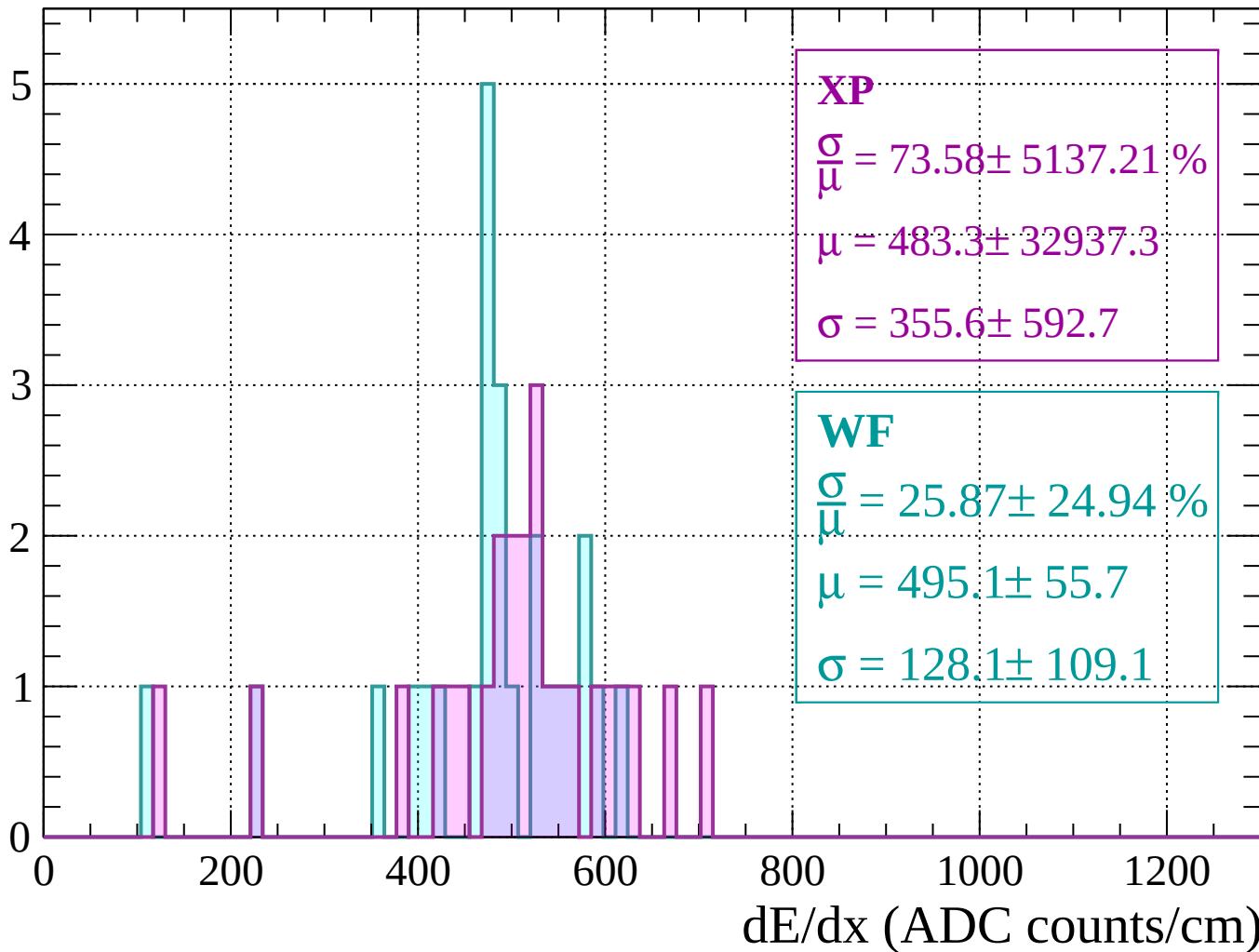
Energy loss | $1160 < p < 1200$

Count



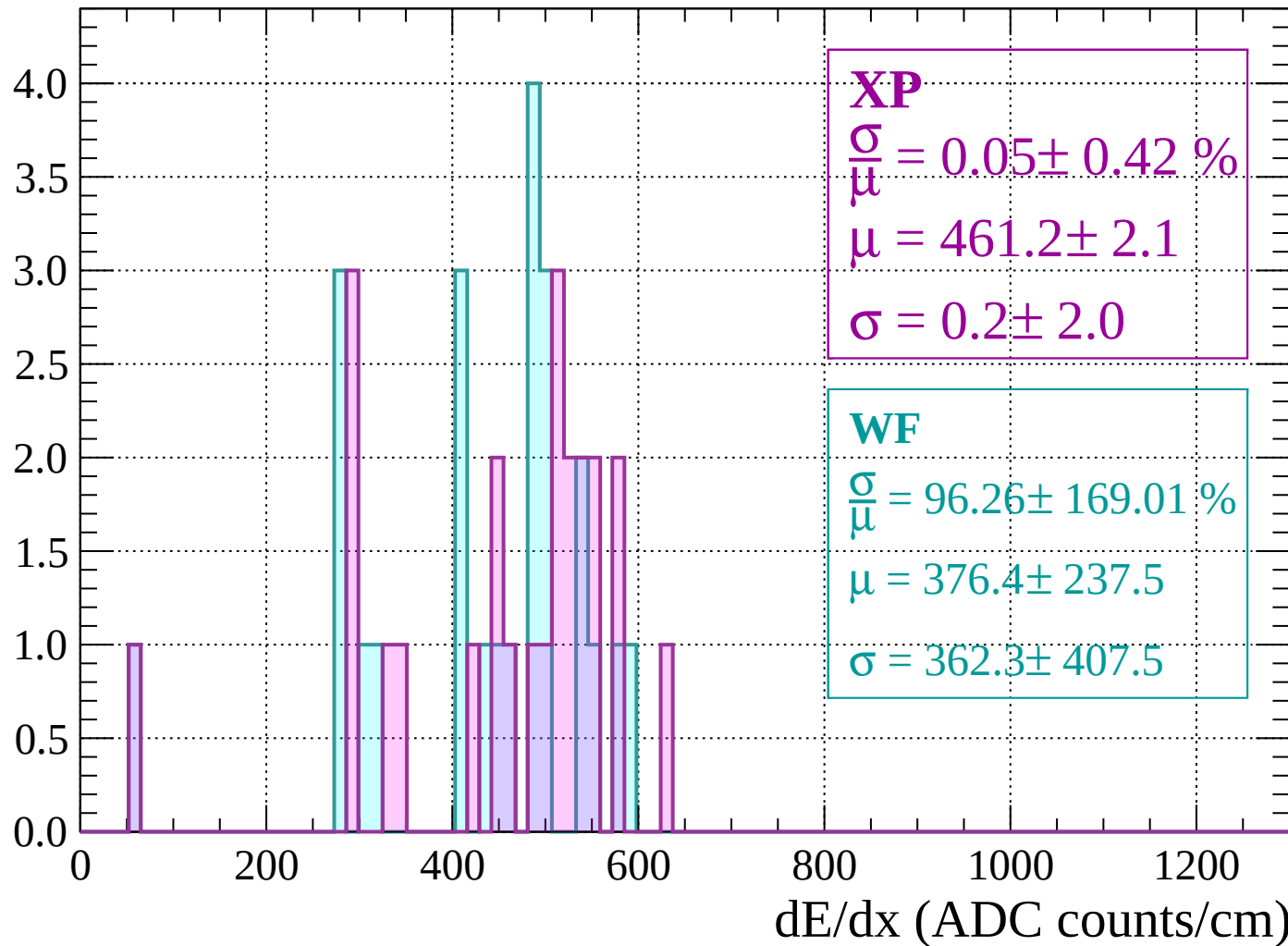
Energy loss | $1200 < p < 1240$

Count



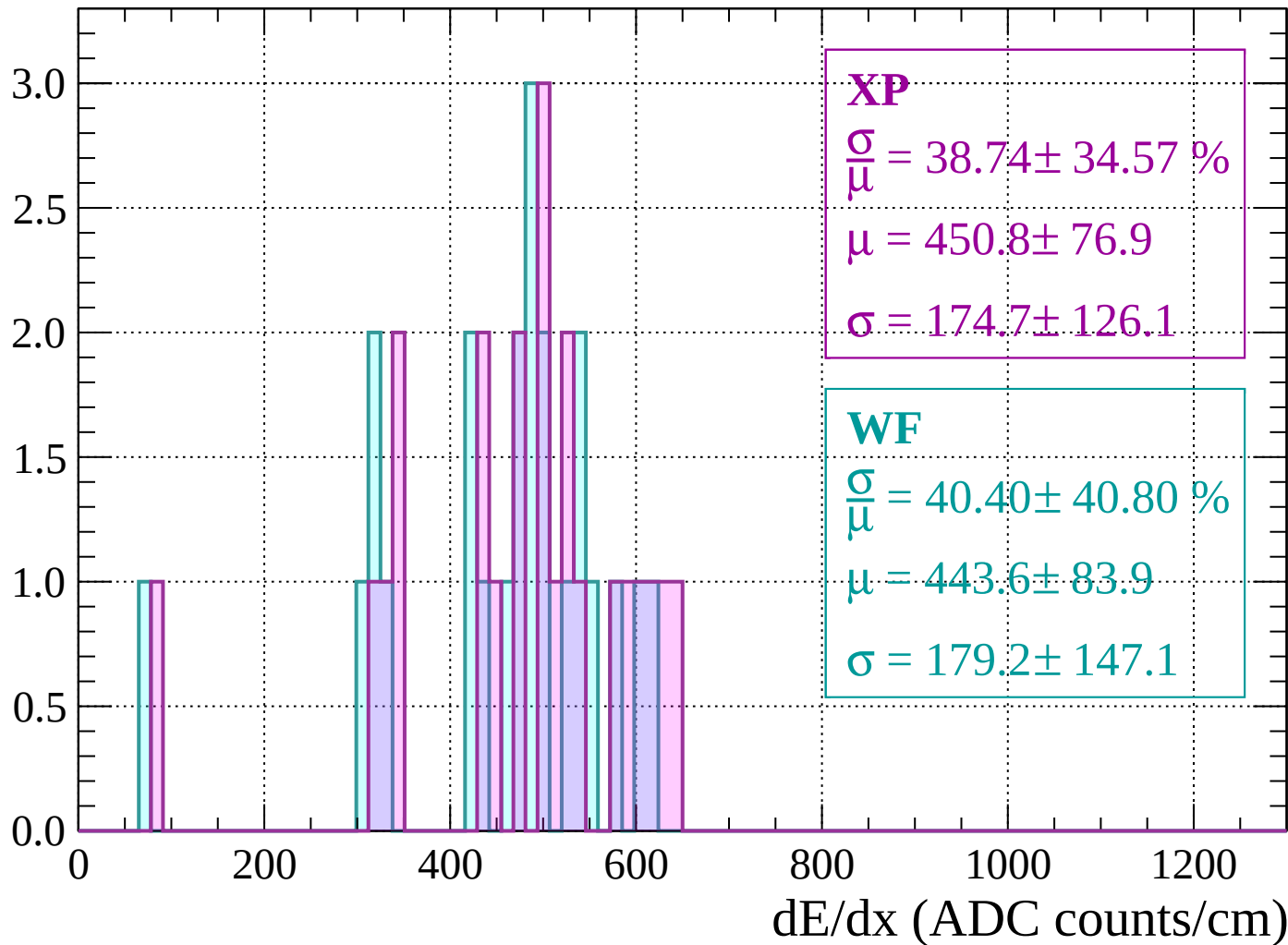
Energy loss | $1240 < p < 1280$

Count



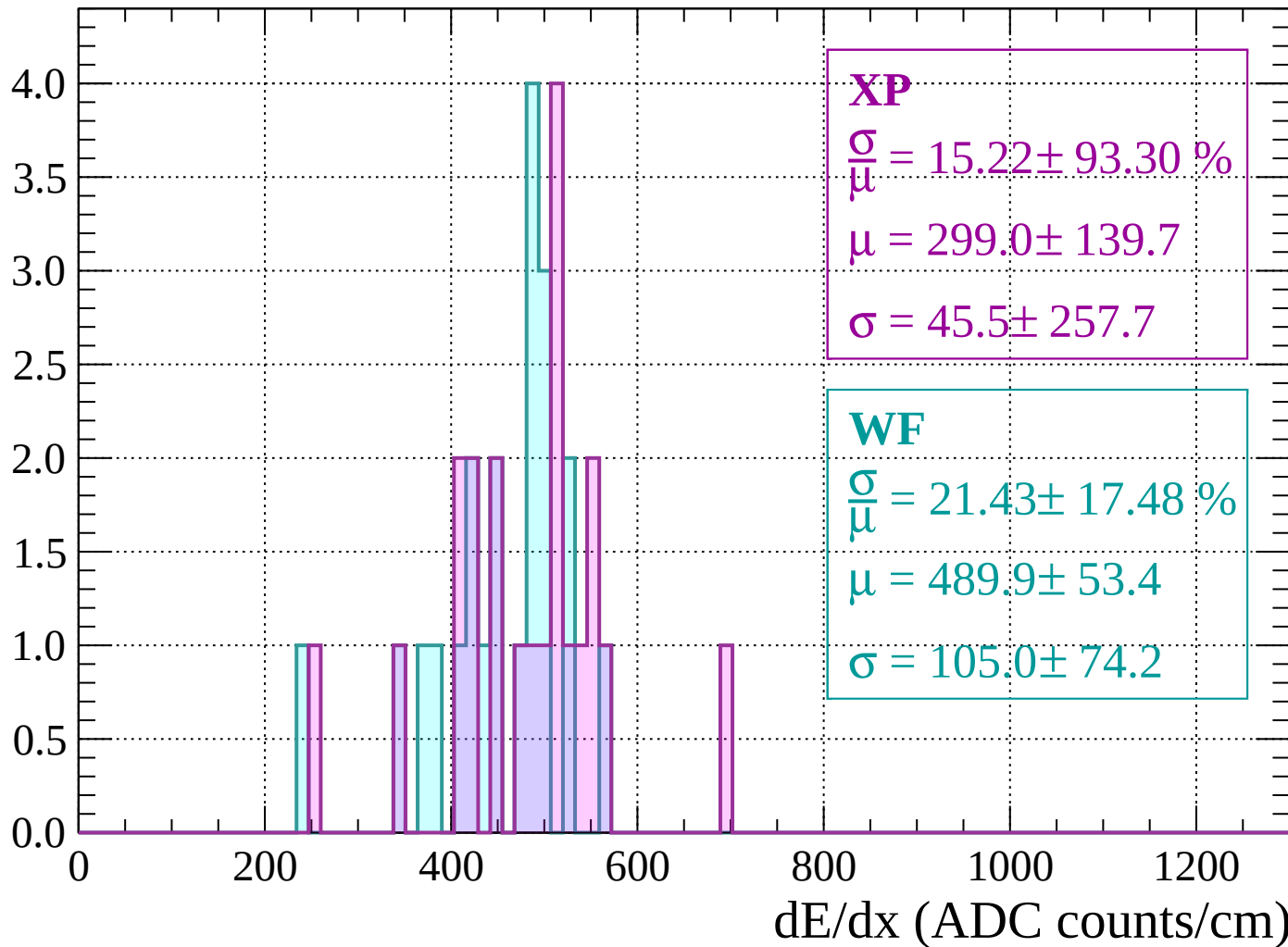
Energy loss | $1280 < p < 1320$

Count



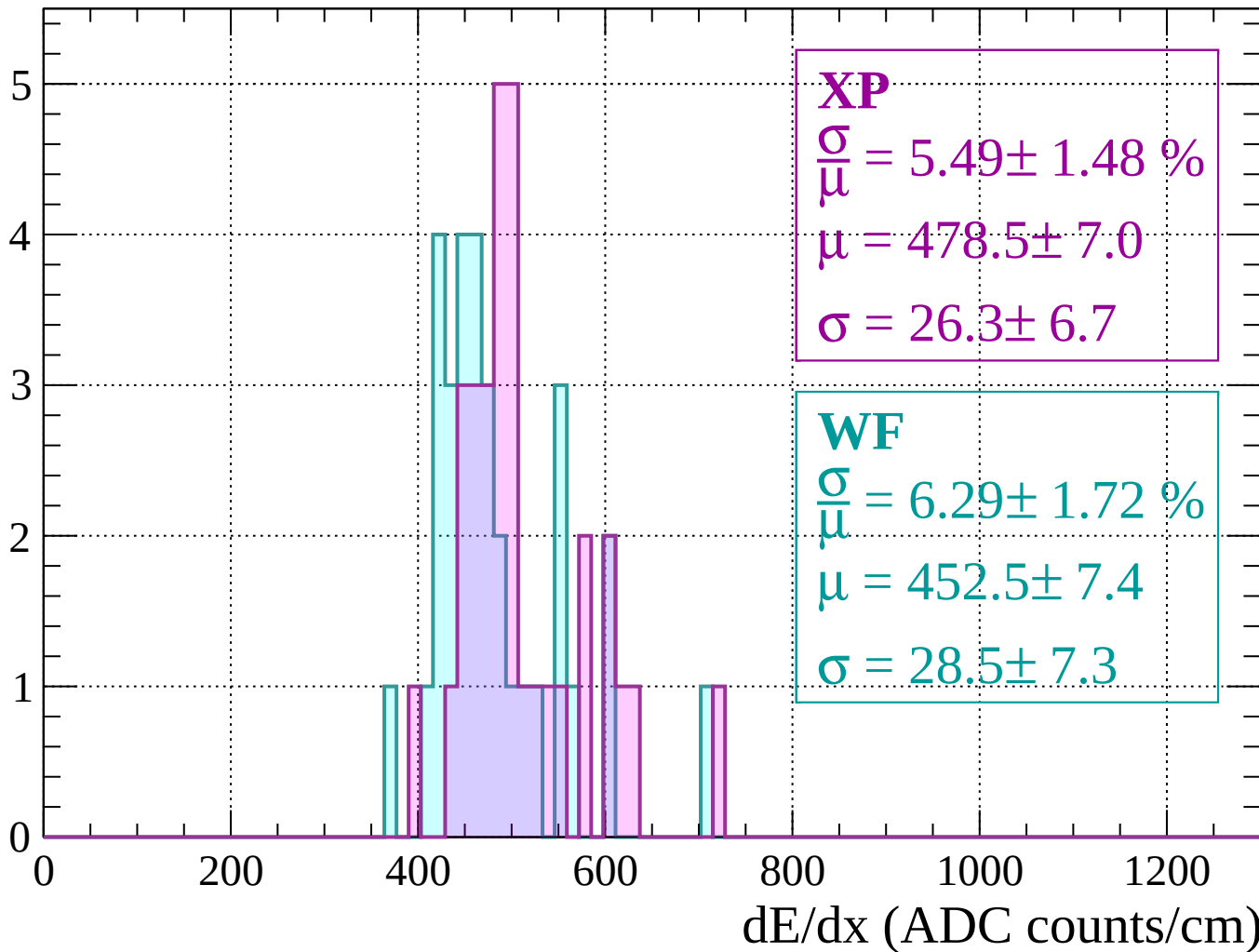
Energy loss | $1320 < p < 1360$

Count



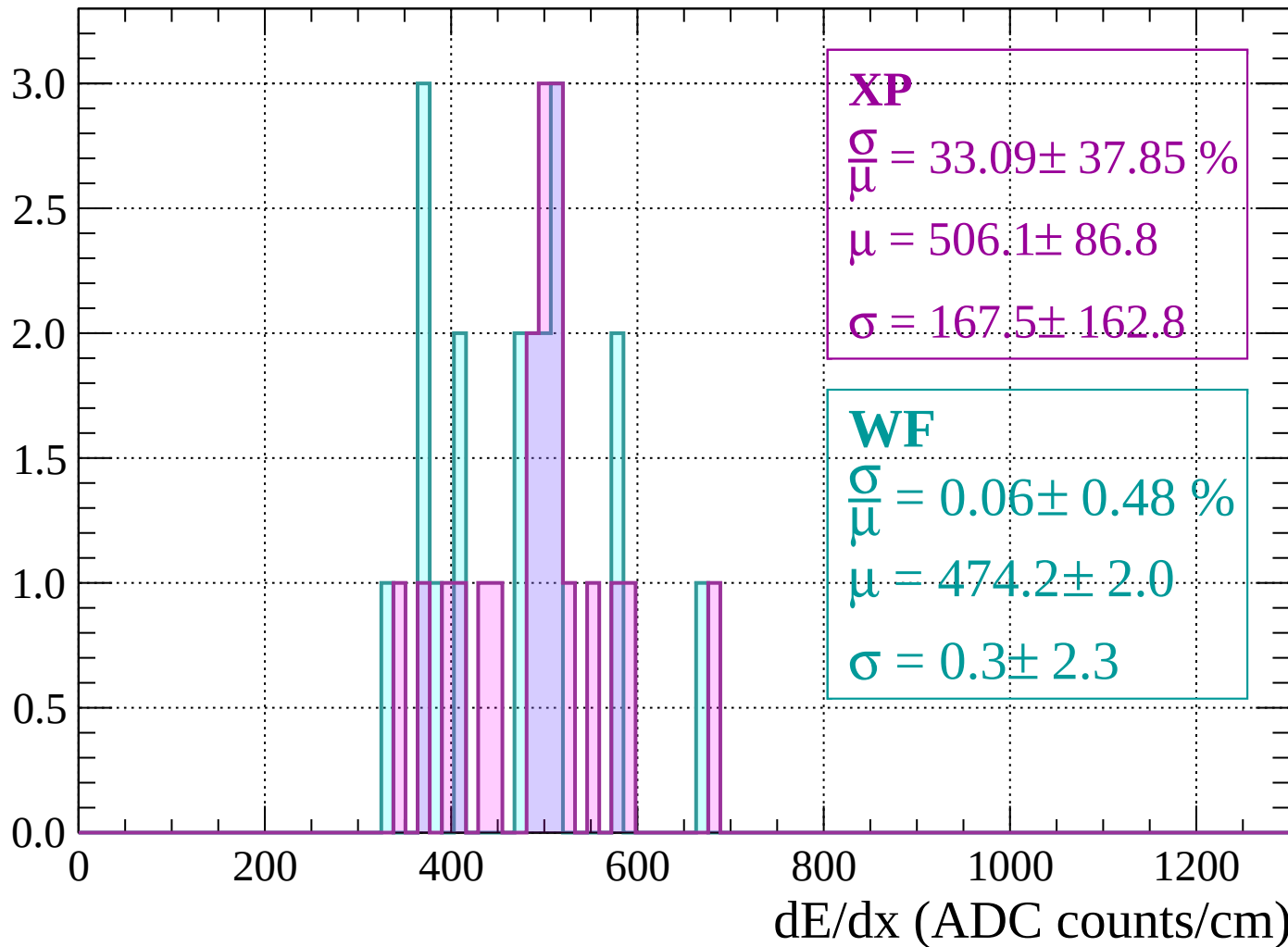
Energy loss | $1360 < p < 1400$

Count



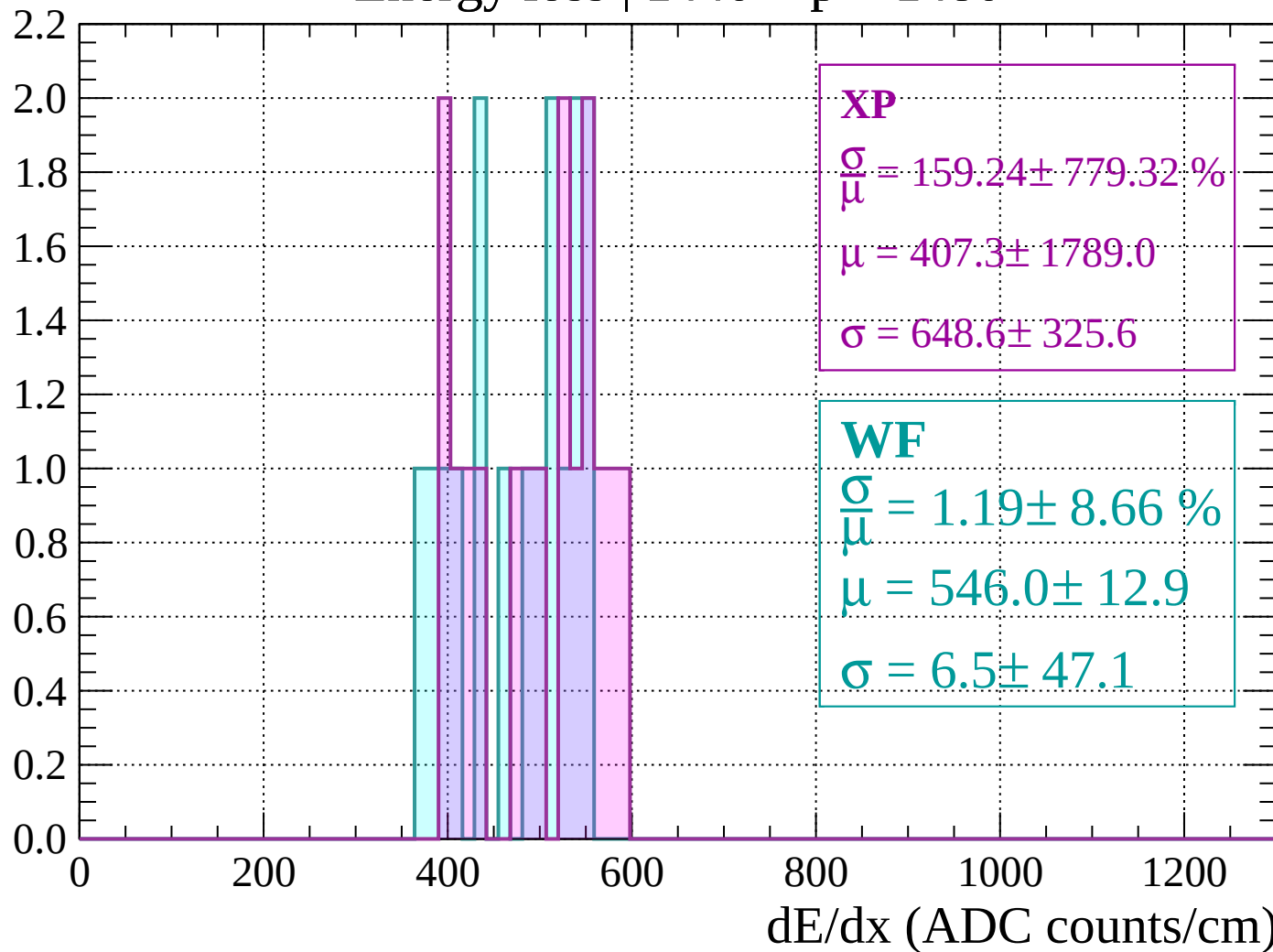
Energy loss | $1400 < p < 1440$

Count



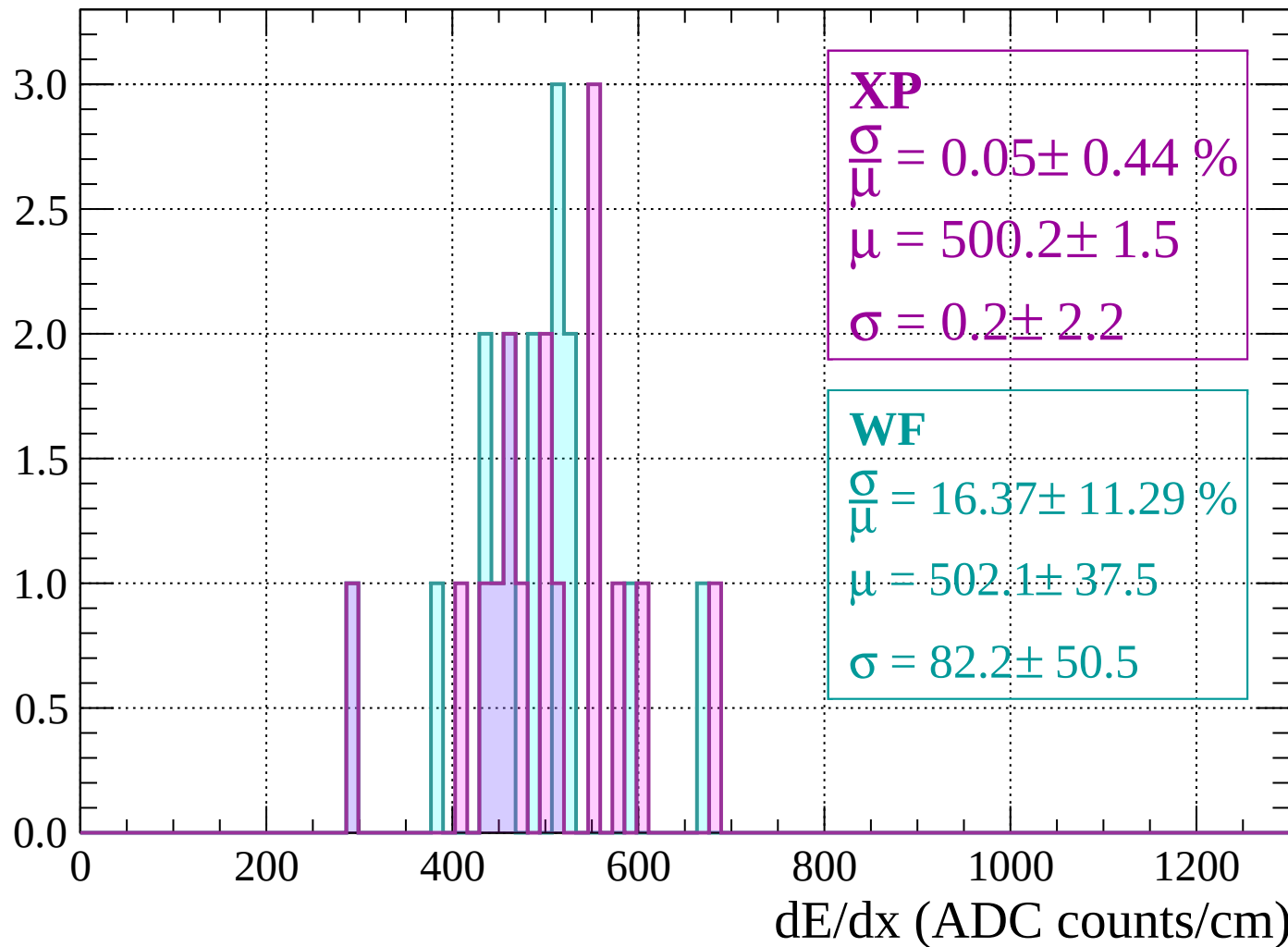
Energy loss | $1440 < p < 1480$

Count



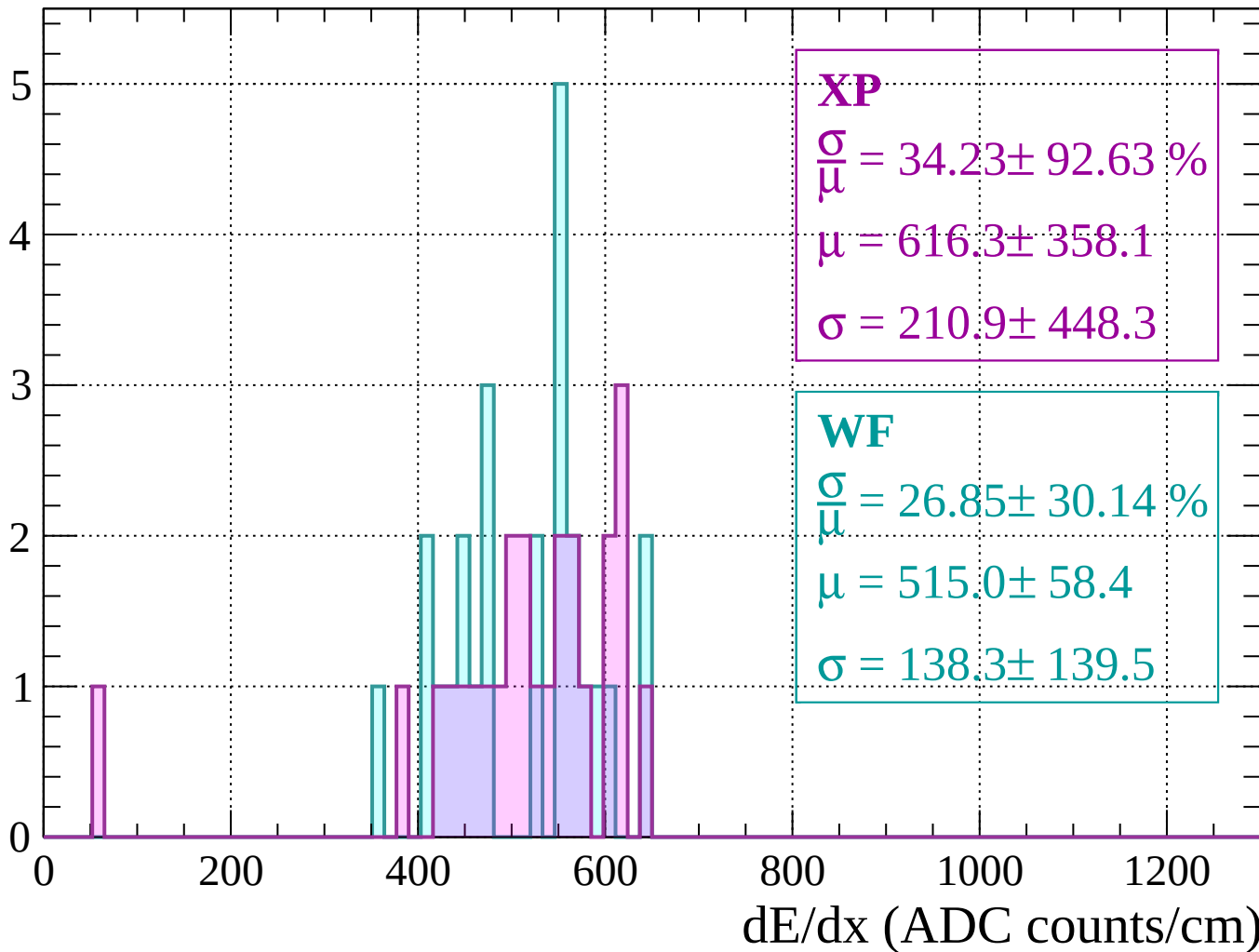
Energy loss | $1480 < p < 1520$

Count



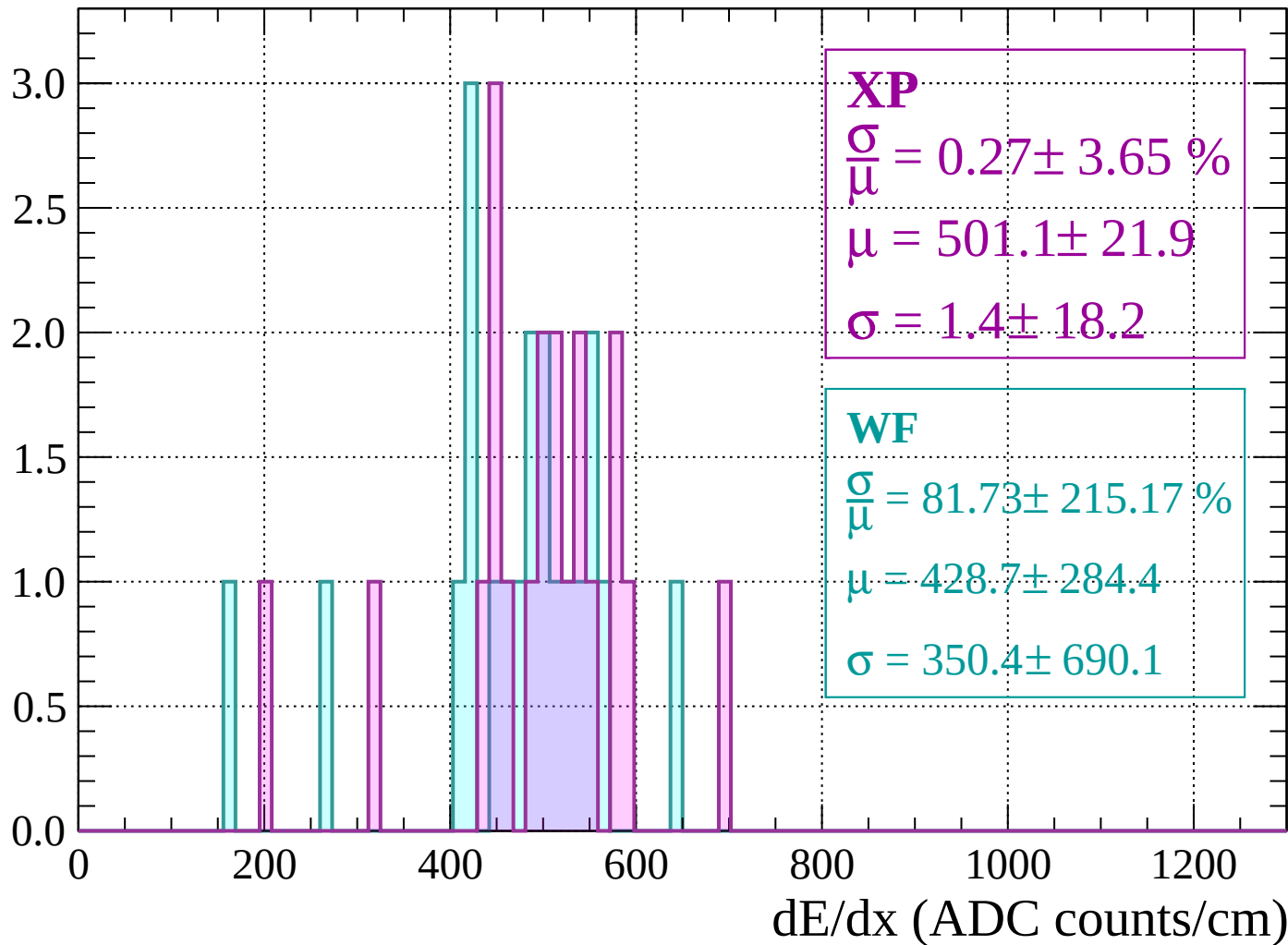
Energy loss | $1520 < p < 1560$

Count



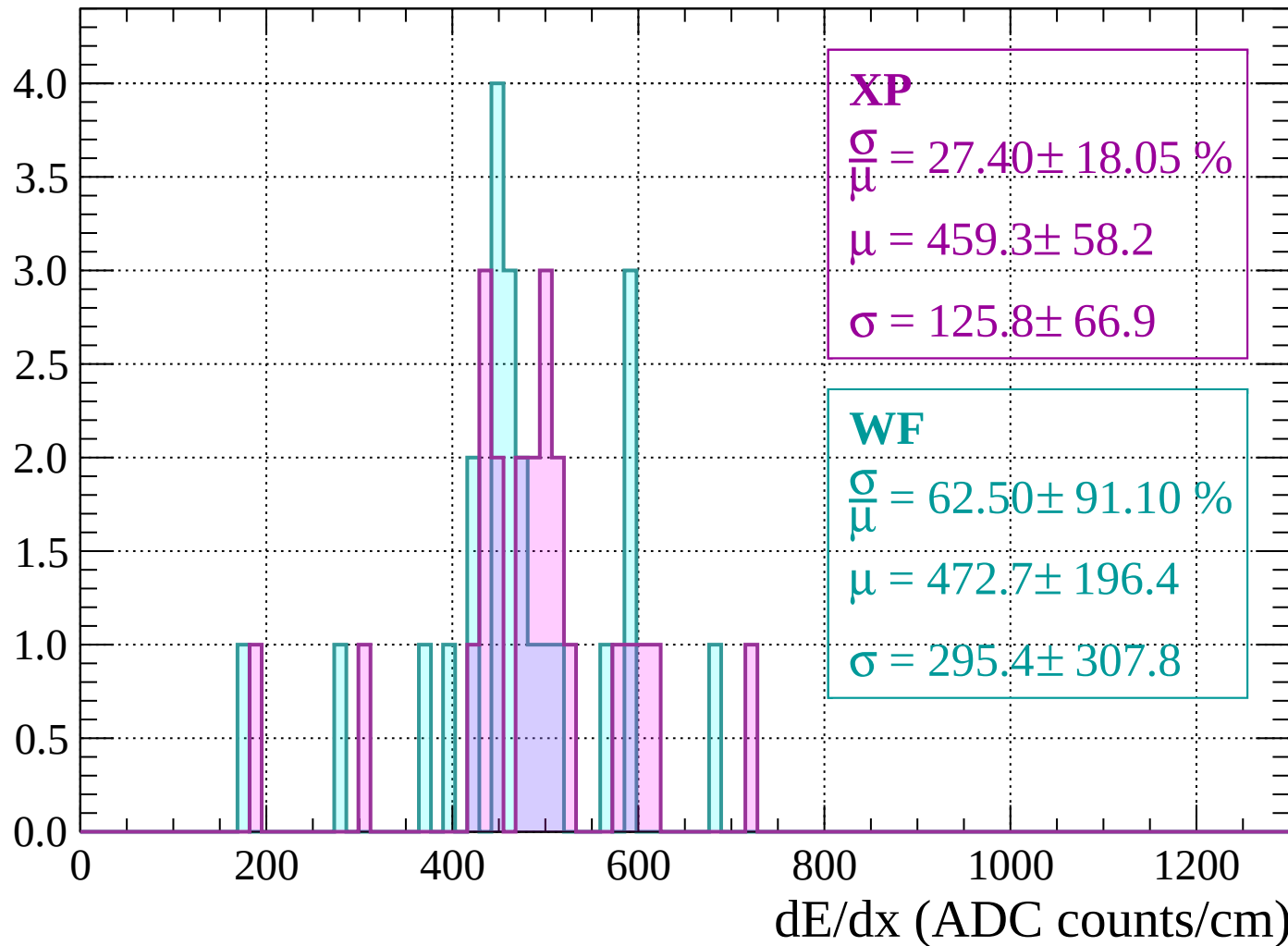
Energy loss | $1560 < p < 1600$

Count



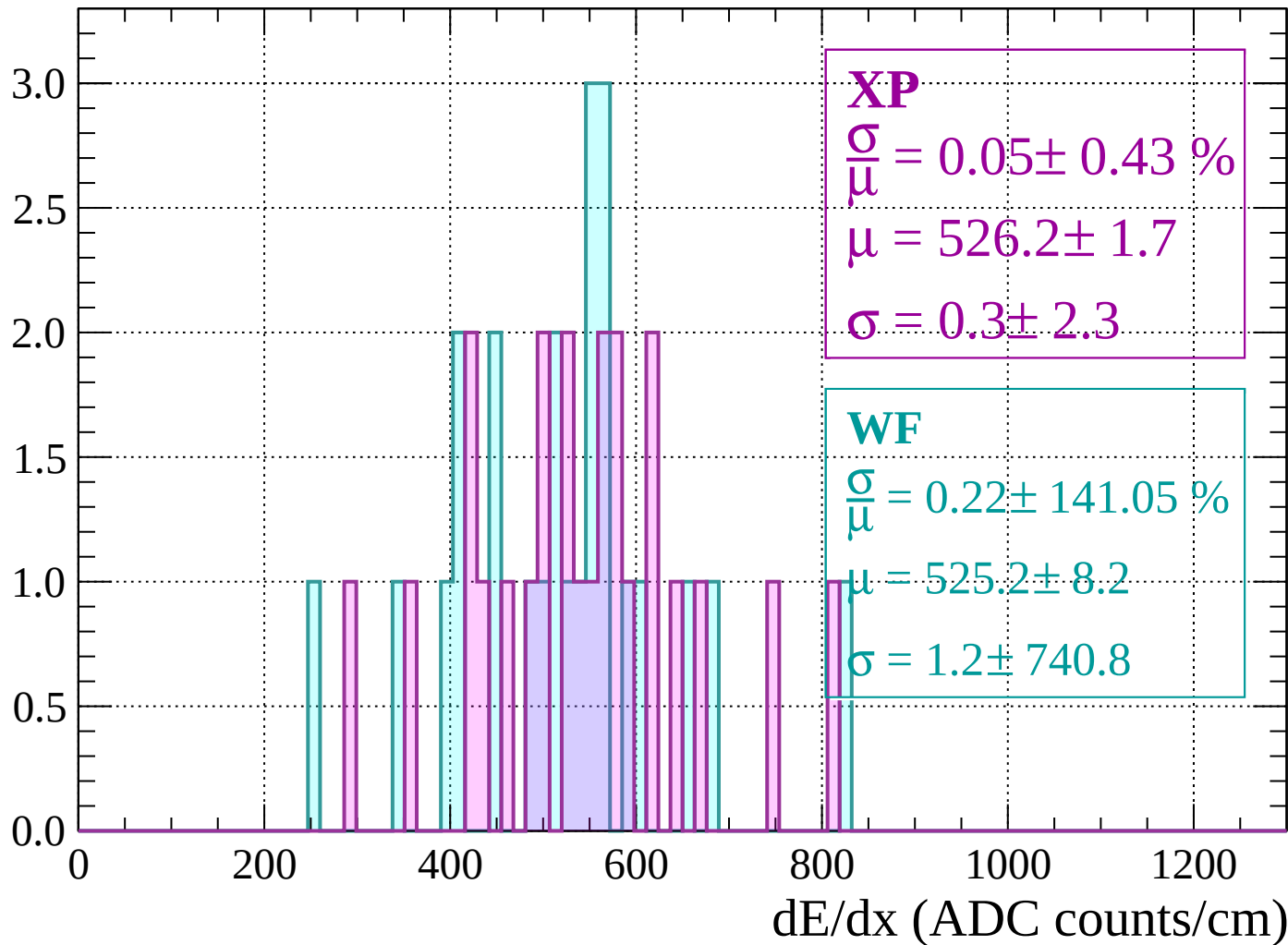
Energy loss | $1600 < p < 1640$

Count



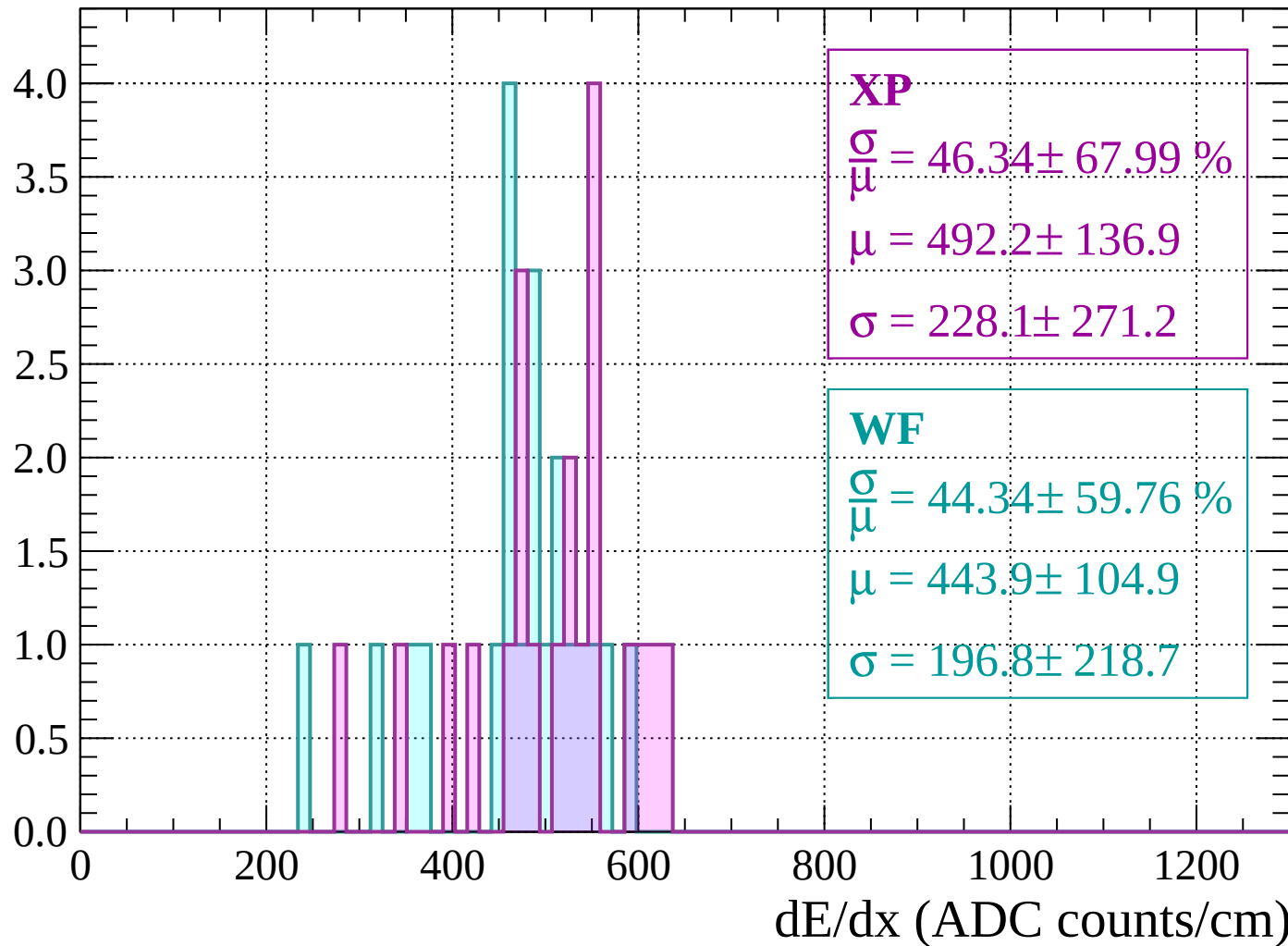
Energy loss | $1640 < p < 1680$

Count



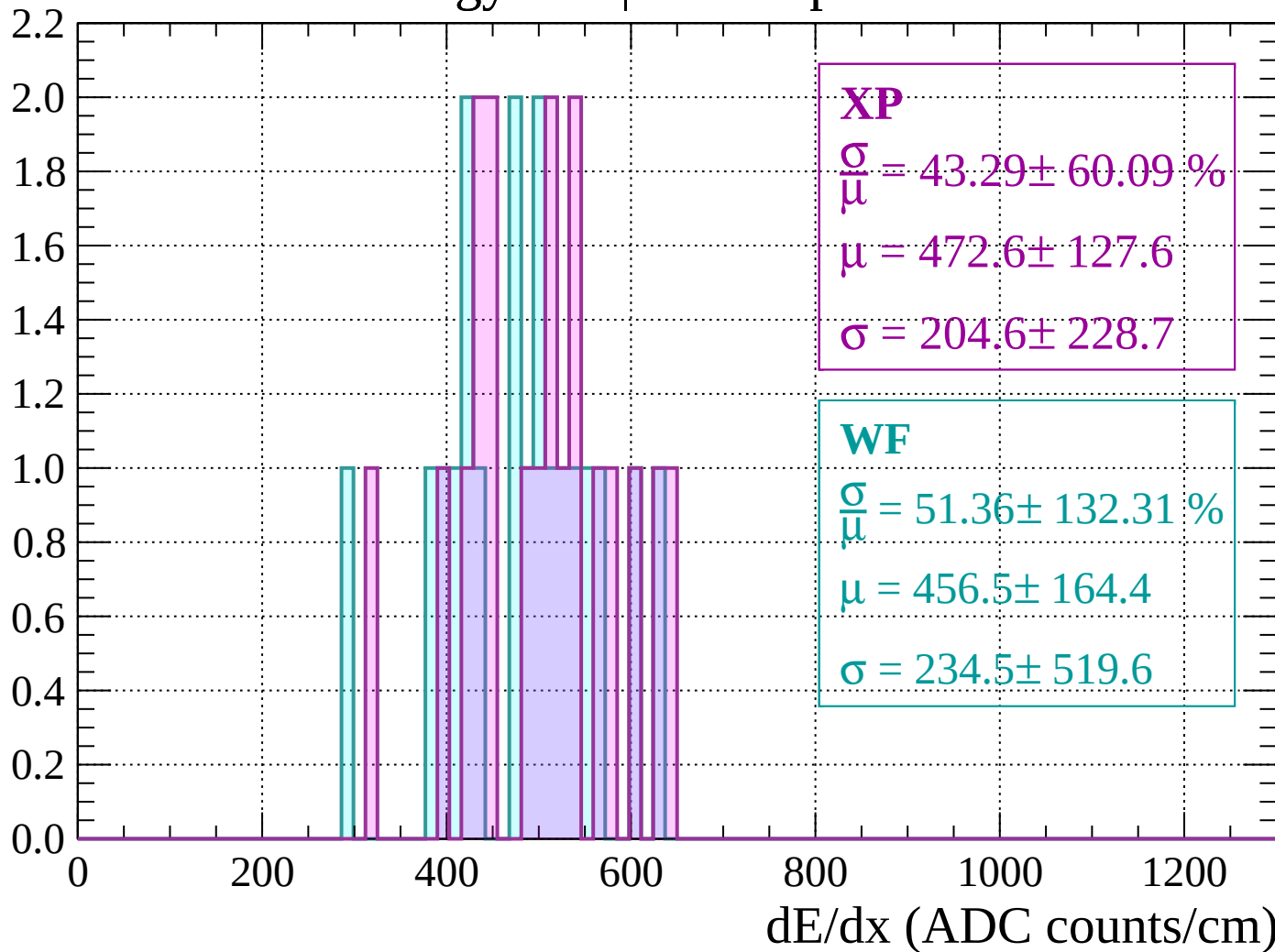
Energy loss | $1680 < p < 1720$

Count



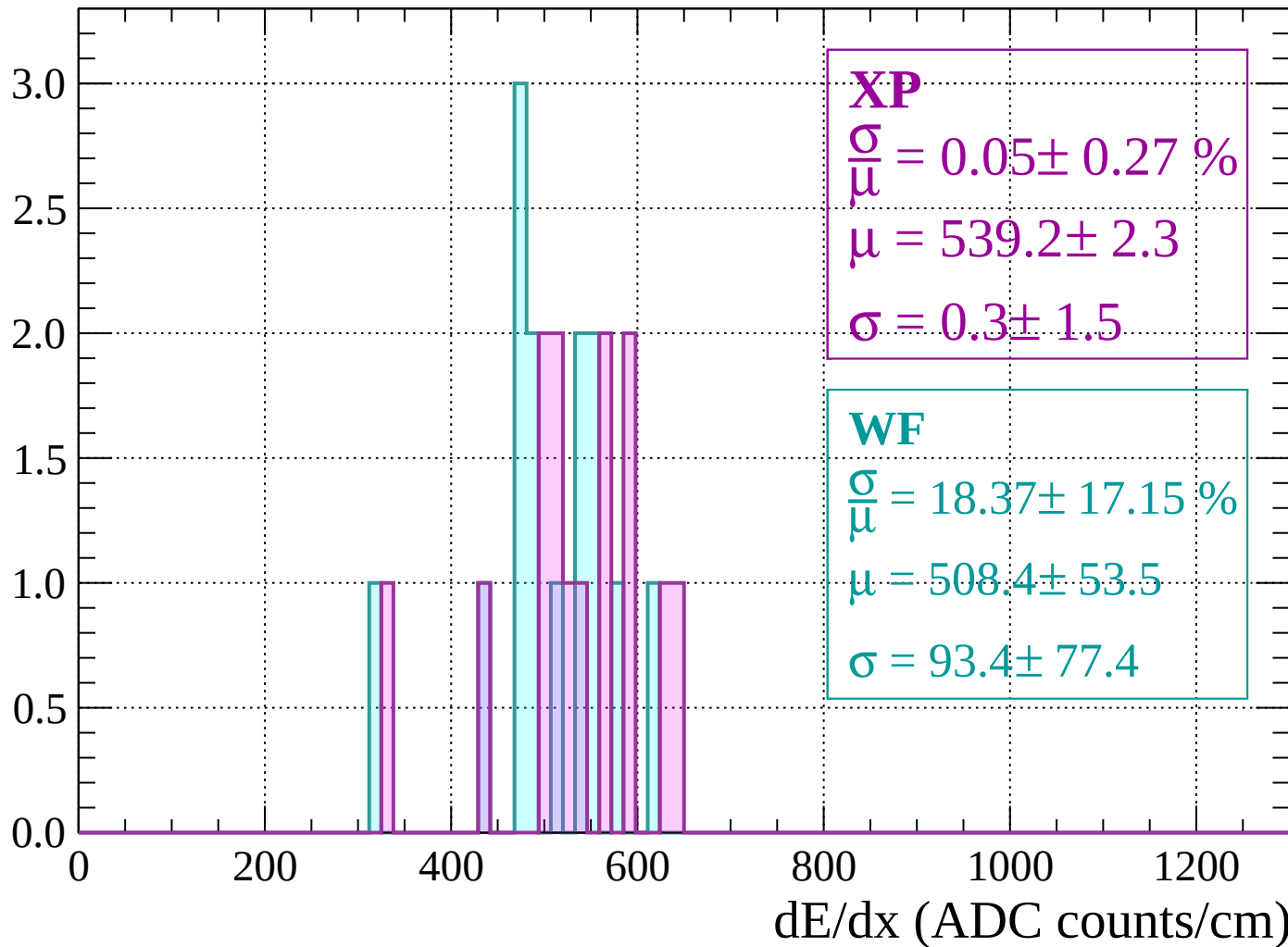
Energy loss | $1720 < p < 1760$

Count



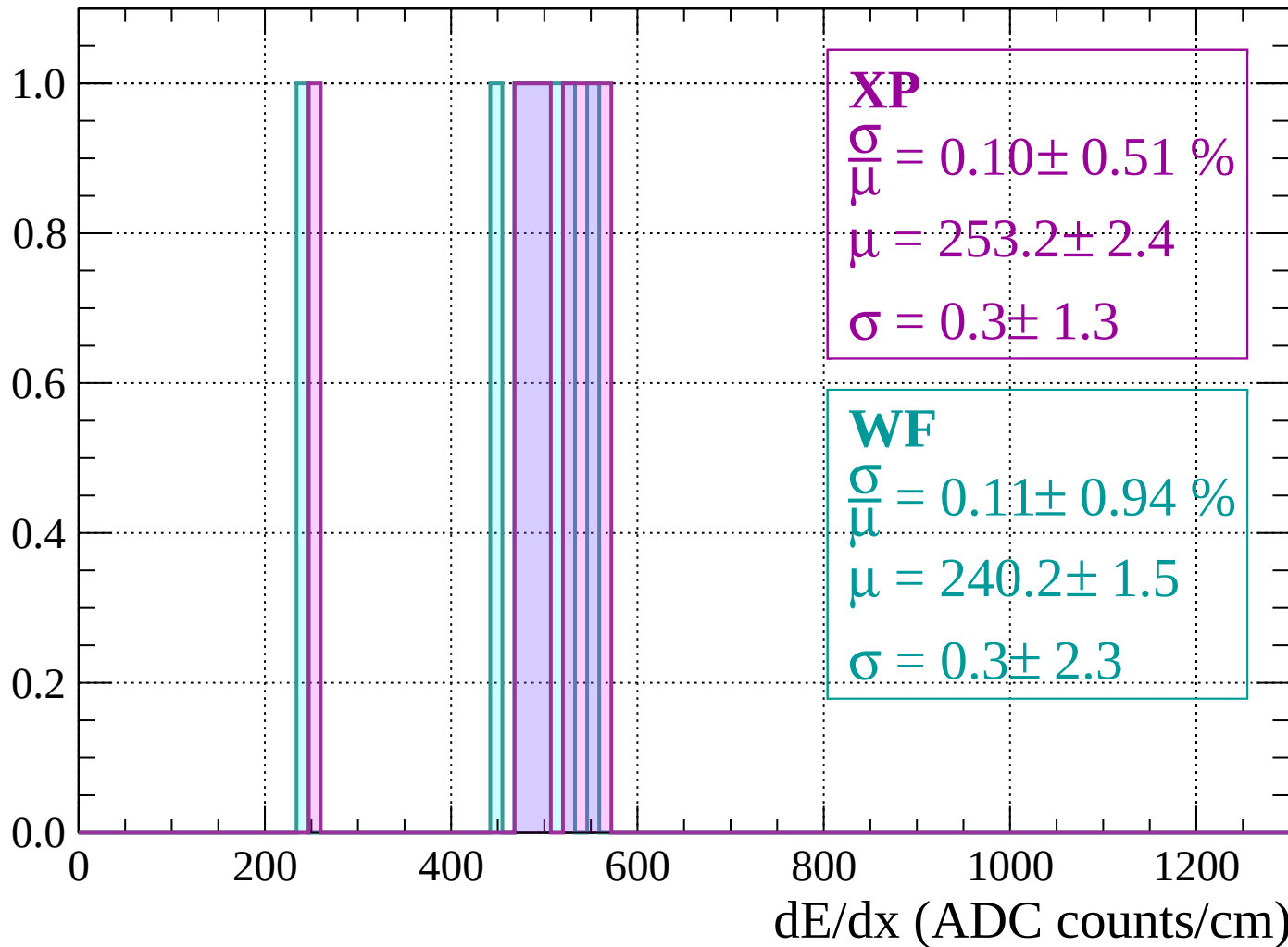
Energy loss | $1760 < p < 1800$

Count



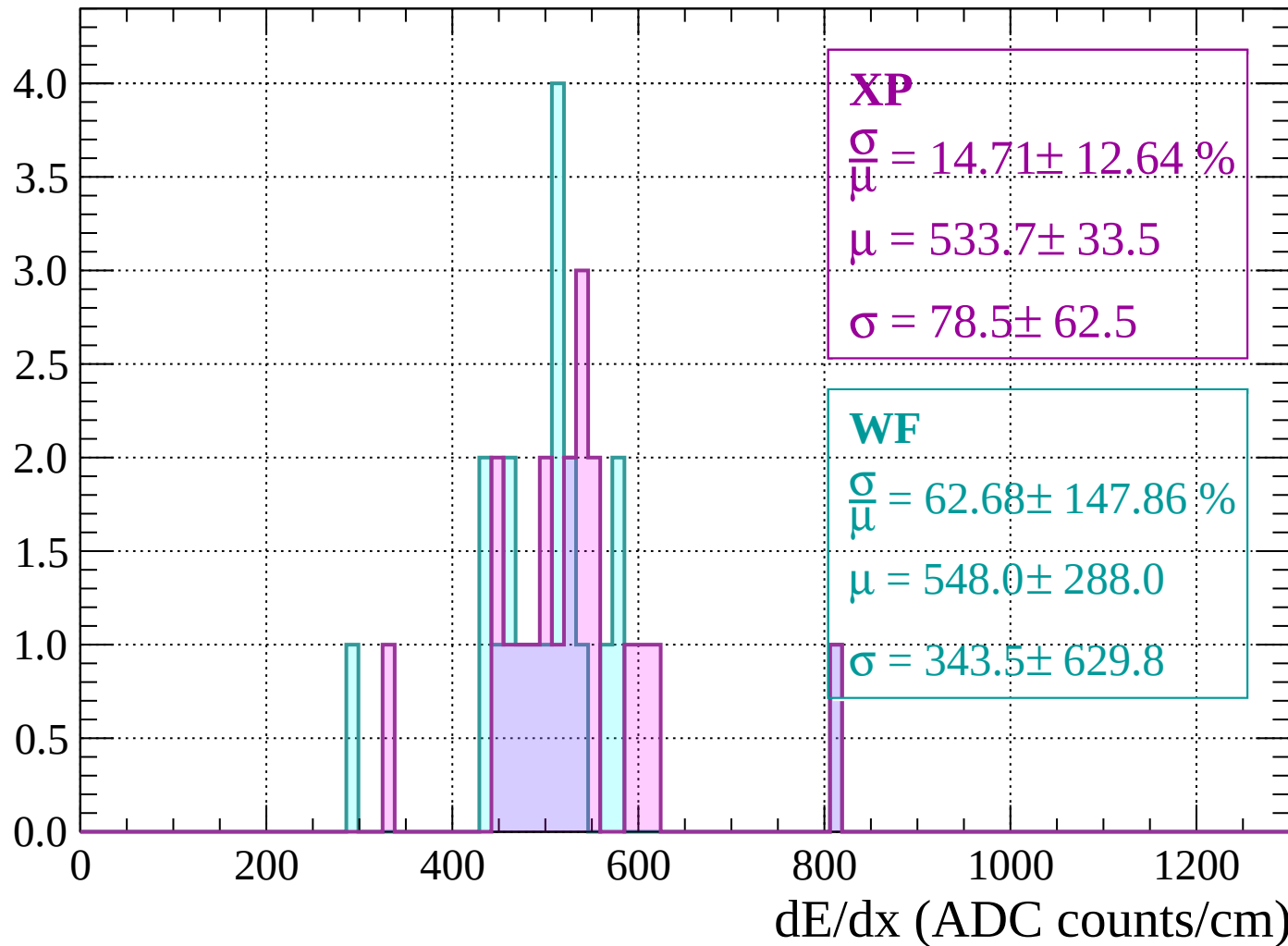
Energy loss | $1800 < p < 1840$

Count



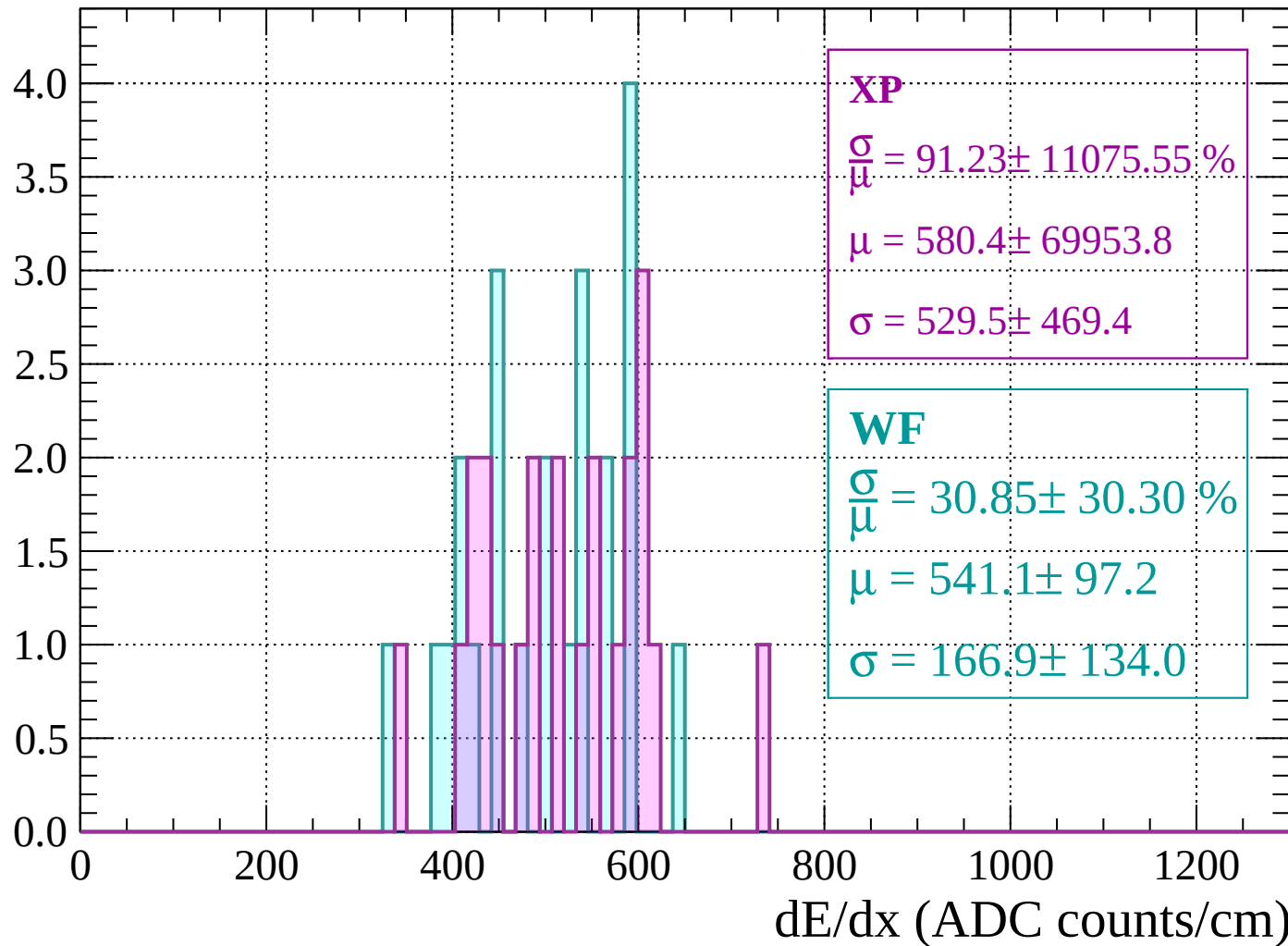
Energy loss | $1840 < p < 1880$

Count



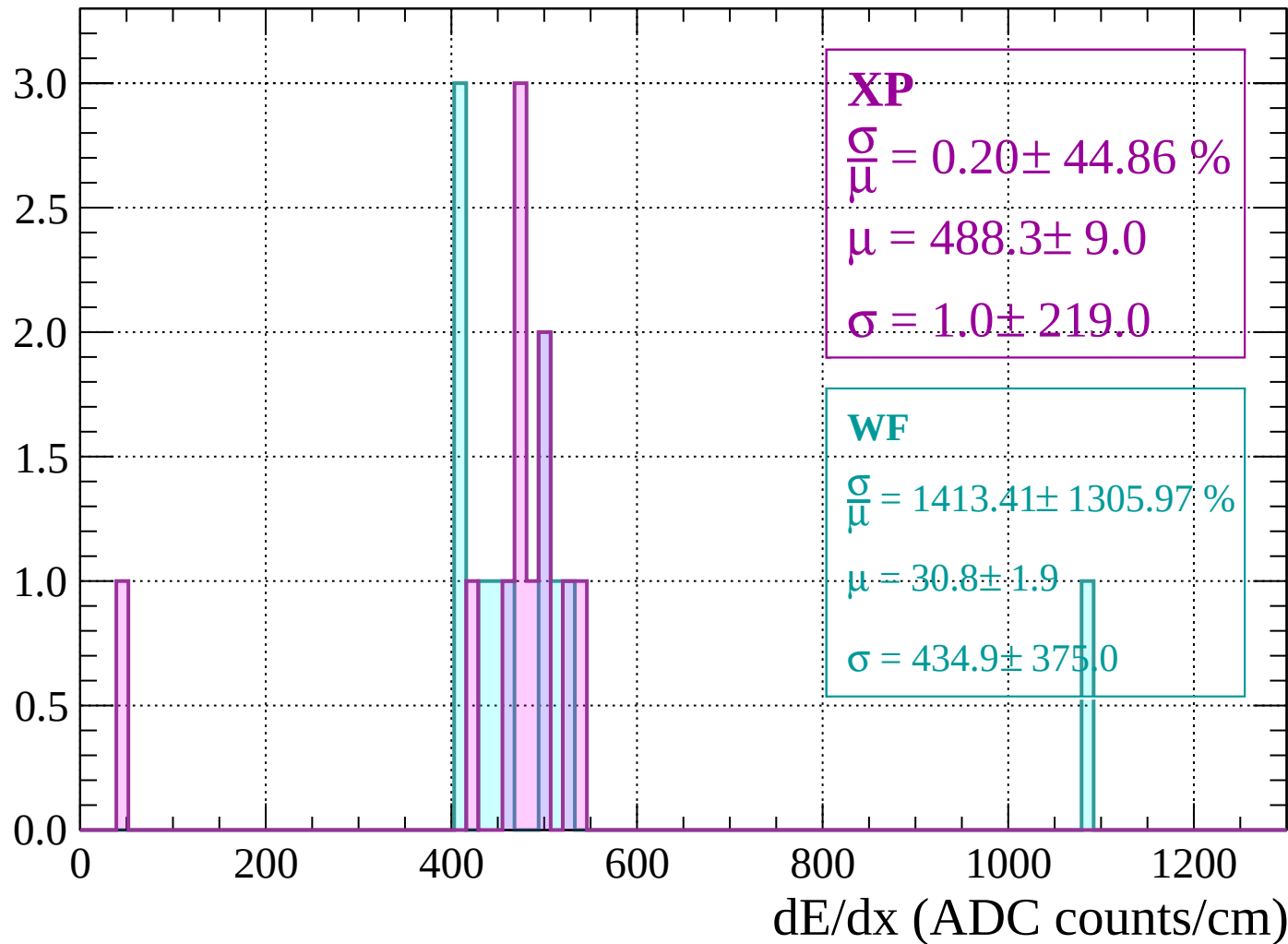
Energy loss | $1880 < p < 1920$

Count



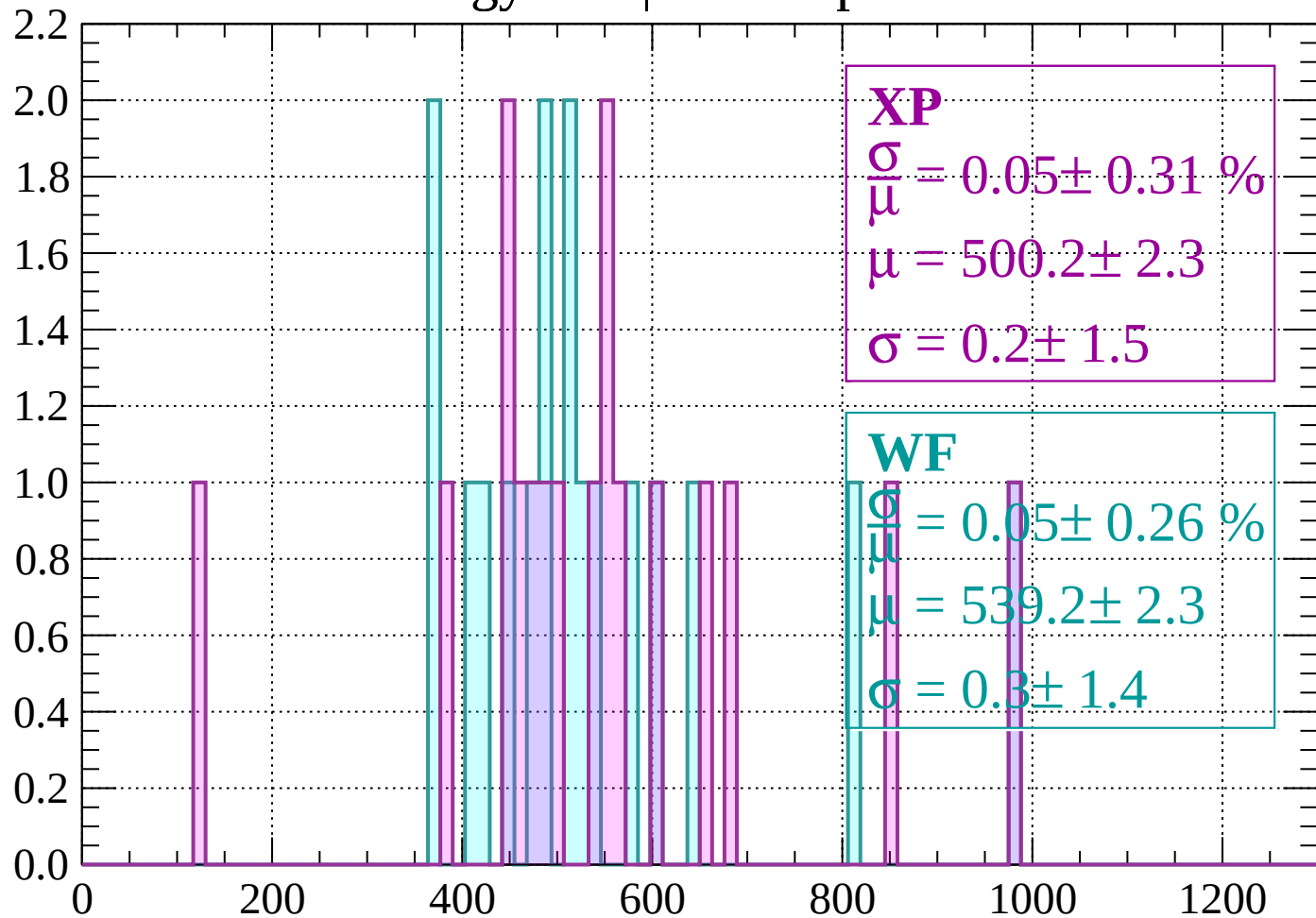
Energy loss | $1920 < p < 1960$

Count



Energy loss | 1960 < p < 2000

Count



XP

$$\frac{\sigma}{\mu} = 0.05 \pm 0.31 \%$$

$$\mu = 500.2 \pm 2.3$$

$$\sigma = 0.2 \pm 1.5$$

WF

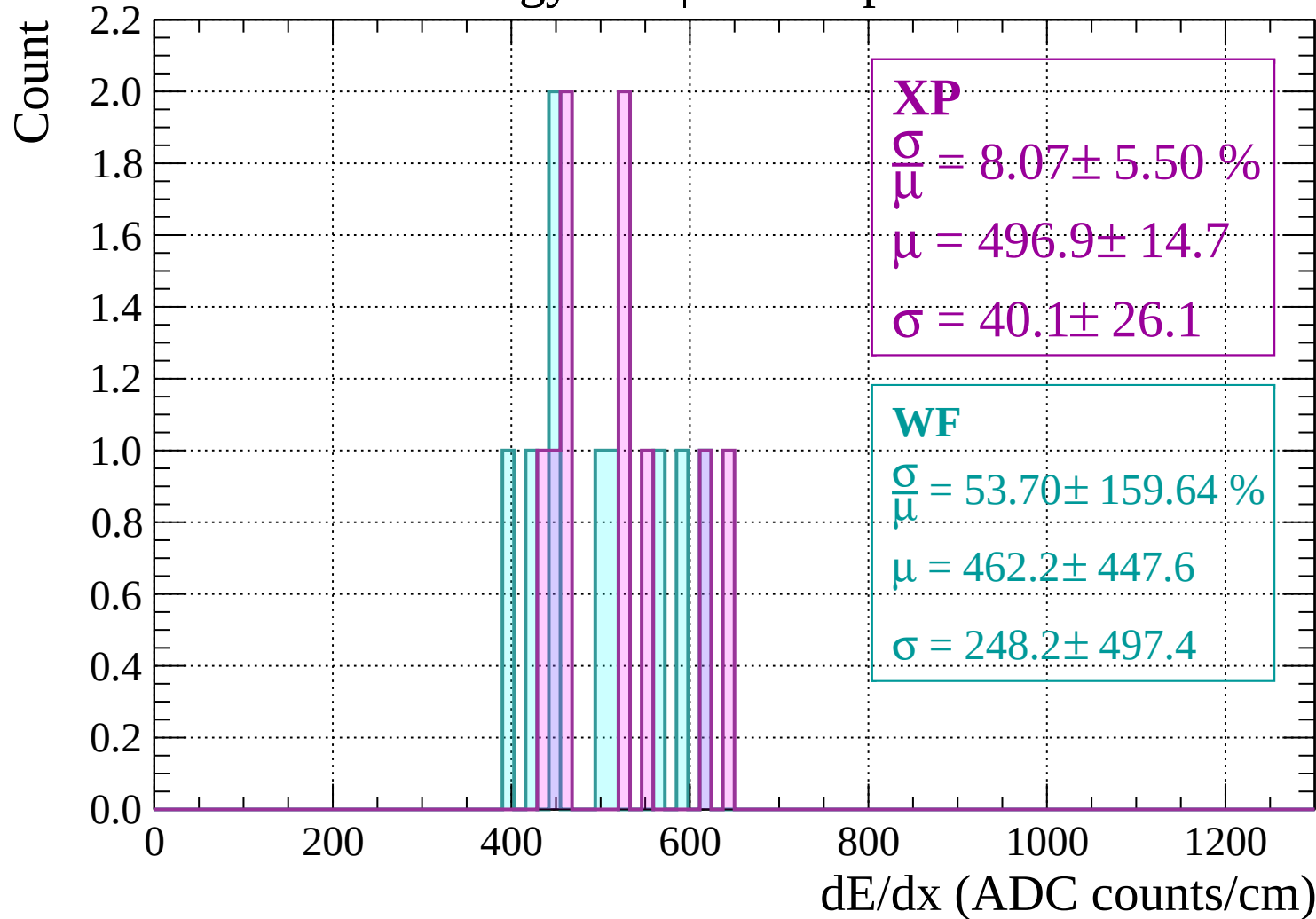
$$\frac{\sigma}{\mu} = 0.05 \pm 0.26 \%$$

$$\mu = 539.2 \pm 2.3$$

$$\sigma = 0.3 \pm 1.4$$

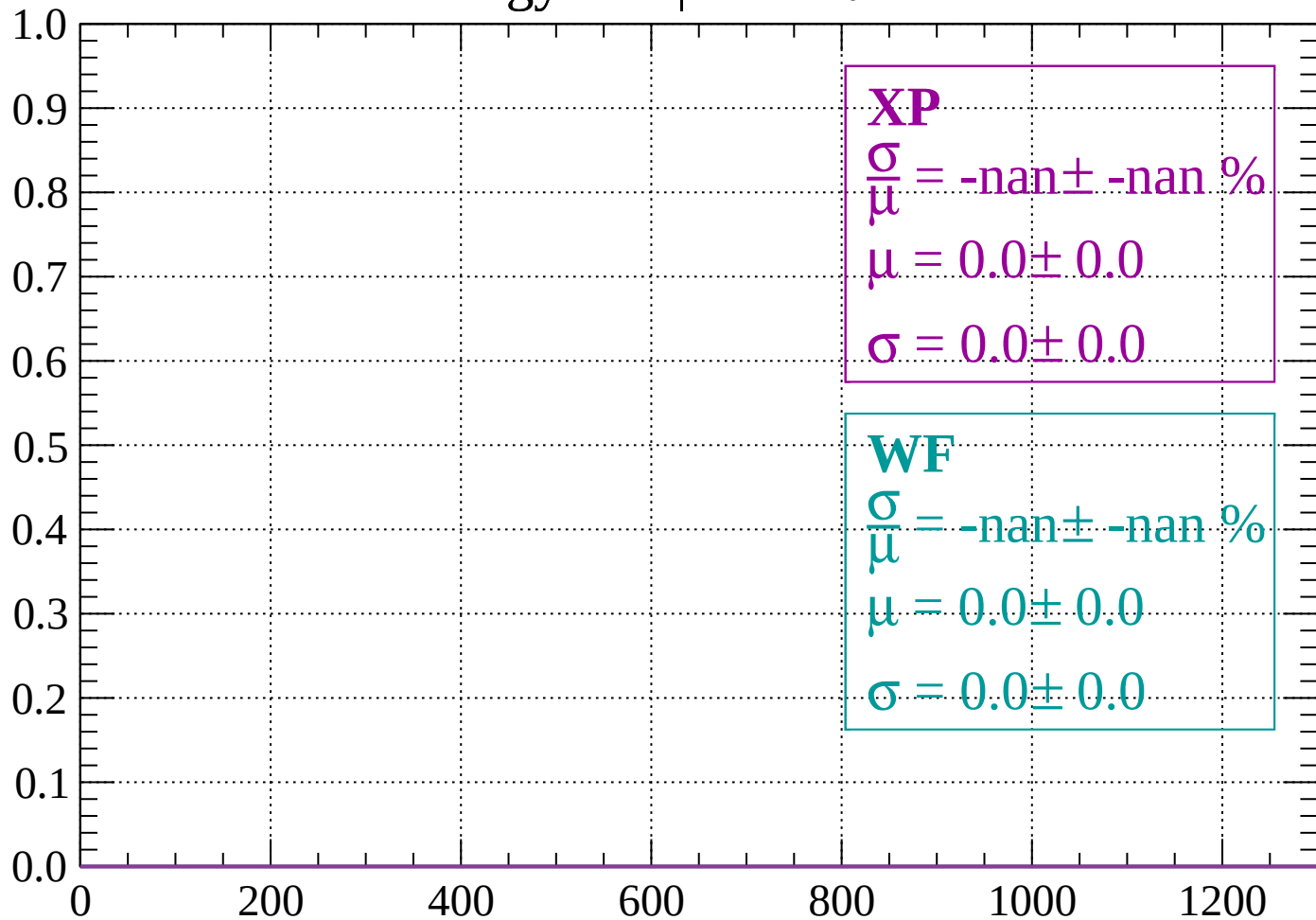
dE/dx (ADC counts/cm)

Energy loss | $2000 < p < 2040$



Energy loss | $-90 < \theta < -88$

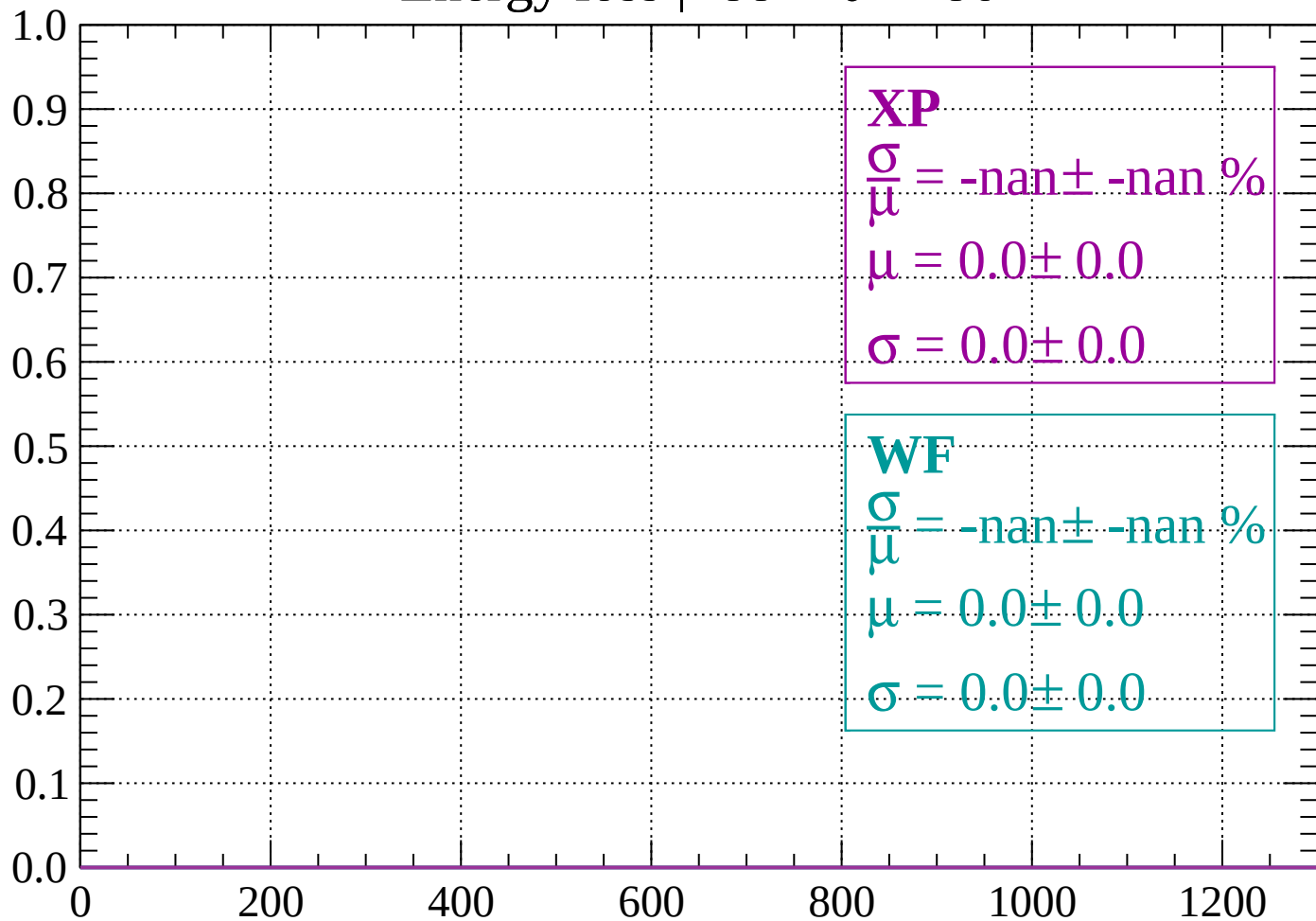
Count



dE/dx (ADC counts/cm)

Energy loss | $-88 < \theta < -86$

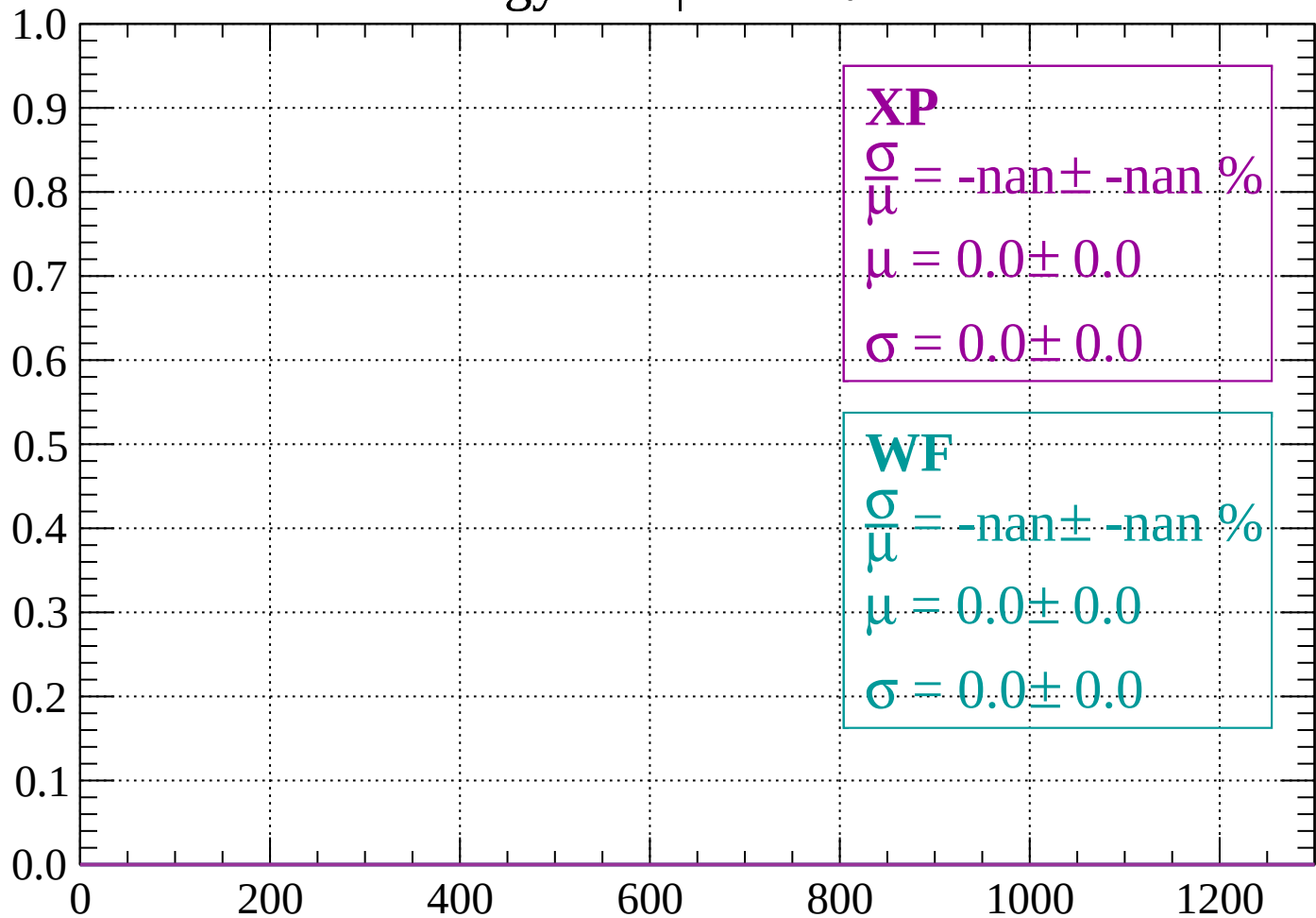
Count



dE/dx (ADC counts/cm)

Energy loss | $-86 < \theta < -84$

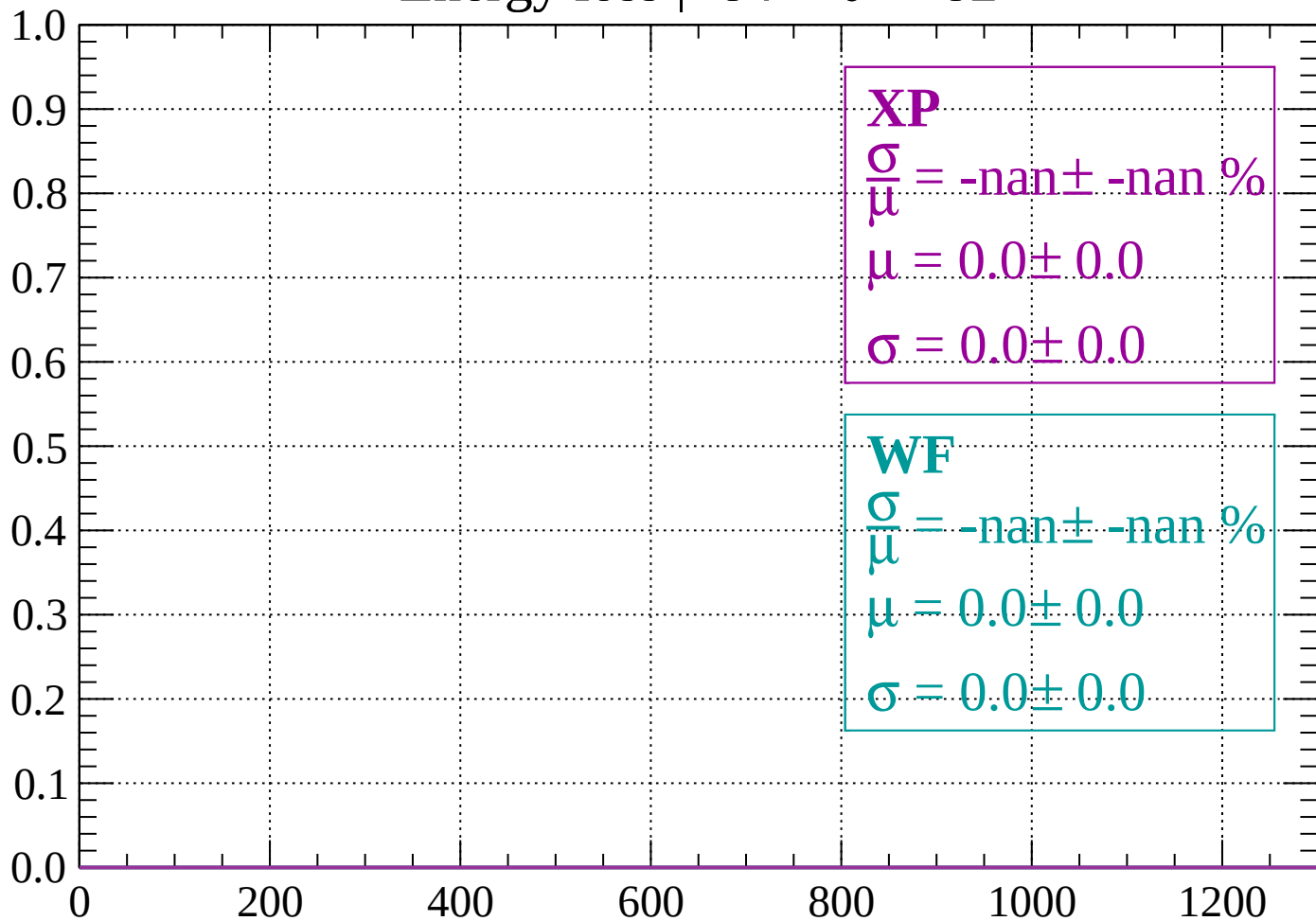
Count



dE/dx (ADC counts/cm)

Energy loss | $-84 < \theta < -82$

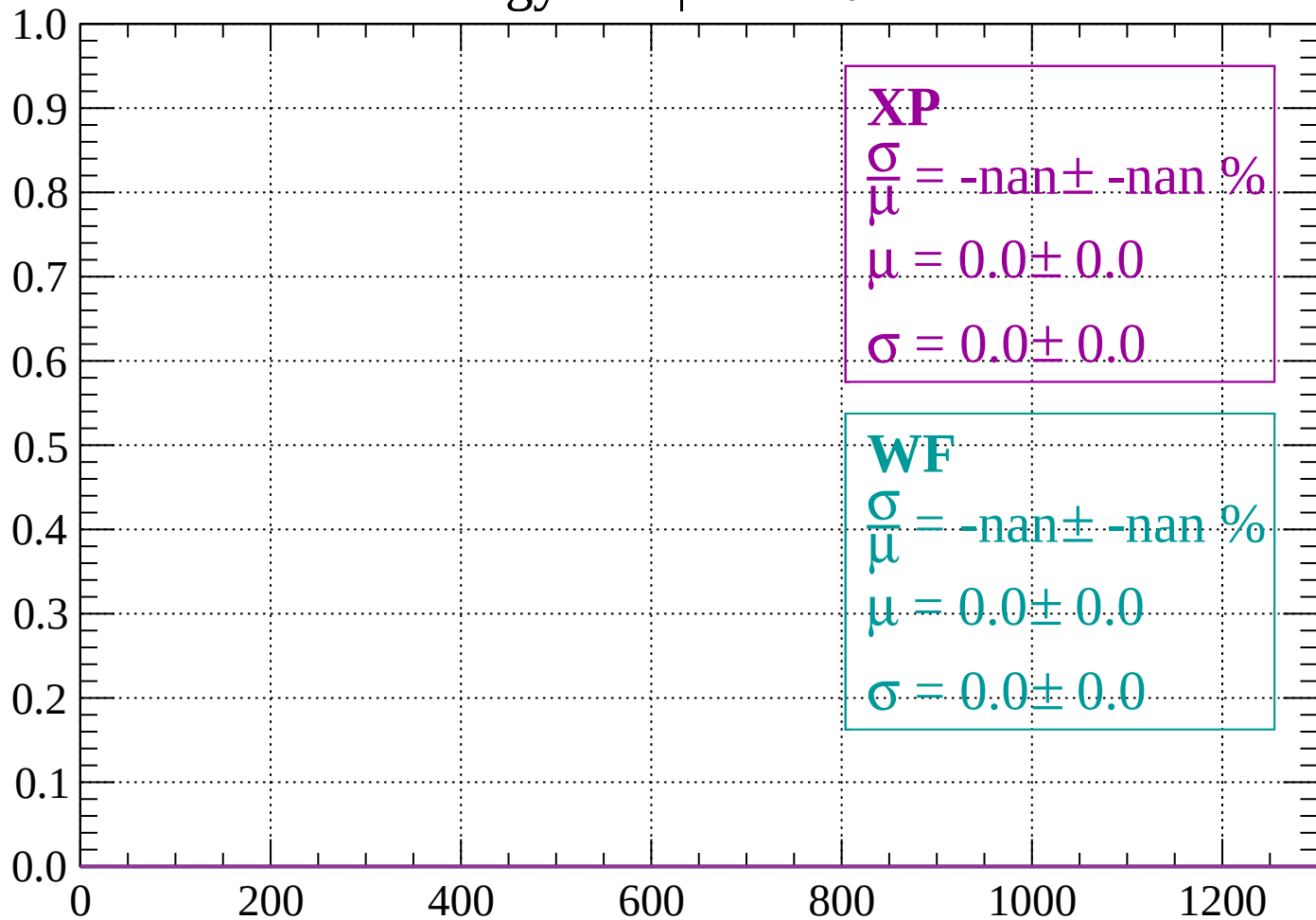
Count



dE/dx (ADC counts/cm)

Energy loss | $-82 < \theta < -80$

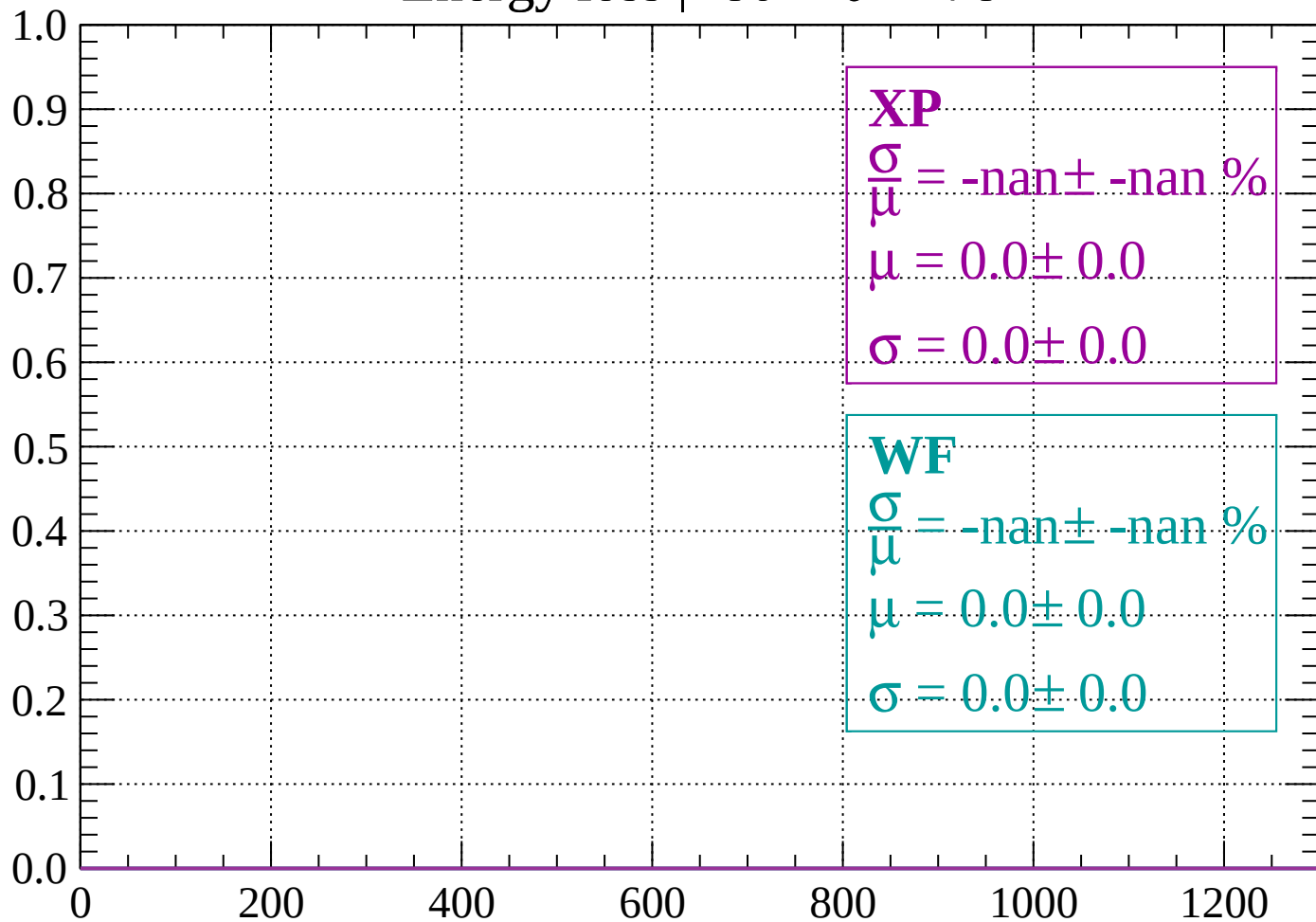
Count



dE/dx (ADC counts/cm)

Energy loss | $-80 < \theta < -78$

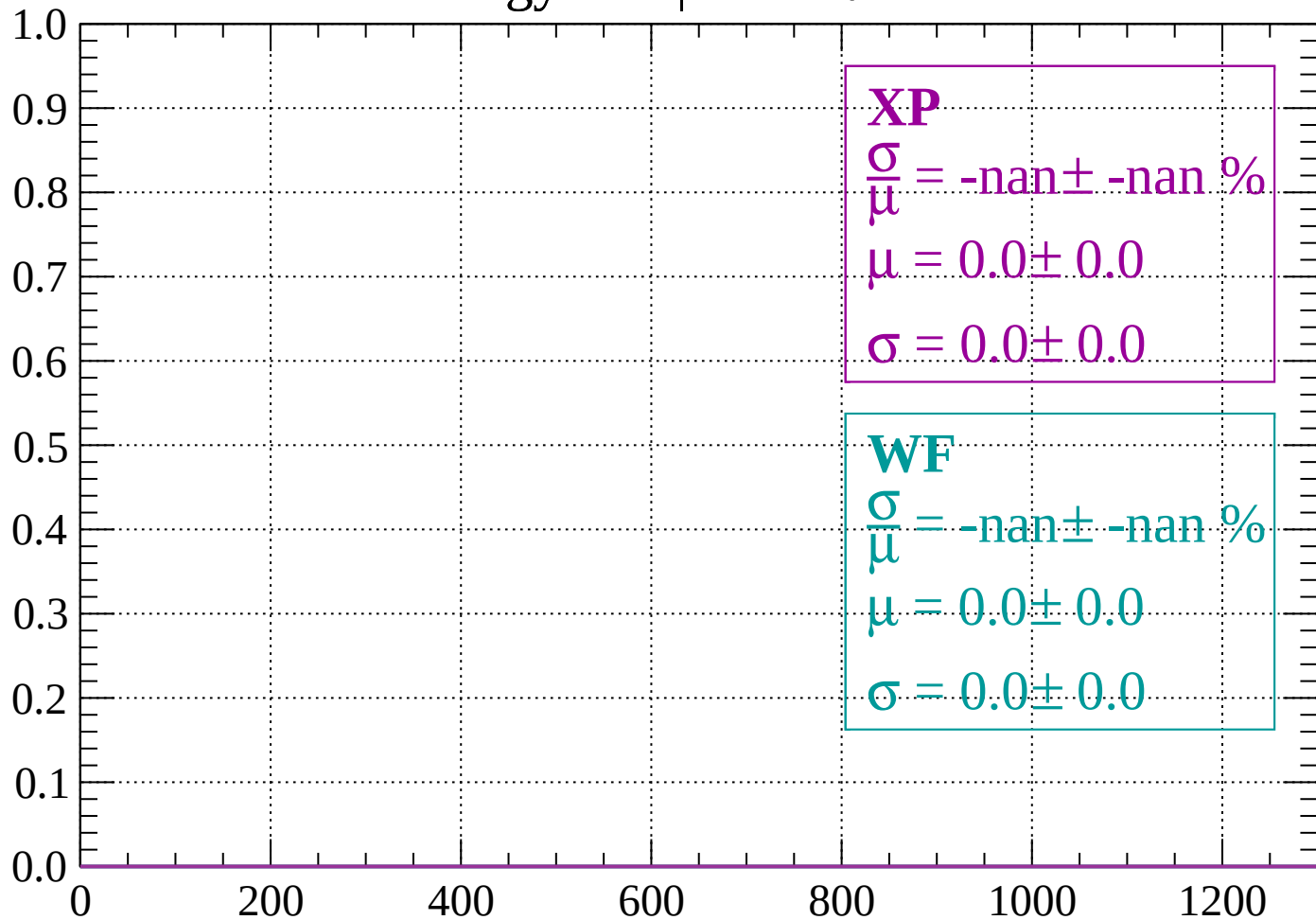
Count



dE/dx (ADC counts/cm)

Energy loss | $-78 < \theta < -76$

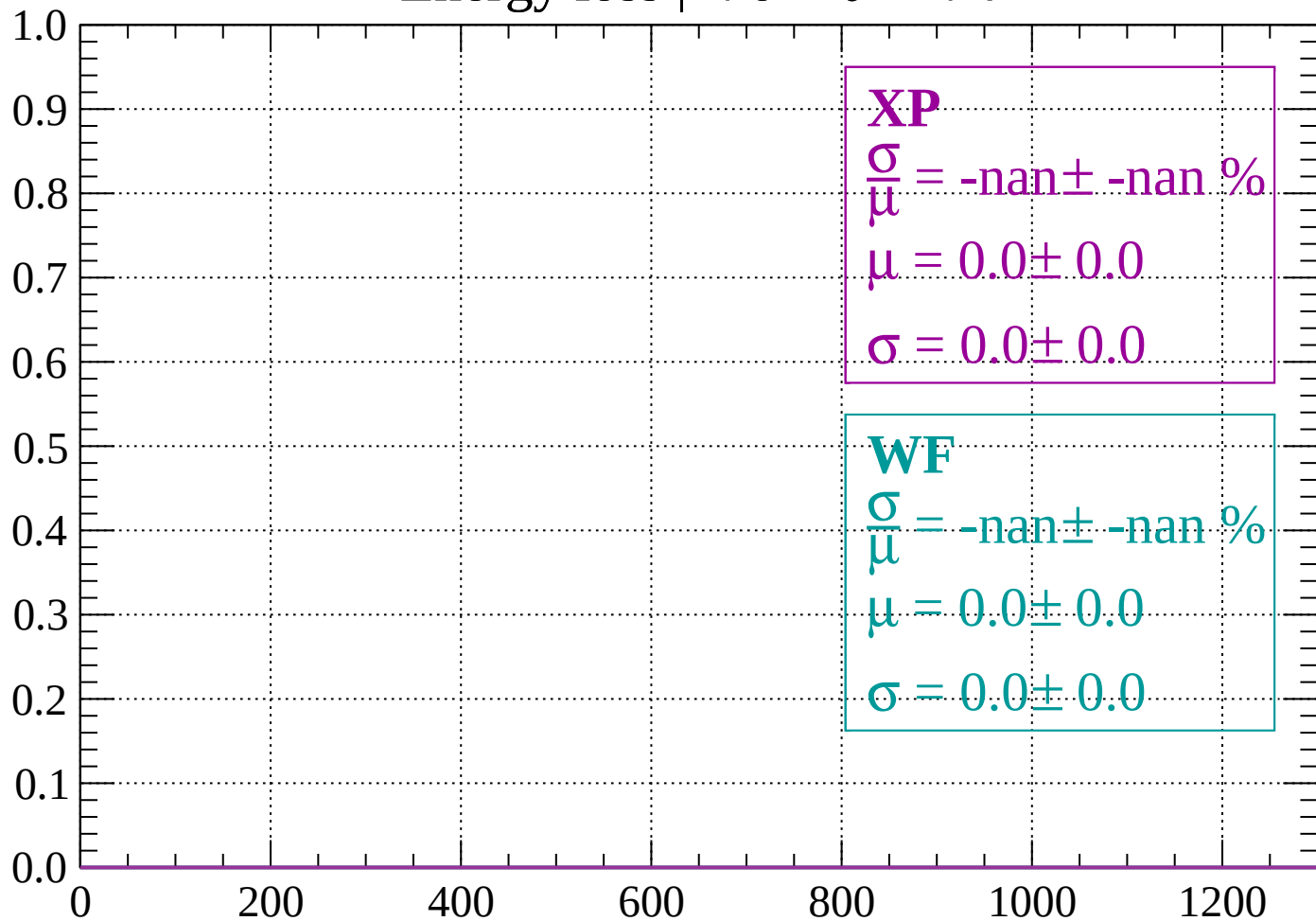
Count



dE/dx (ADC counts/cm)

Energy loss | $-76 < \theta < -74$

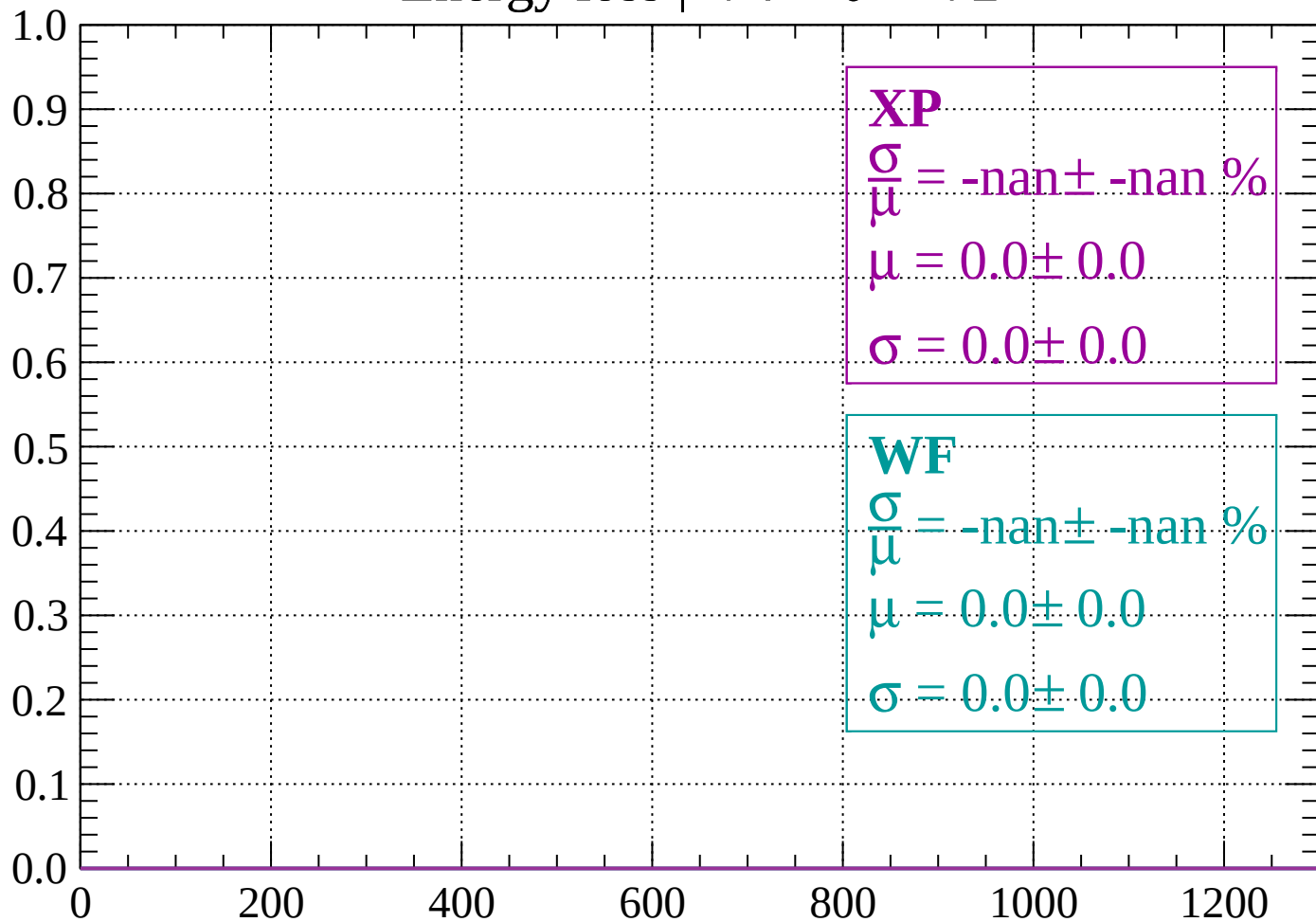
Count



dE/dx (ADC counts/cm)

Energy loss | $-74 < \theta < -72$

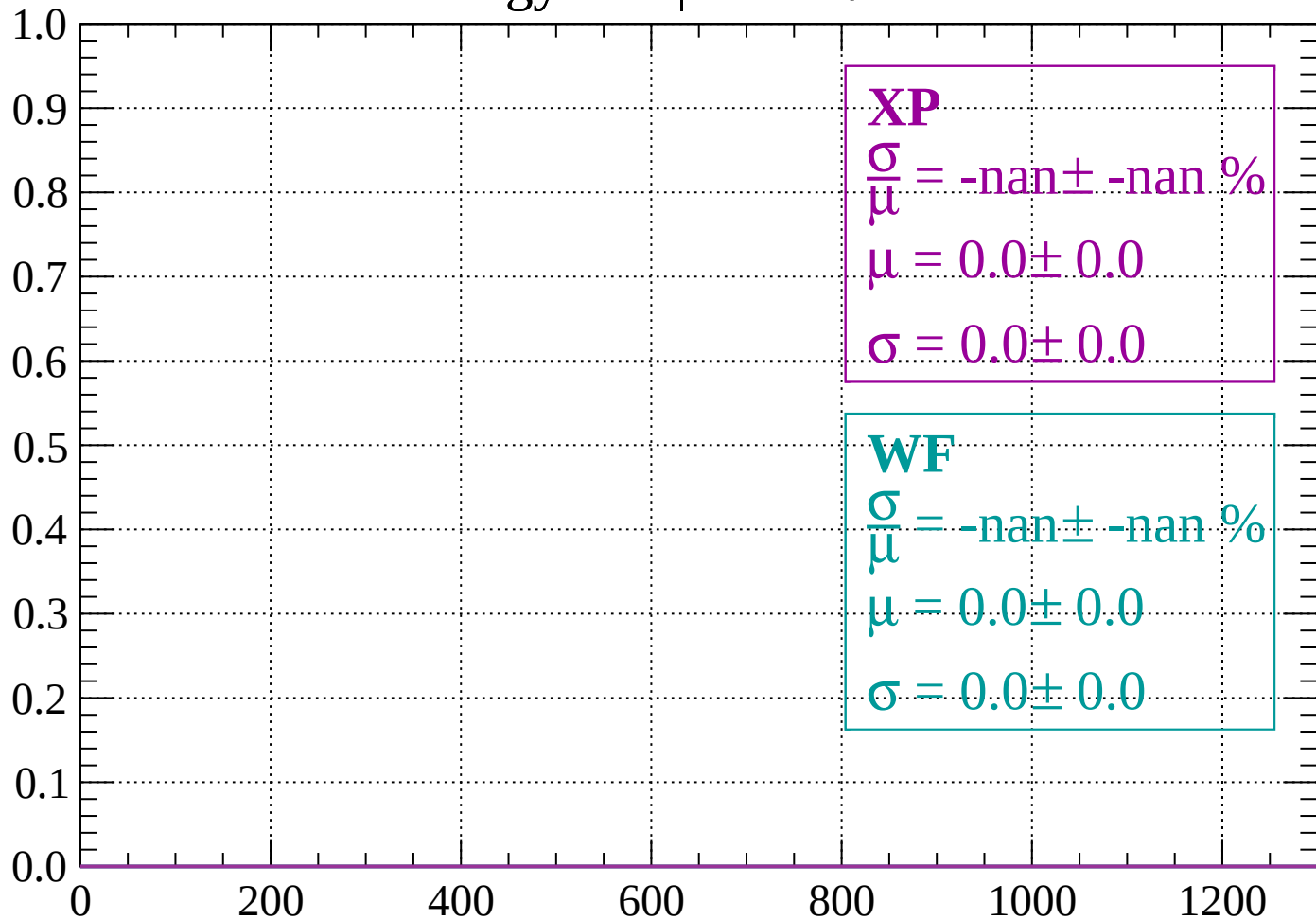
Count



dE/dx (ADC counts/cm)

Energy loss | $-72 < \theta < -70$

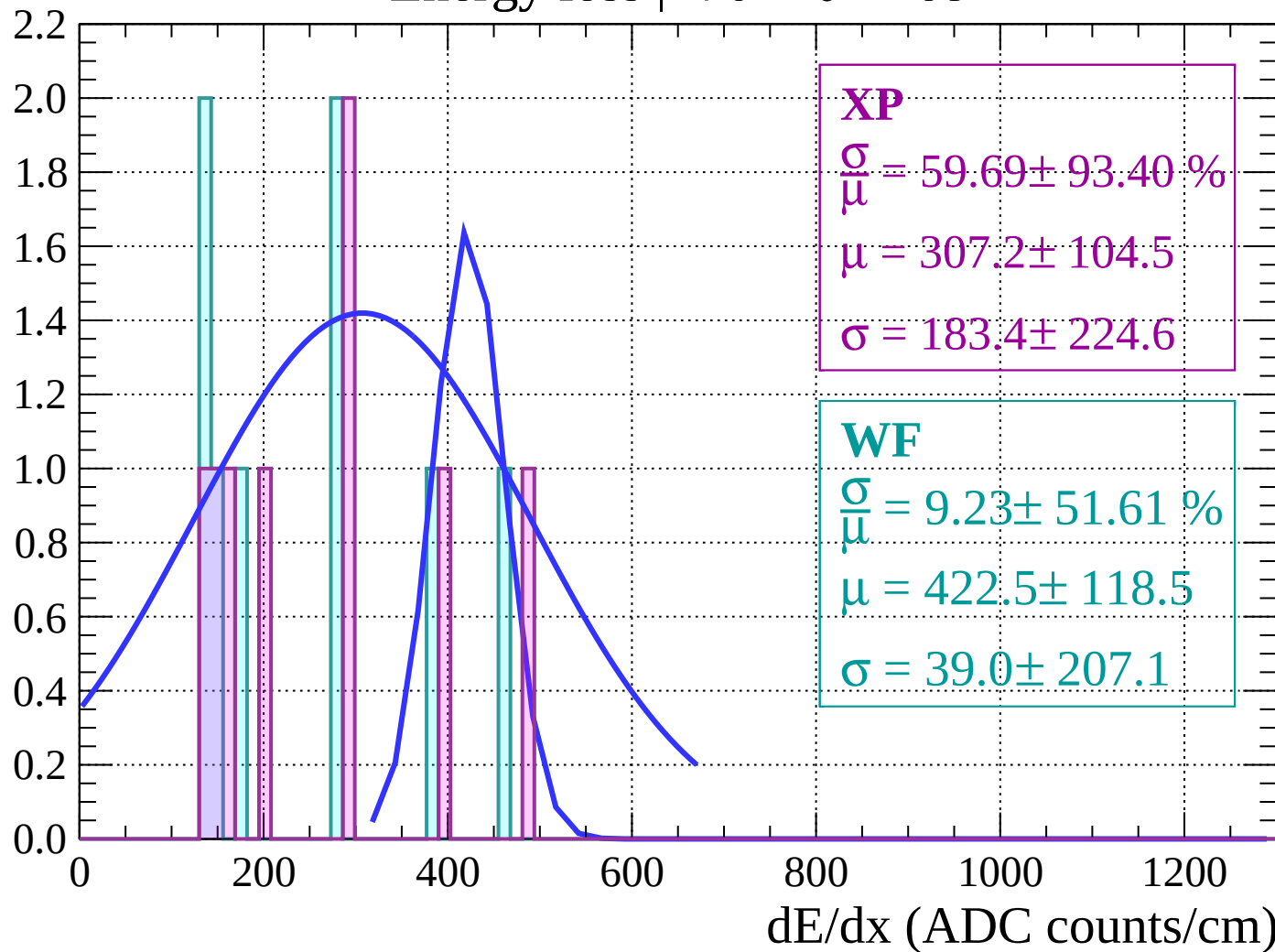
Count



dE/dx (ADC counts/cm)

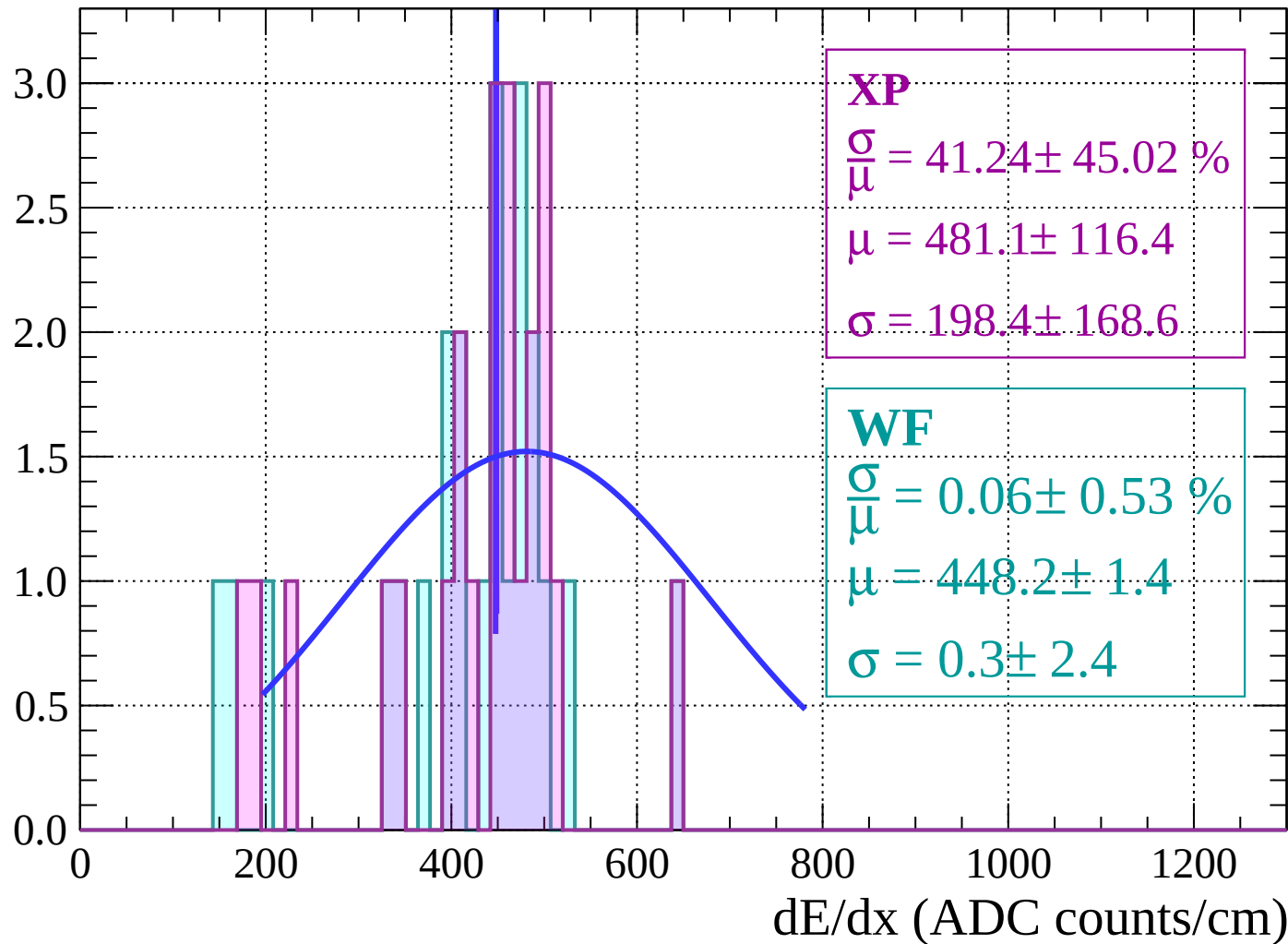
Energy loss | $-70 < \theta < -68$

Count



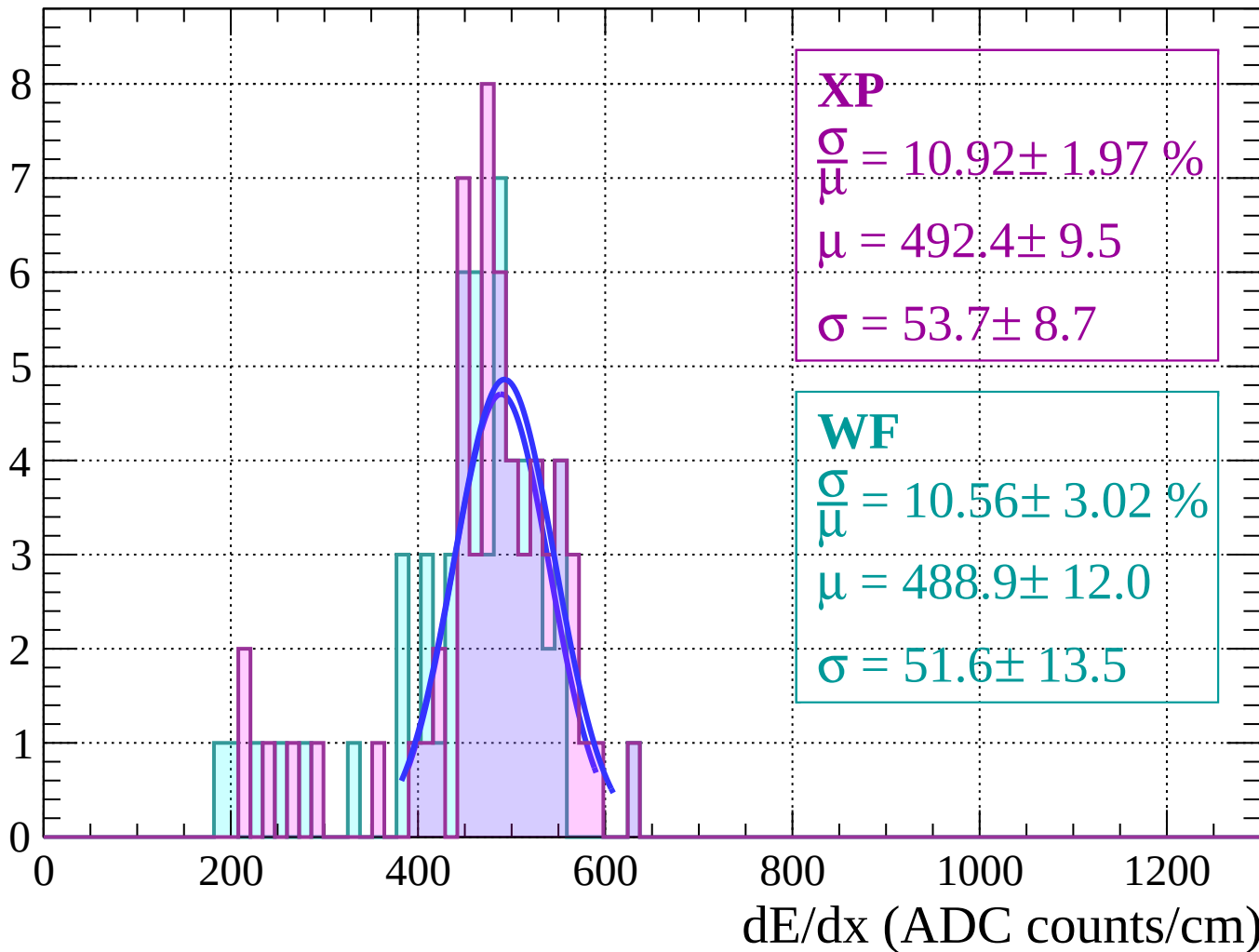
Energy loss | $-68 < \theta < -66$

Count



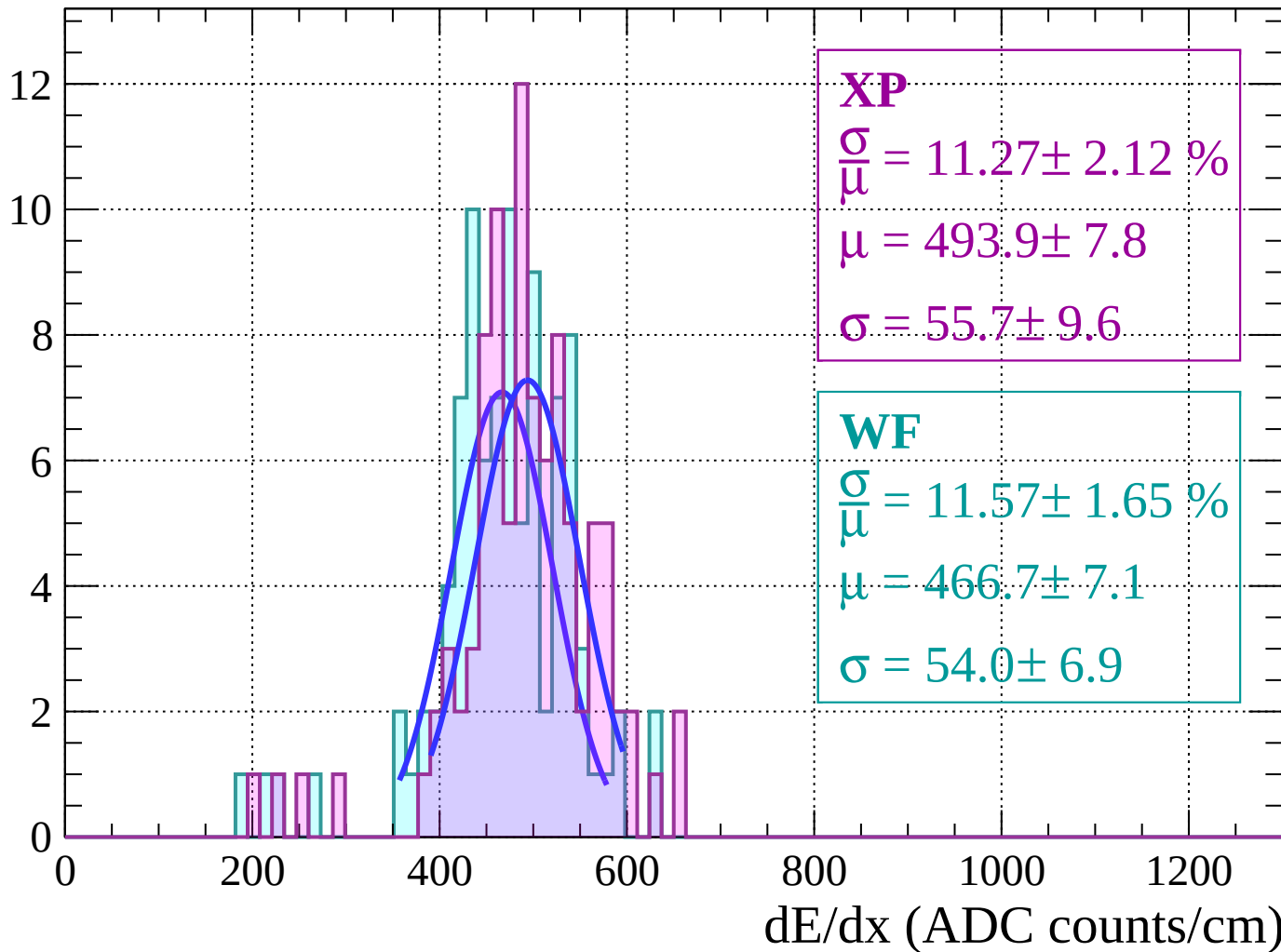
Energy loss | $-66 < \theta < -64$

Count



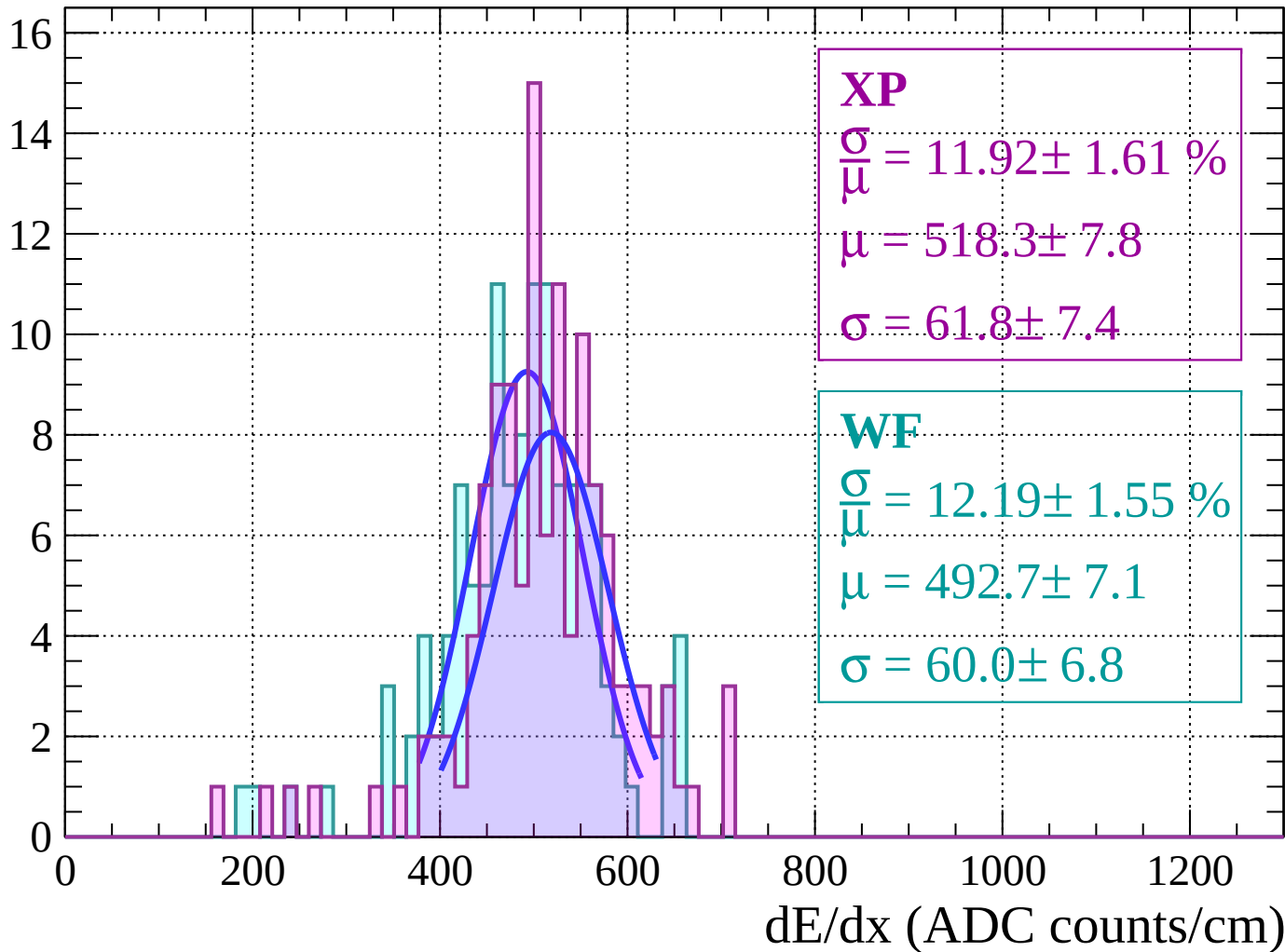
Energy loss | $-64 < \theta < -62$

Count



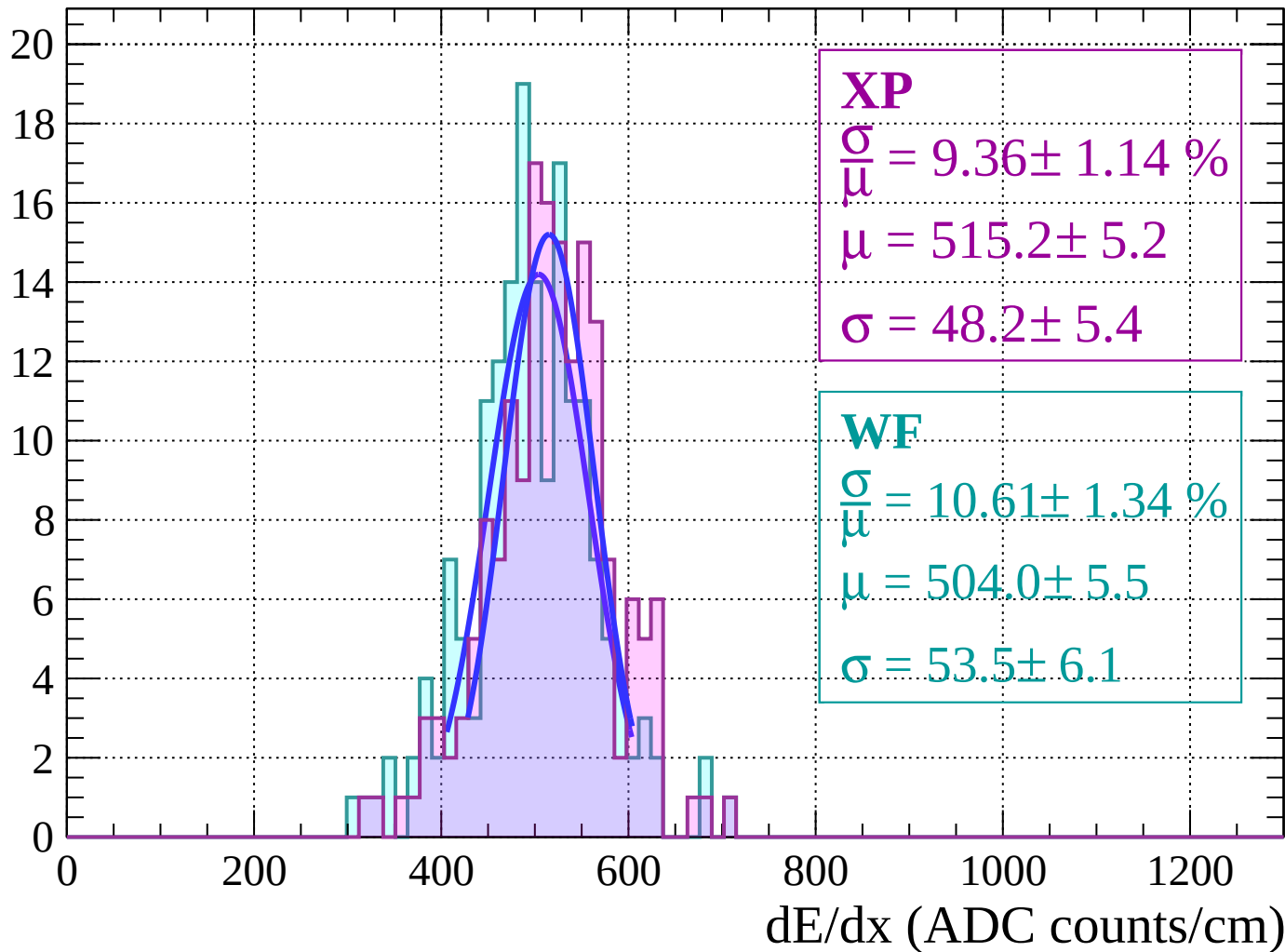
Energy loss | $-62 < \theta < -60$

Count



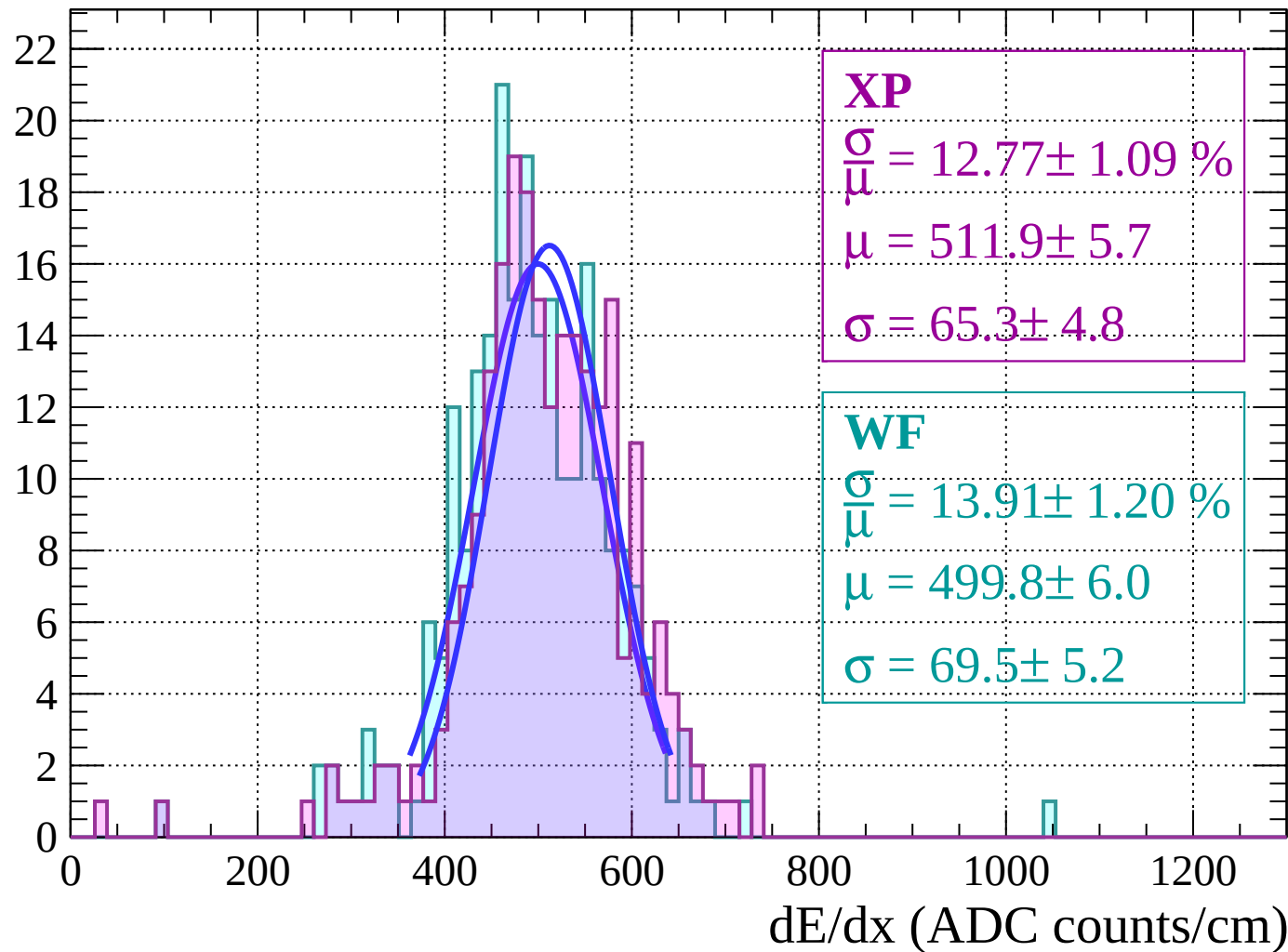
Energy loss | $-60 < \theta < -58$

Count



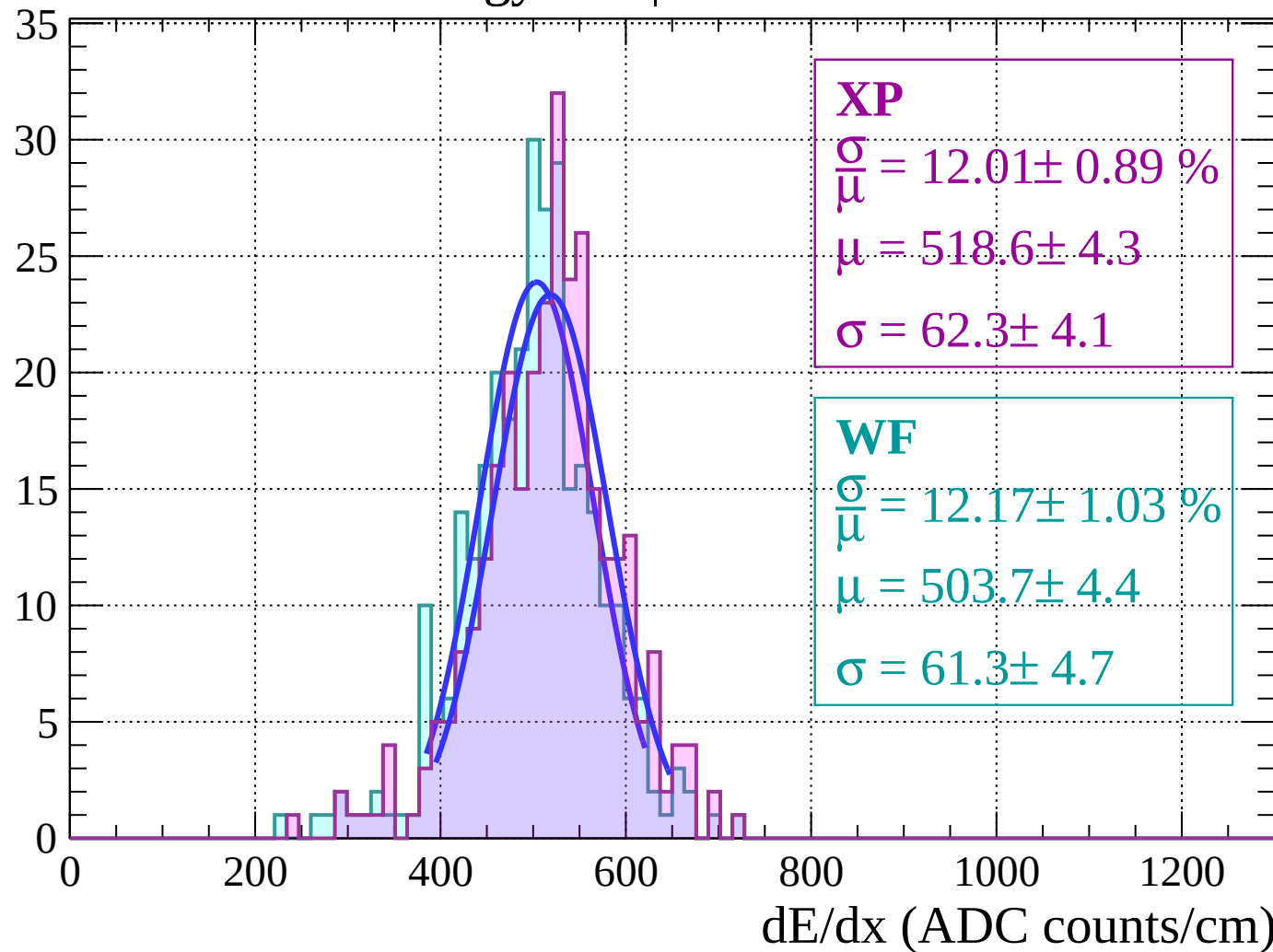
Energy loss $|-58 < \theta < -56$

Count



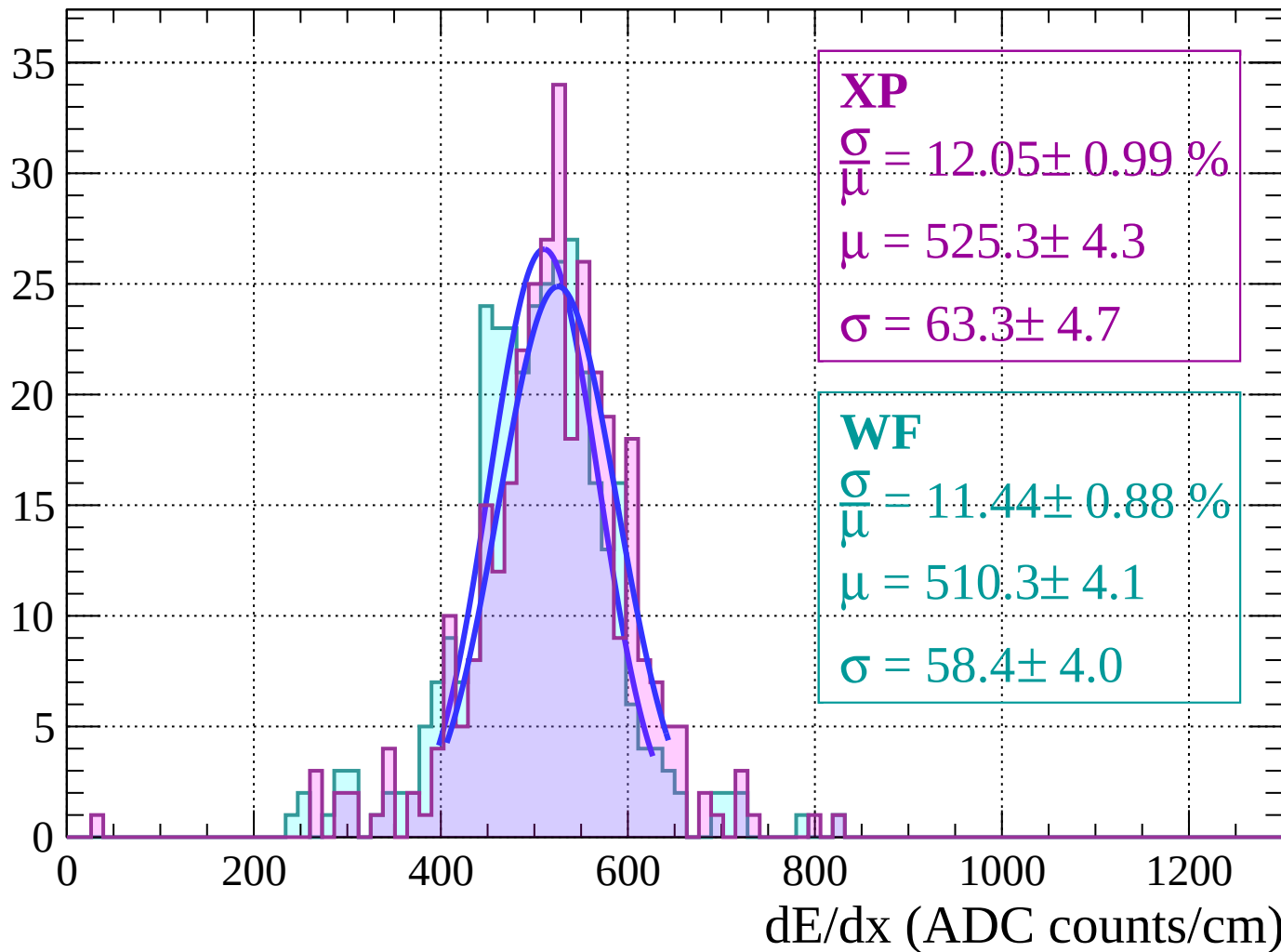
Energy loss $|-56 < \theta < -54$

Count



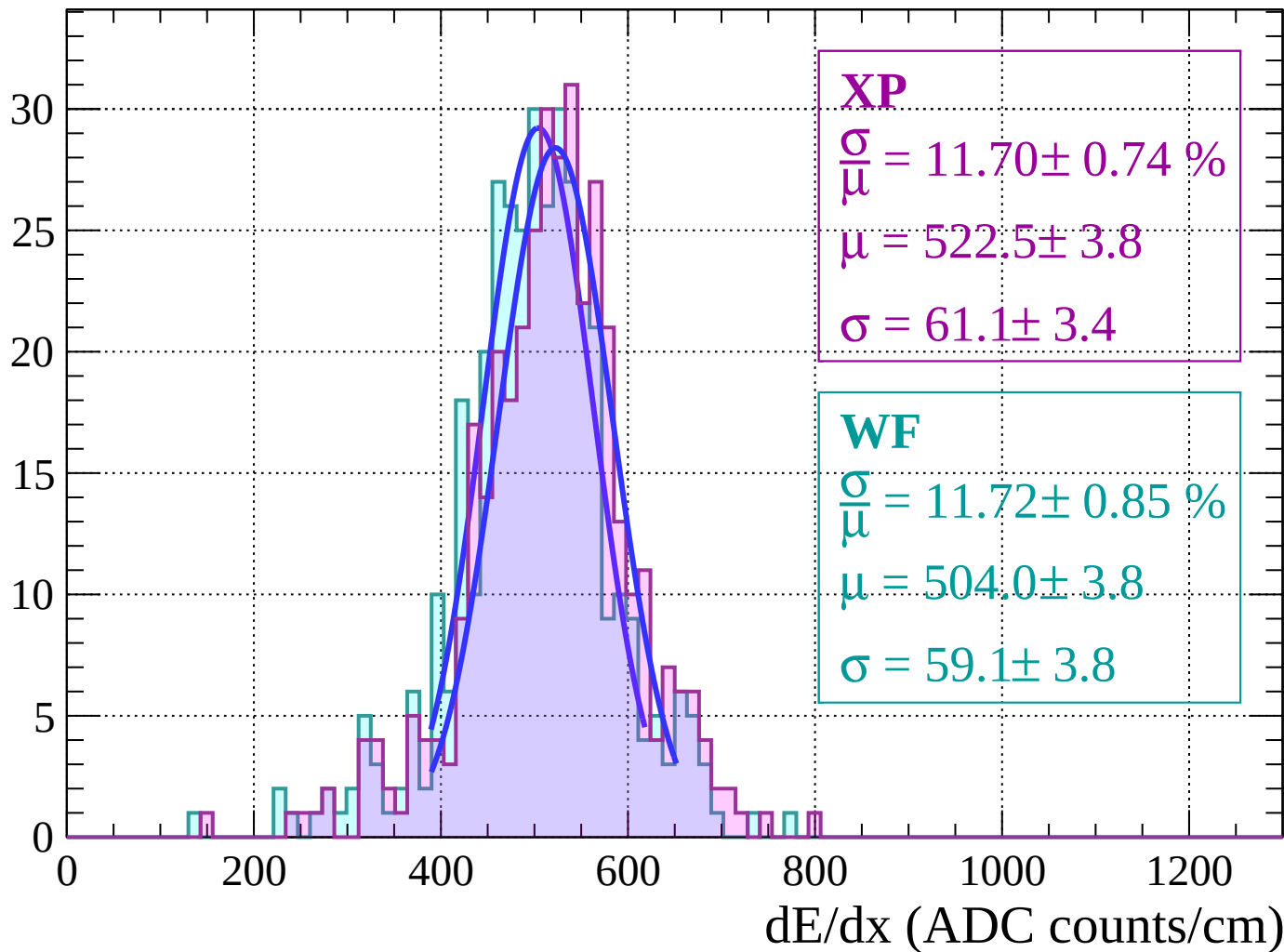
Energy loss $|-54 < \theta < -52$

Count



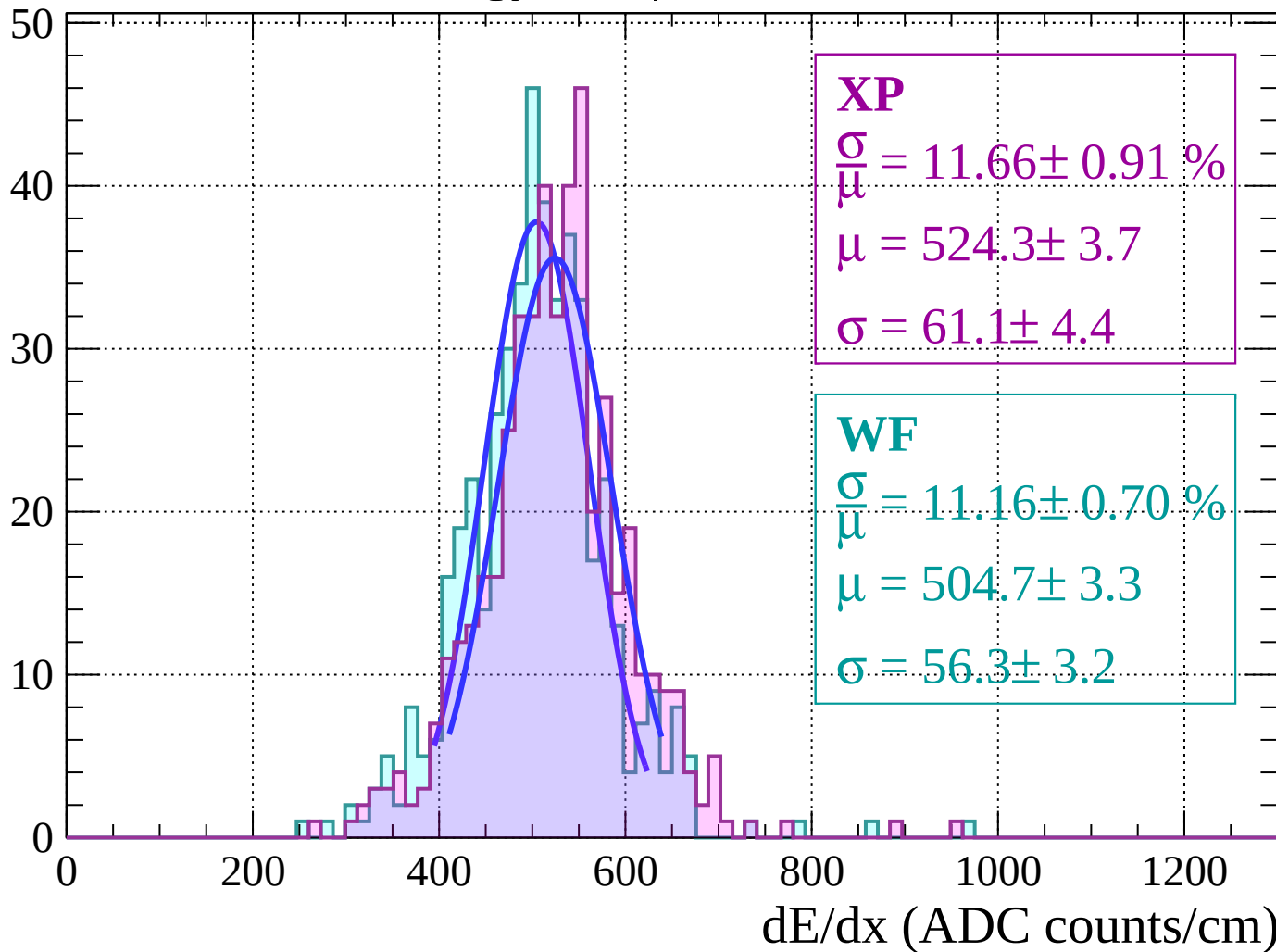
Energy loss $|-52 < \theta < -50$

Count



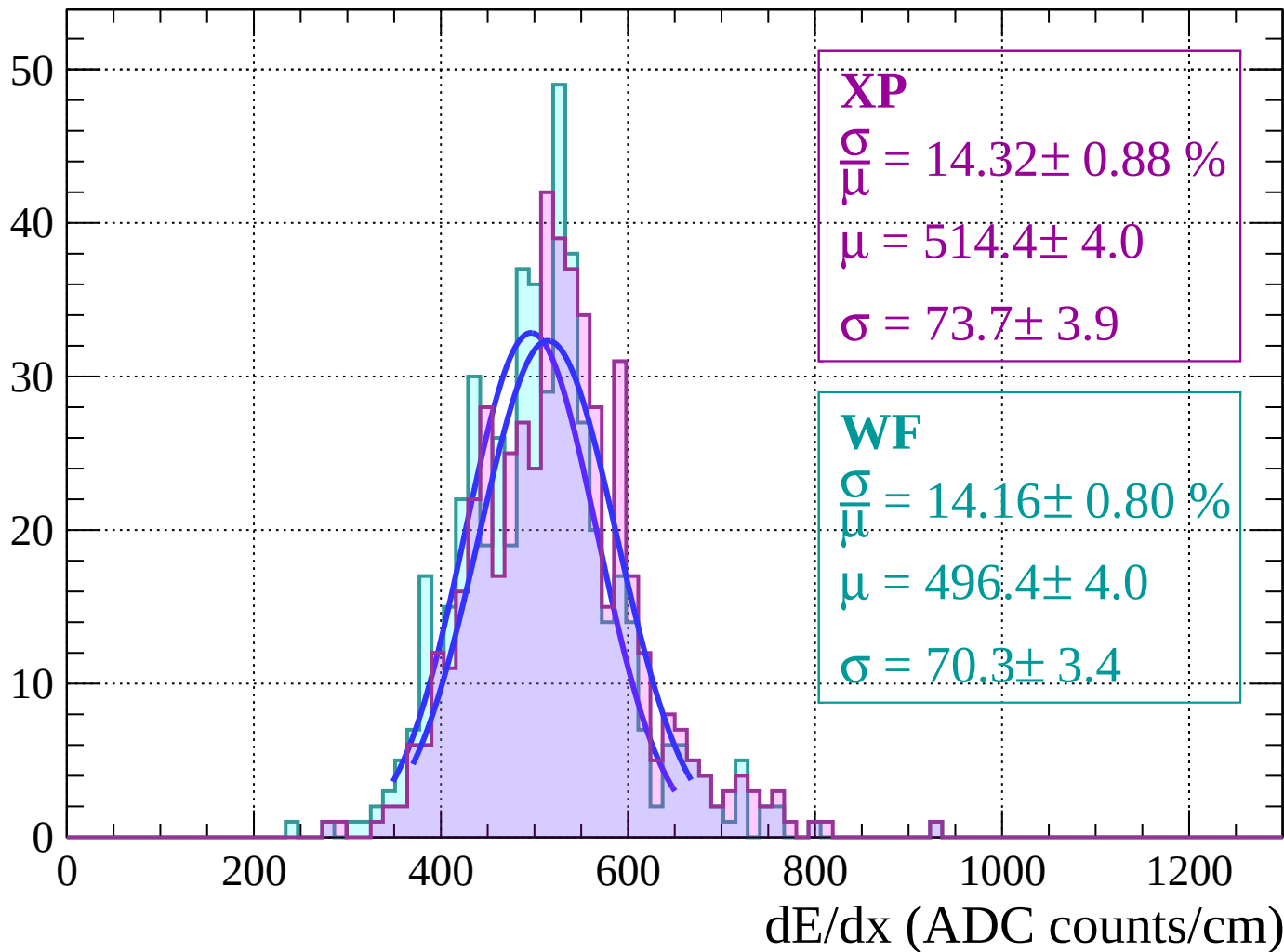
Energy loss | $-50 < \theta < -48$

Count



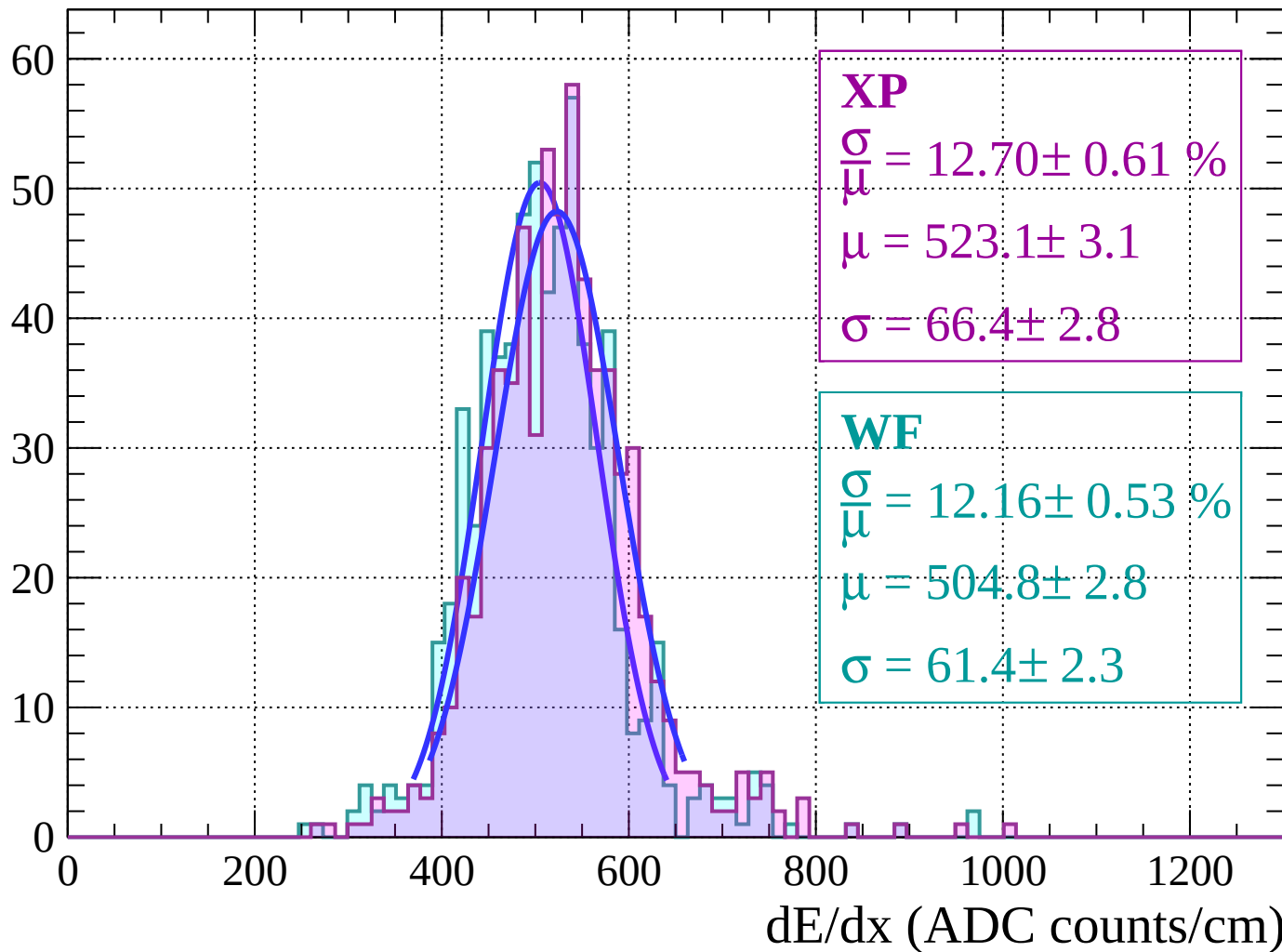
Energy loss | $-48 < \theta < -46$

Count



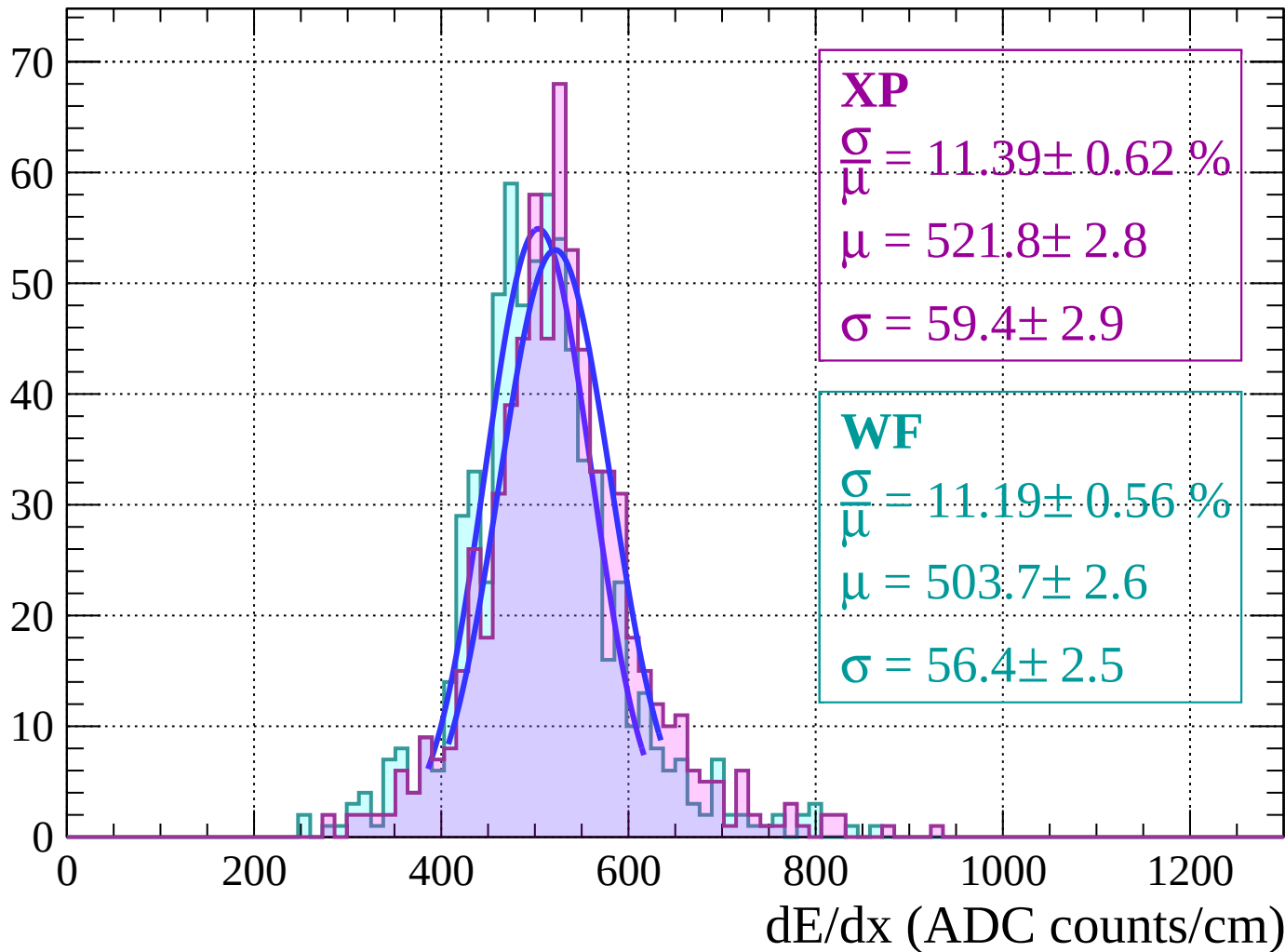
Energy loss $|-46 < \theta < -44$

Count



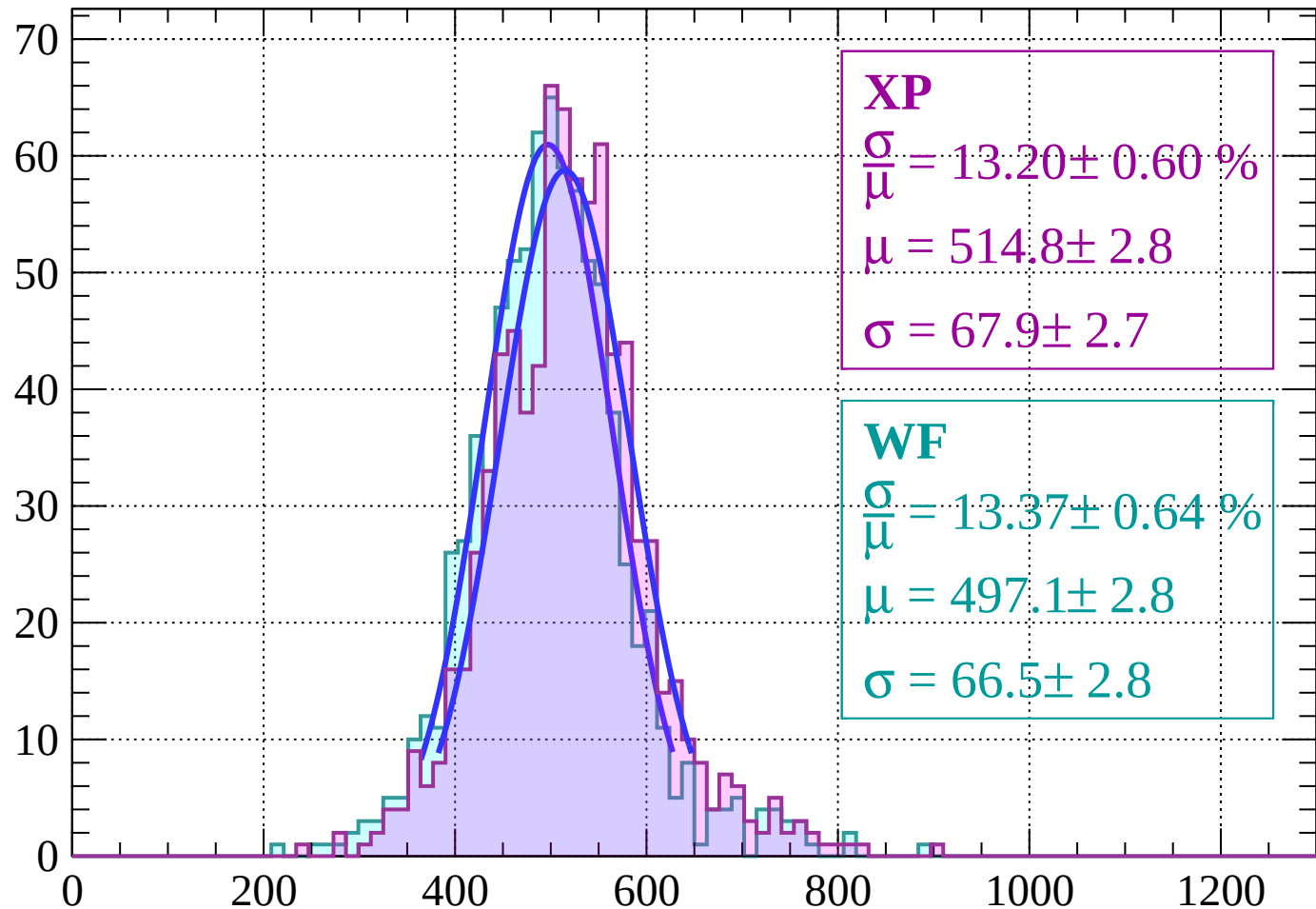
Energy loss | $-44 < \theta < -42$

Count



Energy loss | $-42 < \theta < -40$

Count



XP

$$\frac{\sigma}{\mu} = 13.20 \pm 0.60 \%$$

$$\mu = 514.8 \pm 2.8$$

$$\sigma = 67.9 \pm 2.7$$

WF

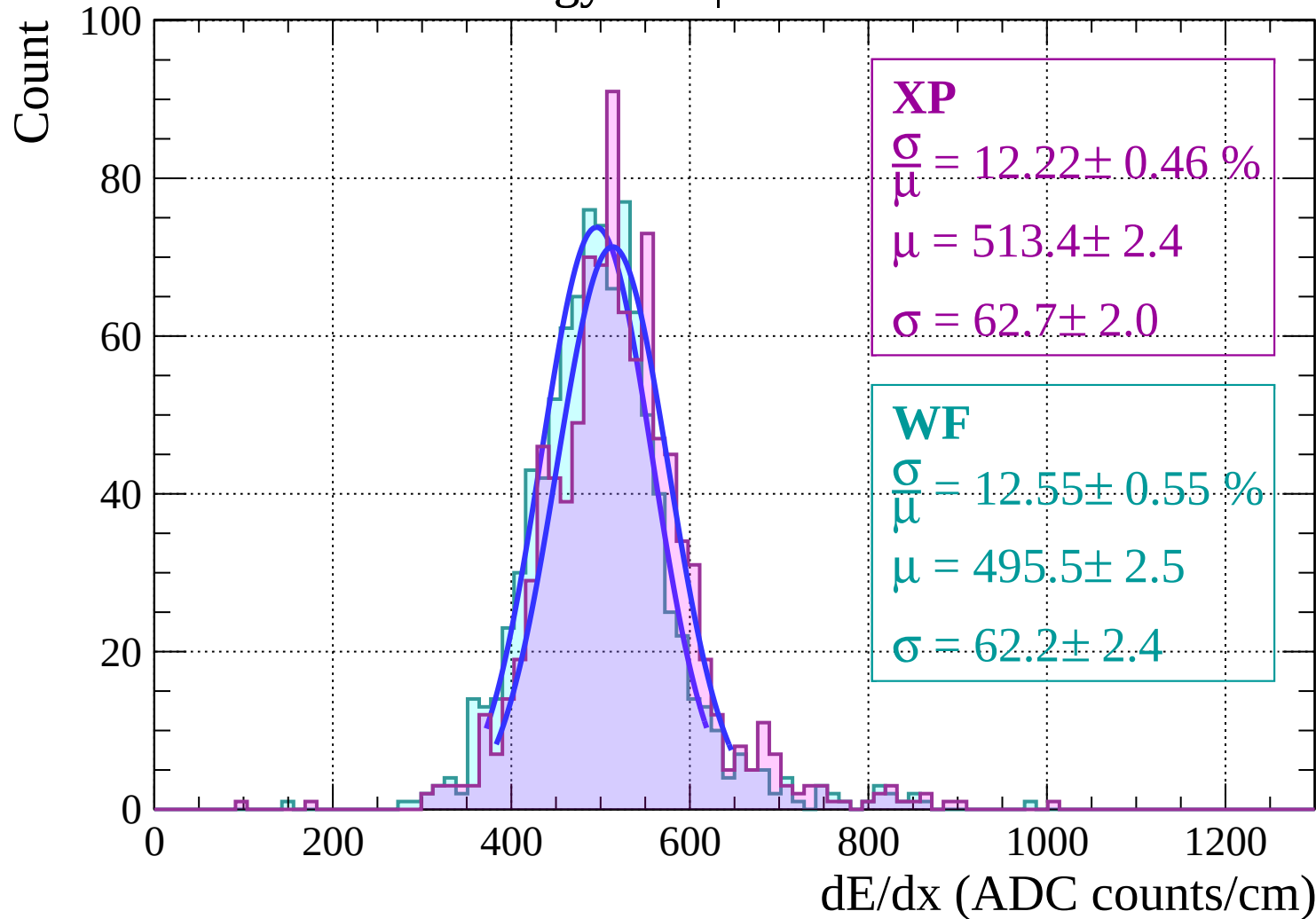
$$\frac{\sigma}{\mu} = 13.37 \pm 0.64 \%$$

$$\mu = 497.1 \pm 2.8$$

$$\sigma = 66.5 \pm 2.8$$

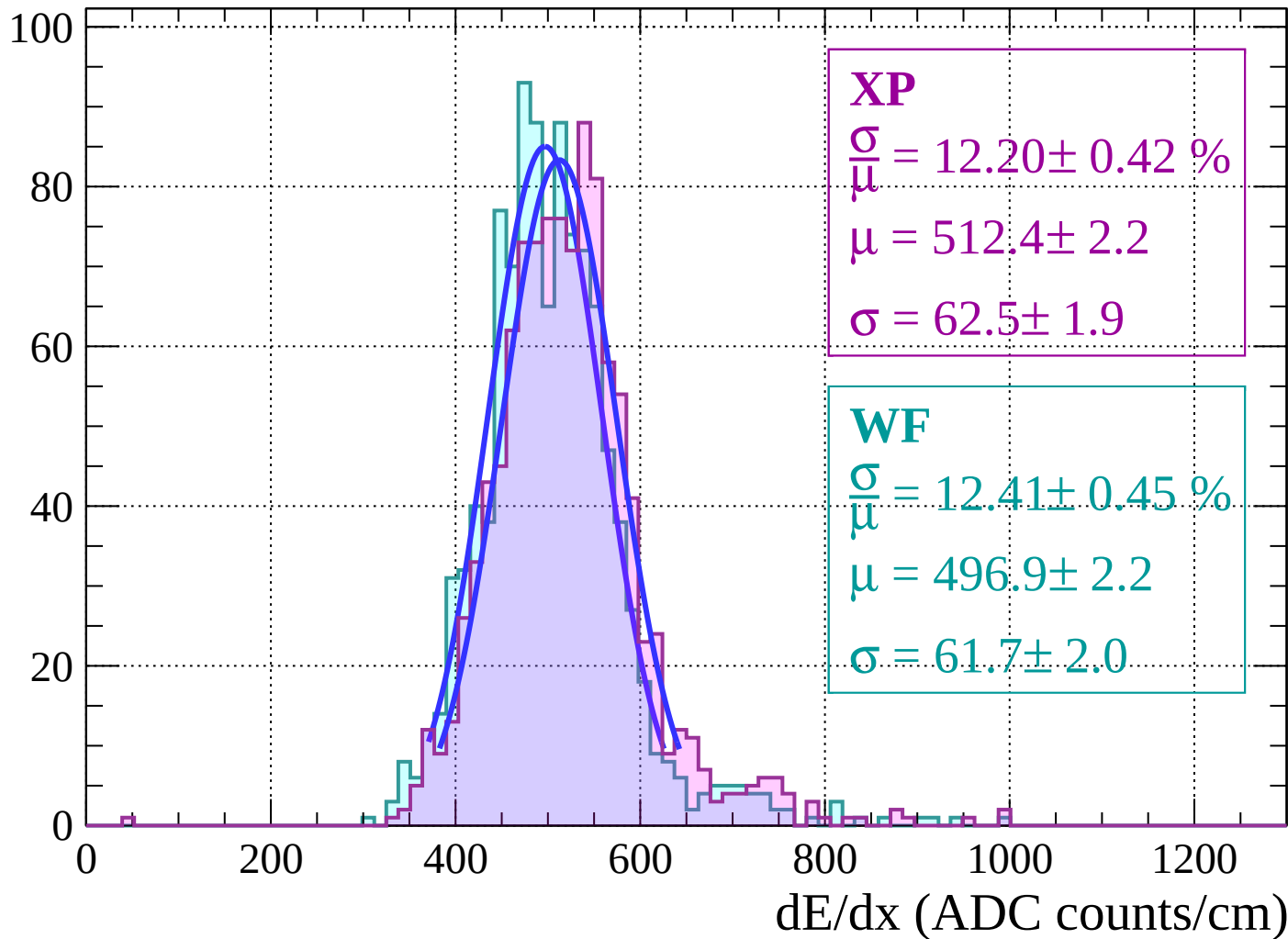
dE/dx (ADC counts/cm)

Energy loss | $-40 < \theta < -38$



Energy loss | $-38 < \theta < -36$

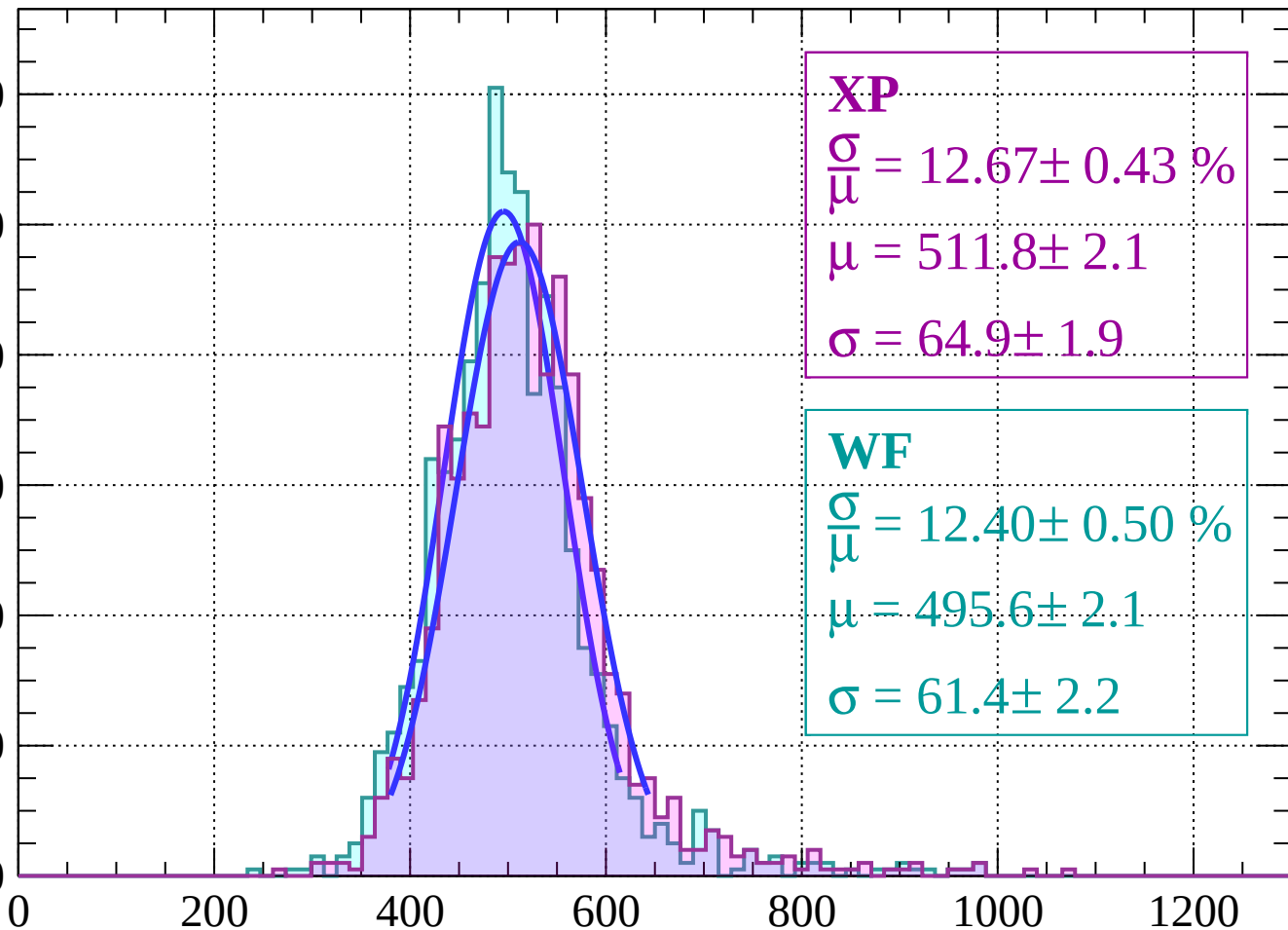
Count



Energy loss $|-36 < \theta < -34$

Count

120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 12.67 \pm 0.43 \%$$

$$\mu = 511.8 \pm 2.1$$

$$\sigma = 64.9 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 12.40 \pm 0.50 \%$$

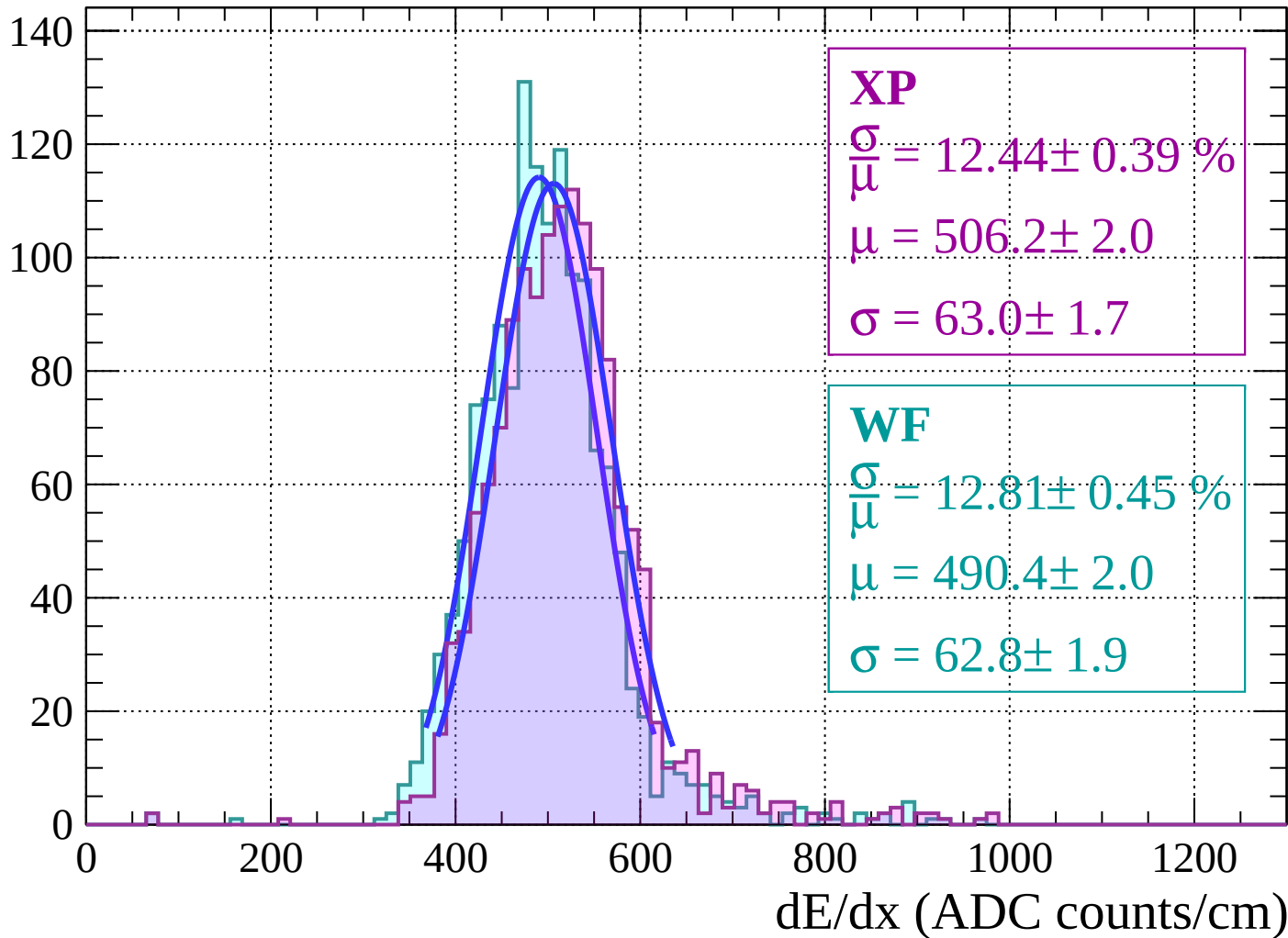
$$\mu = 495.6 \pm 2.1$$

$$\sigma = 61.4 \pm 2.2$$

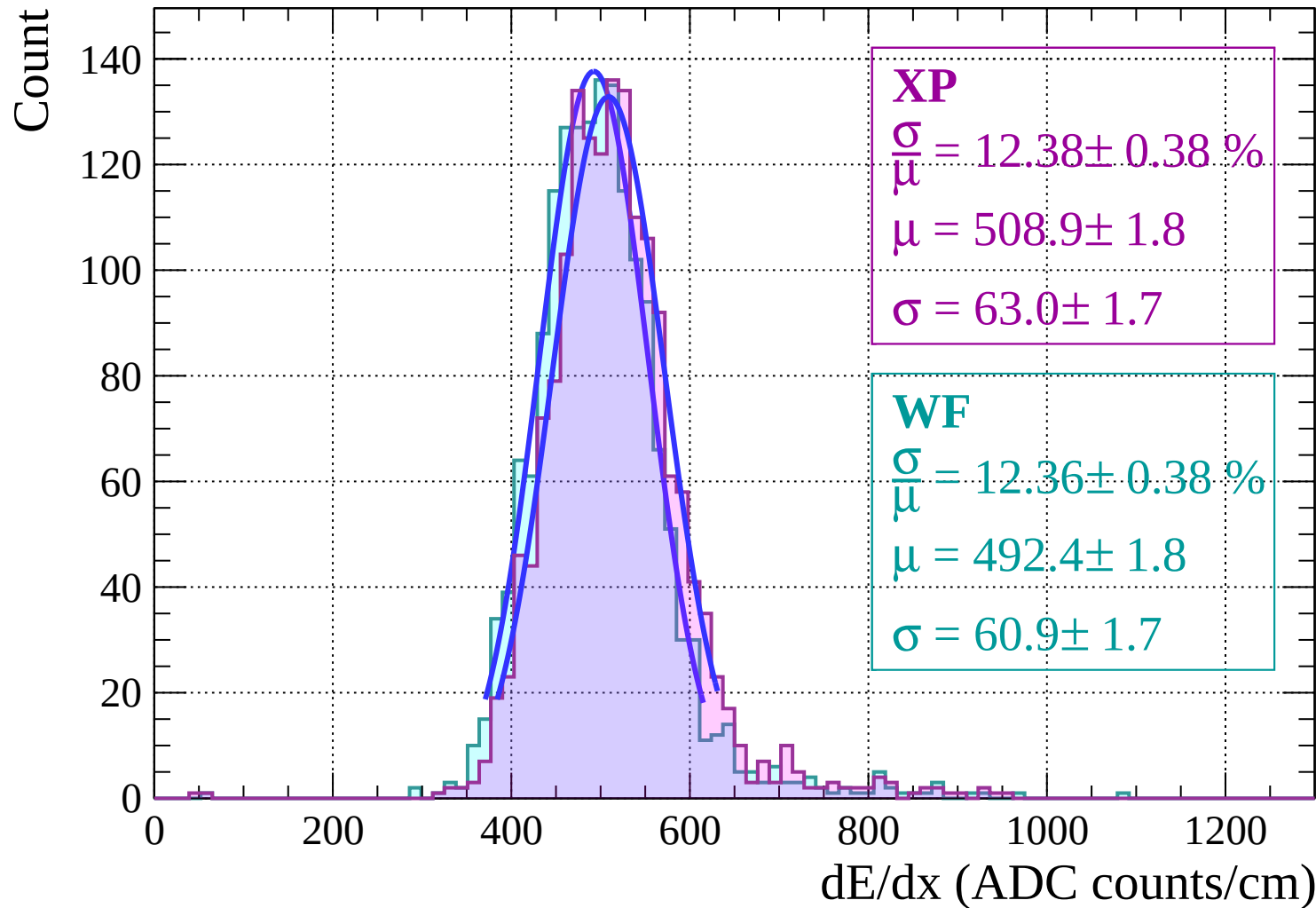
dE/dx (ADC counts/cm)

Energy loss | $-34 < \theta < -32$

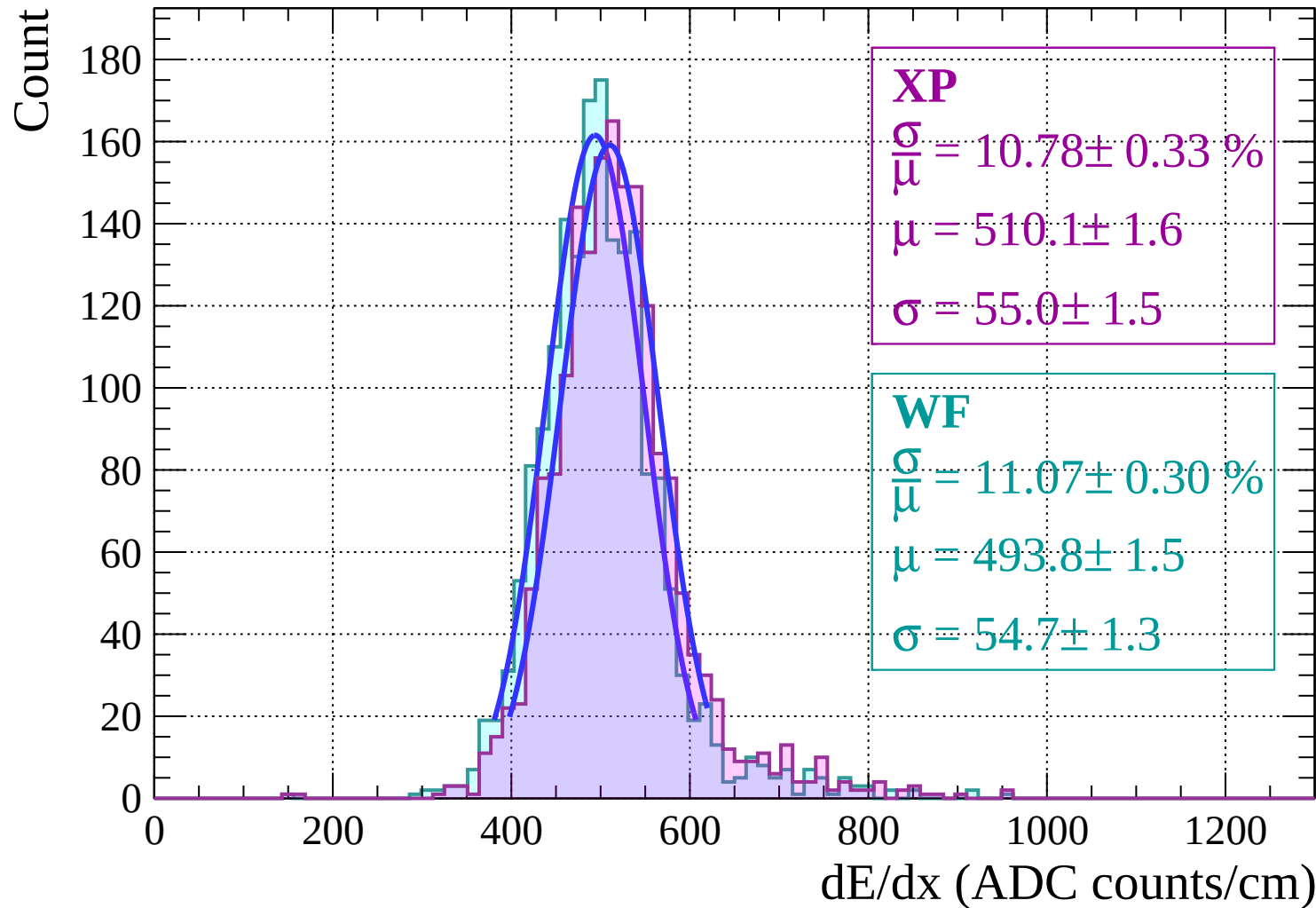
Count



Energy loss | $-32 < \theta < -30$

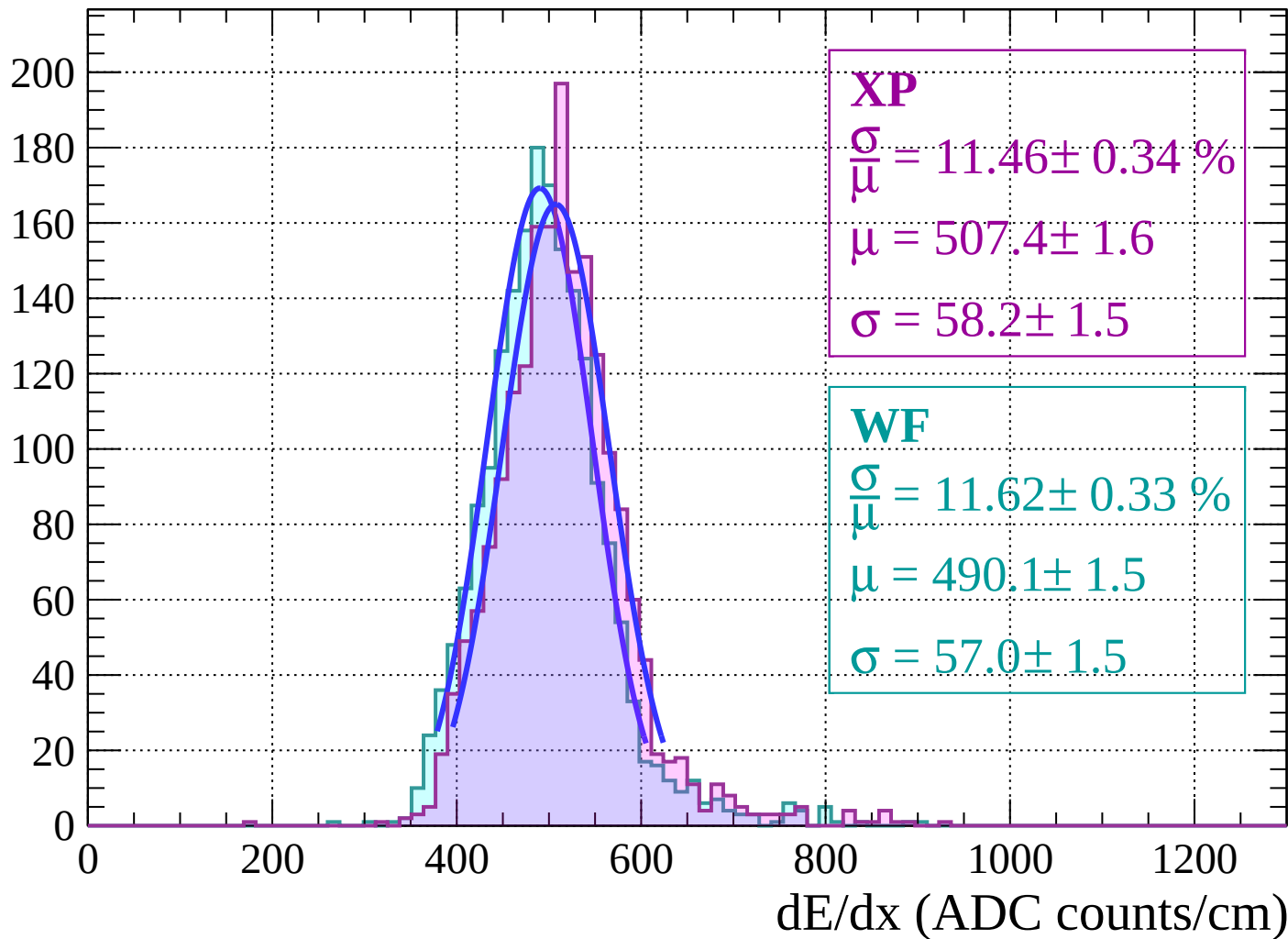


Energy loss | $-30 < \theta < -28$



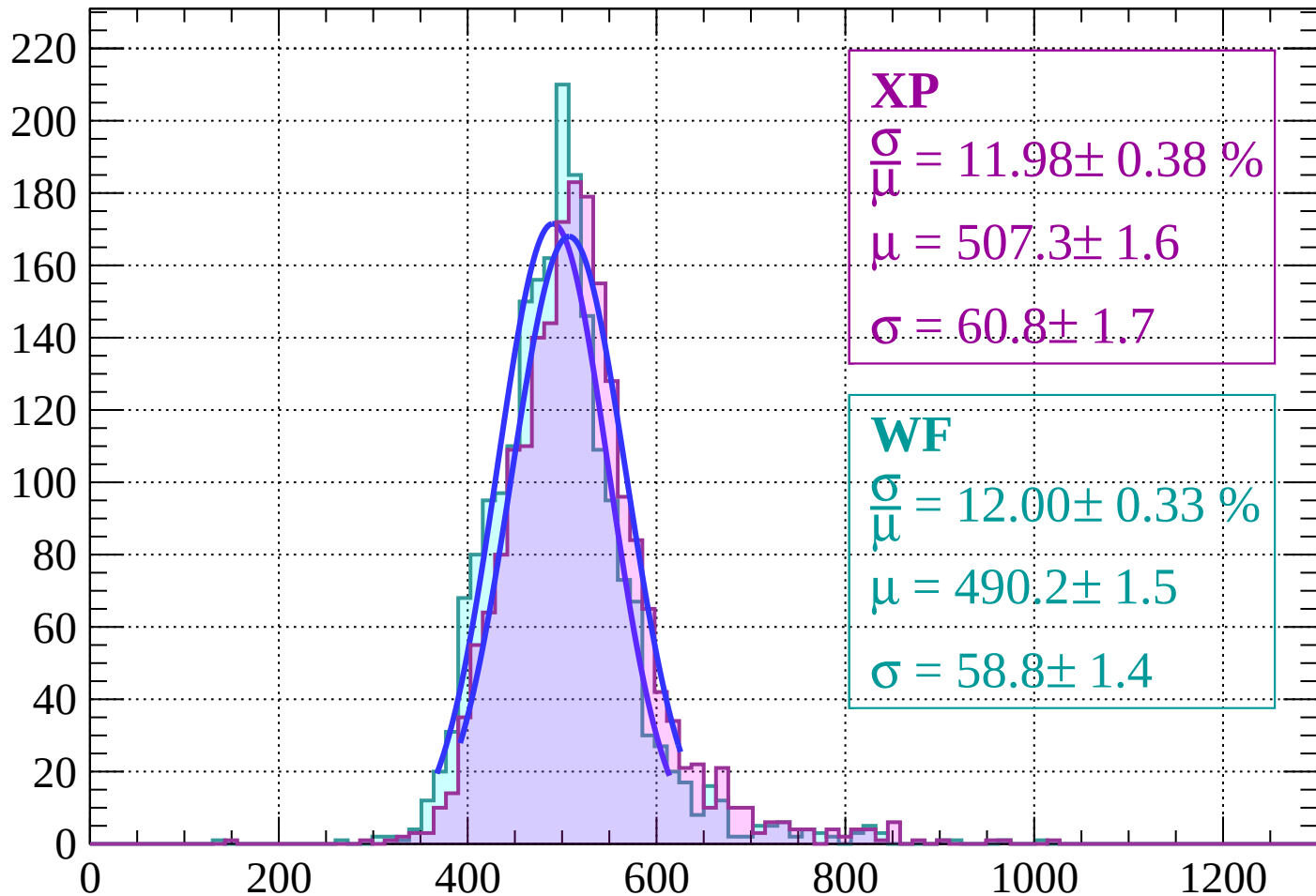
Energy loss | $-28 < \theta < -26$

Count



Energy loss $|-26 < \theta < -24$

Count



XP

$$\frac{\sigma}{\mu} = 11.98 \pm 0.38 \%$$

$$\mu = 507.3 \pm 1.6$$

$$\sigma = 60.8 \pm 1.7$$

WF

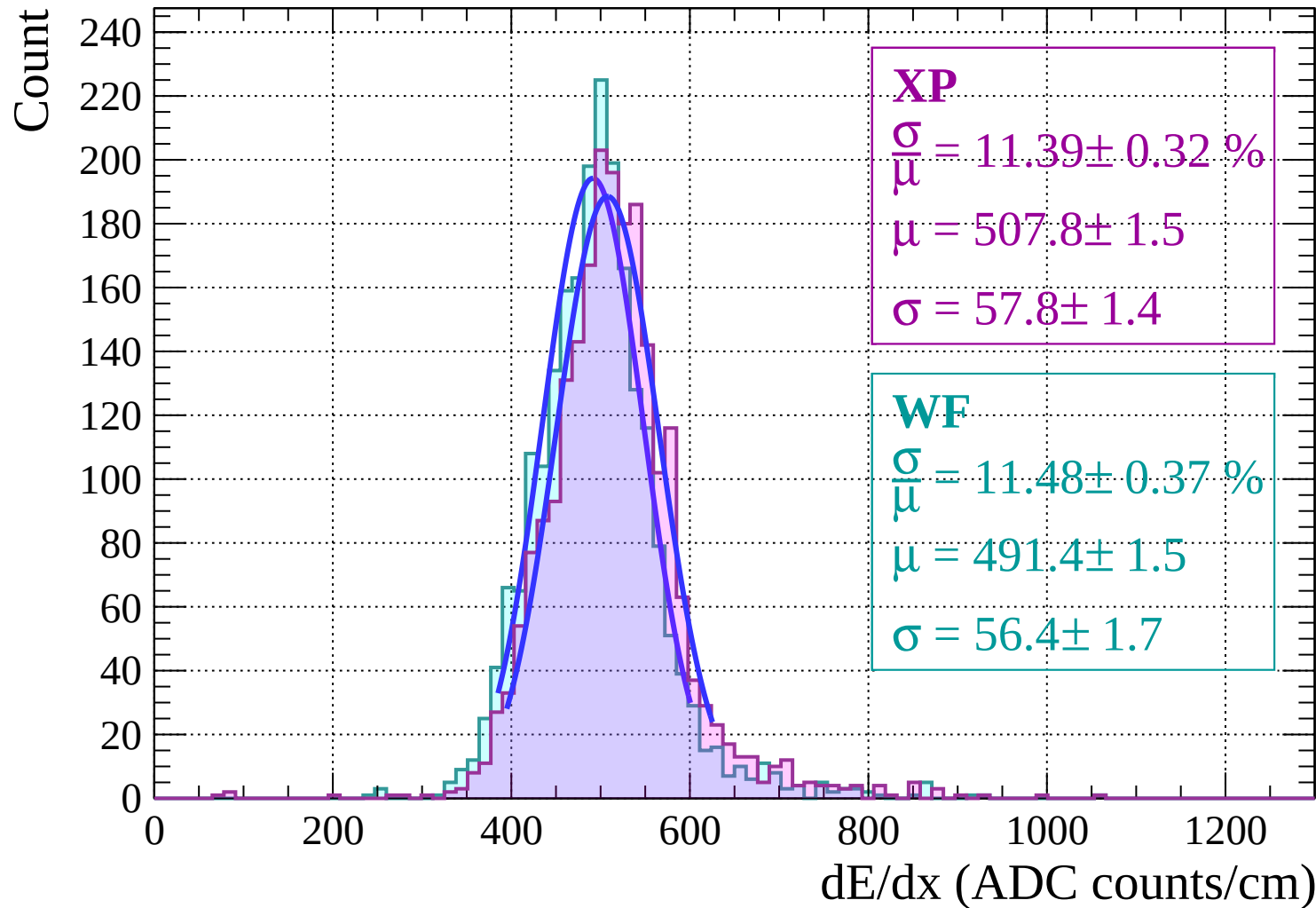
$$\frac{\sigma}{\mu} = 12.00 \pm 0.33 \%$$

$$\mu = 490.2 \pm 1.5$$

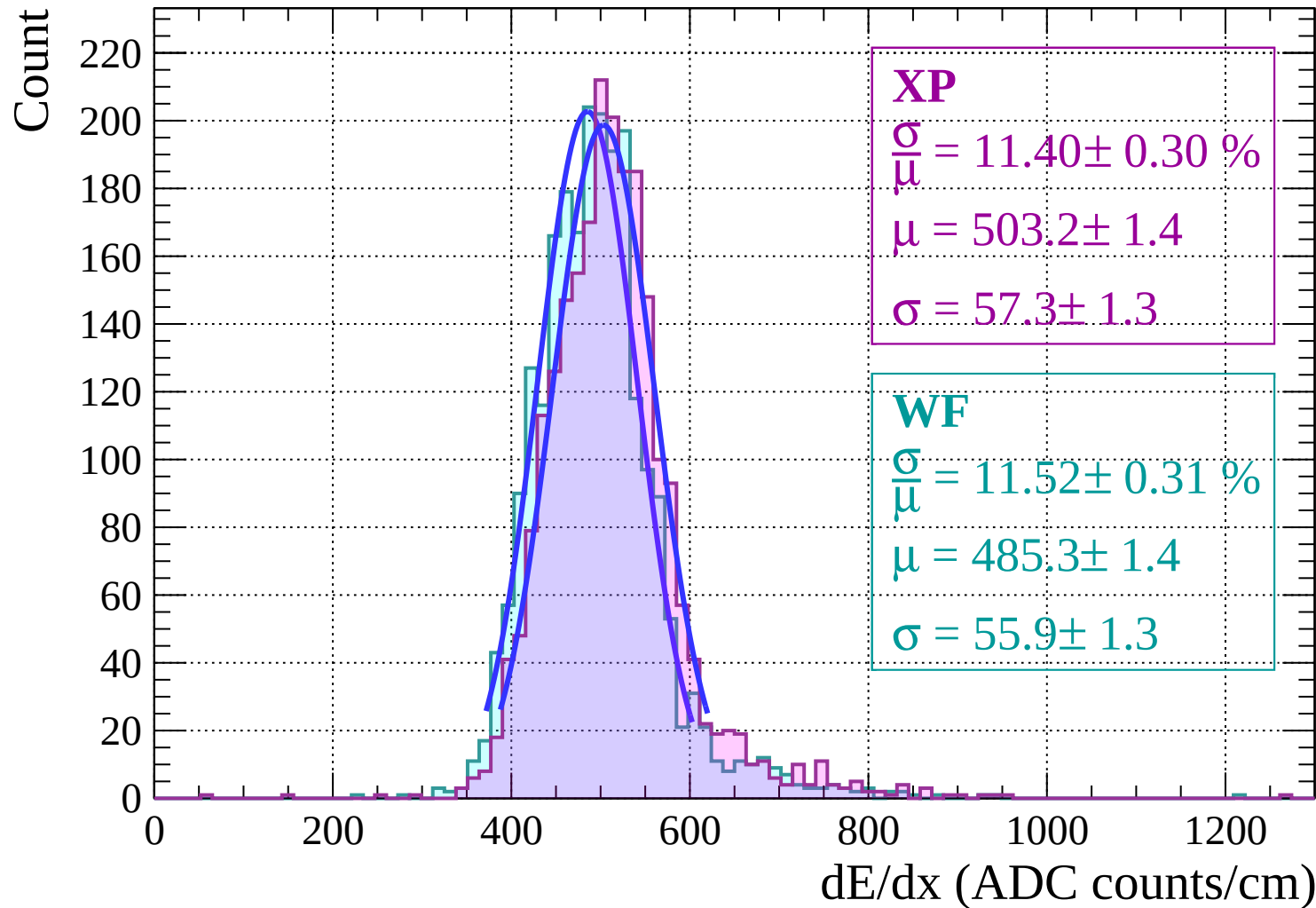
$$\sigma = 58.8 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-24 < \theta < -22$

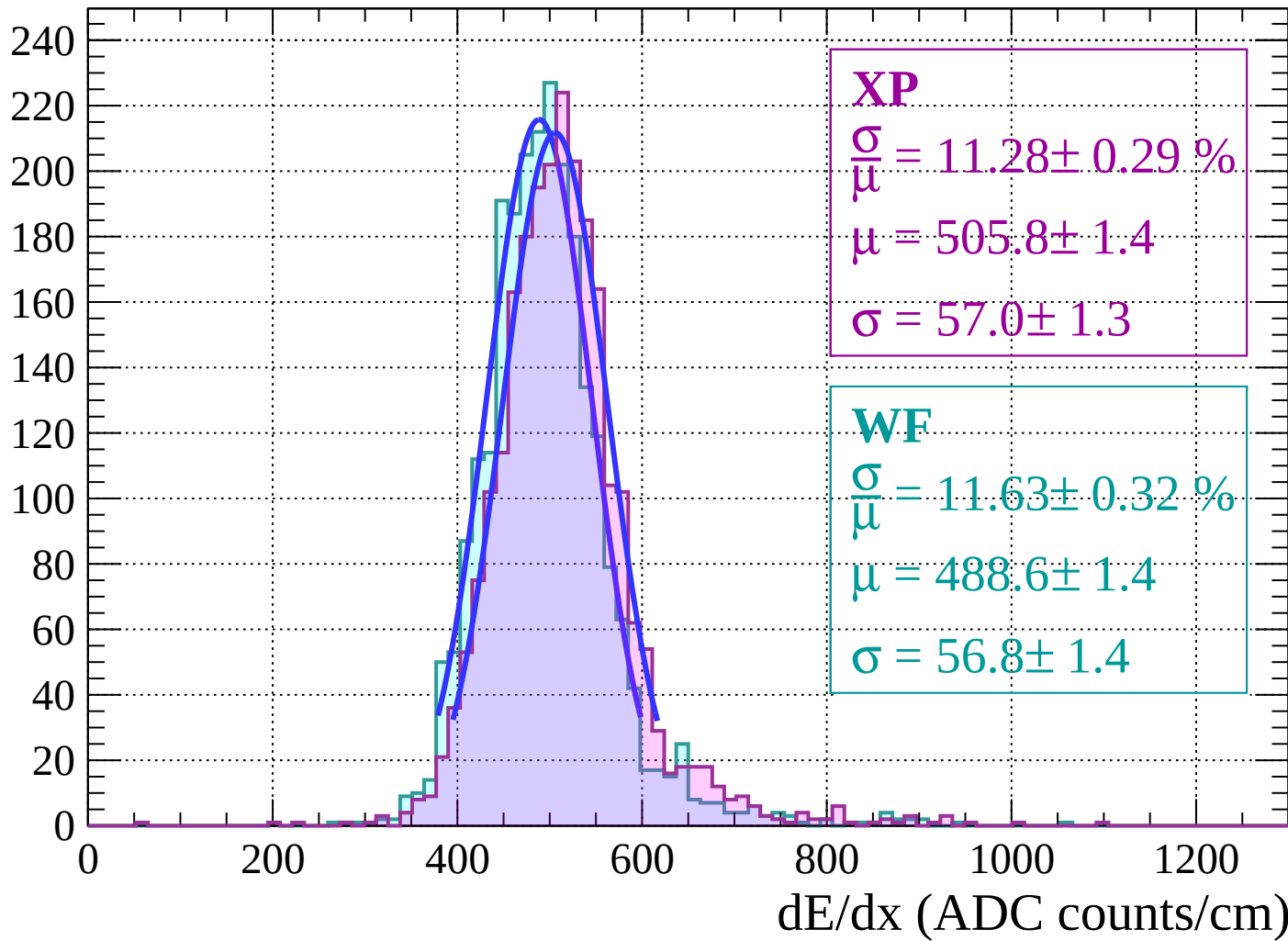


Energy loss | $-22 < \theta < -20$



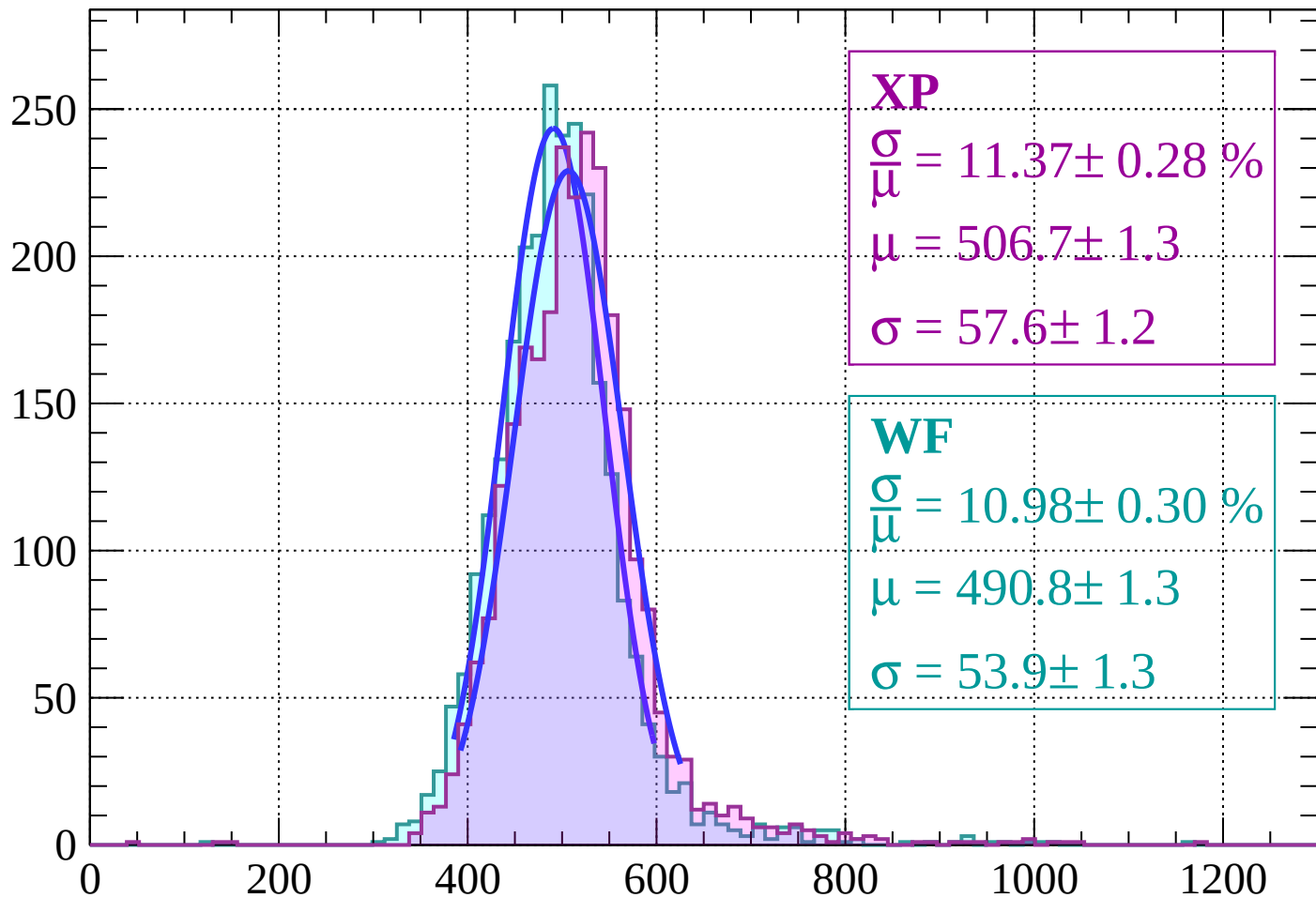
Energy loss | $-20 < \theta < -18$

Count



Energy loss $|-18 < \theta < -16$

Count



XP

$$\frac{\sigma}{\mu} = 11.37 \pm 0.28 \%$$

$$\mu = 506.7 \pm 1.3$$

$$\sigma = 57.6 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 10.98 \pm 0.30 \%$$

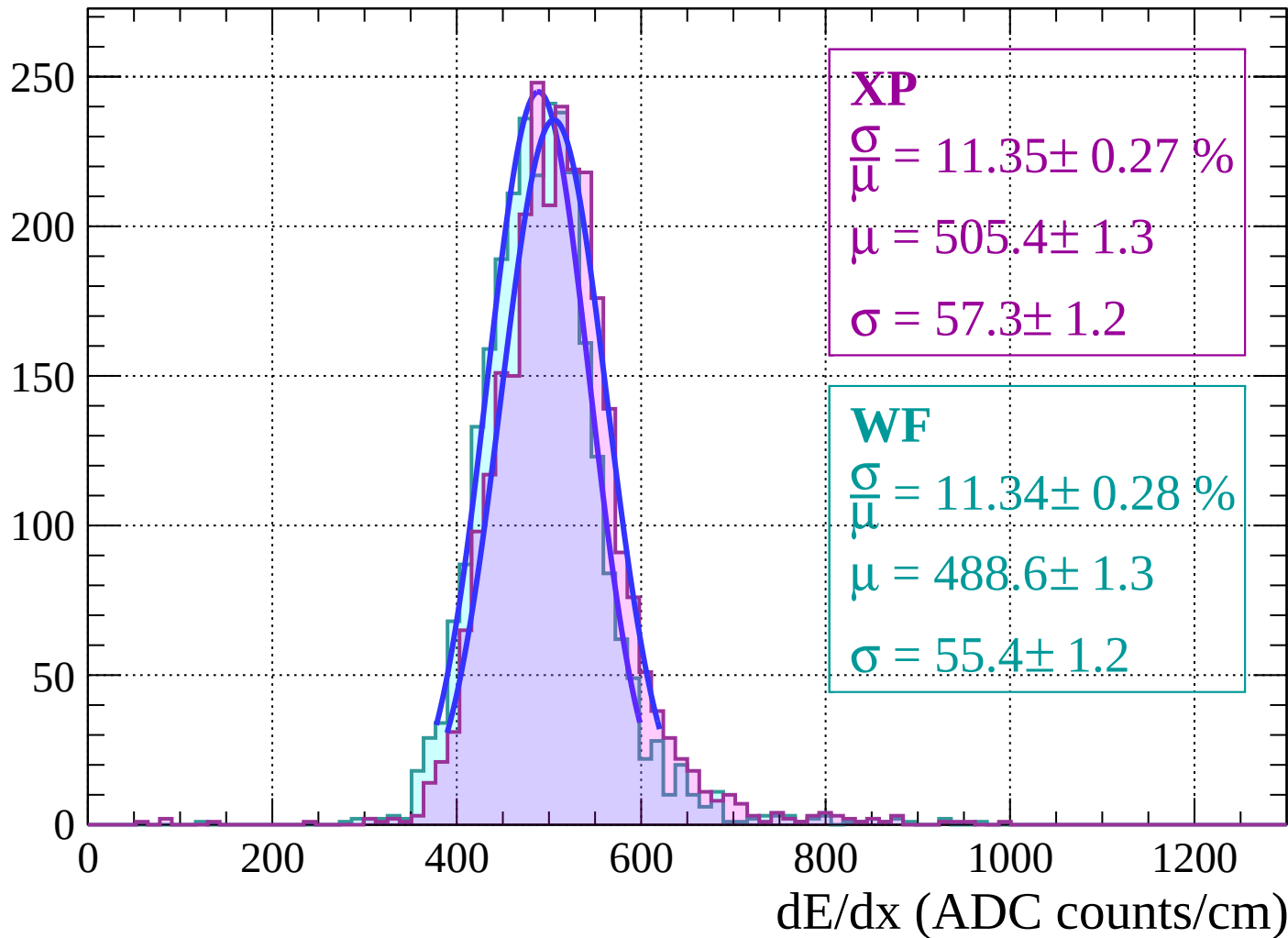
$$\mu = 490.8 \pm 1.3$$

$$\sigma = 53.9 \pm 1.3$$

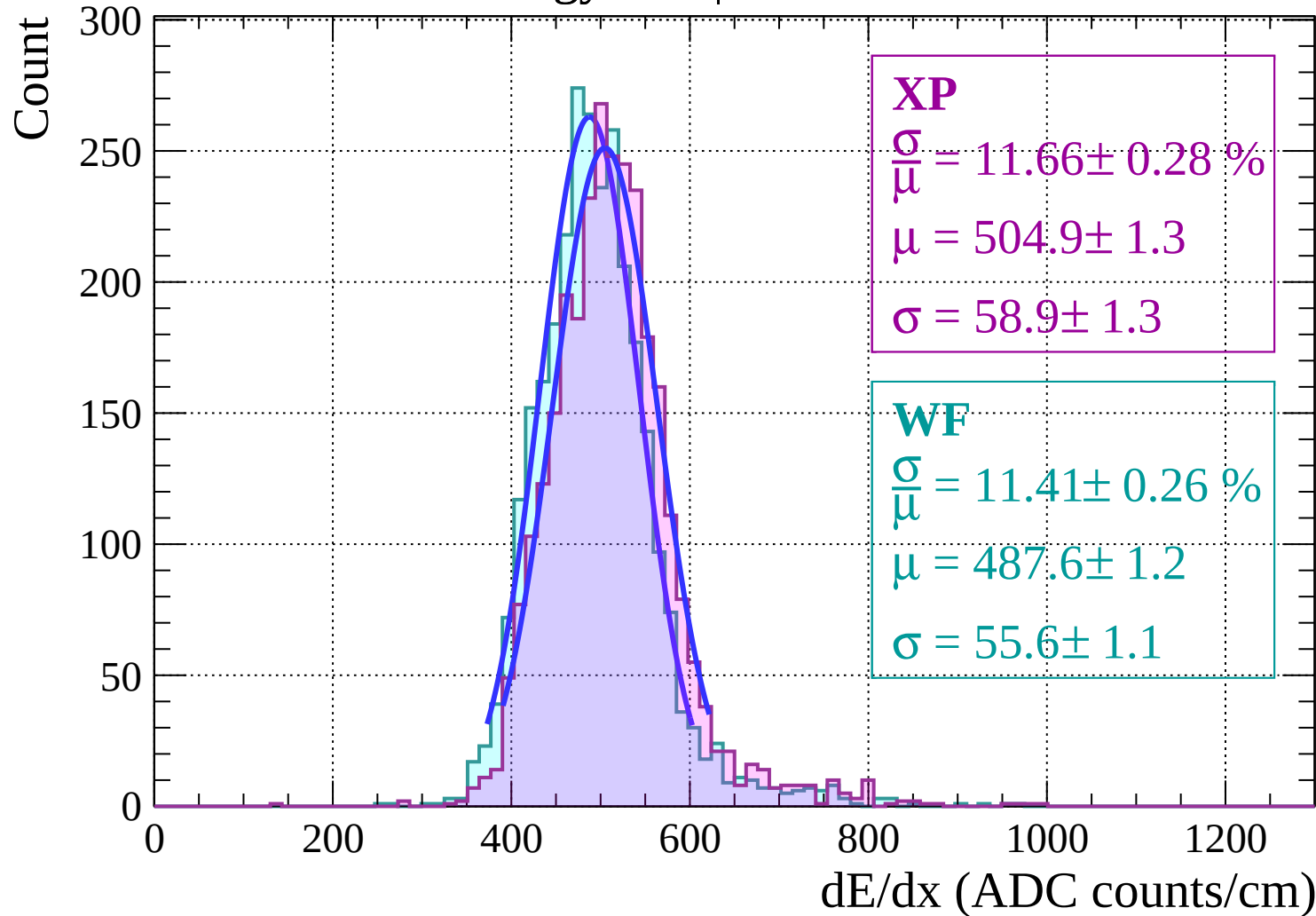
dE/dx (ADC counts/cm)

Energy loss $|-16 < \theta < -14$

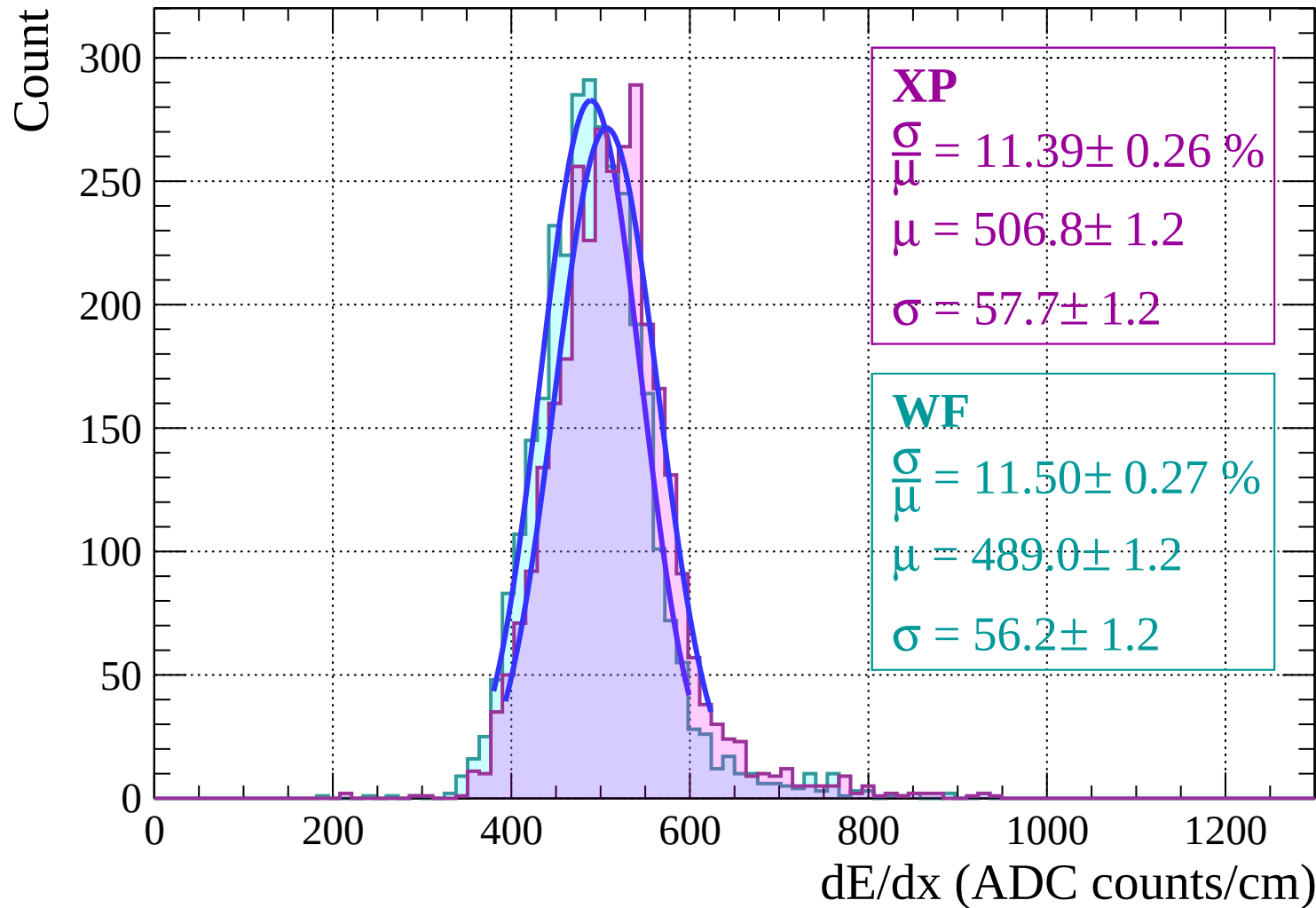
Count



Energy loss $|-14 < \theta < -12$



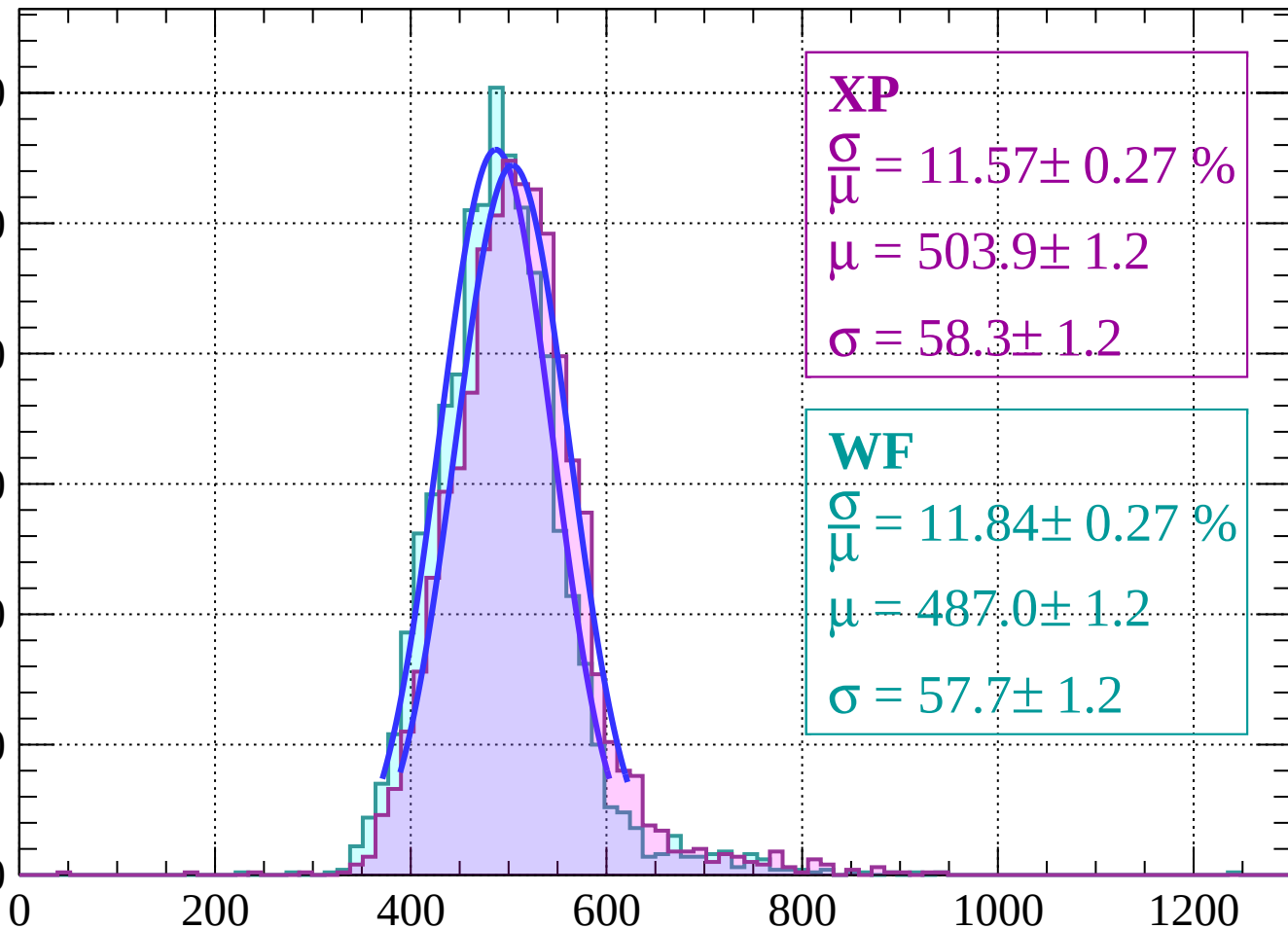
Energy loss | $-12 < \theta < -10$



Energy loss | $-10 < \theta < -8$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 11.57 \pm 0.27 \%$$

$$\mu = 503.9 \pm 1.2$$

$$\sigma = 58.3 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 11.84 \pm 0.27 \%$$

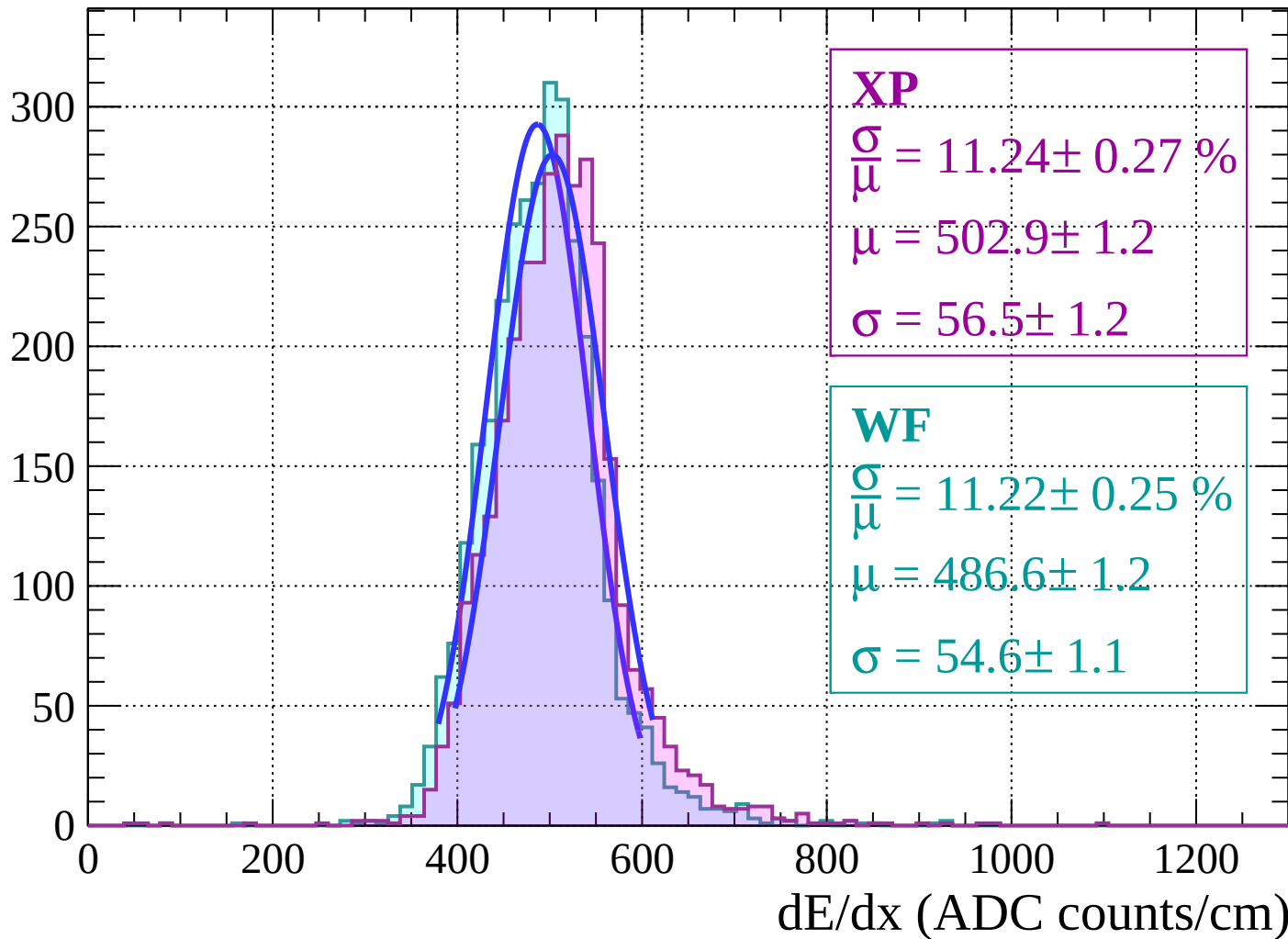
$$\mu = 487.0 \pm 1.2$$

$$\sigma = 57.7 \pm 1.2$$

dE/dx (ADC counts/cm)

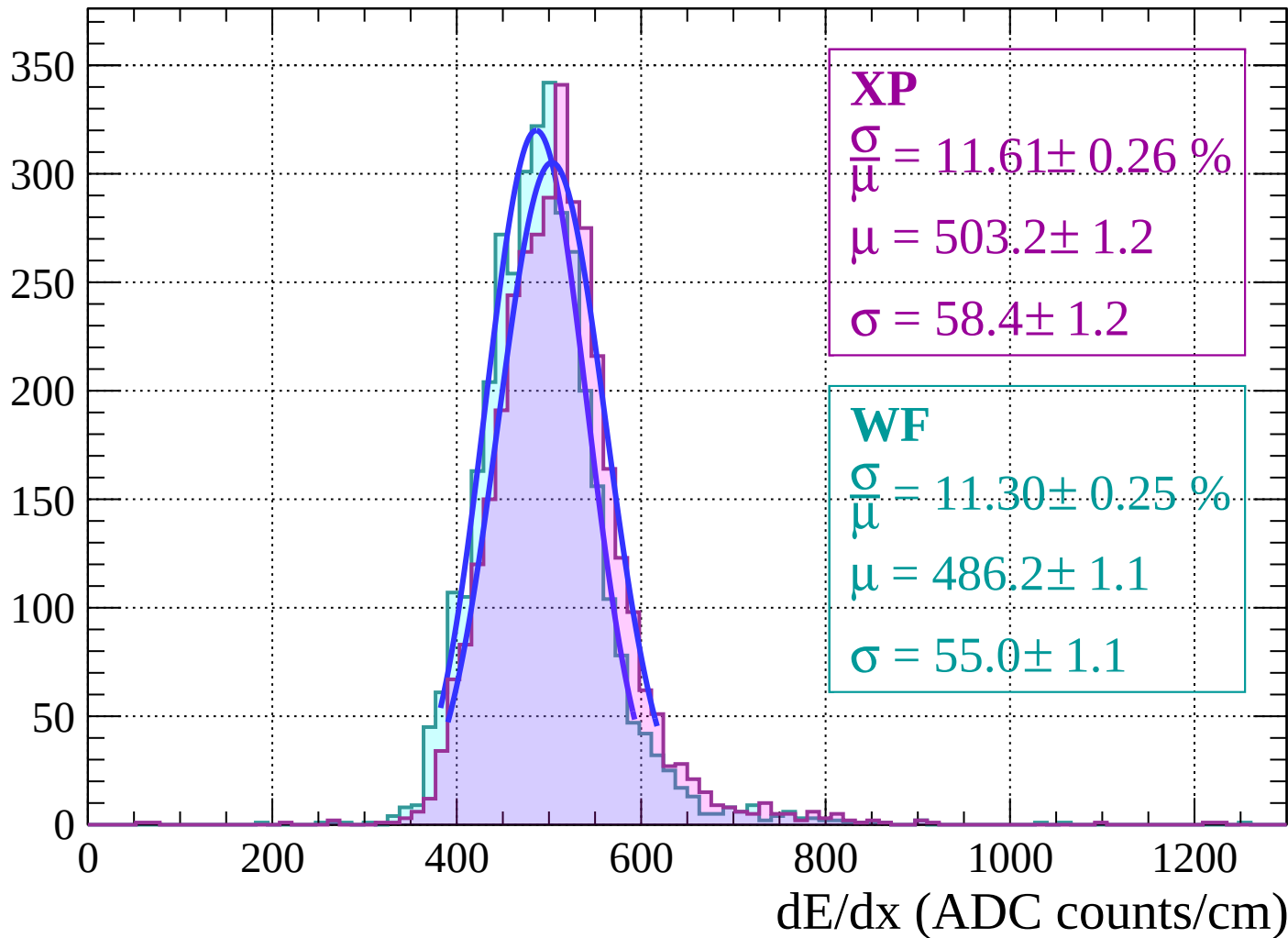
Energy loss $-8 < \theta < -6$

Count



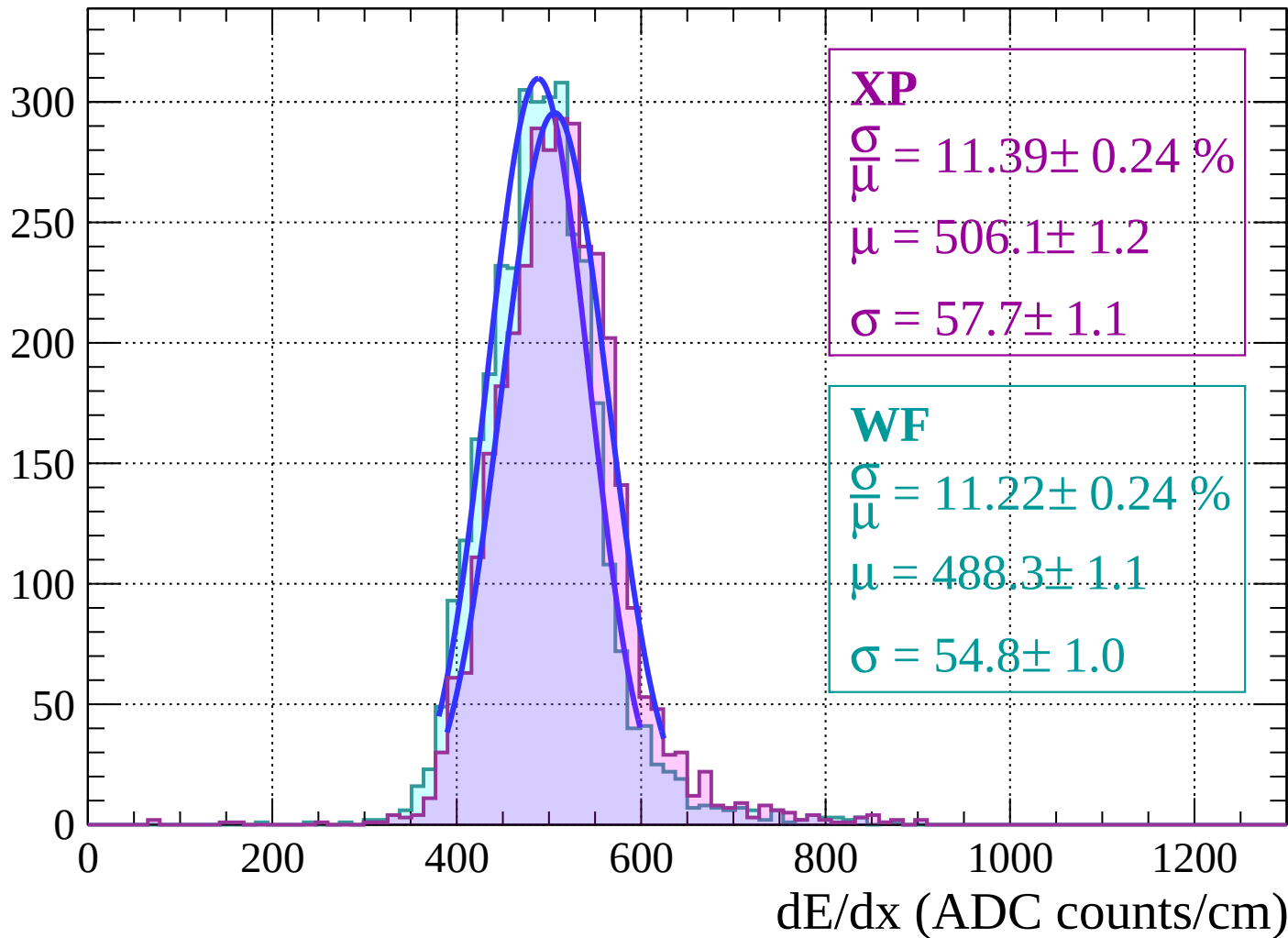
Energy loss | $-6 < \theta < -4$

Count



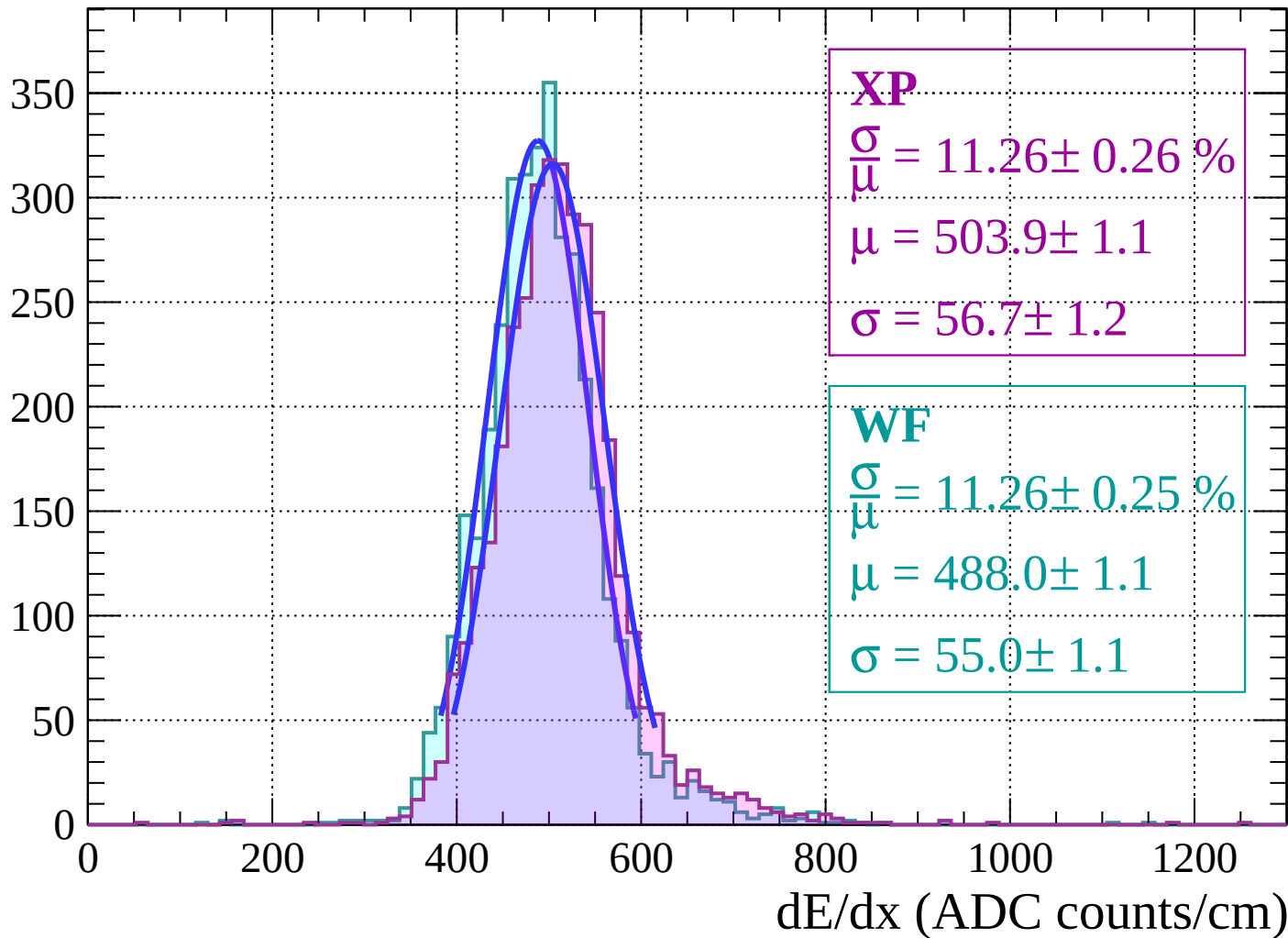
Energy loss $|-4 < \theta < -2|$

Count

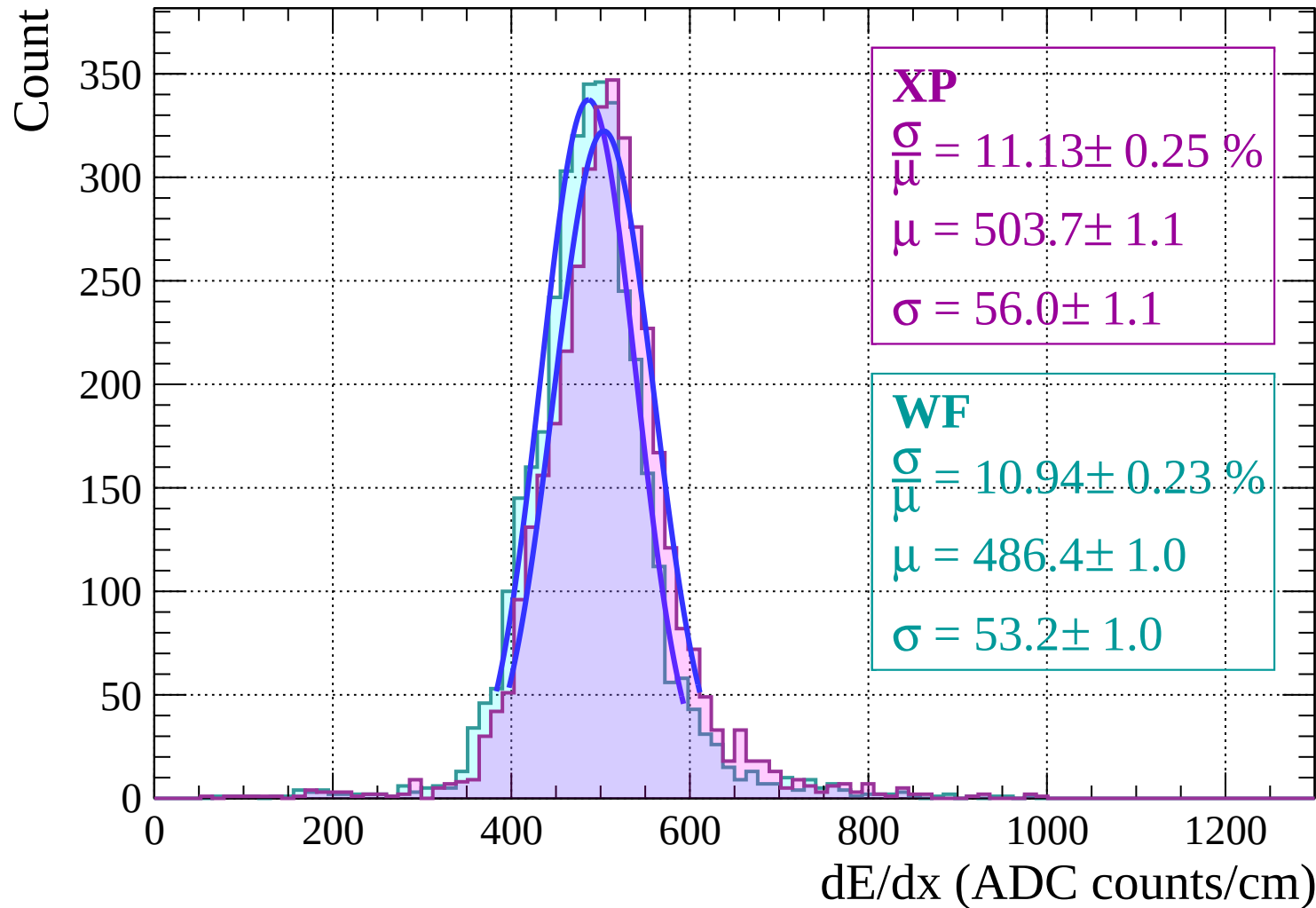


Energy loss | $-2 < \theta < 0$

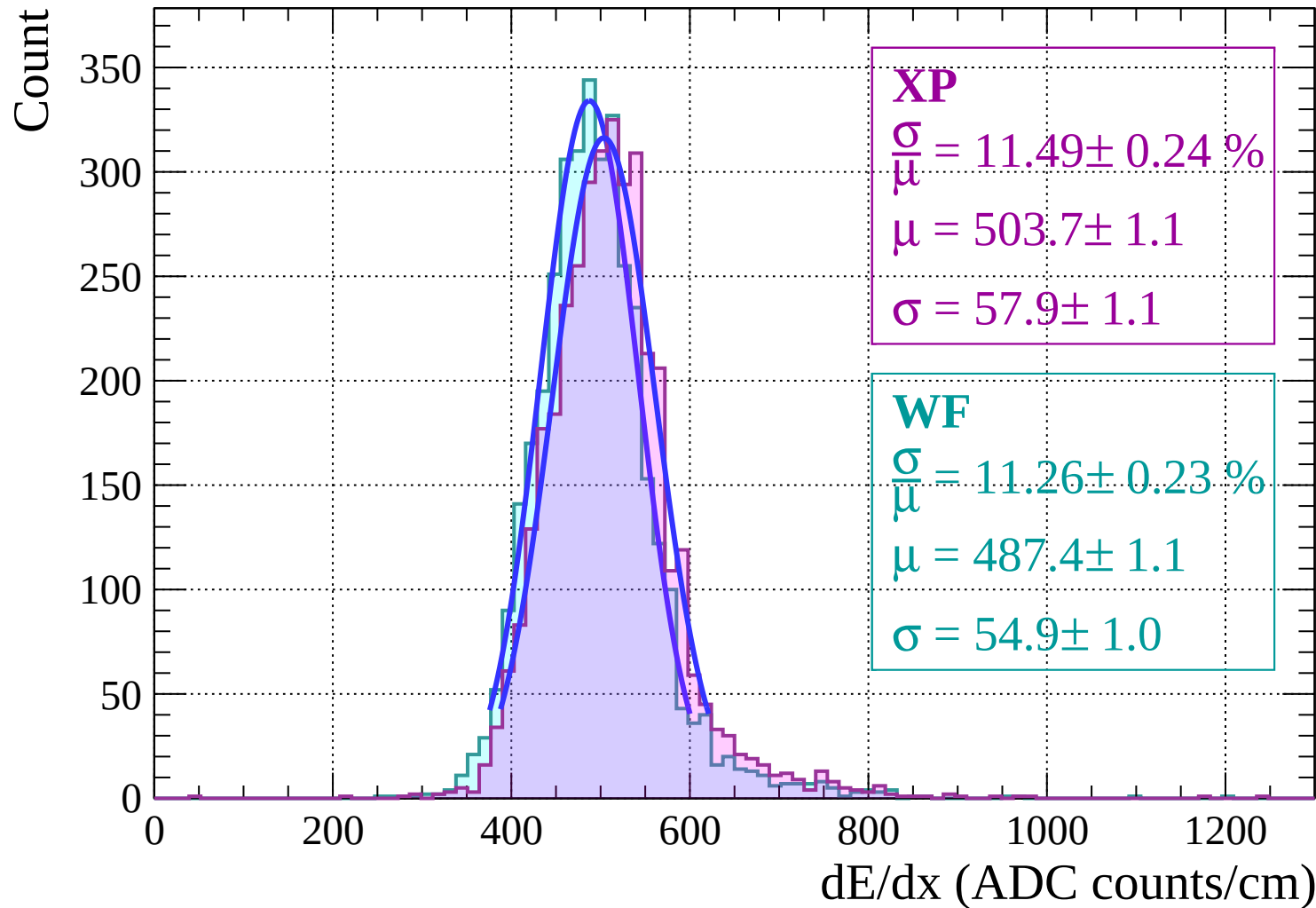
Count



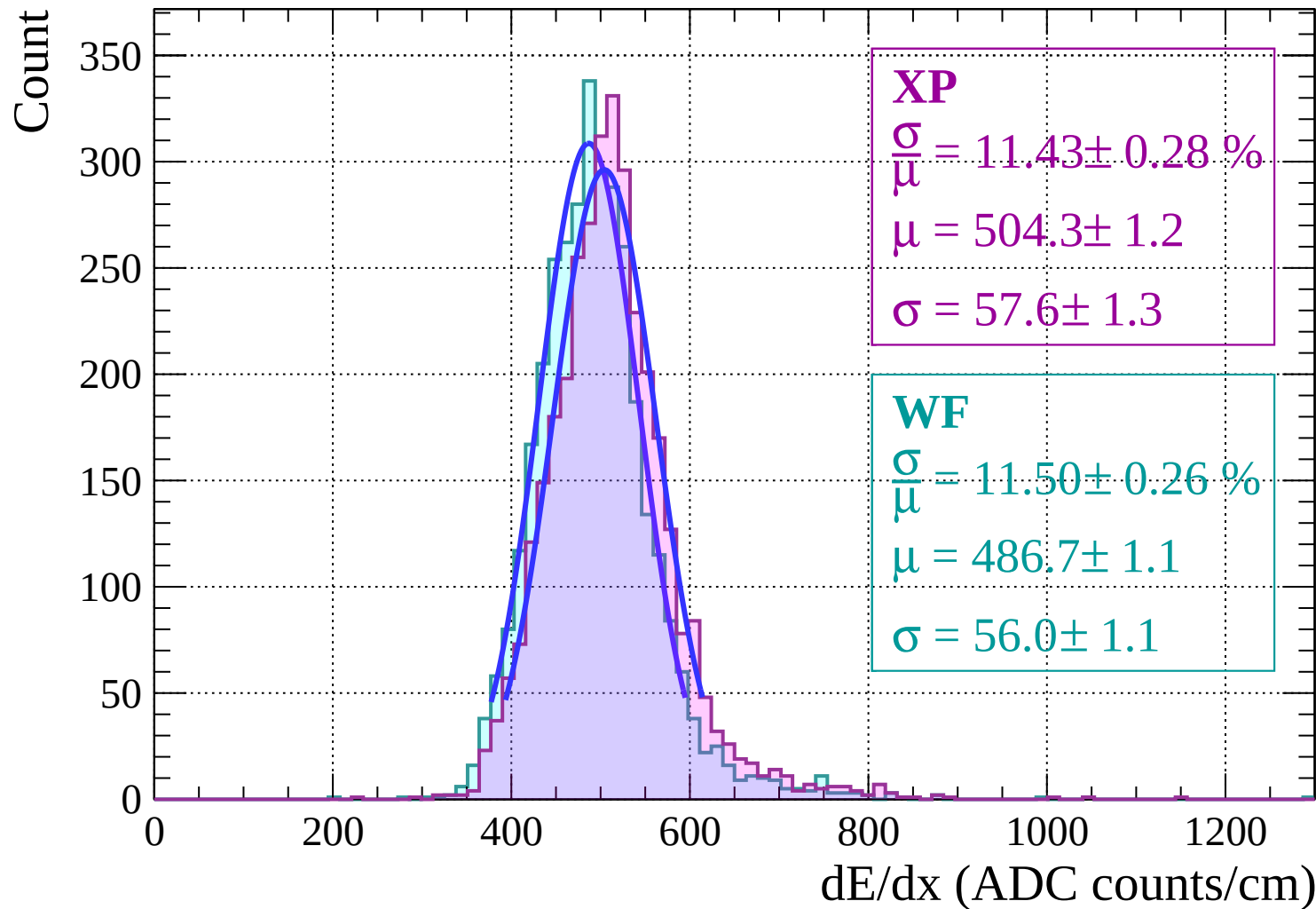
Energy loss | $0 < \theta < 2$



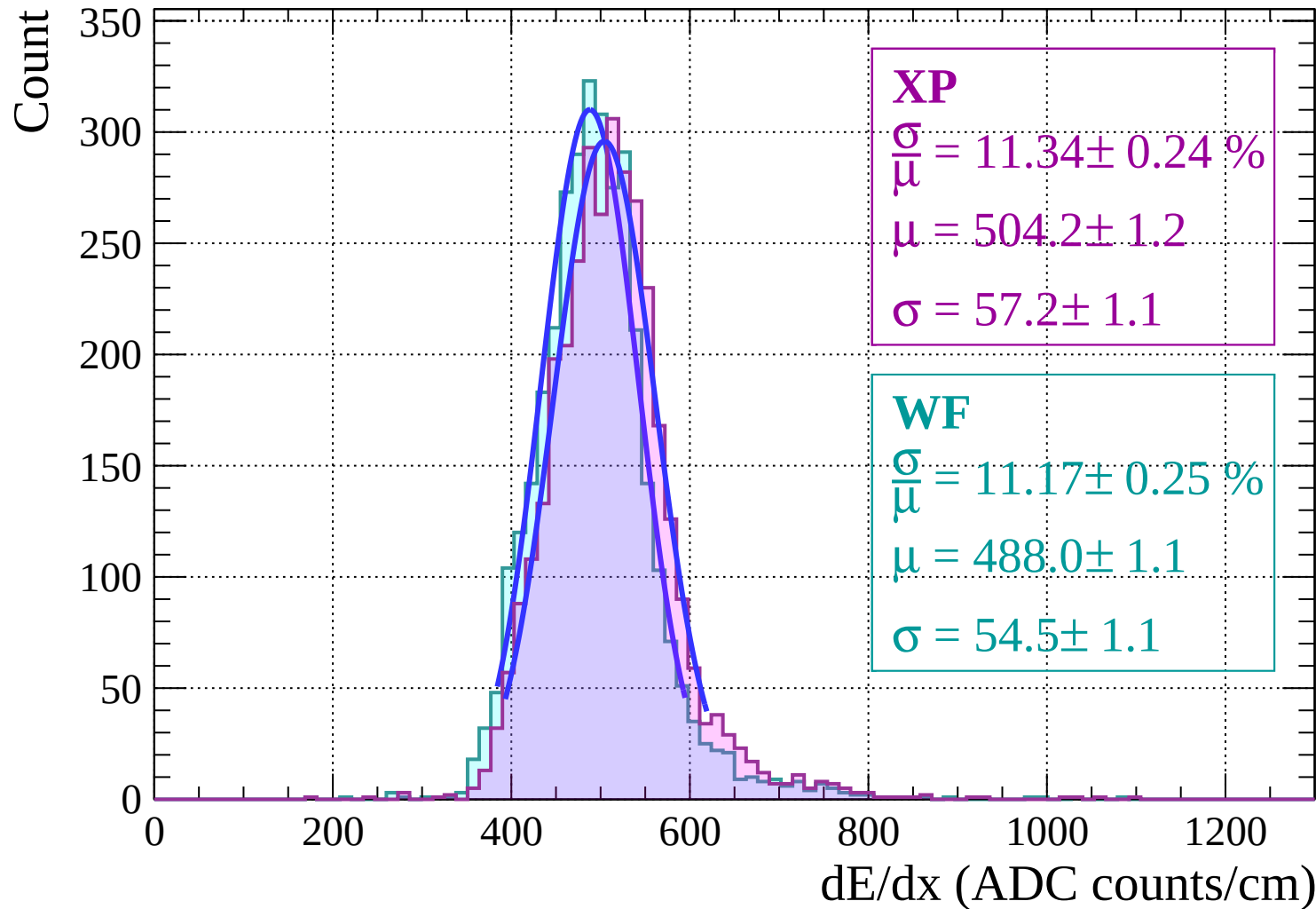
Energy loss | $2 < \theta < 4$



Energy loss | $4 < \theta < 6$



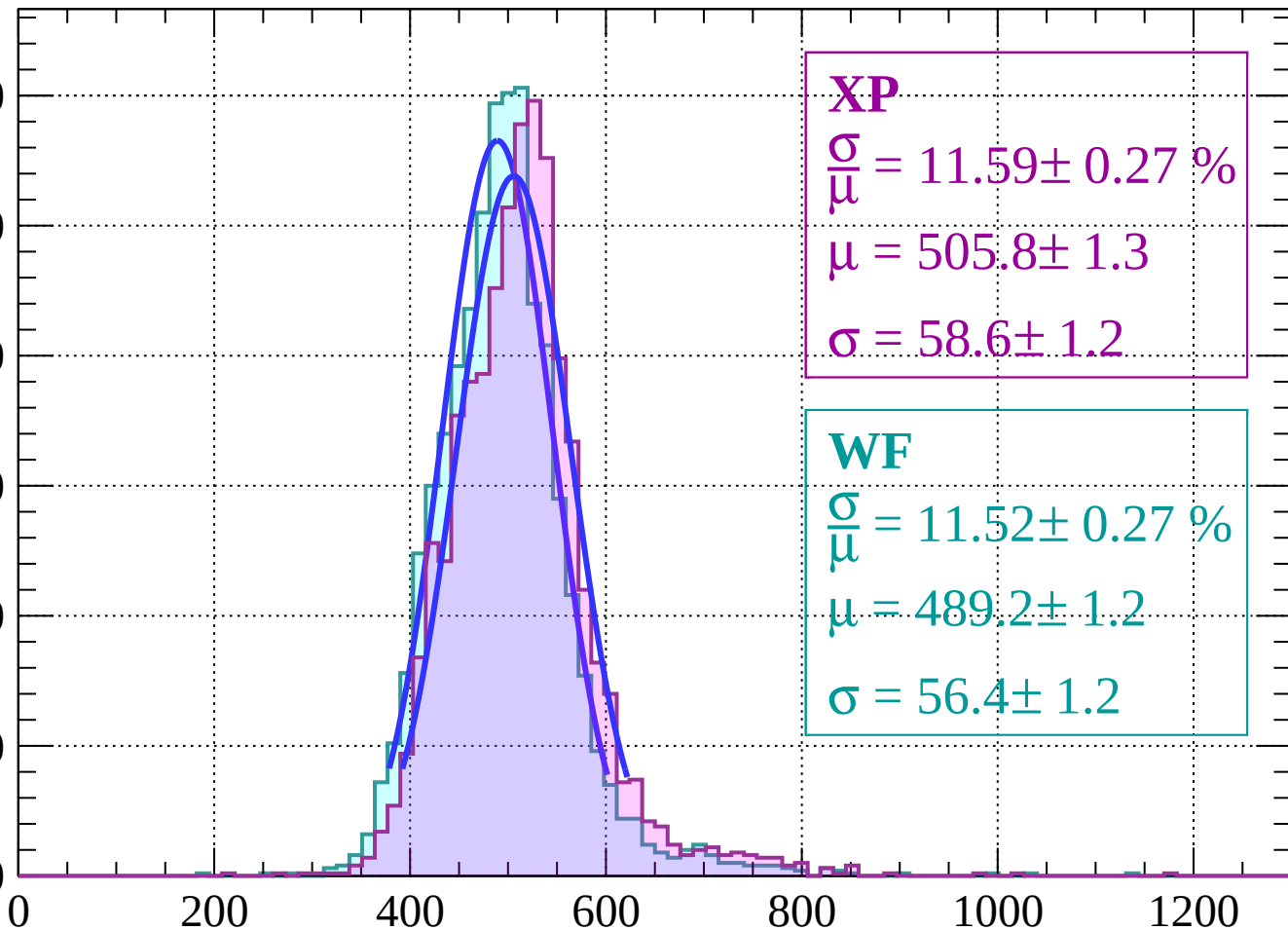
Energy loss | $6 < \theta < 8$



Energy loss | $8 < \theta < 10$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 11.59 \pm 0.27 \%$$

$$\mu = 505.8 \pm 1.3$$

$$\sigma = 58.6 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 11.52 \pm 0.27 \%$$

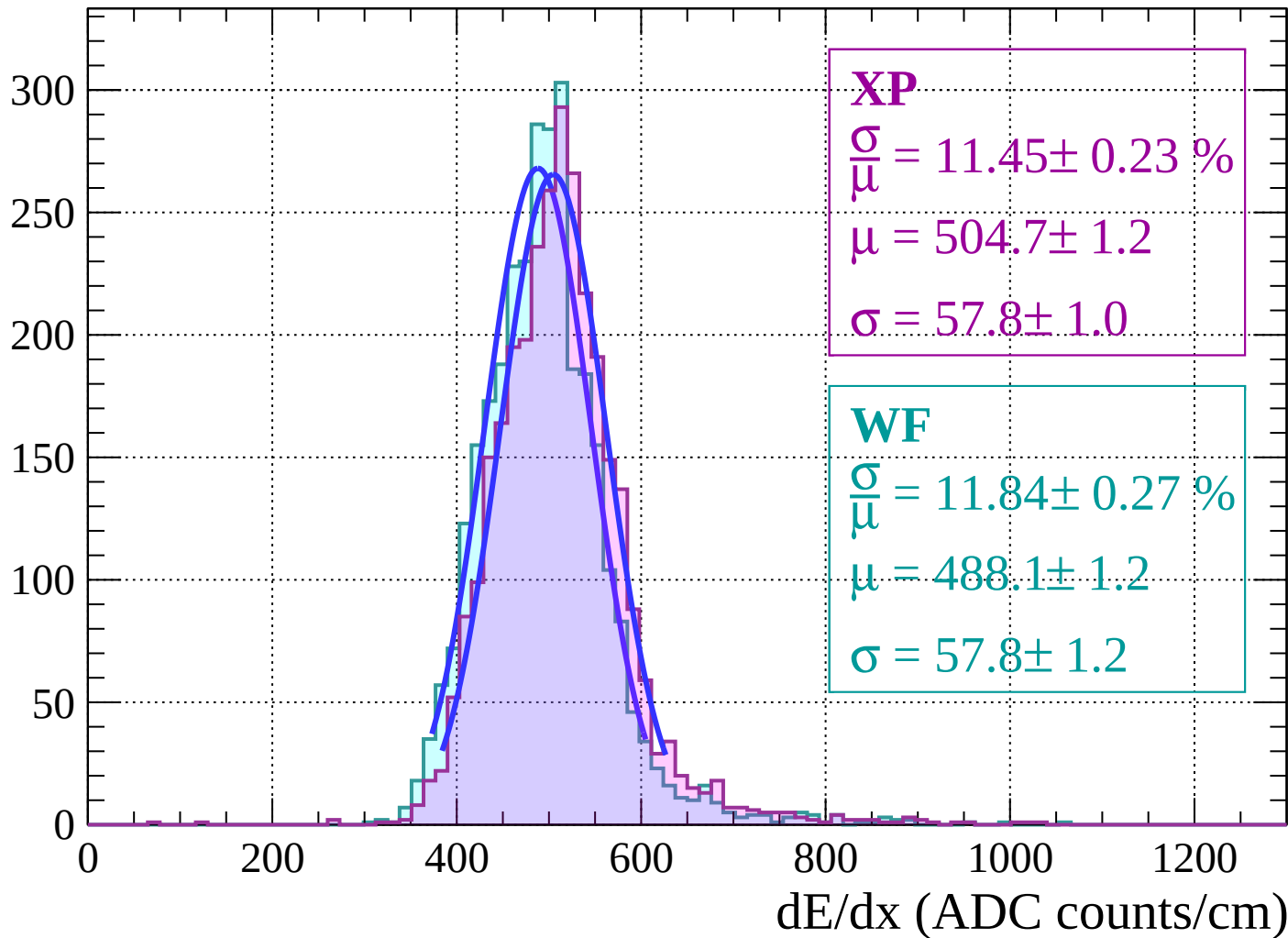
$$\mu = 489.2 \pm 1.2$$

$$\sigma = 56.4 \pm 1.2$$

dE/dx (ADC counts/cm)

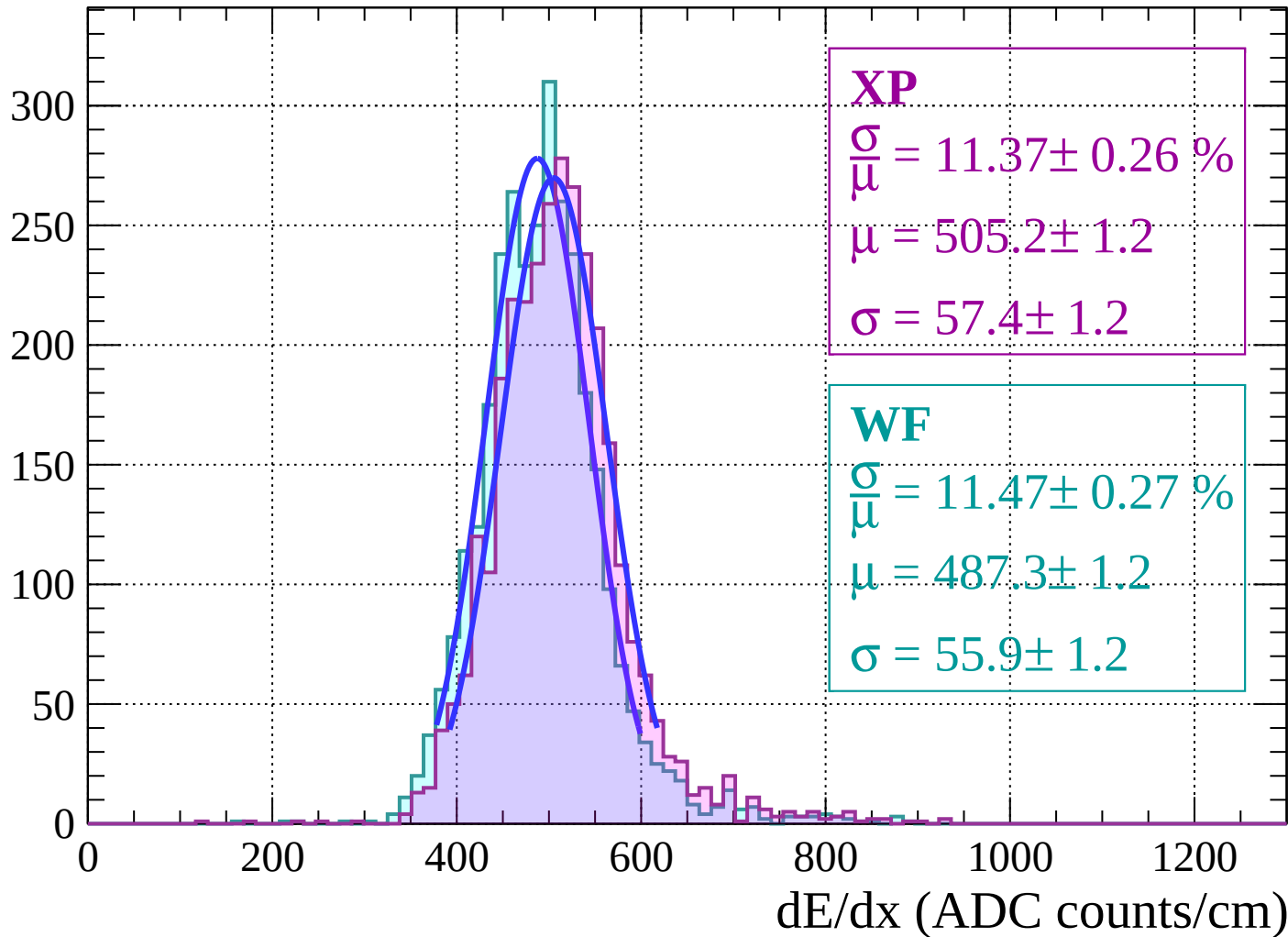
Energy loss | $10 < \theta < 12$

Count



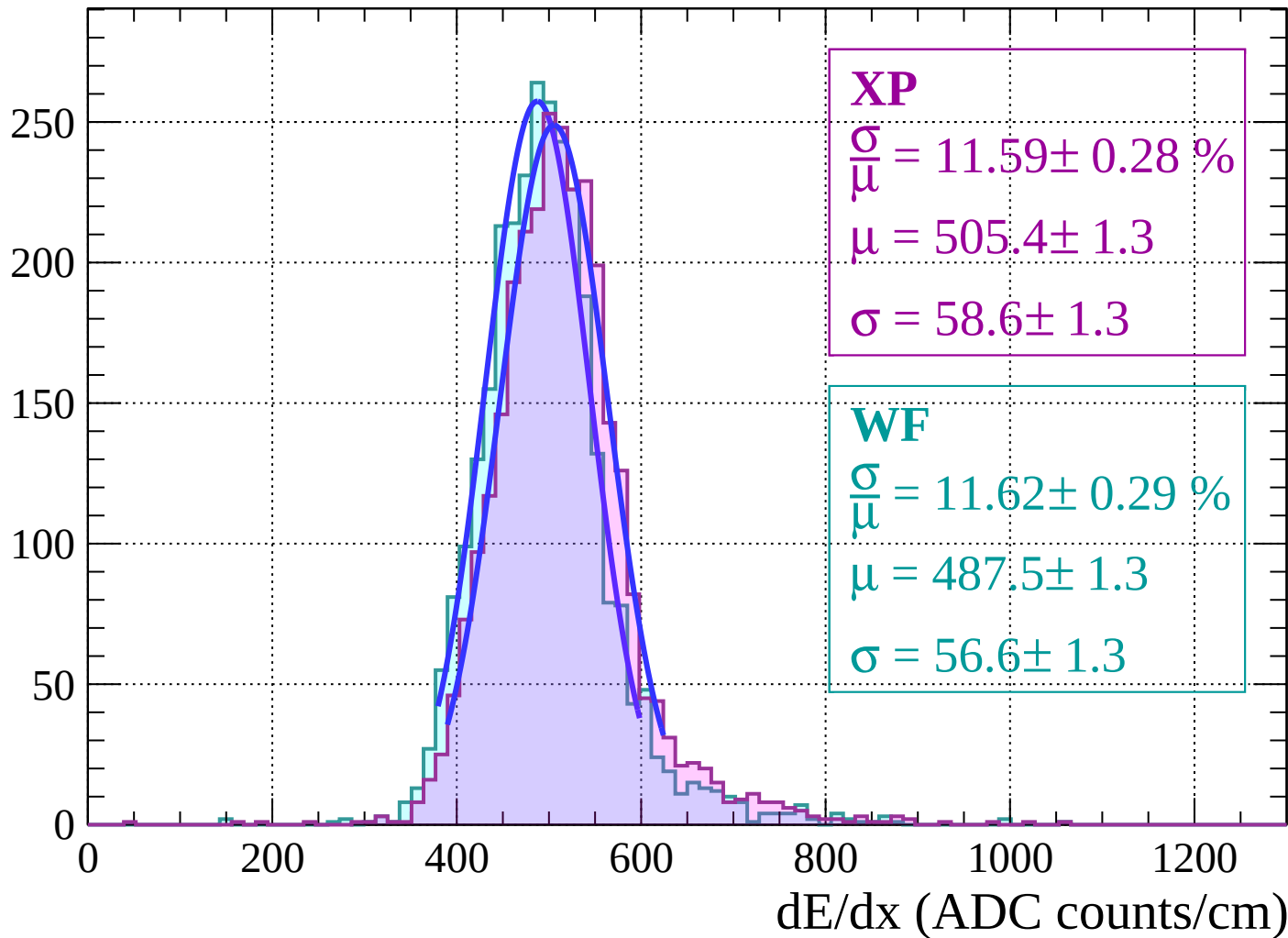
Energy loss | $12 < \theta < 14$

Count



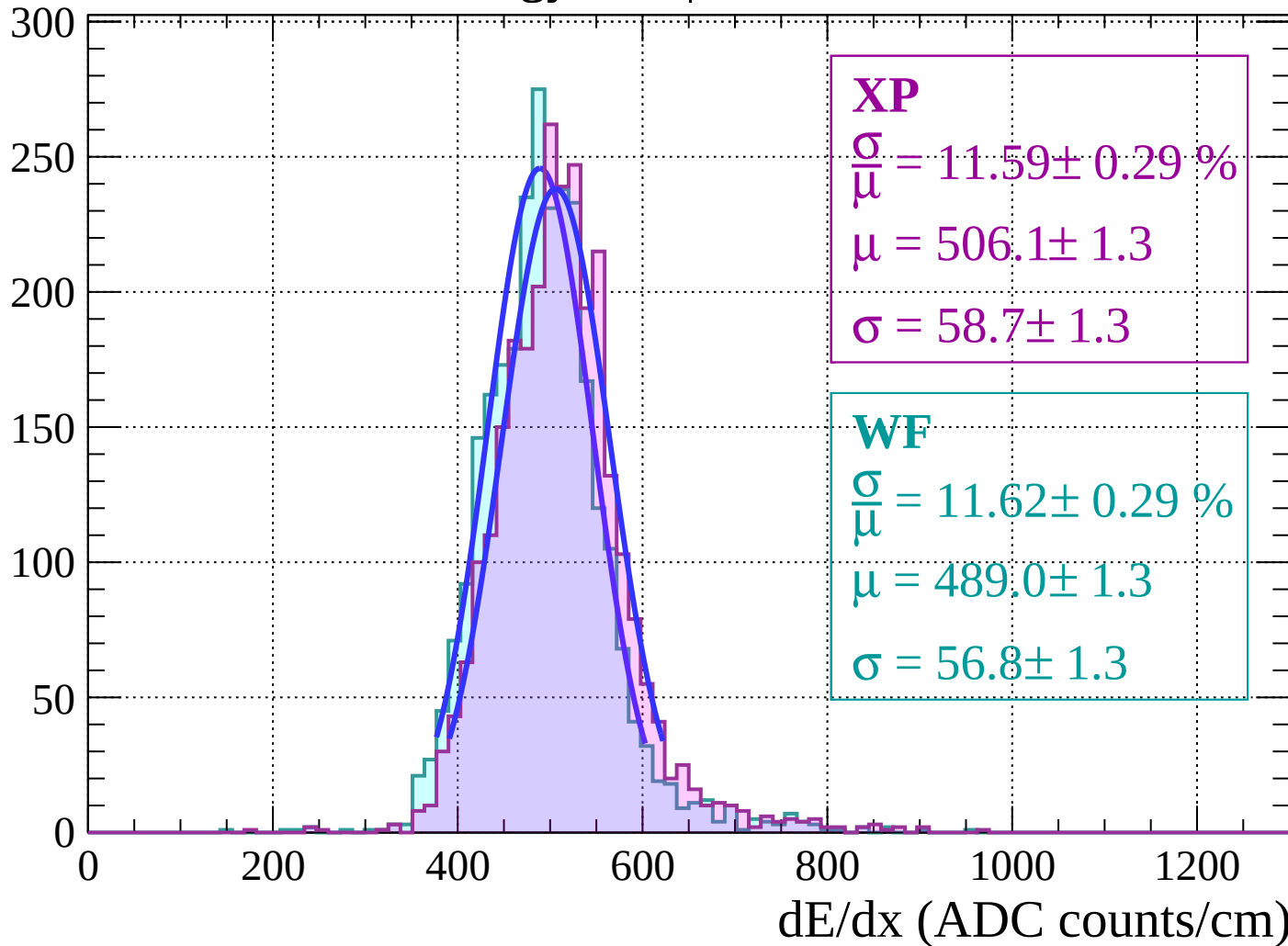
Energy loss | $14 < \theta < 16$

Count



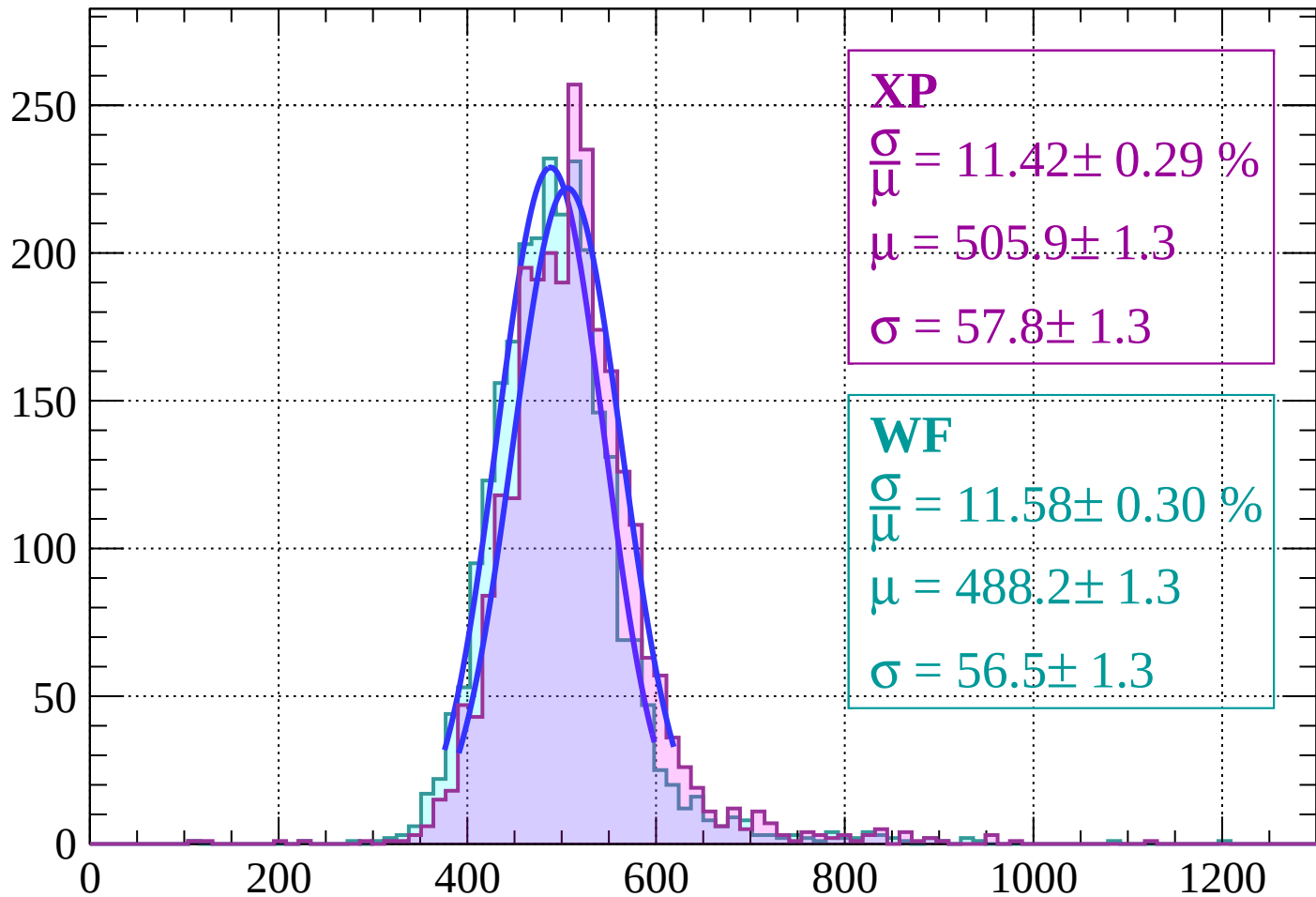
Energy loss | $16 < \theta < 18$

Count



Energy loss | $18 < \theta < 20$

Count



XP

$$\frac{\sigma}{\mu} = 11.42 \pm 0.29 \%$$

$$\mu = 505.9 \pm 1.3$$

$$\sigma = 57.8 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 11.58 \pm 0.30 \%$$

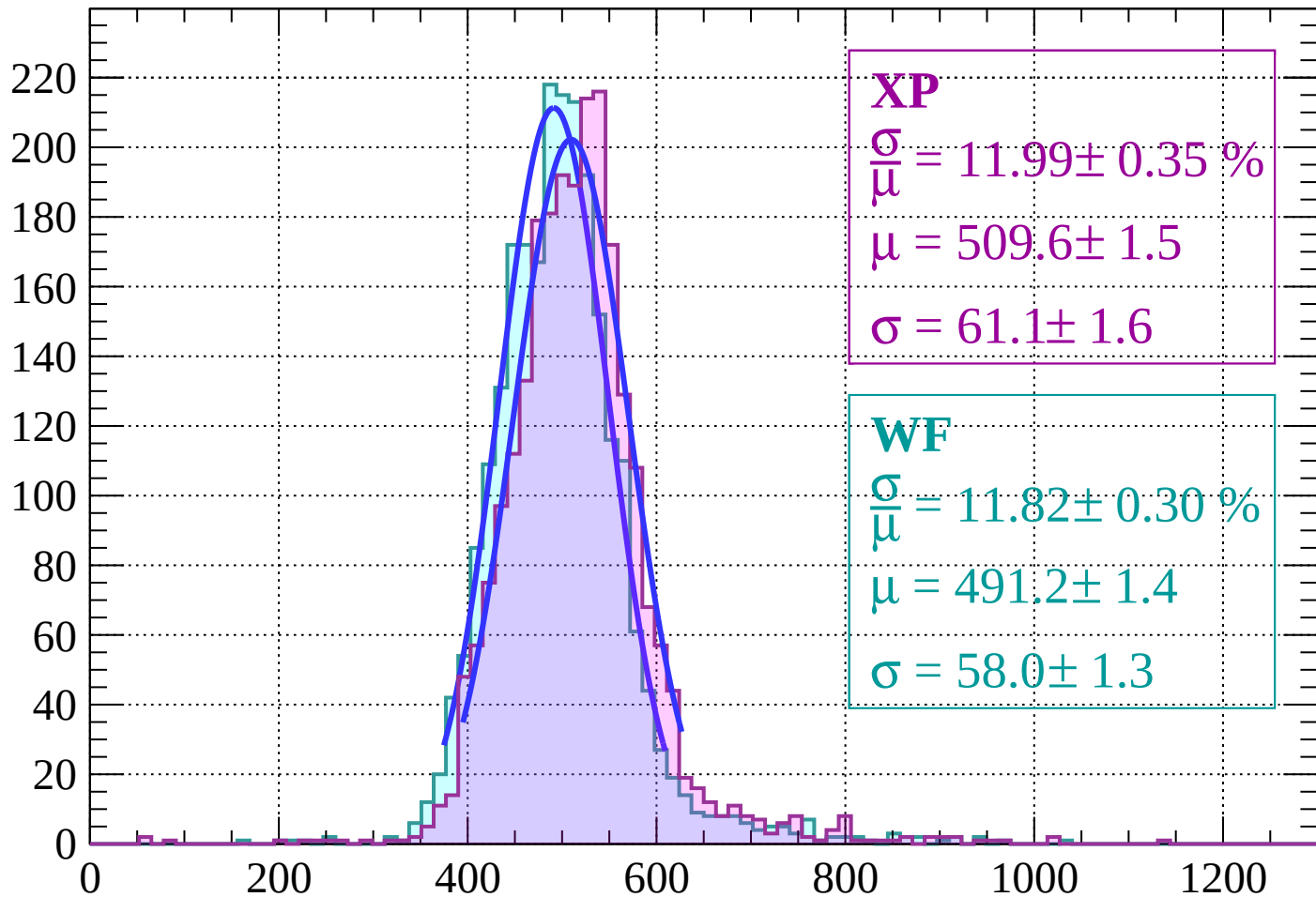
$$\mu = 488.2 \pm 1.3$$

$$\sigma = 56.5 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $20 < \theta < 22$

Count



XP

$$\frac{\sigma}{\mu} = 11.99 \pm 0.35 \%$$

$$\mu = 509.6 \pm 1.5$$

$$\sigma = 61.1 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 11.82 \pm 0.30 \%$$

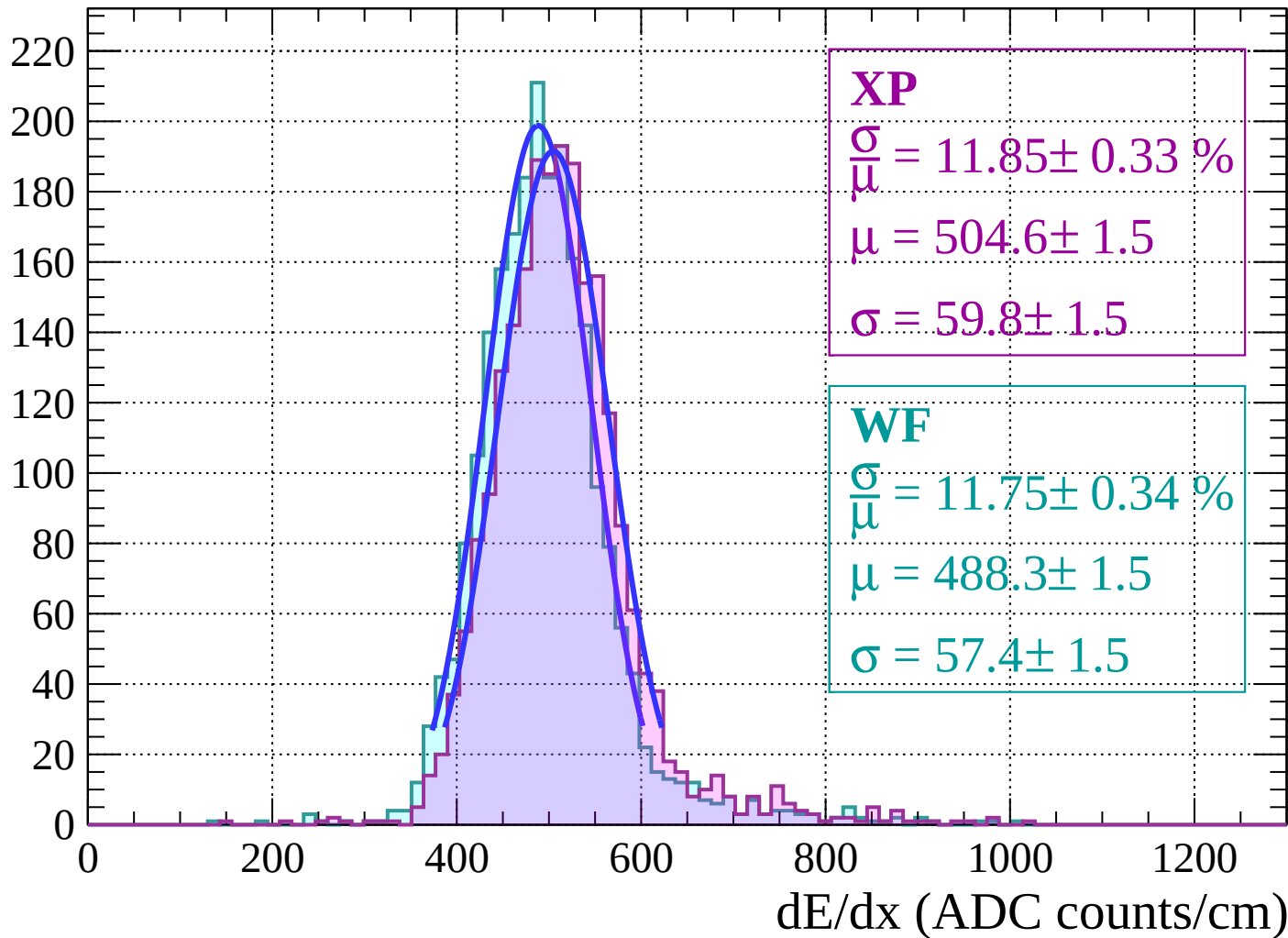
$$\mu = 491.2 \pm 1.4$$

$$\sigma = 58.0 \pm 1.3$$

dE/dx (ADC counts/cm)

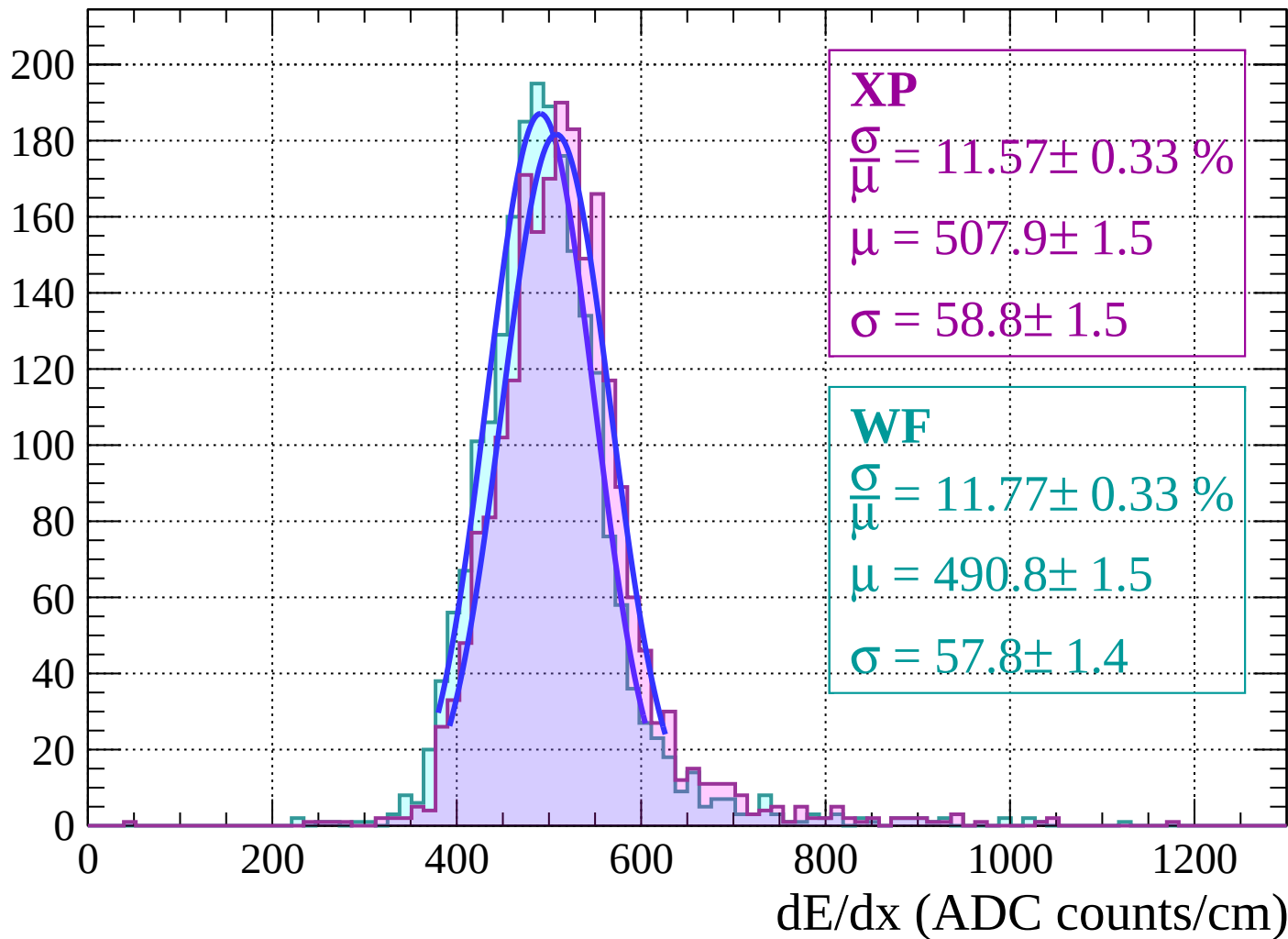
Energy loss | $22 < \theta < 24$

Count



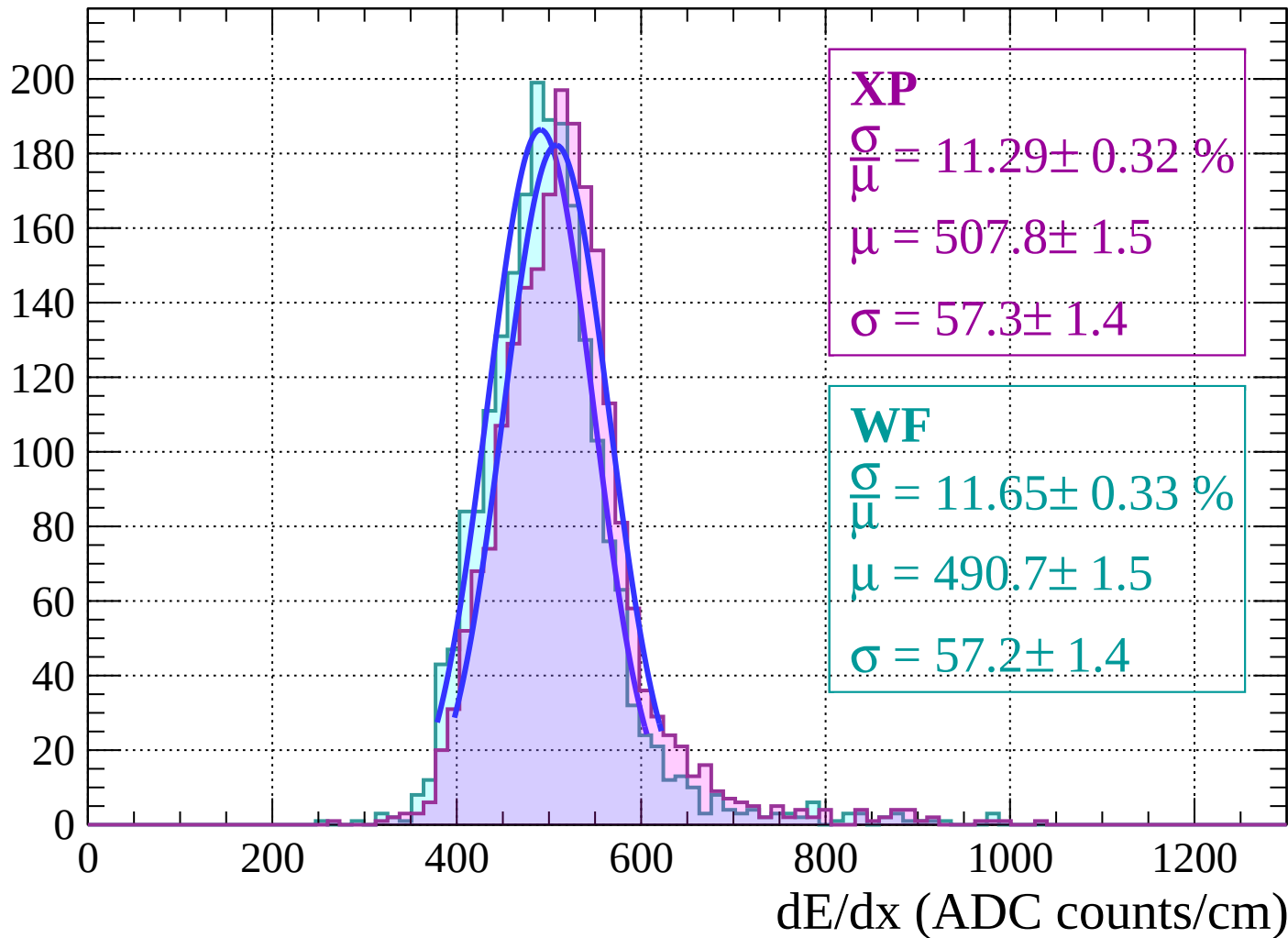
Energy loss | $24 < \theta < 26$

Count

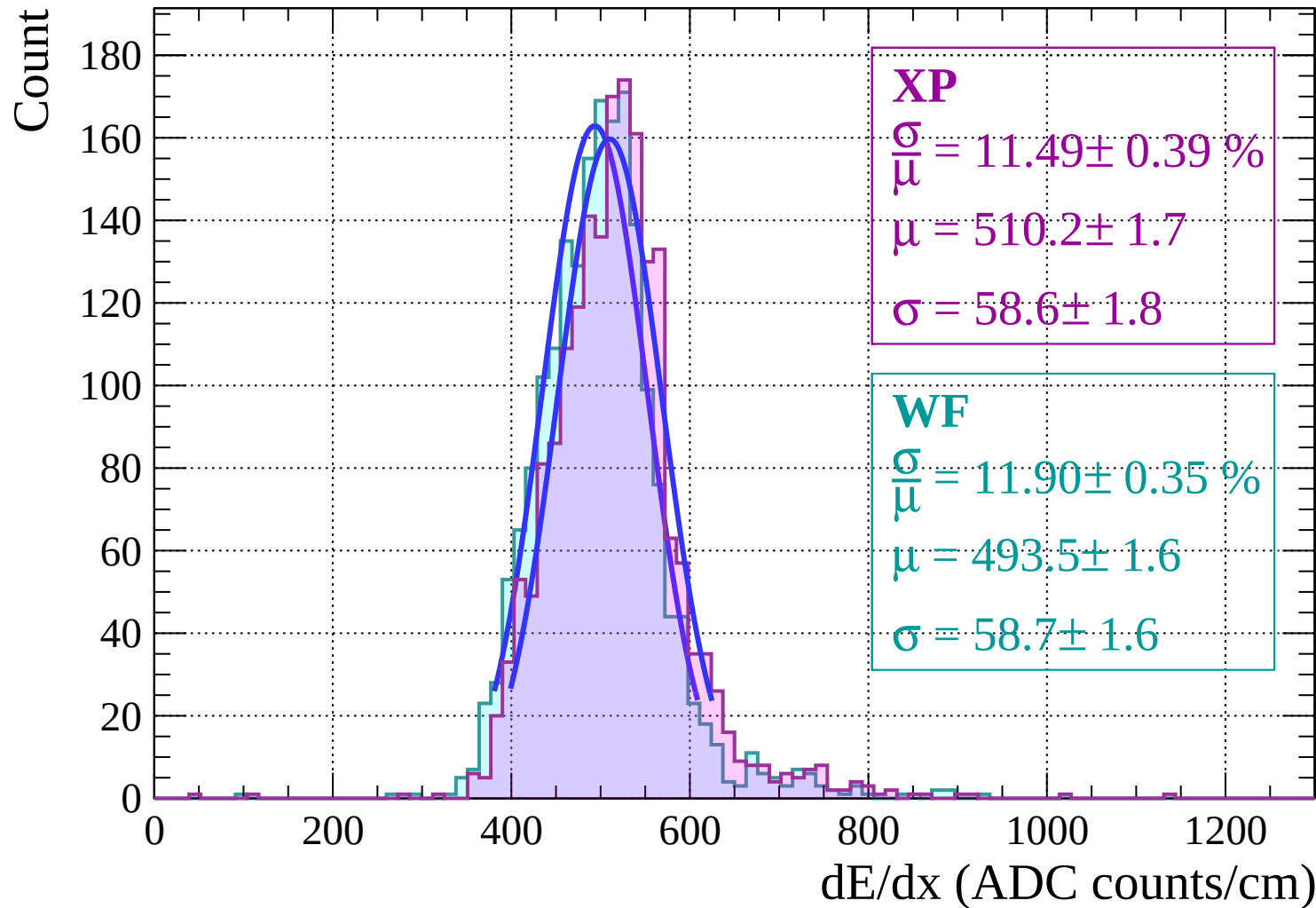


Energy loss | $26 < \theta < 28$

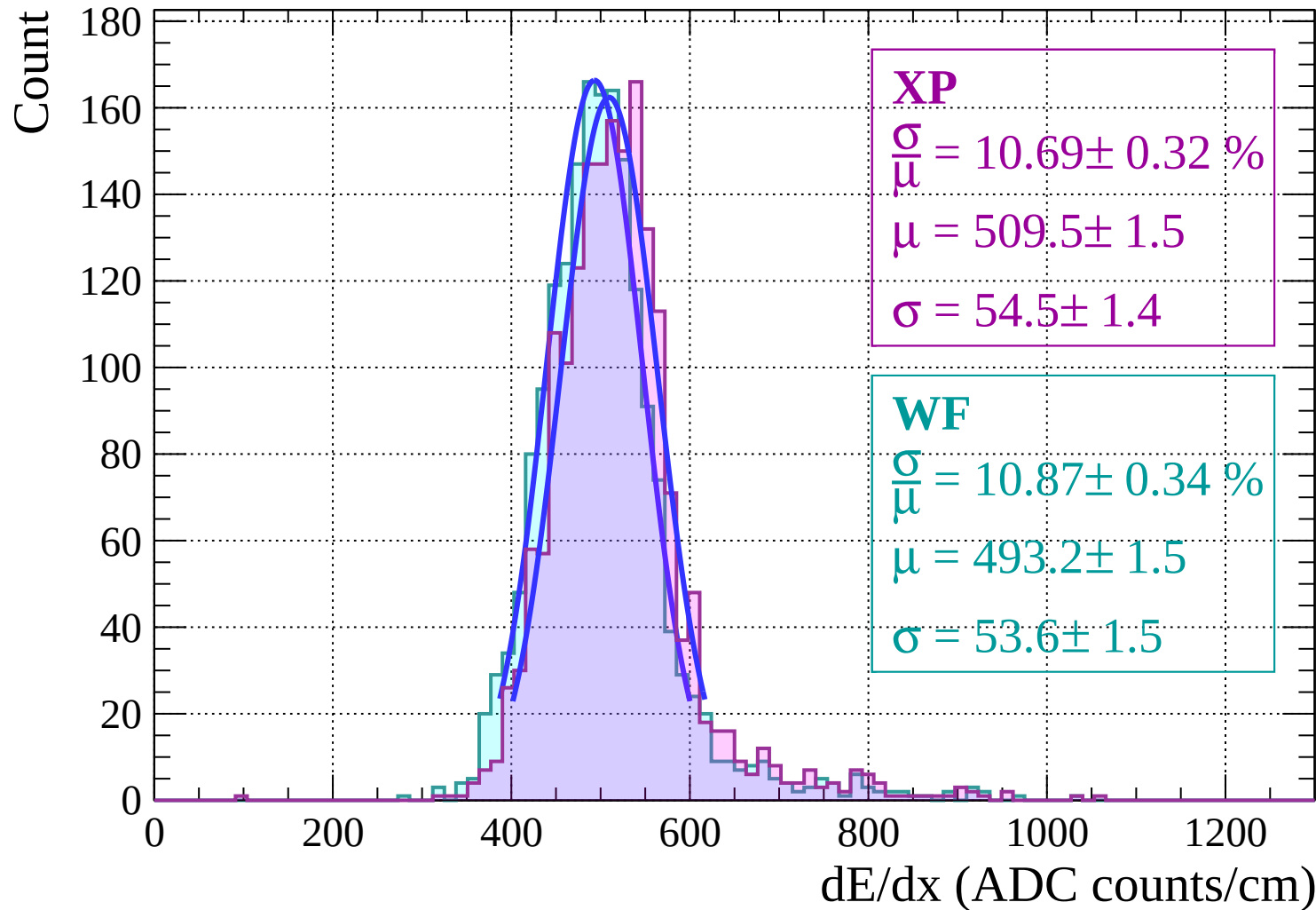
Count



Energy loss | $28 < \theta < 30$

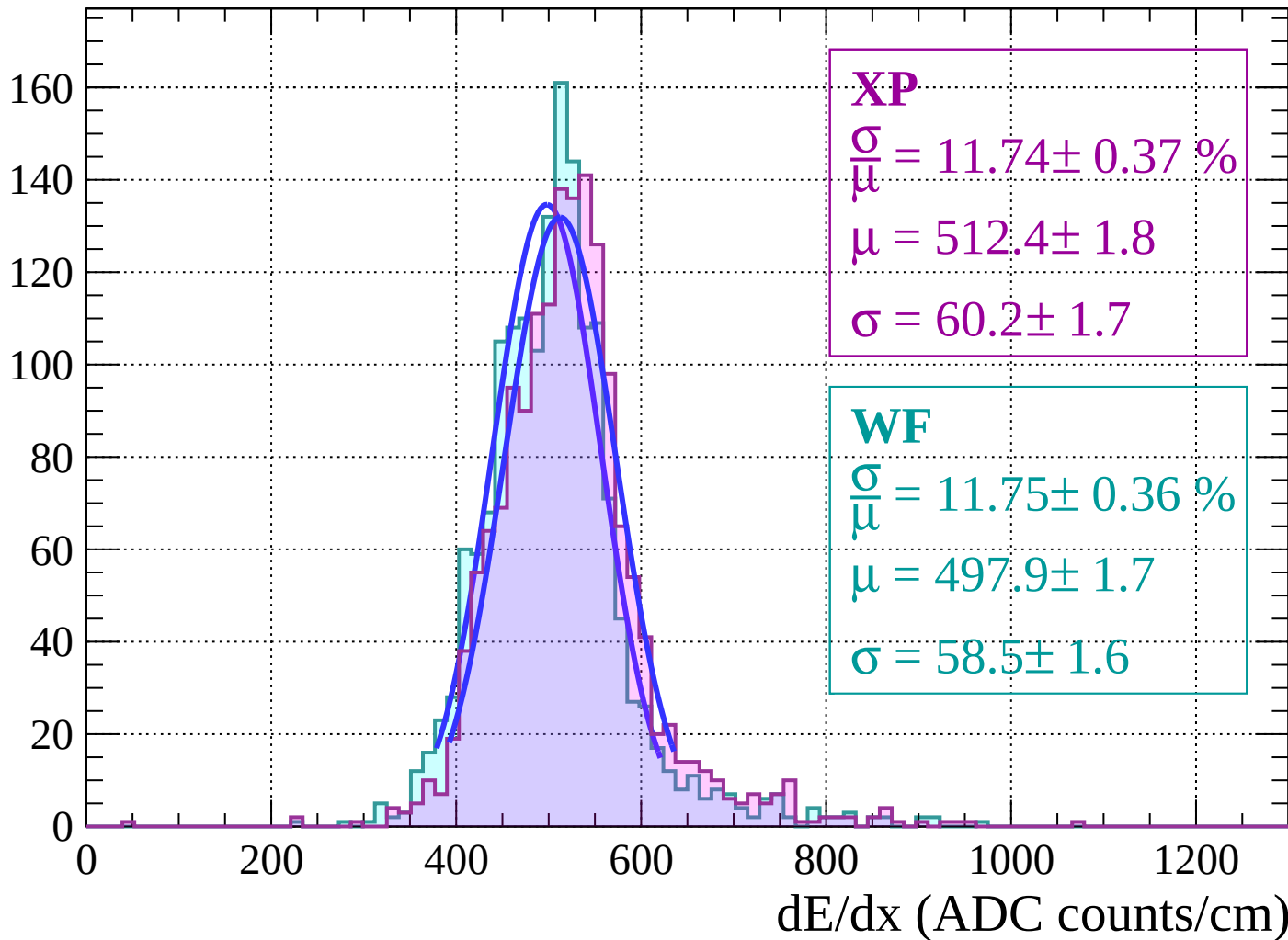


Energy loss | $30 < \theta < 32$

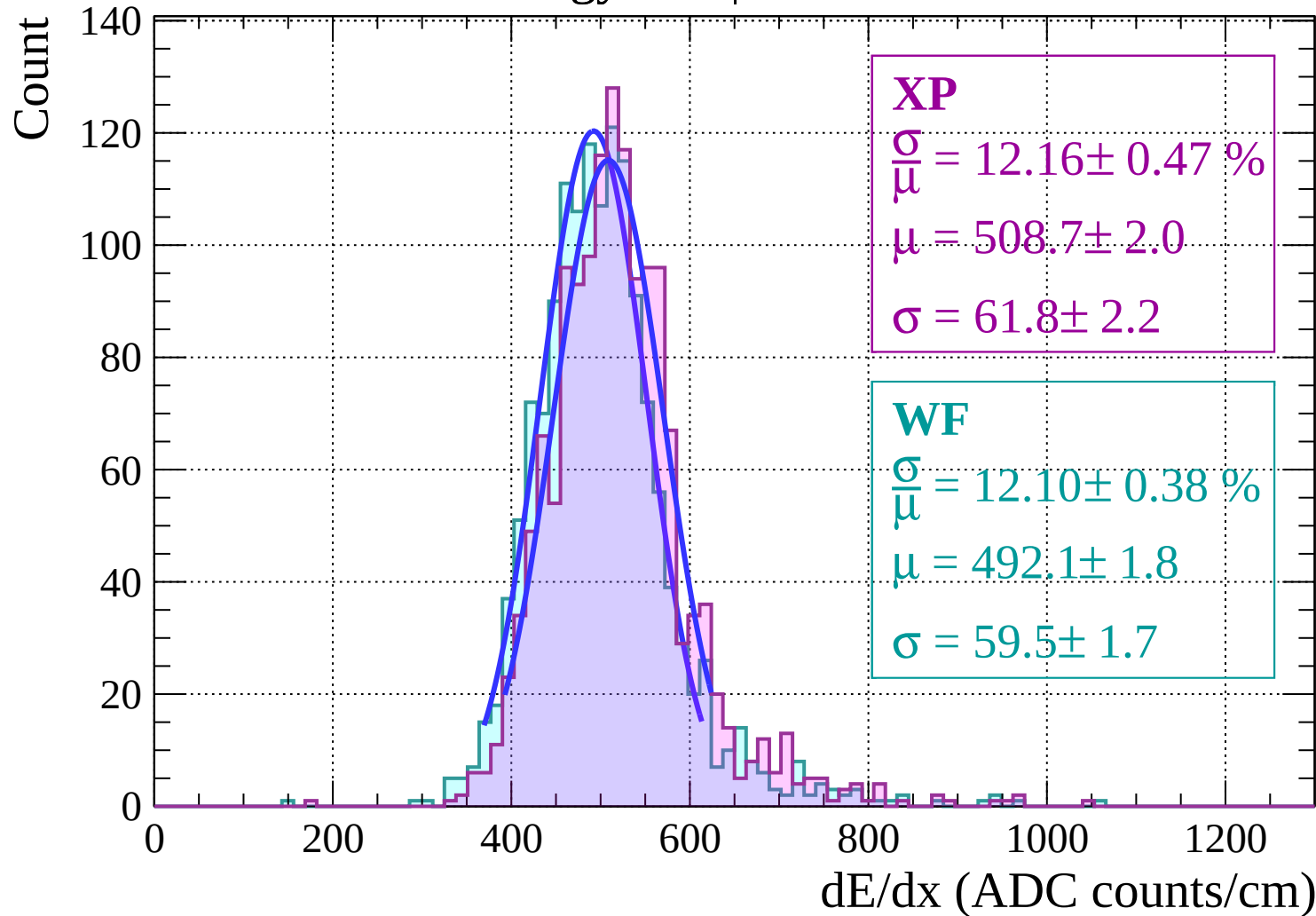


Energy loss | $32 < \theta < 34$

Count

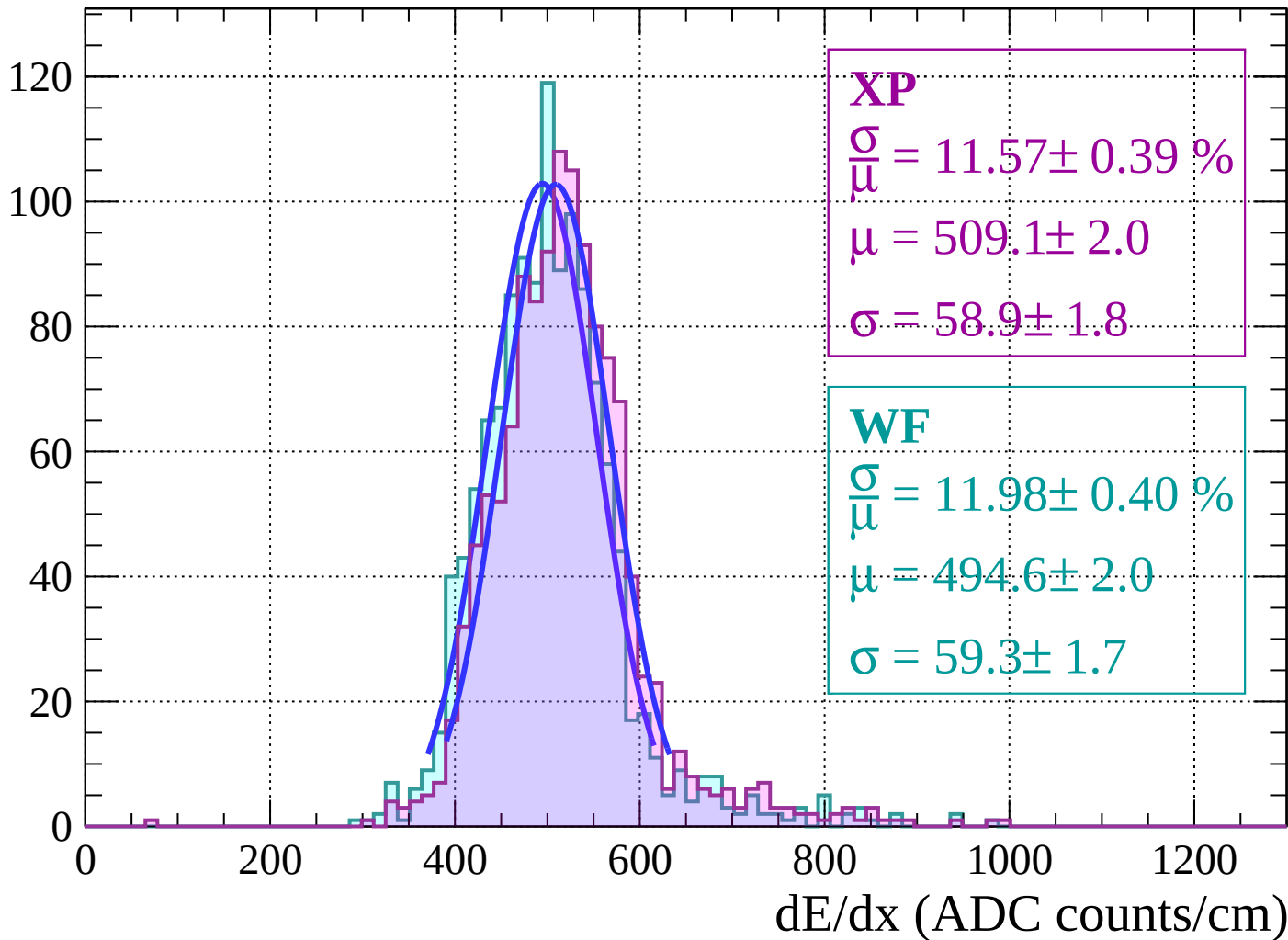


Energy loss | $34 < \theta < 36$



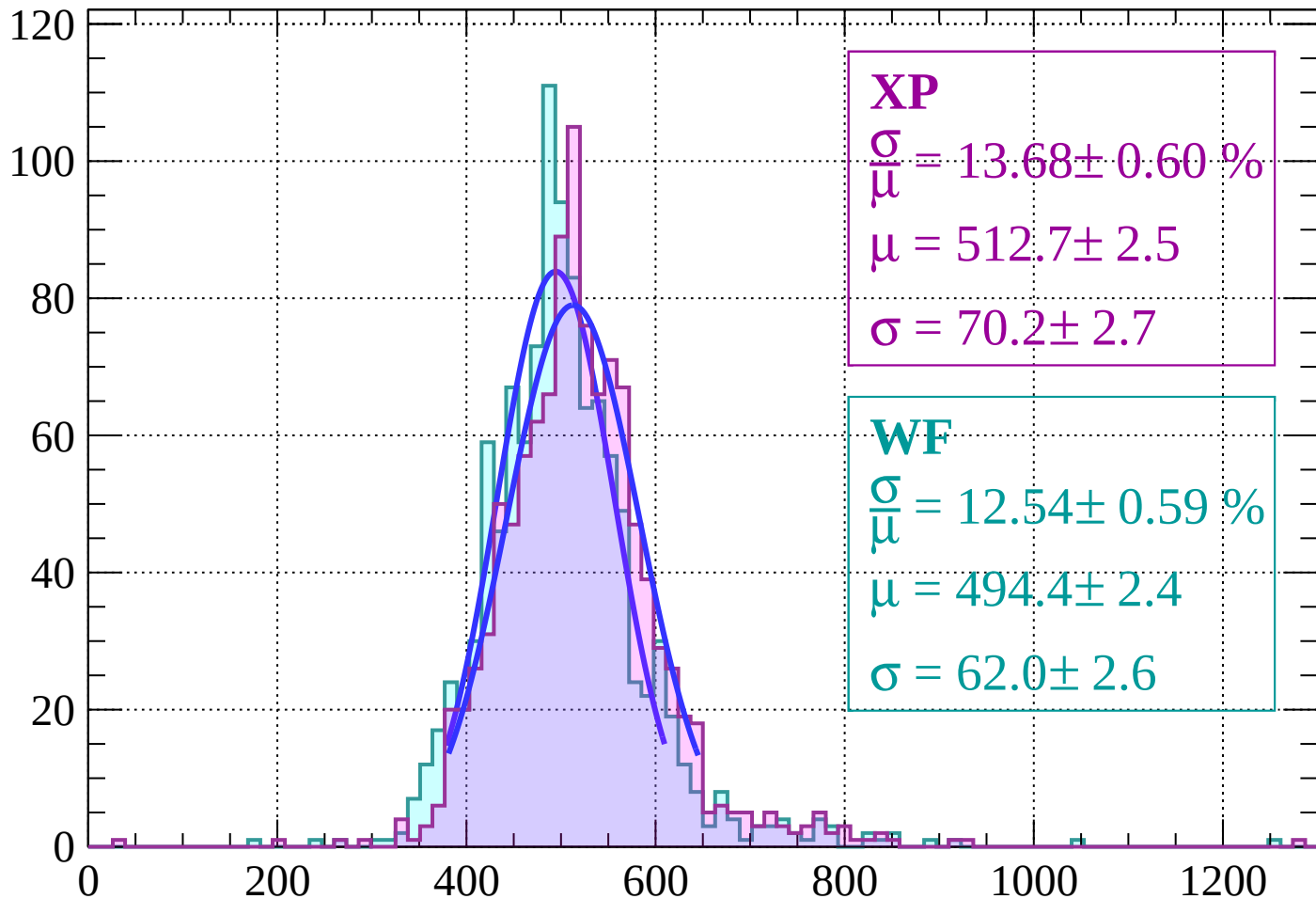
Energy loss | $36 < \theta < 38$

Count



Energy loss | $38 < \theta < 40$

Count



XP

$$\frac{\sigma}{\mu} = 13.68 \pm 0.60 \%$$

$$\mu = 512.7 \pm 2.5$$

$$\sigma = 70.2 \pm 2.7$$

WF

$$\frac{\sigma}{\mu} = 12.54 \pm 0.59 \%$$

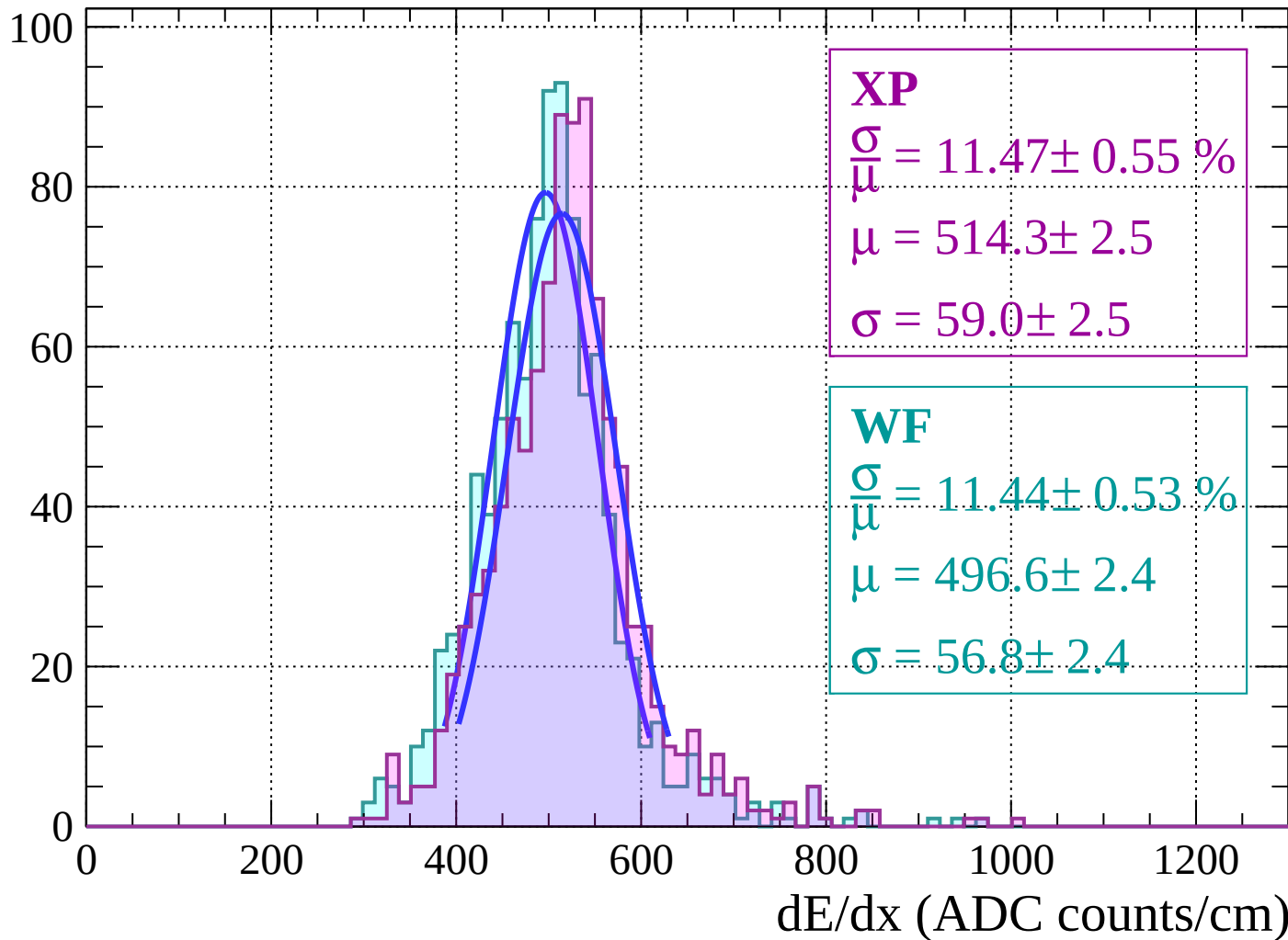
$$\mu = 494.4 \pm 2.4$$

$$\sigma = 62.0 \pm 2.6$$

dE/dx (ADC counts/cm)

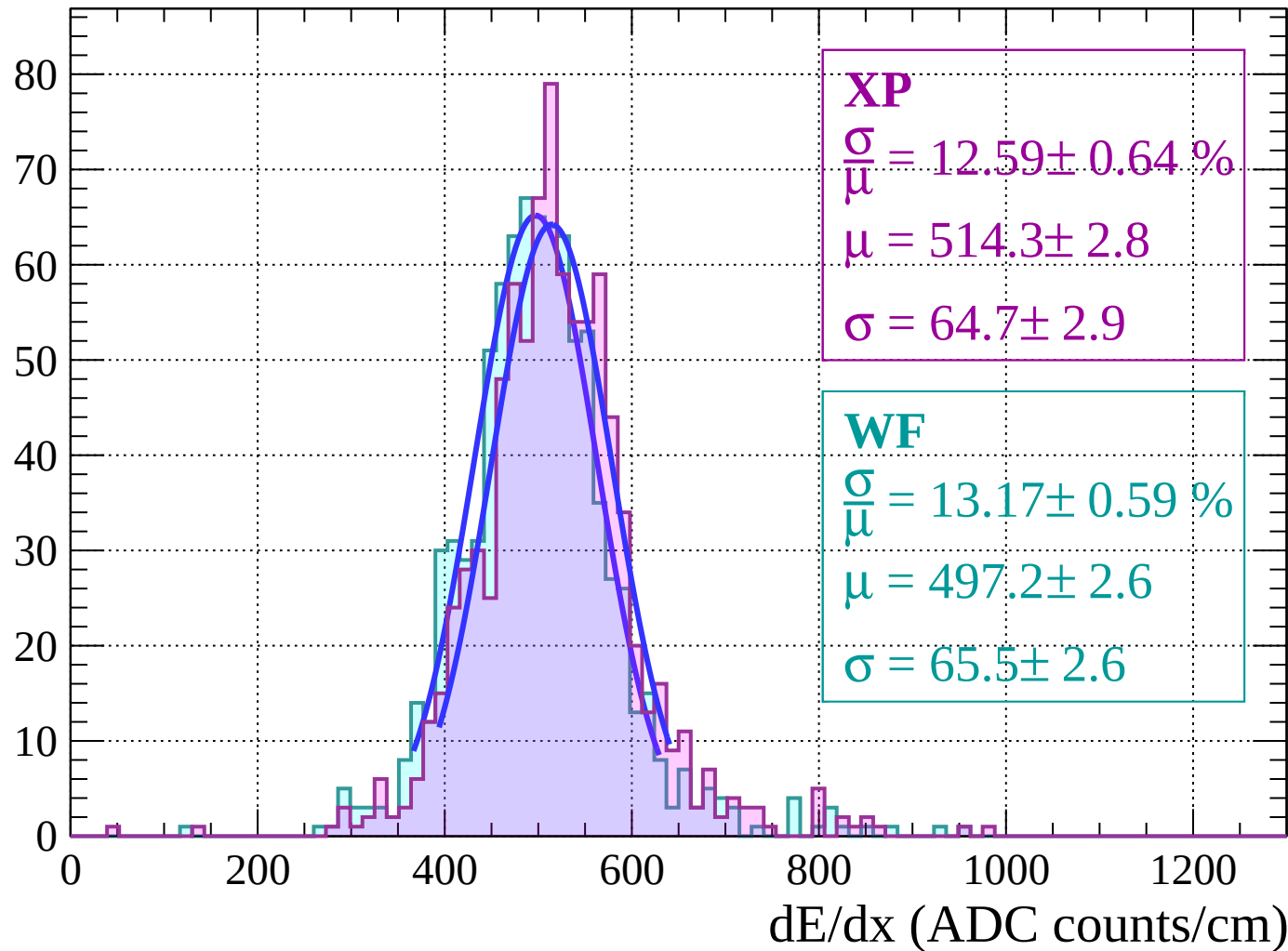
Energy loss | $40 < \theta < 42$

Count



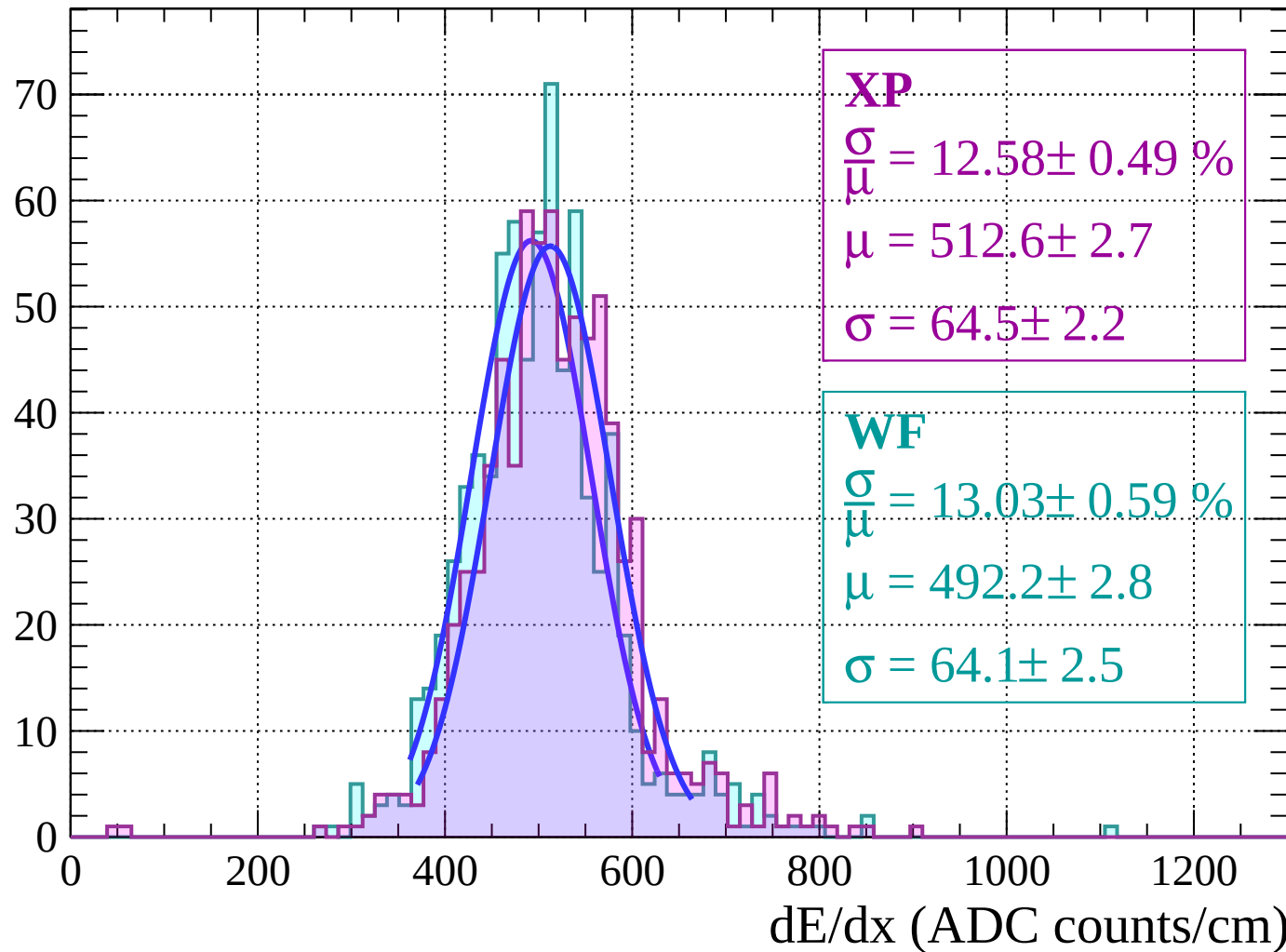
Energy loss | $42 < \theta < 44$

Count



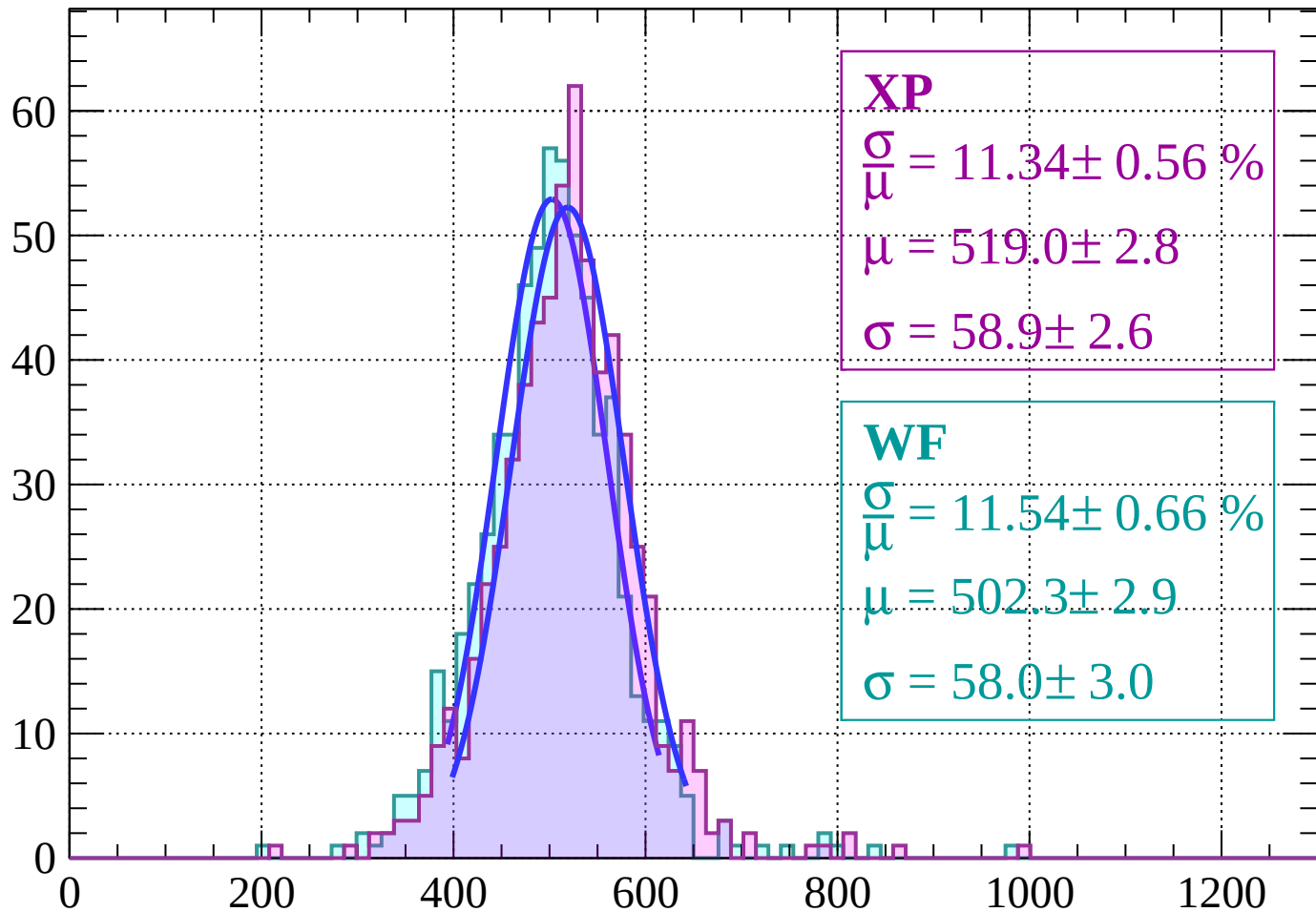
Energy loss | $44 < \theta < 46$

Count



Energy loss | $46 < \theta < 48$

Count



XP

$$\frac{\sigma}{\mu} = 11.34 \pm 0.56 \%$$

$$\mu = 519.0 \pm 2.8$$

$$\sigma = 58.9 \pm 2.6$$

WF

$$\frac{\sigma}{\mu} = 11.54 \pm 0.66 \%$$

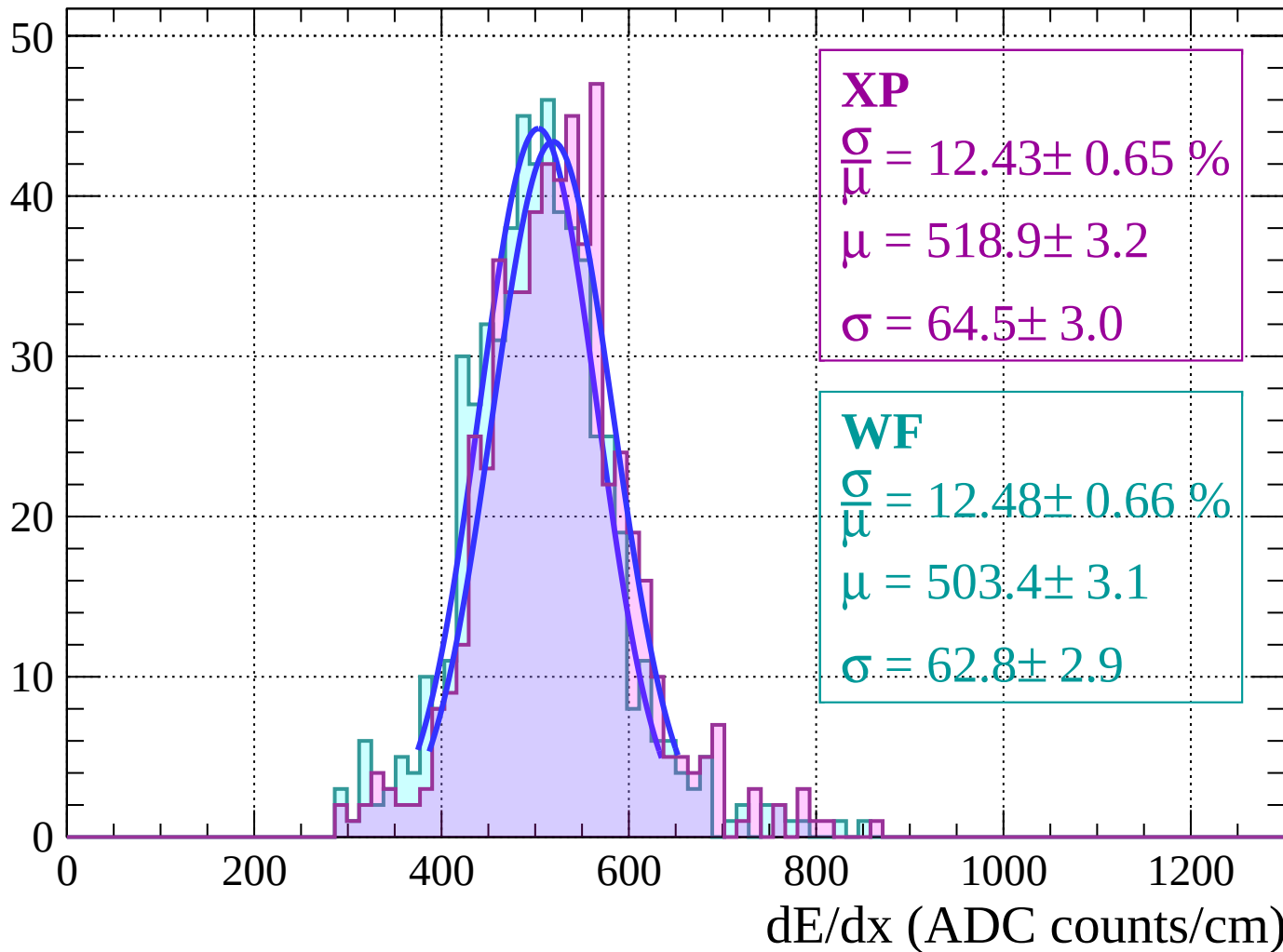
$$\mu = 502.3 \pm 2.9$$

$$\sigma = 58.0 \pm 3.0$$

dE/dx (ADC counts/cm)

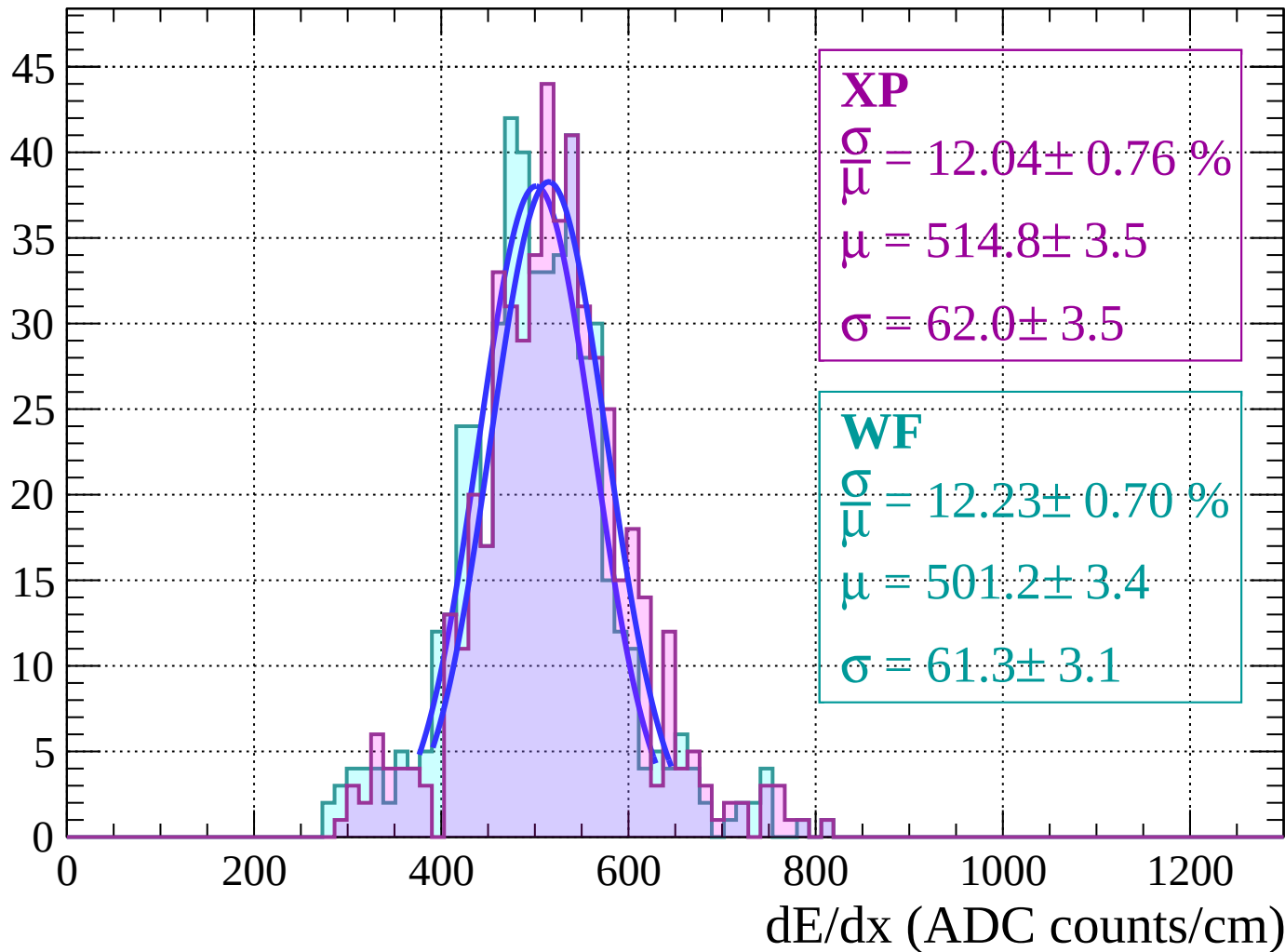
Energy loss | $48 < \theta < 50$

Count



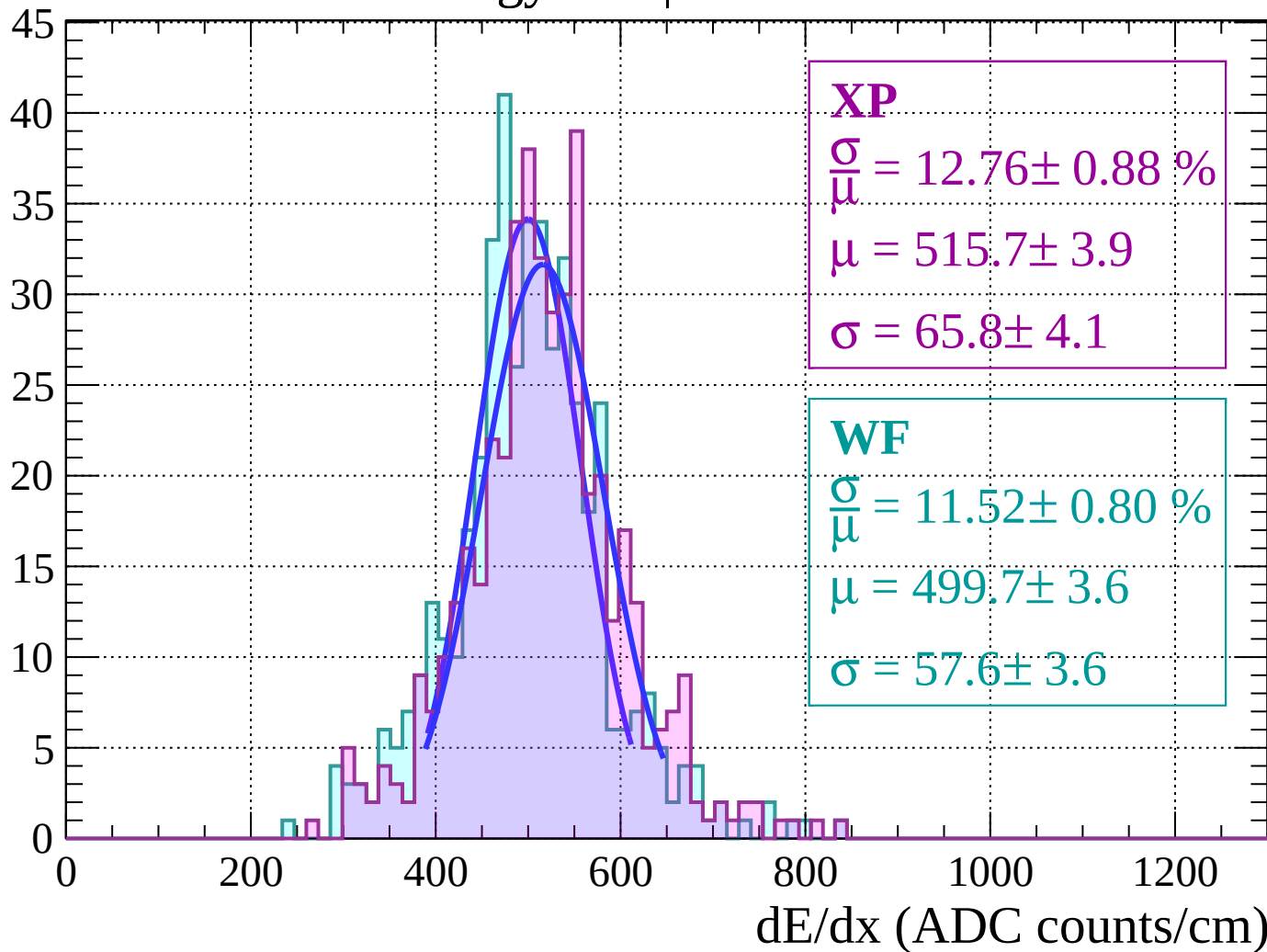
Energy loss | $50 < \theta < 52$

Count



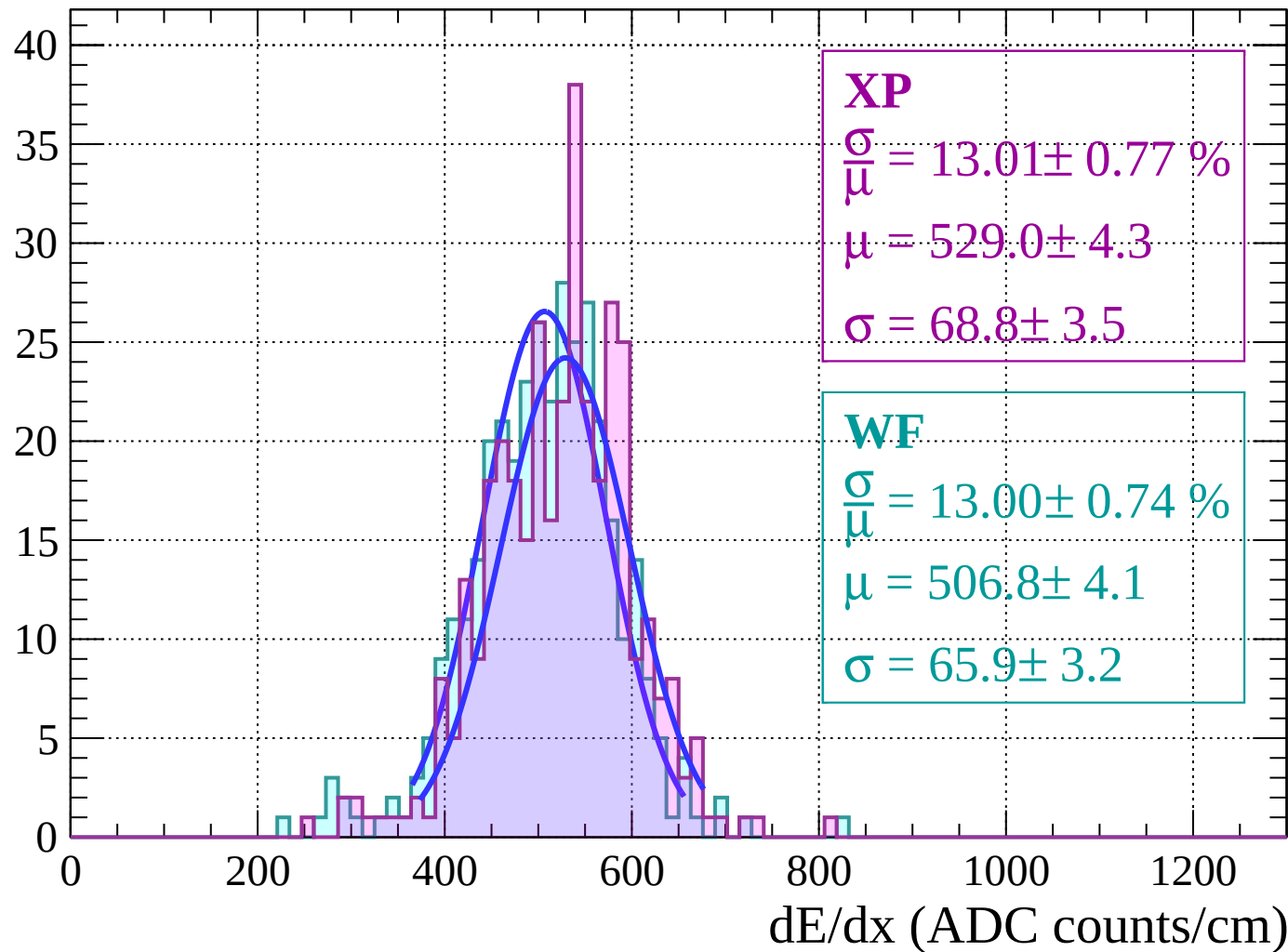
Energy loss | $52 < \theta < 54$

Count



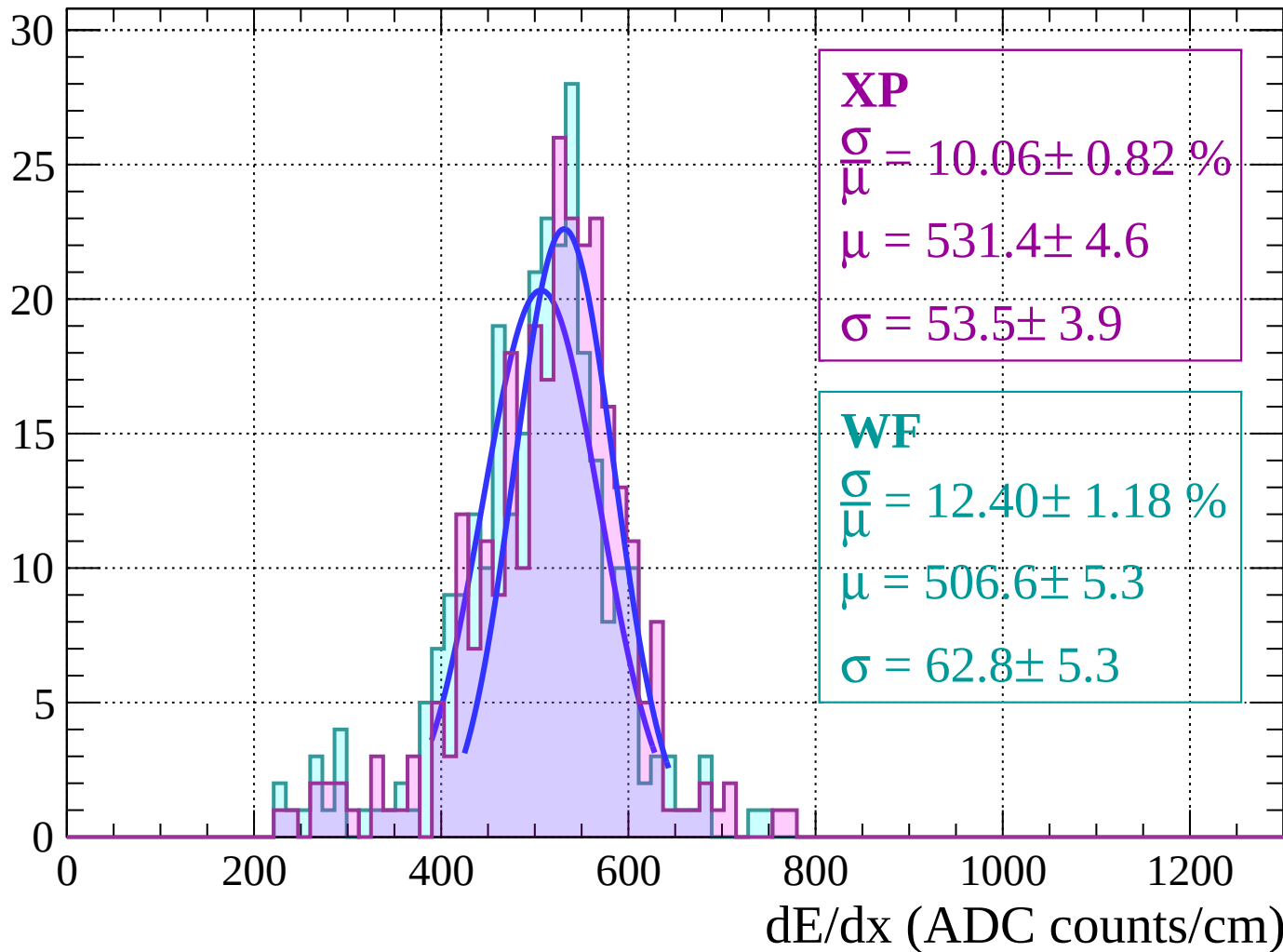
Energy loss | $54 < \theta < 56$

Count



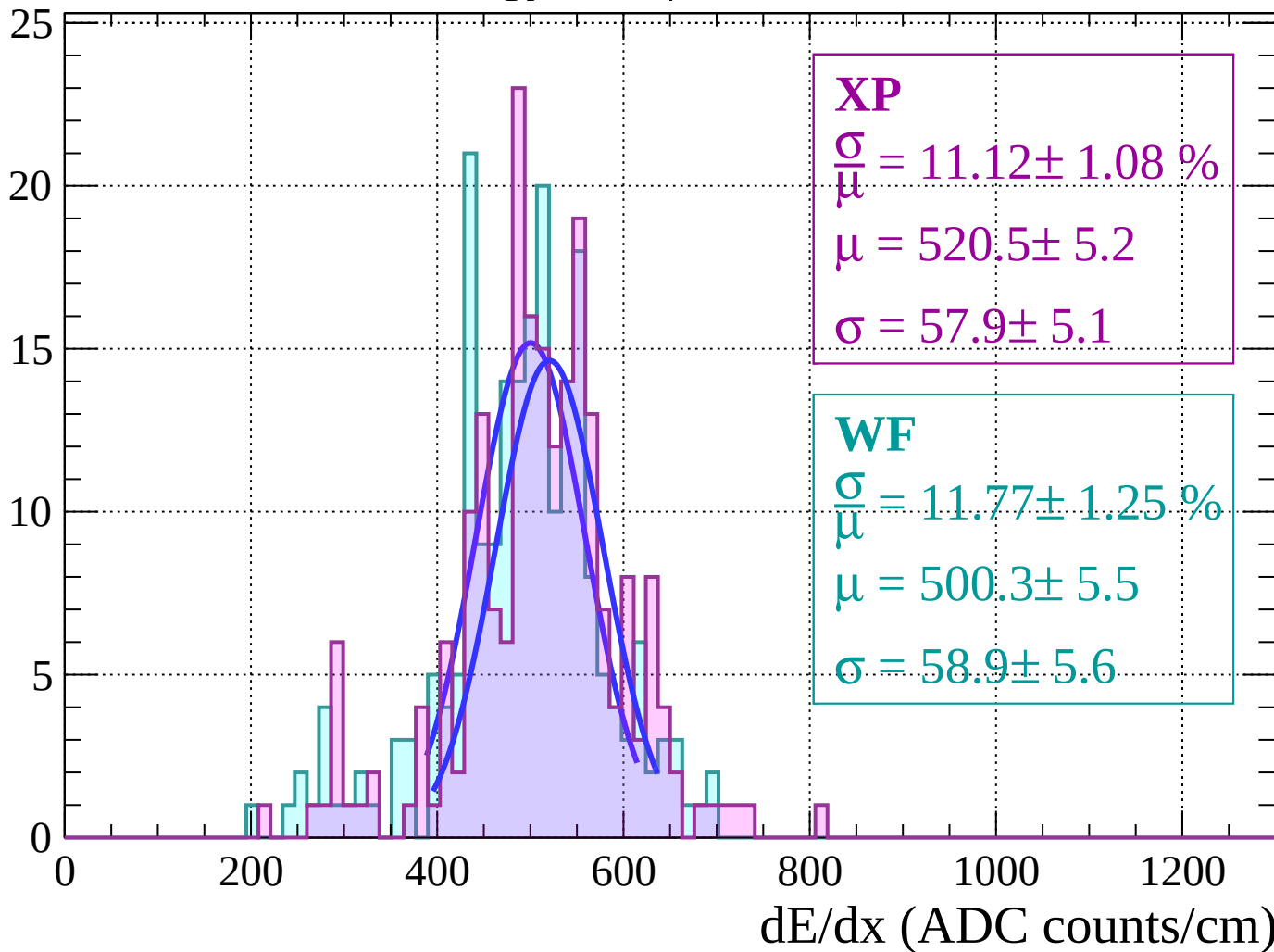
Energy loss | $56 < \theta < 58$

Count



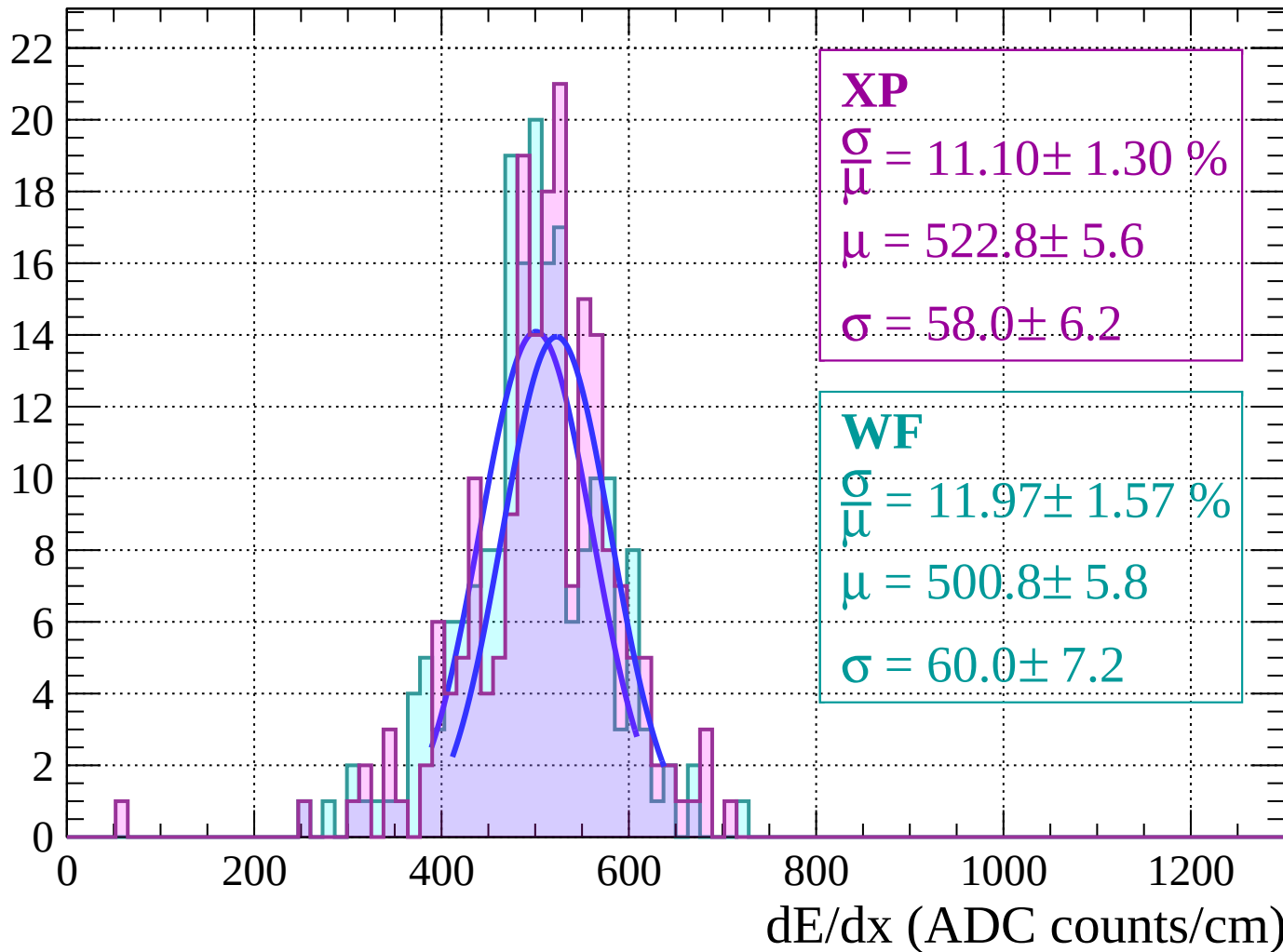
Energy loss | $58 < \theta < 60$

Count



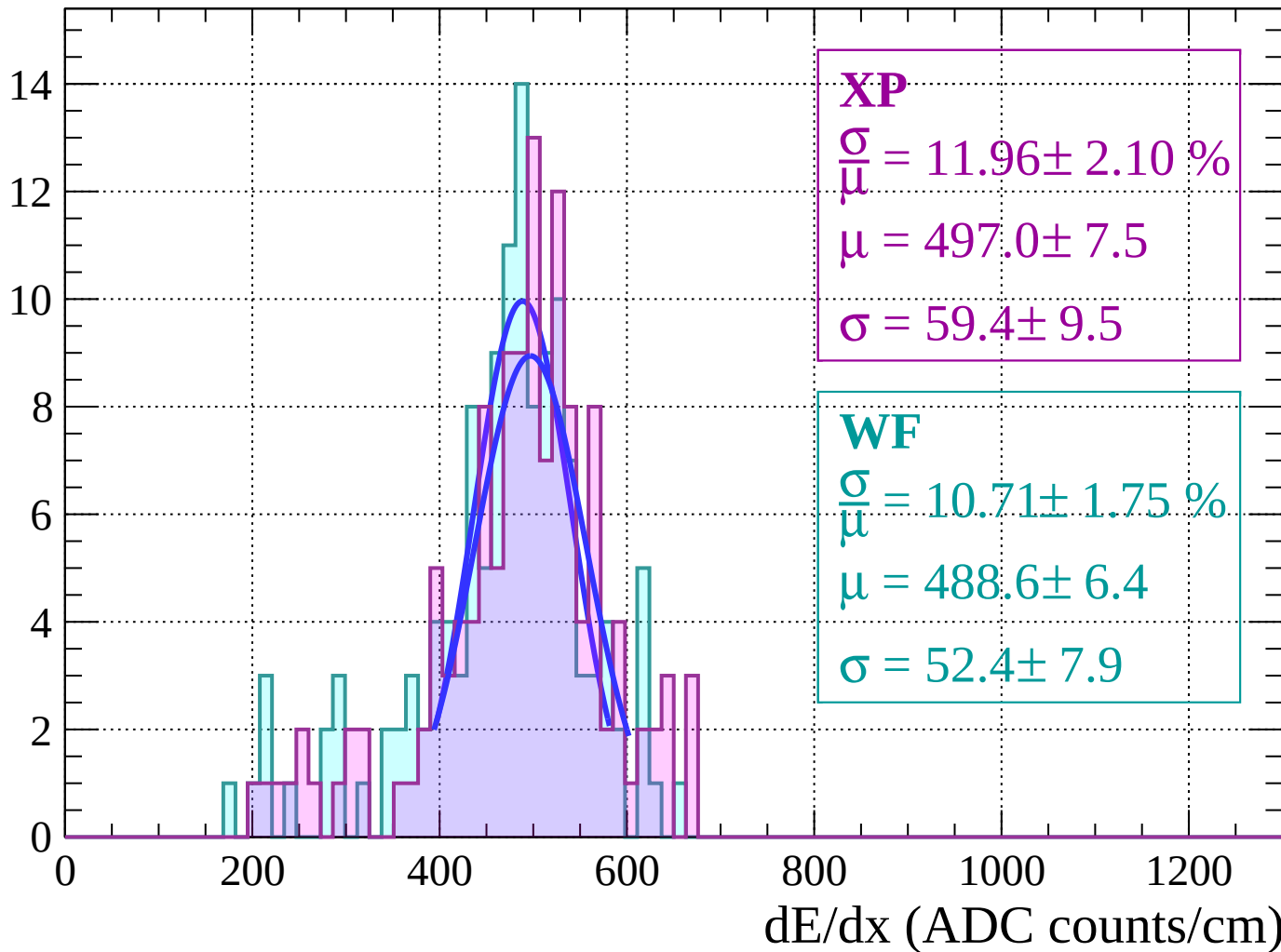
Energy loss | $60 < \theta < 62$

Count



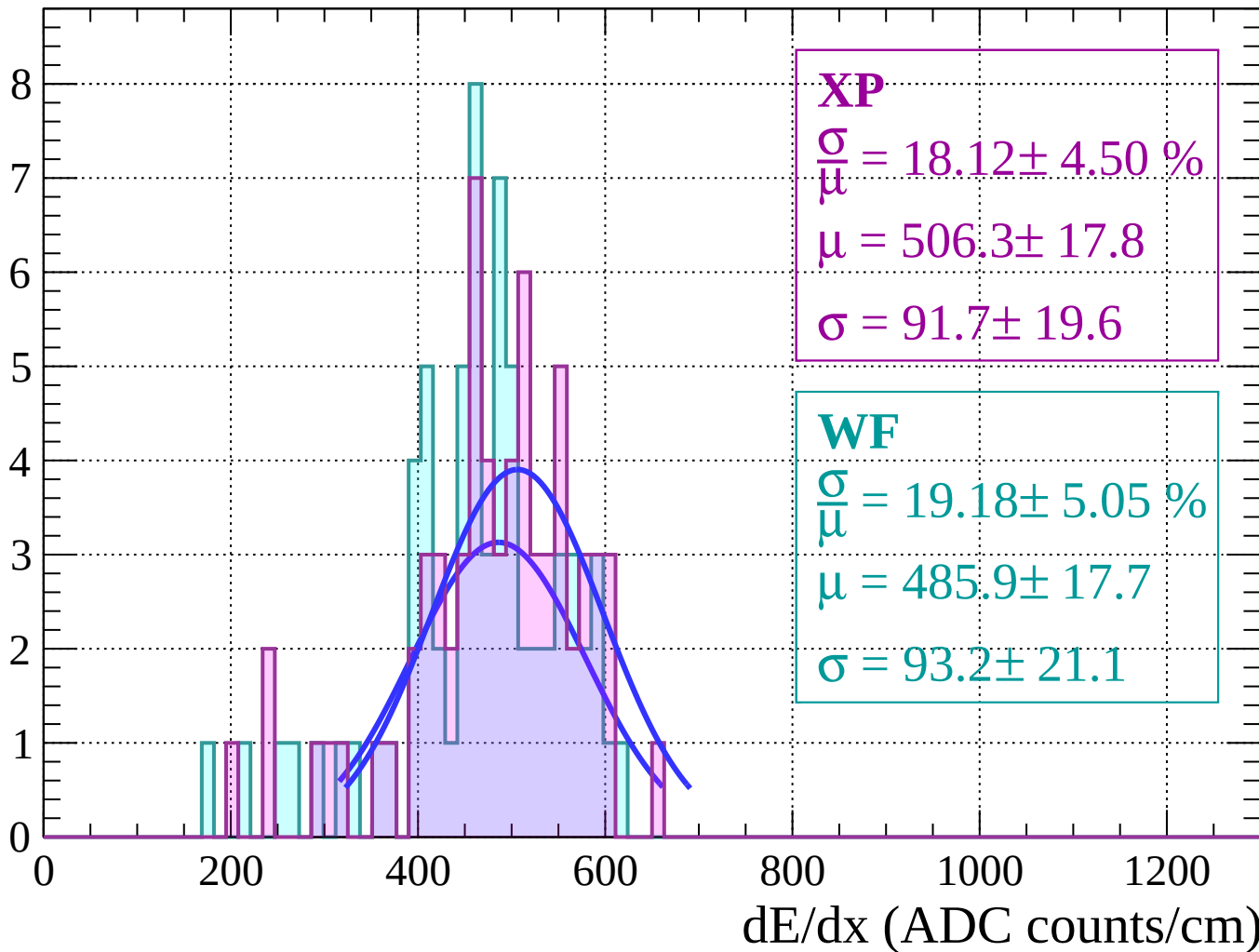
Energy loss | $62 < \theta < 64$

Count



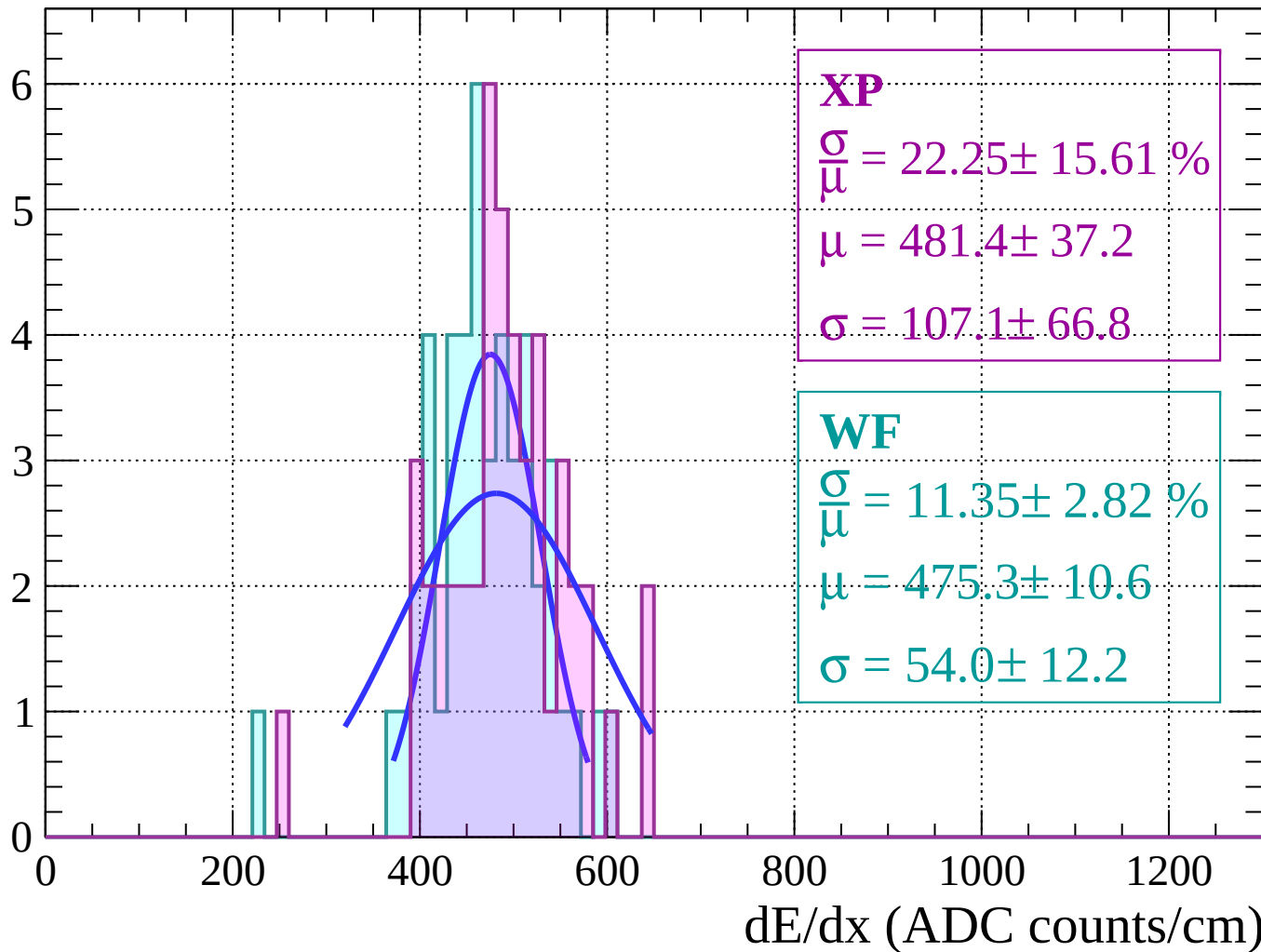
Energy loss | $64 < \theta < 66$

Count



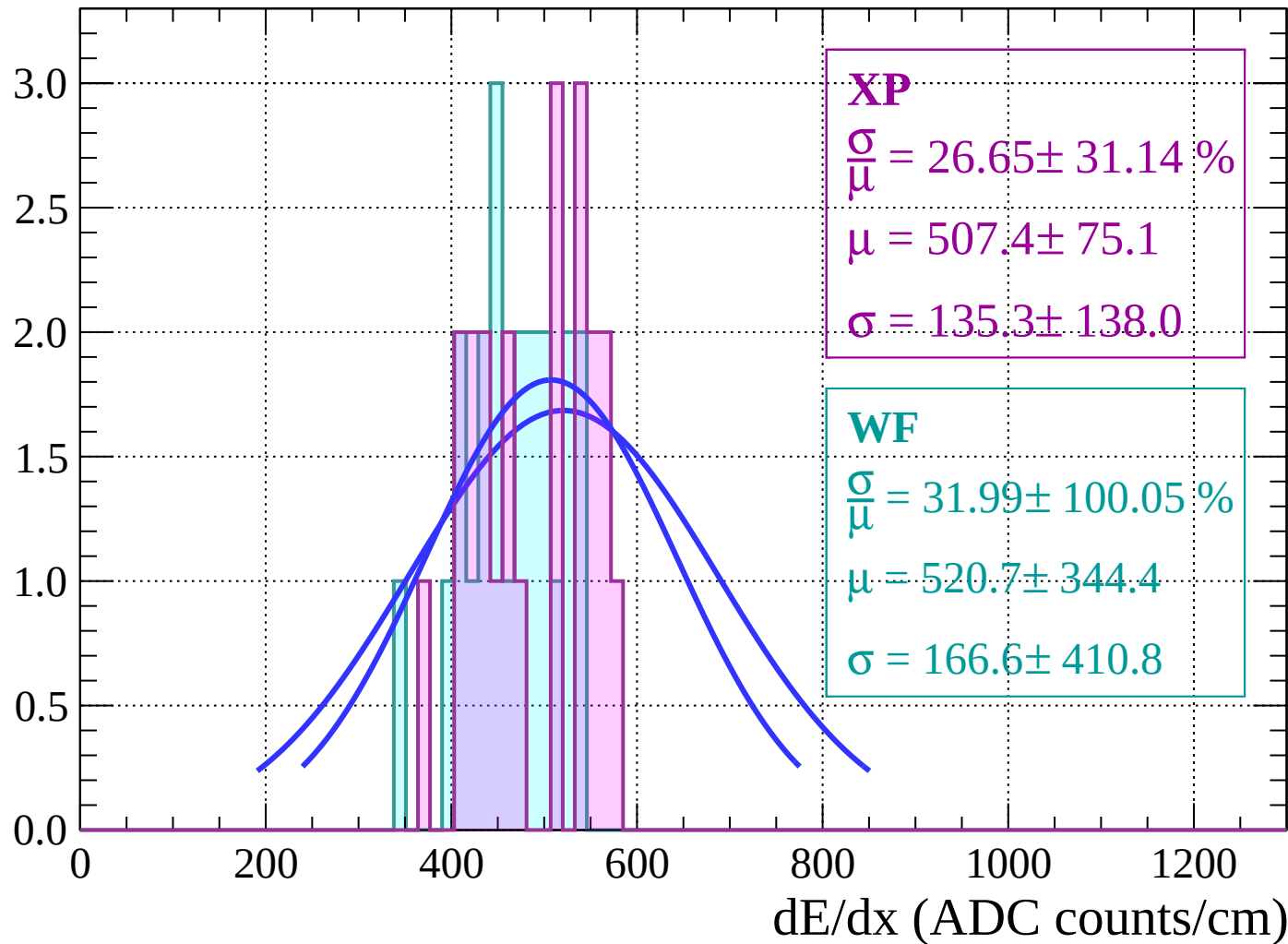
Energy loss | $66 < \theta < 68$

Count



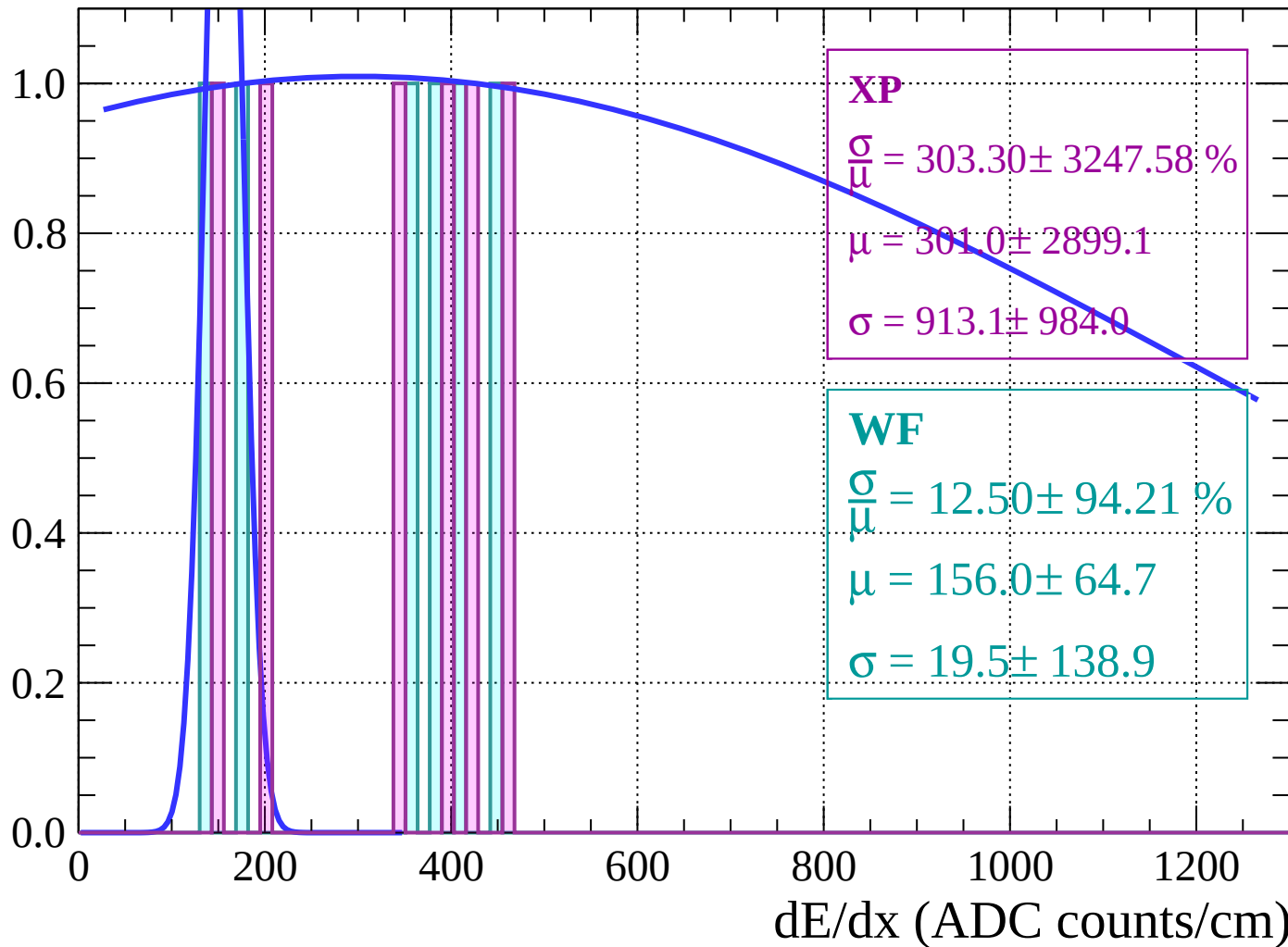
Energy loss | $68 < \theta < 70$

Count



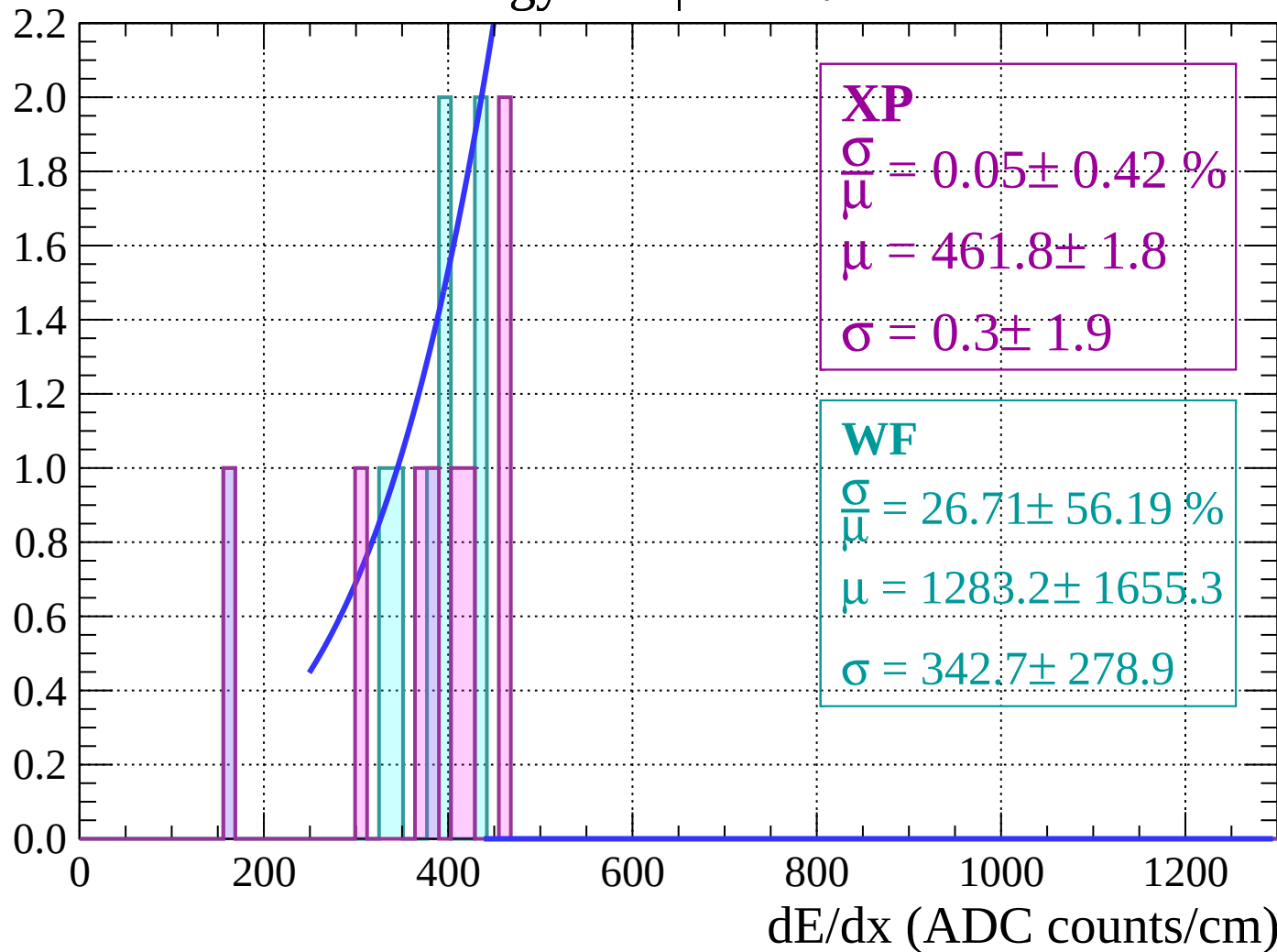
Energy loss | $70 < \theta < 72$

Count



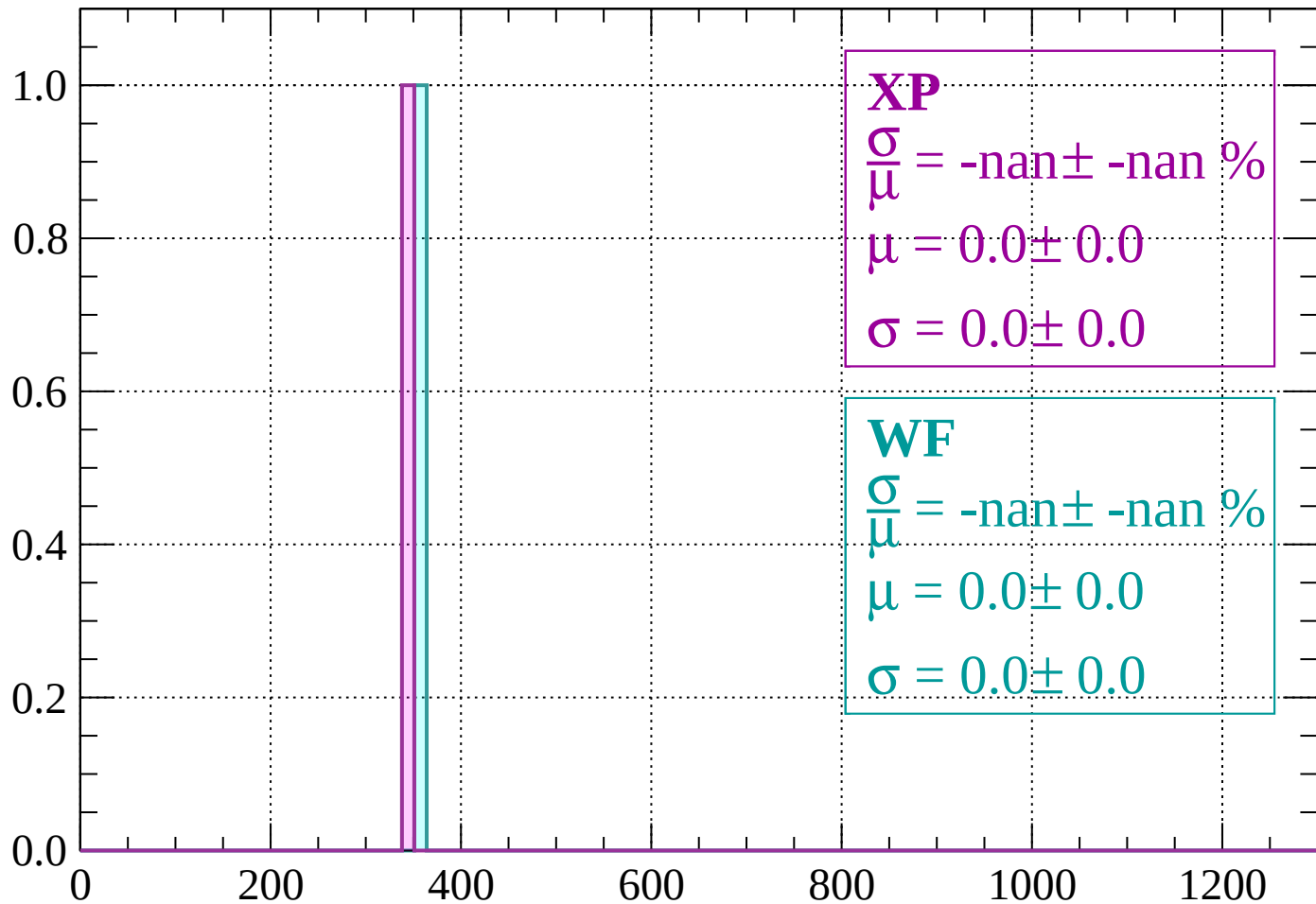
Energy loss | $72 < \theta < 74$

Count



Energy loss | $74 < \theta < 76$

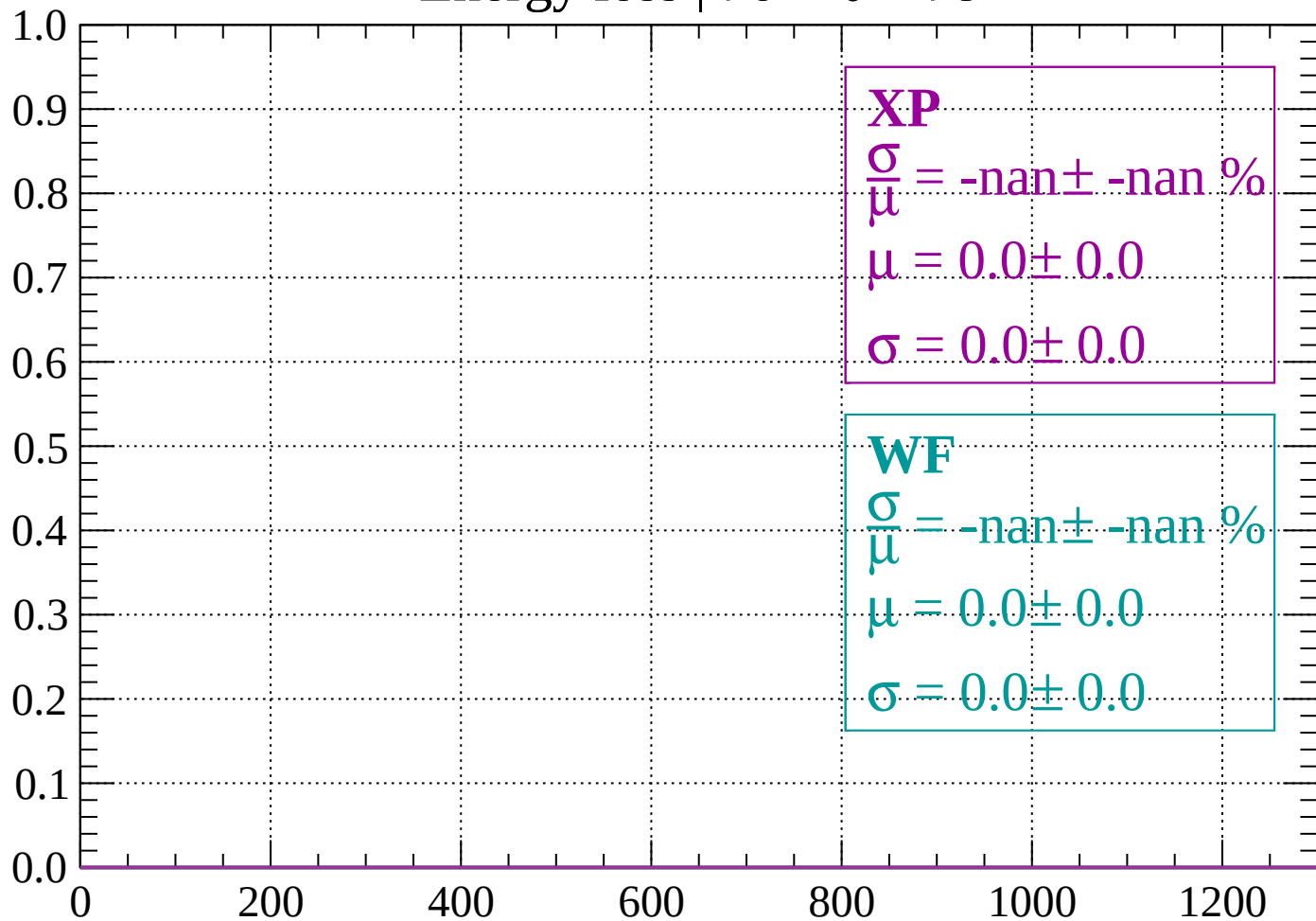
Count



dE/dx (ADC counts/cm)

Energy loss | $76 < \theta < 78$

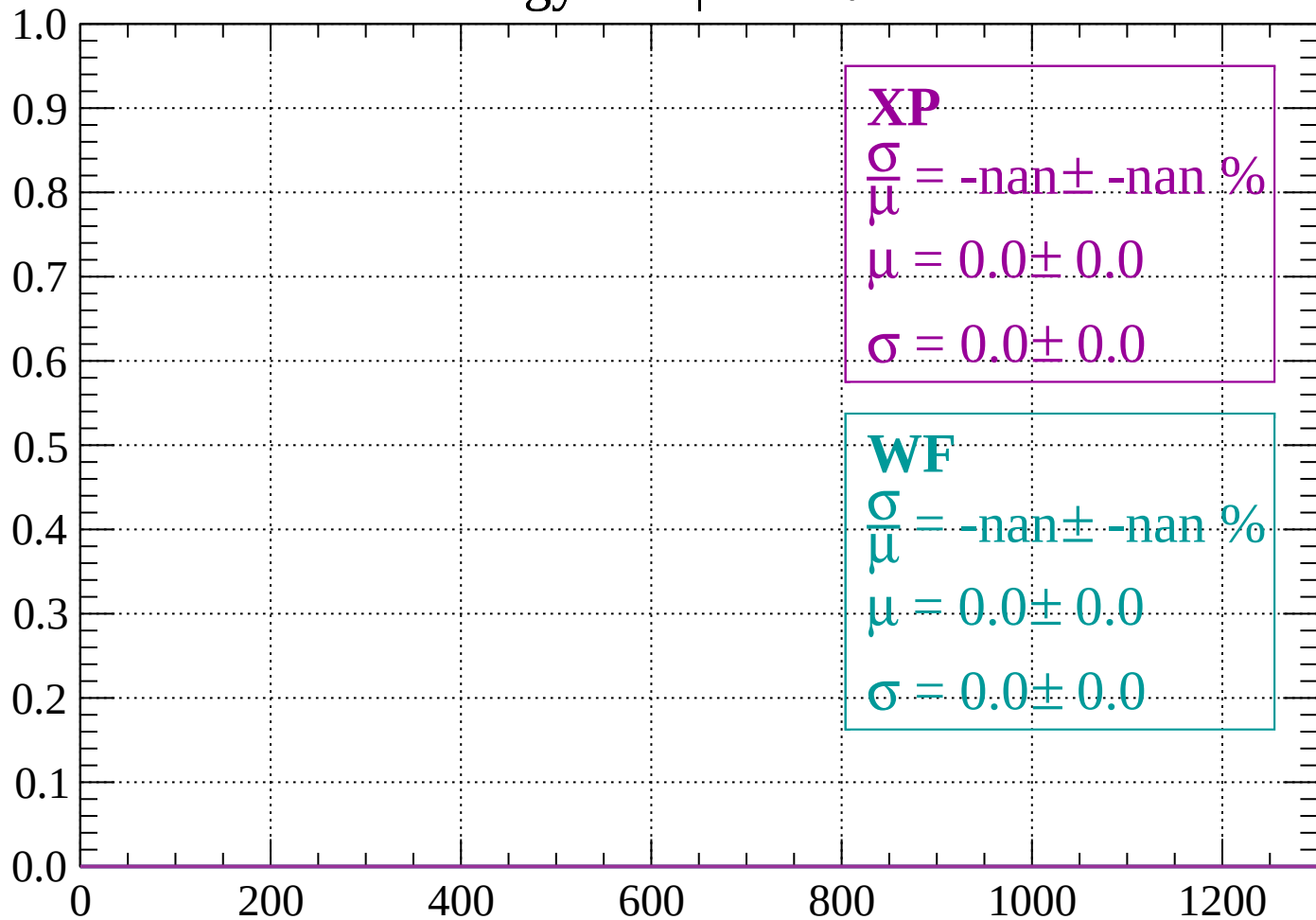
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 80$

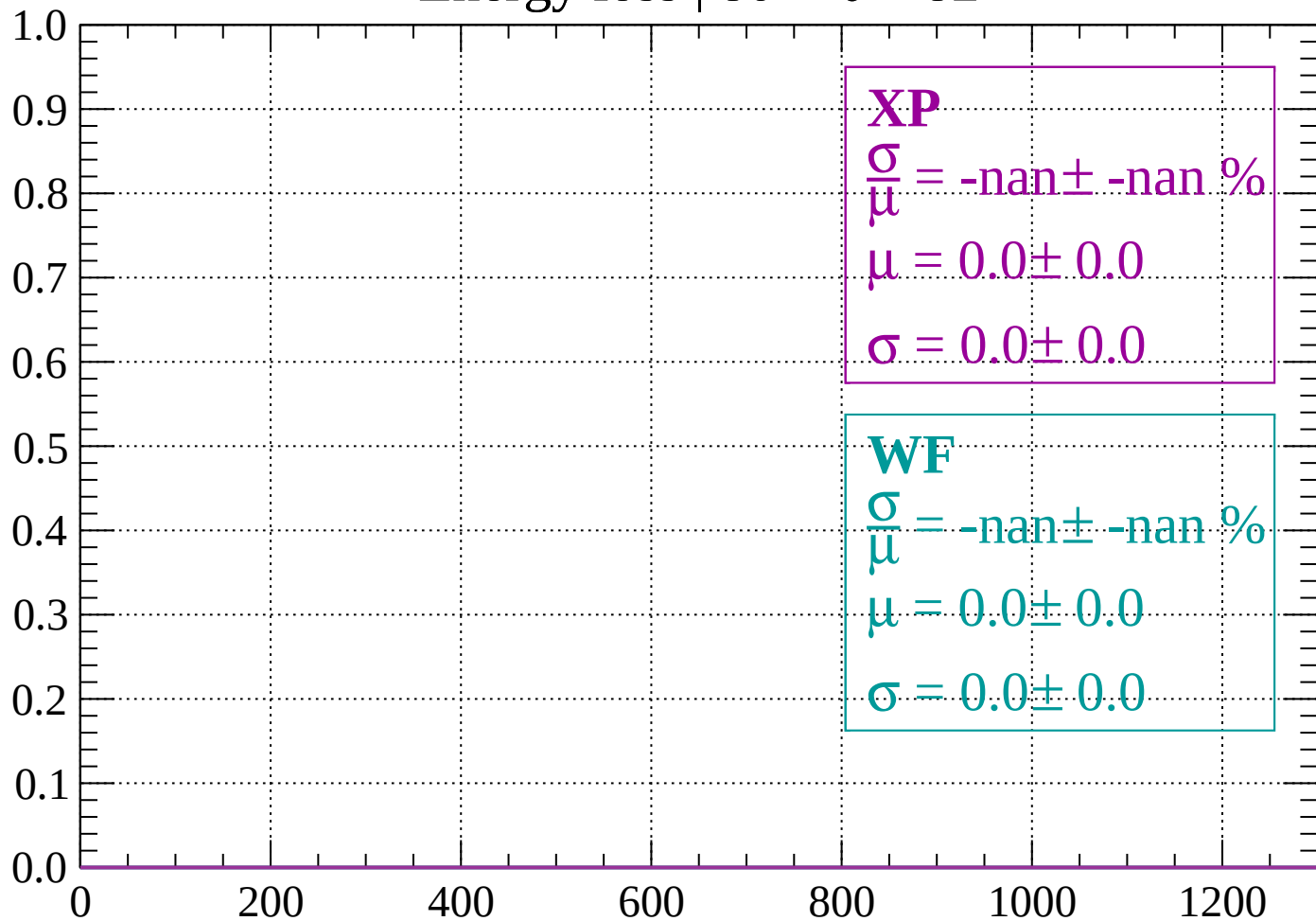
Count



dE/dx (ADC counts/cm)

Energy loss | $80 < \theta < 82$

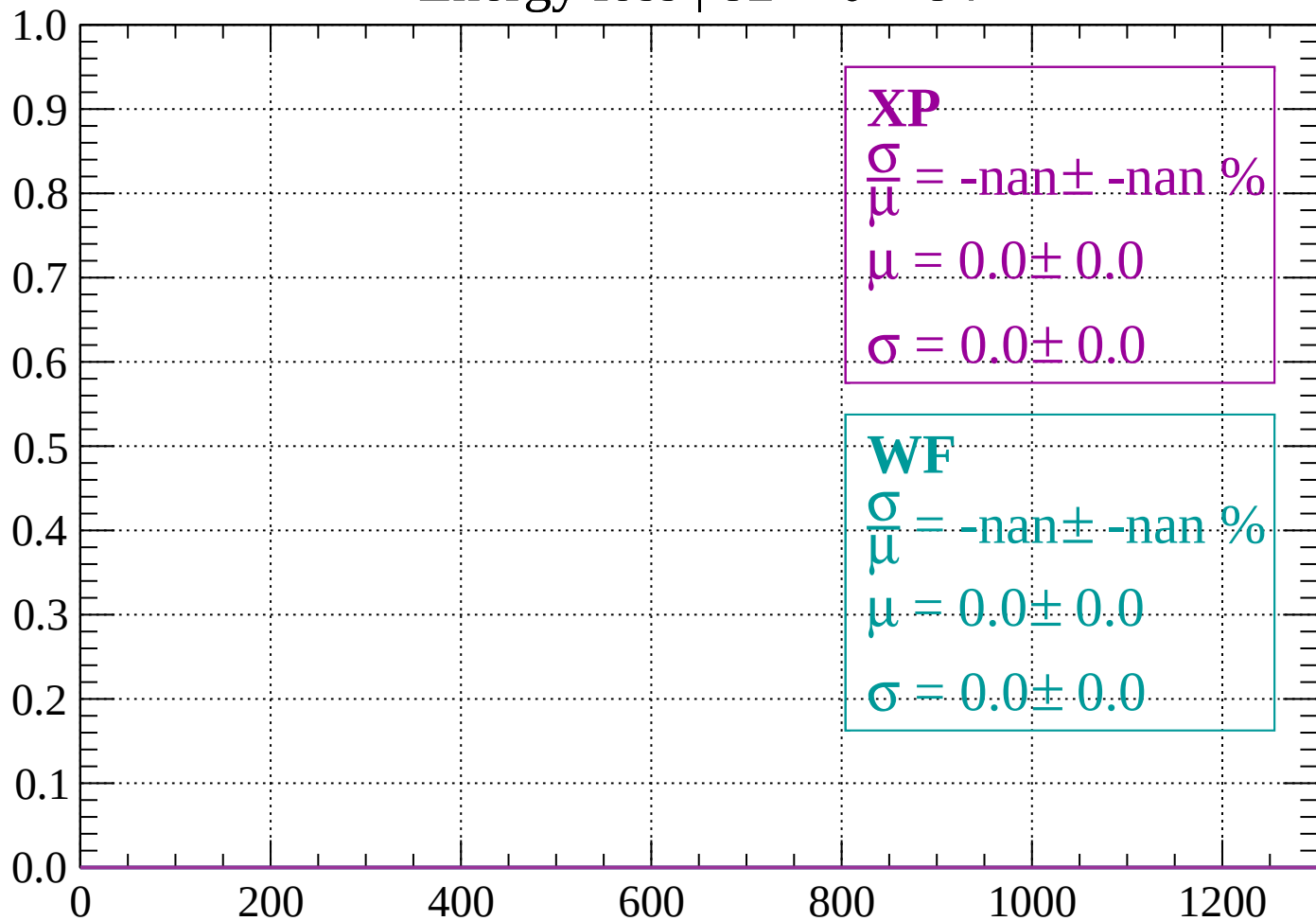
Count



dE/dx (ADC counts/cm)

Energy loss | $82 < \theta < 84$

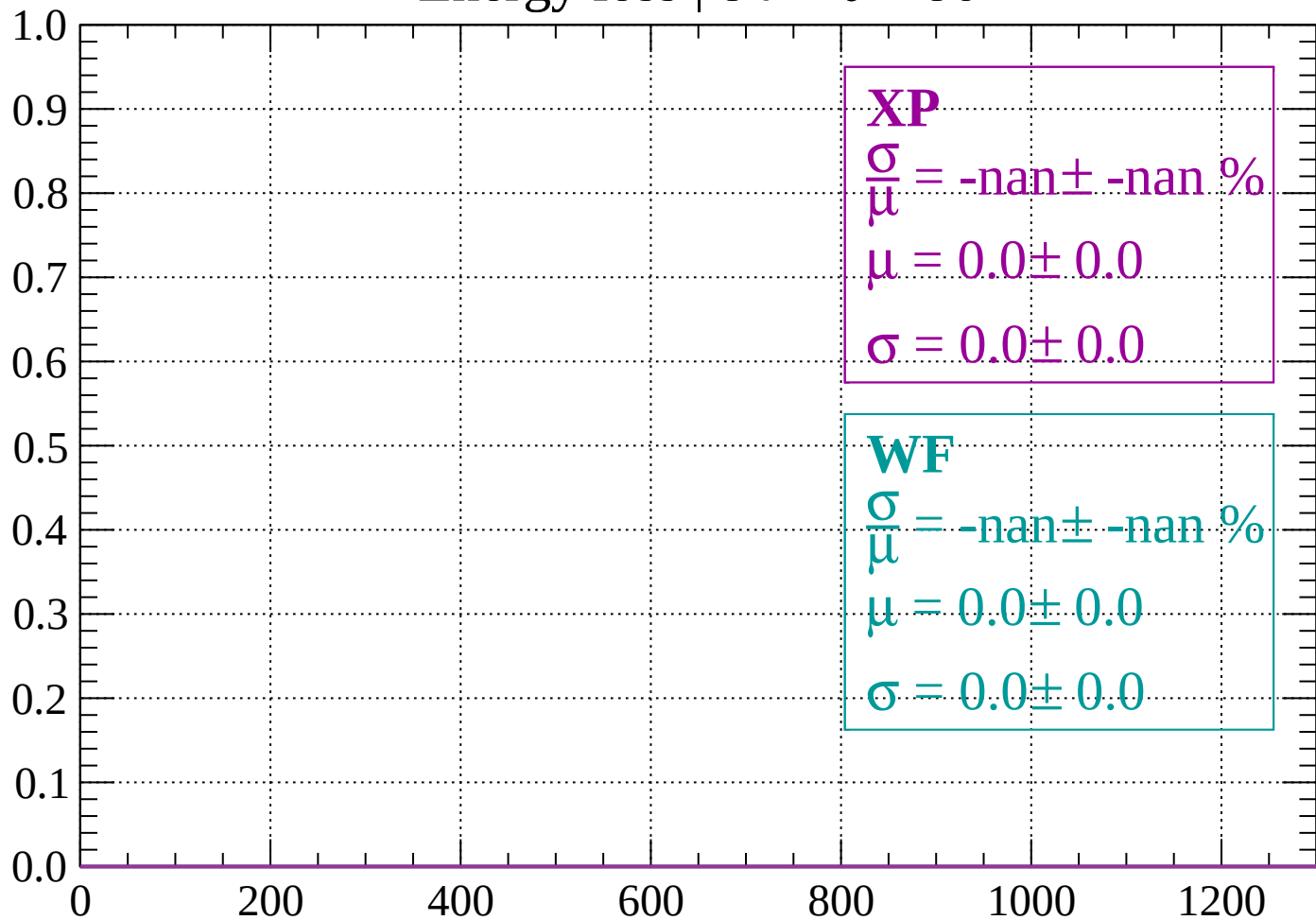
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 86$

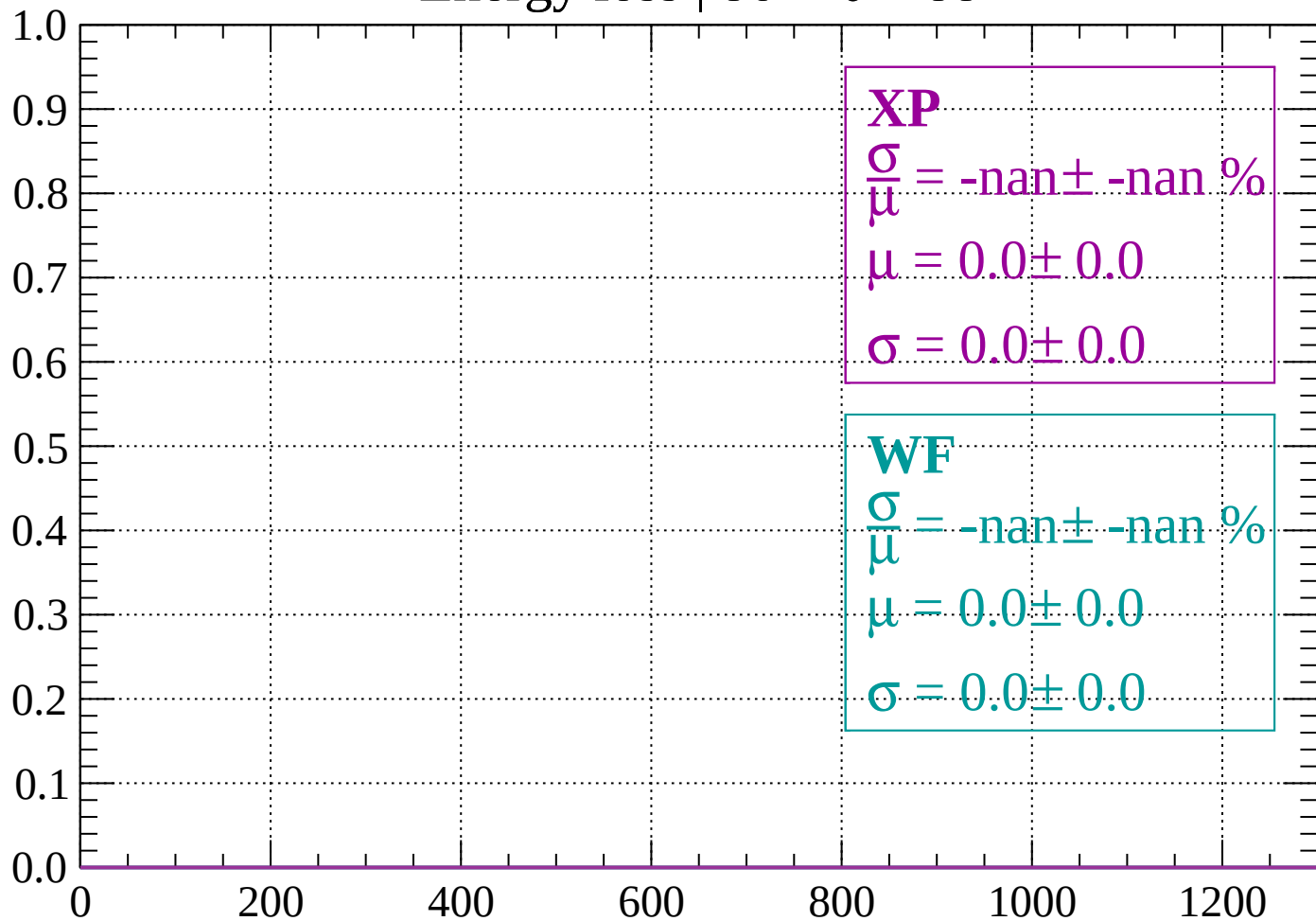
Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

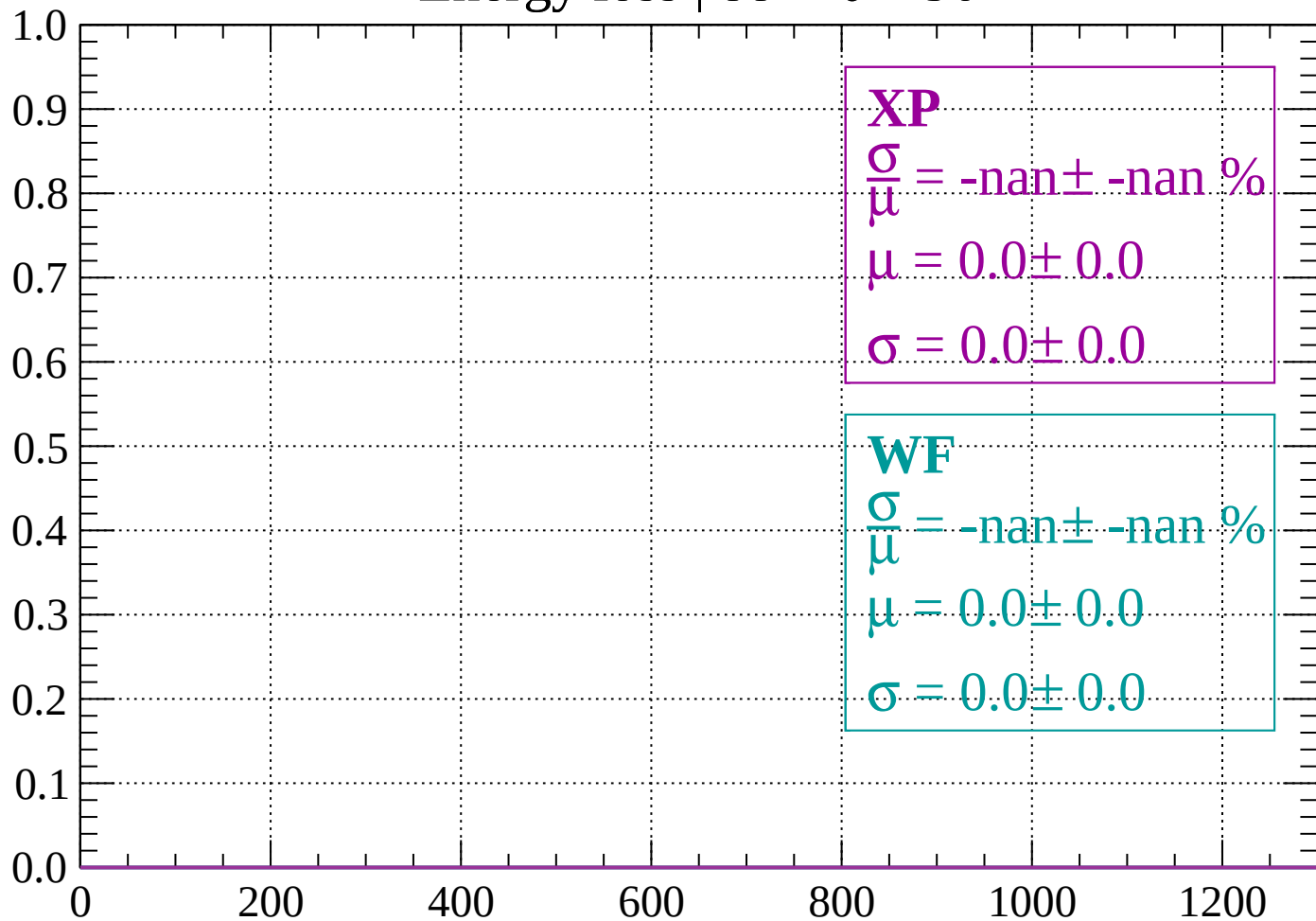
Count



dE/dx (ADC counts/cm)

Energy loss | $88 < \theta < 90$

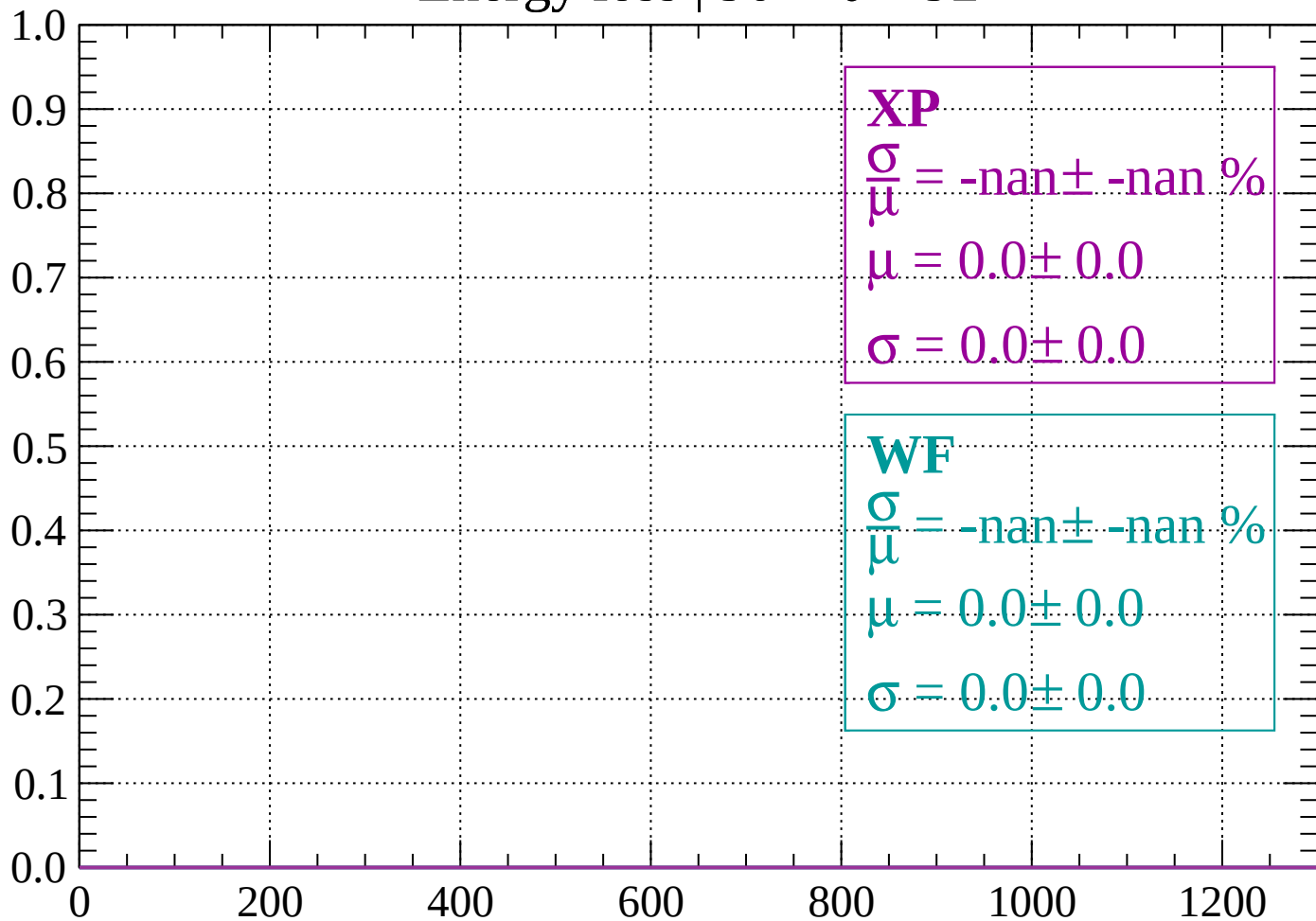
Count



dE/dx (ADC counts/cm)

Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)

